

MODERN FRACTAL GEOMETRY

Volume IV

Scenery, Arithmetic Dynamics, and Rigidity

Fractal distributions, intersections, embeddings,
equidistribution, and Diophantine geometry

Reader conventions

This volume develops scenery methods, arithmetic intersections, embedding rigidity, arithmetic sumsets, equidistribution from fractal measures, and missing-digit Diophantine approximation. Its central theme is that information revealed under repeated magnification can constrain global arithmetic configurations.

Proof conventions

The reader distinguishes three kinds of mathematical assertions. A *local result* has its intended proof owner here, immediately after its statement. A *source theorem* is quoted with its full hypotheses and a precise citation; its proof remains with the cited source, while the deductions made from it here are displayed locally. An *earlier-volume theorem* is stated again at the point of use, but its proof is cross-referenced rather than duplicated. These boundaries are marked explicitly. In this working edition, adjacency of a proof records ownership but is not alone a proof-completeness certificate.

Definitions are introduced where they first become necessary. Informal discussion explains the role of a result, but never replaces a proof or alters its hypotheses. Hausdorff dimension is denoted by $\dim_H$, packing dimension by $\dim_P$, and upper box dimension by $\dim_B$.

Proof scope

Chapter IV.6 is closed by the exact affine-intersection package proved in Volume II, Chapter II.11. The scenery construction, realization, unilateral factor, CP selection, and equidistribution arguments have local proofs. Earlier-volume inputs and the standard analysis and probability background are identified at their points of use. The quoted comparison results in Chapters IV.4, IV.8 and IV.11 are illustrative, not premises in those proofs. An unused fixed-law criterion in Chapter IV.7 is explicitly unproved; its conditional consequence is separate from the proved unilateral factor theorem.

Some literal source statements require special care. A numbered *suspicious source formula* preserves its exact version and hypotheses; an adjacent example checks those hypotheses and the failure of the printed conclusion. Such a record is not a theorem available for use. A qualified replacement, when proved, is stated separately. In particular, a defect in one printed formula is not a refutation of every theorem in the same article.

Earlier-volume inputs

How the main routes fit together. The scenery foundations supply the scaling and realization arguments and the independent CP route to arithmetic intersections. Chapter IV.7 proves the required unilateral Bernoulli factor, including its relative complement construction, before using it for entropy selection. Chapter IV.10 uses a fixed- q pressure estimate and a separate limiting argument; it does not use a Shannon–Frostman endpoint. Illustrative source comparisons and suspicious records have no role as unproved premises in these main routes.

Two theorems proved in Volume II are used in this volume.

- The homogeneous irrational-convolution and affine-intersection theorem used in Chapter IV.6 is proved in Volume II, Chapter II.11, under the title “Affine intersections from irrational homogeneous convolutions.”
- The one-dimensional L^q inverse theorem used in Chapter IV.10 is proved in Volume II, Chapter II.11, under the title “Inverse theorem for near-nonflattening.”

Both statements are reproduced explicitly before use so that their hypotheses can be checked locally. Chapter IV.8 now proves the renormalization obstruction from the local CP, Orponen projection and signed nonconformal magnification chains. The algebraic-contraction companion and other retained comparison results still have explicit source boundaries. Chapter IV.11’s intrinsic counting theorem has a local computer-assisted proof; its extrinsic companion remains source context. The independent scenery route of Chapter IV.7 has local proofs of the future factor, CP typicality, entropy selection and phase closure. Its independence from the L^q route is mathematical, not merely a change in attribution.

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Part I

Scenery and Scaling

Chapter IV.1

CP Distributions and Magnification

Measurable infrastructure, discrete zooms, and CP dimension

This chapter builds the discrete CP language from measurable disintegration through symbolic magnification and the entropy/dimension formula.

1.1 Measurable and dynamical infrastructure

Definition IV.1.1 (*standard Borel space*).

A measurable space is standard Borel if it is measurably isomorphic to the Borel σ -algebra of a Polish space.

Definition IV.1.2 (*probability kernel*).

A probability kernel $\kappa : Y \rightarrow \mathcal{P}(X)$ is a family $(\kappa_y)_{y \in Y}$ such that, for every Borel set $A \subset X$, the map $y \mapsto \kappa_y(A)$ is measurable.

Lemma IV.1.3 (*integration against a kernel*).

If $f : Y \times X \rightarrow [0, \infty]$ is jointly measurable, then

$$y \mapsto \int f(y, x) d\kappa_y(x)$$

is measurable. The same holds for bounded real-valued measurable f .

Proof.

Indicators of measurable rectangles have the property by definition. The class of nonnegative functions with the property is closed under nonnegative linear combinations and monotone limits. The monotone-class theorem gives all nonnegative measurable f . Split a bounded real f into positive and negative parts.

□

Lemma IV.1.4 (*pushforward kernel*).

If $F : Y \times X \rightarrow Z$ is measurable and Z is standard Borel, then

$$(F_*\kappa)_y = (F(y, \cdot))_*\kappa_y$$

is a probability kernel.

Proof.

For a Borel set $C \subset Z$,

$$(F_*\kappa)_y(C) = \int 1_C(F(y, x)) d\kappa_y(x),$$

which is measurable by Lemma IV.1.3. □

Lemma IV.1.5 (*barycenter measure*).

If ν is a probability measure on Y , then

$$\mu(A) = \int \kappa_y(A) d\nu(y)$$

defines a probability measure on X .

Proof.

Countable additivity follows from monotone convergence applied to disjoint unions. □

Definition IV.1.6 (*disintegration along a factor map*).

Let X, Y be standard Borel spaces, let μ be a probability measure on X , let $\pi : X \rightarrow Y$ be Borel, and put $\nu = \pi_*\mu$. A *disintegration of μ over π* is a kernel $y \mapsto \mu_y$ such that

1. $\mu_y(\pi^{-1}\{y\}) = 1$ for ν -almost every y ;
2. $\mu(A) = \int \mu_y(A) d\nu(y)$ for every Borel set $A \subset X$.

Existence is proved constructively in Theorem IV.1.27 below.

Theorem IV.1.7 (*uniqueness a.e.*).

If $y \mapsto \mu_y$ and $y \mapsto \mu'_y$ are two disintegrations of the same μ over π , then there is a ν -null set N such that $\mu_y = \mu'_y$ for every $y \notin N$.

Proof.

Choose a countable π -system $\mathcal{G}$ generating $\mathcal{B}_X$. For $A \in \mathcal{G}$ and every Borel set $C \subset Y$,

$$\int_C \mu_y(A) d\nu(y) = \mu(A \cap \pi^{-1}C) = \int_C \mu'_y(A) d\nu(y).$$

Hence the two measurable functions agree almost everywhere. Take the union of the exceptional sets over the countable family $\mathcal{G}$. Outside it the two probability measures agree on $\mathcal{G}$, and hence on all Borel sets by the π - λ theorem. □

Corollary IV.1.8 (*simultaneous countable versions*).

After modifying a disintegration on one null set, fibre support and every property associated with a prescribed countable family of Borel sets may be imposed simultaneously.

Proof. Take the union of the countably many exceptional null sets and redefine the kernel arbitrarily on that union. $\square$

Caution. The countability is essential: an uncountable intersection of conull sets need not be conull.

Lemma IV.1.9 (*positive-cell selector*).

Let $\{P_1, \dots, P_m\}$ be a finite Borel partition and κ a probability kernel. Define

$$j(y) = \min\{i : \kappa_y(P_i) > 0\}.$$

Then j is measurable.

Proof.

$$\{j = i\} \text{ is } \{\kappa_y(P_i) > 0\} \cap \bigcap_{\ell < i} \{\kappa_y(P_\ell) = 0\}.$$

$\square$

Lemma IV.1.10 (*normalized restriction kernel*).

For a Borel P put $Y_P = \{y : \kappa_y(P) > 0\}$. Then

$$\kappa_y^P(A) = \kappa_y(A \cap P) / \kappa_y(P)$$

defines a kernel on Y_P .

Proof.

Both numerator and denominator are measurable and the denominator is positive.

$\square$

Lemma IV.1.11 (*selected normalized restriction*).

With Lemma IV.1.9, $y \mapsto \kappa_y^{P_{j(y)}}$ is a kernel.

Proof.

Write it as a finite sum over the measurable sets $\{j = i\}$.

$\square$

Thus finite CP-cell selection requires no general measurable-selection theorem.

Definition IV.1.12 (*completion*).

$\mathcal{B}_X^\mu$ denotes the completion of $\mathcal{B}_X$ under μ .

Lemma IV.1.13 (*Borel representative*).

If X and Z are standard Borel spaces, every $\mathcal{B}_X^\mu$ -measurable map $f : X \rightarrow Z$ has a Borel representative equal to f almost everywhere.

Proof.

Take a countable point-separating generator of $B(Z)$, replace each completed preimage by a Borel representative, take one countable union of exceptional null sets, and extend arbitrarily on the complement. □

Lemma IV.1.14 (*kernel modification on a null set*).

Changing κ_y arbitrarily on a ν -null Borel set does not change any barycentre identity against ν .

Proof. The integral of any bounded measurable function over the modified parameter set is zero. □

Definition IV.1.15 (*inverse-limit set*).

For $T : X \rightarrow X$ define

$$\hat{X} = \{(x_0, x_{-1}, \dots) : Tx_{-n} = x_{-n+1} \text{ for } n \geq 1\}$$

and

$$\hat{T}(x_0, x_{-1}, \dots) = (Tx_0, x_0, x_{-1}, \dots).$$

The intended measure is determined on finite cylinders by

$$\begin{aligned} \hat{\mu}\{x_0 \in A_0, \dots, x_{-n} \in A_n\} \\ = \mu(A_n \cap T^{-1}A_{n-1} \cap \dots \cap T^{-n}A_0). \end{aligned} \tag{1.1}$$

Lemma IV.1.16 (*finite-law consistency*).

If $T_*\mu = \mu$, the values in (1.1) form a consistent family of finite-dimensional probability distributions.

Proof.

Deleting the oldest coordinate is immediate. Deleting or shifting the newest one uses $T_*\mu = \mu$. □

The construction and its ergodicity properties are proved in Theorem IV.1.29 below.

Definition IV.1.17 (*constant-roof suspension*).

Let (X, S, μ) be an invertible probability-preserving system and let $L > 0$. On $X \times [0, L)$ with measure $\mu \times \mathcal{L}^1/L$, for $t \in \mathbb{R}$ let $n = n(t, s)$ be the unique integer satisfying $0 \leq s + t - nL < L$, and define

$$\Phi_t(x, s) = (S^n x, s + t - nL). \quad (1.2)$$

Theorem IV.1.18 (*constant-roof suspension*).

$(\Phi_t)_{t \in \mathbb{R}}$ is a measurable measure-preserving flow. It is ergodic if and only if S is ergodic.

Proof.

The integer $n(t, s) = \lfloor (s + t)/L \rfloor$ is measurable. Its elementary cocycle identity gives $\Phi_{t+u} = \Phi_t \Phi_u$. For fixed t , partition $[0, L)$ by the finitely many values of n ; on each piece Φ_t is a product of S^n and vertical translation, so it preserves product measure.

If a measurable set A is invariant under the flow, invariance under all small vertical translations forces $1_A(x, s)$ to be almost everywhere constant in s for μ -almost every x ; write this constant as $g(x)$. Since $\Phi_L(x, s) = (Sx, s)$, the function g is S -invariant, hence constant when S is ergodic. Conversely, an S -invariant set B of intermediate measure makes $B \times [0, L)$ a flow-invariant set of intermediate measure.

□

Corollary IV.1.19 (*suspension factor identity*).

If $\pi : (X, S, \mu) \rightarrow (Y, R, \nu)$ is a factor, then $\Pi(x, s) = (\pi(x), s)$ factors the constant-roof suspension flows and

$$\Pi \Phi_t^S = \Phi_t^R \Pi.$$

Proof.

Both sides use the same $n(t, s)$, and $\pi(S^n x) = R^n \pi(x)$.

□

If scenery time is $t = -\log(\text{scale})$, a b -adic suspension has roof $L = \log b$.

Definition IV.1.20 (*finite generator filtration*).

For a finite partition $\mathcal{A}$ let

$$\mathcal{A}_0^{n-1} = \mathcal{A} \vee T^{-1}\mathcal{A} \vee \dots \vee T^{-(n-1)}\mathcal{A},$$

and denote by $\mathcal{A}_n(x)$ its atom containing x .

Definition IV.1.21 (*factor disintegration*).

Let $\pi : (X, T, \mu) \rightarrow (Y, S, \nu)$ be a factor and let $y \mapsto \mu_y$ be its disintegration.

Lemma IV.1.22 (*uniform zero-relative-entropy estimate*).

Assume the zero-relative-entropy conclusion

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log \mu_{\pi(x)}(\mathcal{A}_n(x)) = 0, \quad (1.3)$$

which is proved in Corollary IV.1.43 below. For every $\delta > 0$ there are G_δ with $\mu(G_\delta) > 1 - \delta$ and n_0 such that, for $x \in G_\delta$ and $n \geq n_0$,

$$\mu_{\pi(x)}(\mathcal{A}_n(x)) \geq \exp(-\delta n). \quad (1.4)$$

Proof.

Apply Egorov to the a.e. convergence in (1.3). □

This is exactly the entropy estimate used in Wu Lemma 6.4(i).

Lemma IV.1.23 (*boundary-null inheritance*).

If $y \mapsto \mu_y$ disintegrates μ and $\mu(B) = 0$, then $\mu_y(B) = 0$ for ν -almost every y . For a countable family of null sets B , one exceptional null set works simultaneously.

Proof.

$0 = \mu(B) = \int \mu_y(B) d\nu$, and the integrand is nonnegative. □

Lemma IV.1.24 (*finite Portmanteau step*).

If $\lambda_k \rightarrow \lambda$ weakly on a compact metric space and $\mathcal{P}$ is a finite Borel partition with $\lambda(\partial P_i) = 0$ for every atom, then

$$\lambda_k(P_i) \rightarrow \lambda(P_i)$$

simultaneously for all i .

Proof.

Each atom is a continuity set of λ ; apply Portmanteau finitely many times. □

Theorem IV.1.25 (*shrinking-cylinder realization – local proof*).

Let

$$\pi : (X, \mu) \rightarrow (\Sigma, \nu)$$

be a measurable factor between compact metric probability spaces, where Σ is a finite-alphabet one-sided shift. Let $C_n(y)$ be the length- n cylinder containing y , and put

$$\mathcal{G}_n = \pi^{-1} \sigma \{C_n(y) : y \in \Sigma\}.$$

Assume $y \mapsto \mu_y$ is the disintegration from Theorem IV.1.27 below. Then, for ν -almost every y ,

$$\frac{\mu|_{\pi^{-1}C_n(y)}}{\mu(\pi^{-1}C_n(y))} \longrightarrow \mu_y \quad (1.5a)$$

weakly as $n \rightarrow \infty$.

Proof.

For $f \in C(X)$, the conditional expectation $E_\mu(f \mid \mathcal{G}_n)(x)$ is constant on the atom $\pi^{-1}C_n(\pi(x))$ and equals

$$\frac{1}{\mu(\pi^{-1}C_n(\pi(x)))} \int_{\pi^{-1}C_n(\pi(x))} f d\mu. \quad (1.5b)$$

The σ -algebras $\mathcal{G}_n$ increase to

$$\mathcal{G}_\infty = \pi^{-1}\mathcal{B}_\Sigma,$$

because cylinders generate the shift Borel σ -algebra. By the martingale convergence theorem,

$$E_\mu(f \mid \mathcal{G}_n) \longrightarrow E_\mu(f \mid \mathcal{G}_\infty) \quad (1.5c)$$

μ -almost everywhere and in L^1 .

The disintegration identity makes

$$E_\mu(f \mid \mathcal{G}_\infty)(x) = \int f d\mu_{\pi(x)}. \quad (1.5d)$$

Choose a countable uniformly dense subset $\{f_j\}$ of $C(X)$. Intersect the conull convergence sets for all j and disintegrate this conull set over π . For ν -a.e. y , (1.5b) converges to $\int f_j d\mu_y$ for every j . Uniform density then gives convergence for every continuous f , hence weak convergence (1.5a).

□

Thus ordinary martingale convergence and the disintegration constructed above give the required limit.

Corollary IV.1.26 (*Wu Lemma 6.4(ii)*).

Fix a finite partition $\mathcal{A}_n$ whose atom boundaries are μ -null, and let $\delta > 0$. Then there are a set $B_\delta^n \subset \Sigma$ with $\nu(B_\delta^n) > 1 - \delta$ and a common sufficiently small cylinder radius r such that, for $y \in B_\delta^n$ and $A \in \mathcal{A}_n$,

$$\mu_{\pi^{-1}B(y,r)}(A) \geq (1 - \delta)\mu_y(A). \quad (1.5)$$

Proof.

Lemma IV.1.23 makes all finitely many atoms μ_y -continuity sets for almost every y . Theorem IV.1.25 and Lemma IV.1.24 give simultaneous convergence. Egorov makes it uniform on a set of measure greater than $1 - \delta$. Zero target masses make (1.5) automatic; positive masses permit a relative one-sided tolerance.

□

We next construct the standard-Borel disintegration used throughout the volume.

Theorem IV.1.27 (*regular conditional probability / disintegration*).

Let X, Y be standard Borel spaces, let μ be a probability on X , let

$$\pi : X \rightarrow Y$$

be Borel, and put $\nu = \pi_*\mu$. Then there is a probability kernel

$$y \mapsto \mu_y$$

such that:

- (i) for every Borel set $A \subset X$, the map $y \mapsto \mu_y(A)$ is Borel;
- (ii)

$$\mu(A) = \int \mu_y(A) d\nu(y); \quad (1.6)$$

- (iii) for ν -almost every y ,

$$\mu_y(\pi^{-1}\{y\}) = 1. \quad (1.7)$$

The kernel is unique ν -almost everywhere.

Proof.

Uniqueness is already contained in Theorem IV.1.7 and Corollary IV.1.8. We prove existence.

Because X is standard Borel, choose a Borel isomorphism

$$\iota : X \rightarrow B$$

onto a Borel subset B of $[0, 1]$. This can be obtained by scaling a standard Borel embedding into $[0, 1]$ by $1/2$. Write $Z = \iota(x)$; in particular, $Z < 1$ everywhere.

For $q \in \mathbb{Q} \cap [0, 1]$, define a finite measure on Y by

$$\lambda_q(C) = \mu(\{Z \leq q\} \cap \pi^{-1}C).$$

It is absolutely continuous with respect to ν . By Radon–Nikodym choose a Borel density

$$F_q = \frac{d\lambda_q}{d\nu}. \quad (1.8)$$

After modifying all countably many F_q on one ν -null set, enforce

$$0 \leq F_q \leq 1,$$

$$F_q \leq F_r \quad \text{for } q < r,$$

$$\lim_{r \downarrow q} F_r = F_q \quad (q \in \mathbb{Q} \cap [0, 1]),$$

$$\lim_{q \uparrow 1} F_q = 1$$

in the rational order. For each rational q , continuity from above of λ_r along a fixed rational sequence $r \downarrow q$ and monotonicity give the right-limit identity by Radon–Nikodym uniqueness. Likewise $Z < 1$ gives the limit at 1 by continuity from below. There are only countably many such identities, so one exceptional null set suffices.

For y outside that null set put

$$F_y(t) = \inf\{F_q(y) : q \in \mathbb{Q}, q > t\}. \quad (1.9)$$

with the usual endpoint conventions. Then F_y is a right-continuous distribution function on $[0, 1]$. Let λ_y be its associated probability measure.

For a half-line $H_q = [0, q]$,

$$y \mapsto \lambda_y(H_q)$$

is a version of

$$E_\mu(1_{\{Z \in H_q\}} \mid \pi)(y).$$

By a monotone-class argument, for every Borel set $D \subset [0, 1]$,

$$y \mapsto \lambda_y(D)$$

is Borel and

$$\mu(Z \in D, \pi \in C) = \int_C \lambda_y(D) d\nu(y) \quad (1.10)$$

for every Borel $C \subset Y$.

Taking $D = B$ gives $\lambda_y(B) = 1$ for ν -almost every y . On the common conull set define

$$\mu_y(A) = \lambda_y(\iota(A))$$

for Borel $A \subset X$; define an arbitrary probability on the exceptional y . Equation (1.10) gives (i)–(ii).

To obtain (iii), choose a countable Borel family $\{C_m\}$ separating points of Y (for example after a Borel embedding of Y into $[0, 1]$). Apply (1.10) to $A = \pi^{-1}C_m$. For ν -a.e. y ,

$$\mu_y(\pi^{-1}C_m) = 1_{C_m}(y) \quad (1.11)$$

simultaneously in m . The intersection of all matching membership conditions is exactly $\pi^{-1}\{y\}$, proving (1.7).

□

Theorem IV.1.28 (*measurable kernels, disintegration, and finite selection*).

Under the standard-Borel hypotheses used in this volume, probability kernels are stable under measurable integration, pushforward, barycenters, null-set modification, and normalized restriction to a selected positive cell. If $\pi : X \rightarrow Y$ is Borel and $\nu = \pi_*\mu$, there is a disintegration $y \mapsto \mu_y$, unique ν -a.e., supported on the fibres of π . All countably many fibrewise versions needed later may be imposed on one common conull set.

Proof. The kernel calculus is given by Lemmas IV.1.3–IV.1.5 and IV.1.9–IV.1.14; uniqueness is Theorem IV.1.7 and Corollary IV.1.8; and existence with fibre support is Theorem IV.1.27. The finite positive-cell selector is supplied by Lemmas IV.1.9–IV.1.11. These results use only countable intersections of exceptional sets, so the conclusion has the simultaneous-version property stated above. $\square$

We next give the natural-extension construction in the form used later.

Theorem IV.1.29 (*natural extension of a standard-Borel endomorphism*).

Let $(X, \mathcal{B}, \mu, T)$ be a probability-preserving Borel endomorphism of a standard Borel space. There is a standard-Borel probability system

$$(\hat{X}, \hat{\mathcal{B}}, \hat{\mu}, \hat{T})$$

and a factor $p_0 : \hat{X} \rightarrow X$ such that:

- (i) $\hat{T}$ is invertible mod zero;
- (ii)

$$p_0 \circ \hat{T} = T \circ p_0; \tag{1.12}$$

- (iii) $\hat{\mathcal{B}}$ is generated by

$$\{p_0 \circ \hat{T}^{-n} : n \geq 0\};$$

- (iv) if T is ergodic, then $\hat{T}$ is ergodic.

Proof. For $n \geq 0$ let

$$\Phi_n(x) = (x, Tx, \dots, T^n x)$$

and let $\rho_n = (\Phi_n)_*\mu$ on X^{n+1} . The family (ρ_n) is projectively consistent: dropping the last coordinate is immediate; dropping the first coordinate gives the same law by T -invariance.

On the cylinder algebra of $X^{\mathbb{Z}_{\leq 0}}$, define the premeasure by these finite-dimensional laws. The ordinary Carathéodory/Kolmogorov extension theorem for countable standard Borel products gives a probability $\tilde{\mu}$. For every adjacent pair,

$$\tilde{\mu}\{x_j = Tx_{j-1}\} = 1.$$

Intersecting countably many such full-measure events gives the inverse-limit space

$$\hat{X} = \{(\dots, x_{-1}, x_0) : Tx_{j-1} = x_j \text{ for all } j \leq 0\} \tag{1.13}$$

with its restricted measure $\hat{\mu}$.

Define

$$\hat{T}(\dots, x_{-1}, x_0) = (\dots, x_{-1}, x_0, Tx_0), \tag{1.14}$$

with the indices shifted one place to the left, so the new -1 coordinate is the old x_0 . Its inverse drops the 0 -coordinate. Consistency of the finite-dimensional laws makes $\hat{T}$ measure preserving. Projection to x_0 is p_0 and gives (1.12). Every finite coordinate block is determined by its earliest coordinate, so the past coordinates generate $\hat{\mathcal{B}}$, proving (iii).

For ergodicity, let $F \in L^2(\hat{\mu})$ satisfy $F \circ \hat{T} = F$. Given $\varepsilon > 0$, density of finite-coordinate functions gives m and a bounded g with

$$\|F - g(x_{-m})\|_2 < \varepsilon.$$

Apply $\hat{T}^m$ and use invariance of F :

$$\|F - g(x_0)\|_2 < \varepsilon. \quad (1.15)$$

Thus F is $\sigma(x_0)$ -measurable, say $F = h(x_0)$. The invariance identity gives

$$h(Tx_0) = h(x_0)$$

$\hat{\mu}$ -a.e., hence μ -a.e. If T is ergodic, h is constant. Therefore $\hat{T}$ is ergodic. $\square$

Corollary IV.1.30 (*natural extension of a factor*).

If

$$\pi : (X, T, \mu) \longrightarrow (Y, S, \nu)$$

is a factor, then the coordinatewise map

$$\hat{\pi}((x_j)_{j \leq 0}) = (\pi(x_j))_{j \leq 0} \quad (1.16)$$

is a factor from the natural extension of X to the natural extension of Y .

Proof.

The defining inverse-limit relation and $\hat{\pi} \circ \hat{T} = \hat{S} \circ \hat{\pi}$ hold coordinatewise. $\square$

Theorem IV.1.31 (*natural extensions and suspensions*).

Every probability-preserving Borel endomorphism of a standard-Borel probability space admits the natural extension of Theorem IV.1.29; factor maps lift coordinatewise as in Corollary IV.1.30. For every invertible probability-preserving system and roof length $L > 0$, the constant-roof suspension of Theorem IV.1.18 is a measurable measure-preserving flow, ergodic exactly when the base is ergodic, and factor maps lift to suspension factors by Corollary IV.1.19.

Proof. Combine Theorem IV.1.29, Corollary IV.1.30, Theorem IV.1.18, and Corollary IV.1.19. The constructions use the same factor map and the same roof cocycle, so they are compatible with the CP-centering applications developed later. $\square$

The relative form needed later is worth isolating before we develop the ergodic-theoretic tools used in its proof.

Relative Shannon–McMillan–Breiman target. Assume that (X, T, μ) is ergodic, that $\mathcal{A}$ is a finite generator, that $h_\mu(T) < \infty$, and that $\pi : (X, T, \mu) \rightarrow (Y, S, \nu)$ is a factor satisfying $h_\nu(S) = h_\mu(T)$. If $(\mu_y)_{y \in Y}$ is the disintegration over π , then for μ -almost every x ,

$$-\frac{1}{n} \log \mu_{\pi(x)}(\mathcal{A}_n(x)) \longrightarrow 0. \quad (1.17)$$

Here the conditional probability is taken with respect to the entire factor fibre, so ordinary Shannon–McMillan–Breiman alone does not imply the claim. Theorem IV.1.42 proves precisely this relative statement.

The following form of Birkhoff's theorem will be used repeatedly.

Lemma IV.1.32 (*maximal ergodic theorem*).

Let T preserve μ , and let $f \in L^1(\mu)$. For $n \geq 1$, put

$$S_n f = \sum_{j=0}^{n-1} f \circ T^j,$$

$$E_n = \left\{ \max_{1 \leq k \leq n} S_k f > 0 \right\}.$$

Then

$$\int_{E_n} f \, d\mu \geq 0. \quad (1.18)$$

Consequently, with

$$E = \left\{ \sup_{k \geq 1} S_k f > 0 \right\},$$

one has

$$\int_E f \, d\mu \geq 0$$

whenever f is integrable.

Proof.

Put

$$h_n = \max\{0, S_1 f, \dots, S_n f\}.$$

If $x \in E_n$, then

$$h_n(x) \leq f(x) + h_n(Tx),$$

and hence $f(x) \geq h_n(x) - h_n(Tx)$. If $x \notin E_n$, then $h_n(x) = 0$ and $0 \geq -h_n(Tx)$. Thus, on all of X ,

$$\mathbf{1}_{E_n} f \geq h_n - h_n \circ T. \quad (1.19)$$

Integration and T -invariance give the finite- n assertion.

As n increases, E_n increases to E . Since $\mathbf{1}_{E_n}f \rightarrow \mathbf{1}_E f$ pointwise and $|\mathbf{1}_{E_n}f| \leq |f|$, dominated convergence gives the limiting assertion.

□

Theorem IV.1.33 (*Birkhoff pointwise ergodic theorem*).

Let T be probability preserving, let $f \in L^1(\mu)$, and let $\mathcal{I}_T$ be the invariant σ -algebra. Then

$$A_n f(x) := \frac{1}{n} \sum_{j=0}^{n-1} f(T^j x) \longrightarrow E(f \mid \mathcal{I}_T)(x) \quad (1.20)$$

for μ -almost every x , and in L^1 when f is bounded.

In particular, if T is ergodic, the limit is

$$\int f d\mu.$$

Proof.

First,

$$\frac{|f(T^n x)|}{n} \longrightarrow 0 \quad (1.21)$$

almost everywhere. Indeed, for $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} \mu\{|f| > \varepsilon n\} < \infty,$$

because the decreasing tail function $t \mapsto \mu\{|f| > t\}$ is integrable. Borel–Cantelli gives (1.21). Consequently the upper and lower Cesàro limits

$$\bar{f} = \limsup_{n \rightarrow \infty} A_n f, \quad \underline{f} = \liminf_{n \rightarrow \infty} A_n f$$

are T -invariant almost everywhere.

Let $g = E(f \mid \mathcal{I}_T)$, which is invariant. For $\varepsilon > 0$, put

$$E_\varepsilon = \{\bar{f} > g + \varepsilon\}.$$

This set is invariant. On E_ε ,

$$\sup_{n \geq 1} S_n((f - g - \varepsilon)\mathbf{1}_{E_\varepsilon}) > 0,$$

whereas outside E_ε all of these partial sums vanish. The maximal ergodic theorem therefore gives

$$\int_{E_\varepsilon} (f - g - \varepsilon) d\mu \geq 0. \quad (1.22)$$

Since $E_\varepsilon \in \mathcal{I}_T$,

$$\int_{E_\varepsilon} (f - g) d\mu = 0.$$

It follows that $-\varepsilon\mu(E_\varepsilon) \geq 0$, and hence $\mu(E_\varepsilon) = 0$. Letting ε range over the positive rationals yields $\bar{f} \leq g$ almost everywhere. Applying the same argument to $\{\bar{f} < g - \varepsilon\}$, with $g - f - \varepsilon$ in place of $f - g - \varepsilon$, yields $\bar{f} \geq g$ almost everywhere. Thus $A_n f \rightarrow g$ almost everywhere. If f is bounded, dominated convergence gives convergence in L^1 . $\square$

Corollary IV.1.34 (*simultaneous genericity*).

On a compact metric probability system, there is one conull set on which Birkhoff convergence holds simultaneously for a fixed countable convergence-determining family of bounded continuous functions.

If the system is ergodic, every point in this conull set is generic for μ .

Proof.

Intersect the countably many Birkhoff conull sets. $\square$

Theorem IV.1.35 (*continuous-time Birkhoff for bounded tests*).

Let $(S_t)_{t \in \mathbb{R}}$ be a jointly measurable probability-preserving flow. For $f \in L^\infty(\mu)$,

$$\frac{1}{T} \int_0^T f(S_t x) dt \longrightarrow E(f \mid I_{\text{flow}})(x) \quad (1.22a)$$

for μ -a.e. x , where I_{flow} is the σ -algebra of sets invariant under every S_t modulo μ .

If the flow is ergodic, the limit is $\int f d\mu$.

Proof. Put

$$g(x) = \int_0^1 f(S_s x) ds.$$

For integer N ,

$$\frac{1}{N} \int_0^N f(S_t x) dt = \frac{1}{N} \sum_{n=0}^{N-1} g(S_n x). \quad (1.22b)$$

Apply the discrete Birkhoff theorem, Theorem IV.1.33, to the time-one map S_1 . The right side converges a.e. to some bounded H .

Because the averages are bounded by $\|f\|_\infty$, convergence also holds in L^1 . For fixed $r > 0$,

$$A_N f(S_r x) - A_N f(x) = \frac{1}{N} \left[\int_N^{N+r} f(S_t x) dt - \int_0^r f(S_t x) dt \right], \quad (1.22c)$$

so

$$\|A_N f \circ S_r - A_N f\|_1 \leq \frac{2r\|f\|_1}{N}.$$

Pass to the L^1 limit to obtain $H \circ S_r = H$ a.e. for every r . The same holds for negative r , hence H is I_{flow} -measurable.

If $A \in I_{\text{flow}}$, invariance gives

$$\int_A A_N f d\mu = \int_A f d\mu$$

for every N . Passing to the bounded L^1 limit gives

$$\int_A H d\mu = \int_A f d\mu,$$

which characterizes $H = E(f \mid I_{\text{flow}})$.

For arbitrary $T \in [N, N+1)$, the difference between the T -average and the N -average is $O(\|f\|_\infty/N)$, so the full real-time limit follows. $\square$

Corollary IV.1.36 (*continuous-flow genericity*).

On a compact metric flow, one conull set works simultaneously for a countable convergence-determining family of bounded continuous tests. If the flow is ergodic, every point in that conull set generates the invariant probability under continuous-time empirical averages.

Proof. Intersect the conull sets from the continuous-time ergodic theorem over a countable dense family of tests. Density then identifies weak convergence of the empirical measures. $\square$

Theorem IV.1.37 (*ergodic decomposition*).

Let T be a probability-preserving Borel map on a standard-Borel probability space $(X, \mathcal{B}, \mu)$. There are a standard-Borel probability space (Z, τ) , a Borel map $\zeta : X \rightarrow Z$ with $\zeta \circ T = \zeta$ μ -a.e., and a probability kernel $z \mapsto \mu_z$ such that

$$\mu = \int \mu_z d\tau(z), \quad (1.23)$$

and for τ -a.e. z :

- (i) μ_z is T -invariant;
- (ii) μ_z is T -ergodic;
- (iii) μ_z is supported on $\zeta^{-1}\{z\}$.

Moreover, for $f \in L^1(\mu)$,

$$E_\mu(f \mid I_T)(x) = \int f d\mu_{\zeta(x)} \quad (1.24)$$

a.e.

Proof. Step 1: make the invariant σ -algebra countably generated mod μ . Because X is standard Borel, choose a countable algebra $\mathcal{C}$ generating $\mathcal{B}$. It is dense in the measure algebra under

$$d(A, B) = \mu(A \triangle B).$$

Hence the whole measure algebra is separable, and so is the subspace corresponding to I_T .

Choose invariant sets $I_1, I_2, \dots$ whose classes are dense in I_T , and let

$$I_0 = \sigma(I_1, I_2, \dots).$$

Every $I \in I_T$ agrees mod μ with an I_0 -set: choose I_{n_k} with

$$\sum_k \mu(I \Delta I_{n_k}) < \infty;$$

Borel–Cantelli makes $1_{I_{n_k}}$ converge a.e. to 1_I , an I_0 -measurable limit.

Step 2: realize I_0 as a standard-Borel factor. Set

$$Z = \{0, 1\}^{\mathbb{N}}, \quad \zeta(x) = (1_{I_m}(x))_m. \quad (1.25)$$

After changing one common null set, $\zeta(Tx) = \zeta(x)$. Put $\tau = \zeta_*\mu$. Apply the locally proved disintegration theorem, Theorem IV.1.27, to ζ :

$$\mu = \int \mu_z d\tau(z),$$

with μ_z supported on the fiber.

Step 3: invariance of the components. Let A range over the countable algebra $\mathcal{C}$. For every Borel $D \subset Z$, using $\zeta \circ T = \zeta$ and invariance of μ ,

$$\begin{aligned} \int_D \mu_z(T^{-1}A) d\tau &= \mu(T^{-1}A \cap \zeta^{-1}D) \\ &= \mu(T^{-1}(A \cap \zeta^{-1}D)) \\ &= \mu(A \cap \zeta^{-1}D) \\ &= \int_D \mu_z(A) d\tau. \end{aligned} \quad (1.26)$$

Therefore $\mu_z(T^{-1}A) = \mu_z(A)$ for τ -a.e. z . Intersect the countably many exceptional sets and extend from $\mathcal{C}$ by the monotone-class theorem. Thus μ_z is T -invariant for almost every z .

Step 4: ergodicity. Take a countable $\mathbb{Q}$ -linear set $\mathcal{F}$ of bounded $\mathcal{C}$ -simple functions dense in $L^1(\rho)$ for every probability ρ on $(X, \mathcal{B})$.

By the Birkhoff theorem, Theorem IV.1.33, and (1.24) for the factor I_0 , for every $f \in \mathcal{F}$ there is a conull set on which

$$A_n f(x) \longrightarrow \int f d\mu_{\zeta(x)}. \quad (1.27)$$

Disintegrate the countable intersection. For τ -a.e. z , for μ_z -a.e. x , (1.27) holds for every $f \in \mathcal{F}$.

Suppose B is T -invariant modulo μ_z . Choose $f_k \in \mathcal{F}$ with $\|f_k - 1_B\|_{L^1(\mu_z)} \rightarrow 0$. Since B is invariant, $A_n 1_B = 1_B$ μ_z -a.e. The contraction property of Cesàro averages in L^1 gives

$$\|A_n f_k - 1_B\|_1 \leq \|f_k - 1_B\|_1. \quad (1.28)$$

For fixed k , (1.27), Fatou and bounded truncation give

$$A_n f_k \longrightarrow \int f_k d\mu_z \quad \text{in } L^1(\mu_z).$$

Let $n \rightarrow \infty$ in (1.28), then $k \rightarrow \infty$:

$$\|1_B - \mu_z(B)\|_1 = 0.$$

A $\{0, 1\}$ -valued function equal a.e. to the constant $\mu_z(B)$ forces $\mu_z(B) \in \{0, 1\}$. Thus μ_z is ergodic.

Finally, since I_T and I_0 agree mod μ , disintegration over ζ is exactly conditional expectation onto I_T , giving (1.24). $\square$

The flow version needed for scenery distributions is supplied next.

Theorem IV.1.38 (*ergodic decomposition for a measurable $\mathbb{R}$ -action*).

Let $(S_t)_{t \in \mathbb{R}}$ be a jointly Borel probability-preserving $\mathbb{R}$ -action on a standard-Borel probability space $(X, \mathcal{B}, \mu)$. Let I_S be the σ -algebra of sets invariant mod μ under every S_t . Then there are a standard-Borel probability space (Z, τ) , an I_S -measurable factor map $\zeta : X \rightarrow Z$, and a probability kernel $z \mapsto \mu_z$ such that

$$\mu = \int \mu_z d\tau(z),$$

and for τ -a.e. z :

- (i) μ_z is invariant under S_t for every $t \in \mathbb{R}$;
- (ii) μ_z is ergodic for the full $\mathbb{R}$ -action;
- (iii) μ_z is supported on $\zeta^{-1}\{z\}$.

Moreover, for every $f \in L^1(\mu)$,

$$E_\mu(f \mid I_S)(x) = \int f d\mu_{\zeta(x)}. \quad (1.28F)$$

Proof. For $T > 0$ write

$$A_T f(x) = \frac{1}{T} \int_0^T f(S_t x) dt.$$

For integer N define

$$g(x) = \int_0^1 f(S_s x) ds.$$

Joint measurability and invariance give $g \in L^1(\mu)$, and

$$A_N f = \frac{1}{N} \sum_{k=0}^{N-1} g \circ S_k.$$

Apply the discrete Birkhoff theorem, Theorem IV.1.33, to the single map S_1 . Thus $A_N f$ converges a.e. to some $h \in L^1$.

Fix $r \in \mathbb{R}$. The difference $A_N f(S_r x) - A_N f(x)$ is the difference of two boundary integrals of total length at most $2|r|$. Put

$$G(x) = \int_0^1 |f(S_s x)| ds.$$

Then $G \in L^1$ and $G(S_n x)/n \rightarrow 0$ a.e.: for every $\varepsilon > 0$,

$$\sum_n \mu\{G > \varepsilon n\} < \infty$$

by the layer-cake estimate, so Borel–Cantelli applies. Splitting the boundary pieces into finitely many unit intervals gives

$$A_N f(S_r x) - A_N f(x) \rightarrow 0$$

for a.e. x . Hence $h \circ S_r = h$ a.e. for every fixed r , so h is I_S -measurable.

If $E \in I_S$, invariance gives

$$\int_E A_N f d\mu = \int_E f d\mu.$$

Passing to the limit shows $\int_E h d\mu = \int_E f d\mu$, hence $h = E(f | I_S)$. The same boundary estimate replaces an arbitrary T by $\lfloor T \rfloor$, so $A_T f$ has the same a.e. limit as $T \rightarrow \infty$.

Step 2: disintegrate over the full flow-invariant σ -algebra. As in the proof of Theorem IV.1.37, separability of the measure algebra gives invariant sets $I_1, I_2, \dots$ whose classes generate $I_S \bmod \mu$. Put

$$\zeta(x) = (1_{I_j}(x))_j \in \{0, 1\}^{\mathbb{N}}$$

and disintegrate μ over ζ using Theorem IV.1.27:

$$\mu = \int \mu_z d\tau(z).$$

For every fixed t and every A in a fixed countable generating algebra $\mathcal{C}$,

$$\begin{aligned} \int_D \mu_z(S_t^{-1} A) d\tau(z) &= \mu(S_t^{-1} A \cap \zeta^{-1} D) \\ &= \mu(A \cap \zeta^{-1} D) \\ &= \int_D \mu_z(A) d\tau(z) \end{aligned}$$

for every Borel $D \subset Z$. Thus for each fixed t , $(S_t)_* \mu_z = \mu_z$ for τ -a.e. z .

To remove the uncountable exceptional-set problem, for each z let

$$H_z = \{t \in \mathbb{R} : (S_t)_* \mu_z = \mu_z\}.$$

Using the countable algebra $\mathcal{C}$, the set $\{(z, t) : t \in H_z\}$ is measurable. The preceding fixed- t conclusion and Fubini imply that H_z has full Lebesgue measure for τ -a.e. z . It is

a subgroup of $\mathbb{R}$ by the action law. A full-measure subgroup equals $\mathbb{R}$: for any r , the full-measure sets H_z and $r + H_z$ intersect, so r is a difference of two elements of H_z . Hence μ_z is simultaneously invariant under every S_t .

Step 3: ergodicity of the components. Choose a countable $\mathbb{Q}$ -linear family $\mathcal{F}$ of bounded simple functions dense in $L^1(\rho)$ for every probability ρ on $(X, \mathcal{B})$. By Step 1 and (1.28F), for every $f \in \mathcal{F}$ there is a μ -conull set on which

$$A_T f(x) \longrightarrow \int f d\mu_{\zeta(x)}.$$

Disintegrate the countable intersection. For τ -a.e. z this convergence holds for μ_z -a.e. x and every $f \in \mathcal{F}$. Let B be invariant mod μ_z under the full flow. Choose $f_k \in \mathcal{F}$ with $\|f_k - 1_B\|_{L^1(\mu_z)} \rightarrow 0$. By invariance and Fubini,

$$A_T 1_B = 1_B$$

μ_z -a.e. Time averaging is an L^1 contraction, so

$$\|A_T f_k - 1_B\|_1 \leq \|f_k - 1_B\|_1.$$

Let $T \rightarrow \infty$ and then $k \rightarrow \infty$. The displayed convergence for f_k gives

$$\|1_B - \mu_z(B)\|_1 = 0,$$

and therefore $\mu_z(B)$ is 0 or 1. Thus μ_z is flow-ergodic. The support-on-fiber and conditional-expectation assertions follow exactly as in the proof of Theorem IV.1.37, because ζ generates I_S mod μ . $\square$

No time-one ergodicity assumption is made.

We now prove the exact one-sided relative form used in the intersection argument.

Lemma IV.1.39 (*triangular ergodic lemma*).

Let (X, T, μ) be ergodic. Suppose $f_j \rightarrow f$ μ -a.e. and in L^1 , and

$$F = \sup_j |f_j| + |f| \in L^1.$$

Then

$$\frac{1}{n} \sum_{j=0}^{n-1} [f_j(T^j x) - f(T^j x)] \longrightarrow 0 \quad (1.29)$$

for μ -a.e. x . Consequently

$$\frac{1}{n} \sum_{j=0}^{n-1} f_j(T^j x) \longrightarrow \int f d\mu. \quad (1.30)$$

Proof. For $K \geq 1$ put

$$G_K = \sup_{j \geq K} |f_j - f|.$$

Then $G_K \rightarrow 0$ a.e. and in L^1 by dominated convergence.

Split the sum in (1.29) into $j < K$ and $j \geq K$. For the tail,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{j=K}^{n-1} |f_j - f|(T^j x) \leq \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} G_K(T^j x) = \int G_K d\mu$$

by the Birkhoff theorem, Theorem IV.1.33.

For the first K terms, each fixed integrable random variable $|f_j - f|(T^j x)$ divided by n tends to zero trivially because j is fixed. Thus the finite initial contribution is $o(1)$. Let $K \rightarrow \infty$. Then apply Birkhoff to f for (1.30). $\square$

Definition IV.1.40 (*conditional information and finite-partition entropy rate*).

On a probability space with measure ρ , for a finite partition $\mathcal{A} = \{A_1, \dots, A_r\}$ and a σ -algebra $\mathcal{H}$, write $p_i = E_\rho(1_{A_i} | \mathcal{H})$ and set

$$I_\rho(\mathcal{A} | \mathcal{H}) = - \sum_{i=1}^r 1_{A_i} \log p_i, \quad H_\rho(\mathcal{A} | \mathcal{H}) = \int I_\rho(\mathcal{A} | \mathcal{H}) d\rho.$$

The value on $A_i \cap \{p_i = 0\}$ may be chosen arbitrarily: that set has measure $\int_{\{p_i=0\}} p_i d\rho = 0$. Equivalently, $H_\rho(\mathcal{A} | \mathcal{H}) = \int \sum_i -p_i \log p_i d\rho$, with $0 \log 0 = 0$. We write $H_\rho(\mathcal{A})$ for conditioning on the trivial σ -algebra. If $\mathcal{B}$ is another finite partition, then

$$I_\rho(\mathcal{A} \vee \mathcal{B} | \mathcal{H}) = I_\rho(\mathcal{B} | \mathcal{H}) + I_\rho(\mathcal{A} | \mathcal{H} \vee \mathcal{B}) \quad \rho\text{-a.e.}$$

Indeed, on a $\mathcal{B}$ -atom B with positive conditional probability, the conditional probability of an $\mathcal{A}$ -atom A given $\mathcal{H} \vee \mathcal{B}$ is $E_\rho(1_{A \cap B} | \mathcal{H}) / E_\rho(1_B | \mathcal{H})$. Taking logarithms proves the formula; its integral is the entropy chain rule. Concavity of $-\sum_i p_i \log p_i$ and conditional Jensen give monotonicity under increasing conditioning algebras, and $0 \leq H_\rho(\mathcal{A} | \mathcal{H}) \leq \log r$.

For a measure-preserving map R , put

$$h_\rho(R, \mathcal{A}) = \lim_{n \rightarrow \infty} \frac{1}{n} H_\rho\left(\bigvee_{k=0}^{n-1} R^{-k} \mathcal{A}\right), \quad h_\rho(R) = \sup_{\mathcal{B} \text{ finite}} h_\rho(R, \mathcal{B}).$$

The first limit exists because the block entropies are subadditive. For an invertible R we use $\mathcal{A}_a^b = \bigvee_{k=a}^b R^{-k} \mathcal{A}$ and interpret an empty join as the trivial partition. Thus positive indices denote the future and negative indices the past.

Lemma IV.1.41 (*relative entropy-rate identity for two finite processes*).

Let (Z, R, ρ) be an ergodic invertible probability system. Let $\mathcal{A}, \mathcal{C}$ be finite partitions such that the two-sided R -orbit of $\mathcal{A}$ generates Z , and the two-sided R -orbit of $\mathcal{C}$ generates a factor σ -algebra $\mathcal{G}_\infty$.

Put

$$\mathcal{G}_j = \bigvee_{m=-j}^{\infty} R^{-m} \mathcal{C}, \quad \mathcal{H}_j = \mathcal{G}_j \vee \bigvee_{i=1}^j R^i \mathcal{A}, \quad (1.31)$$

and

$$f_j = I_\rho(\mathcal{A} \mid \mathcal{H}_j).$$

Here $j \geq 0$ and the join over i is empty when $j = 0$. Put

$$\mathcal{H}_\infty = \mathcal{G}_\infty \vee \bigvee_{i=1}^{\infty} R^i \mathcal{A}, \quad f_\infty = I_\rho(\mathcal{A} \mid \mathcal{H}_\infty).$$

Then $f_j \rightarrow f_\infty$ a.e. and in L^1 , and $F = \sup_{j \geq 0} f_j$ has an integrable exponential tail. The conditional entropies $\int f_j d\rho$ decrease to $\int f_\infty d\rho$; the information functions themselves need not be pointwise decreasing. Moreover,

$$\int f_\infty d\rho = h_\rho(R, \mathcal{A} \vee \mathcal{C}) - h_\rho(R, \mathcal{C}). \quad (1.32)$$

Proof. The algebras $\mathcal{H}_j$ increase to $\mathcal{H}_\infty$. For each $A \in \mathcal{A}$, the conditional probabilities $p_j^A = E_\rho(1_A \mid \mathcal{H}_j)$ are bounded martingales and converge a.e. to $p_\infty^A = E_\rho(1_A \mid \mathcal{H}_\infty)$. The limit is positive at ρ -a.e. point of A , since $\rho(A \cap \{p_\infty^A = 0\}) = 0$. Hence $f_j \rightarrow f_\infty$ a.e.

For completeness, fix $t \geq 0$ and, for each A , let

$$E_j^A = \{p_j^A < e^{-t}\} \cap \bigcap_{\ell < j} \{p_\ell^A \geq e^{-t}\}.$$

These disjoint first-hitting events belong to $\mathcal{H}_j$, so

$$\rho\left(A \cap \bigcup_j E_j^A\right) = \sum_j \int_{E_j^A} p_j^A d\rho \leq e^{-t} \sum_j \rho(E_j^A) \leq e^{-t}.$$

Summing over the atoms gives

$$\rho\{\sup_j f_j > t\} \leq (\#\mathcal{A})e^{-t}. \quad (1.33)$$

Thus F is integrable. Since $0 \leq f_\infty \leq F$ a.e., dominated convergence gives L^1 convergence. Monotonicity of conditional entropy under increasing conditioning algebras gives the assertion about the integrals.

For the entropy identity, for $N \geq 1$ write

$$\mathcal{A}_{-N}^N = \bigvee_{k=-N}^N R^{-k} \mathcal{A}, \quad \mathcal{C}_{-N}^N = \bigvee_{k=-N}^N R^{-k} \mathcal{C}.$$

Then

$$H_\rho(\mathcal{A}_{-N}^N \mid \mathcal{C}_{-N}^N) = \sum_{k=-N}^N H_\rho(R^{-k} \mathcal{A} \mid \mathcal{C}_{-N}^N \vee \mathcal{A}_{-N}^{k-1}). \quad (1.34)$$

This is the forward chain rule, revealing the $\mathcal{A}$ -names from time $-N$ to time N . Shift the k th summand to time zero:

$$H_\rho(\mathcal{A} \mid \mathcal{C}_{-N-k}^{N-k} \vee \mathcal{A}_{-N-k}^{-1}). \quad (1.35)$$

Let $e = H_\rho(\mathcal{A} \mid \mathcal{H}_\infty)$ and, for $M \geq 1$, let

$$b_M = H_\rho(\mathcal{A} \mid \mathcal{C}_{-M}^M \vee \mathcal{A}_{-M}^{-1}).$$

For $-N + M \leq k \leq N - M$, the conditioning algebra in (1.35) contains $\mathcal{C}_{-M}^M \vee \mathcal{A}_{-M}^{-1}$ and is contained in $\mathcal{H}_\infty$. Its conditional entropy is therefore between e and b_M . The same lower bound e holds for every summand, and each of the at most $2M$ boundary summands is at most $\log(\#\mathcal{A})$. Consequently, for $N \geq M$,

$$e \leq \frac{H_\rho(\mathcal{A}_{-N}^N \mid \mathcal{C}_{-N}^N)}{2N+1} \leq b_M + \frac{2M \log(\#\mathcal{A})}{2N+1}.$$

The algebras defining b_M increase to $\mathcal{H}_\infty$. Finite-partition martingale convergence gives $b_M \downarrow e$.

Therefore, first $N \rightarrow \infty$ and then $M \rightarrow \infty$,

$$\frac{1}{2N+1} H_\rho(\mathcal{A}_{-N}^N \mid \mathcal{C}_{-N}^N) \longrightarrow \int f_\infty d\rho. \quad (1.36)$$

On the other hand,

$$H_\rho(\mathcal{A}_{-N}^N \mid \mathcal{C}_{-N}^N) = H_\rho((\mathcal{A} \vee \mathcal{C})_{-N}^N) - H_\rho(\mathcal{C}_{-N}^N).$$

Divide by $2N+1$ and use the finite-partition entropy-rate definition. This gives (1.32). $\square$

Theorem IV.1.42 (*conditional SMB for a one-sided factor*).

Let (X, T, μ) be ergodic, $\mathcal{A}$ a finite generator and $h_\mu(T) < \infty$. Let

$$\pi : (X, T, \mu) \rightarrow (Y, S, \nu)$$

be a factor onto a finite-alphabet one-sided shift whose zero-coordinate partition $\mathcal{C}$ is generating. Let $y \mapsto \mu_y$ be the disintegration from Theorem IV.1.27. Then for μ -a.e. x ,

$$-\frac{1}{n} \log \mu_{\pi(x)}(\mathcal{A}_n(x)) \longrightarrow h_\mu(T) - h_\nu(S). \quad (1.37)$$

In particular, if $h_\mu(T) = h_\nu(S)$, then

$$-\frac{1}{n} \log \mu_{\pi(x)}(\mathcal{A}_n(x)) \longrightarrow 0. \quad (1.38)$$

Proof. Pass to the natural extensions $(\widehat{X}, \widehat{T}, \widehat{\mu})$ and $(\widehat{Y}, \widehat{S}, \widehat{\nu})$ using Theorem IV.1.29 and Corollary IV.1.30. Let

$$\widehat{\mathcal{A}} = p_0^{-1} \mathcal{A}.$$

Let $\mathcal{G}_0$ be the σ -algebra generated by the nonnegative factor coordinates $y_0, y_1, \dots$ on $\widehat{Y}$, pulled back to $\widehat{X}$. Conditioning on $\mathcal{G}_0$ is exactly conditioning on the original one-sided factor point $\pi(x_0)$. Thus

$$I_{\widehat{\mu}}(\widehat{\mathcal{A}}_0^{n-1} \mid \mathcal{G}_0)(\widehat{x}) = -\log \mu_{\pi(x_0)}(\mathcal{A}_n(x_0)). \quad (1.39)$$

Use the conditional-information chain rule:

$$I(\hat{\mathcal{A}}_0^{n-1} \mid \mathcal{G}_0) = \sum_{j=0}^{n-1} I(\hat{T}^{-j} \hat{\mathcal{A}} \mid \mathcal{G}_0 \vee \hat{\mathcal{A}}_0^{j-1}). \quad (1.40)$$

By covariance of conditional expectation under the invertible $\hat{T}$, the j th term equals $f_j(\hat{T}^j \hat{x})$, where

$$f_j = I\left(\hat{\mathcal{A}} \mid \hat{T}^j \mathcal{G}_0 \vee \bigvee_{i=1}^j \hat{T}^i \hat{\mathcal{A}}\right). \quad (1.41)$$

Now $\hat{T}^j \mathcal{G}_0$ is the σ -algebra of factor coordinates y_m , $m \geq -j$. Hence the conditioning algebras in (1.41) increase exactly as in Lemma IV.1.41 toward the full two-sided factor together with the full $\mathcal{A}$ -past, namely $\bigvee_{i \geq 1} \hat{T}^i \hat{\mathcal{A}}$.

The finite-partition information convergence in Lemma IV.1.41 gives $f_j \rightarrow f_\infty$ a.e./ L^1 with an integrable maximal envelope. The natural extension is ergodic by Theorem IV.1.29. Apply Lemma IV.1.39:

$$\frac{1}{n} I(\hat{\mathcal{A}}_0^{n-1} \mid \mathcal{G}_0) \longrightarrow \int f_\infty d\hat{\mu}. \quad (1.42)$$

Finally apply the entropy-rate identity in Lemma IV.1.41 with $\mathcal{A} = \hat{\mathcal{A}}$ and $\mathcal{C}$ the zero-coordinate partition of the two-sided factor. Because $\mathcal{A}$ is a generator of the original endomorphism, the two-sided orbit of $\hat{\mathcal{A}}$ generates the natural extension. We justify the generator entropy assertion explicitly. If $\mathcal{B}$ is any finite partition of the natural extension, then

$$H_{\hat{\mu}}(\mathcal{B} \mid \hat{\mathcal{A}}_{-r}^r) \longrightarrow 0 \quad (r \rightarrow \infty)$$

by finite-partition martingale convergence. The chain rule and conditioning monotonicity give, for fixed r ,

$$H_{\hat{\mu}}(\mathcal{B}_0^{n-1}) \leq H_{\hat{\mu}}(\hat{\mathcal{A}}_{-r}^{n-1+r}) + nH_{\hat{\mu}}(\mathcal{B} \mid \hat{\mathcal{A}}_{-r}^r).$$

Divide by n , let $n \rightarrow \infty$, and then let $r \rightarrow \infty$. This proves that every finite-partition entropy rate is at most the rate of $\hat{\mathcal{A}}$. The same argument with the increasing forward blocks $\mathcal{A}_0^r$ proves $h_\mu(T) = h_\mu(T, \mathcal{A})$. The forward block laws of $\hat{\mathcal{A}}$ are those of $\mathcal{A}$, so adding the factor partition does not change this rate:

$$h_{\hat{\mu}}(\hat{T}, \hat{\mathcal{A}} \vee \mathcal{C}) = h_\mu(T).$$

The $\mathcal{C}$ -process is the natural extension of the factor, so

$$h_{\hat{\mu}}(\hat{T}, \mathcal{C}) = h_\nu(S).$$

Thus (1.32) makes the right side of (1.42) equal to $h_\mu(T) - h_\nu(S)$. Pull back through p_0 to obtain (1.37). $\square$

Corollary IV.1.43 (*Wu's zero-relative-entropy estimate*).

Under the hypotheses of Theorem IV.1.42, and assuming in addition that $h_\mu(T) = h_\nu(S)$,

$$-\frac{1}{n} \log \mu_{\pi(x)}(\mathcal{A}_n(x)) \longrightarrow 0$$

for μ -a.e. x . Thus Lemma IV.1.22 may be applied whenever the Bernoulli factor exists.

Proof. In Theorem IV.1.42 the limit equals $h_\mu(T) - h_\nu(S)$; under the stated hypotheses this difference is zero. $\square$

Theorem IV.1.44 (*conditional entropy and shrinking fibres*).

Let (X, T, μ) be ergodic, let $\mathcal{A}$ be a finite generator with $h_\mu(T) < \infty$, and let $\pi : (X, T, \mu) \rightarrow (Y, S, \nu)$ be a factor onto a finite-alphabet one-sided shift. Then

$$-\frac{1}{n} \log \mu_{\pi(x)}(\mathcal{A}_n(x)) \longrightarrow h_\mu(T) - h_\nu(S)$$

for μ -a.e. x . In the zero-relative-entropy case this convergence is to 0, uniformly on arbitrarily large measurable sets after an Egorov reduction. If the atoms of a fixed $\mathcal{A}_n$ have μ -null boundaries, shrinking factor cylinders realize the conditional measures simultaneously on those atoms, with the one-sided relative estimate used in the ergodic selection theorem, Theorem IV.7.49.

Proof. The conditional Shannon–McMillan–Breiman statement is Theorem IV.1.42, with the zero-relative-entropy specialization Corollary IV.1.43 and Egorov consequence Lemma IV.1.22. Boundary-null inheritance is Lemma IV.1.23, finite Portmanteau is Lemma IV.1.24, shrinking-cylinder realization is Theorem IV.1.25, and the uniform finite-atom consequence is Corollary IV.1.26. $\square$

1.2 Discrete CP construction and the tree dictionary

1. Grid Conventions And Pointed Magnification

Fix $b \geq 2$ and an integer $d \geq 1$. Put $B_1 = [-1, 1]^d$.

For $d = 1$, partition $[-1, 1]$ into b congruent intervals of length $2/b$, left closed and right open except that the final interval contains the endpoint 1. For $d = 2$, use products of these intervals. Denote this finite partition by D_b . The convention makes $D_b(x)$ unique for every $x \in B_1$.

If D is a cell, let T_D be the orientation-preserving affine homothety from D onto B_1 . For a probability μ on B_1 and $x \in B_1$ with $\mu(D_b(x)) > 0$, set

$$M_b(\mu, x) = (\mu^{D_b(x)}, T_{D_b(x)}x), \tag{3.1}$$

where

$$\mu^D = \frac{(T_D)_*(\mu|_D)}{\mu(D)}.$$

Definition IV.1.45 (*positive-mass state*).

A pair (μ, x) is admissible if $x \in B_1$ and $\mu(D_b(x)) > 0$.

Definition IV.1.46 (*adapted law*).

A probability Q on $\mathcal{P}(B_1) \times B_1$ is adapted if

$$Q(d\mu, dx) = Q_1(d\mu)\mu(dx), \quad (3.2)$$

equivalently, for every bounded Borel f ,

$$\int f(\mu, x) dQ(\mu, x) = \int \left[\int f(\mu, x) d\mu(x) \right] dQ_1(\mu).$$

Definition IV.1.47 (*restricted CP distribution*).

A base- b CP distribution is an adapted M_b -invariant probability Q .

Definition IV.1.48 (*intensity*).

The intensity of Q is

$$[Q] = \int \mu dQ_1(\mu). \quad (3.3)$$

For adapted Q this is also the second-coordinate marginal.

Definition IV.1.49 (*extended CP distribution*).

Let $\mathcal{M}_{\text{fr}}$ be the Borel space of nonzero Radon measures ν on $\mathbb{R}^d$ with $\nu(B_1) = 1$; the origin is *not* required to lie in $\text{supp } \nu$. For a framed state (ν, x) with $\nu(D_b(x)) > 0$, put $D = D_b(x)$ and define the full-space pointed magnification

$$M_b^*(\nu, x) = \left(\frac{(T_D)_* \nu}{\nu(D)}, T_D x \right) \quad \text{on } \mathcal{M}_{\text{fr}} \times B_1. \quad (3.3a)$$

An extended CP distribution is a probability Q^e on $\mathcal{M}_{\text{fr}} \times B_1$, invariant under this map, whose push-forward by

$$(\nu, x) \longmapsto (\nu^\square, x)$$

is a restricted CP distribution. No uniqueness is asserted here. The canonical two-sided extension used later is constructed in Theorem IV.2.54.

Lemma IV.1.50 (*Borel measurability and zero-mass states*).

The admissible-state set is Borel and M_b is Borel there. If Q is adapted, then Q gives zero mass to non-admissible states.

Proof. For a fixed Borel cell D , $\mu \mapsto \mu(D)$ is Borel on $\mathcal{P}(B_1)$. Restriction, normalization by the positive scalar $\mu(D)$, and affine push-forward are Borel. There are finitely many cells, so the piecewise definition (3.1) is Borel. For adapted Q ,

$$\begin{aligned} Q\{(\mu, x) : \mu(D_b(x)) = 0\} &= \int \sum_{D \in D_b} 1_{\{\mu(D)=0\}} \mu(D) dQ_1(\mu) \\ &= 0. \end{aligned}$$

□

Lemma IV.1.51 (*continuity set*).

M_b is continuous at every admissible (μ, x) such that (i) x is not on an internal grid boundary, and (ii) $\mu(\partial D_b(x)) = 0$.

Proof. For all nearby x_n the selected cell is the same D . Weak convergence $\mu_n \rightarrow \mu$ and $\mu(\partial D) = 0$ imply

$$\mu_n|_D \longrightarrow \mu|_D, \quad \mu_n(D) \longrightarrow \mu(D) > 0$$

weakly. Affine push-forward and normalization are then continuous. The point coordinate $T_D x_n$ also converges. $\square$

2. Adaptedness And Intensity Under One Magnification

Let $F_b : B_1 \rightarrow B_1$ be the piecewise affine map

$$F_b(x) = T_{D_b(x)}x. \quad (3.4)$$

Lemma IV.1.52 (*adaptedness is preserved*).

If Q is adapted, then $(M_b)_*Q$ is adapted.

Proof. Sample $\mu \sim Q_1$ and then $X \sim \mu$. Condition on the selected cell $D = D_b(X)$. The probability of choosing D is $\mu(D)$, and conditional on D , X has law $\mu|_D/\mu(D)$. After T_D , the new point therefore has law μ^D , exactly the new measure component. This is (3.2) for the image law. $\square$

Lemma IV.1.53 (*intensity evolution*).

If Q is adapted, then

$$[(M_b)_*Q] = (F_b)_*[Q]. \quad (3.5)$$

Hence the intensity of every CP distribution is F_b -invariant. It need not be Lebesgue measure.

Proof. The second coordinate of $M_b(\mu, X)$ is $F_b(X)$. Under adaptedness, X has marginal $[Q]$. Apply the definition of push-forward. $\square$

3. Compact Symbolic Mass-Tree Model

Let $A = \{0, \dots, b-1\}^d$. A mass tree is a family

$$p = (p_v)_{v \in A^*}, \quad p_v \in \Delta(A), \quad (3.6)$$

where $\Delta(A)$ is the probability simplex. The compact product

$$\mathcal{T} = \Delta(A)^{A^*}$$

is metrizable. Every p defines a probability m_p on $A^{\mathbb{N}}$ by

$$m_p([a_1 \dots a_n]) = \prod_{j=1}^n p_{a_1 \dots a_{j-1}}(a_j). \quad (3.7)$$

For $a \in A$, let p^a be the subtree rooted at a ,

$$(p^a)_v = p_{av}.$$

On $\mathcal{T} \times A^{\mathbb{N}}$ define

$$Z(p, \omega) = (p^{\omega_1}, \sigma\omega). \quad (3.8)$$

Lemma IV.1.54 (*continuity of the symbolic kernel*).

Z is continuous. The kernel $p \mapsto m_p$ is weakly continuous.

Proof. Every finite coordinate of p^a is a coordinate projection of p , and σ is continuous. Cylinder probabilities (3.7) are finite products of coordinates of p , hence continuous; cylinders determine weak convergence on $A^{\mathbb{N}}$. $\square$

Definition IV.1.55 (*symbolically adapted law*).

A probability $\hat{Q}$ on $\mathcal{T} \times A^{\mathbb{N}}$ is symbolically adapted if

$$\hat{Q}(dp, d\omega) = \hat{Q}_{\mathcal{T}}(dp) m_p(d\omega). \quad (3.9)$$

Lemma IV.1.56 (*symbolic adaptedness is weakly closed*).

If $\hat{Q}_n$ are symbolically adapted and $\hat{Q}_n \Rightarrow \hat{Q}$, then $\hat{Q}$ is symbolically adapted.

Proof. For continuous f on $\mathcal{T} \times A^{\mathbb{N}}$ define

$$\mathcal{K}f(p) = \int f(p, \omega) dm_p(\omega).$$

By Lemma IV.1.54, $\mathcal{K}f$ is continuous. Thus

$$\int f d\hat{Q}_n = \int \mathcal{K}f d(\hat{Q}_n)_{\mathcal{T}}$$

passes to the weak limit, giving (3.9). $\square$

Lemma IV.1.57 (*symbolic invariant adapted accumulation law*).

Fix a mass tree p and sample $\Omega \sim m_p$. For $N \geq 1$ let

$$\hat{Q}_N = \frac{1}{N} \sum_{n=0}^{N-1} \text{Law}(Z^n(p, \Omega)). \quad (3.10)$$

Every weak accumulation point $\hat{Q}$ of $\hat{Q}_N$ is symbolically adapted and Z -invariant.

Proof. For each n , conditional on the current subtree p_n , the tail $\sigma^n \Omega$ has law m_{p_n} ; hence each term in (3.10) is adapted, and so is $\hat{Q}_N$. Lemma IV.1.56 gives adaptedness of the limit.

For every bounded continuous f ,

$$\int f d(Z_* \hat{Q}_N) - \int f d\hat{Q}_N = \frac{\mathbb{E}f(Z^N(p, \Omega)) - \mathbb{E}f(p, \Omega)}{N} \longrightarrow 0.$$

Continuity of Z allows passage to the limit. $\square$

4. Geometric/Symbolic Dictionary And Endpoint Compatibility

Definition IV.1.58 (*probability-measure quotient and positive paths*).

Write $\Sigma = A^{\mathbb{N}}$ and let

$$\Psi : \mathcal{T} \longrightarrow \mathcal{P}(\Sigma), \quad \Psi(p) = m_p.$$

We identify $p \sim q$ precisely when $m_p = m_q$. The quotient forgets transition vectors below zero-mass prefixes, not the positive-prefix conditional probabilities. Put

$$\Omega_\infty = \{(m, \omega) \in \mathcal{P}(\Sigma) \times \Sigma : m([\omega|n]) > 0 \text{ for every } n \geq 1\}.$$

For $m([\omega|1]) > 0$, define the conditional tail measure

$$m^a = \frac{\sigma_*(m|_{[\omega|1]})}{m([\omega|1])}, \quad \bar{Z}(m, \omega) = (m^{\omega_1}, \sigma\omega) \quad \text{when } m([\omega|1]) > 0. \quad (3.10a)$$

A law $\bar{Q}$ on $\mathcal{P}(\Sigma) \times \Sigma$ is adapted if $\bar{Q}(dm, d\omega) = \bar{Q}_1(dm)m(d\omega)$; it is stationary if $\bar{Z}_*\bar{Q} = \bar{Q}$ on this conull domain. No quotient shift is asserted when the selected child has zero mass: there the raw shift can reveal arbitrary transitions forgotten by Ψ .

Lemma IV.1.59 (*quotient, section, and conditional shift*).

The quotient $\mathcal{T}/\sim$ is homeomorphic to $\mathcal{P}(\Sigma)$. There is a specified Borel section $K : \mathcal{P}(\Sigma) \rightarrow \mathcal{T}$ of Ψ . Every adapted law is concentrated on Ω_∞ ; this set is Borel and $\bar{Z}$ -invariant. The map $\bar{Z}$ is continuous on its positive-child domain. With $Y(p, \omega) = (m_p, \omega)$, the identities

$$YZ = \bar{Z}Y, \quad Z(K(m), \omega) = (K(m^{\omega_1}), \sigma\omega) \quad (3.10b)$$

hold whenever $m_p([\omega|1]) > 0$, respectively $m([\omega|1]) > 0$. Thus every stationary adapted quotient law has a stationary adapted raw lift $\hat{Q} = (K, \text{id})_*\bar{Q}$. This specified lift is measurably conjugate to the quotient law and is ergodic exactly when that law is ergodic.

Proof. Fix a probability vector $u \in \Delta(A)$, for example the uniform vector, and specify every transition of $K(m)$ by

$$(K(m))_v(a) = \begin{cases} m([va])/m([v]), & m([v]) > 0, \\ u(a), & m([v]) = 0, \end{cases} \quad (3.10c)$$

where $[\emptyset] = \Sigma$. Cylinder evaluations are continuous, so each coordinate in (3.10c) is Borel. Induction on the prefix length shows $m_{K(m)}([v]) = m([v])$, also when this mass is zero. Hence $\Psi K = \text{id}$ and Ψ is onto. By Lemma IV.1.54, Ψ is continuous. Its fibers define a closed equivalence relation on the compact space $\mathcal{T}$. The induced continuous bijection from the compact quotient to the Hausdorff space $\mathcal{P}(\Sigma)$ is a homeomorphism. If two trees encode the same measure, their transitions at every positive-mass prefix are the ratios in (3.10c); conversely these ratios and the zero-prefix masses determine all cylinder probabilities. This proves the stated scope of the quotient.

For each n , the map $(m, \omega) \mapsto m([\omega|n])$ is continuous, since the prefix is locally constant. Thus Ω_∞ is Borel. For an adapted law the probability of a zero-mass selected length- n cylinder is

$$\int \sum_{v \in A^n: m([v])=0} m([v]) d\bar{Q}_1(m) = 0.$$

The countable intersection over n has full measure. Moreover $m^a([v]) = m([av])/m([a])$, so $\bar{Z}$ preserves Ω_∞ . On each positive-child branch the numerator $\int_{[a]} f(\sigma\omega) dm(\omega)$ is continuous for every $f \in C(\Sigma)$, since $[a]$ is clopen. Division by $m([a]) > 0$ proves the asserted continuity.

The product formula (3.7) gives $m_{pa} = m_p^a$ whenever $m_p([a]) > 0$, proving the first identity in (3.10b). Formula (3.10c) gives $K(m)^a = K(m^a)$: positive descendants have the same ratios, and zero descendants use the same fixed vector u . This proves the second identity. The lift is adapted because, conditional on $K(m)$, its point has law $m = m_{K(m)}$; m can be recovered by Ψ . The second identity proves stationarity. The maps (K, id) and Y are Borel inverses on the lifted state space and intertwine the maps on the conull domain. Invariant sets therefore correspond, which proves the ergodicity assertion. $\square$

Let $\pi : A^\mathbb{N} \rightarrow B_1$ be the standard base- b coding associated with the nested grid cells. The half-open convention determines a canonical address $a(x)$ for every x . The only non-canonical symbolic addresses are those which, in at least one coordinate, eventually use the maximal digit $b - 1$ to describe an internal b -adic boundary point that the half-open convention assigns to the adjacent cell.

The coding π is continuous, the address map $a : B_1 \rightarrow \Sigma$ is Borel, and

$$\pi a(x) = x, \quad \sigma a(x) = a(F_b x). \quad (3.10d)$$

Indeed, $a^{-1}[v]$ is the level- $|v|$ half-open grid cell indexed by v , and deleting the first digit corresponds to affine rescaling of that cell, including the final-cell outer-face convention. Let $\Sigma_{\text{can}} = a(B_1)$; equivalently it is the Borel set $\{\omega : a(\pi\omega) = \omega\}$. On this set $a\pi$ is the identity.

Lemma IV.1.60 (*stationarity kills non-canonical tails*).

Let m be a σ -invariant probability on $A^\mathbb{N}$. Then m -almost every address is canonical, except possibly addresses representing the outer face $x_j = 1$ by the constant maximal digit in coordinate j ; those addresses are canonical under our final-cell convention.

Proof. For a coordinate j , let E_j be the event that the j th digit is eventually always $b - 1$, including the constant maximal sequence representing the outer face. Then $\sigma^{-1}E_j = E_j$, so if $m(E_j) > 0$ the normalized restriction $m|_{E_j}/m(E_j)$ is shift invariant. On E_j ,

$$1_{\{\omega_n^{(j)} \neq b-1\}} \longrightarrow 0.$$

Shift invariance makes the expectation of this indicator independent of n , so that expectation is zero. Thus on E_j the coordinate is $b - 1$ at every time. This codes the outer face $x_j = 1$, which is canonical by convention, not an internal double address. Taking the finite union over coordinates proves the claim. $\square$

Lemma IV.1.61 (*a.e. semiconjugacy*).

Let $\hat{Q}$ be symbolically adapted and Z -invariant. Push p to the geometric measure

$$\mu_p = \pi_* m_p$$

and ω to $x = \pi(\omega)$. Then, outside a $\hat{Q}$ -null set,

$$\Phi \circ Z = M_b \circ \Phi, \quad (3.11)$$

where $\Phi(p, \omega) = (\mu_p, x)$. The map Φ is continuous on the whole compact raw state space; the factor identity is only asserted on the adapted conull domain.

Proof. The point marginal of $\hat{Q}$ is σ -invariant, because the second coordinate of Z is σ . Lemma IV.1.60 therefore gives $\omega \in \Sigma_{\text{can}}$ almost surely. Adaptedness then gives $m_p(\Sigma_{\text{can}}) = 1$ for $\hat{Q}_{\mathcal{T}}$ -a.e. p . This measure-level conclusion is essential: it removes noncanonical boundary mass from the conditional measure as well as from the sampled point.

By Lemma IV.1.59, the sampled path has positive mass at every prefix. For such a p and a canonical ω with first digit c , the canonical coding gives $\mu_p(D_b(\pi\omega)) = m_p([c]) > 0$. The conditional tail measure is m_{p^c} , and its geometric image is precisely $\mu_p^{D_b(\pi\omega)}$. Identity (3.10d) proves equality of the point coordinates as well. These are the two coordinates of (3.11). Taking all forward preimages of this full-measure set gives one forward-invariant conull domain for every iterate. Finally, continuity of $p \mapsto m_p$ and of π gives continuity of Φ . $\square$

Corollary IV.1.62 (*adapted invariance of the geometric law*).

The geometric push-forward $Q = \Phi_* \hat{Q}$ is an adapted M_b -invariant distribution.

Proof. Adaptedness follows from (3.9) and $x = \pi(\omega)$. Invariance follows from (3.11). $\square$

5. Explicit One- And Two-Dimensional Cp Distributions

Proposition IV.1.63 (*Bernoulli grid models*).

Let $(p_a)_{a \in A}$ be any probability vector. There is a unique grid self-similar probability μ_p satisfying

$$\mu_p = \sum_{a \in A} p_a (f_a)_* \mu_p, \quad (3.12)$$

where f_a maps B_1 onto the child cell indexed by a . Then

$$Q_p = \delta_{\mu_p} \times \mu_p \quad (3.13)$$

is a base- b CP distribution. This works for $d = 1$ and $d = 2$. Its intensity is μ_p , showing explicitly that CP intensity need not be Lebesgue.

Proof. The coding construction gives μ_p . Under μ_p the first digit has law p , and conditional on that digit, affine rescaling leaves the tail law unchanged, so the new measure component is again μ_p and the new point has law μ_p . Therefore (3.13) is adapted and invariant. $\square$

6. Tree Dictionary - Exact Scope And Non-Equivalence Examples

Theorem IV.1.64 (*discrete CP construction*).

The base- b pointed magnification is a well-defined Borel map on admissible states; adaptedness and intensity obey Lemmas IV.1.52 and IV.1.53. In dimensions one and two, adapted invariant distributions exist, including all Bernoulli grid models. More generally, every symbolic invariant adapted mass-tree law pushes through the canonical coding to a geometric CP distribution.

Proof. Combine Lemmas IV.1.50–IV.1.61, Corollary IV.1.62, and Proposition IV.1.63. $\square$

Theorem IV.1.65 (*tree/geometric dictionary*).

The raw map $\Phi(p, \omega) = (\pi_* m_p, \pi\omega)$ is a continuous geometric factor for every stationary adapted raw law, on a forward-invariant conull domain. It need not be one-to-one. For every stationary adapted law on the probability-measure quotient $(\Omega_\infty, \overline{Z})$, its canonical-address conull restriction is measurably conjugate to its geometric b -adic CP law. The conjugacy and its inverse are

$$\overline{\Phi}(m, \omega) = (\pi_* m, \pi\omega), \quad \Lambda(\mu, x) = (a_* \mu, a(x)). \quad (3.14)$$

Every geometric CP distribution Q has the specified stationary adapted raw lift

$$\hat{Q} = ((\mu, x) \mapsto (K(a_* \mu), a(x)))_* Q. \quad (3.15)$$

This lift is measurably conjugate to Q , so it preserves ergodicity. In the quotient dictionary, positive-prefix conditional probabilities are geometric cell masses, conditional tail shift is geometric restriction/normalization/rescaling, adaptedness corresponds in both directions, and stationarity corresponds to CP invariance.

The following distinctions are essential.

(1) A bare symbolic tree with an arbitrarily chosen noncanonical boundary address is not literally the same pointed geometric state under a fixed half-open convention. Stationarity/adaptedness is what removes this ambiguity a.e.; one may not ignore it before taking limits.

(2) The measure marginal does not by itself specify the base b . Lebesgue measure and atomic fixed-point measures give CP distributions for several bases.

(3) Invariance does not force Lebesgue intensity. The unequal-weight models (3.13) have non-Lebesgue invariant intensity.

(4) The canonical full-space extension is constructed from two-sided past information, but the resulting geometric factor need not retain that whole past or be invertible. Its existence requires the two-sided stationary natural-extension construction in Theorem IV.2.54. No extension uniqueness is asserted here.

(5) Raw transitions below zero-mass prefixes are invisible to m_p . Only the quotient, or a specified raw section such as (3.15), gives the conjugacy; canonical addresses alone do not remove this ambiguity.

Proof. Lemma IV.1.59 identifies the tree quotient with $\mathcal{P}(\Sigma)$ and supplies the section K . On the Borel set

$$\mathcal{X}_{\text{can}} = \{(m, \omega) \in \Omega_{\infty} : m(\Sigma_{\text{can}}) = 1, \omega \in \Sigma_{\text{can}}\},$$

the maps in (3.14) are inverse: πa is the identity on B_1 and $a\pi$ is the identity on Σ_{can} . In particular, $a_*\pi_*m = m$ there. Both maps are Borel, since integration against the Borel maps a and π is Borel. A geometric state with all selected prefix masses positive is sent by Λ to this set. Conversely, $m([\omega|n]) = (\pi_*m)(D_{b^n}(\pi\omega))$ on $\mathcal{X}_{\text{can}}$, so its geometric state has all those positive masses.

For a positive first digit c , canonical coding identifies $[c]$ with the selected geometric cell and identifies deletion of c with its affine rescaling. Hence

$$\bar{\Phi}\bar{Z} = M_b\bar{\Phi}, \quad \bar{Z}\Lambda = \Lambda M_b$$

on these positive-path domains. The same coding identities show that $\mathcal{X}_{\text{can}}$ is forward invariant. Adaptedness is preserved in either direction because sampling $\omega \sim m$ corresponds exactly to sampling $x = \pi\omega \sim \pi_*m$, and its inverse is sampling $a(x) \sim a_*\mu$.

For a stationary adapted raw law, $Y_*\hat{Q}$ is adapted: conditional sampling of ω given p uses m_p , and conditioning further only on m_p leaves that same law. The first identity in (3.10b) gives stationarity. Its point marginal is shift invariant, so Lemma IV.1.60 and adaptedness put it on $\mathcal{X}_{\text{can}}$. The same point-marginal argument applies directly to every stationary adapted quotient law. Thus the Borel inverse maps on $\mathcal{X}_{\text{can}}$ prove all the stated quotient conjugacies; Lemma IV.1.61 proves the raw factor assertion, including continuity. Conversely, for a geometric CP law Q , Λ_*Q is adapted and hence has full mass on Ω_{∞} by Lemma IV.1.59. The inverse-map identity proves stationarity. Lifting it by K gives (3.15); the last assertion of that lemma proves its conjugacy and the ergodicity equivalence.

Distinctions (1)–(3) follow from the double-address convention and the Lebesgue/atomic and unequal-weight examples already given. For (4), restriction forgets the past used by the natural extension; even an invertible path system can have a noninvertible geometric factor. For (5), take $b = 2$, $d = 1$. Put $p_v = \delta_0$ at the empty and all-zero prefixes, and put $p_v = r$ at every prefix containing 1. Make another tree p' by replacing r with a different vector r' . Then $p_{0v} = p_v$ and $p'_{0v} = p'_v$, while $m_p = m_{p'} = \delta_{000\dots}$. Thus both raw pointed states with point $000\dots$ are fixed and canonical. The law giving mass 1/2 to each is stationary and adapted; its geometric image is the single fixed state $(\delta_{-1}, -1)$. No conull subset of this two-atom law can make Φ injective. This proves why the raw conjugacy formulation must be replaced by the quotient formulation. $\square$

Raw lifts in block constructions. A law on geometric CP states is not a law on raw trees. When a block-splicing construction tests transition coordinates, each input Q^i must first be replaced by the specified raw law $\hat{Q}^i$ of (3.15), and a raw test F must be integrated against $\hat{Q}^i$, not Q^i . Such tests may see invisible coordinates. The section K is Borel, not claimed continuous. Thus geometric weak convergence alone does not assert weak convergence of the specified raw lifts.

For passing a raw construction back to geometry, no inverse raw map is needed: a continuous geometric test G pulls back to the continuous test $G \circ \Phi$ on the compact raw product. Continuous finite-coordinate tests are uniformly dense there, by the Stone–Weierstrass theorem. Consequently raw weak approximation passes through the continuous factor Φ , and ergodicity passes through its a.e. factor identity. These are the required interfaces for the local-test and sampling uses in Lemmas IV.3.33–IV.3.38; those uses must specify their raw lifts and distinguish the raw and geometric laws.

1.3 Ergodic decomposition and dimension

1. Invariant-Set Saturation And Ergodic Components

Let Q be a base- b CP distribution on $\mathcal{P}(B_1) \times B_1$, with magnification M_b . For an M_b -invariant Borel set U define

$$h_U(\mu) = \mu\{x : (\mu, x) \in U\}. \quad (4.1)$$

Lemma IV.1.66 (*CP invariant sets are fiber-saturated*).

For Q -a.e. (μ, x) ,

$$1_U(\mu, x) = h_U(\mu), \quad (4.2)$$

and $h_U(\mu) \in \{0, 1\}$ for Q_1 -a.e. μ .

Proof. For fixed μ sample $X \sim \mu$ and reveal successively the b -adic cells

$$I_n(X) = D_{b^n}(X).$$

Because U is M_b -invariant, after n magnifications membership in U is unchanged. Conditioned on $I_n(X)$, the normalized location has law equal to the current zoomed measure μ_n . Hence the conditional probability of U given the first n cells is

$$h_U(\mu_n). \quad (4.3)$$

Thus $(h_U(\mu_n))$ is a bounded martingale and

$$h_U(\mu_n) \longrightarrow 1_U(\mu, X) \quad (4.4)$$

almost surely.

Now integrate over the adapted stationary law Q . M_b -invariance makes every $h_U(\mu_n)$ have the same distribution. A bounded martingale has nondecreasing second moments; since the second moments here are constant, every martingale increment has L^2 norm zero. Therefore $h_U(\mu_n) = h_U(\mu_0)$ for all n a.s. Taking the limit in (4.4) gives (4.2). Since the left side is $\{0, 1\}$ -valued, so is $h_U(\mu)$ a.s. $\square$

Corollary IV.1.67 (*restriction to an invariant set remains adapted*).

If $Q(U) > 0$, then $Q_U = Q|_U/Q(U)$ is an adapted CP distribution.

Proof. By adaptation,

$$Q(d\mu, dx) = Q_1(d\mu) \mu(dx).$$

By Lemma IV.1.66, $1_U(\mu, x) = h_U(\mu)$ with $h_U \in \{0, 1\}$. Therefore

$$Q|_U(d\mu, dx) = h_U(\mu) Q_1(d\mu) \mu(dx),$$

so after normalization the conditional law of x given μ is still μ . Invariance is inherited from U . $\square$

Theorem IV.1.68 (*ergodic CP components*).

In the ergodic decomposition

$$Q = \int Q_\xi d\tau(\xi) \tag{4.5}$$

with respect to M_b , τ -a.e. Q_ξ is again a CP distribution.

Proof. Choose a countable generating algebra for the invariant σ -algebra. By Lemma IV.1.66, every generating invariant set is, modulo Q , determined by the measure coordinate alone. Hence the component label ξ may be chosen as a measurable function of μ alone.

Conditioning the adapted law $Q_1(d\mu)\mu(dx)$ on ξ therefore changes only the marginal law of μ and leaves the conditional law $x \mid \mu$ equal to μ . Thus Q_ξ is adapted. Standard ergodic decomposition gives M_b -invariance and ergodicity. $\square$

2. One-Step Information And Exact Telescoping

For an admissible state set

$$I(\mu, x) = -\log \mu(D_b(x)). \tag{4.6}$$

Because the selected cell has positive mass, I is finite Q -a.s. Moreover,

$$\int I dQ = \int H(\mu, D_b) dQ_1(\mu) \leq d \log b. \tag{4.7}$$

For $n \geq 1$ let $D_{b^n}(x)$ be the level- n grid cell containing x .

Definition IV.1.69 (*all-level admissibility*).

A state (μ, x) is all-level admissible if $\mu(D_{b^n}(x)) > 0$ for every $n \geq 1$. This is stronger than the first-step admissibility of Definition IV.1.45.

Lemma IV.1.70 (*information telescoping*).

For every all-level admissible (μ, x) and every $n \geq 1$,

$$-\log \mu(D_{bn}(x)) = \sum_{j=0}^{n-1} I(M_b^j(\mu, x)). \quad (4.8)$$

For a fixed n , it suffices to assume $\mu(D_{bn}(x)) > 0$. Every adapted law gives full mass to all-level admissible states.

Proof. Put $D_{b0}(x) = B_1$. If $\mu(D_{bn}(x)) > 0$, every ancestor cell through level n has positive mass. Induction on $j < n$ therefore defines the zoom $(\mu_j, x_j) = M_b^j(\mu, x)$, and the nested grid rescaling gives

$$\mu_j(D_b(x_j)) = \frac{\mu(D_{bj+1}(x))}{\mu(D_{bj}(x))} > 0.$$

Multiplication telescopes to $\mu(D_{bn}(x))$, and taking minus logarithms gives (4.8). For an adapted law, the probability of a zero-mass selected level- n cell is

$$\int \sum_{D \in D_{bn} : \mu(D)=0} \mu(D) dQ_1(\mu) = 0.$$

Taking the countable intersection over n proves the last assertion. In particular every iterate used in the CP information formula is defined on this common conull domain. $\square$

Corollary IV.1.71 (*ergodic b -adic dimension*).

If Q is ergodic, then for Q -a.e. (μ, x) ,

$$\lim_{n \rightarrow \infty} \frac{-\log \mu(D_{bn}(x))}{n \log b} = \delta_Q, \quad (4.9)$$

where

$$\delta_Q = \frac{1}{\log b} \int I dQ = \frac{1}{\log b} \int H(\mu, D_b) dQ_1(\mu). \quad (4.10)$$

Proof. Apply Birkhoff to the integrable function I in (4.8). $\square$

3. Grid Cells Versus Euclidean Balls

The next lemma removes all hidden assumptions about distance from grid boundaries.

Lemma IV.1.72 (*neighbor-mass Borel–Cantelli estimate*).

Fix a probability μ on B_1 and $\varepsilon > 0$. Let $I_n(x) = D_{bn}(x)$. Then for μ -a.e. x , for all sufficiently large n , every level- n grid cell J whose closure meets the closure of $I_n(x)$ satisfies

$$\mu(J) \leq \exp(\varepsilon n) \mu(I_n(x)). \quad (4.11)$$

Proof. Each level- n cell has at most C_d neighboring cells, with C_d depending only on d . Let E_n be the union of cells I for which some neighbor J satisfies

$$\mu(J) > \exp(\varepsilon n) \mu(I).$$

For every such ordered pair,

$$\mu(I) < \exp(-\varepsilon n)\mu(J).$$

Summing over all ordered neighboring pairs gives

$$\mu(E_n) \leq C_d \exp(-\varepsilon n) \sum_J \mu(J) \leq C_d \exp(-\varepsilon n).$$

The series is summable, so Borel–Cantelli proves (4.11). $\square$

Lemma IV.1.73 (*cell exponent equals local dimension*).

If for a probability μ and point x the limit

$$\lim_{n \rightarrow \infty} \frac{-\log \mu(I_n(x))}{n \log b} = \delta \quad (4.12)$$

exists, and x satisfies Lemma IV.1.72 for every rational $\varepsilon > 0$, then

$$\lim_{r \downarrow 0} \frac{\log \mu(B(x, r))}{\log r} = \delta. \quad (4.13)$$

Proof. A level- n cell has diameter $C_1 b^{-n}$, hence

$$I_n(x) \subset B(x, C_1 b^{-n}). \quad (4.14)$$

Conversely, a ball $B(x, C_2 b^{-n})$ meets only $O_d(1)$ level- n cells, all neighbors of $I_n(x)$ for a suitable fixed C_2 . By Lemma IV.1.72,

$$\mu(B(x, C_2 b^{-n})) \leq C_d \exp(\varepsilon n) \mu(I_n(x)) \quad (4.15)$$

eventually. Equations (4.14)–(4.15) squeeze the logarithmic mass exponent at the discrete radii b^{-n} ; the factor $\exp(\varepsilon n)$ changes the exponent by at most $\varepsilon / \log b$. Let $\varepsilon \downarrow 0$.

For an arbitrary r between two consecutive b -adic scales, monotonicity of $r \mapsto \mu(B(x, r))$ and the bounded ratio of the consecutive radii give the same limit. $\square$

4. Ergodic Dimension Formula

Theorem IV.1.74 (*ergodic CP exact dimensionality*).

Let Q be an ergodic base- b CP distribution. Then for Q_1 -a.e. μ , the measure μ is exact dimensional and

$$\dim \mu = \delta_Q, \quad (4.16)$$

where δ_Q is (4.10).

Proof. By adaptation, a Q -a.e. statement about (μ, x) is equivalent to a statement holding for Q_1 -a.e. μ and μ -a.e. x . Corollary IV.1.71 gives the b -adic limit (4.12) with $\delta = \delta_Q$ for Q -a.e. pair. For each fixed μ , Lemma IV.1.72 holds for μ -a.e. x ; Fubini allows us to intersect this with the conull set from Corollary IV.1.71. Lemma IV.1.73 converts the b -adic limit to Euclidean local dimension. The value is constant because Q is ergodic. $\square$

5. Nonergodic Consequences

Let

$$Q = \int Q_\xi d\tau(\xi)$$

be the ergodic decomposition from Theorem IV.1.68 and set

$$\delta(\xi) = \frac{1}{\log b} \int I dQ_\xi. \quad (4.17)$$

Theorem IV.1.75 (*dimension/entropy formula and nonergodic consequences*).

For τ -a.e. ξ , Q_ξ is a CP distribution and $(Q_\xi)_{1\text{-a.e.}}$ measure is exact dimensional of dimension $\delta(\xi)$. For Q_1 -a.e. μ there is a well-defined component label $\xi(\mu)$, and

$$\dim \mu = \delta(\xi(\mu)). \quad (4.18)$$

There need not be one Q -a.s. constant dimension in the nonergodic case. The mean one-step entropy satisfies

$$\frac{1}{\log b} \int I dQ = \int \delta(\xi) d\tau(\xi). \quad (4.19)$$

Proof. Theorem IV.1.68 gives CP ergodic components and Theorem IV.1.74 applies to each. The component label depends only on μ by the proof of Theorem IV.1.68, giving (4.18). Equation (4.19) is Fubini applied to the ergodic decomposition. $\square$

6. Worked Examples

Example IV.1.76 (*Bernoulli CP law*).

Let $A = \{0, \dots, b-1\}^d$ and $p = (p_a)$. For the Bernoulli grid measure μ_p of Proposition IV.1.63,

$$Q_p = \delta_{\mu_p} \times \mu_p$$

is ergodic whenever the digit process is the usual iid Bernoulli shift. Its one-step information mean is Shannon entropy

$$H(p) = - \sum_a p_a \log p_a,$$

so

$$\dim \mu_p = \frac{H(p)}{\log b}. \quad (4.20)$$

For $d = 2$ this ranges from 0 to 2 because $|A| = b^2$.

Example IV.1.77 (*nonergodic mixture*).

Choose Bernoulli laws p, q with $H(p) \neq H(q)$, and $0 < t < 1$. Then

$$Q = tQ_p + (1 - t)Q_q$$

is a nonergodic CP distribution. A Q -typical measure is exact dimensional, but its dimension equals $H(p)/\log b$ on one component and $H(q)/\log b$ on the other. The average entropy dimension is

$$\frac{tH(p) + (1 - t)H(q)}{\log b},$$

which is not a pointwise dimension shared by both components.

Chapter IV.2

Scenery Flow and Tangent Distributions

Fractal distributions, centering, and the CP/scenery bridge

The discrete zoom process is converted into the continuous scenery flow, with quasi-Palm structure, tangent distributions, and the converse CP realization.

2.1 Full-space sceneries and quasi-Palm laws

Let M be the space of nonzero Radon measures on $\mathbb{R}^d$ with the vague topology:

$$\mu_n \rightarrow \mu$$

iff

$$\int f d\mu_n \rightarrow \int f d\mu$$

for every $f \in C_c(\mathbb{R}^d)$.

Definition IV.2.1 (*normalized measures centered at the origin*).

Put

$$M^* = \{\mu \in M : 0 \in \text{supp}(\mu)\}.$$

Throughout this chapter, in agreement with the b -adic frame fixed in Chapter IV.1, $B_r = [-r, r]^d$ and $B(x, r) = x + B_r$. Thus B_1 is the closed unit cube (the closed unit ball for the ℓ^∞ norm), not the Euclidean unit ball. All sceneries and normalizations in this chapter use this same frame. The Besicovitch covering input below is used in its equivalent fixed-norm, axis-parallel-cube form.

For $\mu \in M^*$ define

$$\mu^* = \mu / \mu(B_1).$$

Because $0 \in \text{supp}(\mu)$,

$$0 < \mu(B_1) < \infty,$$

so normalization is well-defined.

Let

$$M_1^* = \{\mu \in M^* : \mu(B_1) = 1\}.$$

Definition IV.2.2 (*restricted normalization*).

For every nonzero Radon measure μ with $0 < \mu(B_1) < \infty$ define

$$\mu^\square$$

$$\mu^\square = \frac{\mu|_{B_1}}{\mu(B_1)},$$

viewed as a probability measure on B_1 . We use two different state spaces:

$$M^\square = \mathcal{P}(B_1), \quad M_0^\square = \{\nu \in M^\square : 0 \in \text{supp}(\nu)\}.$$

The ambient space $M^\square$ is compact metrizable. Its admissible origin-supported subset $M_0^\square$ is Borel but need not be closed: $n^{-1}\delta_0 + (1 - n^{-1})\delta_z \rightarrow \delta_z$ when $z \neq 0$. The restriction $\mu^\square$ of a centered full-space measure belongs to $M_0^\square$; an uncentered CP component need not. In particular, the canonical extended CP measure component lives in the separate framed space $\mathcal{M}_{\text{fr}}$ of Definition IV.1.49, not in M_1^* .

The two normalizations are different: μ^* retains the full support in $\mathbb{R}^d$, whereas $\mu^\square$ forgets everything outside B_1 .

Lemma IV.2.3 (*evaluation and normalization are Borel*).

For every Borel set A , the map

$$\mu \rightarrow \mu(A)$$

is Borel on the Radon-measure space. Consequently

$$\mu \rightarrow \mu^*$$

and

$$\mu \rightarrow \mu^\square$$

are Borel on M^* .

Proof.

For open U , vague lower semicontinuity gives

$$\mu(U) \leq \liminf \mu_n(U),$$

and for compact K , vague upper semicontinuity gives

$$\limsup_{n \rightarrow \infty} \mu_n(K) \leq \mu(K).$$

A monotone-class argument from open sets gives Borel measurability for all Borel A . For $f \in C_c(\mathbb{R}^d)$,

$$\begin{aligned} & \int f d\mu^* \\ &= [\int f d\mu] / \mu(B_1), \end{aligned}$$

a quotient of Borel functions with positive denominator. A countable convergence-determining family of test functions gives the measure-valued Borel claim. The restriction case is analogous, using bounded continuous functions on B_1 extended to bounded Borel functions on $\mathbb{R}^d$. □

Boundary warning. Neither normalization map is globally continuous: weak/vague convergence can move mass onto or off ∂B_1 .

Lemma IV.2.4 (*continuity at boundary-null measures*).

If $\mu_n \rightarrow \mu$ vaguely and

$$\mu(\partial B_1) = 0,$$

then

$$\mu_n(B_1) \rightarrow \mu(B_1)$$

and

$$\mu_n^\square \rightarrow \mu^\square$$

weakly.

Proof.

Portmanteau gives convergence of B_1 -mass at a continuity set. For a continuous f on B_1 , extend it continuously to a compact neighborhood of B_1 and apply the same boundary-null argument to $f 1_{B_1}$. Divide by convergent positive normalizing masses. □

Definition IV.2.5 (*translation*).

For $x \in \mathbb{R}^d$ define

$$T_x\mu(A) = \mu(A + x).$$

Thus translating by x moves the point x of the original measure to the origin.

If $x \in \text{supp}(\mu)$, define the normalized recentering

$$T_x^*\mu = (T_x\mu)^*.$$

Definition IV.2.6 (*scaling*).

For $t \in \mathbb{R}$ define

$$S_t\mu(A) = \mu(e^{-t}A).$$

This zooms spatial scale e^{-t} up to scale one.

Define the normalized full-space scenery flow

$$S_t^*\mu = (S_t\mu)^*.$$

Lemma IV.2.7 (*joint measurability and continuity before normalization*).

The maps

$$(\mu, x) \mapsto T_x\mu, \quad (\mu, t) \mapsto S_t\mu$$

are jointly continuous in the vague topology.

Proof.

For $f \in C_c(\mathbb{R}^d)$,

$$\int f d(T_x\mu) = \int f(y - x) d\mu(y),$$

and

$$\int f d(S_t\mu) = \int f(e^t y) d\mu(y).$$

When $x_n \rightarrow x$ or $t_n \rightarrow t$, the transformed test functions converge uniformly and have support in one common compact set for large n . Combine with vague convergence.

□

Corollary IV.2.8 (*Borel regularity of normalized translation and scaling*).

The normalized maps

$$\begin{aligned} (\mu, x) &\longmapsto T_x^* \mu \quad \text{on } \{(\mu, x) : x \in \text{supp}(\mu)\}, \\ (\mu, t) &\longmapsto S_t^* \mu \quad \text{on } M^* \times \mathbb{R} \end{aligned}$$

are Borel. They are continuous at points where the corresponding translated/scaled measure gives zero mass to ∂B_1 .

Proof. Each normalized map is the quotient of the corresponding continuous unnormalized map by the mass of B_1 . The numerator and denominator are Borel; at boundary-null measures, Portmanteau gives continuity of the denominator and of the restricted integrals.

□

Lemma IV.2.9 (*flow identity*).

For all $s, t \in \mathbb{R}$,

$$S_s^* S_t^* \mu = S_{s+t}^* \mu. (5.1)$$

Proof.

Scalar normalization commutes with S_s and cancels at the final normalization:

$$\begin{aligned} S_s^* \left(\frac{S_t \mu}{S_t \mu(B_1)} \right) \\ = S_{s+t} \mu / S_{s+t} \mu(B_1). \end{aligned}$$

□

Thus S_t^* is a genuine measurable $\mathbb{R}$ -action on M_1^* .

Definition IV.2.10 (*restricted scaling*).

For $\nu \in M_0^\square$ and $t \geq 0$, define

$$\begin{aligned} S_t^\square \nu \\ = [(S_t \nu)|_{B_1}] / (S_t \nu)(B_1) \\ = S_t(\nu|_{B_{e^{-t}}}) \text{ normalized on } B_1. \end{aligned}$$

Since $0 \in \text{supp}(\nu)$,

$$\nu(B_{e^{-t}}) > 0.$$

Lemma IV.2.11 (*restricted semigroup*).

For $\nu \in M_0^\square$ and $s, t \geq 0$,

$$S_s^\square S_t^\square \nu = S_{s+t}^\square \nu.$$

Proof.

Both sides restrict the original ν to $B_{e^{-(s+t)}}$, dilate that ball to B_1 , and normalize by its mass. Intermediate scalar normalization cancels. □

Definition IV.2.12 (*push-down*).

Let

$$\pi_\square : M_1^* \rightarrow M_0^\square,$$

$$\pi_\square(\mu) = \mu^\square.$$

Lemma IV.2.13 (*intertwining*).

For $t \geq 0$,

$$\pi_\square(S_t^* \mu) = S_t^\square(\pi_\square(\mu)). \quad (5.2)$$

Proof.

Both sides are the normalized restriction of $S_t \mu$ to B_1 . □

Corollary IV.2.14 (*scale invariance after restriction*).

If P is S_t^* -invariant for every $t \in \mathbb{R}$, then

$$P^\square = (\pi_\square)_* P$$

is $S_t^\square$ -invariant for every $t \geq 0$.

Proof. Push the full-space invariance forward through the intertwining identity in Lemma IV.2.13. □

The converse requires care: a single restricted measure contains no information outside B_1 . Invariance of a *distribution*, however, determines all larger-window laws from its restricted law.

Lemma IV.2.15 (*uniqueness of an invariant full-space lift*).

If P_1, P_2 are probabilities on M_1^* invariant under every S_t^* and $(\pi_\square)_*P_1 = (\pi_\square)_*P_2$, then $P_1 = P_2$.

Proof. Fix $a > 0$. For a probability θ on B_1 with $\theta(B_{e^{-a}}) > 0$, define a finite measure on B_{ea} by

$$\mathcal{R}_a(\theta)(A) = \frac{\theta(e^{-a}A)}{\theta(B_{e^{-a}})}, \quad A \subset B_{ea} \text{ Borel.} \quad (5.2a)$$

If $\mu \in M_1^*$, put $\theta = \pi_\square(S_{-a}^*\mu)$. Then $\theta(C) = \mu(e^a C)/\mu(B_{ea})$ for $C \subset B_1$, and $\theta(B_{e^{-a}}) = 1/\mu(B_{ea}) > 0$. Consequently

$$\mathcal{R}_a(\theta) = \mu|_{B_{ea}}. \quad (5.2b)$$

By invariance, the law of θ under either P_i is the common restricted law $(\pi_\square)_*P_i$. Formula (5.2b) therefore gives the same law for $\mu|_{B_{ea}}$ under P_1 and P_2 . Taking a through positive integers, the increasing windows exhaust $\mathbb{R}^d$; their restriction maps generate the Borel sigma-algebra of the vague Radon-measure space. Hence $P_1 = P_2$. $\square$

We call a law on $M^\square$ a *restricted FD* only when it is the push-down $P^\square = (\pi_\square)_*P$ of a full-space FD P . This terminology includes the existence of the lift; the lemma says that the lift, when it exists, is unique.

Definition IV.2.16 (*scale-invariant distribution*).

A probability P on M_1^* is scale invariant if

$$(S_t^*)_*P = P$$

for every $t \in \mathbb{R}$.

Definition IV.2.17 (*quasi-Palm distribution*).

A probability P on M_1^* is quasi-Palm if for every Borel $A \subset M_1^*$,

$$P(A) = 1$$

if and only if, for P -almost every μ ,

$$T_x^*\mu \in A \quad \text{for } \mu\text{-almost every } x \in \mathbb{R}^d. \quad (5.3)$$

The inner " μ -a.e." quantifier ranges over the full Radon measure, not merely $x \in B_1$.

Equivalently, the recentering sampling law defined below has the same null sets as P for every bounded neighborhood of the origin. For a *scale-invariant* P , one such neighborhood suffices: scaling it through a countable expanding family recovers the global condition. The scale-invariance hypothesis is essential for that last reduction.

The global form (5.3) is the stable definition used here.

Lemma IV.2.18 (*full-set and null-set formulations*).

Condition (5.3) is equivalent to:

$$P(A) = 0$$

if and only if, for P -almost every μ ,

$$T_x^* \mu \notin A \quad \text{for } \mu\text{-almost every } x. \quad (5.4)$$

Proof.

Apply (5.3) to A^c . □

Lemma IV.2.19 (*quasi-Palm is equivalence of null classes, not a Palm formula*).

Let U be a bounded Borel neighborhood of 0 and assume $0 < \mu(U) < \infty$ for P -almost every μ . Define the recentered sampling measure

$$R_U(P) = \int \frac{1}{\mu(U)} \left(\int_U \delta_{T_x^* \mu} d\mu(x) \right) dP(\mu). \quad (5.5)$$

If P is quasi-Palm, then

$$R_U(P) \sim P,$$

meaning they have the same null sets.

Proof.

Let A be Borel. If $P(A) = 0$, (5.4) gives $T_x^* \mu \notin A$ for P -almost every μ and μ -almost every x , so $R_U(P)(A) = 0$.

Conversely, suppose $R_U(P)(A) = 0$ and put

$$E = \left\{ \eta : \int_U \mathbf{1}_A(T_z^* \eta) d\eta(z) = 0 \right\}.$$

The integrand is Borel by Corollary IV.2.8, so E is Borel, and $P(E) = 1$. The global quasi-Palm property applied to E says that, for P -almost every μ , $T_x^* \mu \in E$ for μ -almost every x . Normalization only multiplies a measure by a positive scalar and $T_z^* T_x^* \mu = T_{x+z}^* \mu$. Consequently

$$\mathbf{1}_A(T_y^* \mu) = 0 \quad \text{for } \mu \times \mu\text{-almost every } (x, y) \text{ with } y - x \in U. \quad (5.5a)$$

Fubini applies to this sigma-finite product measure, for example first on bounded cubes and then on their countable exhaustion. Choose $r > 0$ with $B(0, r) \subset U$. If $G = \{y : T_y^* \mu \in A\}$ had positive μ -mass, some bounded $G \cap B_R$ would have positive finite mass. At almost every such y , $\mu(B(y, r)) > 0$ because y is in the support of μ . Therefore

$$\int_{G \cap B_R} \mu(B(y, r)) d\mu(y) > 0,$$

contradicting (5.5a). Thus $\mu(G) = 0$ for P -almost every μ . The reverse implication in (5.4) now gives $P(A) = 0$. □

Important distinction. Quasi-Palm asserts

$$R_U(P) \sim P,$$

not

$$R_U(P) = P.$$

Equality with an intensity normalization is the stronger classical Palm identity and is not part of the definition of a fractal distribution.

For completeness, if $R_U(P) \sim P$ for every bounded neighborhood U , apply its forward implication to $U = B_m$, $m \in \mathbb{N}$, and intersect the conull events. This proves the global forward implication in (5.4); the converse follows already from one neighborhood. If P is scale invariant, equivalence for one U suffices instead: for each integer $j \geq 0$ apply it to the P -null set S_{-j}^*A , then change variables $\mu = S_{-j}^*\nu$. The identity $T_x^*S_{-j}^*\nu = S_{-j}^*T_{e^jx}^*\nu$ gives avoidance of A for ν -almost every center in e^jU . Those neighborhoods exhaust $\mathbb{R}^d$.

Definition IV.2.20 (*fractal distribution*).

A fractal distribution (FD) is a probability P on M_1^* that is:

scale invariant under S_t^* ,

quasi-Palm.

An ergodic fractal distribution (EFD) is an FD whose scale-flow action is ergodic.

Theorem IV.2.21 (*scenery formalism*).

The preceding definitions and results give a consistent full-space and restricted-space scenery formalism satisfying:

- (i) translation, scaling, restriction and normalization are Borel on their stated domains;
- (ii) S_t^* is a full $\mathbb{R}$ -action while $S_t^\square$ is only intrinsically a forward semigroup on restricted measures;
- (iii) push-down intertwines the two actions for $t \geq 0$;
- (iv) all normalizing masses are positive and finite on the stated domains;
- (v) boundary discontinuities are explicit and no continuity is asserted without boundary-null hypotheses;
- (vi) quasi-Palm is stated with its complete distribution-a.e./measure-a.e. quantifiers and is not confused with stationarity or Palm equality.

Proof.

Items (i)–(vi) are exactly the results and definitions above. □

Example IV.2.22 (*atomic FD*).

Let $\mu = \delta_0$. Then $S_t^* \mu = \mu$ and $T_0^* \mu = \mu$. Since μ -almost every x equals 0,

$$P = \delta_{\delta_0}$$

is scale invariant and quasi-Palm, hence an EFD.

Example IV.2.23 (*scale invariant but not quasi-Palm*).

In dimension one, let η be normalized Lebesgue measure on $(-\infty, 0]$, viewed in full-space normalized form η^* . Because the half-line is invariant under positive dilations,

$$P = \delta_{\eta^*}$$

is fixed by S_t^* .

However, for η -almost every $x < 0$, recentering the half-line at x produces a measure whose support extends to positive coordinates, while η^* itself assigns no mass to $(0, \infty)$. Let

$$A = \{\nu : \nu((0, \infty)) = 0\}.$$

Then $P(A) = 1$, but $T_x^* \eta \notin A$ for η -almost every x .

Thus P is not quasi-Palm.

This proves directly that scale invariance alone does not imply quasi-Palm.

We now prove the quasi-Palm part of Hochman's theorem. The only covering input is the following standard form of the Besicovitch covering lemma [14, Chapter 3, Problem 3*, p. 154]: there is an integer N_d , depending only on d , such that if $E \subset \mathbb{R}^d$ is bounded and $\mathcal{B}$ is a collection of balls with a ball centered at each point of E , then there are N_d subcollections

$$\mathcal{B}_1, \dots, \mathcal{B}_{N_d}$$

of the original family, each consisting of pairwise disjoint balls, and

$$E \subset \bigcup_{B \in \mathcal{B}_1 \cup \dots \cup \mathcal{B}_{N_d}} B. \quad (5.5a)$$

No differentiation theorem for an arbitrary Radon measure is assumed; the maximal and differentiation estimates needed below are proved from this covering lemma.

Lemma IV.2.24 (*Radon-measure weak maximal inequality*).

Let μ be a Radon measure on $\mathbb{R}^d$ and $f \in L^1(\mu)$. Define the centered maximal function

$$M_\mu f(x) = \sup_{r>0, \mu(B(x,r))>0} \frac{1}{\mu(B(x,r))} \int_{B(x,r)} |f| d\mu. \quad (5.5b)$$

Then, for every $\alpha > 0$,

$$\mu\{M_\mu f > \alpha\} \leq \frac{N_d}{\alpha} \|f\|_{L^1(\mu)}. \quad (5.5c)$$

The same inequality holds when the supremum is restricted to $0 < r < \delta$.

Proof.

Let

$$E_\alpha = \{M_\mu f > \alpha\}.$$

Because μ is Radon, it is enough to bound $\mu(K)$ for compact $K \subset E_\alpha$.

For every $x \in K$, choose a ball B_x centered at x with

$$\int_{B_x} |f| d\mu > \alpha \mu(B_x). \quad (5.5d)$$

For the truncated maximal function, choose $r_x < \delta$.

Apply the Besicovitch covering lemma to the bounded set K and the family $\{B_x : x \in K\}$. It gives a covering subfamily split into N_d pairwise disjoint families. Every family of pairwise disjoint positive-radius balls in $\mathbb{R}^d$ is countable, so all sums below are legitimate.

Since the selected balls cover K and each $\mathcal{B}_i$ is disjoint,

$$\begin{aligned} \mu(K) &\leq \sum_i \sum_{B \in \mathcal{B}_i} \mu(B) \\ &< \frac{1}{\alpha} \sum_i \sum_{B \in \mathcal{B}_i} \int_B |f| d\mu \\ &\leq \frac{N_d}{\alpha} \|f\|_1. \end{aligned} \quad (5.5e)$$

Take the supremum over compact $K \subset E_\alpha$ and use inner regularity.

□

Lemma IV.2.25 (*differentiation in $L^1(\mu)$*).

Let μ be a Radon measure on $\mathbb{R}^d$ and

$f \in L^1_{\text{loc}}(\mu)$.

Then, for μ -almost every x ,

$$\lim_{r \downarrow 0} \frac{1}{\mu(B(x, r))} \int_{B(x, r)} |f(y) - f(x)| d\mu(y) = 0. \quad (5.5f)$$

where radii with zero denominator are omitted.

Proof.

Localize to a large compact ball and multiply f by a compact cutoff; it is enough to treat $f \in L^1(\mu)$.

Define

$$Df(x) = \limsup_{r \downarrow 0} \frac{1}{\mu(B(x, r))} \int_{B(x, r)} |f(y) - f(x)| d\mu(y). \quad (5.5g)$$

Continuous compactly supported functions are dense in $L^1(\mu)$: first truncate/localize f using Radon regularity, approximate indicator functions of finite-measure Borel sets from

inside by compact sets and outside by open sets, then use Urysohn functions between them; pass from simple functions to f .

Choose $g \in C_c(\mathbb{R}^d)$ and put $h = f - g$. Uniform continuity of g gives $Dg = 0$ everywhere. Therefore

$$Df(x) \leq \limsup_{r \downarrow 0} \frac{1}{\mu(B(x, r))} \int_{B(x, r)} |h(y)| d\mu(y) + |h(x)| \leq M_\mu h(x) + |h(x)|. \quad (5.5h)$$

For $\alpha > 0$,

$$\{Df > 2\alpha\} \subset \{M_\mu h > \alpha\} \cup \{|h| > \alpha\}.$$

Thus by Lemma IV.2.24 and Markov's inequality,

$$\mu\{Df > 2\alpha\} \leq \frac{N_d + 1}{\alpha} \|h\|_1. \quad (5.5i)$$

Choose g with $\|h\|_1$ arbitrarily small. Hence $\mu\{Df > 2\alpha\} = 0$. Intersect over positive rational α to obtain $Df = 0$ a.e.

□

Theorem IV.2.26 (*Besicovitch density theorem for Radon measures*).

If μ is Radon on $\mathbb{R}^d$ and A is μ -measurable, then for μ -a.e. x ,

$$\frac{\mu(A \cap B(x, r))}{\mu(B(x, r))} \longrightarrow 1_A(x). \quad (5.5j)$$

as $r \downarrow 0$.

Proof.

Apply Lemma IV.2.25 locally to $f = 1_A$. Writing $B = B(x, r)$, the average absolute difference is $1 - \mu(A \cap B)/\mu(B)$ when $x \in A$, and is $\mu(A \cap B)/\mu(B)$ when $x \notin A$.

□

Thus the Radon-measure maximal and differentiation theorems have been derived locally from the stated Besicovitch covering lemma.

Theorem IV.2.27 (*ergodic decomposition of a measurable flow*).

$$P = \int P_\xi d\tau(\xi),$$

with P_ξ scale-invariant and flow-ergodic for τ -a.e. ξ .

Proof. This is the measurable-flow ergodic decomposition theorem, Theorem IV.1.38, applied to the scenery flow.

□

Definition IV.2.28 (*unit-ball recentering kernel*).

For $\mu \in M_1^*$, define the probability kernel
 $K(\mu, A)$

$$K(\mu, A) = \int_{B_1} 1_A(T_x^* \mu) d\mu(x). \quad (5.6)$$

This is a probability kernel because $\mu(B_1) = 1$.

For a finite measure Q on M_1^* define

$$R(Q)(A) = \int K(\mu, A) dQ(\mu). \quad (5.7)$$

For a quasi-Palm P , Lemma IV.2.19 says

$$R(P) \sim P. \quad (5.8)$$

Lemma IV.2.29 (*density identity for an invariant set*).

Use Theorem IV.2.26. Let $U \subset M_1^*$ be Borel and S_t^* -invariant for all t . Define

$$f_U(\mu) = K(\mu, U).$$

Then, for P -almost every μ ,

$$\lim_{t \rightarrow \infty} f_U(S_t^* \mu) = 1_U(\mu). \quad (5.9)$$

Proof.

Fix an arbitrary Radon measure μ and define the spatial set

$$A_\mu = \{x \in \mathbb{R}^d : T_x^* \mu \in U\}.$$

For $x \in \text{supp}(\mu)$, a direct change of variables using the translation and scaling identities gives

$$\begin{aligned} & f_U(S_t^* T_x^* \mu) \\ &= \frac{\mu(A_\mu \cap B(x, e^{-t}))}{\mu(B(x, e^{-t}))}. \end{aligned} \quad (5.10)$$

The only point requiring comment is membership in U : recentering after the zoom differs from recentering the original measure by the common scaling S_t^* , and U is scale invariant, so the indicator is unchanged.

By Theorem IV.2.26, for μ -almost every x , the right side of (5.10) converges to

$$1_{A_\mu}(x)$$

$$= 1_U(T_x^* \mu). (5.11)$$

Let

$$G_U = \left\{ \nu : \lim_{t \rightarrow \infty} f_U(S_t^* \nu) = 1_U(\nu) \right\}.$$

Equation (5.11) says that, for every μ , one has $T_x^* \mu \in G_U$ for μ -almost every x . Since P is quasi-Palm, the full-set direction of Definition IV.2.17 gives

$$P(G_U) = 1.$$

This is (5.9). □

Lemma IV.2.30 (*invariant sets are spatially saturated*).

For every invariant Borel U ,

$$f_U(\mu) = 1_U(\mu) (5.12)$$

for P -almost every μ .

Consequently, for P -almost every $\mu \in U$,

$$T_x^* \mu \in U$$

$$\text{for } \mu - a.e. x \in \mathbb{R}^d, \text{ and for } P - a.e. \mu \text{ outside } U,$$

$$T_x^* \mu \notin U$$

for μ -almost every x .

Proof.

Because P is scale invariant, for every fixed t , the random variables

$$(f_U(S_t^* \mu), 1_U(S_t^* \mu))$$

and

$$(f_U(\mu), 1_U(\mu))$$

have the same distribution. Since U is invariant,

$$1_U(S_t^* \mu) = 1_U(\mu).$$

By Lemma IV.2.29 and dominated convergence,

$$\int |f_U(\mu) - 1_U(\mu)| dP(\mu) = \int |f_U(S_t^* \mu) - 1_U(S_t^* \mu)| dP(\mu)$$

$\rightarrow 0$.

Hence (5.12).

This proves the assertion for $x \in B_1$. To extend it to all of $\mathbb{R}^d$, note that if $f_U(\mu) = 1$, then the same is true for every sufficiently positive zoom $S_t^* \mu$; hence A_μ has full μ -mass in every sufficiently small centered ball. More directly, the set

$$E = \{\mu : f_U(\mu) = 1\}$$

is forward scale invariant: full A_μ -mass in B_1 implies full mass in every smaller centered ball. Since P is invariant under the invertible scale flow, forward invariance implies that E is invariant modulo P . Therefore, for P -almost every $\mu \in E$, every integer backward scale $S_{-n}^* \mu$ also belongs to E . Thus A_μ has full mass in $B(0, e^n)$ for every n , and these balls exhaust $\mathbb{R}^d$. The case $f_U(\mu) = 0$ is identical. □

Lemma IV.2.31 (*restriction to an invariant set remains quasi-Palm*).

Let U be invariant with $P(U) > 0$ and write

$$P_U = P|U/P(U).$$

Then P_U is a fractal distribution.

Proof.

Scale invariance is immediate from invariance of P and U .

By Lemma IV.2.30, for P -almost every $\mu \in U$, the recentering kernel $K(\mu, \cdot)$ is supported on U ; for P -almost every $\mu \notin U$, it is supported on U^c . Therefore

$$R(P)|U = R(P|U), \quad (5.13)$$

and similarly on U^c .

Since $R(P) \sim P$ by quasi-Palm, restriction of the two equivalent measures to U gives

$$R(P|U) \sim P|U. \quad (5.14)$$

After normalization,

$$R(P_U) \sim P_U,$$

which is exactly the quasi-Palm null-class formulation. □

Lemma IV.2.32 (*conditional nonsingularity passes to ergodic components*).

Use Theorem IV.1.38. Suppose a probability P and a Markov recentering operator R satisfy: (a) $R(P) \sim P$; (b) for every invariant Borel U ,

$$R(P|U) = R(P)|U.$$

Then for τ -a.e. ergodic component P_ξ ,

$$R(P_\xi) \sim P_\xi. \quad (5.15)$$

Proof.

Let

$$h = \frac{dR(P)}{dP}.$$

By equivalence, $h > 0$ P -a.e.

Disintegrate P over the invariant σ -algebra:

$$P = \int P_\xi d\tau(\xi).$$

For every invariant U , (b) and the Radon-Nikodym formula give

$$R(P|U) = hP|U. \quad (5.16)$$

Choose a countable π -system G generating the Borel σ -algebra of M_1^* . For $A \in G$, define

$$F_A(\xi) = R(P_\xi)(A),$$

$$G_A(\xi) = \int_A h dP_\xi.$$

Integrating over any invariant event in the component parameter and using (5.16) gives equality of the integrals of F_A and G_A . Hence

$$F_A = G_A$$

for τ -almost every ξ . Intersect the resulting conull sets over the countable class G . For such ξ , the two finite measures $R(P_\xi)$ and hP_ξ agree on G and hence on all Borel sets.

Also $h > 0$ for P_ξ -almost every point and τ -almost every ξ , because $h > 0$ P -almost everywhere. Therefore

$$R(P_\xi) = hP_\xi \sim P_\xi.$$

□

Theorem IV.2.33 (*ergodic decomposition of fractal distributions*).

If P is a fractal distribution, then τ -almost every scale-ergodic component P_ξ is an ergodic fractal distribution.

Proof.

By Theorem IV.1.38, P_ξ is scale invariant and ergodic for τ -almost every ξ . Lemma IV.2.30 gives hypothesis (b) of Lemma IV.2.32, while the global quasi-Palm property gives (a). Hence Lemma IV.2.32 shows $R(P_\xi) \sim P_\xi$, so P_ξ is quasi-Palm and therefore an EFD. □

This proves the stated form of Hochman's Theorem 1.3: Lemma IV.2.24, Lemma IV.2.25, and Theorem IV.2.26 supply differentiation from the Besicovitch covering lemma, while Theorem IV.1.38 supplies the flow decomposition.

Corollary IV.2.34 (*invariant restrictions remain fractal distributions*).

Using Theorem IV.2.26 and Theorem IV.1.38, every invariant positive- P -measure subset U supports the normalized fractal distribution P_U .

This is stronger bookkeeping than merely knowing the final ergodic components: it is the finite/invariant-set approximation used in the source proof.

Proof. Normalize the restriction $P|_U$. Flow invariance is inherited from the invariance of U , and Theorem IV.2.26 transfers the quasi-Palm equivalence to the restriction. □

Corollary IV.2.35 (*quasi-Palm ergodic components*).

Consequently quasi-Palm preservation under ergodic decomposition follows from Lemmas IV.2.29–IV.2.32, together with the differentiation and flow-decomposition results already proved.

Proof. Apply Lemma IV.2.32 to almost every component supplied by the flow decomposition, using Lemmas IV.2.29–IV.2.31 to verify its two hypotheses. □

2.2 Centering CP distributions

1. Canonical extended CP law

Let Q be a restricted base- b CP distribution on $M^\square \times B_1$.

Lemma IV.2.36 (*stationary CP laws avoid internal grid boundaries*).

Let E_{int} be the union, over all levels, of the internal b -adic grid hyperplanes of B_1 . Then

$$[Q](E_{\text{int}}) = 0, \quad \mu(E_{\text{int}}) = 0 \quad \text{for } Q_1\text{-almost every } \mu.$$

Proof. The second marginal $[Q]$ is invariant under the point map F_b of Chapter IV.1 because Q is invariant under pointed magnification. A point whose j -th coordinate is an

internal level- n boundary reaches an outer endpoint in that coordinate after at most n iterates of F_b , and that coordinate remains at the endpoint thereafter. It therefore cannot return arbitrarily often to the set $E_{\text{int},j}$ of points with an internal boundary in coordinate j . Poincaré recurrence gives $[Q](E_{\text{int},j}) = 0$ for every j ; take their finite union. Adaptedness gives $[Q] = \int \mu dQ_1(\mu)$, so nonnegativity proves the second claim. $\square$

Use the natural-extension theorem, Theorem IV.1.29, to realize a two-sided stationary process

$$(\mu_n, x_n)_{n \in \mathbb{Z}}. \quad (6.1)$$

with one-time marginal Q and

$$(\mu_{n+1}, x_{n+1}) = M_b^\square(\mu_n, x_n). \quad (6.2)$$

for every n outside one common null set.

Definition IV.2.37 (*the invertible CP path system*).

Let Ω be the Borel space of two-sided paths in (6.1) satisfying (6.2), equipped with their stationary law $\widehat{Q}$. The shift $\sigma : \Omega \rightarrow \Omega$ is $((\sigma\omega)_n) = \omega_{n+1}$ and is invertible; its inverse shifts the index backwards. The evaluation at time zero pushes $\widehat{Q}$ to Q . We use the conull invariant subset on which all positive selected-cell masses and all reconstructions below are defined. The natural-extension theorem supplies this path system and its ergodicity when Q is ergodic.

For $k \geq 1$ put $H_k = T_{D_b(x_{-k})}$, the one-step affine map at time $-k$, and set $A_0 = \text{id}$ and $A_n = H_1 \circ \cdots \circ H_n$ for $n \geq 1$. Thus A_n maps the selected level- n past cell $D_n = A_n^{-1}(B_1)$ onto B_1 ; its dilation factor is exactly b^n . Put $p_n = \mu_{-n}(D_n) > 0$ (with $p_0 = 1$) and define

$$\nu_n = \frac{(A_n)_* \mu_{-n}}{p_n}. \quad (6.3)$$

This is the *full* affine image of μ_{-n} , not the measure restricted to D_n . Its support is contained in

$$E_n = A_n(B_1). \quad (6.4)$$

The half-open cell convention is harmless here: by Lemma IV.2.36, every μ_{-n} gives zero mass to all internal cell boundaries, so its restriction to the closed affine preimage $A_n^{-1}(B_1)$ agrees with restriction to the selected half-open cell. For $n \geq m$ put $B_{n,m} = H_{m+1} \circ \cdots \circ H_n$ and $C_{n,m} = B_{n,m}^{-1}(B_1)$. The CP recursion gives

$$\mu_{-m} = \frac{(B_{n,m})_*(\mu_{-n}|_{C_{n,m}})}{\mu_{-n}(C_{n,m})}, \quad p_n = \mu_{-n}(C_{n,m})p_m. \quad (6.4a)$$

Because $C_{n,m} \subset B_1$, the cubes E_n increase:

$$B_1 = E_0 \subset E_1 \subset E_2 \subset \cdots. \quad (6.5)$$

If $A \subset E_m$ is Borel, then $A_n^{-1}A \subset C_{n,m}$, and (6.4a) gives

$$\nu_n(A) = \frac{\mu_{-n}(A_n^{-1}A)}{p_n} = \frac{\mu_{-m}(A_m^{-1}A)}{p_m} = \nu_m(A). \quad (6.6)$$

Lemma IV.2.38 (*canonical full measure*).

There is a unique Radon measure ν on $\bigcup_n E_n$, extended by zero outside that union, such that $\nu|_{E_m} = \nu_m$ for every m . It satisfies $\nu(B_1) = 1$, so it lies in the framed space $\mathcal{M}_{\text{fr}}$; no origin-support condition is asserted here.

Proof. Extend each ν_n by zero outside E_n . By (6.6) and the nestedness of the frames, $\nu_n \leq \nu_{n+1}$ as measures. Hence $\nu(A) = \lim_n \nu_n(A)$ defines a Borel measure, uniquely characterized by the prescribed restrictions, and $\nu|_{B_1} = \mu_0$.

We verify local finiteness; it does not follow merely from the nestedness of closed sets E_n . Write $a_{-k,j} \in \{0, \dots, b-1\}$ for the digit of x_{-k} in coordinate j . In that coordinate the left and right endpoints of E_n are respectively

$$-1 - 2 \sum_{k=1}^n a_{-k,j} b^{k-1}, \quad 1 + 2 \sum_{k=1}^n (b-1 - a_{-k,j}) b^{k-1}. \quad (6.6b)$$

Disintegrate the stationary path law into shift-ergodic components using Theorem IV.1.37. On each component, Birkhoff's theorem implies: if the event $a_{0,j} > 0$ has positive probability, then $a_{-k,j} > 0$ infinitely often and the left endpoints tend to $-\infty$; if it has probability zero, stationarity makes every digit in that coordinate zero almost surely. In the latter case, adaptedness implies that μ_n is supported on the left face $\{x_j = -1\}$: a point sampled from μ_n has every future digit zero, and the nested leftmost cells intersect in that face. Similarly, either the right endpoints tend to $+\infty$ or every μ_n is supported on the right face $\{x_j = 1\}$.

Let F be the intersection of the fixed coordinate faces arising in this way. Every ν_n is supported on F . In each coordinate not fixed to a face, both relevant endpoints of E_n escape whenever needed; in a fixed coordinate the face itself lies in every E_n . Thus for every compact $K \subset \mathbb{R}^d$ there is an $N = N(K)$ such that $K \cap F \subset E_N$. Compatibility now gives $\nu_n(K) = \nu_N(K)$ for all $n \geq N$, so $\nu(K) < \infty$. The locally finite Borel measure ν is therefore Radon on $\mathbb{R}^d$ and is carried by $\bigcup_n E_n$.

Finally, ν_n is a Borel function of the two-sided path by its finite affine formula (6.3). For every nonnegative $f \in C_c(\mathbb{R}^d)$, the finite limit $\int f d\nu = \lim_n \int f d\nu_n$ is Borel. A countable convergence-determining family of such tests shows that the reconstruction map from paths to $\mathcal{M}_{\text{fr}}$ is Borel; the same monotone-limit identity gives vague convergence $\nu_n \rightarrow \nu$.

□

Let Q^e be the law of (ν, x_0) .

Lemma IV.2.39 (*restriction and invariance*).

The push-forward of Q^e under

$$(\nu, x) \mapsto (\nu^\square, x)$$

is Q , and Q^e is invariant under the full-space pointed magnification M_b^* .

Proof.

The first statement is $\nu|_{B_1} = \mu_0$ together with preservation of the point coordinate x_0 . For equivariance, let σ shift a two-sided path forward one time unit, put $D = D_b(x_0)$,

$h = T_D$, and $q = \mu_0(D) > 0$. If primed symbols denote the reconstruction from $\sigma\omega$, the affine maps and selected-cell masses satisfy, for $n \geq 1$,

$$A'_n = h \circ A_{n-1}, \quad p'_n = qp_{n-1}, \quad E'_n = h(E_{n-1}), \quad \nu'_n = \frac{h_*\nu_{n-1}}{q}. \quad (6.6c)$$

For example, the mass identity is the chain rule for the nested selected cells, and the last identity follows directly from (6.3). Passing to the locally finite monotone limits from Lemma IV.2.38, (6.6c) yields

$$\nu(\sigma\omega) = \frac{h_*\nu(\omega)}{\nu(\omega)(D)}, \quad x_0(\sigma\omega) = h(x_0(\omega)),$$

which is exactly the full-space pointed magnification M_b^* of Definition IV.1.49. The path law is stationary, so its Borel image Q^e is M_b^* -invariant. $\square$

This Q^e is the canonical extended CP law used in the rest of Volume IV. No claim is made that an arbitrary invariant full-space lift has the same past coupling.

Write $J(\omega) = (\nu(\omega), x_0(\omega))$ for the reconstruction map. The preceding proof gives the precise relations

$$J_*\widehat{Q} = Q^e, \quad J \circ \sigma = M_b^* \circ J. \quad (6.6d)$$

The map J need not be injective. In particular, its geometric image (Q^e, M_b^*) is a factor of the natural extension, not necessarily an invertible realization of it. The distinction matters for negative times in the suspension flow.

Lemma IV.2.40 (*adaptedness survives canonical extension*).

For the canonical law Q^e , the conditional distribution of x_0 given the full framed measure ν is $\nu|_{B_1}$ (recall that $\nu(B_1) = 1$). In particular, for every nonnegative Borel H on $\mathcal{M}_{\text{fr}} \times B_1$,

$$\int H(\nu, x) dQ^e(\nu, x) = \int \int_{B_1} H(\nu, x) d\nu(x) d(Q^e)_1(\nu). \quad (6.6a)$$

Proof. Fix $n \geq 0$ and let D_n be the level- n past cell containing x_{-n} . Conditional on μ_{-n} and D_n , adaptedness of the one-time marginal Q says that x_{-n} has law $\mu_{-n}|_{D_n}/\mu_{-n}(D_n)$. The present coordinate is the affine image $x_0 = T_{D_n}x_{-n}$, so its conditional law is $\mu_0 = (\nu_n)|_{B_1}$. The finite measure ν_n is determined by (μ_{-n}, D_n) ; hence, for bounded continuous F on the vague Radon-measure space and $g \in C(B_1)$,

$$\mathbb{E}[F(\nu_n)g(x_0)] = \mathbb{E}\left[F(\nu_n) \int g d\mu_0\right].$$

By the compatible construction, $\nu_n \rightarrow \nu$ vaguely and $\mu_0 = \nu|_{B_1}$. Bounded convergence gives the same identity with ν in place of ν_n . A monotone-class argument first in F and then in g gives (6.6a). $\square$

2. Discrete and continuous centering

Definition IV.2.41 (*discrete centering*).

For $x \in \text{supp}(\nu)$, set

$$\text{cent}_0(\nu, x) = T_x^* \nu. \quad (6.7)$$

For an extended CP law Q^e write

$$R = (\text{cent}_0)_* Q^e. \quad (6.8)$$

for the push-forward distribution.

Definition IV.2.42 (*continuous centering*).

Put $L = \log b$ and define

$$\text{cent } Q = \frac{1}{L} \int_0^L (S_t^*)_* R \, dt. \quad (6.9)$$

Lemma IV.2.43 (*one-step factor identity*).

If

$$(\nu', x') = M_b^*(\nu, x),$$

then

$$S_L^* T_x^* \nu = T_{x'}^* \nu'. \quad (6.10)$$

Proof.

Both sides first translate the selected point to the origin and then magnify the selected b-adic cell by factor b. The only scalar discrepancy is removed by the $*$ normalization.

□

Corollary IV.2.44 (*centering as a suspension factor*).

cent_0 is a factor map from (Q^e, M_b^*) to (R, S_L^*) . In particular

$$(S_L^*)_* R = R. \quad (6.11)$$

The suspension is formed over the actual invertible path system $(\Omega, \hat{Q}, \sigma)$. It has probability $\hat{Q} \times (dt/L)$ on $\Omega \times [0, L]$, with (ω, L) identified with $(\sigma\omega, 0)$. Its time-s map is

$$S_s(\omega, t) = (\sigma^k \omega, t + s - kL), \quad k = \left\lfloor \frac{t+s}{L} \right\rfloor \in \mathbb{Z}. \quad (6.12a)$$

Then

$$\Theta(\omega, t) = S_t^* T_{x_0(\omega)}^* \nu(\omega) \quad (6.12)$$

is a factor onto $(\text{cent } Q, S^*)$. No inverse of the pointed geometric map M_b^* is needed or asserted.

Proof. Iterating Lemma IV.2.43 identifies the discrete time- L maps. This gives (6.11). Let $c(\omega) = T_{x_0(\omega)}^* \nu(\omega)$. Equations (6.6d) and (6.10) say $c(\sigma\omega) = S_L^* c(\omega)$. Since σ and the full scaling flow are invertible, replacing ω by $\sigma^{-1}\omega$ also gives the identity at negative integers. Thus $c(\sigma^k\omega) = S_{kL}^* c(\omega)$ for every $k \in \mathbb{Z}$. For the k in (6.12a),

$$\Theta(\mathcal{S}_s(\omega, t)) = S_{t+s-kL}^* c(\sigma^k\omega) = S_{t+s}^* c(\omega) = S_s^* \Theta(\omega, t).$$

The law of c is R , so the law of Θ is exactly (6.9). The path shift preserves $\widehat{Q}$ and the roof pieces in (6.12a) translate onto a partition of $[0, L]$; hence the suspension maps preserve $\widehat{Q} \times (dt/L)$. These formulas prove all factor assertions. $\square$

3. Scale invariance

Lemma IV.2.45 (*scale invariance of the centered law*).

$P = \text{cent } Q$ is invariant under S_s^* for every $s \in \mathbb{R}$.

Proof.

Let $R_t = (S_t^*)_* R$. By (6.11), $R_{t+L} = R_t$. Hence for every s ,

$$\begin{aligned} & (S_s^*)_* P \\ &= (1/L) \int_0^L R_{t+s} dt \\ &= (1/L) \int_s^{L+s} R_u du \\ &= P \end{aligned}$$

by L -periodicity. $\square$

4. Quasi-Palm property of the discrete centering

For a Radon measure η and a bounded Borel set B with positive η -mass, write

$$\begin{aligned} & \langle \eta \rangle_B^* \\ &= \frac{1}{\eta(B)} \int_B \delta_{T_y^* \eta} d\eta(y). \end{aligned} \tag{6.13}$$

For the canonical Q^e , the extended adaptedness identity (6.6a) gives

$$R = \int \langle \nu \rangle_{B_1}^* dQ_1^e(\nu). \quad (6.14)$$

Fix a bounded open neighborhood B of 0 and put

$$R_B = \int \langle \eta \rangle_B^* dR(\eta). \quad (6.15)$$

Lemma IV.2.46 ($R_B \ll R$).

If $R(A) = 0$ for a Borel set $A \subset M_1^*$, then $R_B(A) = 0$.

Proof.

Let $L = \log b$ and take the two-sided realization used to construct Q^e . Write η_{-n} for the full canonical measure at time $-n$, D_n for its selected level- n cell, and $p_n = \mu_{-n}(D_n) > 0$. The present full measure is the normalized affine image of η_{-n} under T_{D_n} , and its truncation to $E_n = T_{D_n}B_1$ is ν_n . For $u \in B_1$ and $y = T_{D_n}u$, the translation–scaling identity gives

$$T_y^* \nu = S_{nL}^* T_u^* \eta_{-n}.$$

Consequently, for the nonnegative random variable $I_n = \int_{E_n} \mathbf{1}_A(T_y^* \nu) d\nu_n(y)$, one has the exact *random* identity

$$I_n = \frac{1}{p_n} \int_{B_1} \mathbf{1}_{S_{-nL}^* A}(T_u^* \eta_{-n}) d\mu_{-n}(u). \quad (6.16)$$

There is no pointwise identity between I_n and the deterministic number $R(A)$; the factor p_n^{-1} is essential.

By Lemma IV.2.40, stationarity, and the discrete scale invariance (6.11), the expectation of the nonnegative integral on the right before multiplication by p_n^{-1} is

$$R(S_{-nL}^* A) = R(A) = 0.$$

Thus that integral, and hence I_n , vanish almost surely. Intersect these conull events over n . Since $\nu_n|_{E_m} = \nu|_{E_m}$ for $n \geq m$ and ν is carried by $\bigcup_m E_m$, it follows that

$$\int \mathbf{1}_A(T_y^* \nu) d\nu(y) = 0 \quad \text{for } Q^e\text{-almost every } (\nu, x).$$

If $\theta = T_x^* \nu$ is sampled from R , recentering θ at any of its points is the same as recentering ν there, after translation. The displayed global null integral therefore implies $\int_B \mathbf{1}_A(T_z^* \theta) d\theta(z) = 0$ for R -almost every θ . This is $R_B(A) = 0$.

□

Lemma IV.2.47 (*two-point reverse domination*).

For every Radon measure ν with $\nu(B_1) > 0$ and every bounded open neighborhood B of 0,

$$\langle \nu \rangle_{B_1}^* \ll \int \langle \theta \rangle_B^* d\langle \nu \rangle_{B_1}^*(\theta). \quad (6.17)$$

Proof.

Let A be a Borel set whose right side in (6.17) assigns mass zero. Writing $\theta = T_x^* \nu$ and then $y = z - x$, Fubini says

$$T_z^* \nu \notin A$$

for $\nu \times \nu$ -a.e. pair (x, z) with $x \in B_1$ and $z - x \in B$.

Suppose instead that

$$G = \{z \in B_1 : T_z^* \nu \in A\}$$

has positive ν -mass. Choose $r > 0$ with $B(0, r) \subset B$. For ν -a.e. $z \in G$

(which is in the support of $\nu|_{B_1}$ after discarding a null set),

$$\nu(B(z, r) \cap B_1) > 0.$$

Therefore

$$\int_G \nu(B(z, r) \cap B_1) d\nu(z) > 0.$$

This is exactly positive $\nu \times \nu$ mass of pairs (x, z) with $x \in B_1$, $z \in G$, and $z - x \in B$, contradicting the Fubini conclusion. Hence $\nu(G) = 0$, proving (6.17). □

Corollary IV.2.48 ($R \ll R_B$).

$$R \ll R_B.$$

Proof.

Integrate (6.17) against Q_1^c and use (6.14). The double integral is exactly R_B . □

Theorem IV.2.49 (*discrete centering is quasi-Palm*).

For every bounded open neighborhood B of 0,

$$R \sim R_B. \quad (6.18)$$

Proof.

Combine Lemma IV.2.46 and Corollary IV.2.48. This is exactly the null-class quasi-Palm property. □

5. Quasi-Palm property of continuous centering

Lemma IV.2.50 (*scaling preserves quasi-Palm*).

If R is quasi-Palm, then $(S_t^*)_* R$ is quasi-Palm for every t .

Proof.

Translation and scaling satisfy

$$S_t T_x = T_{e^t x} S_t.$$

Thus recentering a scaled measure in a neighborhood B corresponds to recentering the original measure in the scaled neighborhood $e^{-t}B$. The quasi-Palm null-class equivalence therefore transfers under S_t^* . □

Lemma IV.2.51 (*mixtures of quasi-Palm laws are quasi-Palm*).

Let (R_t) be a measurable family of quasi-Palm distributions and let ρ be a probability on the parameter space. Then

$$P = \int R_t d\rho(t)$$

is quasi-Palm.

Proof.

For a Borel A ,

$P(A) = 0$ if and only if $R_t(A) = 0$ for ρ -almost every t .

For each such t , quasi-Palm is equivalent to zero mass under the recentering kernel. Integrating this equivalence in t gives the forward implication for P . The converse is identical because a nonnegative integral is zero only when its integrand is zero almost everywhere. □

Corollary IV.2.52 (*the centered law is quasi-Palm*).

$P = \text{cent } Q$ is quasi-Palm.

Proof.

Use $R_t = (S_t^*)_* R$, Theorem IV.2.49, Lemmas IV.2.50–IV.2.51, and (6.9). □

6. Ergodicity

Lemma IV.2.53 (*ergodicity of the centered distribution*).

If Q is ergodic under $M_b^\square$, the invertible path law $\widehat{Q}$ is ergodic under σ . Its geometric factor Q^e is ergodic under M_b^* , its path suspension (6.12a) is flow-ergodic, and $P = \text{cent } Q$ is ergodic under S^* .

Proof.

Ergodicity of $\widehat{Q}$ follows from the natural-extension theorem. Its image under J is ergodic by (6.6d). To prove the assertion about the suspension, let E be a flow-invariant measurable set and let $f = \mathbf{1}_E$. For every rational s , invariance gives $f(\omega, t + s) = f(\omega, t)$ almost everywhere on the part where $t, t + s \in (0, L)$. By Fubini and countability, for almost every ω this holds for all such rational shifts. Translation continuity in L^1 on compact subintervals extends it to real shifts. An integrable function with these local translation invariances is constant almost everywhere on $(0, L)$: convolving on an interior interval with a smooth approximate identity makes it a continuous translation-invariant function, and then pass to the L^1 limit. Thus $f(\omega, t) = g(\omega)$ almost everywhere. The time- L map gives $g \circ \sigma = g$, so ergodicity of $\widehat{Q}$ makes g constant with value zero or one. This proves flow ergodicity. Finally (6.12) pushes every invariant event for P to an invariant event for the path suspension, so P is ergodic. □

7. Centering theorem

Theorem IV.2.54 (*precise centering formulas*).

Every restricted base- b CP distribution Q has the canonical extended law Q^e constructed in Lemmas IV.2.38–IV.2.39. Its discrete centering and continuous centering are given by (6.8)–(6.9), with roof length exactly $\log b$, and the factor identity is (6.10)–(6.12).

Proof.

The canonical extension is given by Lemmas IV.2.38–IV.2.39, the one-step factor identity is Lemma IV.2.43, and Corollary IV.2.44 passes from discrete to continuous centering. □

Theorem IV.2.55 (*centering produces a fractal distribution*).

For every base- b CP distribution Q , the distribution $P = \text{cent } Q$ is a fractal distribution. The S^* flow on P is a factor of the constant-roof-log b suspension of the invertible two-sided CP path system. If Q is ergodic, then P is an ergodic fractal distribution.

Proof.

Scale invariance is Lemma IV.2.45, and the quasi-Palm property is Corollary IV.2.52. The factor statement is Corollary IV.2.44. The ergodicity conclusion is Lemma IV.2.53. □

2.3 Tangent distributions and the converse bridge

Let μ be a Radon measure on $\mathbb{R}^d$ and let $x \in \text{supp}(\mu)$. For $t \geq 0$, define the restricted scenery

$$\begin{aligned} \mu_{x,t}^\square \\ = (S_t T_x \mu)^\square. \end{aligned}$$

The empirical scenery distribution is

$$\langle \mu \rangle_{x,T}^\square = \frac{1}{T} \int_0^T \delta_{\mu_{x,t}^\square} dt. \quad (7.1)$$

Definition IV.2.56 (*tangent distribution*).

A tangent distribution of μ at x is a weak accumulation point of

$$\langle \mu \rangle_{x,T}^\square$$

as $T \rightarrow \infty$.

Lemma IV.2.57 (*compactness of tangent distributions*).

For every μ and every $x \in \text{supp}(\mu)$, the family

$$\{\langle \mu \rangle_{x,T}^\square : T \geq 1\}$$

has weakly convergent subsequences.

Proof.

Every scenery belongs to $M_0^\square \subset M^\square$, while $M^\square = \mathcal{P}(B_1)$ is compact metrizable. Therefore $\mathcal{P}(M^\square)$ is compact metrizable, and the empirical distributions have subsequential limits *in this ambient space*. Compactness alone does not assert that such a limit is supported on $M_0^\square$; that admissibility is part of the later CP-centering argument. □

No full-space vague compactness is used here.

Main theorem of this section. For every Radon measure μ on $\mathbb{R}^d$, for μ -almost every x , every tangent distribution of μ at x is a fractal distribution. The theorem is stated and proved as Theorem IV.2.66 after the reconstruction lemma.

The reconstruction proceeds in the direction

$$\text{tangent scenery limit} \longrightarrow \text{CP law} \longrightarrow \text{centering} \longrightarrow \text{fractal distribution}. \quad (7.2)$$

This order matters because the restricted scenery action is discontinuous; one cannot infer invariance of an arbitrary weak limit without first controlling the relevant boundary sets.

For integer N define the discrete b -adic CP empirical law

$$\langle \nu, y \rangle_N = \frac{1}{N} \sum_{n=0}^{N-1} \delta_{(M_b^\square)^n(\nu, y)}. \quad (7.3)$$

Lemma IV.2.58 (*adapted accumulation points*).

For every probability ν on B_1 , for ν -almost every y , every weak accumulation point of (7.3) is adapted.

Proof.

Fix ν and sample $Y \sim \nu$. Let $\mathcal{F}_n$ be the finite σ -algebra generated by the level- n b -adic cells, with the half-open convention of Chapter IV.1. Write $(\theta_n, z_n) = (M_b^\square)^n(\nu, Y)$. Outside the null union of zero- ν -mass selected cells, θ_n is $\mathcal{F}_n$ -measurable and the conditional law of z_n given $\mathcal{F}_n$ is θ_n .

For $F \in C(M^\square \times B_1)$ put

$$A_F(\theta) = \int F(\theta, z) d\theta(z), \quad D_n(F) = F(\theta_n, z_n) - A_F(\theta_n). \quad (7.3a)$$

We prove $N^{-1} \sum_{n < N} D_n(F) \rightarrow 0$ almost surely. For a fixed $k \geq 1$, partition B_1 into the level- k cells D , and choose one point $z_D \in D$ for each cell. Uniform continuity gives

$$F_k(\theta, z) = \sum_D F(\theta, z_D) \mathbf{1}_D(z), \quad \|F - F_k\|_\infty \rightarrow 0 \quad (k \rightarrow \infty). \quad (7.3b)$$

The random variable $D_n(F_k)$ is $\mathcal{F}_{n+k}$ -measurable and has conditional expectation zero given $\mathcal{F}_n$. For each residue $r \in \{0, \dots, k-1\}$, the variables $D_{r+jk}(F_k)$ therefore form a bounded martingale-difference sequence for the filtration $\mathcal{F}_{r+(j+1)k}$. Indeed the conditional expectation of its j th term with respect to the preceding filtration $\mathcal{F}_{r+jk}$ is zero. The martingale strong law gives

$$\frac{1}{N} \sum_{n < N} D_n(F_k) \rightarrow 0 \quad \text{almost surely.} \quad (7.3c)$$

For completeness, that strong law follows by applying the L^2 martingale convergence theorem to the series of differences divided by $j+1$, whose squared increments have summable expectations, and then applying Kronecker's lemma. Since $|D_n(F) - D_n(F_k)| \leq 2\|F - F_k\|_\infty$, equations (7.3b)–(7.3c) prove the same limit for F .

Choose a countable uniformly dense family of continuous F and intersect its conull sets. If a subsequence of the empirical laws (7.3) converges weakly to Q , continuity of A_F on the compact space $M^\square$ permits passage to the limit in (7.3a). We obtain

$$\int F(\theta, z) dQ(\theta, z) = \int \left(\int F(\theta, z) d\theta(z) \right) dQ_1(\theta)$$

for every continuous F . This is adaptedness. □

This proves the adaptedness half of the CP reconstruction in source Proposition 5.5.

Normalize μ to be supported on $B_{1/2}$. Choose y Lebesgue-typically in $B_{1/2}$, and set

$$\mu' = T_{-y}\mu, \quad x' = x + y.$$

Put $L = \log b$. Choose integers $N_i \rightarrow \infty$ such that the original restricted scenery laws at physical times $N_i L$ converge to the chosen tangent law. This is possible from any convergent time sequence because changing its endpoint by at most L changes the empirical law by at most $2L/T$ against tests bounded by one. After a further subsequence, suppose

$$Q_i = \langle \mu', x' \rangle_{N_i} \rightarrow Q. \quad (7.4)$$

in $\mathcal{P}(M^\square \times B_1)$.

By the adapted accumulation-point argument just recalled, Q is adapted for μ' -a.e. x' .

Lemma IV.2.59 (*Lebesgue ergodicity of the base- b map*).

Let $T_b(z) = bz \pmod{1}$ on $[0, 1)$, with $b \geq 2$. Lebesgue measure is T_b -invariant and ergodic. Consequently, by the locally proved Birkhoff theorem, Theorem IV.1.33, for Lebesgue-a.e. z the empirical measures of the orbit $(T_b^n z)$ converge to Lebesgue measure.

Proof. Let $f \in L^2([0, 1))$ satisfy $f \circ T_b = f$. Its Fourier coefficients obey

$$\widehat{f}(k) = \begin{cases} 0, & b \nmid k, \\ \widehat{f}(k/b), & b \mid k. \end{cases}$$

For every nonzero integer k , repeatedly divide by b until an integer not divisible by b is reached. Hence $\widehat{f}(k) = 0$ for all $k \neq 0$. Thus every invariant L^2 function is constant. Applying this to indicators of invariant sets proves ergodicity. Birkhoff then gives the asserted genericity. $\square$

Thus Theorem IV.1.33 and Lemma IV.2.59 provide the required pointwise ergodic input.

Lemma IV.2.60 (*random-grid Lebesgue intensity*).

For Lebesgue-a.e. y and μ -a.e. x , every Q in (7.4) satisfies

$$\int \theta dQ(\theta, z) = \lambda^\square. \quad (7.5)$$

Proof.

The second coordinate of

$$(M_b^\square)^n(\mu', x')$$

is determined only by the b -adic orbit of x' , not by the measure μ' . For fixed x , translating a Lebesgue-typical y by x preserves the Lebesgue null sets relevant to base- b genericity. Hence Lemma IV.2.59, together with Theorem IV.1.33, gives

$$\frac{1}{N_i} \sum_{n < N_i} \delta_{z_n} \longrightarrow \lambda^\square. \quad (7.6)$$

where z_n is the second coordinate.

Therefore the second marginal of Q is $\lambda^\square$. Since Q is adapted,

$$\int \theta dQ(\theta, z)$$

is exactly the second marginal, proving (7.5).

□

Lemma IV.2.61 (*boundary-null support and CP invariance*).

Assume the conclusion of Lemma IV.2.60. Then

$$(M_b^\square)_* Q = Q. \quad (7.7)$$

Proof.

Let

$$E = \bigcup_{D \in \mathcal{D}_b} \partial D.$$

By (7.5),

$$\int \theta(E) dQ(\theta, z) = \lambda^\square(E) = 0,$$

hence

$$\theta(E) = 0$$

for Q -a.e. state.

Adaptedness also gives

$$Q\{(\theta, z) : \theta(D_b(z)) = 0\} = 0 :$$

conditioned on θ , the union of zero- θ -mass cells has θ -mass zero.

Thus Q is supported on the domain of $M_b^\square$ and on its continuity set described in Lemma IV.1.51.

For the empirical laws Q_i ,

$$(M_b^\square)_* Q_i - Q_i = \frac{1}{N_i} \left[\delta_{(M_b^\square)^{N_i}(\mu', x')} - \delta_{(\mu', x')} \right].$$

so this difference tends weakly to zero. Since $M_b^\square$ is Q -a.e. continuous, the continuous-mapping version of Portmanteau gives

$$(M_b^\square)_* Q_i \rightarrow (M_b^\square)_* Q.$$

Taking limits yields (7.7).

□

Hence Q is a restricted CP distribution.

Put $L = \log b$. The next construction is a *local, restricted* centering of an arbitrary pointed law H on $M^\square \times B_1$; it is not the full-space centering $\text{cent} H$, which is defined only after a canonical extension of a CP distribution. For $m \in \mathbb{N}_0$ and H supported on $\{(\theta, z) : z \in \text{supp } \theta\}$, set

$$C_m(H) = \int \int_0^1 \delta_{(S_{(m+t)L} T_z \theta)^\square} dt dH(\theta, z). \quad (7.8)$$

Here $T_z \theta$ is the finite measure translated so that z becomes the origin; the final restriction-normalization makes the expression well-defined. Define

$$P_m = C_m(Q), \quad P_{m,i} = C_m(Q_i). \quad (7.9)$$

Lemma IV.2.62 (*fixed- m continuity*).

For each fixed m ,

$$P_{m,i} \rightarrow P_m. \quad (7.10)$$

Proof.

$P_{m,i}$ is the image of Q_i under the probability-kernel-valued map

$$(\theta, z) \mapsto \int_0^1 \delta_{(S_{(m+t)L} T_z \theta)^\square} dt. \quad (7.11)$$

For a fixed θ , the restricted scaling map can be discontinuous only at times t for which θ charges the boundary of the corresponding centered ball. A finite measure charges the boundaries for at most countably many radii. Therefore, for every convergent sequence of states in the Q -full continuity set, the integrand in (7.11) converges for Lebesgue-almost every t . Bounded convergence against continuous test functions makes (7.11) continuous there. Since Q gives full mass to that set by Lemmas IV.2.60–IV.2.61, weak convergence $Q_i \rightarrow Q$ implies (7.10). □

Lemma IV.2.63 (*restriction-defect estimate*).

Let

$$E = \bigcup_{D \in \mathcal{D}_b} \partial D$$

and

$$c_m = \lambda^\square\{x : \text{dist}(x, E) < b^{-m}\}.$$

Fix a compatible metric $\rho \leq 1$ on the compact space $M^\square$ and let $d = W_\rho$ be the corresponding bounded Wasserstein metric on $\mathcal{P}(M^\square)$. Then $c_m \rightarrow 0$ and

$$\limsup_{i \rightarrow \infty} d(P_{m,i}, \langle \mu' \rangle_{x', N_i L}^\square) \leq c_m. \quad (7.12)$$

Proof.

Write

$$(\theta_n, z_n) = (M_b^\square)^n(\mu', x').$$

If

$$\text{dist}(z_n, E) \geq b^{-m},$$

then the centered cube of radius $b^{-(m+t)}$, $0 \leq t \leq 1$, remains inside B_1 for each such t . Restriction before centering therefore causes no loss at this deeper scale. The one-roof-time scenery segment obtained from the n -th CP state agrees exactly with the segment of the continuous cube scenery at physical times $(n + m + t)L$.

Thus the two empirical distributions differ only:

in the first m/N_i fraction of terms, which tends to zero;

at indices n for which $\text{dist}(z_n, E) < b^{-m}$. Couple the two laws by using the same index n and same roof parameter t . Their d -distance is at most the probability of these bad indices plus m/N_i , since the cost ρ is at most one.

By Lemmas IV.2.60 and IV.2.59, together with Theorem IV.1.33, the asymptotic frequency of the latter indices is exactly c_m . This gives (7.12).

Finally E is a finite union of grid hyperplanes, so its b^{-m} -neighborhood has Lebesgue measure $c_m \rightarrow 0$.

□

Lemma IV.2.64 (*centering identifies the tangent law*).

Assume the canonical extension from Theorem IV.2.54. Then, as an equality of probabilities on $M^\square$,

$$P = (\pi_\square)_*(\text{cent } Q). \quad (7.13)$$

Proof.

By Lemma IV.2.62, for fixed m ,

$$P_{m,i} \rightarrow P_m.$$

By the assumed tangent convergence,

$$\langle \mu' \rangle_{x', N_i L}^\square \rightarrow P.$$

(the random translation changes the base point and measure together, so the centered Euclidean scenery is unchanged up to the harmless initial normalization). Hence Lemma IV.2.63 gives

$$d(P_m, P) \leq c_m.$$

Let $m \rightarrow \infty$ to obtain

$$P_m \rightarrow P. \quad (7.14)$$

To compare P_m with the full centering, couple $(\theta, z) \sim Q$ to $(\nu, z) \sim Q^e$ through the canonical extension, so that $\theta = \nu|_{B_1}$, and use the same uniform $t \in [0, 1]$ in both centerings. The full-space variable is

$$(\pi_{\square})(S_{(m+t)L}^* T_z^* \nu).$$

Its law is $(\pi_{\square})_*(\text{cent } Q)$: the discrete centering law R is invariant under S_L^* by (6.11), and averaging t over one roof period gives (6.9). If $\text{dist}(z, \partial B_1) \geq b^{-m}$, then for every $0 \leq t \leq 1$ the centered window $z + B_{b^{-(m+t)}}$ is contained in B_1 . On that event the local variable in (7.8) and the full-space variable agree exactly. Their disagreement probability is at most

$$Q\{(\theta, z) : \text{dist}(z, \partial B_1) < b^{-m}\} = \lambda_{\square}\{z : \text{dist}(z, \partial B_1) < b^{-m}\} \leq c_m,$$

where the equality uses the Lebesgue second marginal. The coupling inequality for W_{ρ} therefore gives

$$d(P_m, (\pi_{\square})_*(\text{cent } Q)) \leq c_m \longrightarrow 0. \quad (7.15)$$

Comparing (7.14) and (7.15) proves (7.13). □

Theorem IV.2.65 (*CP representation of a tangent law*).

For μ -almost every x , every restricted tangent distribution P of μ at x is the *restricted centering* of a base- b CP distribution Q :

$$P = (\pi_{\square})_*(\text{cent } Q).$$

Proof.

Choose the random translation y outside the null exceptional sets above. Compactness gives Q along a further subsequence. The adapted accumulation-point argument gives adaptedness, Lemma IV.2.60 gives Lebesgue intensity, Lemma IV.2.61 gives CP invariance, and Lemma IV.2.64 identifies the centering with P . □

Theorem IV.2.66 (*almost-everywhere tangent-distribution theorem*).

For every Radon measure μ , for μ -a.e. x every restricted tangent distribution is a restricted FD. Its full-space FD lift is the centering of the CP law in Theorem IV.2.65, and is unique by Lemma IV.2.15. The required Birkhoff and natural-extension inputs are supplied locally by Theorems IV.1.33 and IV.1.29.

Proof.

Apply Theorem IV.2.65 and then Theorem IV.2.55. □

This closes the difficult upgrade from an empirical limit to an FD without asserting that arbitrary weak limits of arbitrary Borel dynamics are invariant.

Lemma IV.2.67 (*EFD typical measures are generic*).

Let P be a full-space EFD. Then, for P -almost every ν and ν -almost every x ,

$$\langle \nu \rangle_{x,T}^{\square} \longrightarrow P^{\square} = (\pi_{\square})_* P.$$

Proof.

Birkhoff for the scenery flow gives genericity at the origin for P -almost every ν for a countable dense family of continuous tests. Quasi-Palm transfers the conull set of measures with this property to $T_x^* \nu$ for ν -almost every x . Translation of the base point identifies that scenery with the scenery of ν at x . □

Theorem IV.2.68 (*ergodic converse*).

Every EFD P and every integer $b \geq 2$ admit a base- b CP distribution Q with $\text{cent } Q = P$.

Proof.

Choose ν, x as in Lemma IV.2.67. Then $P^{\square}$ is the restricted tangent/generation limit at x , and Theorem IV.2.65 constructs Q with $P^{\square} = (\pi_{\square})_*(\text{cent } Q)$. Both P and $\text{cent } Q$ are scale-invariant full-space laws, so the unique-lift Lemma IV.2.15 gives $P = \text{cent } Q$. □

We avoid the source's abstract probability-lifting lemma by exploiting the actual compact CP state space.

Definition IV.2.69 (*Lebesgue-intensity CPD class*).

Let

$$\Xi = \mathcal{P}(B_1) \times B_1.$$

with its compact weak/product topology. Let CPD_{λ} be the set of probabilities Q on Ξ satisfying:

- (a) adaptedness;
- (b) base- b magnification invariance;
- (c) Lebesgue intensity

$$\int \theta dQ(\theta, z) = \lambda_{B_1}. \tag{7.16}$$

Lemma IV.2.70 (*adaptedness is weakly closed*).

If Q_n are adapted and $Q_n \rightarrow Q$ weakly, then Q is adapted.

Proof.

For $F \in C(\Xi)$ put

$$AF(\theta) = \int F(\theta, z) d\theta(z).$$

If $\theta_n \rightarrow \theta$ weakly, uniform continuity of F on compact Ξ and weak convergence give $AF(\theta_n) \rightarrow AF(\theta)$.

Adaptedness is the identity

$$\int F dQ_n = \int AF(\theta) d(Q_n)_1(\theta).$$

Pass to the limit.

□

Lemma IV.2.71 (*Lebesgue intensity is weakly closed*).

If Q_n have intensity λ and $Q_n \rightarrow Q$, then Q has intensity λ .

Proof.

For $f \in C(B_1)$,

$$\int \left(\int f d\theta \right) dQ_n(\theta, z) = \int f d\lambda.$$

The bracket is continuous in θ .

□

Lemma IV.2.72 (*closedness of CP invariance*).

If $Q_n \in \text{CPD}_\lambda$ and $Q_n \rightarrow Q$, then Q is magnification invariant.

Proof.

Lemmas IV.2.70–IV.2.71 give adaptedness and Lebesgue intensity for Q . Let E be the finite union of first-level b -adic boundaries. Then

$$\int \theta(E) dQ(\theta, z) = \lambda(E) = 0,$$

so $\theta(E) = 0$ for Q -almost every state. Adaptedness also gives zero Q -mass to states whose selected cell has zero θ -mass. Thus the magnification map is Q -almost everywhere continuous, exactly on the continuity set described by Lemma IV.1.51 and used in Lemma IV.2.61.

Hence

$$M_* Q_n \rightarrow M_* Q.$$

by the a.e.-continuous mapping theorem. Since $M_* Q_n = Q_n$, the limit is Q .

□

Corollary IV.2.73 (*compactness and convexity of CP distributions*).

CPD_λ is compact and convex.

Proof.

$\mathcal{P}(\Xi)$ is compact metrizable. Lemmas IV.2.70–IV.2.72 make CPD_λ closed. All three defining properties are affine. □

Lemma IV.2.74 (*centering is continuous on CPD_λ*).

Let

$$c(Q) = (\pi_\square)_*(\text{cent } Q). \quad (7.17)$$

The centering uses the canonical extension Q^e from Theorem IV.2.54. Then

$$c : \text{CPD}_\lambda \rightarrow \mathcal{P}(M^\square)$$

is continuous and affine.

Proof. The finite-window map C_m of (7.8) is affine in its input law. It is continuous at every $Q \in \text{CPD}_\lambda$: for a bounded continuous test on $M^\square$, the proof of Lemma IV.2.62 shows that its time-averaged integrand is continuous at every state (θ, z) with $z \in \text{supp } \theta$, except for a null set of roof times. Adaptedness makes this a Q -full set, so the a.e.-continuous mapping theorem applies to any sequence $Q_j \rightarrow Q$ in CPD_λ .

The coupling in the second half of the proof of Lemma IV.2.64 uses only the Lebesgue second marginal, which is the same for every member of CPD_λ . With the bounded weak metric $d = W_\rho$ fixed there it gives, uniformly in Q ,

$$d(C_m(Q), c(Q)) \leq c_m, \quad c_m \longrightarrow 0. \quad (7.18)$$

Thus c is the uniform limit of the continuous maps C_m on the compact metric domain CPD_λ , and so is continuous. Because each C_m is affine, passing to the same limit also gives $c(sQ + (1-s)Q') = sc(Q) + (1-s)c(Q')$ for $0 \leq s \leq 1$. □

Lemma IV.2.75 (*elementary compact-fiber Borel selector*).

Let X, Y be compact metric spaces and

$$f : X \rightarrow Y$$

continuous and onto. Then there is a Borel map

$$s : Y \rightarrow X$$

with

$$f(s(y)) = y. \quad (7.20)$$

Proof.

Construct a finitely branching tree of nonempty compact sets K_σ :

$$K_\emptyset = X,$$

and for every node σ , choose finitely many compact children

$$K_{\sigma,1}, \dots, K_{\sigma,m(\sigma)}$$

covering K_σ , each contained in K_σ and of diameter at most

$$2^{-|\sigma|-1}.$$

This is possible by compactness.

For $y \in Y$ define $\sigma_1(y)$ to be the smallest child index i for which $y \in f(K_i)$.

Recursively, once $K_{\sigma_n(y)}$ is selected, choose the smallest child whose image contains y .

Every $f(K_\sigma)$ is compact, hence closed. Therefore the set on which a fixed child is the first admissible child is a finite Boolean combination of closed sets. Thus every finite-valued map $\sigma_n(y)$ is Borel.

Choose one point q_σ in each K_σ . Then

$$q_{\sigma_n(y)}$$

is Borel and Cauchy, because the K_σ are nested and their diameters tend to zero. Let $s(y)$ be its limit. Pointwise limits of Borel maps into a metric space are Borel. The nested compact sets all meet the closed fiber $f^{-1}\{y\}$; their unique intersection point is $s(y)$. Hence $f(s(y)) = y$.

□

Corollary IV.2.76 (*Borel CP realization selector*).

Let $Y_b = c(\text{CPD}_\lambda)$, a set of restricted distribution laws on $M^\square$.

Then Y_b is compact and there is a Borel map

$$s_b : Y_b \rightarrow \text{CPD}_\lambda$$

such that

$$c(s_b(P)) = P. (7.21)$$

Proof.

Corollary IV.2.73 and Lemma IV.2.74 make c a continuous map from a compact metric space. Its image Y_b is compact. Apply Lemma IV.2.75.

□

Theorem IV.2.77 (*general converse realization*).

Let P be any FD and $b \geq 2$. Then there exists a base- b CP distribution Q , with Lebesgue intensity, such that

$$P = \text{cent } Q. \quad (7.22)$$

Proof.

Use Theorem IV.1.38 and Theorem IV.2.33 to decompose

$$P = \int P_\xi d\tau(\xi). \quad (7.23)$$

into EFDs.

By Theorem IV.2.68 and the random-grid construction in the preceding results, the restricted law $P_\xi^\square = (\pi_\square)_* P_\xi$ of every EFD component belongs to $Y_b = c(\text{CPD}_\lambda)$.

Use the Borel selector in Corollary IV.2.76 and set

$$Q = \int s_b(P_\xi^\square) d\tau(\xi). \quad (7.24)$$

CPD_λ is convex by Corollary IV.2.73, so Q is a CP distribution with Lebesgue intensity. By affinity of c ,

$$(\pi_\square)_*(\text{cent } Q) = c(Q) = \int c(s_b(P_\xi^\square)) d\tau(\xi) = \int P_\xi^\square d\tau(\xi) = (\pi_\square)_* P. \quad (7.25)$$

Both $\text{cent } Q$ and P are scale-invariant full-space laws; Lemma IV.2.15 upgrades (7.25) to $\text{cent } Q = P$.

□

This is the content of Hochman's Theorem 1.15 (equivalently, Theorem 5.2), including the choice of Lebesgue intensity. The CP distribution is invariant for the discrete magnification M_b . Its continuous centering is the factor of the constant-roof

$$L = \log b$$

suspension proved in Theorem IV.2.54. Under the single time map S_L , a constant-roof suspension has phase fibers: even when the full flow is ergodic, S_L need not be. Thus the discrete CP dynamics should be viewed as the base of the suspension, not as an assertion that an arbitrary flow-ergodic FD is automatically S_L -ergodic. This is why continuous centering averages one roof period.

Theorem IV.2.78 (*base-independent realization*).

Let P be a full-space FD and, for each integer $b \geq 2$, let Q_b be a base- b realization as in Theorem IV.2.77. Write Q_b^e for its canonical full-space extension and $q_b = (\text{pr}_1)_* Q_b^e$ for the measure marginal on $\mathcal{M}_{\text{fr}}$. Let A be a Borel property of nonzero Radon measures such that, for every nonzero Radon ν , $x \in \mathbb{R}^d$, $t \in \mathbb{R}$, and $c > 0$,

$$\mathbf{1}_A(T_x \nu) = \mathbf{1}_A(S_t \nu) = \mathbf{1}_A(c\nu) = \mathbf{1}_A(\nu).$$

The restrictions of A to M_1^* and $\mathcal{M}_{\text{fr}}$ are both denoted by A . Then

$$q_b(A) = P(A) = q_{b'}(A) \quad (b, b' \geq 2). \quad (7.26)$$

The marginal $(Q_b)_1$ instead lives on $\mathcal{P}(B_1)$; in general it cannot be evaluated on the full-space event A . Formula (7.26) is a statement about the canonical extensions, not the unextended CP measures.

Proof. Apply Theorem IV.2.77 separately for each integer base $b \geq 2$. For $L = \log b$, the definition of centering gives the exact identity

$$P(A) = \frac{1}{L} \int_0^L \int \mathbf{1}_A(S_t^* T_x^* \nu) dQ_b^e(\nu, x) dt.$$

The two invariances of A make the integrand $\mathbf{1}_A(\nu)$ for Q_b^e -almost every state. Thus this integral is $q_b(A)$; repeat in base b' . $\square$

This is the base-change principle summarized in the source as Corollary 1.17; no stronger assertion is needed here.

The preceding argument uses Birkhoff's theorem, natural extension, and flow decomposition from Chapter IV.1; the ergodic decomposition of fractal distributions, Theorem IV.2.33; and the canonical centering theorem, Theorem IV.2.54. Lemma IV.2.75 and Corollary IV.2.76 give the compact selector required in this CP setting.

Chapter IV.3

Uniformly Scaling Measures

Generation, smooth covariance, and the structure of fractal distributions

Fractal distributions are converted into uniformly scaling measures; smooth covariance and the global structure of the FD simplex are developed.

3.1 Uniform scaling and smooth changes of coordinates

1. Scenery generation and uniform scaling

Let μ be a Radon measure on $\mathbb{R}^d$ and $x \in \text{supp } \mu$. Write

$$\mu_{x,t}^\square = (S_t T_x \mu)^\square,$$

and

$$\langle \mu \rangle_{x,T}^\square = \frac{1}{T} \int_0^T \delta_{\mu_{x,t}^\square} dt. \quad (8.1)$$

Definition IV.3.1 (*generation by scenery averages*).

The measure μ *generates* a restricted law p at x if (8.1) converges weakly to p in $\mathcal{P}(M^\square)$ as $T \rightarrow \infty$. When P is a full-space FD, saying that μ generates P means that it generates $P^\square = (\pi_\square)_* P$. This shorthand never identifies the two state spaces. The full invariant lift is unique by Lemma IV.2.15; Lemma IV.3.26 below also proves the corresponding full-scenery empirical convergence.

Definition IV.3.2 (*finite-window full-Radon tests*).

Let $\mathcal{R}$ be the space of nonnegative Radon measures on $\mathbb{R}^d$, including zero, with the vague topology. We view full FDs as probabilities on this ambient space, carried by M_1^* . A finite-window test is

$$G(\nu) = g\left(\int f_1 d\nu, \dots, \int f_k d\nu\right), \quad f_i \in C_c(\mathbb{R}^d), \quad g \in C_b(\mathbb{R}^k). \quad (8.1a)$$

Such tests are bounded and vague-continuous. Choose countably many f_i uniformly dense among the compactly supported continuous functions on each fixed compact window, including nonnegative cutoffs that dominate 1 on each B_j . The metric $\sum_i 2^{-i} \min\{1, |\int f_i d\nu - \int f_i d\eta|\}$ metrizes vague convergence: convergent cutoff coordinates give the local mass bounds needed to extend convergence to every C_c

test. Their integral coordinates therefore generate the Borel σ -algebra of $\mathcal{R}$. Countable convergence-determining families of bounded continuous g on the finite coordinate spaces therefore give a countable family (G_j) of tests (8.1a) determining probabilities on $\mathcal{R}$.

Definition IV.3.3 (*scaling and uniformly scaling measures*).

The measure μ is a *scaling measure* if it generates some P_x at μ -a.e. x . It is a *uniformly scaling measure* (USM) if the same P is generated at μ -a.e. x .

Lemma IV.3.4 (*origin genericity inside an FD*).

Let P be a fractal distribution. There is a measurable map

$$\nu \longmapsto P_\nu$$

from a P -conull subset of M_1^* to the full-space scale-ergodic components of P such that, for P -a.e. ν ,

$$\langle \nu \rangle_{0,T}^\square \longrightarrow P_\nu^\square. \quad (8.2)$$

Proof. Use the full-space ergodic decomposition from Theorem IV.1.38. Choose a countable dense family (φ_j) in $C(M^\square)$ and apply the continuous-time Birkhoff theorem to the bounded Borel functions $\varphi_j \circ \pi_\square$ and to all the full-window tests G_j from Definition IV.3.2. On one common conull set,

$$\frac{1}{T} \int_0^T \varphi_j(\pi_\square S_t^* \nu) dt \longrightarrow \int \varphi_j dP_\nu^\square. \quad (8.3)$$

The corresponding limits for G_j are $h_j(\nu) = \int G_j dP_\nu = E_P(G_j \mid \mathcal{I}_{\text{flow}})(\nu)$ and determine the full component. The restricted limits in (8.3) determine $P_\nu^\square$ and give (8.2), because $\pi_\square S_t^* \nu$ is the restricted scenery at the origin. $\square$

Lemma IV.3.5 (*quasi-Palm preserves the component label spatially*).

For P -a.e. ν and ν -a.e. x ,

$$P_{T_x^* \nu} = P_\nu. \quad (8.4)$$

Proof. Every coordinate

$$h_j(\nu) = E_P(G_j \mid \mathcal{I}_{\text{flow}})(\nu)$$

is taken in the version supplied by the Birkhoff limit for G_j , with value zero off its convergence domain. A time shift by any $s \in \mathbb{R}$ changes the finite average by at most $2\|s\|\|G_j\|_\infty/T$. Hence that domain is Borel and exactly flow invariant, and this version of h_j is exactly invariant. Apply the invariant-set saturation theorem, Lemma IV.2.30, to every rational level set $\{h_j < q\}$. Intersect the resulting spatial conull sets over j, q . Recentering at ν -typical x preserves every h_j . The countable family (h_j) determines the component, giving (8.4). The common Birkhoff domain in Lemma IV.3.4 is also preserved at almost every recentering point by the global quasi-Palm full-set implication. Thus the component label and its genericity are available on one common spatial conull set. $\square$

Theorem IV.3.6 (*FD-typical measures are scaling measures*).

If P is an FD, then for P -a.e. ν and ν -a.e. x ,

$$\langle \nu \rangle_{x,T}^\square \longrightarrow P_\nu^\square. \quad (8.5)$$

Consequently, P -a.e. ν is a USM generating its ergodic component. If P is ergodic, P -a.e. ν is a USM generating P .

Proof. By Lemma IV.3.5, $T_x^* \nu$ belongs to the same Birkhoff component as ν for ν -a.e. x , and belongs to the common genericity domain. Scalar normalization cancels exactly, so $\pi_\square S_t^* T_x^* \nu = (S_t T_x \nu)^\square$ for every $t \geq 0$. Apply Lemma IV.3.4 to obtain (8.5). $\square$

2. Dimension of ergodic and nonergodic fractal distributions

Lemma IV.3.7 (*exact dimension on reconstructed past windows*).

Let Q be an ergodic CP distribution with CP dimension δ_Q . Its canonical framed full measure ν is exact dimensional of dimension δ_Q almost surely. The same is true of every full measure obtained from ν by centering and scalar dilation.

Proof. In the stationary two-sided path construction, each measure μ_{-n} has the marginal Q_1 . Theorem IV.1.74 and a countable intersection give exact dimension δ_Q for all these measures simultaneously. Each compatible measure $\nu_n = (A_n)_* \mu_{-n} / p_n$ in (6.3) is a non-singular affine image and a positive scalar multiple, hence has the same local dimension at ν_n -almost every point.

Lemma IV.2.38 supplies a Radon measure ν carried by $\bigcup_n E_n$ with $\nu|_{E_n} = \nu_n$. By Radon differentiation, Theorem IV.2.26, for ν -a.e. $y \in E_n$,

$$\frac{\nu_n(B(y,r))}{\nu(B(y,r))} \longrightarrow 1 \quad (r \downarrow 0).$$

Thus the logarithmic local dimensions of ν_n and ν agree there. The exceptional set where ν_n lacks its dimension is ν -null inside E_n , since their restrictions agree. Intersect these conull sets for all n and use the full-mass exhaustion $\bigcup_n E_n$ to obtain exact dimension δ_Q for ν . Translation, nonsingular scalar dilation and positive normalization preserve this dimension by direct comparison of ball radii and masses. Those are precisely the full-space operations in (6.7)–(6.9). $\square$

Lemma IV.3.8 (*ergodic FD exact dimensionality through CP representation*).

Let P be an EFD. Then there is a number $\delta(P)$ such that P -a.e. ν is exact dimensional of dimension $\delta(P)$.

Proof. By Theorem IV.2.77, choose a base- b CP distribution Q with $\text{cent } Q = P$. Decompose Q into M_b -ergodic CP components Q_η . By Theorem IV.2.55, $\text{cent } Q_\eta$ is a scale-invariant fractal distribution, and

$$P = \int \text{cent } Q_\eta \, d\kappa(\eta). \quad (8.6)$$

Since P is ergodic, it is an extreme point of the convex set of scale-invariant probabilities. Therefore

$$\text{cent } Q_\eta = P$$

for κ -a.e. η . Fix such an ergodic component.

Theorem IV.1.75 says that $(Q_\eta)_1$ -a.e. measure θ is exact dimensional with

$$\dim \theta = \frac{1}{\log b} \int H(\theta, D_b) d(Q_\eta)_1(\theta). \quad (8.7)$$

Lemma IV.3.7 first proves this dimension on the full measure reconstructed from the past, and only then transfers it through centering. Since $\text{cent } Q_\eta = P$, P -a.e. ν is exact dimensional with the constant value in (8.7). $\square$

Definition IV.3.9 (*dimension of an ergodic fractal distribution*).

For an EFD P , $\dim P$ denotes the constant $\delta(P)$ from Lemma IV.3.8.

Theorem IV.3.10 (*nonergodic dimension statement*).

Let

$$P = \int P_\xi d\tau(\xi) \quad (8.8)$$

be the scale-ergodic decomposition of an FD. Then for P -a.e. ν , the measure ν is exact dimensional and

$$\dim \nu = \dim P_{\xi(\nu)}. \quad (8.9)$$

There is in general no single P -a.s. constant dimension. Moreover,

$$\int \dim \nu dP(\nu) = \int \dim P_\xi d\tau(\xi). \quad (8.10)$$

Proof. Theorem IV.2.33 says that τ -a.e. P_ξ is an EFD. Apply Lemma IV.3.8 component-wise. $\square$

Corollary IV.3.11 (*dimension of FD-typical USMs*).

If P is an EFD, then the P -a.e. USM supplied by Theorem IV.3.6 is exact dimensional of dimension $\dim P$. If P is nonergodic, a P -typical measure generates one ergodic component and its dimension is that component's dimension.

Proof. This is Theorem IV.3.10 specialized to an ergodic distribution; the nonergodic assertion is the componentwise statement already proved there. $\square$

This distinction will be important when differentiable coordinate changes are introduced.

3. Invertible linear changes of coordinates

For $A \in \text{GL}_d(\mathbb{R})$ define

$$A^* \nu = (A_* \nu)^* \quad (8.11)$$

on normalized full-space measures, and let $A^* P$ denote the push-forward distribution.

Lemma IV.3.12 (*projective identities*).

For $A \in \text{GL}_d(\mathbb{R})$,

$$A^* S_t^* = S_t^* A^*, \quad (8.12)$$

and

$$A^* T_x^* = T_{Ax}^* A^*. \quad (8.13)$$

Proof. Scalar dilations commute with A , and $A(y-x) = Ay - Ax$. The two sides can differ before the final star operation only by a positive scalar factor, which star-normalization removes. $\square$

Theorem IV.3.13 (*linear covariance*).

If P is an FD, then A^*P is an FD. If P is an EFD, then A^*P is an EFD. If μ generates the restricted law $P^\square$ at a typical point x , then $A_*\mu$ generates $(A^*P)^\square$ at Ax . Hence an invertible linear image of a USM generating P is a USM generating A^*P .

Proof. Scale invariance follows from (8.12). The quasi-Palm null-class equivalence transfers under (8.13). The map $P \mapsto A^*P$ is a measurable conjugacy between the full-space scenery flows, so ergodicity is preserved. A restricted unit-window scenery does not itself determine its linear image if $A^{-1}B_1$ extends beyond B_1 . The generation assertion follows from the larger-window reconstruction in the proof of Theorem IV.3.17 below, applied to $f = A$; that proof does not use the present theorem. $\square$

4. Boundary-safe linearization for C^1 diffeomorphisms

The main subtlety is normalization on B_1 . We isolate it before using differentiability.

Lemma IV.3.14 (*ellipsoid boundaries are P -null*).

Let P be scale invariant and $A \in \text{GL}_d(\mathbb{R})$. Then

$$P\{\nu : \nu(\partial(A^{-1}B_1)) > 0\} = 0. \quad (8.14)$$

More generally, for every fixed $c > 0$,

$$P\{\nu : \nu(\partial(A^{-1}B_c)) > 0\} = 0. \quad (8.15)$$

Proof. For a fixed Radon measure ν , the ellipsoid boundaries

$$\partial(A^{-1}B_{e^{-s}}), \quad s \in \mathbb{R},$$

are pairwise disjoint. A locally finite measure can give positive mass to only countably many members of a pairwise disjoint family. Hence for every ν ,

$$\text{Leb}\{s \in [0, 1] : \nu(\partial(A^{-1}B_{e^{-s}})) > 0\} = 0. \quad (8.16)$$

Integrate (8.16) against P . By scale invariance, the P -probability in the integrand is independent of s , so it must be zero. Rescaling gives (8.15). $\square$

Let $f : U \rightarrow V$ be a C^1 diffeomorphism between open subsets of $\mathbb{R}^d$ and let $x \in U$. Put $A = D_x f$ and define

$$F_{x,t}(y) = e^t[f(x + e^{-t}y) - f(x)]. \quad (8.17)$$

Lemma IV.3.15 (*uniform local linearization*).

For every $R < \infty$,

$$F_{x,t} \longrightarrow A$$

uniformly on B_R as $t \rightarrow \infty$. If f is C^1 , the derivatives also converge uniformly on B_R .

Proof. This is differentiability at x , uniformly on bounded y after writing $h = e^{-t}y$. Uniform derivative convergence follows from continuity of Df near x . $\square$

Lemma IV.3.16 (*boundary-safe scenery comparison*).

Let $A \in \mathrm{GL}_d(\mathbb{R})$ and choose $a > 0$ so that $A^{-1}B_1 \Subset B_{ea}$. Suppose $t_n \rightarrow \infty$, $\theta_n \Rightarrow \theta$ in $M^\square$, and

$$\theta(B_{e^{-a}}) > 0, \quad \theta(\partial B_{e^{-a}}) = 0, \quad \mathcal{R}_a(\theta)(A^{-1}B_1) > 0, \quad \mathcal{R}_a(\theta)(\partial(A^{-1}B_1)) = 0. \quad (8.18)$$

Here $\mathcal{R}_a$ is the larger-window reconstruction from (5.2a). Set $\nu_n = \mathcal{R}_a(\theta_n)$ and $\nu = \mathcal{R}_a(\theta)$. If $F_{x,t_n} \rightarrow A$ uniformly on B_{ea} , then

$$[(F_{x,t_n})_* \nu_n]^\square \Longrightarrow (A_* \nu)^\square. \quad (8.19)$$

In the left side, only the restriction to B_1 is used; the maps may be defined on a neighborhood of B_{ea} .

Proof. The first two conditions in (8.18) and the Portmanteau theorem give $\theta_n(B_{e^{-a}}) \rightarrow \theta(B_{e^{-a}}) > 0$. Formula (5.2a) then implies weak convergence $\nu_n \Rightarrow \nu$ on the fixed larger window B_{ea} . Since $A^{-1}B_1$ lies in its interior, uniform convergence of F_{x,t_n} to A and the last condition in (8.18) give convergence of the restricted push-forwards on B_1 . Their normalizing masses converge to $\nu(A^{-1}B_1) > 0$. Division proves (8.19). The positivity follows whenever $0 \in \mathrm{supp} \nu$, since $A^{-1}B_1$ contains a neighborhood of the origin. $\square$

Theorem IV.3.17 (C^1 *diffeomorphism covariance*).

Let μ be supported on an open U and let $f : U \rightarrow V$ be a C^1 diffeomorphism. Fix a point $x \in U$ at which every restricted tangent distribution of μ is a restricted FD (this holds for μ -almost every x by Theorem IV.2.66), and put $A = D_x f$. If along $T_i \rightarrow \infty$,

$$\langle \mu \rangle_{x, T_i}^\square \Longrightarrow P^\square, \quad (8.20)$$

where P is the unique full-space FD lift, then

$$\langle f_* \mu \rangle_{f(x), T_i}^\square \Longrightarrow (A^* P)^\square. \quad (8.21)$$

Consequently: (a) a scaling measure is sent to a scaling measure; (b) if μ is a USM generating P , then at μ -a.e. x the image scenery generates $(D_x f)^* P$; (c) $f_* \mu$ is a USM exactly in situations where this derivative-pushforward distribution is essentially independent of x . This is automatic for an invertible linear f , but not for a general diffeomorphism.

Proof. Choose $a > 0$ so large that $A^{-1}B_1 \Subset B_{ea}$ and e^a exceeds a local Lipschitz constant of f^{-1} near $f(x)$. For $t > a$ put $\theta_t = \mu_{x, t-a}^\square$. The larger-window reconstruction identity (5.2b) gives, exactly,

$$\mathcal{R}_a(\theta_t) = \frac{(S_t T_x \mu)|_{B_{ea}}}{(S_t T_x \mu)(B_1)}. \quad (8.22)$$

For all sufficiently large t , continuity and local Lipschitzness of f^{-1} imply that the entire preimage $F_{x,t}^{-1}B_1$ lies inside B_{ea} . Consequently the image scenery is the normalized restriction of $(F_{x,t})_*\mathcal{R}_a(\theta_t)$.

The time shift by a does not change empirical limits, so (8.20) implies that the empirical laws of θ_t converge to $P^\square$ along T_i . By Lemma IV.3.14, a P -typical full-space state gives zero mass both to ∂B_{e-a} and to $\partial(A^{-1}B_1)$. The origin-support property supplies the positive masses in (8.18). Thus Lemma IV.3.16 applies at $P^\square$ -almost every restricted state. It says that whenever $t_n \rightarrow \infty$ and $\theta_{t_n} \rightarrow \theta$ at such a state, the image scenery tends to

$$\Psi_A(\theta) := (A_*\mathcal{R}_a(\theta))^\square. \quad (8.22a)$$

The same sequential statement holds with F_{x,t_n} replaced by A , so Ψ_A is continuous at those states. The almost-everywhere continuous mapping theorem, applied to the empirical laws in (8.20), now shows that the image-scenery empirical laws converge to $(\Psi_A)_*P^\square$. To see explicitly why the varying maps cause no loss, take a compact subset of the continuity set with $P^\square$ -mass above $1 - \varepsilon$. A contrary sequence of bad times whose θ_t approach this compact set would have a convergent subsequence contradicting Lemma IV.3.16; the remaining time fraction is at most $\varepsilon + o(1)$. Let $\varepsilon \downarrow 0$.

Finally, scale invariance of the full-space lift P and reconstruction (5.2b) imply that $\mathcal{R}_a(\theta)$, for $\theta \sim P^\square$, has the law of a P -state restricted to B_{ea} . Since $A^{-1}B_1 \subseteq B_{ea}$, applying A and restricting to B_1 gives $(\Psi_A)_*P^\square = (A^*P)^\square$. This proves (8.21). $\square$

Limitation. A non-invertible C^1 map is not covered; $D_x f$ may be singular and the normalization/coordinate-conjugacy argument breaks. A linear projection is therefore not an instance of this theorem.

5. Example I - strongly separated homogeneous Bernoulli measure

Fix $b \geq 3$ and a digit set $D \subset \{0, \dots, b-1\}$ such that distinct selected first-level intervals have disjoint closures. For $a \in D$ let

$$f_a(x) = \frac{x+a}{b} \quad (8.23)$$

on $[0, 1]$ (affinely conjugated to B_1 if desired). Let $p_a > 0$, $\sum_a p_a = 1$, and let μ_p be the self-similar probability

$$\mu_p = \sum_{a \in D} p_a (f_a)_* \mu_p. \quad (8.24)$$

The SSC is automatic from the digit-gap assumption.

Lemma IV.3.18 (*Bernoulli sceneries*).

The discrete b -adic pointed process of μ_p is the Bernoulli shift with digit law p . Its CP distribution is ergodic and its continuous centering P_p is an EFD. Moreover μ_p is a USM generating P_p .

Proof. Conditioning on the first selected digit a and rescaling by f_a^{-1} leaves the tail product law unchanged. Thus the pointed discrete scenery is exactly the Bernoulli shift. The

Bernoulli shift is ergodic, so its two-sided natural extension and that extension's constant-roof suspension are ergodic. The centering theorem, Theorem IV.2.55, identifies that suspension's factor as P_p . Birkhoff along the symbolic coding, with the phase variable in $[0, \log b)$, gives continuous scenery generation at μ_p -a.e. point. $\square$

Corollary IV.3.19 (*dimension of a Bernoulli self-similar measure*).

The measure μ_p is exact dimensional and

$$\dim \mu_p = \frac{H(p)}{\log b}, \quad H(p) = - \sum_{a \in D} p_a \log p_a. \quad (8.25)$$

Proof. Apply the one-step entropy formula in Theorem IV.1.75, or the ordinary Bernoulli cylinder calculation. SSC converts cylinder exponents to Euclidean local dimension. $\square$

6. Example II - a non-Bernoulli finite-state Markov/Gibbs measure

Take $b = 5$ and digits $D = \{0, 4\}$. Let

$$P = \begin{pmatrix} 3/4 & 1/4 \\ 1/3 & 2/3 \end{pmatrix}, \quad (8.26)$$

whose stationary vector is

$$\pi = (4/7, 3/7). \quad (8.27)$$

Let (X_n) be the stationary Markov chain with transition matrix P and let μ_M be the image of its digit process under

$$x = \sum_{n \geq 1} X_n 5^{-n}, \quad X_n \in \{0, 4\}. \quad (8.28)$$

This is a finite-symbol locally constant Gibbs/Markov measure on the strongly separated two-branch Cantor system.

For $i \in \{0, 4\}$, let μ_i be the future coding measure conditional on current state i . Define a pointed CP law by choosing i with law π , choosing x with law μ_i , and, after the first digit j , replacing the measure state by μ_j .

Lemma IV.3.20 (*Markov sceneries*).

This CP law is adapted, invariant and ergodic. Its continuous centering P_M is an EFD and μ_M is a USM generating P_M .

Proof. Adaptedness is built into the sampling rule. Conditioning on the first digit j has probability P_{ij} and shifts the future Markov chain to state j , so one magnification has exactly the stationary Markov transition. Stationarity of π gives invariance. The finite-state chain is irreducible and aperiodic, hence its two-sided shift and constant-roof suspension are ergodic. The same coding/Birkhoff argument as in Lemma IV.3.18 yields generation by μ_M . $\square$

Lemma IV.3.21 (*entropy-rate dimension*).

The measure μ_M is exact dimensional with

$$\dim \mu_M = \frac{h(P, \pi)}{\log 5}, \quad (8.29)$$

where

$$\begin{aligned} h(P, \pi) &= - \sum_i \pi_i \sum_j P_{ij} \log P_{ij} \\ &= -\frac{4}{7} \left(\frac{3}{4} \log \frac{3}{4} + \frac{1}{4} \log \frac{1}{4} \right) - \frac{3}{7} \left(\frac{1}{3} \log \frac{1}{3} + \frac{2}{3} \log \frac{2}{3} \right). \end{aligned} \quad (8.30)$$

Proof. The level- n cylinder mass is

$$\pi_{x_1} \prod_{k=1}^{n-1} P_{x_k x_{k+1}}.$$

Birkhoff for the finite Markov shift gives the exponent $h(P, \pi)$; SSC converts the cylinder exponent to local dimension. $\square$

This example is genuinely non-Bernoulli: the conditional law of the next digit depends on the current digit because the two transition rows in (8.26) are different.

7. Headline results

Theorem IV.3.22 (*generation, uniform scaling, and dimension*).

For every FD P , P -a.e. measure is a USM generating its scale-ergodic component. For an EFD this generated distribution is P itself. EFD-typical measures are exact dimensional with a constant dimension $\dim P$; for a nonergodic FD the dimension is componentwise and need not be globally constant.

Proof. Theorem IV.3.6, Lemma IV.3.8, Theorem IV.3.10, and Corollary IV.3.11. $\square$

Corollary IV.3.23 (*smooth covariance of generated distributions*).

Invertible linear maps send EFDs to EFDs and preserve uniform scaling with the pushed-forward generated distribution. More generally, for a C^1 diffeomorphism the exact derivative covariance (8.21) holds along every convergent scenery subsequence.

Proof. The results from Lemma IV.3.12 through Lemma IV.3.16. $\square$

Theorem IV.3.24 (*separated Bernoulli and finite-state Gibbs examples*).

The separated Bernoulli family (8.23)–(8.25) and the non-Bernoulli finite-state Markov/Gibbs example (8.26)–(8.30) are USMs generating EFDs, with their stated entropy-rate dimension formulas.

Proof. Lemma IV.3.18, Corollary IV.3.19, and Lemmas IV.3.20–IV.3.21. $\square$

3.2 Closedness, ergodic density, and realization

1. Finite-window topology of invariant full lifts

Lemma IV.3.25 (*continuity of invariant full lifts*).

Let P_n, P be scale-invariant probabilities on M_1^* , regarded as probabilities on the ambient vague Radon space $\mathcal{R}$ of Definition IV.3.2. Then

$$P_n \implies P \text{ in } \mathcal{P}(\mathcal{R}) \iff P_n^\square \implies P^\square \text{ in } \mathcal{P}(M^\square). \quad (9.0a)$$

In particular restriction and its unique invariant-lift inverse have the same weak topology on their respective classes. No compactness of the whole normalized Radon state space is asserted.

Proof. Write $p_n = P_n^\square$ and $p = P^\square$. Suppose first that $p_n \rightarrow p$. For a test G in (8.1a), choose $a > 0$ with every $\text{supp } f_i \subseteq B_{ea}$. Reconstruction (5.2b) and scale invariance give the exact identity

$$\int G dP_n = \int G(\mathcal{R}_a(\theta)) dp_n(\theta), \quad \int G dP = \int G(\mathcal{R}_a(\theta)) dp(\theta). \quad (9.0b)$$

Here $\mathcal{R}_a(\theta)$ is the finite measure on B_{ea} defined in (5.2a), extended by zero when evaluated by G ; assign an arbitrary fixed value on the zero-denominator set. For p -a.e. θ , $\theta(B_{e-a}) > 0$ and $\theta(\partial B_{e-a}) = 0$. Positivity follows from origin support, and boundary-nullness follows from Lemma IV.3.14 with A the identity. At these states $\mathcal{R}_a$ is weakly continuous on its finite window: its numerator is a continuous dilated push-forward, and its denominator converges by Portmanteau. Thus the bounded map $G \circ \mathcal{R}_a$ is p -a.e. continuous. The a.e.-continuous mapping theorem proves convergence in (9.0b) for every finite-window test.

We give the tightness step explicitly. Choose nonnegative $\chi_j \in C_c(\mathbb{R}^d)$ with $\chi_j \geq 1$ on B_j . The just-proved test convergence shows that each scalar law $\int \chi_j d\nu$ under P_n converges to its law under P . These scalar laws are therefore uniformly tight. Given $\varepsilon > 0$, choose finite M_j so that, simultaneously for P and every P_n ,

$$\Pr \left\{ \int \chi_j d\nu > M_j \right\} < \varepsilon 2^{-j}.$$

Then the closed set

$$K = \left\{ \nu \in \mathcal{R} : \int \chi_j d\nu \leq M_j \text{ for every } j \right\} \quad (9.0c)$$

has probability at least $1 - \varepsilon$ under all these laws and is vague-compact. Indeed its members have uniformly bounded mass on every compact window. A diagonal subsequence on a countable dense family of C_c tests gives limits for all such tests, since the local mass bounds control the approximation error. The limit is a positive locally bounded functional on C_c , hence a Radon measure by the Riesz representation theorem. The cutoff inequalities persist in the limit. Thus K is sequentially compact, and the metrizable vague topology makes it compact.

Finite-window tests form an algebra separating points of K and containing constants. Stone–Weierstrass approximates any bounded continuous F on $\mathcal{R}$, uniformly on K , by one such test G . Clipping its finite-coordinate function g also ensures $\|G\|_\infty \leq \|F\|_\infty + \varepsilon$. The two integrals outside K contribute at most $2\varepsilon(2\|F\|_\infty + \varepsilon)$; inside K the approximation error is at most 2ε . Convergence for G followed by $\varepsilon \downarrow 0$ proves $P_n \rightarrow P$ for every bounded continuous F .

Conversely, if $P_n \rightarrow P$ on $\mathcal{R}$, then $\nu \mapsto \nu^\square$ is continuous at P -a.e. state: the denominator is positive and $\nu(\partial B_1) = 0$ by Lemma IV.3.14. Lemma IV.2.4 and the a.e.-continuous mapping theorem give $p_n \rightarrow p$. Uniqueness of the invariant lift is Lemma IV.2.15. $\square$

Lemma IV.3.26 (*full-scenery consequence of restricted generation*).

If $\langle \mu \rangle_{x, T_i}^\square \rightarrow P^\square$ for a full scale-invariant law P on M_1^* , then

$$\frac{1}{T_i} \int_0^{T_i} \delta_{(S_t T_x \mu)^*} dt \implies P \quad \text{in } \mathcal{P}(\mathcal{R}). \quad (9.0d)$$

Conversely full convergence in (9.0d) implies restricted convergence. The same assertions hold for a limit as $T \rightarrow \infty$.

Proof. For a finite-window test G , choose a as in the preceding proof. For $t \geq a$, put $\theta_{t-a} = (S_{t-a} T_x \mu)^\square$. The exact reconstruction identity gives

$$\mathcal{R}_a(\theta_{t-a}) = (S_t T_x \mu)^*|_{B_{ea}}.$$

The time shift changes a bounded-test average by at most $2a\|G\|_\infty/T_i$. The a.e.-continuity of $G \circ \mathcal{R}_a$ at $P^\square$ therefore proves convergence of every full finite-window test average to $\int G dP$. The cutoff tightness and compact-set approximation in the proof of Lemma IV.3.25 apply to these empirical laws without requiring them to be invariant. This proves (9.0d). The converse uses the P -a.e. continuity of restriction exactly as in that lemma. Applying the subsequence statement to every sequence $T_i \rightarrow \infty$ gives the assertion for the full limit. $\square$

2. The compact CPD model

Fix $b \geq 2$. Let CPD_λ be the class of restricted base- b CP distributions with Lebesgue intensity, as laws on the compact geometric state space $\mathcal{P}(B_1) \times B_1$. Raw symbolic lifts are different laws and will be specified below.

We use the following facts from Chapter IV.2: CPD_λ is compact by Corollary IV.2.73; every FD P has a representative $Q \in \text{CPD}_\lambda$ with $\text{cent } Q = P$ by Theorem IV.2.77; the centering map

$$\text{cent} : \text{CPD}_\lambda \longrightarrow \text{FD} \quad (9.1)$$

has restricted version $c(Q) = (\text{cent } Q)^\square$ continuous by Lemma IV.2.74. Its full version (9.1) is continuous by Lemma IV.3.25. If Q is ergodic for M_b , then $\text{cent } Q$ is an EFD by the suspension-factor argument in Theorem IV.2.55.

No direct weak-closedness of the quasi-Palm property is assumed.

3. Weak closedness of fractal distributions

Theorem IV.3.27 (*weak closedness of fractal distributions*).

The class FD is compact and weakly closed in $\mathcal{P}(\mathcal{R})$, with weak convergence of laws for the vague Radon topology. Its restricted image is compact in $\mathcal{P}(M^\square)$, and restriction is an affine homeomorphism between these two classes.

Proof. The compact class CPD_λ maps continuously into $\mathcal{P}(\mathcal{R})$ by the centering theorem, Lemma IV.2.74, and Lemma IV.3.25. Theorem IV.2.77 makes its image exactly FD. A continuous image of a compact space is compact, and is closed in this Hausdorff space. Explicitly, if $P_n \rightarrow P$ in $\mathcal{P}(\mathcal{R})$, choose $Q_n \in \text{CPD}_\lambda$ with

$$\text{cent } Q_n = P_n. \quad (9.2)$$

Compactness from Corollary IV.2.73 gives a subsequence $Q_{n_k} \Rightarrow Q \in \text{CPD}_\lambda$. Continuity from Lemma IV.2.74 followed by the full-lift continuity Lemma IV.3.25 gives

$$P = \lim_k P_{n_k} = \lim_k \text{cent } Q_{n_k} = \text{cent } Q. \quad (9.3)$$

Theorem IV.2.55 says $\text{cent } Q$ is an FD. Thus P is an FD. The restricted image is the compact image of c . Restriction is affine and one-to-one by invariant-lift uniqueness; Lemma IV.3.25 proves the homeomorphism assertion. $\square$

This proof deliberately avoids the invalid shortcut “quasi-Palm is obviously weakly closed.”

4. Convexity, ergodic components, and the simplex structure

Lemma IV.3.28 (*convexity*).

FD is convex.

Proof. Let $P = tP_1 + (1-t)P_2$. Scale invariance is linear. If A is P -null, then it is P_1 - and P_2 -null, so the quasi-Palm null-set implication holds for both components and therefore for P . Conversely, if the recentering kernel of P assigns zero mass to A , then the two nonnegative component integrals vanish separately, and quasi-Palm for P_i gives $P_i(A) = 0$. Hence $P(A) = 0$. $\square$

Lemma IV.3.29 (*ergodic decomposition stays inside FD*).

If P is an FD and

$$P = \int P_\xi d\tau(\xi) \quad (9.4)$$

is its scenery-flow ergodic decomposition, then P_ξ is an EFD for τ -a.e. ξ .

Proof. This is Theorem IV.2.33. $\square$

Lemma IV.3.30 (*extreme points*).

An FD is an extreme point of the convex set FD if and only if it is ergodic for the scenery flow.

Proof. If P is not ergodic, choose an invariant Borel set U with $0 < P(U) < 1$. Corollary IV.2.34 gives two distinct FDs P_U and P_{U^c} with

$$P = P(U)P_U + (1 - P(U))P_{U^c},$$

so P is not extreme.

Conversely, let P be ergodic and $P = tP_1 + (1 - t)P_2$ with $0 < t < 1$ and $P_i \in \text{FD}$. All P_i are scenery-flow invariant. Since an ergodic invariant probability is extreme in the convex set of invariant probabilities, $P_1 = P_2 = P$. $\square$

Lemma IV.3.31 (*unique barycentric decomposition*).

Every $P \in \text{FD}$ has the unique barycentric decomposition (9.4) over the extreme points EFD.

Proof. Existence is Lemma IV.3.29. Uniqueness is the ordinary uniqueness of the ergodic decomposition of an invariant probability on a standard-Borel dynamical system. Since Lemma IV.3.30 identifies the ergodic measures with the extreme points inside FD, this is the required barycentric uniqueness. $\square$

Thus FD is a compact metrizable Choquet simplex. We do not invoke a separate abstract uniqueness theorem for the Poulsen simplex; only density of extreme points remains.

5. A precise stationary block-splicing lemma**Definition IV.3.32** (*specified raw CP lifts and local tests*).

Let $\widehat{\text{CPD}}_\lambda$ denote the stationary adapted laws on the raw compact product $\mathcal{T} \times \Sigma$ whose geometric image under Φ belongs to CPD_λ . For every geometric Q we may choose its specified lift (3.15), but this Borel section is not claimed continuous. Other raw laws produced by splicing are kept as their own chosen lifts, not replaced by that section. A K -local test is a bounded continuous function of the first K point digits and the transition vectors p_v with $|v| < K$. The algebra of these tests over all K separates raw states and is uniformly dense in $C(\mathcal{T} \times \Sigma)$ by Stone–Weierstrass. A continuous geometric test G pulls back to the continuous raw test $G \circ \Phi$ by Theorem IV.1.65.

Lemma IV.3.33 (*finite stationary splicing*).

Let $Q^1, \dots, Q^m \in \text{CPD}_\lambda$ and fix stationary adapted raw lifts $\widehat{Q}^i \in \widehat{\text{CPD}}_\lambda$ with $\Phi_* \widehat{Q}^i = Q^i$. Let $t_i > 0$, $\sum_i t_i = 1$. For every block length N there is an ergodic $\widehat{Q}_N \in \widehat{\text{CPD}}_\lambda$, with ergodic geometric image $Q_N = \Phi_* \widehat{Q}_N \in \text{CPD}_\lambda$, such that for every raw K -local test F with $K < N$,

$$\left| \int F d\widehat{Q}_N - \sum_i t_i \int F d\widehat{Q}^i \right| \leq 2\|F\|_\infty \frac{K}{N}. \quad (9.5)$$

Proof. Step 1: block variables. At the root choose a type I with $\Pr(I = i) = t_i$ and then a full source tree with marginal $(\hat{Q}^i)_{\mathcal{T}}$. Retain only its transitions at depths $0, \dots, N - 1$. At every depth- N terminal node attach an independent block variable with this same law, and repeat recursively. This uses a countable product of independent block variables. Products of the retained transitions give consistent cylinder masses and hence the realized tree measure m_p .

Step 2: adapted point. Conditional on the realized tree p , sample $\omega \sim m_p$. Denote this boundary-rooted law by ρ_N . It is raw adapted. The depth- N word is chosen using only the retained root block, not the independent subtrees attached at its terminal nodes. Conditional on this word, the chosen fresh subtree therefore has the original block-tree law, and its point tail has its own tree measure. Consequently $Z_*^N \rho_N = \rho_N$.

Step 3: stationary phase. Put $\rho_{N,r} = Z_*^r \rho_N$ and

$$\hat{Q}_N = \frac{1}{N} \sum_{r=0}^{N-1} \rho_{N,r}.$$

The identity $Z_*^N \rho_N = \rho_N$ proves $Z_* \hat{Q}_N = \hat{Q}_N$. Adaptedness is preserved under subtree shift by conditional sampling, so every summand and their average are adapted.

Step 4: ergodicity. Keep the phase as an auxiliary coordinate: on phase r use the law $\rho_{N,r}$, and send $((p, \omega), r)$ to $(Z(p, \omega), r + 1 \bmod N)$. This extended law is invariant. On each phase its N th power is mixing. To verify this, take two finite-depth local tests. After sufficiently many block shifts, the current block root lies beyond every tree coordinate and point digit used by the initial test. Selection of that root uses ancestor transitions only. Its independent block decorations and conditionally sampled tail are therefore independent of the initial finite window, with law $\rho_{N,r}$ after rerooting r levels. The two test expectations factor. Uniform approximation by local tests, followed by approximation of Borel indicators in L^2 , proves mixing for all bounded measurable tests on that phase. An invariant event for the extended one-step map has probability zero or one on each phase, because it is invariant under this mixing N th power. One-step invariance makes these probabilities equal on the cyclic phases. The extended system is ergodic, and its raw factor $\hat{Q}_N$ is ergodic.

Step 5: Lebesgue intensity. Let ρ_b be the uniform Bernoulli measure on Σ , namely $\rho_b([u]) = b^{-d|u|}$. The point marginal of each source raw law is ρ_b : it is canonical by stationarity/adaptedness and its geometric marginal is Lebesgue. At every interior phase, stationarity of the source law gives the same expected mass for a cylinder contained in that block. If a cylinder $[u] = [vw]$ crosses its next boundary, independence of the fresh subtree gives

$$\mathbb{E}m_{\text{current}}([vw]) = \mathbb{E}m_{\text{source}}([v]) \mathbb{E}m_{\text{fresh}}([w]) = b^{-d|v|} b^{-d|w|} = b^{-dK}.$$

Iterate this factorization if u crosses more boundaries. Thus the raw point intensity is ρ_b , whose geometric image is λ . Theorem IV.1.65 gives an adapted invariant geometric image Q_N , with this intensity, and passes ergodicity to that image. Hence $Q_N \in \text{CPD}_\lambda$.

Step 6: approximation. If $r + K \leq N$, the entire K -local window belongs to the retained source block. Conditional on its type i , its law is the corresponding window of $Z_*^r \hat{Q}^i = \hat{Q}^i$.

At most K of the N phases fail this condition; their test discrepancy is at most $2\|F\|_\infty$. This proves (9.5) on the explicitly chosen raw laws. $\square$

6. Density of ergodic CP distributions

Lemma IV.3.34 (*density of ergodic CP distributions*).

Every $\widehat{Q} \in \widehat{\text{CPD}}_\lambda$ is the raw weak limit of ergodic laws in that class. Consequently ergodic elements of CPD_λ are dense in CPD_λ .

Proof. Fix a raw $\widehat{Q}$, finitely many continuous raw tests $F_1, \dots, F_r$, and $\varepsilon > 0$. Approximate each F_j uniformly by a K -local test G_j within $\varepsilon/3$, using one common depth K . Apply Lemma IV.3.33 with the *single* raw input $\widehat{Q}^1 = \widehat{Q}$, $t_1 = 1$. Choose $N > K$ so large that

$$2 \max_j \|G_j\|_\infty \frac{K}{N} < \frac{\varepsilon}{3}.$$

The lemma constructs an ergodic raw $\widehat{Q}_N$, not merely a geometric approximant. For each j ,

$$\left| \int F_j d\widehat{Q}_N - \int F_j d\widehat{Q} \right| \leq 2\|F_j - G_j\|_\infty + 2\|G_j\|_\infty K/N < \varepsilon.$$

Finite-test neighborhoods give $\widehat{Q}_N \rightarrow \widehat{Q}$. For a geometric Q , first fix its raw lift (3.15), then push these approximants through the continuous factor Φ . Their geometric images are ergodic CPDs of Lebesgue intensity and converge to Q . No continuity of the Borel section K is used. $\square$

7. Density of EFDs and the Poulsen assertion

Theorem IV.3.35 (*Poulsen-simplex structure and density of EFDs*).

FD is a compact metrizable Choquet simplex whose extreme points are the EFDs, and EFDs are dense. In this precise sense FD is a Poulsen simplex.

Proof. Compactness is Theorem IV.3.27; convexity and unique extreme-point decomposition are Lemmas IV.3.28–IV.3.31.

Let P be any FD. Choose $Q \in \text{CPD}_\lambda$ with $\text{cent } Q = P$. By Lemma IV.3.34, choose ergodic $Q_n \rightarrow Q$. Restricted centering continuity in Lemma IV.2.74 and full-lift continuity in Lemma IV.3.25 give

$$\text{cent } Q_n \longrightarrow \text{cent } Q = P.$$

By Theorem IV.2.55, $\text{cent } Q_n$ is an EFD. Thus EFDs are dense in FD. $\square$

No external theorem about the uniqueness or homogeneity of the abstract Poulsen simplex is used or needed.

8. Variable long-block splicing for realization

We now prove the realization theorem rather than infer it from density alone.

Definition IV.3.36 (*discrete CP generation*).

A probability μ on B_1 CP-generates Q if for μ -a.e. x ,

$$\frac{1}{N} \sum_{n=0}^{N-1} \delta_{M_b^n(\mu, x)} \implies Q. \quad (9.6)$$

By the centering/scenery comparison in Lemma IV.3.39, if Q has Lebesgue intensity and μ CP-generates Q , then μ is a USM generating cent Q in continuous scenery.

Lemma IV.3.37 (*uniform good prefixes for an ergodic CPD*).

Let $\hat{Q}$ be an ergodic stationary adapted raw law and let $F_1, \dots, F_r$ be bounded continuous raw local tests. For every $\varepsilon > 0$ there is m such that the set G of raw pointed states z for which

$$\left| \frac{1}{n} \sum_{k=0}^{n-1} F_j(Z^k z) - \int F_j d\hat{Q} \right| < \varepsilon \quad (9.7)$$

for every $j \leq r$ and every $n \geq m$ satisfies $\hat{Q}(G) > 1 - \varepsilon$. The analogous statement holds for an ergodic geometric CPD and bounded continuous geometric tests, with $Z, \hat{Q}$ replaced by M_b, Q .

Proof. Birkhoff gives convergence for almost every state simultaneously for the finitely many tests. Egorov makes it uniform outside a set of measure less than ε . This argument applies to either of the two stated systems on its invariant conull domain. $\square$

Lemma IV.3.38 (*long-block realization of a CPD*).

Every $Q \in \text{CPD}_\lambda$ is CP-generated by some probability μ on B_1 .

Proof. Step 1: ergodic approximants. Fix the raw lift $\hat{Q}$ of Q in (3.15). By the raw form of Lemma IV.3.34, choose ergodic $\hat{Q}_i \in \widehat{\text{CPD}}_\lambda$ with

$$\hat{Q}_i \implies \hat{Q}, \quad Q_i = \Phi_* \hat{Q}_i \implies Q. \quad (9.8)$$

Choose a countable dense family (F_j) in the unit ball of $C(\mathcal{T} \times \Sigma)$ consisting of raw local tests. By inserting zero tests if necessary, index it so that F_j depends on depth at most j . After passing to a subsequence, require

$$\left| \int F_j d\hat{Q}_i - \int F_j d\hat{Q} \right| < \varepsilon_i \quad (9.9)$$

for $j \leq i$, where $\varepsilon_i > 0$ is summable. This is convergence of the chosen raw laws, not continuity of a Borel lifting section applied to geometric convergence.

Step 2: source thresholds. Apply Lemma IV.3.37 to $\hat{Q}_i$, the tests $F_1, \dots, F_i$, and tolerance ε_i . Obtain a threshold m_i and a good set G_i of $\hat{Q}_i$ -measure greater than $1 - \varepsilon_i$ such that

every prefix average of length $n \geq m_i$ is ε_i -close to $\widehat{Q}_i$ on those tests. Enlarge m_i so that $m_i \geq i$.

Step 3: choose macroblock lengths. Inductively choose n_i and put $S_i = n_1 + \cdots + n_i$ so that

$$\frac{S_{i-1}}{n_i} \rightarrow 0, \quad (9.10)$$

$$\frac{i}{n_i} \rightarrow 0, \quad (9.11)$$

and

$$\frac{m_{i+1}}{S_i} \rightarrow 0. \quad (9.12)$$

All source thresholds are chosen first. It suffices to choose the integer $n_i \geq i(S_{i-1} + m_i + m_{i+1} + i)$; then each of (9.10)–(9.12) follows directly, and $n_i \geq m_i$.

Step 4: random recursive measure. At every level- S_{i-1} node independently sample a full source tree with marginal $(\widehat{Q}_i)_\mathcal{T}$ and retain its transitions at depths $0, \dots, n_i - 1$. Repeat with the next source law at its terminal nodes. This countable collection of independent decorations defines a raw tree p . Conditional on it, sample $\omega \sim m_p$.

The node entered by the sampled path at level S_{i-1} is selected using earlier transitions only. Its source tree p_i therefore has the stated marginal, independent of that selection. Its length- n_i point word has conditional distribution given by m_{p_i} . Complete this word to an auxiliary source address ξ_i with the conditional tail law of m_{p_i} on that cylinder. Zero-mass words have probability zero. Then (p_i, ξ_i) has law $\widehat{Q}_i$ and shares the actual block word with ω . The auxiliary completions are supplied by Borel conditional kernels on a countable product. Thus the probability that this source state fails G_i is at most ε_i . Borel–Cantelli gives only finitely many bad source blocks almost surely; no independence of these good events is needed.

We also establish canonical addresses for the *nonstationary* realized measure. Each source raw point intensity is the uniform Bernoulli ρ_b . A fixed finite cylinder word splits into consecutive block pieces; independence of the new decorations gives the product of their expected masses. Each piece contributes b^{-d} per digit, as in Step 5 of Lemma IV.3.33. Hence

$$\mathbb{E}m_p([v]) = b^{-d|v|} \quad \text{for every finite } v, \quad \mathbb{E}m_p = \rho_b.$$

Since $\rho_b(\Sigma \setminus \Sigma_{\text{can}}) = 0$, $m_p(\Sigma_{\text{can}}) = 1$ for almost every realization. All estimates below are made on the common conull coupled event with this property, positive selected prefixes, and eventually good source blocks. The deterministic realization will be selected after proving the actual raw-orbit convergence.

Step 5: averages at block endpoints. For a fixed test F_j , all sufficiently large $i \geq j$ have a good source prefix of length n_i . Its local windows agree with the actual ones except at at most j positions crossing the end of the retained block. Hence

$$\frac{1}{n_i} \sum_{k=S_{i-1}}^{S_i-1} F_j(Z^k(p, \omega)) = \int F_j d\widehat{Q}_i + O(\varepsilon_i) + O(j/n_i). \quad (9.13)$$

Indeed, before the last j positions, the actual local tree and point window equal those of $Z^{k-S_{i-1}}(p_i, \xi_i)$; this includes off-path raw transition coordinates. Goodness controls the source average for length $n_i \geq m_i$, and the remaining discrepancy is at most $2j/n_i$ since $\|F_j\|_\infty \leq 1$. Using (9.9), the right side tends to $\int F_j d\hat{Q}$. Because $S_{i-1}/n_i \rightarrow 0$, the newest block dominates the endpoint average, so

$$\frac{1}{S_i} \sum_{k=0}^{S_i-1} F_j(Z^k(p, \omega)) \longrightarrow \int F_j d\hat{Q}. \quad (9.14)$$

Step 6: all intermediate times. Let $S_{i-1} < N < S_i$. If $N - S_{i-1} < m_i$, then by (9.12), with the index shifted, the new partial piece has length $o(S_{i-1})$, so its contribution cannot disturb the already good average at S_{i-1} .

If $N - S_{i-1} \geq m_i$, the source uniform-good-prefix property says its current prefix average is ε_i -close to $\hat{Q}_i$ and hence close to $\hat{Q}$. The actual and source prefixes differ at at most j positions, only if the prefix reaches the end of the retained block. In the total average this error is at most $2j/N$. The total average is therefore a convex combination of the preceding endpoint average and the good source prefix average, plus a vanishing error. Both terms tend to $\int F_j d\hat{Q}$.

Therefore the average converges for every N , not only at the endpoints.

Step 7: all tests. Intersect the conull sets over the countable dense family (F_j) . This proves actual raw empirical convergence to $\hat{Q}$ on the coupled conull set. The convergence event depends only on (p, ω) , not on auxiliary source completions. Projecting away those completions and applying Fubini therefore gives a deterministic realized tree p with $m_p(\Sigma_{\text{can}}) = 1$ for which convergence holds for m_p -a.e. ω .

Fix this tree and put $\mu = \pi_* m_p$. For every positive selected prefix, its conditional tail also has canonical full mass, because deleting a canonical prefix leaves a canonical address. The direct coding argument of Lemma IV.1.61 thus identifies every actual raw iterate with $M_b^k(\mu, \pi\omega)$, without assuming stationarity of the realized tree. Pushing raw empirical convergence through the continuous Φ gives (9.6) with $\Phi_* \hat{Q} = Q$, for μ -a.e. x . $\square$

9. From CP generation to continuous scenery generation

Lemma IV.3.39 (from CP generation to continuous scenery generation).

If $Q \in \text{CPD}_\lambda$ and μ CP-generates Q , then for μ -a.e. x ,

$$\langle \mu \rangle_{x,T}^\square \implies (\text{cent } Q)^\square. \quad (9.15)$$

Proof. Fix a point x at which (9.6) holds and all selected cells have positive mass. For μ -a.e. x these conditions hold simultaneously. Put $L = \log b$, write $(\theta_n, z_n) = M_b^n(\mu, x)$, and set

$$H_N = \frac{1}{N} \sum_{n=0}^{N-1} \delta_{(\theta_n, z_n)}, \quad A_N = \langle \mu \rangle_{x, NL}^\square, \quad c(Q) = (\text{cent } Q)^\square.$$

Every selected state has $z_n \in \text{supp } \theta_n$ on a common μ -conull set: condition on a positive-mass level- n cell, where the rescaled point has law θ_n , and intersect over n . For each fixed $m \in \mathbb{N}$, the finite-window kernel C_m of (7.8) is Q -a.e. continuous, by Lemma IV.2.62; therefore $C_m(H_N) \rightarrow C_m(Q)$.

Let $D_m = \{z \in B_1 : \text{dist}_\infty(z, \partial B_1) \leq b^{-m}\}$ and $e_m = \lambda(D_m)$. Its boundary has Lebesgue measure zero, so the Lebesgue point marginal of Q and (9.6) give

$$\frac{1}{N} \#\{0 \leq n < N : z_n \in D_m\} \longrightarrow e_m, \quad e_m \longrightarrow 0.$$

If $z_n \notin D_m$, the centered cube of radius $b^{-(m+t)}$ lies strictly inside B_1 for all $0 \leq t \leq 1$. Thus restricting to the selected cell before centering loses no mass on this window, and the exact affine coordinate change gives

$$(S_{(m+t)L} T_{z_n} \theta_n)^\square = (S_{(n+m+t)L} T_x \mu)^\square.$$

Couple $C_m(H_N)$ with the continuous average over $[mL, (N+m)L]$ using the same n, t . With the bounded Wasserstein metric $d = W_\rho$, $\rho \leq 1$, from Lemma IV.2.63, the cost is at most the bad-index frequency. Moving this averaging interval to $[0, NL]$ costs at most $2m/N$. Hence

$$\limsup_{N \rightarrow \infty} d(A_N, C_m(Q)) \leq e_m.$$

The canonical-extension coupling in Lemma IV.2.64, equivalently (7.18), gives $d(C_m(Q), c(Q)) \leq c_m$ with $c_m \rightarrow 0$. Consequently

$$\limsup_{N \rightarrow \infty} d(A_N, c(Q)) \leq e_m + c_m.$$

Letting $m \rightarrow \infty$ proves convergence at integer roof times. The terminal interval for arbitrary T has length less than L , and changes any bounded-test average by $O(L/T)$. This proves (9.15). In particular the boundary-error frequency need not be zero at a fixed depth m ; the proof uses first $N \rightarrow \infty$ and then $m \rightarrow \infty$. Lemma IV.3.26 also yields the full-space empirical convergence to $\text{cent } Q$. $\square$

Theorem IV.3.40 (*realization of every fractal distribution by a USM*).

Every fractal distribution P is generated by a uniformly scaling measure.

Proof. By Theorem IV.2.77, choose $Q \in \text{CPD}_\lambda$ with

$$\text{cent } Q = P. \tag{9.16}$$

By Lemma IV.3.38, choose μ which CP-generates Q . Lemma IV.3.39 gives, for μ -a.e. x ,

$$\langle \mu \rangle_{x,T}^\square \Longrightarrow P^\square.$$

Thus μ is a USM generating P . $\square$

10. Relation to the structure theorem

These results recover the three corresponding conclusions of Käenmäki–Sahlsten–Shmerkin: weak closedness, the Poulsen-simplex structure, and realization by uniformly scaling measures. Here weak closedness is proved through compact CP representation and centering, the simplex theorem through stationary block splicing, and realization through variable long-block recursive splicing.

Chapter IV.4

Scaling Limits of Self-Conformal Measures

Research theorems and explicit overlapping models

Overlaps prevent us from identifying a small ball with one symbolic cylinder. We instead compare its scenery with a probability mixture of rescaled cylinder images. Entropy matching and a dimension lock make the mixture and its constituents agree on average at small scales. This leads to a full proof of the no-separation uniform-scaling theorem of Bárány, Käenmäki, Pyörälä, and Wu, followed by source-delimited extensions and two concrete overlapping models.

Definition IV.4.1 (*conformal IFS in the source convention*).

Let $U \subset \mathbb{R}^d$ be bounded, open, and convex. A finite family

$$\Phi = (f_i)_{i \in \Lambda}$$

is a conformal iterated function system if each $f_i : U \rightarrow U$ is injective and $C^{1+\alpha}$ on U for some $\alpha > 0$, if

$$\sup_{x \in U} \|Df_i(x)\| < 1, \quad Df_i(x) \neq 0,$$

and if

$$\|Df_i(x)\|^{-1} Df_i(x)$$

is orthogonal for every $x \in U$. The attractor K is the unique nonempty compact set satisfying

$$K = \bigcup_i f_i(K).$$

We work with a common domain for which $K \subseteq U$; equivalently, one may shrink an admissible source domain to a compact invariant neighborhood of the attractor. In dimension one this is simply a finite $C^{1+\alpha}$ family of injective contracting maps with derivative never zero.

Definition IV.4.2 (*quasi-Bernoulli symbolic measure*).

A probability $\bar{\mu}$ on $\Lambda^{\mathbb{N}}$ is quasi-Bernoulli if there is $C \geq 1$ such that for all finite words $\mathbf{i}, \mathbf{j}$,

$$C^{-1} \bar{\mu}([\mathbf{i}]) \bar{\mu}([\mathbf{j}]) \leq \bar{\mu}([\mathbf{ij}]) \leq C \bar{\mu}([\mathbf{i}]) \bar{\mu}([\mathbf{j}]). \quad (10.1)$$

If $C = 1$ it is Bernoulli. If $\Pi : \Lambda^{\mathbb{N}} \rightarrow K$ is the canonical coding map, the projected measure

$$\mu = \Pi_* \bar{\mu}$$

is also called quasi-Bernoulli, even when Π is many-to-one because of overlap.

The definition does not require $\bar{\mu}$ to be shift invariant. The next lemma is the bridge to the invariant Feng–Hu theorem proved in Volume I.

Lemma IV.4.3 (*stationary representative of a quasi-Bernoulli law*).

For every quasi-Bernoulli probability $\bar{\mu}$ on the finite full shift there is a shift-invariant ergodic probability m with $m \sim_{C'} \bar{\mu}$, where C' depends only on the quasi-Bernoulli constant. Consequently, the projection $\Pi_* \bar{\mu}$ under a conformal IFS is exact dimensional.

Proof. Let σ be the left shift. The quasi-Bernoulli inequalities and the partition into words of length n give, first for cylinders and then for Borel sets A ,

$$C^{-1} \bar{\mu}(A) \leq (\sigma_*^n \bar{\mu})(A) = \sum_{|u|=n} \bar{\mu}([u] \cap \sigma^{-n} A) \leq C \bar{\mu}(A). \quad (10.1a)$$

Any weak limit m of the Cesàro averages of $\sigma_*^n \bar{\mu}$ is shift invariant. Testing (10.1a) on continuous nonnegative functions shows $C^{-1} \bar{\mu} \leq m \leq C \bar{\mu}$ as measures. The same inequalities and (10.1) imply

$$C^{-4} m([u]) m([v]) \leq m([uv]) \leq C^4 m([u]) m([v]). \quad (10.1b)$$

Summing (10.1b) twice over a gap word gives a constant $c = C^{-8} > 0$ such that, for cylinder events A determined by an initial block of length k and B determined by another finite block,

$$m(A \cap \sigma^{-(k+n)} B) \geq c m(A) m(B) \quad (n \geq 0). \quad (10.1c)$$

If E were a shift-invariant set with $0 < m(E) < 1$, approximate E and E^c in measure by such finite-block events A, B . Invariance makes $m(E \cap \sigma^{-(k+n)} E^c) = 0$ for every n , whereas (10.1c) and the approximation errors give a strict positive lower bound when the errors are small. Thus m is ergodic.

The fully proved Feng–Hu conformal theorem (Volume I, T-I.9.08) makes $\Pi_* m$ exact dimensional. Bounded mutual absolute continuity passes from m and $\bar{\mu}$ to their projections and preserves their almost-everywhere local dimensions: differentiation of the Radon–Nikodym derivative makes the ratio of the two small-ball masses tend to a finite positive number. Therefore $\Pi_* \bar{\mu}$ is exact dimensional as well. $\square$

Definition IV.4.4 (*uniformly scaling measure and EFD*).

A measure is *uniformly scaling* if, at almost every point, its scenery empirical distributions converge to one common distribution P . An *ergodic fractal distribution* (EFD) is a fractal distribution that is ergodic under the scenery flow, in the sense of Chapters IV.2–IV.3. In the proofs, P denotes a full-space EFD and ξ the unit restriction of a state sampled from it. The restricted presentation has a unique invariant full-space lift by Lemma IV.2.15; at spatial points in the interior of the unit window the small sceneries of the lift and of ξ coincide. We keep the fixed cube window of Chapters IV.2–IV.3; orthogonal maps act on the full-space lift before restriction. Thus saying that a measure generates P means that its restricted empirical laws converge to $P^\square$; the full-space law is recovered by the unique invariant lift. We mark restricted scenery states and empirical laws explicitly by the superscript $\square$ in the proof below.

The proof of uniform scaling without separation has a different shape from the usual cylinder argument. At a small scale many overlapping symbolic cylinders can contribute to the same ball. Their conditional images form a mixture. The following entropy calculation explains why, when the mixture has no more entropy than its typical components, its small-scale conditionals must resemble those components.

Lemma IV.4.5 (*entropy and conditional-mixture deficit*).

Let ζ be a probability on a measurable family of compactly supported probabilities (ρ_h) on $\mathbb{R}^d$, and put $\rho = \int \rho_h d\zeta(h)$. Let $\mathcal{D}_k$ be the half-open dyadic partition at depth k . For y with positive ρ_h - and ρ -mass in $D_k(y)$, write $p_{h,k}(y)$ and $q_k(y)$ for the vectors of conditional masses of its 2^d children under ρ_h and ρ , respectively. Then

$$H(\rho, \mathcal{D}_n) = H(\rho, \mathcal{D}_0) + \int \int \sum_{k=0}^{n-1} [H(p_{h,k}(y)) + D_{\text{KL}}(p_{h,k}(y) \| q_k(y))] d\rho_h(y) d\zeta(h). \quad (10.2a)$$

If for some $a \geq 0$ and $\delta > 0$ one has

$$H(\rho, \mathcal{D}_n) \leq (a + \delta)n, \quad \int \int \frac{1}{n} \sum_{k < n} H(p_{h,k}(y)) d\rho_h(y) d\zeta(h) \geq a - \delta, \quad (10.2b)$$

then

$$\int \int \frac{1}{n} \sum_{k < n} \|p_{h,k}(y) - q_k(y)\|_1 d\rho_h(y) d\zeta(h) \leq 2\sqrt{\delta}. \quad (10.2c)$$

Let $p_{h,k,\ell}(y)$ and $q_{k,\ell}(y)$ be the vectors of conditional masses of the level- $(k + \ell)$ descendants of $D_k(y)$. For every fixed $1 \leq \ell < n$,

$$\int \int \frac{1}{n - \ell} \sum_{k < n - \ell} \|p_{h,k,\ell}(y) - q_{k,\ell}(y)\|_1 d\rho_h(y) d\zeta(h) \leq 2\ell\sqrt{\delta} + O(\ell/n). \quad (10.2c1)$$

Proof. For a finite partition $\mathcal{P}$, the chain rule says that its entropy along the successive dyadic refinements is the entropy of its level-zero cell plus the sum of the expected entropies of the child vectors. Apply this to ρ . Because ρ is the mixture of the ρ_h , its expectation can be computed by first drawing h and then $y \sim \rho_h$. For each child vector the elementary identity

$$-\sum_j p_j \log q_j = H(p) + D_{\text{KL}}(p \| q)$$

gives (10.2a) after summation. All terms are well-defined almost everywhere for the following countable reason. For every dyadic atom Q with $\rho(Q) = 0$, the mixture identity gives $\rho_h(Q) = 0$ for ζ -almost every h . Intersect over the countably many atoms in all dyadic levels. For each remaining h , a positive-mass selected parent has positive ρ -mass, and a child of zero ρ -mass also has zero ρ_h -mass. The conditional relative-entropy terms are therefore defined almost everywhere. This does not assert $\rho_h \ll \rho$ for an uncountable mixture. Subtract the second inequality in (10.2b) from the first and use nonnegativity of relative entropy to obtain a mean relative-entropy bound 2δ . Pinsker's finite-alphabet inequality $\|p - q\|_1^2 \leq 2D_{\text{KL}}(p \| q)$ and Jensen's inequality give (10.2c). Finally the conditional

law of an ℓ -level descendant is a product of the successive conditional kernels. Insert and subtract one kernel at a time; total variation changes by at most the ℓ^1 distance of that kernel. Summing over the ℓ levels and averaging over k gives the last assertion. The missing terminal ℓ values of k contribute at most $2\ell/n$. $\square$

Corollary IV.4.6 (*zero entropy deficit gives matching dyadic sceneries*).

For the mixture in Lemma IV.4.5, suppose there is $a \geq 0$ such that, for every fixed translated dyadic grid used below,

$$H(\rho, \mathcal{D}_n) = an + o(n), \quad \int \int \sum_{k < n} H(p_{h,k}(y)) d\rho_h(y) d\zeta(h) = an + o(n). \quad (10.2c2)$$

Then the mean dyadic discrepancy in (10.2d) tends to zero for that grid.

Proof. Apply Lemma IV.4.5 with an arbitrarily small fixed $\delta > 0$ and then take $n \rightarrow \infty$. For a fixed depth ℓ , the conditional minimeasures are supported in the unit cube. Couple their masses cellwise across the level- ℓ descendants. The matched part moves by at most $\sqrt{d} \cdot 2^{-\ell}$, while the unmatched part is at most the child-vector ℓ^1 distance. Hence their Prokhorov distance is at most

$$\|p_{h,k,\ell}(y) - q_{k,\ell}(y)\|_1 + \sqrt{d} 2^{-\ell}. \quad (10.2c3)$$

Equation (10.2c1) gives a limsup at most $2\ell\sqrt{d} + \sqrt{d} 2^{-\ell}$. First let $\delta \downarrow 0$, then $\ell \rightarrow \infty$. $\square$

Lemma IV.4.7 (*mean dyadic-to-ball transfer*).

Retain the mixture $\rho = \int \rho_h d\zeta(h)$ from Lemma IV.4.5. Assume that ρ and ρ_h for ζ -almost every h are compactly supported, exact dimensional, and have local dimensions bounded by one common finite number D . For a translation a of the dyadic grid, let $\rho^{D_k^a(y)}$ and $\rho_h^{D_k^a(y)}$ denote the two normalized measures in the same level- k cell, both rescaled to $[0, 1]^d$. If for Lebesgue-almost every grid translation a the mean dyadic discrepancy satisfies

$$\lim_{N \rightarrow \infty} \int \int \frac{1}{N} \sum_{k < N} d_{\text{LP}}(\rho^{D_k^a(y)}, \rho_h^{D_k^a(y)}) d\rho_h(y) d\zeta(h) = 0, \quad (10.2d)$$

then

$$\lim_{T \rightarrow \infty} \int \int \frac{1}{T} \int_0^T d_{\text{LP}}(\rho_{y,t}^\square, (\rho_h)_{y,t}^\square) dt d\rho_h(y) d\zeta(h) = 0. \quad (10.2e)$$

The same conclusion holds if (10.2d) is known for one grid translation whose doubling orbit is Lebesgue-generic at ρ -almost every point.

Proof. Fix $0 < \varepsilon < 1$. Choose an integer m so large that the subset of $[0, 1]^d$ within distance 2^{-m+2} of the boundary has Lebesgue measure less than ε . For almost every grid translation a , the doubling orbit of $y + a$ is Lebesgue-generic for ρ -almost every y : for each fixed y the assertion holds for Lebesgue-almost every a , and Fubini gives the claimed common choice. Consequently, at all but an upper-density ε fraction of integers k , the level- k cell $D_k^a(y)$ contains $B(y, 2^{-k-m+1})$.

We next control normalization. Let ξ be either ρ or ρ_h , and let $f_j = -\log \xi(B(y, 2^{-j}))$. Exact dimensionality gives $f_j/j \rightarrow d_\xi(y) \log 2$ at ξ -almost every y , where $0 \leq d_\xi(y) \leq D$. Choose a fixed integer c with $2^c > \sqrt{d}$. Since $D_k^a(y) \subset B(y, 2^c 2^{-k})$,

$$\frac{\xi(B(y, 2^{-k-m}))}{\xi(D_k^a(y))} \geq \exp[-(f_{k+m} - f_{k-c})]. \quad (10.2f)$$

The nonnegative differences on the right obey

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{k < N} (f_{k+m} - f_{k-c}) \leq (m+c)D \log 2, \quad (10.2g)$$

because the sum telescopes into two boundary blocks of $m+c$ terms and f_j/j has the stated limit. Choose $\tau > 0$ so small that $(m+c)D \log 2 / \log(1/\tau) < \varepsilon$. Markov's inequality applied to (10.2f)–(10.2g) shows that at all but upper-density ε of the levels, the inner ball has at least a τ fraction of $\xi(D_k^a(y))$. Apply this separately to ρ and ρ_h . By dominated convergence and Fubini, the mean frequency of levels failing any of the three conditions (grid interior and two mass bounds) is at most 3ε .

On a good level k , rescale the common cell to $[0, 1]^d$. The sceneries of ρ and ρ_h at times in $[(k+m-1) \log 2, (k+m) \log 2]$ are normalized restrictions of the two cell minimeasures to balls with the same center and radii $r \in [2^{-m}, 2^{-m+1}]$; both denominators are at least τ . Put $e_k = d_{\text{LP}}(\rho^{D_k^a(y)}, \rho_h^{D_k^a(y)})$. The defining Prokhorov inequalities, applied before and after restriction, show that the distance of the normalized restrictions is bounded by

$$C_{m,\tau}(e_k + \rho^{D_k^a(y)}(\{z : \|z - z_k\|_\infty - r\} \leq 2e_k)) \quad (10.2h)$$

when $e_k < \tau/8$; the bound 1 suffices otherwise. Here z_k is the rescaled location of y . To see (10.2h), enlarge every test set by e_k , use the two Prokhorov inequalities to compare its mass inside the ball, discard the displayed annulus, and divide by denominators at least τ ; rescaling the ball costs at most 2^m . For any probability ξ and any $e > 0$, Fubini in the radius gives

$$\int_{2^{-m}}^{2^{-m+1}} \xi(\{z : \|z - z_k\|_\infty - r\} \leq 2e) dr \leq 4e. \quad (10.2i)$$

Integrating (10.2h) over the stated time interval (whose logarithmic Jacobian is bounded above and below by constants depending on m) therefore bounds its mean scenery discrepancy by $C_{m,\tau}e_k$.

Average over $k < N$, then over (h, y) . The bad levels contribute at most 3ε ; the good ones tend to zero by (10.2d). The initial and terminal m time units are negligible as $N \rightarrow \infty$. Thus the limsup of the left side of (10.2e) is at most 3ε . Let $\varepsilon \downarrow 0$. $\square$

Lemma IV.4.8 (*overlap-safe tangent-mixture representation*).

Let Φ and $\mu = \Pi_* \tilde{\mu}$ satisfy Definitions IV.4.1–IV.4.2. Assume $\dim \mu > 0$. There are a compact family $\mathcal{H}$ of C^1 conformal embeddings on a fixed neighborhood V of K , with derivatives bounded above and away from zero, and $C_* < \infty$ such that the following holds. For μ -almost every x , every tangent distribution Q of μ at x in its full-space presentation, and $Q^\square$ -almost every restricted state ξ , there is $\zeta \in \mathcal{P}(\mathcal{H})$ for which, writing

$$M_\zeta = \int_{\mathcal{H}} h_* \mu d\zeta(h),$$

put $W = (-1, 1)^d$, the interior of the closed scenery window $B(0, 1) = [-1, 1]^d$. One has $M_\zeta(W) > 0$, the set $\{h \in \mathcal{H} : h(V) \subset W\}$ has positive ζ -mass, and

$$C_*^{-1} \frac{M_\zeta|_W}{M_\zeta(W)} \leq \xi \leq C_* \frac{M_\zeta|_W}{M_\zeta(W)}. \quad (10.2j)$$

The mixture, rather than an individual cylinder image, is essential when the IFS overlaps. The open restriction is important: weak limits of the mixtures may acquire mass on the boundary, whereas a fractal-distribution scenery state gives that boundary zero mass. We make no assertion that the mixture itself is boundary-null.

Proof. Fix a compact invariant reference neighborhood V of K inside the common domain; its interior contains K . Let $q < 1$ bound the contraction norms and let L be a common Hölder constant for $\log \|Df_i\|$. The chain rule and conformality give, for every word $u = u_1 \cdots u_n$ and $y, z \in V$,

$$|\log \|Df_u(y)\| - \log \|Df_u(z)\|| \leq L|y - z|^\alpha \sum_{j=0}^{n-1} q^{j\alpha} \leq \frac{L}{1 - q^\alpha} |y - z|^\alpha. \quad (10.2j1)$$

Telescoping the matrix products in the derivative chain rule, using the positive minimum of the one-step derivative norms, also gives

$$\|Df_u(y) - Df_u(z)\| \leq C_0 \sup_{v \in V} \|Df_u(v)\| |y - z|^\alpha. \quad (10.2j2)$$

Here is the secant version of bounded distortion needed for compactness. Injectivity and compactness imply that the one-step secant ratio $|f_i(y) - f_i(z)|/|y - z|$ divided by $\|Df_i(z)\|$ is bounded above and away from zero on $V \times V$ off the diagonal. The $C^{1+\alpha}$ estimate and conformality show that its logarithm has absolute value at most $C|y - z|^\alpha$ close to the diagonal; after enlarging C this holds for all distinct $y, z \in V$. Apply this estimate successively to the pairs $(f_{u_{j+1} \cdots u_n}(y), f_{u_{j+1} \cdots u_n}(z))$. Since their separation is at most $q^{n-j}|y - z|$, the errors sum to a geometric series. Thus

$$e^{-C_1|y-z|^\alpha} \|Df_u(z)\| |y - z| \leq |f_u(y) - f_u(z)| \leq e^{C_1|y-z|^\alpha} \|Df_u(z)\| |y - z|. \quad (10.2j3)$$

Consequently, for a word u , $\text{diam } f_u(V)$ is comparable to $\|Df_u(y)\|$ uniformly in u and $y \in V$. The one-step derivative norms have a positive minimum and a maximum below one.

At radius r , take the stopping antichain $\mathcal{W}(r)$ of first words u with $\text{diam } f_u(V) \leq r/16$. Let $\mathcal{W}(x, r)$ be the words in this antichain for which $f_u(K)$ meets $B(x, r)$. Set $p_u = \bar{\mu}([u])$ and $A_{x,r} = \sum_{u \in \mathcal{W}(x,r)} p_u$. The quasi-Bernoulli inequalities, first on symbolic tail cylinders and then on all Borel sets by a monotone-class argument, give on $B(x, r)$ the measure comparison

$$C^{-1} \sum_{u \in \mathcal{W}(x,r)} p_u (f_u)_* \mu \leq \mu \leq C \sum_{u \in \mathcal{W}(x,r)} p_u (f_u)_* \mu. \quad (10.2k)$$

No multiplicity bound for the geometric overlaps is used.

Let $g_{x,r}(z) = r^{-1}(z - x)$ and $h_{u,x,r} = g_{x,r} \circ f_u$. Stopping geometry gives $cr \leq \text{diam } f_u(V) \leq r/16$, so the maps $h_{u,x,r}$ have uniformly bounded translations and derivatives bounded above and below. Their derivatives have a uniform Hölder modulus by the same distortion estimate. Their C^1 closure $\mathcal{H}$ is compact by Arzelà–Ascoli. The normalized version of (10.2j3) persists in the limit and supplies a common positive inverse-Lipschitz constant on V ; therefore its elements remain conformal embeddings. Put

$$\zeta_{x,r} = A_{x,r}^{-1} \sum_{u \in \mathcal{W}(x,r)} p_u \delta_{h_{u,x,r}}.$$

Normalizing (10.2k) on $B(x, r)$ shows that $\mu_{x,-\log r}^\square$ is C^2 -equivalent to the normalized restriction of $M_{\zeta_{x,r}}$ to $B(0, 1)$.

The restriction mass cannot vanish at a positive proportion of typical scales. Every retained image $f_u(K)$ lies inside $B(x, 3r)$, so (10.2k) applied to the larger ball gives $A_{x,r} \leq C\mu(B(x, 3r))$. Applied to $B(x, r)$, it also gives

$$M_{\zeta_{x,r}}(B(0, 1)) \geq C^{-2} \frac{\mu(B(x, r))}{\mu(B(x, 3r))}. \quad (10.2l)$$

There is also a uniformly interior part of this mixture. Let $\mathcal{W}_{\text{cen}}(x, r)$ consist of the stopping words whose images $f_u(K)$ meet $B(x, r/4)$. Each such word has $f_u(V) \subset B(x, r/2)$, and $\mathcal{W}_{\text{cen}}(x, r) \subset \mathcal{W}(x, r)$. Applying the lower half of the quasi-Bernoulli comparison to $B(x, r/4)$ gives

$$\zeta_{x,r}\{h : h(V) \subset B(0, 1/2)\} \geq C^{-2} \frac{\mu(B(x, r/4))}{\mu(B(x, 3r))}. \quad (10.2ll)$$

By Lemma IV.4.3, μ is exact dimensional of some dimension $D \leq d$. Write $F_x(t) = -\log \mu(B(x, e^{-t}))$. Then $F_x(t)/t \rightarrow D$ at μ -almost every x . With $a = \log 3$, monotonicity and a change of variables give

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T [F_x(t) - F_x(t - a)] dt \leq aD. \quad (10.2m)$$

The same telescoping bound holds with the fixed-ratio increment $F_x(t + \log 4) - F_x(t - \log 3)$. For every $\epsilon > 0$, choose $b > 0$ large enough that Markov's inequality and (10.2l)–(10.2m) make both the restriction mass in (10.2l) and the interior fraction in (10.2ll) at least b^{-1} outside an upper-density ϵ set of times.

It remains to pass to scenery limits without losing the mixture through a restriction discontinuity. Fix b and take a convergent sequence of good-time joint parameters

$(\mu_{x,t}^\square, \zeta_{x,e^{-t}})$; compactness gives a limit (ξ, ζ) . The closed condition $h(V) \subset \overline{B}(0, 1/2)$ and (10.211) imply that ζ assigns at least b^{-1} mass to maps whose images of V lie in W . Pass to a further subsequence so that the closed-window normalizing masses $m_j = M_{\zeta_{x,e^{-t_j}}}(B(0, 1))$ converge to $m_0 > 0$; the good-time lower bound ensures positivity. If ξ gives zero mass to $\partial B(0, 1)$, then testing (10.2k) against nonnegative continuous functions supported in the open ball and passing to the limit gives, as measures on W ,

$$C^{-2} \frac{M_\xi|_W}{m_0} \leq \xi|_W \leq C^2 \frac{M_\xi|_W}{m_0}.$$

The interior mixing mass makes $m_W = M_\zeta(W) > 0$, and boundary-nullity gives $\xi(W) = 1$. Integrating the last inequalities yields $C^{-2} \leq m_W/m_0 \leq C^2$. Substitution therefore gives (10.2j) with $C_* = C^4$. Thus the closure of the good-time parameter set has property (10.2j) at every boundary-null state.

By Theorem IV.2.66, every tangent law Q at a typical x is a fractal distribution. Its restricted states give zero mass to the fixed unit-sphere boundary by Lemma IV.3.14. The good times have lower density at least $1 - \epsilon$, so the closed-set Portmanteau inequality gives $Q^\square$ -mass at least $1 - \epsilon$ to states with (10.2j). Let $\epsilon \downarrow 0$ through a countable sequence; the union over the corresponding b has full $Q^\square$ -mass. $\square$

Lemma IV.4.9 (*upper tangent-law dimension at an exact-dimensional point*).

Let σ be an exact-dimensional compactly supported probability of dimension d_σ , and let x be a σ -typical point where this local dimension holds and where all tangent distributions are fractal distributions. If Q is the invariant full-space lift of any tangent distribution at x and $Q = \int P_\omega d\kappa(\omega)$ is its scale-ergodic decomposition, then

$$\int \dim P_\omega d\kappa(\omega) \leq d_\sigma. \quad (10.2n)$$

Proof. Fix $s > 0$. On origin-supported full-space states normalized to have mass one on $B(0, 1)$, and on their unit restrictions, put

$$\varphi_s(\xi) = -\log \xi(B(0, e^{-s})) \in [0, \infty].$$

The two presentations give the same value: $\varphi_s(\xi) = \varphi_s(\xi^\square)$. For a scenery of σ at x ,

$$\varphi_s(\sigma_{x,t}^\square) = F_x(t + s) - F_x(t), \quad F_x(t) = -\log \sigma(B(x, e^{-t})). \quad (10.2o)$$

Exact dimensionality at x says $F_x(t)/t \rightarrow d_\sigma$. Consequently

$$\frac{1}{T} \int_0^T \varphi_s(\sigma_{x,t}^\square) dt = \frac{1}{T} \left(\int_T^{T+s} F_x(u) du - \int_0^s F_x(u) du \right) \longrightarrow s d_\sigma. \quad (10.2p)$$

Let $T_j \rightarrow \infty$ give the tangent law Q . The truncated function $\min\{\varphi_s, M\}$ is bounded and continuous at every state which gives no mass to $\partial B(0, e^{-s})$ and positive mass to the inner

ball. The latter positivity holds for fractal-distribution states at the origin, and the boundary-null property is Lemma IV.3.14. Weak convergence of the empirical laws, followed by (10.2p) and monotone convergence in M , gives

$$\int \varphi_s dQ \leq s d_\sigma. \quad (10.2q)$$

In particular $\varphi_s \in L^1(Q)$.

The scenery cocycle and Birkhoff's theorem for the time- s map give, for Q -almost every ξ ,

$$\lim_{n \rightarrow \infty} \frac{-\log \xi(B(0, e^{-ns}))}{ns} = \frac{1}{s} \mathbb{E}_Q(\varphi_s | \mathcal{I}_s)(\xi), \quad (10.2r)$$

where $\mathcal{I}_s$ is its invariant sigma-algebra. By quasi-Palm and Theorem IV.3.10, the left side is the exact dimension of the scale-ergodic component containing ξ . Integrating (10.2r) and using (10.2q) yields (10.2n). $\square$

Proposition IV.4.10 (*dimension locking and component-scene transfer*).

Let μ be an exact-dimensional quasi-Bernoulli self-conformal measure of dimension $d_\mu > 0$. Let Q be the invariant full-space lift of a tangent distribution at a typical point, with ergodic decomposition $Q = \int P_\omega d\kappa(\omega)$. Then

$$\dim P_\omega = d_\mu \quad \text{for } \kappa\text{-a.e. } \omega. \quad (10.2s)$$

For $Q^\square$ -almost every restricted state ξ , choose a mixture representation from Lemma IV.4.8, and let

$$\rho = \frac{M_\xi|_W}{M_\xi(W)} = \int \rho_h d\zeta_W(h), \quad \rho_h = \frac{(h_*\mu)|_W}{h_*\mu(W)}, \quad (10.2t)$$

where $W = (-1, 1)^d$ and ζ_W is the probability obtained by weighting ζ by $h_*\mu(W)$; zero-weight components are discarded. Then ρ and ζ_W -almost every ρ_h are exact dimensional of dimension d_μ , and

$$\lim_{T \rightarrow \infty} \int \int \frac{1}{T} \int_0^T d_{\text{LP}}(\rho_{y,t}^\square, (\rho_h)_{y,t}^\square) dt d\rho_h(y) d\zeta_W(h) = 0. \quad (10.2u)$$

Proof. By Theorem IV.3.10, P_ω -typical full-space states $\hat{\xi}$ are exact dimensional of dimension $\dim P_\omega$. Their unit restrictions $\xi = \hat{\xi}^\square$ have the same exact dimension. For such a restricted state ξ , Lemma IV.4.8 gives $\xi \sim_{C_*} \rho$ as in (10.2t). Bounded mutual absolute continuity preserves local dimension, so ρ is exact dimensional of dimension $\dim P_\omega$. Every positive restriction of $h_*\mu$ is exact dimensional of dimension d_μ , since a conformal C^1 embedding with bounded nonzero derivative is locally bi-Lipschitz.

For a compactly supported exact-dimensional probability λ of dimension d_λ , the fully proved entropy-dimension theorem (Volume I, C-I.4.12) gives

$$n^{-1}H(\lambda, \mathcal{D}_n) \longrightarrow d_\lambda \log 2. \quad (10.2v)$$

The result applies to every fixed translated dyadic grid: translation of λ preserves exact dimensionality and entropy-dimension, while translating the grid is the same as translating the measure in the opposite direction.

Entropy is concave under mixtures. Apply (10.2v) to ρ and to ρ_h in (10.2t). The component entropy rates are uniformly bounded by $d \log 2 + O(1/n)$ because all supports lie in one fixed compact set, so dominated convergence permits integration over h . Therefore

$$\dim P_\omega \log 2 = \lim_{n \rightarrow \infty} \frac{H(\rho, \mathcal{D}_n)}{n} \geq \lim_{n \rightarrow \infty} \int \frac{H(\rho_h, \mathcal{D}_n)}{n} d\zeta_W(h) = d_\mu \log 2. \quad (10.2w)$$

The upper bound in Lemma IV.4.9 says $\int \dim P_\omega d\kappa \leq d_\mu$. Since every component dimension is at least d_μ by (10.2w), equality holds componentwise almost everywhere, proving (10.2s).

Now both sides of the mixture in (10.2t) have dyadic entropy rate $d_\mu \log 2$, on every fixed translated grid. The integrated component child-entropy rate is the same, by the entropy chain rule and (10.2v). Corollary IV.4.6 gives mean matching dyadic sceneries. Lemma IV.4.7 then gives (10.2u). $\square$

The equality of entropy rates used above can be applied again after one expands every symbolic cylinder by a fixed finite prefix code. We record the transfer separately from the final group argument.

Lemma IV.4.11 (*bounded-equivalence and equal-dimension scene transfer*).

Suppose $\rho = \int \rho_h d\zeta(h)$ is a compactly supported mixture on $\mathbb{R}^d$, and ρ and ζ -almost every ρ_h are exact dimensional of the same dimension D . If ρ is a USM generating P , then

$$\lim_{T \rightarrow \infty} \int \int d_{\text{LP}} \left(P^\square, \frac{1}{T} \int_0^T \delta_{(\rho_h)_{y,t}^\square} dt \right) d\rho_h(y) d\zeta(h) = 0. \quad (10.2x)$$

Moreover, if two Radon measures σ, τ are mutually absolutely continuous on an open set W , then at their common almost every point of W their sceneries have Prokhorov distance tending to zero as $t \rightarrow \infty$. Thus a boundedly equivalent positive restriction of a USM generates the same distribution on that restriction.

Proof. For the final assertion write $d\sigma = w d\tau$ on W . Differentiation of w with respect to τ , proved in Lemma IV.2.25, gives at almost every y

$$\frac{1}{\tau(B(y, r))} \int_{B(y, r)} |w(z) - w(y)| d\tau(z) \longrightarrow 0. \quad (10.2y)$$

Since $0 < w(y) < \infty$, normalizing the two restrictions on $B(y, r)$ shows directly that their total-variation distance is at most

$$\frac{2 \int_{B(y, r)} |w(z) - w(y)| d\tau(z)}{\int_{B(y, r)} w(z) d\tau(z)},$$

which tends to zero. Scenery rescaling preserves this bound.

For (10.2x), the entropy-dimension theorem (Volume I, C-I.4.12) gives, on each fixed translated dyadic grid,

$$H(\rho, \mathcal{D}_n) = Dn \log 2 + o(n), \quad \int H(\rho_h, \mathcal{D}_n) d\zeta(h) = Dn \log 2 + o(n). \quad (10.2z)$$

The integral limit follows by dominated convergence: all components have support in one compact set, so their level- n entropies are at most $dn \log 2 + O(1)$. The entropy chain rule turns the second equality into the second hypothesis of Corollary IV.4.6. That corollary and Lemma IV.4.7 give mean scenery distance zero between ρ and ρ_h . Let G be the fixed Borel set where ρ generates $P^\square$. Since ρ is a USM, $\rho(G^c) = 0$. The mixture identity gives

$$0 = \rho(G^c) = \int \rho_h(G^c) d\zeta(h).$$

Thus for ζ -almost every h , G is ρ_h -conull. This fixed-event Fubini argument, not a universal constituent-AC assertion, supplies the required typical points. The average Prokhorov distance of the two scenery states tends to zero. Split times into those whose state distance is at most η and the complement. The Prokhorov definition bounds the distance of their empirical laws by η plus the complementary time fraction, which is at most the average state distance divided by η . Letting $T \rightarrow \infty$ and then $\eta \downarrow 0$, followed by the triangle inequality and dominated convergence, gives (10.2x). $\square$

The next theorem is the no-separation result of Bárány–Käenmäki–Pyörälä–Wu [4].

Theorem IV.4.12 (*uniform scaling in dimension one and for similarities*).

Let Φ be a conformal IFS on $\mathbb{R}^d$ with attractor K and let μ be a quasi-Bernoulli measure on K . If either

- (a) $d = 1$, or
- (b) every map in Φ is a similarity,

then μ is uniformly scaling and generates an ergodic fractal distribution.

Source. Bárány–Käenmäki–Pyörälä–Wu, Theorem 1.1, p. 3 [4]. The following proof makes the overlap and orientation steps explicit.

Proof. The projected measure is exact dimensional by Lemma IV.4.3, whose invariant representative is covered by the projection-entropy theorem of Volume I. Write $D = \dim \mu$. If $D = 0$, take a typical x and set $F_x(t) = -\log \mu(B(x, e^{-t}))$. Then $F_x(t)/t \rightarrow 0$. For every fixed $s > 0$, telescoping as in (10.2p) gives

$$\frac{1}{T} \int_0^T -\log \mu_{x,t}^\square(B(0, e^{-s})) dt \longrightarrow 0. \quad (10.2aa)$$

Markov's inequality says that the scenery assigns mass at least $1 - \varepsilon$ to $B(0, e^{-s})$ at a density-one set of times. Given a neighborhood of δ_0 in the weak topology, first choose s and then ε so that this mass condition puts a state in that neighborhood. Hence the empirical scenery distributions converge to the EFD δ_{δ_0} .

Assume $D > 0$. For each tangent law at a point in the common conull set of Lemmas IV.4.8–IV.4.10, take the conull set of its scale-ergodic components for which those

lemmas hold. Choose any two P_1, P_2 from these sets, allowing them to come from different tangent laws. Choose for each $i \in \{1, 2\}$ a P_i -typical full-space state $\hat{\xi}_i$ whose unit restriction $\xi_i = \hat{\xi}_i^\square$ has the representation $\xi_i \sim_{C_*} \rho_i$, where

$$\rho_i = \frac{\left(\int h_* \mu d\zeta_i(h) \right) |_W}{\int h_* \mu(W) d\zeta_i(h)}, \quad W = (-1, 1)^d. \quad (10.2ab)$$

By Theorem IV.3.6, $\hat{\xi}_i$ is a USM generating P_i . The same is true of ξ_i at its almost every interior point, since its sufficiently small sceneries there coincide with those of $\hat{\xi}_i$. Lemma IV.4.11 transfers this to ρ_i . By Proposition IV.4.10, ρ_i and its component restrictions all have dimension D . Applying (10.2x) gives, for the reweighted mixing probability $\zeta_{i,W}$ of (10.2t),

$$\int \int d_{\text{LP}} \left(P_i^\square, \frac{1}{T} \int_0^T \delta_{(h_* \mu)_{h(x),t}^\square} dt \right) 1_{h^{-1}(W)}(x) d\mu(x) d\zeta_{i,W}(h) \longrightarrow 0. \quad (10.2ac)$$

Indeed a restriction to the open unit ball has the same small sceneries at each of its interior points. In (10.2ac) the integration against μ with the indicator is dominated by integration against the normalized component restriction ρ_h ; this is why the reweighted law $\zeta_{i,W}$ is sufficient.

We next arrange that the derivatives of the maps in (10.2ab) range over the needed orthogonal directions, without changing the distribution they generate. Let O_i be the orthogonal part of Df_i in dimension one (a sign), or of f_i in the similarity case, for symbols with $\bar{\mu}([i]) > 0$. Let $G \leq O(d)$ be the compact group generated by these parts. Every map in the tangent mixtures of Lemma IV.4.8 is a limit of rescaled positive-weight compositions. Hence its derivative sign in dimension one, or its fixed orthogonal part in the similarity case, belongs to G . The closure of the positive-word semigroup is already G : positive powers of each element in a compact group approach the identity along a subsequence, so its inverse is also approximated by positive powers. For every $\varepsilon > 0$ there is a finite complete prefix code $\mathcal{A}_\varepsilon$ with

$$\{O_u : u \in \mathcal{A}_\varepsilon\} \varepsilon\text{-dense in } G. \quad (10.2ad)$$

To construct it, choose finitely many target directions forming an $\varepsilon/2$ -net and distinct words v_j of a common length. For each j append a word w_j whose orthogonal product makes $O_{v_j w_j}$ lie within $\varepsilon/2$ of the j th target. The $v_j w_j$ are incomparable. Complete them to a finite prefix code by adding all unrepresented words at the maximum chosen length. The assumption $D > 0$ ensures at least two symbols have positive cylinder weight, and quasi-Bernoulli gives positive weight to every word in this support alphabet.

Set $p_u = \bar{\mu}([u])$ and

$$\nu_\varepsilon = \sum_{u \in \mathcal{A}_\varepsilon} p_u (f_u)_* \mu. \quad (10.2ae)$$

Quasi-Bernoulli comparison of the symbolic tail laws gives $\nu_\varepsilon \sim_C \mu$. Replace the mixture

in (10.2ab) by

$$\rho_{i,\varepsilon} = \frac{\left(\int h_* \nu_\varepsilon d\zeta_i(h) \right)|_W}{\int h_* \nu_\varepsilon(W) d\zeta_i(h)}. \quad (10.2af)$$

It is boundedly equivalent to ρ_i , hence is a USM generating P_i and has exact dimension D . Its components are the positive restrictions of $(h \circ f_u)_* \mu$, also of exact dimension D . Lemma IV.4.11 therefore gives the analogue of (10.2ac) for the component maps $h \circ f_u$.

Lemma IV.4.8 gives positive mixing mass to maps h with $h(V) \subset W$. Every $h \circ f_u$ then has its whole image of K inside the unit ball. For each i , the preceding transfer therefore holds with no indicator for a positive-mass family $\mathcal{H}_{i,\varepsilon}^{\text{int}}$ of these component maps. Its orthogonal parts are ε -dense in G : for each interior h , the parts of $h \circ f_u$ run through $O_h O_u$, and left multiplication by $O_h \in G$ preserves density.

Choose one sequence $T_j \rightarrow \infty$ along which the two nonnegative integrals of scene-distribution errors over these interior families are summable. Fubini and Borel–Cantelli imply that, for almost every interior pair (h, g) , at μ -almost every common point x ,

$$\langle h_* \mu \rangle_{h(x), T_j}^\square \longrightarrow P_1^\square, \quad \langle g_* \mu \rangle_{g(x), T_j}^\square \longrightarrow P_2^\square. \quad (10.2ag)$$

Take a further subsequence along which the restricted scenery empirical laws of μ at such an x converge to $Q_x^\square$. The tangent-distribution theorem IV.2.66 makes this a restricted FD; let Q_x be its unique invariant full-space lift. The smooth covariance theorem IV.3.17, applied locally to h and g , gives

$$P_1^\square = ((Dh(x))^* Q_x)^\square, \quad P_2^\square = ((Dg(x))^* Q_x)^\square.$$

Uniqueness of the invariant full-space lift in Lemma IV.2.15 then yields

$$P_1 = (Dh(x))^* Q_x, \quad P_2 = (Dg(x))^* Q_x, \quad P_1 = (Dh(x) Dg(x)^{-1})^* P_2. \quad (10.2ah)$$

The positive scalar part of a conformal derivative acts trivially on a scale-invariant distribution. In dimension one the remaining orthogonal part is its sign; for a similarity it is its fixed orthogonal part. Thus the final relation in (10.2ah) is $P_1 = (O_h O_g^{-1})^* P_2$.

For arbitrary $\varepsilon > 0$, choose positive-mass subfamilies of the two interior mixing laws whose orthogonal parts lie within 2ε of the same element of G . The almost-everywhere relation (10.2ah) holds for a pair in their product. Consequently $P_1 = O^* P_2$ for some $\|O - I\| \leq 4\varepsilon$. The action of $O(d)$ on full-space distributions, followed by restriction to the fixed cube, is continuous at the identity by the boundary-null Lemma IV.3.14. Hence $P_1^\square = P_2^\square$ as $\varepsilon \downarrow 0$. Unique lifting gives $P_1 = P_2$ in full space.

The pair was arbitrary among the qualifying conull sets of components of all tangent laws at typical points. Therefore the ergodic decomposition of every such law is concentrated on one common value P ; every such tangent law equals P . This proves that μ is a USM, and P is ergodic by its construction as an ergodic component. $\square$

Reader note. Important scope: there is no separation hypothesis; in $d > 1$ the theorem does not cover arbitrary conformal IFSs, only the similarity case; and quasi-Bernoulli includes projected Bernoulli measures even with overlaps.

Source theorem: Orthogonal mixtures of tangent distributions

Let Φ be a conformal IFS on $\mathbb{R}^d$ with attractor K and let μ be a quasi-Bernoulli measure on K . Then there exists an EFD P such that for μ -a.e. x and every tangent distribution Q of μ at x , there is a probability ζ on the orthogonal group $O(d)$ satisfying

$$Q = \int_{O(d)} O_* P d\zeta(O). \quad (10.2)$$

Source boundary

Bárány–Käenmäki–Pyörälä–Wu, arXiv:2308.11399v1, Theorem 1.2, p. 3. The proof remains in the cited source.

Reader note. Scope warning: the orthogonal-mixture theorem above is not the assertion that μ itself is a USM in every dimension. It describes the ergodic components and orthogonal mixing of tangent distributions.

Source theorem: Subsequential scaling after a conformal change of coordinates

Let Φ be a conformal IFS on $\mathbb{R}^d$ with attractor K and let μ be a quasi-Bernoulli measure on K . Then there exist an EFD P , a sequence $n_k \rightarrow \infty$, and a map h in the source class $C^{1+\alpha}(\mathbb{R}^d)$, injective and conformal on its defining neighborhood, such that for $h_*\mu$ -a.e. x ,

$$\frac{1}{n_k} \int_0^{n_k} \delta_{(h_*\mu)_{x,t}^\square} dt \longrightarrow P^\square. \quad (10.3)$$

Source boundary

Bárány–Käenmäki–Pyörälä–Wu, arXiv:2308.11399v1, Theorem 1.3, p. 3. The proof remains in the cited source.

Reader note. Scope warning: this is subsequential uniform scaling after a source-class smooth conformal coordinate change; it is not a theorem for an arbitrary C^1 map.

Define on $\mathbb{R}$

$$f_0(x) = \frac{x}{2}, \quad f_1(x) = \frac{x}{2} + \frac{1}{4}, \quad f_2(x) = \frac{x}{2} + \frac{1}{2}. \quad (10.4)$$

Let

$$p = \left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right), \quad \bar{\mu} = p^{\mathbb{N}}.$$

Lemma IV.4.13 (*attractor and overlap*).

The attractor is $K = [0, 1]$, and the system has genuine interval overlap:

$$f_0(K) \cap f_1(K) = [1/4, 1/2], \quad f_1(K) \cap f_2(K) = [1/2, 3/4].$$

Proof. Each image of $[0, 1]$ is one of the three intervals

$$[0, 1/2], \quad [1/4, 3/4], \quad [1/2, 1].$$

Their union is $[0, 1]$, so Hutchinson invariance gives the attractor $K = [0, 1]$. The two displayed intersections are immediate from these intervals and both have positive length. $\square$

Lemma IV.4.14 (*regularity and symbolic measure class*).

The family $\Phi_A = (f_0, f_1, f_2)$ is a similarity IFS and $\bar{\mu}$ is Bernoulli, hence quasi-Bernoulli with constant $C = 1$. The projected measure

$$\mu_A = \Pi_* \bar{\mu}$$

is therefore a quasi-Bernoulli measure in the BKPW sense.

Proof. Every f_i is an affine similarity of contraction ratio $1/2$. For the product symbolic law, cylinder probabilities multiply exactly under concatenation, so the quasi-Bernoulli inequalities hold with $C = 1$. Projection by the coding map is precisely the BKPW projected quasi-Bernoulli construction. $\square$

Lemma IV.4.15 (*explicit distribution and non-atomicity*).

Let U, V be independent uniform random variables on $[0, 1]$. Then

$$\mu_A = \text{Law}\left(\frac{U + V}{2}\right). \quad (10.5)$$

In particular μ_A is absolutely continuous and non-atomic, with density

$$\rho(x) = \begin{cases} 4x, & 0 \leq x \leq 1/2, \\ 4(1 - x), & 1/2 \leq x \leq 1. \end{cases} \quad (10.6)$$

Proof. Let B_n, C_n be independent Bernoulli($1/2$) bits and put $D_n = B_n + C_n$. Then

$$\mathbb{P}(D_n = 0, 1, 2) = \left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right).$$

The random fixed-point series for (10.4) is

$$X = \sum_{n \geq 1} 2^{-(n-1)} \frac{D_n}{4} = \sum_{n \geq 1} \frac{B_n + C_n}{2^{n+1}} = \frac{U + V}{2},$$

where

$$U = \sum_{n \geq 1} \frac{B_n}{2^n}, \quad V = \sum_{n \geq 1} \frac{C_n}{2^n}$$

are independent uniform $[0, 1]$ variables. Convolution of the two uniform densities is triangular; scaling by $1/2$ gives (10.6), hence no atom occurs. $\square$

Theorem IV.4.16 (*the overlapping measure μ_A is uniformly scaling*).

Proof. The preceding lemmas show that Φ_A is a similarity IFS and that μ_A is a projected quasi-Bernoulli measure. The uniform-scaling source theorem therefore applies. The explicit interval overlap also confirms that no separation hypothesis has entered the argument.

□

Additional checked fact. By (10.6), μ_A is equivalent to Lebesgue measure on every compact subinterval of $(0, 1)$, and has local dimension one at μ_A -a.e. point. This is a direct computation, not a conclusion imported from the source theorem on uniform scaling.

Fix $\varepsilon = 1/10$ and define

$$h(x) = x + \varepsilon x(1 - x). \quad (10.7)$$

On $U = (-1, 2)$,

$$h'(x) = 1.1 - 0.2x, \quad 0.7 < h'(x) < 1.3. \quad (10.8)$$

Thus h is a real-analytic increasing diffeomorphism from U onto the open interval $V = h(U)$.

Define

$$g_i = h \circ f_i \circ h^{-1}, \quad i = 0, 1, 2, \quad (10.9)$$

with f_i from Model A, and let $\Phi_B = (g_0, g_1, g_2)$.

Lemma IV.4.17 (*conformal contraction and genuine nonlinearity*).

The family Φ_B is a real-analytic conformal IFS on V . For every $y = h(x)$,

$$|g'_i(y)| = \frac{1}{2} \frac{h'(f_i(x))}{h'(x)} \leq \frac{1}{2} \frac{1.3}{0.7} = \frac{13}{14} < 1. \quad (10.10)$$

Moreover the system is genuinely nonlinear: for example

$$g'_0(h(x)) = \frac{1}{2} \frac{h'(x/2)}{h'(x)}$$

depends on x .

Proof. Analyticity and injectivity follow from conjugating the affine contractions by the analytic diffeomorphism h . The derivative formula is the chain rule, and (10.8) gives the uniform bound (10.10). The displayed expression for g'_0 is nonconstant, so the system is not affine-linear. □

Lemma IV.4.18 (*attractor and overlap*).

The attractor is

$$K_B = h([0, 1]) = [0, 1],$$

and

$$g_0(K_B) \cap g_1(K_B) = h([1/4, 1/2]),$$

which is a nondegenerate interval.

Proof. Conjugacy sends the Model-A attractor and every intersection to their h -images. Since $h(0) = 0$ and $h(1) = 1$, we have $K_B = [0, 1]$; strict monotonicity of h preserves nondegeneracy of the overlap interval. $\square$

Definition IV.4.19 (*the Model B measure*).

Use the same Bernoulli symbolic law

$$\bar{\mu} = p^{\mathbb{N}}, \quad p = \left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right),$$

and set

$$\mu_B = (\Pi_B)_* \bar{\mu}.$$

Lemma IV.4.20 (*measure class and non-atomicity*).

The measure μ_B is Bernoulli/quasi-Bernoulli in the BKPW sense and

$$\mu_B = h_* \mu_A. \tag{10.11}$$

Hence it is non-atomic; indeed it is absolutely continuous.

Proof. The coding maps satisfy $\Pi_B = h \circ \Pi_A$ because the two IFSs are conjugate. Thus (10.11) holds. The symbolic measure is Bernoulli with $C = 1$, and an injective C^1 diffeomorphism sends the atomless absolutely continuous measure μ_A to an atomless absolutely continuous measure. $\square$

Theorem IV.4.21 (*the nonlinear overlapping measure μ_B is uniformly scaling*).

Proof. The three preceding lemmas verify, respectively, one-dimensional conformal $C^{1+\alpha}$ regularity with strict contraction, genuine interval overlap, and the projected quasi-Bernoulli property. The one-dimensional clause of the uniform-scaling source theorem applies without a separation hypothesis. $\square$

Remark. Φ_B is smoothly conjugate to a linear IFS and therefore is not an example of the “totally non-linear” alternative in the normality theorem below. It is designed to illustrate the scaling theorem itself.

Theorem IV.4.22 (*two explicit overlapping uniformly scaling systems*).

Models A and B are explicit overlapping systems to which the uniform-scaling source theorem applies. Every local hypothesis has been verified above, and no separation assumption has been inserted.

Proof. For Model A, the first three lemmas verify conformality, overlap, quasi-Bernoulli structure, and non-atomicity. The next three verify the same hypotheses for Model B. Apply the uniform-scaling source theorem stated at the start of the chapter; it imposes no separation condition. $\square$

Definition IV.4.23 (*asymptotic contraction ratio*).

For a contracting one-dimensional source map f with fixed point p , the literal BKPW convention is

$$\lambda(f) = f'(p) \in (-1, 1) \setminus \{0\}.$$

This is a signed derivative, not its absolute value. The positive contraction size $|\lambda(f)|$ is a different quantity. For a word $\mathbf{i}$, $f_{\mathbf{i}}$ denotes the composition and $\lambda(f_{\mathbf{i}})$ is its derivative at its own fixed point; it is not in general the product of the generators' fixed-point derivatives. The source defines λ at the end of printed p. 3 of arXiv:2308.11399v1. No positivity is imposed by the general one-dimensional conformal-IFS definition.

Definition IV.4.24 (*the two normality conventions*).

For a real number t , write $\{t\} = t - [t]$ for its fractional part, where $[t]$ is the greatest integer not exceeding t . Let $\beta > 1$ and let λ_{β} denote the absolutely continuous invariant probability of the piecewise map

$$T_{\beta} : [0, 1) \rightarrow [0, 1), \quad T_{\beta}(x) = \{\beta x\}.$$

The literal definition printed in BKPW v1, p. 4, calls x β -normal when

$$\frac{1}{N} \sum_{n=1}^N \delta_{\{\beta^n x\}} \longrightarrow \lambda_{\beta}. \quad (10.11a)$$

Here we interpret its invariant target on the half-open representatives $[0, 1)$; noninteger multiplication is not an endomorphism of $\mathbb{R}/\mathbb{Z}$.

By contrast, *iterated* T_{β} -normality means

$$\frac{1}{N} \sum_{n=1}^N \delta_{T_{\beta}^n(\{x\})} \longrightarrow \lambda_{\beta}. \quad (10.11b)$$

This is the convention developed in Chapter IV.10. For an integer $\beta = b \geq 2$, induction gives $T_b^n(\{x\}) = \{b^n x\}$, so the two sequences agree and their target is Lebesgue probability. They need not agree for noninteger β : if $\beta = (1 + \sqrt{5})/2$ and $x = 1/\beta$, then $T_{\beta}^2 x = 0$, while $\{\beta^2 x\} = \beta - 1 > 0$. No equivalence of the two normality conditions is asserted for a noninteger base.

Definition IV.4.25 (*Pisot number*).

A Pisot number is a real algebraic integer $\beta > 1$ whose other Galois conjugates all have absolute value < 1 .

Definition IV.4.26 (*arithmetic independence from β*).

In the positive-derivative subcase of a one-dimensional conformal IFS $\Phi = (f_i)$, the literal source definition of arithmetic independence from $\beta > 1$ is that zero is an accumulation point of

$$\{\log \lambda(f_i) + n \log \beta : i \in \Lambda^*, n \in \mathbb{N}\}. \quad (10.12)$$

A sufficient condition is

$$\frac{\log \lambda(f_i)}{\log \beta} \notin \mathbb{Q}$$

for some generator i . The positivity qualification makes every displayed real logarithm defined. It is a restriction of this illustrative source presentation, not an additional hypothesis printed in the general BKPW IFS definition. Replacing λ by $|\lambda|$, or replacing orientation-reversing generators by iterates, requires a separate statement and argument; neither transposition is silently made here.

Definition IV.4.27 (*total nonlinearity*).

The IFS Φ is totally non-linear if there is no invertible C^1 map g for which

$$(g \circ f_i \circ g^{-1})_i$$

consists entirely of affine-linear maps.

Source theorem: An illustrative normality assertion and its literal-convention boundary

Suspicious source formula IV.4.28. BKPW, arXiv:2308.11399v1, Theorem 1.4, printed p. 4, makes the following assertion, using its power-sequence convention (10.11a):

Let Φ be a conformal IFS on $\mathbb{R}$ with attractor K , let μ be a non-atomic quasi-Bernoulli measure on K , and let $\beta > 1$ be Pisot. If Φ is arithmetically independent of β , then for every C^1 diffeomorphism h , $h_*\mu$ -a.e. x is β -normal.

Furthermore, if Φ is totally non-linear and consists of real-analytic maps, then the conclusion holds for every Pisot $\beta > 1$.

Source boundary

The exact version examined is <https://arxiv.org/pdf/2308.11399v1>, submitted 22 August 2023 (the first page is dated 23 August 2023). The complete 25-page article, including its standing definitions and Section 4, is the scope of this comparison; no published-version concordance is asserted. Its general IFS hypotheses are printed on pp. 2 and 6, its projected quasi-Bernoulli convention on p. 2, its signed derivative on p. 3, and its arithmetic-independence and normality definitions on p. 4. The numbered assertion above is illustrative source material, not a proved or usable local input.

Example IV.4.29 disproves its first conclusion if the printed power sequence and invariant target are read literally. It does not refute the intended iterated-map normality result.

Example IV.4.29 (*a literal power-sequence obstruction*).

The positive-derivative, non-atomic, quasi-Bernoulli, Pisot, and arithmetic-independence hypotheses of the first source assertion above can all hold while its literal conclusion (10.11a) fails at almost every point.

Proof. Let $U = (-1, 2)$ and put

$$f_j(x) = (x + j)/3 \quad (j = 0, 1, 2), \quad \beta = (1 + \sqrt{5})/2.$$

All three maps are injective real-analytic maps $U \rightarrow U$, with nonzero constant derivative $1/3$ and common Lipschitz constant $1/3 < 1$. Their derivatives are Hölder continuous for every exponent, so all the one-dimensional $C^{1+\alpha}$ conformal-IFS hypotheses are met. The three-symbol alphabet is allowed by the source's printed $\#\Lambda \geq 2$ in Section 2.1, p. 6. Their attractor is $K = [0, 1] \Subset U$, since the three images of K are its three closed thirds.

Give the symbolic space $\{0, 1, 2\}^{\mathbb{N}}$ the fair product law $\bar{\mu}$. For every two finite words $\mathbf{i}, \mathbf{j}$,

$$\bar{\mu}([\mathbf{ij}]) = 3^{-|\mathbf{i}|-|\mathbf{j}|} = \bar{\mu}([\mathbf{i}])\bar{\mu}([\mathbf{j}]),$$

so the quasi-Bernoulli inequalities hold with $C = 1$. Its coding projection has mass 3^{-k} on each half-open triadic interval of length 3^{-k} ; the exceptional endpoint codings form a countable set of product-law measure zero. Hence the projected probability μ is Lebesgue probability m on $[0, 1]$. In particular it is non-atomic and is quasi-Bernoulli in the precise projected source sense. Take the globally defined C^1 diffeomorphism $h(x) = x$.

The number β is a Pisot number: it is an algebraic integer with minimal polynomial $t^2 - t - 1$, and its other conjugate $\beta' = (1 - \sqrt{5})/2 = -1/\beta$ has absolute value less than 1. Moreover $\log 3 / \log \beta$ is irrational. Otherwise there would be positive integers a, b with $\beta^a = 3^b$; conjugating in $\mathbb{Q}(\sqrt{5})$ would give $(\beta')^a = 3^b$, impossible after taking absolute values.

For a word of length k , its contraction derivative is 3^{-k} . To verify the source's accumulation-point condition directly, put $\alpha = \log 3 / \log \beta$. Pigeonholing the $M + 1$ fractional parts $0, \alpha, \dots, M\alpha$ into M equal subintervals gives integers $1 \leq k \leq M$ and n with $0 < |k\alpha - n| \leq 1/M$. As $M \rightarrow \infty$, the selected k have an unbounded subsequence, since finitely many k have a positive minimum distance from an integer. On that subsequence $n \geq 1$ eventually. Consequently the nonzero numbers

$$-k \log 3 + n \log \beta$$

tend to zero, exactly as required by (10.12).

We now prove the almost-everywhere power-sequence limit without any normality theorem. For a nonzero integer ℓ define

$$A_{N,\ell}(x) = \frac{1}{N} \sum_{n=1}^N \exp(2\pi i \ell \beta^n x).$$

For $v > u \geq 1$, direct integration gives

$$\left| \int_0^1 \exp(2\pi i \ell (\beta^v - \beta^u)x) dx \right| \leq \frac{1}{\pi |\ell| (\beta^v - \beta^u)} \leq \frac{\beta^{-v}}{\pi |\ell| (1 - \beta^{-1})}.$$

Expanding the square and summing the off-diagonal terms therefore gives a finite constant $C_{\beta, \ell}$ such that

$$\int_0^1 |A_{N, \ell}(x)|^2 dx \leq \frac{1}{N^2} \left(N + \frac{2}{\pi |\ell| (1 - \beta^{-1})} \sum_{v=2}^{\infty} (v-1) \beta^{-v} \right) \leq \frac{C_{\beta, \ell}}{N}.$$

For each $\varepsilon > 0$, Chebyshev's inequality makes

$$\sum_{j=1}^{\infty} m\{x : |A_{j^2, \ell}(x)| > \varepsilon\} < \infty.$$

Borel–Cantelli, first for $\varepsilon = 1/r$ with $r \in \mathbb{N}$, then for all nonzero integers ℓ , gives a common full-measure set on which $A_{j^2, \ell}(x) \rightarrow 0$ for every such ℓ . For $j^2 \leq N < (j+1)^2$, boundedness of each summand gives

$$|A_{N, \ell}(x) - A_{j^2, \ell}(x)| \leq \frac{2(N - j^2)}{N} \leq \frac{2(2j+1)}{j^2} \rightarrow 0.$$

Thus $A_{N, \ell}(x) \rightarrow 0$ for every nonzero integer ℓ on that same full-measure set. Uniform approximation of continuous periodic functions by trigonometric polynomials now shows

$$\frac{1}{N} \sum_{n=1}^N \delta_{\{\beta^n x\}} \rightarrow m$$

as probabilities on $\mathbb{R}/\mathbb{Z}$: the limit for each trigonometric polynomial is its constant Fourier coefficient, and a uniform approximation changes both averages and integrals by at most its uniform error.

Finally $m \neq \lambda_{\beta}$. Indeed choose $0 < a < \beta - 1$. Then

$$T_{\beta}^{-1}([0, a)) = [0, a/\beta) \cup [1/\beta, (1+a)/\beta), \quad m(T_{\beta}^{-1}([0, a))) = \frac{2a}{\beta} \neq a.$$

The two intervals lie in $[0, 1)$ because $1 + a < \beta < 2$. Lebesgue probability is therefore not T_{β} -invariant, whereas λ_{β} is invariant by its definition. The almost-everywhere power-sequence limit is m , not λ_{β} , contradicting only the literal formulation of the source assertion. The identification $[0, 1) \rightarrow \mathbb{R}/\mathbb{Z}$ is a Borel bijection, so the distinct probability targets remain distinct on the circle; convergence to λ_{β} on the representatives would also imply convergence to its circle push-forward. $\square$

Relation to the cited proof and later chapters. BKPW Section 4, printed pp. 17–19, invokes Hochman–Shmerkin's Theorems 1.1–1.2. In their arXiv:1302.5792v3,

printed p. 4, β -normality is genericity for the iterates of T_β and the Parry target. Their separate Corollary 1.3 concerns the power sequence modulo one with the Lebesgue target. These are not interchangeable formulations. The source proof is thus not evidence for the literal power-sequence/Parry-target combination above. This comparison neither proves nor refutes the intended iterated-map BKPW theorem. Chapter IV.10 develops its normality mechanism independently; it does not use this illustrative card as an input.

Source theorem: Retained comparison target: iterated-map normality

To retain the mathematical aim of the source argument, we separately record its intended iterated-map formulation. Let Φ be a positive-derivative conformal IFS on the line, let μ be a non-atomic projected quasi-Bernoulli probability, and let $\beta > 1$ be Pisot. Under the arithmetic-independence condition (10.12), the comparison target is that for every C^1 diffeomorphism h ,

$$\frac{1}{N} \sum_{n=1}^N \delta_{T_\beta^n(\{x\})} \longrightarrow \lambda_\beta \quad \text{for } h_*\mu\text{-almost every } x.$$

The totally non-linear real-analytic alternative asks for this conclusion for every Pisot $\beta > 1$.

Source boundary

This is an illustrative comparison target, not a proved local input. Its provenance is the iterated-map interpretation of BKPW v1 Section 4 and its Hochman–Shmerkin references. The convention check above does not supply a full proof of this interpretation, nor does the counterexample refute it. Its missing local proof remains explicitly in Category 3; no later result invokes this card. The unrestricted signed-log interpretation is also not supplied by the positive-derivative subcase stated here.

Definition IV.4.30 (*resonance and dissonance*).

Exact-dimensional probabilities μ, ν on $\mathbb{R}$ resonate if

$$\dim(\mu * \nu) < \min\{1, \dim \mu + \dim \nu\}. \quad (10.13)$$

Otherwise they dissonate.

Definition IV.4.31 (*independent conformal IFSs*).

Let

$$\Psi = (f_i)_{i \in \Gamma}, \quad \Phi = (g_j)_{j \in \Lambda}$$

be positive-derivative one-dimensional conformal IFSs. In this real-log source subcase they are independent if zero is an accumulation point of

$$\{\log \lambda(f_i) - \log \lambda(g_j) : i \in \Gamma^*, j \in \Lambda^*\}. \quad (10.14)$$

A sufficient condition is

$$\frac{\log \lambda(f_i)}{\log \lambda(g_j)} \notin \mathbb{Q}$$

for some pair of generators.

Source theorem: Illustrative non-resonance in the positive-derivative source subcase

Let Ψ, Φ be one-dimensional conformal IFSs with attractors K_Ψ, K_Φ . If Ψ and Φ are independent, then for every pair of quasi-Bernoulli measures μ on K_Ψ and ν on K_Φ ,

$$\dim(\mu * \nu) = \min\{1, \dim \mu + \dim \nu\}. \quad (10.15)$$

Source boundary

Bárány–Käenmäki–Pyörälä–Wu, arXiv:2308.11399v1, Theorem 1.5, pp. 4–5. The proof remains in the cited source. This card uses the positive-derivative subcase in which the printed real logarithms are defined; it does not certify an absolute-value or orientation-reversing translation.

Scope warning. This is a measure-convolution theorem. It does not automatically give a theorem about every arithmetic sum of the underlying sets.

Definition IV.4.32 (*diagonal affine IFS*).

Let

$$F_i(z) = A_i z + a_i, \quad A_i = \text{diag}(\rho_i, \lambda_i),$$

with $0 < |\rho_i|, |\lambda_i| < 1$.

Definition IV.4.33 (*rectangular strong separation*).

The IFS satisfies rectangular strong separation if

$$F_i([-1, 1]^2) \cap F_j([-1, 1]^2) = \emptyset \quad (i \neq j).$$

Definition IV.4.34 (*independent eigenvalues*).

For a word $\mathbf{i}$, write $\rho_{\mathbf{i}}, \lambda_{\mathbf{i}}$ for the products of the corresponding diagonal entries. The positive-size version of independent eigenvalues means that zero is an accumulation point of

$$\{\log |\lambda_{\mathbf{i}}| - \log |\rho_{\mathbf{j}}| : \mathbf{i}, \mathbf{j} \in \Lambda^*\}. \quad (10.16)$$

The literal v1 source writes the logarithms without absolute values. Consequently the source card below is presented only when all diagonal entries are positive, where the two formulas agree. The displayed absolute-value definition is not a certified signed extension of that source theorem. A common sufficient condition for the displayed accumulation-point condition is an irrational log-ratio of the two positive sizes for one generator.

Definition IV.4.35 (*self-affine Bernoulli measure and simple spectrum*).

Choose $p_i > 0$ with $\sum_i p_i = 1$, and let μ be the unique probability satisfying

$$\mu = \sum_i p_i (F_i)_* \mu.$$

Its two diagonal Lyapunov exponents are

$$\chi_x = - \sum_i p_i \log |\rho_i|, \quad \chi_y = - \sum_i p_i \log |\lambda_i|.$$

The spectrum is simple when $\chi_x \neq \chi_y$.

Definition IV.4.36 (*principal projection*).

A principal orthogonal projection is projection onto one of the coordinate axes. A non-principal orthogonal projection is any one-dimensional orthogonal projection different from those two coordinate projections.

Source theorem: Projections of diagonal self-affine measures

In the illustrative positive-diagonal subcase, let Φ be a diagonal affine IFS on $\mathbb{R}^2$ satisfying rectangular strong separation and having independent eigenvalues. Let μ be a self-affine measure on its attractor with simple Lyapunov spectrum. Then for every non-principal orthogonal projection π ,

$$\dim(\pi_* \mu) = \min\{1, \dim \mu\}. \quad (10.17)$$

Source boundary

Bárány–Käenmäki–Pyörälä–Wu, arXiv:2308.11399v1, Theorem 1.6, p. 5. The proof remains in the cited source; no conclusion for negative diagonal entries is imported by this card.

Companion result. In this positive-diagonal source subcase, BKPW Proposition 6.1 additionally states that under rectangular strong separation every such self-affine measure is uniformly scaling and generates an EFD; in the simple-spectrum case the EFD has additional anisotropic invariances.

The additional source cards are illustrative material, not substitutes for proofs in the later chapters. The literal normality assertion is rejected as stated above; the other application cards retain external proof boundaries and their explicit positive-log scope. The verifications and model deductions made in this chapter are proved locally.

The logical boundaries are important. The normality source card above is not a substitute for the independent normality arguments of Chapter IV.10; the non-resonance theorem does not prove the set-sum results of Chapter IV.9; and the projection theorem does not replace any projection theorem from Volume III. Finally, the absence of separation in the first scaling theorem cannot be transferred to the diagonal projection theorem, whose rectangular strong separation hypothesis is explicit.

Part II

Arithmetic Intersections and Embeddings

Chapter IV.5

Multiplicatively Independent Dynamics

Invariant sets, symbolic approximation, and homogeneous models

The arithmetic branch begins with exact symbolic models for closed $\times m$ -invariant sets and homogeneous outer approximations.

5.1 Arithmetic invariant sets and symbolic approximation

1. Definitions and endpoint conventions

Definition IV.5.1 (*forward invariant closed set*).

A closed set $A \subset \mathbb{T}$ is T_m -invariant if

$$T_m(A) \subset A. \quad (11.1)$$

Equality is not required. Every statement below uses only forward invariance.

Definition IV.5.2 (*m -adic cylinders and occupied language*).

For a word $u = (u_1, \dots, u_n) \in \{0, \dots, m-1\}^n$, put

$$[u]_m = \left[\sum_{j=1}^n u_j m^{-j}, \sum_{j=1}^n u_j m^{-j} + m^{-n} \right) \subset \mathbb{T}.$$

Let

$$L_n(A) = \{u : [u]_m \cap A \neq \emptyset\}, \quad N_n(A) = \#L_n(A). \quad (11.2)$$

The half-open convention removes ambiguity in $N_n(A)$. A point with two ordinary base- m expansions may have two symbolic addresses, but it belongs to one half-open geometric cylinder at each level. All symbolic statements below are therefore made either in the full preimage $\pi_m^{-1}(A)$, where both addresses are harmlessly retained, or in the geometric language $L_n(A)$.

Definition IV.5.3 (*symbolic coding and saturation*).

Let

$$\Sigma_m = \{0, \dots, m-1\}^{\mathbb{N}}$$

with left shift σ , and define

$$\pi_m(\omega) = \sum_{j \geq 1} \omega_j m^{-j} \pmod{1}. \quad (11.3)$$

Then $T_m \pi_m = \pi_m \sigma$. For $A \subset \mathbb{T}$, define

$$X_A = \pi_m^{-1}(A). \quad (11.4)$$

A symbolic set X is called *coding-saturated* when

$$X = \pi_m^{-1}(\pi_m X).$$

Lemma IV.5.4 (*exact symbolic/set correspondence with saturation*).

If A is closed and T_m -invariant, then X_A is closed, shift invariant, and coding-saturated. Conversely, if $X \subset \Sigma_m$ is closed, shift invariant, and coding-saturated, then $A = \pi_m X$ is closed and T_m -invariant, and $X = X_A$.

Proof. Continuity of π_m gives closedness. If $\omega \in X_A$, then

$$\pi_m(\sigma\omega) = T_m(\pi_m\omega) \in A,$$

so $\sigma\omega \in X_A$. Saturation is tautological. Conversely, $\pi_m X$ is compact and

$$T_m(\pi_m X) = \pi_m(\sigma X) \subset \pi_m X.$$

Saturation gives the final identity. □

Remark IV.5.5 (*RM01*).

Without saturation the coding is not one-to-one because of points with two base- m expansions. The geometric invariant set, not a particular choice of endpoint address, is the canonical object.

Definition IV.5.6 (*multiplicative independence*).

Positive numbers a, b are *multiplicatively independent*, written $a \sim b$, when

$$\frac{\log a}{\log b} \notin \mathbb{Q}. \quad (11.5)$$

Lemma IV.5.7 (*integer power criterion and persistence under powers*).

For integers $p, q \geq 2$, the following are equivalent:

1. $\log p / \log q \notin \mathbb{Q}$;
2. $p^a \neq q^b$ for every positive integers a, b .

Moreover, if $p \sim q$, then $p^k \sim q^\ell$ for all $k, \ell \geq 1$.

Proof. A rational equality $\log p / \log q = b/a$ is equivalent to

$$a \log p = b \log q,$$

and hence to $p^a = q^b$. Also

$$\frac{\log(p^k)}{\log(q^\ell)} = \frac{k \log p}{\ell \log q},$$

so multiplying an irrational number by a nonzero rational cannot make it rational. $\square$

2. Word growth and dimension of an invariant closed set

Lemma IV.5.8 (*submultiplicativity of occupied word counts*).

For all $n, k \geq 1$,

$$N_{n+k}(A) \leq N_n(A)N_k(A). \quad (11.6)$$

Proof. Fix an occupied level- n cylinder $[u]_m$. The map T_m^n sends this cylinder bijectively, apart from the harmless identified endpoint, to $\mathbb{T}$. If a level- $(n+k)$ descendant $[uv]_m$ meets A , then $[v]_m$ meets

$$T_m^n(A \cap [u]_m) \subset A$$

by forward invariance. Thus at most $N_k(A)$ descendants of $[u]_m$ are occupied. Sum over at most $N_n(A)$ occupied parents. $\square$

Corollary IV.5.9 (*word-growth exponent exists*).

The limit

$$h_m(A) = \lim_{n \rightarrow \infty} \frac{1}{n} \log N_n(A) = \inf_{n \geq 1} \frac{1}{n} \log N_n(A) \quad (11.7)$$

exists by Fekete's lemma.

Proof. Apply Fekete's lemma to the subadditive sequence $\log N_n(A)$ from Lemma IV.5.8. $\square$

We next prove that this exponent is exactly the geometric dimension. The proof is included because Theorem IV.5.16 needs a scale k at which the homogeneous outer approximation has dimension arbitrarily close to $\dim_H A$.

Lemma IV.5.10 (*maximal-entropy measure from word counts*).

Let $X \subset \Sigma_m$ be a nonempty closed shift-invariant set and let $N_n(X)$ be the number of length- n words occurring in X . Put

$$h = \lim_{n \rightarrow \infty} \frac{1}{n} \log N_n(X). \quad (11.8)$$

There is a shift-invariant probability ν on X with entropy $h_\nu(\sigma) = h$.

Proof. For each N , choose one point $x_w \in X$ from every nonempty length- N cylinder $[w]$ and let μ_N be the uniform measure on these N_N cylinders. Thus

$$H_{\mu_N}(\mathcal{P}_0^{N-1}) = \log N_N, \quad (11.9)$$

where $\mathcal{P}$ is the one-coordinate partition. Set

$$\nu_N = \frac{1}{N} \sum_{j=0}^{N-1} \sigma_*^j \mu_N. \quad (11.10)$$

By compactness, a subsequence converges weakly to ν . The telescoping identity

$$\sigma_* \nu_N - \nu_N = \frac{\sigma_*^N \mu_N - \mu_N}{N}$$

gives $\sigma_* \nu = \nu$.

Fix k . For each residue class modulo k , decompose the N -coordinate partition into length- k blocks plus at most $2k$ boundary coordinates. Entropy subadditivity gives

$$k \log N_N \leq \sum_{j=0}^{N-k} H_{\sigma_*^j \mu_N}(\mathcal{P}_0^{k-1}) + O(k^2 \log m). \quad (11.11)$$

Entropy is concave under averaging, hence

$$H_{\nu_N}(\mathcal{P}_0^{k-1}) \geq \frac{1}{N} \sum_{j=0}^{N-1} H_{\sigma_*^j \mu_N}(\mathcal{P}_0^{k-1}). \quad (11.12)$$

Let $N \rightarrow \infty$ along the chosen subsequence. The partition is clopen, so its finite entropy is continuous in the measure, and (11.11)–(11.12) yield

$$H_\nu(\mathcal{P}_0^{k-1}) \geq kh. \quad (11.13)$$

On the other hand, ν is supported on X , so this entropy is at most $\log N_k(X)$. Dividing by k and taking the infimum over k gives $h_\nu(\sigma) \geq h$, while the support word-count bound gives $h_\nu(\sigma) \leq h$. Hence equality. $\square$

Lemma IV.5.11 (*neighbor-cylinder estimate*).

Let μ be any probability on $\mathbb{T}$, let $I_n(x)$ be the level- n m -adic cylinder containing x , and fix $\varepsilon > 0$. For μ -a.e. x , for all large n , every level- n m -adic cylinder J meeting $B(x, 2m^{-n})$ satisfies

$$\mu(J) \leq e^{\varepsilon n} \mu(I_n(x)). \quad (11.14)$$

Proof. At level n , each I has at most five cylinders meeting its $2m^{-n}$ -neighborhood. Call I bad if one such neighbor J has

$$\mu(J) > e^{\varepsilon n} \mu(I).$$

Charging each bad I to one such J gives

$$\sum_{I \text{ bad}} \mu(I) \leq 5e^{-\varepsilon n} \sum_J \mu(J) \leq 5e^{-\varepsilon n}. \quad (11.15)$$

Borel–Cantelli proves the claim. □

Theorem IV.5.12 (*dimension formula for closed T_m -invariant sets*).

For every nonempty closed T_m -invariant set A ,

$$\dim_H A = \overline{\dim}_B A = \underline{\dim}_B A = \frac{h_m(A)}{\log m} = \lim_{n \rightarrow \infty} \frac{\log N_n(A)}{n \log m}. \quad (11.16)$$

For $A = \emptyset$, all dimensions are taken to be $-\infty$ or 0 according to the book's global empty-set convention; no later argument depends on that choice. Finite nonempty A have dimension 0 and $h_m(A) = 0$.

Proof. The level- n occupied cylinders cover A , so immediately

$$\overline{\dim}_B A \leq \lim_{n \rightarrow \infty} \frac{\log N_n(A)}{n \log m} = \frac{h_m(A)}{\log m},$$

and $\dim_H A \leq \overline{\dim}_B A$.

For the reverse inequality, apply Lemma IV.5.10 to X_A . Choose an ergodic component of ν with entropy $h_m(A)$; this is possible because entropy is affine under ergodic decomposition and no invariant measure on X_A can have entropy exceeding $h_m(A)$. Replace ν by that component and put $\mu = (\pi_m)_* \nu$.

If $h_m(A) = 0$ there is nothing to prove. If $h_m(A) > 0$, then ν is non-atomic: an atom of an ergodic invariant probability has a finite orbit and hence zero entropy. The set of sequences participating in a double base- m expansion is countable, so it has ν -measure zero. Thus π_m is one-to-one ν -a.e.; in particular, the first-symbol partition and the geometric m -adic partition have the same name process modulo a null set, and the factor μ has entropy $h_m(A)$.

The finite generating m -adic partition therefore satisfies Shannon–McMillan–Breiman with entropy $h_m(A)$. Hence for μ -a.e. x ,

$$-\frac{1}{n} \log \mu(I_n(x)) \longrightarrow h_m(A). \quad (11.17)$$

Fix $\varepsilon > 0$. By Lemma IV.5.11, eventually the ball $B(x, 2m^{-n})$ has mass at most a fixed number of level- n neighbor masses, each at most $e^{\varepsilon n} \mu(I_n(x))$. Using (11.17),

$$\mu(B(x, 2m^{-n})) \leq C \exp[-n(h_m(A) - 2\varepsilon)] \quad (11.18)$$

for typical x and large n . Hence the lower local dimension of μ is at least $(h_m(A) - 2\varepsilon)/\log m$ at μ -a.e. x . The mass-distribution/Billingsley lemma gives

$$\dim_H A \geq \frac{h_m(A) - 2\varepsilon}{\log m}.$$

Let $\varepsilon \downarrow 0$. All inequalities in (11.16) are now equalities. □

3. Homogeneous outer approximation

Definition IV.5.13 (*level- k homogeneous outer model*).

Let

$$D_k(A) = \left\{ i \in \{0, \dots, m^k - 1\} : \left[\frac{i}{m^k}, \frac{i+1}{m^k} \right) \cap A \neq \emptyset \right\}. \quad (11.19)$$

For $i \in D_k(A)$ define

$$f_i(x) = m^{-k}x + im^{-k}. \quad (11.20)$$

Let A_k be the attractor of this IFS.

Lemma IV.5.14 (*containment and regularity*).

For every k ,

$$A^{\text{rep}} \subset A_k. \quad (11.21)$$

Here $A^{\text{rep}} \subset [0, 1)$ is the set of unique representatives of the circle set A . Equivalently, the image of A_k in $\mathbb{T}$ contains A ; no real-line endpoint 1 is included by this notation. The IFS $\{f_i : i \in D_k(A)\}$ has common contraction m^{-k} and satisfies the open set condition with open set $(0, 1)$; hence A_k is a homogeneous interval-OSC set. Its grid images may share endpoints, so this is not in general the stricter regular SSC class. It satisfies the OSC hypothesis of Theorem IV.6.2 and the endpoint-safe version of Theorem IV.7.59. Degenerate one-map cases are allowed separately.

Proof. Let $x \in A^{\text{rep}}$ and let i be the level- k half-open m -adic cell containing x . Represent $T_m^k x$ in $[0, 1)$ as well. Then $T_m^k x \in A^{\text{rep}}$ and

$$x = f_i(T_m^k x).$$

Iterating this identity puts x in a nested sequence of IFS cylinder intervals with diameters tending to zero. Hence $x \in A_k$. The images $f_i((0, 1))$ are pairwise disjoint open subintervals of $(0, 1)$, even when their closures share endpoints. □

Lemma IV.5.15 (*dimension of the outer model*).

$$\dim_H A_k = \frac{\log N_k(A)}{k \log m}. \quad (11.22)$$

Proof. At level n , the model has exactly $N_k(A)^n$ cylinders of length m^{-kn} with disjoint interiors. The upper bound follows from this cover. For the lower bound, put equal Bernoulli weights on the $N_k(A)$ maps. A ball of radius comparable to m^{-kn} meets only $O(1)$ level- n cylinders, so its measure is $O(N_k(A)^{-n})$. Frostman's elementary estimate yields (11.22). $\square$

Theorem IV.5.16 (*homogeneous outer approximation*).

Let $A \subset \mathbb{T}$ be closed and T_m -invariant. For every $\varepsilon > 0$, there exist $k \geq 1$ and a homogeneous interval-OSC m^{-k} -self-similar set $\tilde{A} \subset [0, 1]$ such that

$$A^{\text{rep}} \subset \tilde{A}, \quad \dim_H \tilde{A} \leq \dim_H A + \varepsilon. \quad (11.23)$$

The same statement holds simultaneously for a closed T_q -invariant circle set B , with an independent block length ℓ . If a real-line set in $[0, 1]$ is used instead, its possible extra endpoint 1 must be handled separately; it is not silently covered by a circle representative. If m and q are multiplicatively independent, then m^k and q^ℓ remain multiplicatively independent.

Proof. By Theorem IV.5.12, choose k so large that

$$\frac{\log N_k(A)}{k \log m} \leq \dim_H A + \varepsilon. \quad (11.24)$$

Take $\tilde{A} = A_k$. Lemmas IV.5.14–IV.5.15 prove the geometric claims. The final arithmetic statement is Lemma IV.5.7. $\square$

Source comparison. This is the locally expanded version of Wu Proposition 9.3. The source uses the same occupied m -adic cells; here the invariant-set dimension formula and the endpoint and power cases have been proved explicitly.

4. Product/slice dictionary and tractable cases

Lemma IV.5.17 (*product/slice dictionary*).

Let $A, B \subset \mathbb{R}$ be compact, let $u, v \in \mathbb{R}$, and define

$$\ell_{u,v} = \{(x, y) : y = ux + v\}. \quad (11.25)$$

Then

$$(A \times B) \cap \ell_{u,v} = \{(x, ux + v) : x \in A, ux + v \in B\}. \quad (11.26)$$

If $u \neq 0$, either coordinate projection restricts to a bi-Lipschitz affine map between the slice and $(uA + v) \cap B$ (or $A \cap u^{-1}(B - v)$). Consequently all Hausdorff and box dimensions agree. If $u = 0$, the slice is $A \times \{v\}$ when $v \in B$ and empty otherwise, while $(uA + v) \cap B$

is $\{v\}$ or empty. This degenerate coordinate case must be treated separately in intersection theorems.

Proof. Equation (11.26) is direct. For $u \neq 0$, the map $x \mapsto (x, ux+v)$ and either coordinate projection have Lipschitz affine inverses on the line. $\square$

Proposition IV.5.18 (*elementary universal intersection bounds*).

For compact $A, B \subset \mathbb{R}$ and $u \neq 0$,

$$\dim_H((uA + v) \cap B) \leq \min\{\dim_H A, \dim_H B\}, \quad (11.27)$$

and the analogous upper-box bound holds with upper box dimensions. If either set is finite, the intersection has dimension zero.

Proof. The intersection is contained in B and in $uA + v$. Affine maps preserve all the dimensions in question. The finite case is immediate. $\square$

Proposition IV.5.19 (*commensurable dynamics can have large intersections*).

The irrationality assumption $\log p / \log q \notin \mathbb{Q}$ cannot be removed from the sharp Furstenberg target. For example, let $p = q = 3$ and let C be the middle-third Cantor set. Then C is T_3 -invariant, and taking $A = B = C$, $u = 1$, and $v = 0$ gives

$$\dim_H((uA + v) \cap B) = \dim_H C = \frac{\log 2}{\log 3}, \quad (11.28)$$

whereas

$$\max\{0, \dim_H A + \dim_H B - 1\} = \max\left\{0, \frac{2 \log 2}{\log 3} - 1\right\} < \frac{\log 2}{\log 3}. \quad (11.29)$$

Thus commensurability produces genuine resonant exceptions.

Proof. Only the strict inequality needs checking. If $d = \dim_H C < 1$, then $2d - 1 < d$. $\square$

Proposition IV.5.20 (*full-set and coordinate-slice cases*).

If $A = \mathbb{T}$, then for any interval representative and any $u \neq 0$, the intersection $(uA + v) \cap B$ is a subset of B and may equal a nontrivial affine piece of B ; there is no stronger universal loss than the trivial minimum-dimension bound without using arithmetic independence. Coordinate slices $u = 0$ are exactly as in Lemma IV.5.17. These cases explain why all sharp later statements must keep the noncoordinate-slope and multiplicative-independence hypotheses visible.

Proof. Immediate from Lemma IV.5.17 and Proposition IV.5.18. $\square$

Theorem IV.5.21 (*classical tractable cases and dictionary*).

The product/slice equivalence is Lemma IV.5.17; the universal, finite, and coordinate cases are Propositions IV.5.18 and IV.5.20; Proposition IV.5.19 gives an explicit rational-log-ratio resonance counterexample. The sharp incommensurable Furstenberg inequality requires the additional arguments of Chapters IV.6 and IV.7.

Proof. Combine Lemma IV.5.17 and Propositions IV.5.18–IV.5.20. $\square$

5. The invariant-set and symbolic dictionary

Theorem IV.5.22 (*arithmetic invariant sets and symbolic codings*).

For every $m \geq 2$:

1. closed forward T_m -invariant subsets of $\mathbb{T}$ correspond exactly to closed shift-invariant coding-saturated subsets $X_A \subset \Sigma_m$ through π_m ;
2. the geometric language $L_n(A)$ is independent of the two-expansion ambiguity and satisfies submultiplicativity;
3. the common Hausdorff/box dimension is the word-growth exponent

$$\dim_H A = \lim_{n \rightarrow \infty} \frac{\log N_n(A)}{n \log m}; \quad (11.30)$$

4. multiplicative independence $p \approx q$ is equivalent to the absence of a positive-power equality $p^a = q^b$ and survives passage to p^k, q^ℓ ;
5. circle/ $[0, 1]$ endpoint representatives differ only by the identified endpoint and do not affect dimensions.

Proof. Combine the results from Lemma IV.5.4 through Theorem IV.5.12. □

6. Dependence of the argument

The proof uses finite symbolic entropy and compactness, entropy subadditivity and concavity, ergodic decomposition, finite-partition Shannon–McMillan–Breiman, Borel–Cantelli, and the elementary Moran–Frostman estimate for homogeneous finite IFSs. It does not use the later sharp intersection theorem. Wu’s Proposition 9.3 provides a useful comparison, but the occupied-cylinder construction has been proved locally above.

Chapter IV.6

The L^q and Convolution Route

From the Volume II inverse theorem to affine intersections

This chapter develops the L^q route to Furstenberg-type intersections. The one cross-volume theorem needed by the deduction is stated first so that every hypothesis can be checked at the point of use. Its proof is in Volume II, Chapter II.11.

6.1 The Volume-II affine-intersection theorem

Earlier-volume theorem: homogeneous irrational affine intersections

Let $K_j \subset \mathbb{R}$ be attractors of finite homogeneous real self-similar systems with contractions $0 < |\lambda_j| < 1$ and satisfying the open set condition. If

$$\frac{\log |\lambda_1|}{\log |\lambda_2|} \notin \mathbb{Q},$$

then for every invertible affine map $g : \mathbb{R} \rightarrow \mathbb{R}$,

$$\overline{\dim_B}(K_1 \cap g(K_2)) \leq \max\{\dim_H K_1 + \dim_H K_2 - 1, 0\}. \quad (\text{DII11-A})$$

The corresponding measure-level strengthening, also proved in Volume II, is as follows. If η_j are the equal-weight self-similar probabilities on K_j , then for every $a \neq 0$ and $q > 1$,

$$D(\eta_1 * S_a \eta_2, q) = \min\{\dim_H K_1 + \dim_H K_2, 1\}. \quad (\text{DII11-Aq})$$

Consequently, for every

$$\alpha < \min\{\dim_H K_1 + \dim_H K_2, 1\},$$

the projected measure has a small-scale upper Frostman estimate

$$(\eta_1 * S_a \eta_2)(B(y, r)) \leq C_{\alpha, a} r^\alpha,$$

and the associated affine fibres satisfy the corresponding upper-box packing bound. The proof, given in Volume II, Chapter II.11, under the title “Affine intersections from irrational homogeneous convolutions,” passes through the one-dimensional L^q inverse theorem, the pleasant-model formula, irrational-rotation convolution, L^q -to-Frostman conversion, and fibre packing.

6.2 Geometric intersection deduction

Let $p, q \geq 2$ be multiplicatively independent integers, and let A, B be the closed $\times p$ - and $\times q$ -invariant sets in the conventions of Definitions IV.5.1 and IV.5.6. For $u \neq 0$ and $v \in \mathbb{R}$, the target is

$$\overline{\dim_B}((uA + v) \cap B) \leq \max\{0, \dim_H A + \dim_H B - 1\}. \quad (\text{C13.1})$$

The case $u = 0$ is handled separately.

The approximation theorem of Chapter IV.5 gives the following: for every $\varepsilon > 0$ there are regular homogeneous Cantor supersets

$$A \subset A_\varepsilon, \quad B \subset B_\varepsilon,$$

with contractions p^{-k} and $q^{-\ell}$, respectively, satisfying the required separation/regularity and

$$\dim_{\mathrm{H}} A_\varepsilon \leq \dim_{\mathrm{H}} A + \varepsilon, \quad \dim_{\mathrm{H}} B_\varepsilon \leq \dim_{\mathrm{H}} B + \varepsilon. \quad (\text{C13.2})$$

The homogeneous affine-intersection theorem proved in Volume II states that for finite homogeneous real self-similar attractors K_1, K_2 satisfying the open set condition and having irrational logarithmic contraction ratio, every invertible affine map g satisfies

$$\overline{\dim_{\mathrm{B}}}(K_1 \cap g(K_2)) \leq \max\{\dim_{\mathrm{H}} K_1 + \dim_{\mathrm{H}} K_2 - 1, 0\}. \quad (\text{C13.3})$$

Its Volume-II proof supplies the L^q convolution, Frostman, and fibre-packing chain. We use exactly the displayed conclusion here.

Lemma IV.6.1 (*Powered multiplicative independence*).

For every $k, \ell \geq 1$,

$$\frac{\log(p^{-k})}{\log(q^{-\ell})} \notin \mathbb{Q}.$$

Proof. If the ratio were $a/b \in \mathbb{Q}$, then after clearing signs and denominators we would have

$$p^{kb} = q^{\ell a}$$

for nonzero integers a, b , contradicting multiplicative independence of p and q . $\square$

Theorem IV.6.2 (*Homogeneous geometric bridge*).

Let K_1, K_2 be the regular homogeneous approximants supplied by Chapter IV.5, and let ν_i be their uniform self-similar probabilities. For an invertible affine map $g(x) = ax + b$:

1. the product measure $\nu_1 \times \nu_2$ has the lower Frostman/Ahlfors exponent

$$s = \dim_{\mathrm{H}} K_1 + \dim_{\mathrm{H}} K_2;$$

2. the projection $x - ay$ has law $\nu_1 * S_{-a}\nu_2$;
3. the measure-level part of the Volume-II theorem gives L^q dimension $\min\{s, 1\}$ for every $q > 1$, and hence projection decay r^α for every $\alpha < \min\{s, 1\}$;
4. its fibre part gives upper-box dimension at most $s - \alpha$;
5. letting $\alpha \uparrow \min\{s, 1\}$ gives (C13.3).

Proof. The product Frostman exponent is the sum of the two factor exponents. Under the linear map $(x, y) \mapsto x - ay$, the product measure has law $\nu_1 * S_{-a}\nu_2$. The measure-level conclusion of the Volume-II theorem gives item 3, and its fibre-packing conclusion gives item 4. Since the fibre over b is affinely equivalent to $K_1 \cap (aK_2 + b)$, letting α increase to $\min\{s, 1\}$ gives

$$\overline{\dim_{\mathrm{B}}}(K_1 \cap (aK_2 + b)) \leq s - \min\{s, 1\} = \max\{s - 1, 0\}.$$

$\square$

6.3 Affine intersection theorem and scope

Theorem IV.6.3 ($\times p/\times q$ affine intersection).

Let $p, q \geq 2$ be multiplicatively independent, let A, B be closed $\times p$ - and $\times q$ -invariant subsets of $[0, 1]$, and let $u, v \in \mathbb{R}$. Then

$$\overline{\dim}_B((uA + v) \cap B) \leq \max\{0, \dim_H A + \dim_H B - 1\}.$$

Proof. Fix $\varepsilon > 0$ and choose the approximants $A_\varepsilon, B_\varepsilon$ satisfying (C13.2). The preceding lemma shows that their contraction logarithms are irrationally related, so the Volume-II affine-intersection theorem applies.

For $u \neq 0$ set $g(x) = ux + v$. Since

$$(uA + v) \cap B \subset (uA_\varepsilon + v) \cap B_\varepsilon,$$

monotonicity of upper box dimension and (C13.3) give

$$\begin{aligned} \overline{\dim}_B((uA + v) \cap B) &\leq \max\{\dim_H A_\varepsilon + \dim_H B_\varepsilon - 1, 0\} \\ &\leq \max\{\dim_H A + \dim_H B - 1 + 2\varepsilon, 0\}. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ proves (C13.1). If $u = 0$, then $(uA + v) \cap B$ is empty or the singleton $\{v\}$, hence has upper-box dimension zero. This case is recorded separately and is not used to pretend that g is invertible. $\square$

Corollary IV.6.4 (*scope consequences*).

1. **Hausdorff and packing consequences.** For every bounded set E ,

$$\dim_H E \leq \dim_P E \leq \overline{\dim}_B E,$$

so the bound in Theorem IV.6.3 also bounds Hausdorff and packing dimensions of the same affine intersections.

2. **Homogeneous self-similar variant.** For two finite homogeneous real self-similar attractors satisfying the open set condition and irrational logarithmic contraction ratio, the stronger direct statement is precisely the Volume-II theorem; no invariant-set approximation is required.
3. **Measure-level statement actually justified.** For the uniform homogeneous self-similar product measures above, the projected measure obeys, for every

$$\alpha < \min\{\dim_H K_1 + \dim_H K_2, 1\},$$

an upper Frostman estimate

$$\nu(B(y, r)) \leq C_\alpha r^\alpha$$

at sufficiently small scales. The associated fibres satisfy the packing bound in the earlier-volume measure-level affine-intersection theorem stated at the start of this chapter.

4. **Claims not made.** The argument does not infer exact dimensionality of arbitrary $\times p/ \times q$ invariant measures from the set theorem. It does not assert a uniform Frostman constant simultaneously over all slopes without a separate compact-slope argument. It does not replace upper-box dimension by a stronger dimension notion except through the standard inequalities in item 1.

Proof. The first item is the standard chain of dimension inequalities applied to the preceding theorem. The second and third are respectively the set-level and measure-level parts of the Volume-II affine-intersection theorem. The final item records the precise scope of those premises. $\square$

Chapter IV.7

Meng Wu's CP-Distribution Route

One-sided Sinai coding, slice CP laws, ergodic selection, and slicing

This chapter gives the CP-distribution route used in Wu's proof. The argument proceeds from a future-measurable Bernoulli factor, through slice CP distributions and ergodic selection, to the all-scales slicing estimate. It is independent of the L^q route of Chapter IV.6.

7.1 The future-measurable Sinai factor theorem

1. Setup

Let $(Y, \mathcal{C}, \nu, S)$ be the invertible natural extension of the finite-generator endomorphism used below, and let

$$\mathcal{F} = \mathcal{F}_+$$

be its future σ -algebra. We use the following facts proved earlier: $S^{-1}\mathcal{F} \subset \mathcal{F}$; finite $\mathcal{F}$ -measurable partitions descend to the original endomorphism; when the entropy is positive, $\mathcal{F}$ is atomless; and its entropy increment equals the endomorphism entropy h . The lifted finite forward generator is $\mathcal{F}$ -measurable, and the natural-extension identity $\bigvee_{m \geq 0} S^m \mathcal{F} = \mathcal{C}$ holds modulo null sets. The exact-proportion future-measurable tower lemma needed below is proved next; it is stronger than simply invoking an unrestricted Rokhlin tower.

For a partition α , $S^{-j}\alpha$ denotes its *pullback*: its label at y is $\alpha(S^j y)$. This is not the function $\alpha \circ S^{-j}$. In particular $S^{-1}\mathcal{F} \subset \mathcal{F}$ allows forward orbit labels in $\mathcal{F}$, not arbitrary past labels. The constructions below keep that distinction explicit. Finite-block perturbations and a seed do not alone establish the factor theorem. We shall first construct an unrestricted relative Bernoulli process and then shift only its small marker into the future algebra.

Lemma IV.7.1 (*finite entropy refinement and coarsening*).

Let $p = (p_a)_{a \in A}$ be a finite probability law and let $h \geq H(p)$ be finite. There is a finite law q on $A \times \{1, \dots, k\}$ with $H(q) = h$ and with first-coordinate marginal p .

Proof. Put $d = h - H(p) \geq 0$ and choose k with $\log k \geq d$. The laws $r(t) = (1 - t)(1, 0, \dots, 0) + t(1/k, \dots, 1/k)$, $0 \leq t \leq 1$, vary continuously from entropy zero to entropy $\log k$. The intermediate value theorem supplies t with $H(r(t)) = d$. Set $q_{a,j} = p_a r_j(t)$. With $0 \log 0 = 0$, expansion of the finite sum gives

$$H(q) = - \sum_{a,j} p_a r_j \log(p_a r_j) = H(p) + H(r) = h.$$

The map $(a, j) \mapsto a$ pushes q to p . Applying this same deterministic map to independent coordinates preserves independence, as is seen by summing their product probabilities over the separate fibers. $\square$

If $h = 0$, then, after deleting zero-weight symbols, every target law of entropy zero is a point mass. Hence assume $h > 0$. The ergodic natural extension is then aperiodic: a positive-measure finite periodic orbit would, by ergodicity, exhaust the system and force entropy zero. Given a finite probability vector p with $H(p) = h$, the remaining factor contract asks for an $\mathcal{F}$ -measurable partition η of law p for which

$$\eta, S^{-1}\eta, S^{-2}\eta, \dots$$

are independent. If $H(p) < h$, the entropy-refinement lemma first refines p to a finite law q with $H(q) = h$; after constructing the Bernoulli partition for q , we coarsen it deterministically back to p .

2. A future-measurable tower with exact finite proportions

Lemma IV.7.2 (*strong future HRK lemma*).

Let $(Y, \mathcal{C}, \nu, S)$ be an invertible ergodic aperiodic standard probability system. Let $\mathcal{F} \subset \mathcal{C}$ be an atomless σ -algebra such that

$$S^{-1}\mathcal{F} \subset \mathcal{F}, \quad \bigvee_{m \geq 0} S^m \mathcal{F} = \mathcal{C} \quad \text{modulo null sets.} \quad (2.1a)$$

For every integer $n \geq 1$, every finite $\mathcal{F}$ -measurable partition $\mathcal{A}$ with positive-mass atoms, and every $0 < \gamma < 1/n$, there exists $B \in \mathcal{F}$ such that

$$B, S^{-1}B, \dots, S^{-(n-1)}B \quad \text{are pairwise disjoint,} \quad (2.1b)$$

and for every atom E of $\mathcal{A}$,

$$\nu(B \cap E) = \gamma \nu(E). \quad (2.1c)$$

In particular $\nu(B) = \gamma$ and the tower in (2.1b) has measure $n\gamma$.

Proof. Write $R = S^{-1}$, so that $R^j D \in \mathcal{F}$ whenever $D \in \mathcal{F}$ and $j \geq 0$. We first construct, for arbitrary $H \geq n$ and $\zeta > 0$, an $\mathcal{F}$ -measurable base C with

$$C, RC, \dots, R^{H-1}C \text{ disjoint}, \quad \nu\left(\bigcup_{j=0}^{H-1} R^j C\right) > 1 - \zeta. \quad (2.1d)$$

Here is a local proof of the ordinary invertible tower used in this step. Aperiodicity and a countable Borel family separating points give a positive-measure set E_0 whose first H R -images are disjoint: for almost every point its first H iterates are distinct, some finite partition from the separating family distinguishes those iterates, and one of the resulting name cells has positive measure. Atomlessness lets us choose $E \subset E_0$ with $0 < \nu(E) < \zeta/(4H)$. By ergodicity, the first-return columns

$$\{R^j E_r : 0 \leq j < r\}, \quad E_r = \{x \in E : \text{the first return to } E \text{ is at } r\}, \quad r \geq H,$$

partition Y modulo null sets: almost every backward orbit meets E , and the most recent such meeting identifies a unique return column and level. In each column take a base at heights $0, H, 2H, \dots, ([r/H] - 1)H$. The resulting base $C_{\text{full}} \in \mathcal{C}$ has disjoint first H R -images; fewer than H levels are omitted per return column, so their total measure is at most $H\nu(E) < \zeta/4$.

By (2.1a), the increasing algebras $S^m \mathcal{F}$ generate $\mathcal{C}$; conditional-expectation martingale convergence makes their measurable sets dense in measure. Choose m and $A \in S^m \mathcal{F}$ with $\nu(A \triangle C_{\text{full}}) < \xi$, where $\xi > 0$ will be chosen below. Then $A_0 = R^m A \in \mathcal{F}$ is ξ -close to $R^m C_{\text{full}}$, another height- H tower base. Trim the overlaps by

$$C = A_0 \setminus \bigcup_{i=1}^{H-1} R^i A_0 \in \mathcal{F}.$$

This makes $C \cap R^i C = \emptyset$ for $1 \leq i < H$, hence all its first H levels are pairwise disjoint. Since the corresponding levels of $R^m C_{\text{full}}$ are disjoint,

$$\nu(C \triangle R^m C_{\text{full}}) \leq \xi + \sum_{i=1}^{H-1} \nu(A_0 \cap R^i A_0) \leq (2H - 1)\xi.$$

Consequently the coverage of the C -tower is at least $1 - \zeta/4 - H(2H - 1)\xi$. Taking ξ sufficiently small proves (2.1d).

We next record the finite simultaneous-splitting fact used to make the tower independent of $\mathcal{A}$. If $f_1, \dots, f_M$ are nonnegative integrable functions on an atomless finite measure space and $0 < t < 1$, there is a measurable set D satisfying

$$\int_D f_i d\nu = t \int f_i d\nu \quad (1 \leq i \leq M). \quad (2.1e)$$

Let $\mathcal{K}$ be the set of functions $g \in L^\infty$ with $0 \leq g \leq 1$ and $\int g f_i = t \int f_i$ for all i . It is nonempty because it contains the constant function t . The order bounds and the finitely many L^1 pairing constraints are weak-star closed. Alaoglu's theorem therefore makes $\mathcal{K}$

weak-star compact, and the Krein–Milman theorem gives it an extreme point [15, Theorems 4.14(a) and 4.30]. If that extreme point had $0 < g < 1$ on a positive-measure set, then for some $a > 0$ the set $\{a \leq g \leq 1 - a\}$ could be partitioned into $M + 1$ positive-measure pieces. Their M vectors of f_i -integrals are linearly dependent. A nonzero bounded signed linear combination of their indicators would perturb g in both directions while preserving every integral and $0 \leq g \leq 1$, contradicting extremality. Thus the extreme point is an indicator, which proves (2.1e). Applying this fact successively with fractions $1/n, 1/(n-1), \dots, 1/2$ partitions a set into n pieces having equal integrals for every member of a prescribed finite family of nonnegative densities.

Now write the atoms of $\mathcal{A}$ as $E_1, \dots, E_s$. Choose $H > n$ and $\zeta > 0$ so small that

$$\zeta + \frac{n-1}{H} < (1 - n\gamma) \min_{1 \leq i \leq s} \nu(E_i). \quad (2.1f)$$

Take C from (2.1d). For $0 \leq j \leq H - n$ and $1 \leq i \leq s$, define on $\mathcal{F}|_C$ the finite measure

$$m_{j,i}(D) = \nu(R^j D \cap E_i), \quad D \subset C.$$

It satisfies $m_{j,i}(D) \leq \nu(D)$, so it has an $\mathcal{F}|_C$ -measurable $L^1(\nu)$ density. Apply (2.1e) to all these finitely many densities, partitioning $C = C_0 \sqcup \dots \sqcup C_{n-1}$ so that

$$m_{j,i}(C_r) = \frac{1}{n} m_{j,i}(C) \quad \text{for all } j, i, r. \quad (2.1g)$$

Set

$$B_* = \bigcup_{j=0}^{H-n} R^j C_{j \bmod n} \in \mathcal{F}.$$

For $0 \leq a < a' < n$, two putative points in $R^a B_* \cap R^{a'} B_*$ either lie in different levels of the C -tower, which is impossible, or lie in the same level but in $R^k C_r \cap R^k C_{r'}$ with $r \neq r'$, also impossible. Thus $B_*, RB_*, \dots, R^{n-1} B_*$ are disjoint. Using (2.1g) and the disjoint tower levels,

$$\nu(B_* \cap E_i) = \frac{1}{n} \sum_{j=0}^{H-n} \nu(R^j C \cap E_i) \geq \frac{\nu(E_i) - \zeta - (n-1)/H}{n} > \gamma \nu(E_i), \quad (2.1h)$$

where $H\nu(C) \leq 1$ bounds the omitted top $n-1$ levels. Since $\mathcal{F}$ is atomless, choose $B_i \subset B_* \cap E_i$, $B_i \in \mathcal{F}$, with $\nu(B_i) = \gamma \nu(E_i)$, and put $B = \bigcup_i B_i$. The tower levels of B remain disjoint because $B \subset B_*$, and (2.1c) holds by construction. $\square$

3. A tower measurability lemma

Lemma IV.7.3 (*tower pullback measurability*).

Let $\mathcal{F}$ be a σ -algebra with $S^{-1}\mathcal{F} \subset \mathcal{F}$. Let $B \in \mathcal{F}$ and suppose that

$$B, S^{-1}B, \dots, S^{-(n-1)}B$$

are pairwise disjoint. For $i = 0, \dots, n-1$, let $g_i : B \rightarrow A$ be a finite-valued function measurable for $\mathcal{F}|_B$. Define g on the tower U by

$$g(S^{-i}x) = g_i(x), \quad x \in B,$$

and define it off U by any finite-valued $\mathcal{F}$ -measurable rule. Then g is $\mathcal{F}$ -measurable.

Proof.

For $a \in A$ and fixed i ,

$$\{y \in S^{-i}B : g(y) = a\} = S^{-i}\{x \in B : g_i(x) = a\}.$$

The inner set belongs to $\mathcal{F}$, so its preimage belongs to $S^{-i}\mathcal{F} \subset \mathcal{F}$. The tower levels and the off-tower part are $\mathcal{F}$ -measurable. Taking the finite union over i proves that every level set $\{g = a\}$ belongs to $\mathcal{F}$. □

Corollary IV.7.4 (*whole-block maps of measurable base data*).

Let $Z : B \rightarrow E$ be any finite $\mathcal{F}|_B$ -measurable variable and

$$\Gamma : E \rightarrow A^n$$

any deterministic map. Put

$$g_i(x) = \Gamma(Z(x))_i.$$

Then the function obtained by placing these names on the tower as in Lemma IV.7.3 is $\mathcal{F}$ -measurable. No triangularity condition on Γ is needed.

In particular, let θ be a finite $\mathcal{F}$ -measurable partition and set on B

$$\Theta_n^+(x) = (\theta(x), \theta(Sx), \dots, \theta(S^{n-1}x)).$$

Every coordinate of this *forward* name is $\mathcal{F}$ -measurable because $\{\theta \circ S^j = a\} = S^{-j}\{\theta = a\}$. Hence any map

$$\Gamma : K^n \rightarrow A^n$$

may be applied to Θ_n^+ on the base and assembled down the tower while remaining $\mathcal{F}$ -measurable. This does not assert closeness to the old labels $\theta(S^{-i}x)$ on the tower levels.

Proof. The variable Z makes every coordinate g_i measurable on B , so apply Lemma IV.7.3. For the final assertion take $Z = \Theta_n^+$. □

The required measurability of the base data is substantive. For example, on the fair two-sided binary shift, $\mathcal{F} = \sigma(x_j : j \geq 0)$ and $\theta(x) = x_0$, the label $\theta(S^{-1}x) = x_{-1}$ is independent of $\mathcal{F}$ and is not $\mathcal{F}$ -measurable. Restricting to a positive $\mathcal{F}$ -measurable base does not change this fact. Arbitrary past-block codes therefore cannot be justified by this corollary.

4. Atomless cellwise law correction

Lemma IV.7.5 (*exact conditional law correction at optimal mismatch cost*).

Let $\mathcal{F}$ be atomless. Let $\mathcal{P}$ be a finite $\mathcal{F}$ -measurable partition, let X be a finite A -valued $\mathcal{F}$ -measurable function, and let q be a probability vector on A . Then there exists an A -valued $\mathcal{F}$ -measurable function Y such that, on every positive-measure cell C of $\mathcal{P}$,

$$\text{Law}(Y \mid C) = q,$$

and

$$\nu\{X \neq Y\} = \sum_{C \in \mathcal{P}} \nu(C) \text{TV}(\text{Law}(X \mid C), q).$$

Consequently, Y is independent of $\mathcal{P}$ and has law q .

Proof.

Fix a positive-measure cell C . Write

$$r_a = \frac{\nu(C \cap \{X = a\})}{\nu(C)}.$$

For each a , retain inside $C \cap \{X = a\}$ a subset of relative mass $\min\{r_a, q_a\}$. This is possible because the restricted measure algebra of every positive-measure $\mathcal{F}$ -set is atomless. The total retained mass is

$$\sum_a \min\{r_a, q_a\} = 1 - \text{TV}(r, q).$$

The remaining pieces have total relative mass $\text{TV}(r, q)$, and the deficits $(q_a - r_a)_+$ have the same total mass. Split the excess pieces atomlessly into pieces of the deficit masses and relabel them by the corresponding deficit symbols. The resulting label has law q on C and changes X on exactly the proportion $\text{TV}(r, q)$ of C . Perform the construction on the finitely many cells of $\mathcal{P}$ and take their union.

□

5. Entropy deficiency gives the uniform approximate-independence input

Let α be a finite A -valued partition and write

$$X_i = \alpha \circ S^i.$$

Let

$$\text{Def}(\alpha) = H(\alpha) - h(S, \alpha).$$

Lemma IV.7.6 (*one coordinate versus an arbitrary predecessor block*).

For every $i \geq 1$,

$$I(X_i; X_0, \dots, X_{i-1}) \leq \text{Def}(\alpha).$$

Hence, using Pinsker with natural logarithms,

$$\text{TV}(\text{Law}(X_i, X_0^{i-1}), \text{Law}(X_i) \otimes \text{Law}(X_0^{i-1})) \leq \sqrt{\text{Def}(\alpha)/2}.$$

Proof.

By stationarity $H(X_i) = H(\alpha)$. The finite conditional entropies

$$H(X_i | X_0, \dots, X_{i-1})$$

are bounded below by the entropy rate $h(S, \alpha)$: by stationarity they are the time-zero entropies conditional on finite pasts, whose decreasing limit is $h(S, \alpha)$. Therefore

$$I(X_i; X_0^{i-1}) = H(\alpha) - H(X_i | X_0^{i-1}) \leq H(\alpha) - h(S, \alpha).$$

Pinsker gives the displayed TV estimate.

□

Lemma IV.7.7 (*future-algebra independent perturbation sequence*).

Assume that α is $\mathcal{F}$ -measurable, that $\mathcal{F}$ is atomless and $S^{-1}\mathcal{F} \subset \mathcal{F}$. Thus all the *forward* labels X_i defined above belong to $\mathcal{F}$. Let q be a probability vector on the same finite alphabet A . For every n , there are $\mathcal{F}$ -measurable partitions

$$\psi_0, \dots, \psi_{n-1}$$

such that:

(a) $\psi_0, \dots, \psi_{n-1}$ are mutually independent; (b) each ψ_i has law q ; (c) for every i ,

$$\nu\{\psi_i \neq X_i\} \leq \sqrt{\text{Def}(\alpha)/2} + \text{TV}(\text{Law}(\alpha), q).$$

Proof.

We construct backwards. The induction invariant after $\psi_{i+1}, \dots, \psi_{n-1}$ have been constructed is:

$$Q_{i+1} := (\psi_{i+1}, \dots, \psi_{n-1}), \quad \text{Law}(Q_{i+1}) = q^{\otimes(n-i-1)}.$$

and Q_{i+1} is independent of

$$(X_0, \dots, X_i).$$

For $i=n-1$ the suffix is empty, so the invariant is vacuous.

Assume the invariant. Let P_i be the finite partition generated by

$$X_0, \dots, X_{i-1}, Q_{i+1}.$$

Since Q_{i+1} is independent of the entire vector $(X_0, \dots, X_i)$, conditioning also on Q_{i+1} does not change the conditional law of X_i given the predecessor block. Hence Lemma IV.7.6 gives

$$\sum_{C \in P_i} \nu(C) \text{TV}(\text{Law}(X_i | C), \text{Law}(\alpha)) \leq \sqrt{\text{Def}(\alpha)/2}.$$

By the triangle inequality,

$$\sum_C \nu(C) \text{TV}(\text{Law}(X_i | C), q) \leq \sqrt{\text{Def}(\alpha)/2} + \text{TV}(\text{Law}(\alpha), q).$$

Apply Lemma IV.7.5 to P_i and X_i . It produces an F -measurable ψ_i whose conditional law is exactly q on every P_i -cell and whose mismatch probability is bounded by the displayed quantity.

Because its conditional law is q on P_i , ψ_i is independent of both the predecessor block and Q_{i+1} . Since Q_{i+1} already has product law and is independent of the predecessor block, the enlarged suffix

$$(\psi_i, Q_{i+1})$$

has product law $q^{\otimes(n-i)}$ and is independent of $(X_0, \dots, X_{i-1})$. This is the next induction invariant.

This proves the finite independent perturbation assertion for the named forward variables. It does not identify their coordinates with the past labels occurring on a downward tower. That compatibility is a distinct issue in a future-measurable orbit code.

□

6. Local Hall Matching Lemma

Lemma IV.7.8 (*mass-dominated overlap matching*).

Let $C_1, \dots, C_M$ and $D_1, \dots, D_N$ be finite measurable partitions of a set, and let G be measurable. Suppose the selected source cells C_a satisfy

$$\nu(C_a \cap G) \geq (1/2)\nu(C_a),$$

$$\nu(C_a) \geq m_s > 0,$$

and every target cell D_b meeting G satisfies

$$\nu(D_b) \leq m_t \leq \frac{m_s}{4}.$$

Join C_a to D_b whenever $C_a \cap D_b \cap G$ is nonempty. Then there is an injective matching of the selected C_a into the D_b .

Proof.

For any subfamily E of selected source cells,

$$\nu(G \cap \bigcup E)$$

$$= \sum_{C \in E} \nu(C \cap G)$$

$$\geq (m_s/2)|E|.$$

This set lies in the union of the neighboring target cells. Each such target cell has mass at most $m_s/4$. Therefore the number of neighbors is at least $2|E|$, in particular at least $|E|$. Hall's finite marriage criterion holds. (The finite Hall theorem itself has the standard augmenting-path proof and is used here only in this elementary finite form.) □

7. Phase entropy and recovery lemmas

Lemma IV.7.9 (*the Rokhlin phase partition has small entropy rate*).

Let $B \in F$ satisfy

$$B, S^{-1}B, \dots, S^{-(n-1)}B$$

are pairwise disjoint and $\nu(B) = 1/(n+1)$. Put

$$R = Y \setminus \bigcup_{i=0}^{n-1} S^{-i}B,$$

and let ζ be the finite partition

$$\{B, S^{-1}B, \dots, S^{-(n-1)}B, R\}.$$

Then

$$h(S, \zeta) \leq \frac{2 \log(n+1)}{n+1}. \quad (\text{PH1})$$

Proof. Write Z for the phase label. Under S^{-1} , every tower level $S^{-i}B$ with $0 \leq i < n-1$ moves deterministically to $S^{-(i+1)}B$. Only points in the top level $S^{-(n-1)}B$ or in R can have a non-deterministic next phase. Those two sets have total measure $2/(n+1)$. Therefore

$$H(Z \circ S^{-1} \mid Z) \leq \frac{2}{n+1} \log(n+1).$$

The entropy rate of a finite stationary process is bounded by its one-step conditional entropy, which proves (PH1). □

Lemma IV.7.10 (*injective tower block code recovers the generator modulo small phase entropy*).

Let θ be a finite partition with alphabet K . On the tower above let

$$\Theta_n(x) = (\theta(x), \theta(S^{-1}x), \dots, \theta(S^{-(n-1)}x)), \quad x \in B.$$

Suppose a finite block code Γ is injective on a family $\mathcal{T}$ of Θ_n -cells, and the union of those cells has conditional B -measure at least $1 - \varepsilon$. Assemble an A -valued partition α on the tower by

$$(\alpha(x), \dots, \alpha(S^{-(n-1)}x)) = \Gamma(\Theta_n(x)),$$

and define α arbitrarily off the tower. With

$$A_n = \bigvee_{j=-(n-1)}^{n-1} S^{-j}\alpha,$$

there is a decoder of θ from $A_n \vee \zeta$ whose error probability is at most

$$e := \varepsilon + \frac{1}{n+1}. \quad (\text{PH2})$$

Here H_2 denotes binary entropy with the same logarithm convention as H . Moreover, if $e \leq 1/2$, then

$$h(S, \alpha) \geq h(S, \theta) - H_2(e) - e \log \max\{|K| - 1, 1\} - \frac{2 \log(n+1)}{n+1}. \quad (\text{PH3})$$

Proof. Let $y = S^{-i}x$ lie on tower level i . The phase $\zeta(y)$ gives i . The two-sided α -window A_n contains the n labels

$$\alpha(x), \alpha(S^{-1}x), \dots, \alpha(S^{-(n-1)}x),$$

because relative to y their time offsets lie between $-(n-1)$ and $n-1$. If $\Theta_n(x)$ belongs to $\mathcal{T}$, the output word identifies the unique source Θ_n -cell by injectivity of Γ , and hence determines its i th θ coordinate, namely $\theta(y)$.

Thus decoding fails only when x lies in a discarded source cell or y lies off the tower, giving (PH2). When $|K| \geq 2$ and $e \leq 1/2$, the function $H_2(u) + u \log(|K| - 1)$ is increasing on $0 \leq u \leq 1/2$. Fano's finite-alphabet inequality, with the actual decoder error at most e , therefore yields

$$H(\theta \mid A_n \vee \zeta) \leq H_2(e) + e \log \max\{|K| - 1, 1\}.$$

For $|K| = 1$ the conditional entropy is zero and this bound is automatic. Put

$$c_e = H_2(e) + e \log \max\{|K| - 1, 1\}.$$

For $N \geq 1$, stationarity gives, for every $0 \leq i < N$,

$$H(S^{-i}\theta \mid S^{-i}(A_n \vee \zeta)) \leq c_e.$$

The join of the shifted windows $S^{-i}A_n$, $0 \leq i < N$, is contained in the α -block from time $-(n-1)$ through $N+n-2$; adjoining the corresponding ζ -block gives the analogous block of $\alpha \vee \zeta$. By the chain rule,

$$H\left(\bigvee_{i=0}^{N-1} S^{-i}\theta\right) \leq H\left(\bigvee_{j=-(n-1)}^{N+n-2} S^{-j}(\alpha \vee \zeta)\right) + Nc_e.$$

Divide by N and let $N \rightarrow \infty$. The fixed $2(n-1)$ -site boundary disappears, so

$$h(S, \theta) \leq h(S, \alpha \vee \zeta) + c_e.$$

Since

$$h(S, \alpha \vee \zeta) \leq h(S, \alpha) + h(S, \zeta),$$

Lemma IV.7.9 gives (PH3). □

Lemma IV.7.11 (*entropy-efficient future seed at equality*).

In the natural-extension setup above, assume $H(p) = h > 0$. For every $\tau > 0$ there is an F -measurable finite partition α , on the alphabet of p after adjoining at most one zero-mass symbol, such that

$$\text{TV}(\text{Law}(\alpha), p) < \tau, \quad \text{Def}(\alpha) = H(\alpha) - h(S, \alpha) < \tau. \quad (\text{SEED})$$

Proof. If p is uniform on its positive-mass alphabet, adjoin one zero-mass symbol. Then in all cases there are probability vectors q arbitrarily close to p with $H(q) > h$. Choose q so close that entropy continuity and $\text{TV}(q, p)$ are both within a small fraction of τ , and put

$$g = H(q) - h > 0.$$

Let θ be the finite F -measurable generator with $h(S, \theta) = h$, and write the actual past block as

$$\Theta_n^- = (\theta, \theta \circ S^{-1}, \dots, \theta \circ S^{-(n-1)}).$$

This block need not be F -measurable. It is measurable with respect to $F_n = S^{n-1}F$, where $S^jF = \{S^jE : E \in F\}$. The algebra F_n is atomless, satisfies $S^{-1}F_n \subset F_n$, and its increasing translates generate $\mathcal{C}$. We perform the finite coding in F_n and shift its output back into F only at the end. By SMB and stationarity, for all large n there is a union G_θ of Θ_n^- -cells with measure $> 1 - \delta$ such that every retained cell C satisfies

$$\nu(C) \geq \exp[-n(h + g/4)].$$

Hence the number of retained source cells is at most $\exp[n(h + g/4)]$.

For $q^{\otimes n}$ choose a product-typical set W_n with $q^{\otimes n}(W_n) > 1 - \delta$ such that every $w \in W_n$ has empirical symbol law δ -close to q and

$$q^{\otimes n}(w) \leq \exp[-n(H(q) - g/4)].$$

Therefore

$$|W_n| \geq (1 - \delta) \exp[n(H(q) - g/4)] = (1 - \delta) \exp[n(h + 3g/4)],$$

which is larger than the number of retained source cells for all large n . Choose an injective assignment of the retained Θ_n^- -cells to distinct words in W_n , and extend it arbitrarily to a map Γ on all Θ_n^- -names.

Apply Lemma IV.7.2 with the algebra F_n , height n , $\gamma = 1/(n+1)$, and all positive-mass Θ_n^- -cells. It gives $B \in F_n$ with exact conditional proportions, hence $\nu(B) = 1/(n+1)$ and tower mass $n/(n+1)$. Assemble the output β from Γ on the tower, and use a fixed symbol off the tower. More precisely,

$$\beta(S^{-i}x) = \Gamma_i(\Theta_n^-(x)), \quad x \in B, \quad 0 \leq i < n.$$

Lemma IV.7.3, used with F_n , makes β F_n -measurable.

On retained source cells every output word has empirical law δ -close to q . Exact base proportions imply that averaging over the tower levels makes $\text{Law}(\beta)$ $O(\delta + 1/n)$ -close to q , hence τ -close to p after choosing parameters. The code is injective on source cells of conditional base mass $> 1 - \delta$, so Lemma IV.7.10 gives

$$h(S, \beta) \geq h - o_{\delta, n}(1).$$

At the same time $\text{Law}(\beta) \rightarrow p$, so $H(\beta) \rightarrow H(p) = h$ by finite-alphabet entropy continuity. Finally set

$$\alpha = \beta \circ S^{n-1}.$$

The preimage under S^{n-1} of an F_n -measurable set belongs to F . Invariance of ν gives $\text{Law}(\alpha) = \text{Law}(\beta)$ and $h(S, \alpha) = h(S, \beta)$: every finite block distribution is just a translate of the corresponding β block distribution. Thus choosing first δ small and then n large gives (SEED). $\square$

The finite delay in this proof is harmless for a seed because (SEED) does not ask that α be close to a specified old partition. It cannot simply be repeated in a near-identity upgrade: shifting the new partition also shifts its comparison with the old one. The comparison below isolates that additional requirement; the proof we shall use instead follows a moving entropy-interior path.

8. An unused fixed-law comparison criterion

Unproved interface IV.7.12 (*future near-identity upgrade, equality case*).

Assume that $(Y, \mathcal{C}, \nu, S)$ is invertible and ergodic, that F is atomless with $S^{-1}F \subset F$ and $\bigvee_{m \geq 0} S^m F = \mathcal{C}$ modulo null sets, that there is a finite F -measurable generator of entropy $h > 0$, and that p is a finite law with $H(p) = h$. The comparison assertion, not assumed in our proof, is the following: for every $\varepsilon_1 > 0$ there exists $\delta_1 = \delta_1(\varepsilon_1) > 0$ such that, whenever α_1 is F -measurable and

$$\text{TV}(\text{Law}(\alpha_1), p) < \delta_1, \quad \text{Def}(\alpha_1) < \delta_1,$$

then for every $0 < \delta_2 < \delta_1$ there is an F -measurable α_2 with

$$\text{TV}(\text{Law}(\alpha_2), p) < \delta_2, \quad \text{Def}(\alpha_2) < \delta_2, \quad \nu\{\alpha_2 \neq \alpha_1\} < \varepsilon_1.$$

The target alphabet is fixed throughout; if needed it is enlarged once by one zero-weight symbol. This assertion is not proved here. It is stronger than the finite-delay seed and is not a consequence of the forward-coordinate perturbation lemma alone. It is retained only to distinguish this possible route from the moving-law construction that proves the factor theorem below.

The precise failure of the former proof. The former argument applied the independent perturbation lemma to the past labels $\alpha_1 \circ S^{-i}$ and then treated their entire block as F -measurable on a future-measurable tower base. Both operations require an additional argument. For a concrete test, take

$$Y = \{0, 1\}^{\mathbb{Z}}, \quad \nu = \left(\frac{1}{2}, \frac{1}{2}\right)^{\mathbb{Z}}, \quad (Sy)_j = y_{j+1}, \\ F = \sigma(y_j : j \geq 0), \quad \alpha_1(y) = y_0.$$

This is an invertible ergodic system; F is atomless, $S^{-1}F \subset F$, and its increasing translates generate $\mathcal{C}$. The partition α_1 is a finite generator and $\text{Def}(\alpha_1) = 0$. Nevertheless, for every F -measurable binary label ψ ,

$$\nu\{\psi \neq \alpha_1 \circ S^{-1}\} = \nu\{\psi \neq y_{-1}\} = \frac{1}{2},$$

by independence of y_{-1} from F . The old zero-error perturbation of this past coordinate is therefore impossible. The repaired perturbation lemma uses $\alpha_1 \circ S^i$ instead. That repairs the finite lemma, not the near-identity coding of the downward tower.

Austin's Proposition 6.2 is an unrestricted invertible-system argument, not a proof that all its operations remain in a prescribed future algebra [17]. Wu's Theorem 6.3 invokes the endomorphism version of Sinai's theorem and cites Ornstein–Weiss for that extension [3, Section 6, Theorem 6.3 and reference 28]. No counterexample to those source factor theorems is asserted here: the calculation above refutes the particular future-measurability inference in this textbook reconstruction. The full published unilateral source has now been read. Its construction follows an entropy-interior path of target laws, rather than asserting Interface IV.7.12. We next give that alternative construction, including the relative Bernoulli process that its finite perturbations do not supply by themselves.

9. The unilateral construction: independence defect and continuity

The distinction between a finite independent perturbation and a stationary independent process is important here. A perturbation of finitely many labels does not supply an entire Bernoulli process independent of the old factor. The following reconstruction of Ornstein–Weiss makes that additional input visible. Our constants are deliberately chosen with room for approximation; we do not rely on the source's compressed numerical estimates.

Definition IV.7.13 (*conditional independence defect*).

For an A -valued partition α , a probability vector q on the same alphabet, and $n \geq 1$, put

$$\mathcal{G}_n(\alpha) = \sigma(\alpha \circ S, \dots, \alpha \circ S^n), \quad r_{n,a} = \mathbb{E}_\nu(1_{\{\alpha=a\}} \mid \mathcal{G}_n(\alpha)),$$

and define

$$D_n(q, \alpha) = \int \sum_{a \in A} |r_{n,a} - q_a| d\nu, \quad D(q, \alpha) = \lim_{n \rightarrow \infty} D_n(q, \alpha).$$

Conditional Jensen makes D_n nondecreasing, and $D_n \leq 2$. The conditional-expectation martingale also gives

$$D(q, \alpha) = \int \sum_a |r_{\infty,a} - q_a| d\nu, \quad r_{\infty,a} = \mathbb{E}_\nu(1_{\{\alpha=a\}} \mid \mathcal{G}_\infty(\alpha)), \quad \mathcal{G}_\infty(\alpha) = \bigvee_n \mathcal{G}_n(\alpha).$$

Thus $D(q, \alpha) = 0$ means precisely that α has law q and is independent of its entire strictly forward name. Stationarity then gives product laws for all finite names. We use the partition distance $d(\alpha, \beta) = \nu\{\alpha \neq \beta\}$ and the norm $\|q - q'\|_1 = \sum_a |q_a - q'_a|$; the latter is twice total variation.

Lemma IV.7.14 (*continuity of the conditional defect*).

On a fixed finite alphabet, $D(q, \alpha)$ is continuous in q and in the partition distance. In addition,

$$\|\text{Law}(\alpha) - q\|_1 \leq D(q, \alpha) \leq \|\text{Law}(\alpha) - q\|_1 + \sqrt{2 \text{Def}(\alpha)}. \quad (\text{OW.1})$$

The continuity modulus at α can be chosen from its stationary finite-name laws alone, uniformly over partitions with those same laws.

Proof. The first inequality is conditional Jensen. Stationarity and the entropy chain rule, read in reverse time, give

$$H(\alpha \mid \mathcal{G}_n(\alpha)) \downarrow h(S, \alpha).$$

Indeed the reverse chain expansion of an N -name averages these decreasing conditional entropies, and division by N gives their limit. For the infinite conditional law, its mean relative entropy with respect to the marginal law of α is therefore $H(\alpha) - h(S, \alpha) = \text{Def}(\alpha)$. Pinsker and Jensen prove the second inequality in (OW.1).

More generally set

$$t_n(\alpha) = H(\alpha \mid \mathcal{G}_n(\alpha)) - h(S, \alpha) \downarrow 0.$$

The mean relative entropy of r_∞ with respect to r_n equals $t_n(\alpha)$, by expanding the finite entropy sums and conditioning. Terms with $r_{n,a} = 0$ also have $r_{\infty,a} = 0$ almost surely. Consequently

$$0 \leq D(q, \alpha) - D_n(q, \alpha) \leq \int \|r_\infty - r_n\|_1 d\nu \leq \sqrt{2t_n(\alpha)}. \quad (\text{OW.2})$$

For $e = d(\alpha, \beta) \leq 1/2$ and $k = |A| \geq 2$, write $\omega_k(e) = H_2(e) + e \log(k-1)$. Fano's bound gives $H(\alpha \mid \beta), H(\beta \mid \alpha) \leq \omega_k(e)$. The block chain rule and entropy-rate continuity yield

$$|h(S, \alpha) - h(S, \beta)| \leq 2\omega_k(e), \quad |t_n(\alpha) - t_n(\beta)| \leq (2n+3)\omega_k(e). \quad (\text{OW.3})$$

For the latter inequality, compare the entropy of the $(n+1)$ -name and that of its n -coordinate suffix, then use the former inequality.

There is no conditioning-stability assumption hidden in the finite defect: if $P(a, w)$ denotes the joint law of the current label and its n -name,

$$D_n(q, \alpha) = \sum_{a, w} |P(a, w) - q_a P(w)|.$$

The coupling by the two actual names gives

$$|D_n(q, \alpha) - D_n(q, \beta)| \leq 4(n+1)e. \quad (\text{OW.4})$$

Choose first n with $t_n(\alpha)$ small, and then e small in (OW.3)–(OW.4). Equation (OW.2) proves continuity. The one-symbol case is immediate. Finally $|D(q, \alpha) - D(q', \alpha)| \leq \|q - q'\|_1$. All quantities used to choose n and the modulus depend only on the stationary name laws of α , proving the uniformity assertion. $\square$

10. An entropy-interior path of laws

Lemma IV.7.15 (*summable entropy-slack target path*).

Let p be a finite law with $H(p) = h > 0$, after deleting zero-weight symbols. There are probability vectors $p_i \rightarrow p$ such that

$$H(p_i) < H(p_{i+1}) < h, \quad d_i := \|p_i - p_{i+1}\|_1 > 0, \quad \sum_i d_i < \infty,$$

and for each i there is $c_i > 0$ with

$$\|q - p_{i+1}\|_1 \leq 2d_i \implies H(q) \leq h - c_i. \quad (\text{OW.5})$$

Proof. Choose a with $p_a = \max_b p_b$, put $v = e_a - p$, and let $p(t) = p + tv$ for $0 \leq t \leq t_0$, with $0 < t_0 < 1/2$. All its coordinates are positive. Differentiating the finite entropy sum gives

$$\left. \frac{d}{dt} H(p(t)) \right|_{t=0} = -\log p_a - H(p) \leq 0, \quad \frac{d^2}{dt^2} H(p(t)) = -\sum_b \frac{v_b^2}{p_b(t)} \leq -\|v\|_2^2.$$

Since $h > 0$, $v \neq 0$. With $c = \|v\|_2^2/2$ we have $h - H(p(t)) \geq ct^2$ and $H(p(t))$ strictly decreases for $t > 0$.

Put $m = (1 - t_0) \min_b p_b > 0$. On the simplex region with coordinates at least $m/2$, the gradient of entropy is bounded in maximum norm by some $L \geq 1$. Hence every law q with

$$\|q - p(t)\|_1 \leq \min\{m/2, ct^2/(2L)\}$$

satisfies $H(q) \leq h - ct^2/2$. Write $V = \|v\|_1$ and choose $0 < \lambda \leq 1$ so small that

$$\frac{4V\lambda}{t_0} \leq \frac{c}{2L}, \quad 4V\lambda t_0 \leq m/2.$$

Set $t_i = t_0/(1 + \lambda i)$ and $p_i = p(t_i)$. Then $t_i \downarrow 0$ and

$$2d_i = 2V(t_i - t_{i+1}) \leq \frac{4V\lambda}{t_0} t_{i+1}^2 \leq \min\{m/2, ct_{i+1}^2/(2L)\}.$$

This proves (OW.5) with $c_i = ct_{i+1}^2/2$. Also $\sum_i d_i = Vt_1 < \infty$ by telescoping. $\square$

11. Zero-relative-entropy independence transfer

One part of the source's relative-factor argument can be proved from the relative entropy identity already established in Chapter IV.1. It transfers an existing independent process through an entropy-zero extension; it does not construct that process.

Lemma IV.7.16 (*independence through an entropy-exhausting factor*).

Let β, β_0, γ be finite partitions in an ergodic invertible probability system. Suppose that β_0 is measurable with respect to $\mathcal{B}_\beta = \sigma(\beta \circ S^j : j \in \mathbb{Z})$, that $h(S, \beta_0) = h(S, \beta)$, and that the stationary γ process is iid and its full orbit algebra is independent of $\mathcal{B}_{\beta_0}$. Then its full orbit algebra is independent of $\mathcal{B}_\beta$.

Proof. The factor generated by $\beta \vee \gamma$ contains that generated by $\beta_0 \vee \gamma$. Factor-entropy monotonicity and independence give

$$h(S, \beta \vee \gamma) \geq h(S, \beta_0 \vee \gamma) = h(S, \beta_0) + H(\gamma) = h(S, \beta) + H(\gamma).$$

The reverse inequality follows from the finite-block entropy inequality, because $h(S, \gamma) = H(\gamma)$. Thus equality holds. The factor-monotonicity assertion is valid even though β_0 need not be a single-coordinate function of β : approximate β_0 in partition distance by a finite two-sided β name, apply the block inequality, and let the Fano entropy-rate error tend to zero.

Apply Lemma IV.1.41 on the symbolic factor generated by $\beta \vee \gamma$, with generator $\mathcal{A} = \beta \vee \gamma$ and factor partition $\mathcal{C} = \beta$. Its entropy identity yields

$$H(\gamma | \mathcal{B}_\beta \vee \sigma(\gamma \circ S^{-j} : j \geq 1)) = h(S, \beta \vee \gamma) - h(S, \beta) = H(\gamma).$$

The β coordinate of the generator and all its past coordinates are already in $\mathcal{B}_\beta$. Equality of conditional and marginal entropy means that the mean relative entropy of the conditional law of γ with respect to its marginal is zero. Hence that conditional law equals the marginal almost surely. Stationarity and invariance of $\mathcal{B}_\beta$ now give, by induction over any finite increasing sequence of times, the conditional product law of those γ labels given $\mathcal{B}_\beta$. A monotone-class argument extends this from cylinders to the full γ orbit algebra and proves independence. $\square$

This removes the need to import the source's cited zero-relative-entropy independence-transfer lemma. It does not supply an entropy-exhausting Bernoulli factor β_0 , or a Bernoulli complement to a prescribed β_0 .

12. Finite couplings with a prescribed Bernoulli name

We now reconstruct the complement argument rather than import it. The given Bernoulli partition is kept fixed. Typical source words will be matched to independent model words having exactly the same given-factor name. Entropy provides the number of choices; a tower transfers those finite choices to the original system. For a finite partition V , $|V|$ denotes the number of its positive-mass atoms. We shall use the finite-alphabet Fano modulus

$$\omega_m(t) = H_2(t) + t \log(m-1) \quad (m \geq 2, 0 \leq t \leq 1/2), \quad \omega_1(t) = 0.$$

Partitions on a common m -symbol alphabet at mismatch $t \leq 1/2$ have entropy rates differing by at most $2\omega_m(t)$, by the block chain rule and Fano's inequality. The bound is used only at sufficiently small t .

Lemma IV.7.17 (*relative finite-name coupling*).

Let H be a finite partition whose bilateral stationary process is iid, let P be a finite A -valued partition, and let p be a law on A . Put

$$r = h(S, P \vee H) - H(H).$$

For every $n \geq 1$ there is an atomless auxiliary probability space carrying $(X_i, Z_i, J_i)_{0 \leq i < n}$ such that $(X_i, Z_i)_{i < n}$ has the same law as $(P \circ S^i, H \circ S^i)_{i < n}$, the J_i are iid of law p and independent of the entire vector $(Z_i)_{i < n}$, and

$$\mathbb{P}\{J_i \neq X_i\} \leq \sqrt{\frac{H(P) - r}{2}} + \text{TV}(\text{Law}(P), p). \quad (\text{RC.1})$$

The same assertion holds with S replaced by S^{-1} .

Proof. Apply Lemma IV.1.41 on the symbolic factor generated by $P \vee H$, with generator $\mathcal{A} = P \vee H$ and factor partition $\mathcal{C} = H$. The H coordinate and its past are already in $\mathcal{B}_H$. Since the whole H process has entropy rate $H(H)$,

$$H(P | \mathcal{B}_H \vee \sigma(P \circ S^{-j} : j \geq 1)) = r.$$

By stationarity, finite conditioning by $Z_0, \dots, Z_{n-1}, X_0, \dots, X_{i-1}$ gives conditional entropy at least r . Its mean relative entropy with respect to $\text{Law}(P)$ is therefore at most $H(P) - r$. Pinsker bounds the mean total variation by $\sqrt{(H(P) - r)/2}$.

Take the finite law of the old (X, Z) name and multiply its space by an independent unit interval. Every positive old-name cell in this auxiliary space is atomless. Construct $J_{n-1}, \dots, J_0$ backwards, using Lemma IV.7.5. At step i , condition on

$$(Z_0, \dots, Z_{n-1}), (X_0, \dots, X_{i-1}), (J_{i+1}, \dots, J_{n-1}).$$

The induction invariant is that the constructed suffix is product law $p^{\otimes(n-i-1)}$ and independent of the entire Z name and $X_0, \dots, X_i$. Thus it does not change the conditional law of X_i . The optimal cellwise law correction gives conditional law p for J_i and the mismatch bound (RC.1). The enlarged suffix is independent of the old prefix and the Z name, so the induction continues. All old variables remain unchanged. At the last step the required independence and product law follow. Entropy rates and iid-factor properties are unchanged on reversing time, proving the last assertion. $\square$

Lemma IV.7.18 (*entropy-rich finite refinements*).

Let the ergodic invertible system have a finite generator θ and entropy $0 < h < \infty$. If B is a finite partition with $h(S, B) < h$, then for every $0 < \beta < h - h(S, B)$ there is a finite refinement Q of B such that $h(S, Q) = h - \beta$.

Proof. Positive entropy and ergodicity make the full probability algebra atomless. Choose an increasing family E_t , $0 \leq t \leq 1$, with $\nu(E_t) = t$; an atomless standard probability space admits a uniform $[0, 1]$ variable, whose sublevel sets provide this family. Let ζ_t equal the θ label on E_t and a new symbol outside, and set $Q_t = B \vee \zeta_t$ on one fixed finite alphabet. Then Q_t refines B , and

$$d(Q_t, Q_s) \leq |t - s|, \quad h(S, Q_0) = h(S, B), \quad h(S, Q_1) = h.$$

The last equality holds because Q_1 contains the finite generator. The Fano entropy-rate bound used in Lemma IV.7.14 makes $t \mapsto h(S, Q_t)$ continuous. The intermediate value theorem gives the desired refinement. $\square$

13. A seed preserving the prescribed factor**Lemma IV.7.19** (*exact-law relative entropy seed*).

In the finite-generator ergodic invertible system of entropy h , let H have an iid stationary process, and let p be a finite law with

$$H(p) + H(H) = h, \quad H(p) > 0.$$

For every $\tau > 0$ there is a finite partition P of law exactly p with $h(S, P \vee H) > h - \tau$.

Proof. Choose $0 < \beta < \min\{\tau/4, H(p)/2\}$. Lemma IV.7.18 gives a finite refinement Q of H with entropy $h - \beta$. Use backward n -names, as in Lemma IV.7.10. SMB gives source Q -name cells of total mass $> 1 - \epsilon$ with weights between $\exp[-n(h - \beta + \epsilon)]$ and $\exp[-n(h - \beta - \epsilon)]$. The analogous typical H -name cells have mass $> 1 - \epsilon$ and weights between $\exp[-n(H(H) + \epsilon)]$ and $\exp[-n(H(H) - \epsilon)]$. Here ordinary finite-process SMB is Theorem IV.1.42 applied to the one-sided name law with trivial factor; stationarity identifies the reversed-time name law.

For the iid model of law p , let W_n be a set of words with probability $> 1 - \epsilon$, each word of probability at most $\exp[-n(H(p) - \epsilon)]$ and empirical law at L^1 distance $< \delta$ from p . The iid law of large numbers gives these properties for all large n . In particular

$$|W_n| \geq (1 - \epsilon) \exp[n(H(p) - \epsilon)].$$

For each typical H -name z , the number of typical Q -name cells within that H -cell is at most

$$\frac{\exp[-n(H(H) - \epsilon)]}{\exp[-n(h - \beta + \epsilon)]} = \exp[n(H(p) - \beta + 2\epsilon)].$$

Choose $0 < \epsilon < \beta/4$. For all large n this is smaller than $|W_n|$. Assign different words from W_n to those Q -cells separately within each H -name. Thus the pair consisting of the assigned

p -word and the unchanged H -word determines its retained source Q -cell. Retained cells have total mass $> 1 - 2\epsilon$.

Apply Lemma IV.7.2 with the full algebra, height n , base mass $1/(n+1)$, and all positive Q -name cells. Their conditional proportions on the base equal their global proportions. Place the assigned p -words down the tower to define P' ; use a fixed label on discarded names and off the tower. The partition H is never altered. For $P' \vee H$, Lemma IV.7.10 has error at most $e = 2\epsilon + 1/(n+1)$ and gives

$$h(S, P' \vee H) \geq h - \beta - H_2(e) - e \log \max\{|Q| - 1, 1\} - \frac{2 \log(n+1)}{n+1}. \quad (\text{RC.2})$$

Also, averaging the empirical assigned words over tower levels yields

$$\|\text{Law}(P') - p\|_1 \leq \delta + 4\epsilon + \frac{2}{n+1}. \quad (\text{RC.3})$$

The unretained source mass and the off-tower mass account for the last two terms; no model-word weights are substituted for actual source masses.

Correct the marginal of P' to p by Lemma IV.7.5 on the full atomless algebra with trivial conditioning partition. The corrected P has mismatch $t = \text{TV}(\text{Law}(P'), p)$. The joint partitions $P \vee H$ and $P' \vee H$ also have mismatch t , so their entropy rates differ by at most $2\omega_{|A||H|}(t)$. Choose ϵ, δ sufficiently small, and then n sufficiently large, so the losses in (RC.2) and this correction are together less than $\tau - \beta$. This proves the assertion. $\square$

14. A near-identity upgrade preserving the prescribed factor

The seed just constructed need not be close to any given partition. The next argument adds that requirement. We match many possible model words to each old name, so a refinement carrying almost all the entropy can be encoded without changing most old labels. Bad old names are allowed arbitrary outputs, but their typical refinement cells are still encoded: this is what makes the entropy accuracy independent of the small, fixed movement budget.

Lemma IV.7.20 (*exact-law relative near-identity upgrade*).

Let the ergodic invertible system have a finite generator and entropy $0 < h < \infty$. Let H have an iid bilateral process, and let p be a finite law with $H(p) > 0$ and $H(p) + H(H) = h$. For every $\epsilon > 0$ there is $\delta > 0$ such that, if $\text{Law}(P) = p$ and

$$0 < d := h - h(S, P \vee H) < \delta,$$

then for every $\tau > 0$ there is a partition $\tilde{P}$ of law exactly p satisfying

$$d(P, \tilde{P}) < \epsilon, \quad h(S, \tilde{P} \vee H) > h - \tau. \quad (\text{RC.4})$$

The given partition H is not altered.

Proof. Delete zero-weight symbols of p , and put $H_I = H(p)$, $H_H = H(H)$. Take

$$\kappa = \min\{\epsilon/16, 1/100\}, \quad \delta = \kappa^4/128, \quad \rho = \sqrt{d/2} < \kappa^2/16.$$

Choose $0 < \beta < \min\{\tau/4, d/10\}$. Lemma IV.7.18 gives a finite refinement Q of $P \vee H$ with $h(S, Q) = h - \beta$. Fix positive tolerances η, σ , with

$$\eta < \min\{d/8, \beta/10, \kappa/100\}.$$

They can be made as small as needed after Q is fixed; n will then be chosen sufficiently large.

Names, their weights, and the auxiliary coupling. Use backward n -names. Ordinary SMB, in the internal form used in Lemma IV.7.19, provides typical (P, H) -name cells and typical Q -name cells, each family of mass $> 1 - \eta$, with weights respectively in

$$[e^{-n(h-d+\eta)}, e^{-n(h-d-\eta)}], \quad [e^{-n(h-\beta+\eta)}, e^{-n(h-\beta-\eta)}]. \quad (\text{RC.5})$$

Write a (P, H) -name as $a = (x, z)$ and call it a *parent*; a Q -name cell contained in it is called a *child*. These words describe only the refinement relation between finite partitions. Let W_H be an iid H -word family of probability $> 1 - \eta$, each weight in $[e^{-n(H_H+\eta)}, e^{-n(H_H-\eta)}]$. Let W_I be an iid p -word family of probability $> 1 - \eta$ such that each word has weight at most $e^{-n(H_I-\eta)}$ and empirical law at L^1 distance $< \sigma$ from p . The iid law of large numbers gives these families for all large n , also when H is constant. In particular

$$|W_I| \geq (1 - \eta)e^{n(H_I-\eta)}. \quad (\text{RC.6})$$

The model words are the *product* family $W_I \times W_H$. Under their independent law, every pair has weight at most $e^{-n(h-2\eta)}$.

Lemma IV.7.17, with the exact old marginal p , couples the actual old-name distribution to an iid I -name independent of the whole H -name, keeping the H -name identical. Its expected number of old- P versus model- I mismatches is at most ρn . All the coupling probabilities in the following matching argument are on that auxiliary space; its cells are not being treated as subsets of Y . Only the actual source-name indices will later be used for a tower code on Y .

Many close choices for most parents. A parent is *eligible* if it is typical in (RC.5) and its H -name belongs to W_H . An eligible parent $a = (x, z)$ is *good* if at least half of its conditional coupling mass has I -word in W_I and Hamming distance at most κn from x ; Hamming distance is the number of unequal coordinates. Otherwise it is bad. The ineligible mass is at most 2η . The unconditional coupling failure mass is at most $\eta + \rho/\kappa$, by Markov's inequality. Bad eligible parents have conditional failure greater than $1/2$. Consequently the total mass of parents not good is at most

$$6\eta + 2\rho/\kappa. \quad (\text{RC.7})$$

For a good parent $a = (x, z)$ connect it to each pair (w, z) in $W_I \times W_H$ with Hamming distance from x at most κn . For any subfamily $\mathcal{S}$ of good parents, the coupling mass over these graph edges is at least half their actual source mass. The neighbor set $N(\mathcal{S})$ therefore satisfies

$$|N(\mathcal{S})|e^{-n(h-2\eta)} \geq \frac{1}{2}|\mathcal{S}|e^{-n(h-d+\eta)}, \quad |N(\mathcal{S})| \geq \frac{1}{2}|\mathcal{S}|e^{n(d-3\eta)}. \quad (\text{RC.8})$$

Put $M = \lfloor e^{n(d-4\eta)} \rfloor$. For large n it is positive and $M \leq \frac{1}{2}e^{n(d-3\eta)}$. Replace each good parent by M formal copies, all having its same neighbors. A set of copies supported on $\mathcal{S}$ has at most $M|\mathcal{S}|$ elements, so (RC.8) gives Hall's condition. The elementary finite marriage theorem, in the augmenting-path form used in Lemma IV.7.8, assigns M different prototype pairs to every good parent, with all pairs distinct. Each assigned pair keeps the parent's H -name and changes at most κn of its P labels.

Capacity for all retained children, including bad parents. For each eligible parent the number of its typical Q children is at most

$$e^{-n(h-d-\eta)}e^{n(h-\beta+\eta)} = e^{n(d-\beta+2\eta)} < M \quad (\text{RC.9})$$

for large n , because $\beta > 6\eta$. For a fixed $z \in W_H$, the number of eligible parents with that H -name is at most

$$e^{-n(H_H-\eta)}e^{n(h-d+\eta)} = e^{n(H_I-d+2\eta)}. \quad (\text{RC.10})$$

Thus M slots per such parent require at most $e^{n(H_I-2\eta)}$ different I -words. By (RC.6), the available W_I is larger for all large n . After the Hall assignments, allocate M unused words with the same H -name to every bad eligible parent; (RC.10) ensures enough remain. Pools of pairs are disjoint across all eligible parents. Inject the typical Q children of each eligible parent into its pool, using (RC.9). Retain these children, both in good and in bad parents. Their total source mass is greater than $1 - 3\eta$. The assigned pair (I -word, H -word) now determines its retained Q -name cell uniquely.

Transfer to the actual system and correct the marginal. Apply Lemma IV.7.2 with the full algebra $\mathcal{C}$, height n , base mass $1/(n+1)$, and the actual positive Q -name cells. The full algebra is atomless and the system is aperiodic since $h > 0$. The base has exactly the global proportions of all these cells. For a retained name based at $x \in B$, place its assigned I -word on $x, S^{-1}x, \dots, S^{-(n-1)}x$. This defines P' consistently because the levels are disjoint. Leave H unchanged; its actual name is exactly the second coordinate of the assigned pair. Use any fixed P' label on discarded cells and off the tower. Lemma IV.7.10 applied to this injective pair code gives, with $e = 3\eta + 1/(n+1) \leq 1/2$,

$$h(S, P' \vee H) \geq h - \beta - \omega_{|Q|}(e) - \frac{2 \log(n+1)}{n+1}. \quad (\text{RC.11})$$

The old- P movement on good retained names is at most κ , averaged over tower levels. Using (RC.7) and the discarded and off-tower masses gives the safe bound

$$d(P, P') \leq \kappa + 9\eta + 2\rho/\kappa + \frac{1}{n+1}. \quad (\text{RC.12})$$

Every retained output word has empirical law within σ of p , whether its parent is good or bad. Exact source proportions on the base therefore give

$$\|\text{Law}(P') - p\|_1 \leq \sigma + 6\eta + \frac{2}{n+1}. \quad (\text{RC.13})$$

Use Lemma IV.7.5 on the full algebra, with trivial conditioning, to correct this marginal to p . Its cost is $t = \text{TV}(\text{Law}(P'), p)$. The corrected partition $\tilde{P}$ has

$$d(P, \tilde{P}) \leq d(P, P') + t, \quad h(S, \tilde{P} \vee H) \geq h(S, P' \vee H) - 2\omega_{|A||H|}(t). \quad (\text{RC.14})$$

After Q and β are fixed, choose η, σ sufficiently small and n sufficiently large that $t < \kappa$ and the losses following β in (RC.11), (RC.14) sum to less than $\tau - \beta$. These choices preserve all the preceding large- n comparisons. Since $2\rho/\kappa < \kappa/8$, (RC.12) can at the same time be made less than 2κ . Thus total movement is less than $3\kappa < \varepsilon$, while the joint entropy is greater than $h - \tau$. This proves (RC.4). $\square$

The full-algebra exact-proportion tower is important: we did not shift a previously chosen base to make its frequencies asymptotically independent. That shift is the mixing-dependent step in Ornstein's original Lemma 4', proof step 5 [18, pp. 342–346]. Here the source-name frequencies are exact at the base, by the internal tower lemma. The rest is a finite matching and entropy-recovery argument.

Theorem IV.7.21 (*a prescribed independent Bernoulli complement*).

Let $(Y, \mathcal{C}, \nu, S)$ be ergodic, invertible and have a finite generator, with $0 \leq h < \infty$. Let H be a finite partition whose bilateral process is iid. If p is a finite law with

$$H(p) + H(H) = h,$$

there is a partition P of law p whose bilateral process is iid and whose entire orbit algebra is independent of the entire H orbit algebra.

Proof. If $H(p) = 0$, choose the corresponding constant partition, so assume $H(p) > 0$. First observe the equality criterion. If a partition V of law p satisfies $h(S, V \vee H) = h$, apply Lemma IV.1.41 on the symbolic factor of $V \vee H$ with generator $\mathcal{A} = V \vee H$ and factor partition $\mathcal{C} = H$. Its H coordinate and past lie in $\mathcal{B}_H$, so the identity gives

$$H(V \mid \mathcal{B}_H \vee \sigma(V \circ S^{-j} : j \geq 1)) = h - H(H) = H(p). \quad (\text{RC.15})$$

Conditional and marginal entropies are equal, so the mean relative entropy of the conditional V law with respect to p is zero. Consequently that conditional law is p almost surely. By stationarity the same holds at every integer time, conditioning on the whole H process and all earlier V coordinates. For $t_1 < \dots < t_m$, successive conditional expectations therefore give

$$\mathbb{E} \left(\prod_{i=1}^m 1_{\{V \circ S^{t_i} = a_i\}} \mid \mathcal{B}_H \right) = \prod_{i=1}^m p_{a_i}. \quad (\text{RC.16})$$

The monotone-class theorem extends this from cylinders to the full V orbit algebra. It proves both the iid property and the required independence.

Take a summable sequence $\varepsilon_i > 0$ tending to zero, and let $\delta_i > 0$ be supplied by Lemma IV.7.20 for ε_i . Lemma IV.7.19 gives P_1 of exact law p with $0 \leq h - h(S, P_1 \vee H) < \delta_1$. If this gap is zero, (RC.15)–(RC.16) finish the proof. Otherwise use Lemma IV.7.20 with output tolerance $\tau = \delta_2/2$ to produce P_2 at distance less than ε_1 and gap less than δ_2 . Continue, stopping if a gap becomes zero. If no stage stops, we obtain exact-law partitions with

$$d(P_i, P_{i+1}) < \varepsilon_i, \quad 0 < h - h(S, P_i \vee H) < \delta_i. \quad (\text{RC.17})$$

We may replace δ_i by $\min\{\delta_i, 1/i\}$ at the outset. Borel–Cantelli implies that the finite labels $P_i(y)$ are eventually constant for almost every y . Their limit P has law exactly p , and $d(P_i, P) \leq \sum_{j \geq i} \varepsilon_j \rightarrow 0$. Fano entropy-rate continuity for the joint partitions with the fixed H then yields $h(S, P \vee H) = h$. Apply (RC.15)–(RC.16). $\square$

15. The relative binary process needed for unilateral coding

Theorem IV.7.22 (*strict-slack relative binary Bernoulli process*).

In the invertible finite-generator system above, let α be any finite partition. If $0 < u < 1$ and

$$H_2(u) < h(S) - h(S, \alpha),$$

there is a measurable set $C \in \mathcal{C}$ for which $(1_C \circ S^j)_{j \in \mathbb{Z}}$ are independent Bernoulli(u) variables and their entire orbit algebra is independent of

$$\mathcal{B}_\alpha = \sigma(\alpha \circ S^j : j \in \mathbb{Z}).$$

Independence here means that $\nu(E \cap G) = \nu(E)\nu(G)$ for every set in the first orbit algebra and every $G \in \mathcal{B}_\alpha$. No future measurability is required of C .

Proof. Write $h_\alpha = h(S, \alpha) < h$. On the bilateral symbolic factor generated by α , the coordinate partition is a finite generator and the entropy is h_α . If $h_\alpha > 0$, choose a finite law of entropy h_α by Lemma IV.7.1 starting with a point mass. Theorem IV.7.21 on that symbolic factor, with prescribed H constant, gives an iid partition α_0 of entropy h_α . Pull it back to Y ; its entire orbit algebra is contained in $\mathcal{B}_\alpha$. If $h_\alpha = 0$, let α_0 be constant instead.

Refine the binary law $(u, 1 - u)$ to a finite law q of entropy $h - h_\alpha$, using Lemma IV.7.1. In the original finite-generator system, apply Theorem IV.7.21 with prescribed $H = \alpha_0$ and law q . It gives an iid partition independent of the whole α_0 factor. Coarsen it to the binary law to obtain a set C of measure u . The binary process is still iid and independent of that whole factor. Finally Lemma IV.7.16, with old partition α and entropy-exhausting partition α_0 , transfers independence to the whole $\mathcal{B}_\alpha$. This proves the assertion. $\square$

Source boundary

This is the binary strict-slack case of Ornstein–Weiss Theorem 5, published pp. 161–162 [19]. The proof above supplies its finite matching/complement and zero-relative-entropy transfer inputs internally. It does not identify a finite perturbation with a stationary process, and does not import a Sinai or isomorphism theorem.

16. Improvement by a small independent marker

Lemma IV.7.23 (*future improvement with uniform entropy slack*).

Fix a probability law q on $k \geq 2$ symbols and numbers $0 < a < b$ and $c > 0$ such that

$$\|q' - q\|_1 \leq b \implies H(q') \leq h - c. \quad (\text{OW.6})$$

There is $\Delta = \Delta(a, c, k) > 0$ such that every F -measurable α satisfying $a \leq D(q, \alpha) < b$ admits an F -measurable $\tilde{\alpha}$ with

$$D(q, \tilde{\alpha}) \leq D(q, \alpha) - \Delta, \quad d(\alpha, \tilde{\alpha}) \leq 4(D(q, \alpha) - D(q, \tilde{\alpha})). \quad (\text{OW.7})$$

Proof. By (OW.1) and (OW.6), $h(S, \alpha) \leq H(\alpha) \leq h - c$. Choose

$$0 < u < \min\{1/2, a/(16k), a/8\}, \quad H_2(u) < c.$$

For all sufficiently large n , $D_n(q, \alpha) \geq a/2$. Since $\|r_n - q\|_1 \leq 2$, the atoms B of $\mathcal{G}_n(\alpha)$ with $\|r_n(B) - q\|_1 \geq a/4$ have total measure at least $a/8$. On each such atom choose, in a fixed order, two symbols $i(B), j(B)$ with

$$r_{n,i}(B) - q_i \geq a/(8k) > u, \quad q_j - r_{n,j}(B) \geq a/(8k) > u. \quad (\text{OW.8})$$

Such symbols exist because the signed differences sum to zero.

Theorem IV.7.22 gives a binary process independent of the entire α orbit. For an integer $N \geq 0$ let $C_N = S^{-N}C$, and change the current label from $i(B)$ to $j(B)$ on $B \cap \{\alpha = i(B)\} \cap C_N$; elsewhere leave it unchanged. Call the resulting partition $\hat{\alpha}_N$. Its mismatch with α is

$$d_* = u \sum_{B \text{ chosen}} \nu(B) r_{n,i(B)}(B) \geq u^3. \quad (\text{OW.9})$$

On each chosen atom the moved mass is $ur_{n,i}(B) \leq u$. By (OW.8) it exceeds neither the excess nor the deficit. Consequently the conditional L^1 distance from q decreases there by twice the moved mass:

$$D_n(q, \alpha) - \int \|\text{Law}(\hat{\alpha}_N \mid \mathcal{G}_n(\alpha)) - q\|_1 d\nu = 2d_*. \quad (\text{OW.10})$$

Increase n , if necessary, so that $\int \|r_\infty - r_n\|_1 d\nu < u^3/4$. The relabeling on a fixed atom is a stochastic matrix, so it contracts L^1 distances. Thus replacing the finite old conditional law by the infinite old conditional law costs at most $u^3/4$ in (OW.10). Adjoining the marker variables at strictly positive times does not alter that conditional law: the time-zero marker is independent of those variables and of the entire old orbit algebra. Moreover

$$\mathcal{G}_\infty(\hat{\alpha}_N) \subset \mathcal{G}_\infty(\alpha) \vee \sigma(1_{C_N} \circ S^j : j \geq 1).$$

Conditional Jensen under this inclusion proves

$$D(q, \hat{\alpha}_N) \leq D(q, \alpha) - 2d_* + u^3/4 \leq D(q, \alpha) - 7d_*/4. \quad (\text{OW.11})$$

It remains to put the new partition in F , without shifting the old one. Because $\bigvee_{M \geq 0} S^M F = \mathcal{C}$, there are $A_N \in S^N F$ with $\nu(A_N \triangle C) \rightarrow 0$. The sets $E_N = S^{-N}A_N$ belong to F and approximate C_N . Replace C_N by E_N in the relabeling rule and call the output $\tilde{\alpha}_N$. It belongs to F , since every old forward-name atom and the current old label belong to F . Also $d(\tilde{\alpha}_N, \hat{\alpha}_N) \leq \nu(A_N \triangle C)$.

The stationary process laws of $\hat{\alpha}_N$ are identical for every N : the entire shifted marker process is iid and independent of the entire old process. The uniformity clause of Lemma IV.7.14 therefore applies to all these reference partitions with one modulus. Choose N so large that

$$|D(q, \tilde{\alpha}_N) - D(q, \hat{\alpha}_N)| < d_*/4, \quad d(\tilde{\alpha}_N, \hat{\alpha}_N) < d_*/4.$$

Then its defect gain is at least $3d_*/2$ and its mismatch with the old partition is at most $5d_*/4$. Equations (OW.7) follow, for example with $\Delta = u^3$. Only the new independent marker was shifted; α stayed anchored throughout. $\square$

Theorem IV.7.24 (*summable unilateral factor construction*).

Every finite probability law p with $H(p) = h > 0$ is the law of an F -measurable Bernoulli partition.

Proof. Use Lemma IV.7.15 and choose $0 < e_i < d_i$ with $e_i \rightarrow 0$. There is an F -measurable α_1 with $D(p_1, \alpha_1) < e_1$: refine p_1 to a finite law of entropy h by Lemma IV.7.1, apply the equality seed Lemma IV.7.11 with arbitrarily small error, and coarsen. A deterministic coarsening cannot increase deficiency: its current-label mutual information with its future is bounded, by data processing, by the corresponding mutual information of the uncoarsened process. Its law also converges to p_1 . Equation (OW.1) proves the assertion. After this step all α_i use the fixed alphabet of p .

Suppose α_i has been chosen. Then

$$D(p_{i+1}, \alpha_i) \leq D(p_i, \alpha_i) + d_i < 2d_i.$$

By (OW.5), (OW.6) holds with $q = p_{i+1}$, $b = 2d_i$ and $c = c_i$. Apply Lemma IV.7.23 repeatedly until the defect is less than e_{i+1} . This terminates after finitely many steps: while the defect is at least e_{i+1} there is a uniform positive decrement. If $e_{i+1} \geq 2d_i$, the current partition already suffices. The mismatch bound telescopes, giving an F -measurable α_{i+1} with

$$D(p_{i+1}, \alpha_{i+1}) < e_{i+1}, \quad d(\alpha_i, \alpha_{i+1}) \leq 4D(p_{i+1}, \alpha_i) < 8d_i.$$

Since $\sum_i d_i < \infty$, Borel–Cantelli gives an almost surely eventually constant label sequence. Its limit η is F -measurable after a common null modification, and $d(\alpha_i, \eta) \rightarrow 0$. Lemma IV.7.14 and $p_i \rightarrow p$ imply $D(p, \eta) = 0$. The characterization in Definition IV.7.13 proves that the stationary names of η have product law p . $\square$

The relative complement, independence transfer, future approximation and summable coding have now all been proved. None of this asserts the unused fixed-law upgrade: a sequence of moving entropy-slack laws was essential to the unilateral improvement argument.

17. What the unused fixed-law criterion would imply

Theorem IV.7.25 (*equality future Bernoulli factor*).

Under the hypotheses of Interface IV.7.12, assume additionally that the quantitative upgrade asserted there holds. Then there is an F -measurable partition η of law p such that

$$\eta, S^{-1}\eta, S^{-2}\eta, \dots$$

are independent.

Proof. Choose a summable sequence $\varepsilon_m > 0$. For each m let d_m be a positive input tolerance supplied by Interface IV.7.12 for ε_m . Choose a decreasing sequence $r_m \downarrow 0$ with $r_m < d_m$ and r_{m+1} small enough to be an admissible output tolerance at stage m .

Use Lemma IV.7.11 to choose an initial F -measurable α_1 with

$$\text{TV}(\text{Law}(\alpha_1), p) < r_1, \quad \text{Def}(\alpha_1) < r_1.$$

Recursively apply Interface IV.7.12 so that

$$\text{TV}(\text{Law}(\alpha_{m+1}), p) < r_{m+1}, \quad \text{Def}(\alpha_{m+1}) < r_{m+1}, \quad \nu\{\alpha_{m+1} \neq \alpha_m\} < \varepsilon_m.$$

The sequence is Cauchy in the partition metric and converges, after a common null modification, to an F -measurable finite partition η . Its law converges to p . For finite partitions,

$$|h(S, \alpha) - h(S, \beta)| \leq H(\alpha | \beta) + H(\beta | \alpha),$$

and the right side tends to zero with the partition metric by the finite Fano bound. Hence entropy rate is continuous along this Cauchy sequence. Together with $H(\alpha_m) \rightarrow H(p)$ and $\text{Def}(\alpha_m) \rightarrow 0$, this gives

$$H(\eta) = h(S, \eta) = H(p).$$

For a stationary finite-alphabet process, equality of the one-symbol entropy and entropy rate implies that one coordinate is independent of its entire finite past; by stationarity and induction every finite backward name has product law. Thus

$$\eta, S^{-1}\eta, S^{-2}\eta, \dots$$

are independent. □

This proves the conditional comparison deduction. It does not prove the fixed-law criterion, and neither statement is used in the following endomorphism factor proof.

18. Descent to the Original Endomorphism and Strict Entropy

Theorem IV.7.26 (*future-measurable Bernoulli factor*).

Let $(X, \mathcal{B}, \mu, T)$ be the ergodic finite-generator endomorphism used below, with finite entropy h . Let p be a finite probability vector with

$$H(p) \leq h.$$

There is a finite measurable partition β on X of law p such that

$$\beta, T^{-1}\beta, T^{-2}\beta, \dots$$

are independent. Equivalently the corresponding unilateral Bernoulli shift of base law p is a factor of $(X, \mathcal{B}, \mu, T)$.

Source boundary

This is the endomorphism factor statement invoked in Wu's Theorem 6.3 [3], which refers to Ornstein–Weiss [19]. The unrestricted prescribed complement and unilateral coding were proved above. The following argument completes descent and the strict-entropy reduction without using the fixed-law comparison criterion.

Proof. Theorem IV.7.24 supplies the equality-case future Bernoulli partition used below. If $h = 0$ the claim is trivial as noted above.

First suppose $H(p) = h$. Pass to the natural extension. The construction in the first part of the chapter identifies the future algebra F and shows that finite F -measurable partitions descend exactly to X . Apply the equality-case construction to obtain the F -measurable equality-entropy Bernoulli partition η . Descend η through the future-algebra identification. Its backward-name independence is exactly the required unilateral independence on X .

Now suppose $H(p) < h$. The entropy-adjustment lemma proved above produces a finite refinement q of p on a finite alphabet, with

$$H(q) = h,$$

and a deterministic coarsening c satisfying $c_*q = p$. Apply the equality case to q , then compose its Bernoulli partition with c . Coordinatewise deterministic coarsening preserves independence and gives law p . $\square$

This proof uses no arithmetic-intersection theorem. Its future Bernoulli partition supplies the later entropy selection and slicing arguments.

7.2 Slice-generated CP laws and the skew product

1. Homogeneous interval-OSC models

Let $0 < \alpha, \beta < 1$ and assume

$$\frac{\log \alpha}{\log \beta} \notin \mathbb{Q}. \quad (14.1)$$

If $\log \alpha / \log \beta > 1$, interchange the two coordinate systems and apply the coordinate swap $(x, y) \mapsto (y, x)$. The new ratio is the reciprocal of the old one and lies in $(0, 1)$. The swap is an isometry, carries a non-axis-parallel line of slope m to one of slope m^{-1} , and preserves all slice and covering dimensions used below. We work in these swapped coordinates when necessary and conjugate the final geometric conclusion back. Thus throughout the symbolic construction we have

$$0 < \theta := \frac{\log \alpha}{\log \beta} < 1. \quad (14.1a)$$

Let

$$\Phi = \{\phi_i(x) = \alpha x + a_i\}_{i=1}^m, \quad \Psi = \{\psi_j(y) = \beta y + b_j\}_{j=1}^\ell \quad (14.2)$$

be homogeneous self-similar IFSs on $\mathbb{R}$ with nonsingleton attractors. We use the following interval form of the open set condition.

Definition IV.7.27 (*interval open set condition*).

There are bounded open intervals J_α, J_β such that the images of each interval under its own IFS are contained in that interval and are pairwise disjoint as open intervals. Their closures may share endpoints. The stricter regular SSC model requires these closures to be disjoint. The present argument includes that model but also treats the endpoint contacts of the outer approximants in Chapter IV.5.

Write the attractors as C_α, C_β , and put

$$K = C_\alpha \times C_\beta. \quad (14.3)$$

After affine normalization we may and do regard K as a compact subset of $[0, 1]^2$.

The number θ is fixed by (14.1a); in particular $\beta^\theta = \alpha$ exactly. Since (14.1) holds, the circle rotation

$$R_\theta(t) = t - \theta \pmod{1} \quad (14.5)$$

is irrational and uniquely ergodic.

Assume there is a non-axis-parallel line ℓ_0 such that

$$\gamma := \overline{\dim_B}(K \cap \ell_0) > 0. \quad (14.6)$$

Reflections reduce the negative-slope case to positive slope, so we take the initial slope positive.

Variable symbolic fibres. Let

$$\Lambda_\alpha = \{a_1, \dots, a_m\}, \quad \Lambda_\beta = \{b_1, \dots, b_\ell\}, \quad \Lambda = \Lambda_\alpha \times \Lambda_\beta. \quad (14.7)$$

For $s \in \mathbb{T} = \mathbb{R}/\mathbb{Z}$ define the one-step alphabet by

$$L(s) = \begin{cases} \Lambda, & s \in [0, \theta) \pmod{1}, \\ \Lambda_\alpha \times \{b_1\}, & \text{otherwise.} \end{cases} \quad (14.8)$$

Define

$$X_t = \prod_{n \geq 0} L(R_\theta^n t), \quad (14.9)$$

so the left shift maps

$$\sigma : X_t \longrightarrow X_{R_\theta t}. \quad (14.10)$$

Equip every X_t with the α -symbolic metric

$$d_\alpha(x, x') = \alpha^{\min\{n \geq 0 : x_{n+1} \neq x'_{n+1}\}}. \quad (14.11)$$

Let

$$Z(t) = \{n \geq 0 : R_\theta^n(t) \in [0, \theta)\},$$

and enumerate it as $w_1(t) < w_2(t) < \dots$. For

$$x = ((c_n, d_n))_{n \geq 1} \in X_t$$

define

$$\pi_t(x) = \left(\sum_{n \geq 1} \alpha^{n-1} c_n, \sum_{r \geq 1} \beta^{r-1} d_{w_r(t)+1} \right). \quad (14.12)$$

The second coordinate reads a β -digit exactly at the scheduled times.

Lemma IV.7.28 (*uniform symbolic/geometric comparison*).

There is $C \geq 1$ such that for every t :

- (a) $\pi_t : X_t \rightarrow K$ is onto;
- (b) π_t is C -Lipschitz;
- (c) every depth- k symbolic cylinder has image of diameter comparable, uniformly in t , to α^k ;
- (d) for every $A \subset X_t$,

$$\dim_H A = \dim_H \pi_t(A), \quad (14.13)$$

and the α -cylinder covering exponent of A equals the Euclidean covering exponent of $\pi_t(A)$.

Proof. Let

$$r_k(t) = \#\{0 \leq j < k : R_\theta^j(t) \in [0, \theta)\}.$$

The crossing-number identity for irrational rotations gives

$$|r_k(t) - k\theta| \leq 2 \quad (14.14)$$

uniformly in t and k . Since $\beta^\theta = \alpha$, (14.14) yields

$$C^{-1}\alpha^k \leq \beta^{r_k(t)} \leq C\alpha^k. \quad (14.15)$$

Every point of each attractor has an IFS address, and the infinitely many scheduled second-coordinate positions read every digit of its address. This proves (a). A depth- k cylinder has image a product of cylinder sets with hull side lengths

$$\alpha^k \text{ diam } C_\alpha, \quad \beta^{r_k(t)} \text{ diam } C_\beta.$$

Both are comparable to α^k , proving (b) and (c). The hull rectangles of distinct depth- k cylinders have disjoint interiors, by the interval open set condition. Each contains a disk of radius $c\alpha^k$ and has diameter at most $C\alpha^k$. Thus any ball of radius $C_0\alpha^k$ meets at most $C_1(C_0)$ such rectangles: their centers lie in a ball of radius $C_2(C_0)\alpha^k$, and their disjoint inscribed disks have area at least $c_2\alpha^{2k}$. All constants are uniform in t and k .

Consequently a cover of $\pi_t(A)$ by balls of radius comparable to α^k lifts to at most a fixed number of depth- k symbolic cylinders per ball. The forward Lipschitz bound gives the reverse covering inequality. This proves equality of the covering exponents. For Hausdorff dimension, lift each ball of an arbitrary fine cover separately, choosing k so that its radius is comparable to α^k . The lifted cylinders have symbolic diameter at most a fixed multiple of that radius and fixed multiplicity; their s -cost is at most a fixed multiple of the original cost for each $s > 0$. This and the Lipschitz inequality prove (14.13). No bi-Lipschitz assertion at touching endpoints is used. $\square$

For $t \in \mathbb{T}$ and $z \in K$, let $\ell_{t,z}$ denote the line through z of slope

$$\beta^{-t}. \quad (14.16)$$

Define $\mathcal{F}$ to be the collection of triples (A, x, t) such that $A \subset X_t$ is nonempty compact, $x \in A$, and

$$\pi_t(A) \subset K \cap \ell_{t, \pi_t(x)}. \quad (14.17)$$

Lemma IV.7.29 (*slice family is shift-stable*).

If $(A, x, t) \in \mathcal{F}$, then

$$(\sigma(A \cap [x_1]), \sigma x, R_\theta t) \in \mathcal{F}. \quad (14.18)$$

Moreover, if $(A_n, x_n, t_n) \rightarrow (A, x, t)$, every (A_n, x_n, t_n) belongs to $\mathcal{F}$, and $R_\theta^j(t)$ never hits the scheduling boundary for $j \geq 0$, then $(A, x, t) \in \mathcal{F}$.

Proof. On the first branch the geometric map induced by σ is

$$\Phi_t(u, v) = \begin{cases} (S_\alpha u, S_\beta v), & t \in [0, \theta), \\ (S_\alpha u, v), & \text{otherwise,} \end{cases} \quad (14.19)$$

Here the inverse maps are taken on the branch specified by the first symbol of the symbolic cylinder under consideration. Even at an endpoint shared by two cylinders, each specified affine inverse is well defined. Directly from (14.12), on that specified branch,

$$\pi_{R_\theta t}(\sigma y) = \Phi_t(\pi_t y). \quad (14.20)$$

Each such affine branch sends a line of slope β^{-t} to a line of slope $\beta^{-R_\theta t}$, proving (14.18). A single-valued geometric inverse map will be selected below, after its endpoint exceptional set has been shown null for the slice measures.

For the second assertion, the finite symbolic schedules of X_{t_n} eventually agree with those of X_t on every fixed depth when the limiting orbit avoids the scheduling boundary. Thus $\pi_{t_n} \rightarrow \pi_t$ uniformly on every finite-depth approximation. Pass to Hausdorff limits in (14.17). $\square$

2. The Extended Magnification and Global Adaptedness

Put

$$Y = \mathcal{P}(X) \times X \times \mathbb{T}, \quad (14.21)$$

where $X = \Lambda^\mathbb{N}$ is an ambient compact full shift containing every X_t . For every state (μ, x, t) with $\mu([x_1]) > 0$, define

$$\hat{M}(\mu, x, t) = (\mu^{[x_1]}, \sigma x, R_\theta t), \quad (14.22)$$

where

$$\mu^{[x_1]} = \frac{\sigma_*(\mu|_{[x_1]})}{\mu([x_1])}. \quad (14.23)$$

On the zero-mass set define $\hat{M}$ arbitrarily. Every adapted law gives that set mass zero.

Definition IV.7.30 (*global adaptedness*).

A probability Q on Y is globally adapted if

$$\int f(\mu, x, t) dQ = \int \left[\int f(\mu, x, t) d\mu(x) \right] dQ_{1,3}(\mu, t) \quad (14.24)$$

for every bounded Borel f . Thus, conditionally on (μ, t) , the point coordinate has law μ .

Lemma IV.7.31 (*global adaptedness is stable*).

The map $\hat{M}$ is continuous at every state with $\mu([x_1]) > 0$. Global adaptedness is preserved by $\hat{M}$ and by weak limits. If Q_k are globally adapted, $Q_k \Rightarrow Q$, and $\hat{M}_* Q_k \Rightarrow Q$, then $\hat{M}_* Q = Q$.

Proof. The symbolic cylinder $[x_1]$ is clopen, so restriction and normalization are continuous whenever its mass is positive. The conditional sampling argument used in the discrete CP construction of Theorem IV.1.64 shows that, given (μ, t) , choosing $x \sim \mu$ and zooming to its first cylinder again produces the new point with the new conditional measure. This proves preservation.

For continuous f on the compact space Y , the function

$$(\mu, t) \mapsto \int f(\mu, x, t) d\mu(x)$$

is continuous. Thus (14.24) passes to weak limits for continuous f , and then for all bounded Borel f by a monotone-class argument. In particular, for any globally adapted Q ,

$$Q\{(\mu, x, t) : \mu([x_1]) = 0\} = \int \sum_{a \in \Lambda: \mu([a])=0} \mu([a]) dQ_{1,3}(\mu, t) = 0.$$

The discontinuity set of $\hat{M}$ is therefore Q -null. The bounded continuous mapping theorem gives $\hat{M}_* Q_k \Rightarrow \hat{M}_* Q$ whenever $Q_k \Rightarrow Q$ through adapted laws. Comparing this limit with the assumed $\hat{M}_* Q_k \Rightarrow Q$ proves invariance. $\square$

3. A CP distribution generated by a large slice

Choose $t_0 \in \mathbb{T}$ and a line ℓ of slope β^{-t_0} with

$$\overline{\dim_B}(K \cap \ell) = \gamma. \quad (14.25)$$

Such a pair is obtained from ℓ_0 by the inverse maps S_α, S_β and the density of the multiplicative orbit $\alpha^n \beta^{-m}$; this is the elementary slope-normalization step preceding Wu's Section 4.

Let

$$E = \pi_{t_0}^{-1}(K \cap \ell). \quad (14.26)$$

By Lemma IV.7.28,

$$\overline{\dim_B} E = \gamma. \quad (14.27)$$

Hence there are $n_k \rightarrow \infty$ with

$$\frac{\log N_{\alpha^{n_k}}(E)}{-n_k \log \alpha} \rightarrow \gamma. \quad (14.28)$$

For every occupied depth- n_k cylinder $[u]$, choose one point $x_u \in E \cap [u]$. Define the uniform discrete measure

$$\mu_k = \frac{1}{N_{\alpha^{n_k}}(E)} \sum_{[u] \cap E \neq \emptyset} \delta_{x_u}. \quad (14.29)$$

Define the pointed law

$$P_k = \frac{1}{N_{\alpha^{n_k}}(E)} \sum_{[u] \cap E \neq \emptyset} \delta_{(\mu_k, x_u, t_0)}, \quad (14.30)$$

and the Cesàro magnification law

$$Q_k = \frac{1}{n_k} \sum_{j=0}^{n_k-1} \hat{M}^j P_k. \quad (14.31)$$

Every P_k and Q_k is globally adapted.

Define the one-step information functional

$$\mathcal{H}(Q) = \frac{1}{\log \alpha} \int \log \mu([x_1]) dQ_{1,2}(\mu, x). \quad (14.32)$$

Because $\log \alpha < 0$, one has $\mathcal{H}(Q) \geq 0$.

Lemma IV.7.32 (*entropy telescoping*).

For every k ,

$$\mathcal{H}(Q_k) = \frac{\log N_{\alpha^{n_k}}(E)}{-n_k \log \alpha}. \quad (14.33)$$

Proof. Along a chosen word $u = u_1 \cdots u_{n_k}$,

$$\sum_{j=0}^{n_k-1} \log \frac{\mu_k([u_1 \cdots u_{j+1}])}{\mu_k([u_1 \cdots u_j])} = \log \mu_k([u]) = -\log N_{\alpha^{n_k}}(E).$$

Average first over j and then over the equally weighted occupied cylinders, and divide by $\log \alpha$. $\square$

Take Q to be a weak accumulation point of (Q_k) .

Lemma IV.7.33 (*weak-limit CP law and entropy*).

The law Q is $\hat{M}$ -invariant and globally adapted, its (μ, x) -marginal is a CP distribution, and

$$\mathcal{H}(Q) = \gamma. \quad (14.34)$$

Proof. Lemma IV.7.31 first gives global adaptedness of Q . For continuous F on Y ,

$$\int F d(\hat{M}_* Q_k) - \int F dQ_k = \frac{\int F d(\hat{M}_*^{n_k} P_k) - \int F dP_k}{n_k} \longrightarrow 0.$$

Consequently $\hat{M}_* Q_k \Rightarrow Q$; the final clause of Lemma IV.7.31 passes invariance to the limit. The marginal on (μ, x) is therefore adapted and invariant under the ordinary symbolic magnification, hence is a CP distribution. The information functional is continuous along globally adapted laws for a more precise reason. Let $\mathcal{A}_1 = \{[a] : a \in \Lambda\}$ and put $0 \log 0 = 0$. By adaptedness and (14.32),

$$\mathcal{H}(Q') = \frac{1}{\log \alpha} \int \sum_{a \in \Lambda} \mu([a]) \log \mu([a]) dQ'_{1,3}(\mu, t) \quad (14.34a)$$

for $Q' = Q$ and for every $Q' = Q_k$. The integrand is bounded in absolute value by $\log |\Lambda|$ and is continuous because every $[a]$ is clopen. Hence $\mathcal{H}(Q_k) \rightarrow \mathcal{H}(Q)$. Lemma IV.7.32 and (14.28) give $\mathcal{H}(Q) = \gamma$. $\square$

Let

$$Q = \int Q_\omega d\tau(\omega) \quad (14.35)$$

be the ergodic decomposition under $\hat{M}$.

Lemma IV.7.34 (*global adaptedness of ergodic components*).

For τ -a.e. ω , the law Q_ω is globally adapted. Its first-two marginal is consequently an ergodic symbolic CP distribution.

Proof. It suffices to show that an invariant-set label is, modulo Q , determined by the pair (μ, t) rather than by the sampled point x . Let A be an invariant Borel set and put

$$h_A(\mu, t) = \mu\{x : (\mu, x, t) \in A\}.$$

For fixed (μ, t) sample $X \sim \mu$ and reveal its first n symbols. Conditional on that prefix, the remaining point has the zoomed law $\mu^{[X_1 \cdots X_n]}$, while the phase is $R_\theta^n t$. Invariance of A therefore makes

$$h_A(\mu^{[X_1 \cdots X_n]}, R_\theta^n t) = \mathbb{E}_\mu(1_A(\mu, X, t) \mid X_1, \dots, X_n). \quad (14.35a)$$

These conditional expectations form a bounded martingale and converge to $1_A(\mu, X, t)$. When (μ, X, t) is sampled from the stationary law Q , their distributions do not depend on n . Their second moments are thus constant. Orthogonality of martingale increments forces each increment to vanish in $L^2(Q)$, so (14.35a) equals $h_A(\mu, t)$ for every n and

$$1_A(\mu, x, t) = h_A(\mu, t) \in \{0, 1\} \quad \text{for } Q\text{-a.e. } (\mu, x, t).$$

Apply this to a countable generating family for the invariant σ -algebra. The ergodic-component label ω is a measurable function of (μ, t) modulo Q . Disintegrating the global adaptation identity (14.24) over that label shows that the conditional law of x given (μ, t, ω) is still μ . Hence Q_ω is globally adapted for τ -a.e. ω . Ergodicity and invariance pass to its first-two marginal under the factor map $(\mu, x, t) \mapsto (\mu, x)$. $\square$

Since

$$\gamma = \mathcal{H}(Q) = \int \mathcal{H}(Q_\omega) d\tau(\omega), \quad (14.36)$$

the set

$$\Omega_\gamma = \{\omega : \mathcal{H}(Q_\omega) \geq \gamma\} \quad (14.37)$$

has positive τ -measure.

Lemma IV.7.35 (*symbolic CP dimension formula*).

Let Q^* be an ergodic, globally adapted, $\hat{M}$ -invariant law supported on $\{(\mu, x, t) : \text{supp } \mu \subset X_t\}$. For $Q_{1,3}^*$ -a.e. (μ, t) , the measure μ is exact dimensional in the metric d_α , with dimension $\mathcal{H}(Q^*)$. The measure $(\pi_t)_*\mu$ has the same Euclidean dimension.

Proof. Set $I(\mu, x) = -\log \mu([x_1])$ on the full-measure positive-cylinder domain. Global adaptedness gives

$$\int I dQ^* = \int \sum_{a \in \Lambda} -\mu([a]) \log \mu([a]) dQ_{1,3}^*(\mu, t) = (-\log \alpha) \mathcal{H}(Q^*) < \infty.$$

For Q^* -a.e. state, Birkhoff's theorem and the cylinder telescope give

$$-\frac{1}{n} \log \mu([x_1 \cdots x_n]) = \frac{1}{n} \sum_{j=0}^{n-1} I(\hat{M}^j(\mu, x, t)) \longrightarrow (-\log \alpha) \mathcal{H}(Q^*). \quad (14.37a)$$

Disintegration using global adaptedness makes this statement valid for μ -a.e. x for $Q_{1,3}^*$ -a.e. (μ, t) . A ball in d_α of radius comparable to α^n is a depth- n cylinder, so (14.37a) is precisely the fixed-center local-dimension limit $\log \mu(B_{d_\alpha}(x, r)) / \log r \rightarrow \mathcal{H}(Q^*)$. The mass-distribution characterization of exact dimension then gives $\dim_{\text{H}} \mu = \mathcal{H}(Q^*)$. To transfer the local dimension rather than only the dimension of sets, fix such a pair (μ, t) . Denote by R_u the hull rectangle of a depth- n symbolic cylinder $[u]$, and put

$$\mathcal{N}_n(u) = \{v : \text{dist}(R_u, R_v) \leq \alpha^n\}.$$

The packing argument in Lemma IV.7.28 bounds both the number of neighbors of a rectangle and the number of rectangles for which it is a neighbor by one constant C . For $\varepsilon > 0$, call u bad if

$$\sum_{v \in \mathcal{N}_n(u)} \mu([v]) > e^{\varepsilon n} \mu([u]).$$

The total mass of the bad cylinders is at most

$$\sum_{u \text{ mbad}} \mu([u]) \leq e^{-\varepsilon n} \sum_u \sum_{v \in \mathcal{N}_n(u)} \mu([v]) \leq C e^{-\varepsilon n}.$$

By Borel–Cantelli, for μ -a.e. x its depth- n cylinder is not bad for all sufficiently large n . Set $\eta = (\pi_t)_*\mu$ and $z = \pi_t x$. Every cylinder contributing to $\eta(B(z, \alpha^n))$ is a neighbor of the cylinder $[x_1 \cdots x_n]$. Hence, eventually,

$$\eta(B(z, \alpha^n)) \leq e^{\varepsilon n} \mu([x_1 \cdots x_n]).$$

Conversely $\pi_t([x_1 \cdots x_n]) \subset B(z, C\alpha^n)$ after increasing C , so this latter ball has mass at least $\mu([x_1 \cdots x_n])$. Combining both inequalities with (14.37a), and then taking a countable sequence $\varepsilon \downarrow 0$, proves that η is exact dimensional with the same dimension. $\square$

Lemma IV.7.36 (*slice support survives the limit*).

For Q -a.e. (μ, x, t) ,

$$(\text{supp } \mu, x, t) \in \mathcal{F}. \quad (14.38)$$

The same is true for Q_ω -a.e. state for τ -a.e. ω .

Proof. Each Q_k is supported on the forward orbit of triples originating in E , and Lemma IV.7.29 makes those triples belong to $\mathcal{F}$. Any weak-limit state is a limit of such triples. The t -marginal of Q is R_θ -invariant, hence Lebesgue by unique ergodicity; therefore it gives zero mass to the countable scheduling-boundary orbit. Apply the closedness clause of Lemma IV.7.29. The ergodic-component statement follows by disintegration. $\square$

Theorem IV.7.37 (*slice-generated CP law*).

Under (14.6) there exists an ergodic $\widehat{M}$ -invariant component Q^* such that:

- (a) its first-two marginal $P^* = Q_{1,2}^*$ is an ergodic CP distribution;
- (b)

$$\dim P^* = \mathcal{H}(Q^*) \geq \gamma; \quad (14.39)$$

- (c) for Q^* -a.e. (μ, x, t) , the measure μ is supported on a symbolic preimage of a slice of K , namely

$$\pi_t(\text{supp } \mu) \subset K \cap \ell_{t, \pi_t(x)}. \quad (14.40)$$

Proof. Choose $\omega \in \Omega_\gamma$ satisfying Lemmas IV.7.34 and IV.7.36, and let $Q^* = Q_\omega$. Lemma IV.7.35 gives $\dim P^* = \mathcal{H}(Q^*)$ in the symbolic model and after geometric coding. $\square$

4. The Geometric Skew Product

Choose a single-valued Borel inverse-IFS map S_α on C_α and S_β on C_β , choosing either admissible branch at shared endpoints. For the positive-dimensional slice laws selected above this choice is immaterial almost everywhere: their geometric measures are nonatomic by Lemma IV.7.35, and a non-axis-parallel line meets each vertical or horizontal cylinder boundary in at most one point. The countable union of these boundary intersections therefore has zero mass. Now define

$$U : K \times \mathbb{T} \longrightarrow K \times \mathbb{T}, \quad U(z, t) = (\Phi_t(z), R_\theta t). \quad (14.41)$$

For $z = (z_1, z_2)$,

$$U_t^n(z) = (S_\alpha^n(z_1), S_\beta^{r_n(t)}(z_2)), \quad (14.42)$$

where $r_n(t)$ is the visit count from (14.14).

Let $\mathcal{A}_n^t$ be the finite partition of K into the geometric images under π_t of depth- n cylinders in X_t , assigning shared endpoints consistently with the chosen branches. By the interval-OSC comparison,

$$\text{diam}(A) \asymp \alpha^n$$

uniformly in t for $A \in \mathcal{A}_n^t$; the union of all boundaries of $\mathcal{A}_n^t$ has zero mass for every non-atomic slice measure considered below. For a probability η on K and z in a positive-mass atom $\mathcal{A}_n^t(z)$, define

$$\eta_{\mathcal{A}_n^t(z)} = \frac{(U_t^n)_*(\eta|_{\mathcal{A}_n^t(z)})}{\eta(\mathcal{A}_n^t(z))}. \quad (14.43)$$

If $\eta = \pi_t \mu$ and $z = \pi_t x$, then the symbolic/geometric intertwining gives

$$\eta_{\mathcal{A}_n^t(z)} = \pi_{R_\theta^n t}(\mu^{[x_1 \cdots x_n]}). \quad (14.44)$$

5. From a symbolic CP orbit to a geometric orbit

Global adaptedness makes the symbolic point and the measure coordinate have the same geometric barycenter. This gives a direct comparison of their empirical laws.

Lemma IV.7.38 (*conditional zooms and geometric orbits*).

Let Q^* be as in Theorem IV.7.37. Let (μ, x, t) be a Q^* -generic state for which $\text{supp } \mu \subset X_t$, $x \in X_t$, and every selected cylinder along its orbit has positive mass. These conditions hold for Q^* -a.e. state. Put

$$\eta = \pi_t \mu, \quad z = \pi_t x.$$

Then the two sequences of probabilities on $K \times \mathbb{T}$,

$$\Xi_N = \frac{1}{N} \sum_{n=0}^{N-1} \eta_{\mathcal{A}_n^t(z)} \times \delta_{R_\theta^n t}, \quad (14.45)$$

and

$$\Omega_N = \frac{1}{N} \sum_{n=0}^{N-1} \delta_{U^n(z,t)}, \quad (14.46)$$

have the same weak limits. Moreover,

$$\Xi_N \longrightarrow \nu^* := \int (\pi_s \vartheta) \times \delta_s dQ_{1,3}^*(\vartheta, s). \quad (14.47)$$

Proof. Define two maps on Y by the digit formula (14.12), ignoring the unscheduled second-coordinate symbols when a state is not in X_s :

$$F(\vartheta, y, s) = (\pi_s y, s), \quad G(\vartheta, y, s) = (\pi_s \vartheta) \times \delta_s. \quad (14.48)$$

Here G takes values in $\mathcal{P}(K \times \mathbb{T})$; when discussing its barycenter, it is understood that a probability law on that space is averaged to a probability on $K \times \mathbb{T}$. At every phase outside the countable orbit of the scheduling boundary, the allowed finite alphabets are locally constant through each fixed depth. The coding maps π_s therefore vary continuously on finite-depth approximations, uniformly over symbolic tails. Consequently both F and G are continuous at states with such a phase.

The s -marginal of the ergodic law Q^* is invariant under the irrational rotation R_θ and hence is Lebesgue. It gives the discontinuity set of F and G mass zero. Genericity of the stated (μ, x, t) and the a.e.-continuous mapping theorem thus yield

$$\frac{1}{N} \sum_{n < N} \delta_{F(\widehat{M}^n(\mu, x, t))} \implies F_* Q^*, \quad \frac{1}{N} \sum_{n < N} G(\widehat{M}^n(\mu, x, t)) \implies \int G dQ^*. \quad (14.49)$$

By the symbolic/geometric intertwining and (14.44), the two expressions on the left are respectively Ω_N and Ξ_N . Finally global adaptedness, Lemma IV.7.34, gives for every continuous f on $K \times \mathbb{T}$,

$$\int f d(F_* Q^*) = \int \int f(\pi_s y, s) d\vartheta(y) dQ_{1,3}^*(\vartheta, s) = \int f d\left(\int G dQ^*\right).$$

The common limit is exactly the law ν^* in (14.47). $\square$

Corollary IV.7.39 (*U-invariance*).

The measure ν^* in (14.47) is U -invariant.

Proof. It is a weak limit of the orbit empirical measures Ω_N . For any continuous f at points of continuity of U ,

$$\int f \circ U d\Omega_N - \int f d\Omega_N = O(1/N).$$

Each affine inverse branch is continuous away from the finite first-level endpoint boundaries. Conditional on a phase, ν^* is a mixture of nonatomic measures on non-axis-parallel slices, so those vertical and horizontal boundaries have zero mass. The scheduling boundaries also have zero mass because the phase marginal is Lebesgue. Thus the entire discontinuity set of U has ν^* -mass zero. Approximate $f \circ U$ by bounded continuous functions off a small neighborhood of the discontinuity set and pass to the limit. $\square$

6. The Good-Slice Property

The required good times can be selected from a single measurable set of CP states. The point is that the scale threshold in the entropy bound is chosen *before* the orbit: it does not depend on the state or on the later time k .

Lemma IV.7.40 (*uniform good states of an ergodic slice CP law*).

Let Q^* be the ergodic law of Theorem IV.7.37, and put $h = \mathcal{H}(Q^*) \geq \gamma > 0$. For every $\varepsilon > 0$ there are $\delta > 0$, $n_0 \in \mathbb{N}$, and a Borel set G of triples (μ, x, t) with $Q^*(G) > 1 - \varepsilon$ such that, for every triple in G and $\eta = (\pi_t)_* \mu$,

$$\sup_{y \in \mathbb{R}^2} \eta(B(y, \delta)) \leq \varepsilon, \quad H(\eta, \mathcal{D}_n) \geq n(h \log 2 - \varepsilon) \quad \text{for every } n \geq n_0. \quad (14.50)$$

For Q^* -almost every triple, the set of integers $k \geq 0$ for which $\widehat{M}^k(\mu, x, t) \in G$ has asymptotic density $Q^*(G)$.

Proof. Lemma IV.7.35 and Lemma IV.7.28 give, for $Q_{1,3}^*$ -almost every (μ, t) , an exact-dimensional probability $\eta = (\pi_t)_* \mu$ of dimension $h > 0$. Hence η has no atoms. For every non-atomic probability on the common compact set K ,

$$\lim_{r \downarrow 0} \sup_{y \in \mathbb{R}^2} \eta(\bar{B}(y, r)) = 0. \quad (14.50a)$$

Indeed, a failure would give $r_j \downarrow 0$, centers y_j in a fixed compact neighborhood of K , and mass at least $c > 0$ in each closed ball. After taking $y_j \rightarrow y$, every neighborhood of y would have mass at least c , contradicting $\eta\{y\} = 0$. The map $\eta \mapsto \sup_y \eta(B(y, \delta))$ is Borel: its supremum equals the supremum over rational centers, since for rational $q_j \rightarrow y$, Fatou's lemma gives $\eta(B(y, \delta)) \leq \liminf_j \eta(B(q_j, \delta))$. To choose a uniform δ , bound this supremum by the supremum of masses of closed balls of radius 2δ with rational centers. By (14.50a) the latter tends to zero for every non-atomic η . Hence δ can be chosen so that the first inequality in (14.50) holds on a set of Q^* -measure greater than $1 - \varepsilon/2$.

For the entropy inequality, a dyadic cell $D_n(z)$ lies in $B(z, C2^{-n})$, where C depends only on the ambient dimension. Exact dimensionality therefore gives, for η -almost every z ,

$$\liminf_{n \rightarrow \infty} \frac{-\log \eta(D_n(z))}{n \log 2} \geq h.$$

Fatou's lemma, applied to the nonnegative information functions, yields

$$\liminf_{n \rightarrow \infty} \frac{H(\eta, \mathcal{D}_n)}{n \log 2} \geq h. \quad (14.50b)$$

Thus for each such η the second inequality in (14.50) holds for all sufficiently large n . The Borel sets on which it holds simultaneously for every $n \geq n_0$ increase to a full- Q^* -measure set as $n_0 \rightarrow \infty$. Choose n_0 so that one of them has measure greater than $1 - \varepsilon/2$, and intersect it with the preceding non-concentration set to obtain G with $Q^*(G) > 1 - \varepsilon$.

Finally $\hat{M}$ preserves and is ergodic for Q^* . Birkhoff's theorem applied to $\mathbf{1}_G$ gives the asserted asymptotic frequency on a common conull set. $\square$

The exact symbolic/geometric identity (14.44) now transfers these same fixed parameters to conditional slice zooms.

Lemma IV.7.41 (*geometric good-slice estimate*).

For every $\varepsilon > 0$ there exist $\delta(\varepsilon) > 0$ and $n_0(\varepsilon)$ such that for ν^* -a.e. (z, t) there exists a slice measure $\eta = \pi_t \mu$, with $z \in \text{supp } \eta$, for which, writing

$$E_N = \left\{ 1 \leq k \leq N : \begin{array}{l} \sup_y \eta_{\mathcal{A}_k^t(z)}(B(y, \delta)) \leq \varepsilon, \\ H(\eta_{\mathcal{A}_k^t(z)}, \mathcal{D}_n) \geq n(\gamma \log 2 - 2\varepsilon) \quad \text{for all } n \geq n_0 \end{array} \right\},$$

one has

$$\liminf_{N \rightarrow \infty} \frac{\#E_N}{N} \geq 1 - \varepsilon. \quad (14.51)$$

Proof. Choose the generic representation supplied by Lemma IV.7.38. Apply Lemma IV.7.40 with ε replaced by any $\varepsilon' < \varepsilon$, and retain its δ, n_0 . For a Q^* -generic

triple (μ, x, t) let $\eta = (\pi_t)_* \mu$ and $z = \pi_t(x)$. Equation (14.44) gives, at every k for which $\hat{M}^k(\mu, x, t) \in G$,

$$\eta_{\mathcal{A}_k^t(z)} = (\pi_{R_\theta^k t})_* \mu_k, \quad (\mu_k, x_k, R_\theta^k t) = \hat{M}^k(\mu, x, t).$$

Consequently both inequalities in the definition of E_N hold at these k : the entropy bound follows from $h \geq \gamma$ and $\varepsilon' < \varepsilon$, while the non-concentration bound has the same δ . Lemma IV.7.40 gives lower density strictly greater than $1 - \varepsilon$. The barycenter identity (14.47) transfers this full- Q^* -measure property to ν^* -almost every (z, t) , with the selected measure η as the witness. $\square$

7. An ergodic geometric point factor

The geometric law is not merely an invariant mixture: it is the point factor of the already selected ergodic CP law. The following entropy calculation also supplies the positive-entropy hypothesis needed later.

Lemma IV.7.42 (*positive entropy of the geometric point factor*).

The measure ν^* is U -ergodic, and

$$h_{\nu^*}(U) \geq (-\log \alpha) \mathcal{H}(Q^*) > 0. \quad (14.53a)$$

Proof. The point map $F(\mu, x, t) = (\pi_t x, t)$ intertwines $\hat{M}$ with U , by (14.20), and $F_* Q^* = \nu^*$ by global adaptedness. Thus it is a measurable factor of the ergodic system Q^* and is ergodic.

Let $\mathcal{B}$ record the first-level geometric cylinder in each coordinate, and let $\mathcal{B}_n = \bigvee_{j=0}^{n-1} U^{-j} \mathcal{B}$. Conditionally on a phase t , this partition refines the geometric image of the depth- n scheduled symbolic partition: the first coordinate is read at every step and the second is read at every scheduled step. Outside the cylinder boundaries, which are null for each positive-dimensional slice measure, these images are identified with the symbolic cylinders. Write $\mathcal{A}_n^{\text{sym}}$ for the depth- n symbolic partition and Q_t^* for the disintegration of $Q_{1,3}^*$ over t . The conditional point law at phase t is $\int (\pi_t)_* \mu dQ_t^*(\mu)$. Concavity of finite entropy therefore gives

$$H_{\nu^*}(\mathcal{B}_n) \geq H_{\nu^*}(\mathcal{B}_n | t) \geq \int H(\mu, \mathcal{A}_n^{\text{sym}}) dQ_{1,3}^*(\mu, t).$$

The conditional chain rule, adaptedness and invariance of Q^* give

$$\int H(\mu, \mathcal{A}_n^{\text{sym}}) dQ_{1,3}^* = \sum_{j=0}^{n-1} \int -\log \mu_j([x_{j+1}]) dQ^* = n(-\log \alpha) \mathcal{H}(Q^*),$$

where $(\mu_j, \sigma^j x, R_\theta^j t) = \hat{M}^j(\mu, x, t)$. Divide by n and use the finite-partition entropy-rate definition to obtain (14.53a). Finiteness follows from the finite generator below. $\square$

We may therefore take

$$\nu_\infty = \nu^*. \quad (14.53)$$

No selection of an ergodic component with unverified entropy is necessary.

Theorem IV.7.43 (*ergodic skew-product measure*).

There exists an ergodic U -invariant probability ν_∞ on $K \times \mathbb{T}$ satisfying:

- (a) its t -marginal is Lebesgue measure;
- (b) for ν_∞ -a.e. (z, t) , the point z lies on a slice of K of slope β^{-t} carrying a probability measure η whose conditional zooms satisfy the good-slice property (14.51);
- (c) the finite partition by first-level geometric cylinders and a schedule partition $\{[0, \theta), [\theta, 1)\}$ of $\mathbb{T}$ is a finite generator modulo boundary-null sets;
- (d) $0 < h_{\nu_\infty}(U) < \infty$;
- (e) the geometric and entropy hypotheses of the abstract selection theorem, Theorem IV.7.51, hold. Its Bernoulli-factor supplier is Theorem IV.7.26; no L^q inverse theorem enters.

Proof. Take $\nu_\infty = \nu^*$ and use Lemma IV.7.42 for ergodicity. Its phase marginal is invariant under R_θ and hence is Lebesgue by unique ergodicity, proving (a). Lemma IV.7.41 proves (b).

For (c), take the schedule partition $\{[0, \theta), [\theta, 1)\}$. The forward iterates of its endpoints are dense under the irrational rotation. Distinct phases are consequently separated by some iterated partition. Once the phase is known, the geometric cylinder names determine the first coordinate at every step and the second at the infinitely many scheduled steps. Their cylinder diameters tend to zero, so these names determine the point. All iterated circle boundaries have zero measure because the phase marginal is Lebesgue. The boundaries of every finite geometric join lie, at each fixed phase off the finite scheduling-boundary orbit, in finitely many vertical and horizontal cylinder boundary lines. Each has zero mass under the conditional nonatomic non-axis-parallel slice measures. Fubini therefore proves that every finite join is boundary-null. This proves the finite generator and its boundary-null property.

Lemma IV.7.42 gives the positive lower bound in (d), and the finite generator bounds entropy from above by the logarithm of its cardinality. These verify the geometric and entropy hypotheses in (e). Theorem IV.7.26 then provides precisely the required unilateral Bernoulli factor. $\square$

7.3 The Strict-Factor Form of the Ergodic Selection Theorem

Selection target IV.7.44 (*ergodic selection theorem*).

Let (X, d) be a compact metric space, let μ be a Borel probability measure, and let $T : X \rightarrow X$ be an ergodic measure-preserving map. Let $\mathcal{A}$ be a finite generating Borel partition and

$$\mathcal{A}_n = \bigvee_{j=0}^{n-1} T^{-j} \mathcal{A}.$$

Assume that $\mu(\partial A) = 0$ for every atom $A \in \mathcal{A}_n$ and every $n \geq 1$. Let $\xi \in \mathbb{R} \setminus \mathbb{Q}$ and write

$$R_\xi(t) = t - \xi \pmod{1}.$$

For every $\varepsilon > 0$ there is $n_2 = n_2(\varepsilon)$ such that, for every $n \geq n_2$, there are pairwise disjoint measurable sets

$$C_1, \dots, C_{N(n,\varepsilon)} \subset X$$

and pairwise disjoint measurable supersets

$$\tilde{C}_1, \dots, \tilde{C}_{N(n,\varepsilon)} \subset X$$

with the following properties:

$$1. \mu\left(\bigcup_i C_i\right) \geq 1 - \varepsilon;$$

2. for every i ,

$$\#\{A \in \mathcal{A}_n : C_i \cap A \neq \emptyset\} \leq e^{n\varepsilon};$$

3. $C_i \subset \tilde{C}_i$ and

$$\mu(C_i) \geq (1 - \varepsilon)\mu(\tilde{C}_i),$$

and, for μ -a.e. x , the sequence

$$\{R_\xi^k(0) : k \in \mathbb{N}, T^k x \in \tilde{C}_i\}$$

is uniformly distributed in $[0, 1)$ for every i .

This is the exact finite-generator form of the selection theorem used in the slicing argument below.

Lemma IV.7.45 (*Bernoulli-return rotation sampling*).

Let $\pi : (X, T, \mu) \rightarrow (\Sigma, \sigma, \nu)$ be a finite-alphabet Bernoulli factor, possibly the one-point factor, and let $B \subset \Sigma$ have positive ν -measure. If ξ is irrational, then for μ -a.e. x the sequence

$$\{R_\xi^k(0) : k \in \mathbb{N}, \sigma^k \pi(x) \in B\}$$

is uniformly distributed in $[0, 1)$.

Proof. A nontrivial Bernoulli shift is mixing and hence weakly mixing; the one-point case is immediate. Therefore the product of the factor with the irrational rotation is ergodic. Apply Birkhoff to indicators of $B \times J$ for a countable interval algebra $J \subset [0, 1)$, and divide by the return frequency to B . This gives uniform distribution for almost every factor point and almost every initial rotation phase. Since $R_\xi^k(t) = R_\xi^k(0) + t$ modulo one, translation invariance of Lebesgue measure transfers the conclusion to phase 0. Pull back through π . $\square$

1. Conditional SMB with an Entropy Gap

Lemma IV.7.46 (*conditional SMB with entropy loss*).

Let (X, T, μ) be ergodic and let $\mathcal{A}$ be a finite generator. Let

$$\pi : (X, T, \mu) \longrightarrow (Y, S, \nu)$$

be an ergodic factor onto a finite-alphabet one-sided shift whose zero-coordinate partition is a generator. Let (μ_y) be the disintegration over π and put

$$\rho = h_\mu(T) - h_\nu(S) \geq 0.$$

Then for μ -a.e. x ,

$$-\frac{1}{n} \log \mu_{\pi(x)}(\mathcal{A}_n(x)) \longrightarrow \rho. \quad (4)$$

Consequently, for every $\eta > 0$ and $\kappa > 0$ there are a measurable set $G = G(\eta, \kappa) \subset X$ and n_0 such that

$$\mu(G) > 1 - \kappa$$

and, for $x \in G$ and $n \geq n_0$,

$$\mu_{\pi(x)}(\mathcal{A}_n(x)) \geq \exp[-n(\rho + \eta)]. \quad (5)$$

Proof. The one-sided relative conditional Shannon–McMillan–Breiman theorem, Theorem IV.1.42, proved in Chapter IV.1, applies to exactly this finite-alphabet factor. Since $\mathcal{A}$ is a finite generator, its relative entropy rate over the factor equals

$$h_\mu(T) - h_\nu(S) = \rho.$$

This gives (4). Apply Egorov to obtain uniform convergence on a set of measure $> 1 - \kappa$, and retain only the lower half of the two-sided asymptotic estimate. $\square$

Lemma IV.7.47 (*fiber name count with strict entropy loss*).

Under Lemma IV.7.46, fix $\eta, \kappa > 0$ and let G, n_0 be as there. For ν -a.e. y define

$$E_G(y, n) = \{C \in \mathcal{A}_n : \pi^{-1}(y) \cap G \cap C \neq \emptyset\}.$$

After deleting the standard null exceptional set, for every $n \geq n_0$,

$$\#E_G(y, n) \leq \exp[n(\rho + \eta)]. \quad (6)$$

Proof. If $C \in E_G(y, n)$, choose $x \in \pi^{-1}(y) \cap G \cap C$ outside the common exceptional set. Since $\mathcal{A}_n(x) = C$, (5) gives

$$\mu_y(C) \geq \exp[-n(\rho + \eta)].$$

The distinct C are disjoint and μ_y is a probability. Summing their masses proves (6). $\square$

Remark. This is exactly Wu's estimate (3), with δ replaced by $\rho + \eta$.

Theorem IV.7.48 (*strict-factor version of Wu Theorem 6.1*).

Assume the hypotheses in the selection target IV-C15-TARGET, with $0 < h_\mu(T) < \infty$. Suppose that for every $\zeta > 0$ there is a nontrivial finite-alphabet Bernoulli factor π_ζ whose entropy h_ζ satisfies

$$h_\mu(T) - \zeta < h_\zeta < h_\mu(T). \quad (7)$$

Then the three conclusions of IV-C15-TARGET hold. The zero-entropy case is covered separately by Theorem IV.7.49, using the one-point Bernoulli factor; no positive-entropy strict factor is postulated in that case.

Proof. Fix $\varepsilon > 0$. The strict-factor hypothesis supplies a Bernoulli factor with gap

$$0 < \rho = h_\mu(T) - h_\nu(\sigma) < \varepsilon/4. \quad (8)$$

Choose an SMB tolerance

$$0 < \eta < \varepsilon/4. \quad (9)$$

Independently choose the measure-loss parameter $\kappa > 0$ so small that, with $\kappa_1 = \sqrt{\kappa}$,

$$(1 - \kappa - \kappa_1)(1 - \kappa)(1 - \kappa_1) \geq 1 - \varepsilon. \quad (10)$$

Apply Lemma IV.7.46 with η and κ . It gives a large set G and n_0 with

$$\mu_{\pi(x)}(\mathcal{A}_n(x)) \geq \exp[-n(\rho + \eta)]$$

for $x \in G$ and $n \geq n_0$.

Disintegration gives $\int \mu_y(G) d\nu(y) = \mu(G) > 1 - \kappa$. Hence $\int (1 - \mu_y(G)) d\nu(y) < \kappa$, and Markov's inequality gives $B_1 = \{y : \mu_y(G) > 1 - \kappa_1\}$ with $\nu(B_1) > 1 - \kappa_1$ and

$$\mu_y(G) > 1 - \kappa_1. \quad (11)$$

Fix $n \geq n_0$. For each of the finitely many atoms $C \in \mathcal{A}_n$, the conditional expectation of 1_C on the first r factor coordinates converges ν -a.e. to $\mu_y(C)$ as $r \rightarrow \infty$. This is Theorem IV.1.44, applied simultaneously to all atoms; Egorov's theorem then gives a common r and a set B_κ^n of ν -measure $> 1 - \kappa$ such that for every $y \in B_\kappa^n$ and every $C \in \mathcal{A}_n$,

$$\frac{\mu(\pi^{-1}(B(y, r)) \cap C)}{\mu(\pi^{-1}(B(y, r)))} \geq (1 - \kappa)\mu_y(C). \quad (12)$$

Let Y_0 be the common conull set on which the disintegration is supported on its fibre and Lemma IV.7.47 holds for every $n \geq n_0$. Set $B_2 = B_1 \cap B_\kappa^n \cap Y_0$. For $y \in B_2$ let $E(y, n)$ be the $\mathcal{A}_n$ -atoms meeting $\pi^{-1}(y) \cap G$. By Lemma IV.7.47,

$$\#E(y, n) \leq \exp[n(\rho + \eta)] \leq \exp[n\varepsilon/2] \leq \exp[n\varepsilon]. \quad (13)$$

The radius- r factor balls form a finite partition and any two are equal or disjoint. Enumerate the positive-measure balls meeting B_2 as B_i , choose $y_i \in B_2 \cap B_i$, and define

$$\tilde{C}_i = \pi^{-1}(B_i), \quad C_i = \tilde{C}_i \cap \bigcup_{C \in E(y_i, n)} C.$$

The families C_i and $\tilde{C}_i$ are pairwise disjoint, and C_i meets at most $\#E(y_i, n)$ atoms of $\mathcal{A}_n$. Since $G \cap \pi^{-1}(y_i)$ is contained in their union, (11) implies that the sum of their μ_{y_i} -masses is greater than $1 - \kappa_1$. Summing (12) over these atoms yields

$$\mu(C_i) \geq (1 - \kappa)(1 - \kappa_1)\mu(\tilde{C}_i) \geq (1 - \varepsilon)\mu(\tilde{C}_i). \quad (14)$$

The selected factor balls cover B_2 , so $\sum_i \mu(\tilde{C}_i) = \sum_i \nu(B_i) > 1 - \kappa - \kappa_1$. Consequently the union estimate is

$$\mu\left(\bigcup_i C_i\right) \geq (1 - \kappa - \kappa_1)(1 - \kappa)(1 - \kappa_1) \geq 1 - \varepsilon \quad (15)$$

by (10). Property (2) of Wu Theorem 6.1 follows from (13).

Finally, Lemma IV.7.45 applies to each positive-measure factor ball B_i , giving uniform distribution of the rotation times selected by returns to $\tilde{C}_i$. This proves the three conclusions. $\square$

Theorem IV.7.49 (*ergodic selection theorem*).

Under the hypotheses of the selection target above, the sets C_i and $\tilde{C}_i$ exist with all three stated properties.

Proof. Fix $\varepsilon > 0$. Choose $\delta > 0$ so small that, with $\delta_1 = \sqrt{\delta}$,

$$\delta \leq \varepsilon, \quad (1 - \delta - \delta_1)(1 - \delta)(1 - \delta_1) \geq 1 - \varepsilon. \quad (15.1)$$

By Theorem IV.7.26, choose a finite-alphabet Bernoulli factor $\pi : (X, T, \mu) \rightarrow (\Sigma, \sigma, \nu)$ with entropy $h_\nu(\sigma) = h_\mu(T)$; when $h_\mu(T) = 0$ this is the one-point Bernoulli system. Apply Lemma IV.7.46 with $\rho = 0$, $\eta = \delta$, and $\kappa = \delta$. There are $G \subset X$ and n_2 such that $\mu(G) > 1 - \delta$ and, for $x \in G$, $n \geq n_2$,

$$\mu_{\pi(x)}(\mathcal{A}_n(x)) \geq e^{-n\delta}. \quad (15.2)$$

Disintegration and Markov's inequality give a set $B_1 \subset \Sigma$ with $\nu(B_1) > 1 - \delta_1$ and $\mu_y(G) > 1 - \delta_1$ for $y \in B_1$.

Fix $n \geq n_2$. By Theorem IV.1.44, after shrinking factor cylinders uniformly on a set B_δ^n of ν -measure $> 1 - \delta$, there is a common factor-cylinder radius r such that

$$\frac{\mu(\pi^{-1}(B(y, r)) \cap A)}{\mu(\pi^{-1}(B(y, r)))} \geq (1 - \delta)\mu_y(A) \quad (15.3)$$

for $y \in B_\delta^n$ and every $A \in \mathcal{A}_n$. Let Y_0 be a common conull factor set on which the conditional measures are supported on their fibers and the fiber name-count conclusion of Lemma IV.7.47 holds for every relevant n . Put $B_2 = B_1 \cap B_\delta^n \cap Y_0$. For $y \in B_2$ let

$$E(y, n) = \{A \in \mathcal{A}_n : \pi^{-1}(y) \cap G \cap A \neq \emptyset\}.$$

Equation (15.2) and disjointness of the atoms give $\#E(y, n) \leq e^{n\delta} \leq e^{n\varepsilon}$.

The radius- r balls in the finite-alphabet shift are either disjoint or equal; enumerate the distinct positive-measure balls meeting B_2 by B_i and choose $y_i \in B_2 \cap B_i$. Define

$$\tilde{C}_i = \pi^{-1}(B_i), \quad C_i = \tilde{C}_i \cap \bigcup_{A \in E(y_i, n)} A.$$

These two families are pairwise disjoint. From (15.3) and $\mu_{y_i}(G) > 1 - \delta_1$,

$$\mu(C_i) \geq (1 - \delta)(1 - \delta_1)\mu(\tilde{C}_i) \geq (1 - \varepsilon)\mu(\tilde{C}_i). \quad (15.4)$$

Moreover $\nu(\bigcup_i B_i) \geq \nu(B_2) > 1 - \delta - \delta_1$, so (15.1) gives

$$\mu\left(\bigcup_i C_i\right) \geq (1 - \delta - \delta_1)(1 - \delta)(1 - \delta_1) \geq 1 - \varepsilon. \quad (15.5)$$

The atom-count bound gives property (2), and Lemma IV.7.45, applied to each factor ball B_i , gives property (3). Thus all clauses of Theorem IV.7.49 hold. $\square$

Corollary IV.7.50 (*arbitrarily high strict Bernoulli entropy is enough*).

For a positive-entropy system in the Wu intersection route, replace the dependency on an equal-entropy unilateral Bernoulli factor by the strictly weaker input

$$\text{for every } \zeta > 0 \text{ there is } h' \text{ with } h - \zeta < h' < h \quad (16)$$

realized by a nontrivial finite-alphabet Bernoulli factor. Then Theorem IV.7.49 and the conclusions of the following slicing argument remain valid.

Proof. For each target ε in Theorem IV.7.49, invoke the strict Bernoulli-factor hypothesis above with $\zeta < \varepsilon/4$ and apply Theorem IV.7.48. The factor is allowed to depend on ε ; Wu Theorem 6.1 itself fixes ε before constructing the families C_i, C'_i . $\square$

7.4 Slicing and arithmetic intersection deductions

1. The abstract selection output used here

We isolate the exact consequence of Theorem IV.7.49.

Theorem IV.7.51 (*abstract selection theorem*).

Let Z be a compact metric space, T an ergodic measure-preserving Borel map for a Borel probability ν , and $\mathcal{A}$ a finite generating Borel partition. Assume that every atom of every finite forward join $\mathcal{A}^m$ has ν -null boundary, and that $0 < h_\nu(T) < \infty$. Let R_ξ be an irrational circle rotation.

Then for every $\varepsilon > 0$ and every sufficiently large m there are disjoint measurable sets

$$C_1, \dots, C_N \subset Z$$

and disjoint supersets C'_i with $C_i \subset C'_i$ such that:

1.

$$\nu\left(\bigcup_i C_i\right) \geq 1 - \varepsilon; \quad (16.1)$$

2. each C_i meets at most $\exp(\varepsilon m)$ atoms of $\mathcal{A}^m$;

3.

$$\nu(C_i) \geq (1 - \varepsilon)\nu(C'_i); \quad (16.2)$$

4. for ν -a.e. z , and every i with positive return frequency, the phase sequence

$$\{R_\xi^k(s) : k \geq 0, T^k z \in C'_i\}$$

is uniformly distributed on the circle.

The same conclusion holds with the equal-entropy Bernoulli factor replaced by the arbitrarily high strict Bernoulli factors supplied by Theorem IV.7.26; this is the strict-factor version in Corollary IV.7.50. No hypothesis from the preceding slice construction occurs in this abstract theorem.

Proof. Apply the ergodic selection theorem, Theorem IV.7.49, to (Z, T, ν) and its stated boundary-null finite generator. The geometric good-slice property is an additional input in the later slicing argument, not a hypothesis needed for this abstract selection conclusion. The strict-factor variant is Theorem IV.7.48 and Corollary IV.7.50. $\square$

2. Two elementary lemmas used in the final geometry

Lemma IV.7.52 (*density of a selected UD subsequence*).

Let (x_k) be uniformly distributed in $\mathbb{T}$, and let $F \subset \mathbb{N}$ have upper density d . Then

$$\text{Leb} \overline{\{x_k : k \in F\}} \geq d.$$

Proof. Let $E = \overline{\{x_k : k \in F\}}$. For a finite union J of intervals contained in E^c ,

$$\text{Leb}(J) = \lim_{N \rightarrow \infty} \frac{1}{N} \#\{k < N : x_k \in J\} \leq \liminf_{N \rightarrow \infty} \left(1 - \frac{\#(F \cap [0, N))}{N}\right) = 1 - d.$$

Approximate E^c from inside by finite unions of intervals. $\square$

Lemma IV.7.53 (*entropy survives removal of one small ball*).

For every d there is C_d such that the following holds. Let η be a probability on $\mathbb{R}^d$, let $0 < \delta < 1$, and suppose

$$\sup_y \eta(B(y, \delta)) \leq \varepsilon. \quad (16.3)$$

If $2^{-n} \leq \delta$, then

$$\inf_y H(\eta|_{B(y, \delta)^c}, \mathcal{D}_n) \geq H(\eta, \mathcal{D}_n) - C_d n \varepsilon^{1/2}. \quad (16.4)$$

Here the restriction is not renormalized.

Proof. Fix y . Let E be the union of dyadic n -cubes meeting $B(y, \delta)$. Since $2^{-n} \leq \delta$, the set E lies in a fixed-multiple δ -ball and is covered by $O_d(1)$ balls of radius δ . Thus

$$\eta(E) \leq C_d \varepsilon. \quad (16.5)$$

For a finite measure ρ of mass a on at most M atoms,

$$H(\rho) \leq a \log M + a \log(1/a). \quad (16.6)$$

The entropy lost on E is therefore at most $C_d \varepsilon n + C_d \varepsilon \log(1/\varepsilon)$. For small ε this is at most $C_d n \varepsilon^{1/2}$; enlarging the constant handles the remaining range. $\square$

Lemma IV.7.54 (*entropy gives a covering lower bound*).

If a finite positive measure ρ is supported on a set A and $\mathcal{D}_n$ is the dyadic partition, then

$$\log N_{2^{-n}}(A) \geq \frac{H(\rho, \mathcal{D}_n)}{\rho(\mathbb{R}^d)} + \log \rho(\mathbb{R}^d) - O_d(1), \quad (16.7)$$

provided $\rho(\mathbb{R}^d) > 0$.

Proof. If A meets M dyadic cubes, then the entropy of ρ over those cubes is at most

$$\rho(\mathbb{R}^d) \log M + \rho(\mathbb{R}^d) \log \frac{1}{\rho(\mathbb{R}^d)}.$$

Rearrange. In the applications below $\rho(\mathbb{R}^d) \geq 1 - O(\varepsilon)$, so the normalization contributes only an $O_\varepsilon(1)$ term. $\square$

Lemma IV.7.55 (*closure of fixed-scale rich line witnesses*).

Let $K \subset \mathbb{R}^2$ be compact, $D_1 \subset K$, $r_0 > 0$, $\delta > 0$, and $L \geq 1$. Suppose that for every direction $s \in F_0 \subset \mathbb{T}$ there is a line ℓ_s of direction s through a point of D_1 such that

$$N_\delta((K \cap \ell_s) \setminus B(y, r_0)) \geq L \quad \text{for every } y \in K. \quad (16.7a)$$

Then for every $s \in \overline{F_0}$ there is a line of direction s through $\overline{D_1}$ for which the same inequality holds with cL in place of L , where $c > 0$ is an absolute constant.

Proof. Fix $s \in \overline{F_0}$ and choose $s_j \in F_0$ tending to s . Choose $z_j \in D_1 \cap \ell_{s_j}$. By compactness, pass to a subsequence with $z_j \rightarrow z \in \overline{D_1}$, and let ℓ be the line through z of direction s . This line is fixed for the rest of the argument.

Fix $y \in K$. For each j , choose a maximal δ -separated subset P_j of $(K \cap \ell_{s_j}) \setminus B(y, r_0)$. Its δ -balls cover that set, so $|P_j| \geq L$. Retain the same $[L]$ points from every P_j . Compactness of $K^{[L]}$ gives a further subsequence on which these points converge coordinatewise. Their limits lie in $(K \cap \ell) \setminus B(y, r_0)$ and remain δ -separated. Any ball of radius δ contains at most an absolute number of δ -separated points in the plane. Consequently

$$N_\delta((K \cap \ell) \setminus B(y, r_0)) \geq c[L] \geq cL.$$

The further subsequence may depend on y , but ℓ does not; the argument therefore proves the estimate for every $y \in K$. $\square$

3. A local polar covering lemma

Lemma IV.7.56 (*many directions times many radial points*).

Fix $r_0, m_0, R > 0$. Let $D_1 \subset K \subset [-R, R]^2$ and $\gamma, \eta \geq 0$. For an integer $n \geq n_0(r_0, m_0, R)$, suppose that for a set $F \subset \mathbb{T}$ and every $s \in F$ there is a line ℓ_s of direction s meeting D_1 such that

$$N_{2^{-n}}((K \cap \ell_s) \setminus B(y, r_0)) \geq 2^{n(\gamma-\eta)} \quad (16.8)$$

for every $y \in K$. Assume also that

$$\text{Leb}(F) \geq m_0. \quad (16.9)$$

Then

$$N_{2^{-n}}(K - D_1) \geq c(r_0, m_0, R)2^{n(1+\gamma-C\eta)}. \quad (16.10)$$

Proof. Choose a maximal 2^{-n} -separated subset S of F in the circle metric. Since F has measure at least m_0 ,

$$\#S \geq cm_02^n. \quad (16.11)$$

For each $s \in S$ choose $z_s \in \ell_s \cap D_1$. By (16.8), after deleting $B(z_s, r_0)$ from the slice if necessary, choose a 2^{-n} -separated set $P_s \subset K \cap \ell_s$ with

$$\#P_s \geq c2^{n(\gamma-\eta)}. \quad (16.12)$$

For $p \in P_s$ the difference $p - z_s$ lies on the line through the origin of direction s and has length at least r_0 .

Consider

$$v = p - z_s, \quad v' = p' - z_{s'}.$$

If $|s - s'| \geq C2^{-n}$, $|v|, |v'| \geq r_0$, and both vectors lie in one fixed bounded ball, elementary polar geometry gives

$$|v - v'| \geq cr_0|s - s'|$$

unless their radial coordinates differ by at least $c2^{-n}$, in which case the same separation is immediate. By thinning S and each P_s by fixed factors, we obtain a $c2^{-n}$ -separated subset of $K - D_1$ of cardinality at least

$$c \#S \min_s \#P_s \geq c2^{n(1+\gamma-C\eta)}.$$

A $c2^{-n}$ -separated set requires the same order of 2^{-n} -balls, proving (16.10). $\square$

Lemma IV.7.57 (*difference-set covering inequality*).

For bounded $A, B \subset \mathbb{R}^d$,

$$N_\delta(A - B) \leq C_d N_\delta(A) N_\delta(B). \quad (16.13)$$

Proof. Cover A and B by δ -balls. The difference of two such balls lies in a 2δ -ball, and each 2δ -ball is covered by $O_d(1)$ δ -balls. $\square$

4. Proposition 7.1 in textbook form

Let

$$K = C_\alpha \times C_\beta,$$

and let ν_∞ be the ergodic U -invariant measure from Theorem IV.7.43.

Proposition IV.7.58 (*large phase set, small selector, rich slices*).

For every $\varepsilon > 0$ there are $r_0(\varepsilon) > 0$ and $n_0(\varepsilon)$ such that for every $n \geq n_0$ one can find

$$(z, t) \in K \times \mathbb{T},$$

a slice measure η supported on a line of slope β^{-t} , a measurable set $D \subset K \times \mathbb{T}$, and a set $N \subset \mathbb{N}$ with the following properties:

1. η is supported on $K \cap \ell$ for a line ℓ of slope β^{-t} ;
- 2.

$$\frac{1}{n} \log N_{2^{-n}}(\Pi_1 D) \leq C\varepsilon; \quad (16.14)$$

3. $U^k(z, t) \in D$ for every $k \in N$;
- 4.

$$\overline{\text{Leb}\{R_\theta^k(t) : k \in N\}} \geq 1 - C\varepsilon; \quad (16.15)$$

5. for every $k \in N$, the conditional slice zoom $\eta_{A_k^t(z)}$ is supported on a line through $U_t^k(z) \in \Pi_1 D$ of slope $\beta^{-R_\theta^k t}$ and satisfies

$$\inf_y \frac{1}{n \log 2} H(\eta_{A_k^t(z)}|_{B(y, r_0)^c}, \mathcal{D}_n) \geq \gamma - C\varepsilon^{1/2}. \quad (16.16)$$

Proof. Fix ε . Lemma IV.7.41 supplies r_0 and a full ν_∞ -measure set of points for which a set G of times of lower density at least $1 - \varepsilon$ satisfies small-ball non-concentration and the entropy lower bound.

Apply the abstract selection theorem at a symbolic generation m corresponding to dyadic scale n :

$$\alpha^m \leq 2^{-n} < \alpha^{m-1}. \quad (16.17)$$

The selected families $C_i \subset C'_i$ cover $1 - \varepsilon$ of the measure, each C_i meets at most $\exp(\varepsilon m)$ m -atoms, and return phases through C'_i are uniformly distributed.

Write $H(C) = \{k \geq 0 : U^k(z, t) \in C\}$. Choose the point so that every return set in these finitely many families has its ergodic frequency, and so that it never visits their zero-measure members. Then $w_i = \nu_\infty(C'_i)$ is the density of $H(C'_i)$. The families are disjoint, so $\sum_i w_i \leq 1$. Since G has lower density at least $1 - \varepsilon$ and the union of the $H(C_i)$ has density at least $1 - \varepsilon$, for all sufficiently large N we have

$$\sum_{i: w_i > 0} \#(G \cap H(C_i) \cap [0, N)) \geq (1 - 3\varepsilon)N.$$

For each such N , some i satisfies $\#(G \cap H(C_i) \cap [0, N)) \geq (1 - 3\varepsilon)Nw_i$; otherwise summation would contradict the preceding inequality. There are finitely many indices, so one i_0

satisfies this along an unbounded sequence of common prefixes. Consequently

$$\frac{\overline{\text{dens}}(G \cap H(C_{i_0}))}{\text{dens}(H(C'_{i_0}))} \geq 1 - C\varepsilon. \quad (16.18)$$

Set

$$D = C_{i_0}, \quad N = G \cap H(D).$$

The uniformly distributed phase sequence along returns to C'_{i_0} and Lemma IV.7.52, applied after enumerating $H(C'_{i_0})$, gives (16.15): the selected indices have relative upper density at least $1 - C\varepsilon$ because $\#(H(C'_{i_0}) \cap [0, N])/N \rightarrow w_{i_0} > 0$. No relative lower-density claim is required. The m -cylinder count for D , uniform symbolic/geometric comparability, and (16.17) give (16.14).

Choose the typical base point outside all boundaries of the countably many partitions $\mathcal{A}_k^t$ and in the support of η . Then (14.43) shows that $U_t^k(z) \in \text{supp } \eta_{A_k^t(z)}$; because $U^k(z, t) \in D$, this point lies in $\Pi_1 D$. The geometric skew product makes the supporting line's slope $\beta^{-R_\theta^k t}$. For $k \in N$, the good-slice property gives

$$\sup_y \eta_{A_k^t(z)}(B(y, r_0)) \leq \varepsilon$$

and

$$H(\eta_{A_k^t(z)}, \mathcal{D}_n) \geq n(\gamma \log 2 - C\varepsilon).$$

Apply Lemma IV.7.53 to remove one r_0 -ball, obtaining (16.16) after changing C . □

5. Regular homogeneous Cantor slices

Theorem IV.7.59 (*homogeneous interval-OSC slices*).

Let $C_\alpha, C_\beta \subset \mathbb{R}$ be homogeneous self-similar sets satisfying the interval open set condition in Definition IV.7.27, with

$$\frac{\log \alpha}{\log \beta} \notin \mathbb{Q}.$$

Then for every $u, v \in \mathbb{R}$,

$$\overline{\dim}_B((uC_\alpha + v) \cap C_\beta) \leq \max\{0, \dim_H C_\alpha + \dim_H C_\beta - 1\}. \quad (16.19)$$

Proof. The case $u = 0$, or a singleton attractor in either system, gives an empty intersection or a singleton. Assume $u \neq 0$ and both attractors are nonsingletons. Affine changes preserve upper box dimension, so the intersection is equivalent to a non-axis-parallel line slice of

$$K = C_\alpha \times C_\beta.$$

Let

$$\gamma = \overline{\dim}_B(K \cap \ell).$$

If $\gamma = 0$ there is nothing to prove. Suppose $\gamma > 0$. The slice and selection constructions above apply, hence Proposition IV.7.58 holds for every small ε and all large n .

Set $D_1 = \Pi_1 D$ and $D'_1 = \overline{D_1} \subset K$. For every $k \in N$, (16.16) and Lemma IV.7.54 give, for all $y \in K$,

$$N_{2-n}((K \cap \ell_k) \setminus B(y, r_0)) \geq 2^{n(\gamma - C\varepsilon^{1/2} - O(1/n))},$$

where ℓ_k has slope $\beta^{-R_\theta^k t}$ and passes through $U_t^k(z) \in D_1$. The restriction in (16.16) has mass at least $1 - O(\varepsilon)$, so the $\log \rho(\mathbb{R}^2)$ term in (16.7) contributes only $O_\varepsilon(1)$.

Let $d(q)$ denote the unoriented direction of a line of slope β^{-q} for the representative $0 \leq q < 1$ of a phase, and put

$$F_{\text{dir}} = \overline{\{d(R_\theta^k t) : k \in N\}}.$$

The map $d : (0, 1) \rightarrow \mathbb{T}$ is smooth with its angular derivative bounded above and below by positive constants depending only on β . It has a jump at the phase endpoint; deleting that single phase changes no Lebesgue measure. By (16.15), $\text{Leb}(F_{\text{dir}}) \geq m_0(\beta) > 0$ for sufficiently small ε . Apply the fixed-scale closure Lemma IV.7.55 to the return directions. Its line through D'_1 and its constant packing loss yield, for every $s \in F_{\text{dir}}$,

$$N_{2-n}((K \cap \ell_s) \setminus B(y, r_0)) \geq 2^{n(\gamma - C\varepsilon^{1/2} - O(1/n))} \quad (16.20)$$

uniformly in $y \in K$, after increasing C .

Apply Lemma IV.7.56 with $F = F_{\text{dir}}$, D'_1 in place of D_1 , and $\eta = C\varepsilon^{1/2}$:

$$N_{2-n}(K - D'_1) \geq 2^{n(1 + \gamma - C\varepsilon^{1/2} - O(1/n))}. \quad (16.21)$$

On the other hand, a cover of D_1 by 2^{-n} -balls covers D'_1 after a fixed enlargement, so $N_{2-n}(D'_1) \leq CN_{2-n}(D_1)$. Lemma IV.7.57 and (16.14) give

$$\begin{aligned} N_{2-n}(K - D'_1) &\leq CN_{2-n}(K)N_{2-n}(D'_1) \\ &\leq N_{2-n}(K)2^{C\varepsilon n + O(1)}. \end{aligned} \quad (16.22)$$

Therefore

$$\frac{1}{n \log 2} \log N_{2-n}(K) \geq 1 + \gamma - C\varepsilon^{1/2} - O(1/n). \quad (16.23)$$

Let $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$:

$$\overline{\dim_B} K \geq 1 + \gamma. \quad (16.24)$$

For homogeneous interval-OSC sets,

$$\overline{\dim_B} K = \dim_H K = \dim_H C_\alpha + \dim_H C_\beta. \quad (16.25)$$

Thus $\gamma \leq \dim_H C_\alpha + \dim_H C_\beta - 1$, which is (16.19). $\square$

6. Closed times- p /times- q invariant sets

Theorem IV.7.60 (*closed times- p /times- q invariant sets*).

Let $p, q \geq 2$ be multiplicatively independent integers. Let $A, B \subset [0, 1]$ be closed and invariant under

$$T_p(x) = px \pmod{1}, \quad T_q(x) = qx \pmod{1}.$$

Then for every $u, v \in \mathbb{R}$,

$$\overline{\dim}_B((uA + v) \cap B) \leq \max\{0, \dim_H A + \dim_H B - 1\}. \quad (16.26)$$

Proof. Again $u = 0$ is trivial. Project A, B to circle sets $A_{\mathbb{T}}$ and $B_{\mathbb{T}}$. They are closed and invariant, and identifying the two endpoints preserves Hausdorff dimension. Their unique representatives in $[0, 1)$ contain $A \setminus \{1\}$ and $B \setminus \{1\}$. Fix $\varepsilon > 0$. By Theorem IV.5.16, there are $k, l \geq 1$ and homogeneous interval-OSC supersets

$$A \setminus \{1\} \subset A_\varepsilon, \quad B \setminus \{1\} \subset B_\varepsilon,$$

with contraction ratios

$$\alpha = p^{-k}, \quad \beta = q^{-l},$$

and

$$\dim_H A_\varepsilon \leq \dim_H A + \varepsilon, \quad \dim_H B_\varepsilon \leq \dim_H B + \varepsilon. \quad (16.27)$$

Multiplicative independence survives powers:

$$\frac{\log \alpha}{\log \beta} = \frac{k \log p}{l \log q} \notin \mathbb{Q}. \quad (16.28)$$

Every point of $(uA + v) \cap B$ except possibly $u + v$ and 1 lies in $(uA_\varepsilon + v) \cap B_\varepsilon$. Adding those at most two points does not increase any nonnegative upper box dimension. Apply Theorem IV.7.59:

$$\begin{aligned} \overline{\dim}_B((uA + v) \cap B) &\leq \max\{0, \overline{\dim}_B((uA_\varepsilon + v) \cap B_\varepsilon)\} \\ &\leq \max\{0, \dim_H A_\varepsilon + \dim_H B_\varepsilon - 1\} \\ &\leq \max\{0, \dim_H A + \dim_H B - 1 + 2\varepsilon\}. \end{aligned} \quad (16.29)$$

Let $\varepsilon \downarrow 0$. □

7. Homogeneous planar self-similar slices

Let

$$\mathcal{F} = \{f_i(x) = \lambda O_\xi x + t_i\}_{i=1}^m \quad (16.30)$$

be a homogeneous planar self-similar IFS with $0 < \lambda < 1$, irrational $\xi \in \mathbb{T}$, and strong separation. Let X be its attractor. The coding map

$$\Pi : \Lambda^{\mathbb{N}} \rightarrow X, \quad \Pi((a_n)) = \sum_{n \geq 1} \lambda^{n-1} O_\xi^{n-1} a_n \quad (16.31)$$

is bi-Lipschitz because of SSC. Assume there is a line ℓ_0 with

$$\gamma = \overline{\dim}_B(X \cap \ell_0) > 0. \quad (16.32)$$

Lemma IV.7.61 (*slice CP law in the planar model*).

There is an ergodic CP distribution Q on the full shift $\Lambda^{\mathbb{N}}$ with

$$\dim Q \geq \gamma$$

such that Q_1 -a.e. symbolic measure projects under Π to a measure supported on a line slice of X .

Proof. Let

$$A = \Pi^{-1}(X \cap \ell_0).$$

Bi-Lipschitz coding gives $\overline{\dim_B A} = \gamma$. Choose n_j realizing the word-count exponent, put equal mass on one point in every occupied depth- n_j cylinder, and average the pointed laws under ordinary symbolic magnification. Exactly the entropy-telescoping proof of Lemma IV.7.32 gives a CP limit of dimension γ .

The family “support projects into some line slice” is stable under magnification because f_i^{-1} rotates line directions by $-\xi$, and it is closed under the symbolic Hausdorff limit. Pass to an ergodic component with dimension at least γ . $\square$

Let $W : X \rightarrow X$ be the inverse IFS map, so $W|_{f_i(X)} = f_i^{-1}$. SSC makes this a well-defined expanding map. If $\mathcal{A}_n$ is the partition into level- n cylinders, define, whenever $\eta(\mathcal{A}_n(z)) > 0$,

$$\eta_{\mathcal{A}_n(z)} = \frac{(W^n)_*(\eta|_{\mathcal{A}_n(z)})}{\eta(\mathcal{A}_n(z))}. \quad (16.33)$$

Lemma IV.7.62 (*invariant geometric measure with good slices*).

There is an ergodic W -invariant probability ν on X such that for every $\varepsilon > 0$ there are $\delta > 0$ and n_0 with the following property for ν -a.e. z : there is a slice measure η supported on $\ell \cap X$ and a set of times G of lower density greater than $1 - \varepsilon$ such that, for $k \in G$,

$$\sup_y \eta_{\mathcal{A}_k(z)}(B(y, \delta)) \leq \varepsilon, \quad (16.34)$$

and, for all $n \geq n_0$,

$$H(\eta_{\mathcal{A}_k(z)}, \mathcal{D}_n) \geq n(\gamma \log 2 - C\varepsilon). \quad (16.35)$$

Moreover, the direction of the supporting slice at time k is the original direction rotated by $-k\xi$.

Proof. For a Q -generic pair (μ, x) , Birkhoff gives

$$\frac{1}{N} \sum_{k < N} \Pi_*(\mu^{[x_1 \cdots x_k]}) \longrightarrow \nu_0 := \int \Pi_* \vartheta dQ(\vartheta, y), \quad (16.36)$$

since $\vartheta \mapsto \Pi_* \vartheta$ is continuous. The symbolic/geometric identity says that the k th term is exactly the normalized geometric zoom of $\eta = \Pi_* \mu$ in the atom $\mathcal{A}_k(\Pi x)$. A martingale-difference orbit comparison, identical to Lemma IV.7.38, but with no circle coordinate, shows that the W -orbit empirical measures of Πx have the same limit. Hence ν_0 is W -invariant. Boundary masses vanish by SSC and non-atomicity of positive-dimensional CP measures.

Apply the local CP typical-measure theorem to Q : on a set of k of lower density greater than $1 - \varepsilon$, the conditional symbolic measures have small fixed-scale atoms and entropy at least γ . Bi-Lipschitz Π converts this to (16.34)–(16.35). Pass to a W -ergodic component inheriting this full-measure property. The direction rotation follows from

$$W(f_i(X) \cap \ell) \subset X \cap f_i^{-1}(\ell),$$

and f_i^{-1} rotates by $-\xi$. □

Theorem IV.7.63 (*homogeneous planar self-similar slices*).

Under (16.30), for every line $\ell \subset \mathbb{R}^2$,

$$\overline{\dim}_B(X \cap \ell) \leq \max\{0, \dim_H X - 1\}. \quad (16.37)$$

Proof. Let γ be the upper box dimension of the slice. If $\gamma = 0$, there is nothing to prove. Use Lemmas IV.7.61–IV.7.62 and apply the abstract selection theorem, Theorem IV.7.51, to the ergodic system (X, W, ν) with the finite first-level cylinder generator and the irrational rotation R_ξ . The proof of Proposition IV.7.58 carries over word for word: for every $\varepsilon > 0$ and all large n one obtains a set $D_1 \subset X$ with

$$N_{2^{-n}}(D_1) \leq 2^{C\varepsilon n}, \quad (16.38)$$

and a phase/direction set F of Lebesgue measure at least $1 - C\varepsilon$ such that, for every $s \in F$, there is a line ℓ_s through D_1 with

$$\inf_y N_{2^{-n}}((X \cap \ell_s) \setminus B(y, r_0)) \geq 2^{n(\gamma - o_\varepsilon(1))}. \quad (16.39)$$

Apply Lemma IV.7.56 with $K = X$ and then Lemma IV.7.57:

$$N_{2^{-n}}(X - D_1) \geq 2^{n(1 + \gamma - o_\varepsilon(1))},$$

while

$$N_{2^{-n}}(X - D_1) \leq CN_{2^{-n}}(X)N_{2^{-n}}(D_1).$$

Hence

$$\overline{\dim}_B X \geq 1 + \gamma. \quad (16.40)$$

For a self-similar set satisfying SSC, $\overline{\dim}_B X = \dim_H X$, so $\gamma \leq \dim_H X - 1$. □

Chapter IV.8

Embeddings and Renormalization

Affine inclusion, microsets, and the rigidity frontier

Three embedding notions are separated before the elementary renormalization mechanism and the research-level rigidity frontier are introduced.

8.1 Embedding notions and elementary renormalization

1. Three notions of embedding

The term “positive-measure embedding” is ambiguous unless the source measure is specified. The convention below distinguishes, for example, a Smith–Volterra set with positive Lebesgue overlap from a global affine interval inclusion. It also distinguishes microset containment from either of those notions.

Definition IV.8.1 (*nonsingular affine and C^1 embeddings*).

Let $E, F \subset \mathbb{R}$ be compact. A nonsingular affine map is

$$g(x) = ax + b, \quad a \neq 0.$$

An *affine embedding* of E into F is such a map with $g(E) \subset F$. A C^1 embedding of E into F is a C^1 diffeomorphism from an open neighborhood of E onto its image such that $g(E) \subset F$. A *self-embedding* is an embedding with $E = F$.

Definition IV.8.2 (*positive-source-measure partial embedding*).

Let μ be a Borel probability with $\text{supp } \mu = E$. A *positive-measure affine embedding* of (E, μ) into F means that there are a nonsingular affine map g and a Borel set $E' \subset E$ such that

$$\mu(E') > 0, \quad g(E') \subset F.$$

The corresponding C^1 notion is defined with g a C^1 diffeomorphism near E .

Reader note. This is a partial embedding of positive *source* measure. It is not the assertion that F has positive Lebesgue measure, nor that $\mathcal{H}^s(g(E) \cap F) > 0$ for an automatically selected s . Those are different notions.

Definition IV.8.3 (*centered minisets and microsets*).

Let $F \subset \mathbb{R}$ be nonempty and compact. For $x \in F$ and $\lambda \geq 1$ put

$$\text{Mini}(F; x, \lambda) = \lambda(F - x) \cap [-1, 1]. \quad (17.1)$$

A nonempty compact set $M \subset [-1, 1]$ is a *centered microset* of F if there are $x_n \in F$ and $\lambda_n \rightarrow \infty$ such that

$$\text{Mini}(F; x_n, \lambda_n) \longrightarrow M \quad (17.2)$$

in the Hausdorff metric. This is the convention used in Definition IV.8.3. Other common window conventions are affine variants, but no equivalence is used without an explicit normalization.

Definition IV.8.4 (*local tangent embedding*).

A compact set E *locally tangent-embeds* in F if some centered microset M of F contains a nonsingular affine copy of E :

$$g(E) \subset M \quad (17.3)$$

for some affine map $g(x) = ax + b$ with $a \neq 0$.

Definition IV.8.5 (*logarithmic commensurability*).

For $0 < \alpha, \beta < 1$, say that α and β are multiplicatively commensurable if

$$\frac{\log \alpha}{\log \beta} \in \mathbb{Q}.$$

Equivalently, there are positive integers p, q such that

$$\alpha^p = \beta^q. \quad (17.4)$$

They are multiplicatively independent otherwise.

Definition IV.8.6 (*normalized affine parameter window*).

For constants $0 < c < C < \infty$ and $B < \infty$ define

$$\mathcal{A}[c, C; B] = \{x \mapsto sx + t : c \leq |s| \leq C, |t| \leq B\}. \quad (17.5)$$

This is a compact subset of the two-dimensional affine-parameter space.

2. Relations among the three embedding notions

Lemma IV.8.7 (*global inclusion implies positive-measure partial embedding*).

Let μ be a probability supported on E . If $g(E) \subset F$ for a nonsingular affine map g , then (E, μ) positive-measure affinely embeds in F .

Proof. Take $E' = E$. Then $\mu(E') = 1$ and $g(E') \subset F$. □

Lemma IV.8.8 (*positive-measure partial embedding need not be global*).

There is a compact nowhere-dense set $F \subset [0, 1]$ of positive Lebesgue measure such that $([0, 1], \text{Leb})$ has a positive-measure affine embedding into F , but no nonsingular affine image of $[0, 1]$ is contained in F .

Proof. Take the Smith–Volterra set F . It is compact, nowhere dense, and $\text{Leb}(F) > 0$. With $E = [0, 1]$, $\mu = \text{Leb}|_{[0,1]}$, $g = \text{id}$, and $E' = F$, one has

$$\mu(E') = \text{Leb}(F) > 0, \quad g(E') = F \subset F.$$

Every nonsingular affine image of $[0, 1]$ is a nondegenerate interval. If such an interval were contained in F , then F would have nonempty interior, contradicting nowhere density.

□

The next example separates the tangent notion from both positive source measure and global inclusion.

Lemma IV.8.9 (*a countable compact set with an interval microset*).

There exists a countable compact set $F \subset [0, 1]$ having $[0, 1]$ as a centered microset. Consequently $[0, 1]$ locally tangent-embeds in F , but $([0, 1], \text{Leb})$ has no positive-measure affine embedding into F and $[0, 1]$ has no global affine embedding into F .

Proof. For $n \geq 2$ put

$$c_n = 2^{-n}, \quad s_n = 2^{-n^3}, \quad G_n = \{j/n : 0 \leq j \leq n\},$$

and define

$$F = \{0\} \cup \bigcup_{n \geq 2} (c_n + s_n G_n). \quad (17.6)$$

Each cluster is finite, while $c_n \rightarrow 0$ and $s_n \rightarrow 0$. Hence every sequence in F either remains in finitely many finite clusters or has cluster index tending to infinity and converges to 0. Thus F is compact and countable.

Use the base point $x_n = c_n$ and magnification $\lambda_n = s_n^{-1}$. The n th cluster becomes exactly G_n . Every other cluster, as well as the point 0, leaves every fixed bounded window because

$$\min_{m \neq n} \frac{|c_m - c_n|}{s_n} \longrightarrow \infty. \quad (17.7)$$

Thus, for all sufficiently large n ,

$$\text{Mini}(F; c_n, s_n^{-1}) \cap [0, 1] = G_n, \quad (17.8)$$

and no foreign cluster enters a fixed neighborhood of $[0, 1]$. Since $G_n \rightarrow [0, 1]$ in the Hausdorff metric, $[0, 1]$ is a centered microset of F . The identity therefore witnesses a local tangent embedding.

If $E' \subset [0, 1]$ has positive Lebesgue measure and g is nonsingular affine, then E' and $g(E')$ are uncountable. Since F is countable, $g(E') \not\subset F$. A global affine image of $[0, 1]$ is a nondegenerate interval and is likewise impossible in a countable set. □

Lemma IV.8.10 (*one inclusion does not imply a parameter family*).

The existence of one global affine embedding does not, in general, imply a positive-dimensional or continuous family of affine embeddings.

Proof. Take $E = F = \{0, 1\}$. A nonsingular affine map preserving E as a set is either $x \mapsto x$ or $x \mapsto 1 - x$. Hence the affine self-embedding set is finite. $\square$

Lemma IV.8.11 (*a structured model where all three notions hold*).

Let C be the middle-third Cantor set and let μ be its $(1/2, 1/2)$ Cantor measure. Then C globally affinely self-embeds, positive-measure self-embeds, and locally tangent-embeds in itself.

Proof. The cylinder map $g(x) = x/3$ satisfies $g(C) \subset C$, proving global and hence positive-measure embedding by Lemma IV.8.7.

For the tangent statement take $x_n = 0$ and $\lambda_n = 3^n$. A point of C lying in $[0, 3^{-n}]$ has its first n ternary digits equal to zero, so

$$3^n C \cap [0, 1] = C. \quad (17.9)$$

Thus C is itself a centered microset of C and the identity copy witnesses the local tangent embedding. $\square$

Theorem IV.8.12 (*embedding notions: exact implication diagram*).

With Definitions IV.8.1–IV.8.4:

1. global affine embedding implies positive-source-measure affine embedding for every probability supported on the source;
2. the converse fails, even for source $[0, 1]$ with Lebesgue measure;
3. local tangent embedding does not imply either global affine embedding or positive-source-measure affine embedding;
4. the existence of one affine inclusion does not in general imply a parameter family of inclusions;
5. in structured self-similar examples all three notions can hold simultaneously.

Proof. The five assertions are Lemmas IV.8.7–IV.8.11, respectively. $\square$

3. Fixed separated homogeneous models

We now prove the elementary renormalization mechanism used throughout the rest of the chapter.

Homogeneous separated-model setup. Let $X, Y \subset [0, 1]$ be nontrivial compact self-similar sets generated by orientation-preserving homogeneous similarities

$$\psi_i(x) = \alpha x + a_i \quad (i \in I), \quad \phi_j(x) = \beta x + b_j \quad (j \in J), \quad (17.10)$$

where $0 < \alpha, \beta < 1$. Normalize

$$\text{conv}(X) = \text{conv}(Y) = [0, 1]. \quad (17.11)$$

Assume the first-level closed basic intervals $\psi_i([0, 1])$ are pairwise disjoint, as are the $\phi_j([0, 1])$, with positive minimum gaps

$$\Delta_X > 0, \quad \Delta_Y > 0. \quad (17.12)$$

This is intentionally stronger than an abstract open-set condition, since it rules out boundary touching in the renormalization model.

For a word $v = v_1 \cdots v_m$, write

$$\psi_v = \psi_{v_1} \circ \cdots \circ \psi_{v_m}, \quad |v| = m,$$

and similarly for ϕ_u . Then

$$|\psi'_v| = \alpha^{|v|}, \quad |\phi'_u| = \beta^{|u|}. \quad (17.13)$$

Lemma IV.8.13 (*cylinder identity and level gaps*).

For every word v ,

$$\psi_v(X) = X \cap \psi_v([0, 1]), \quad (17.14)$$

and analogously for Y . Distinct level- n cylinders of Y are separated by at least

$$\beta^{n-1} \Delta_Y. \quad (17.15)$$

Proof. Strong interval separation gives disjoint first-level cylinder intervals. The self-similar equation restricted to one such interval leaves exactly the corresponding cylinder; iteration gives (17.14).

If two length- n words first differ after a common prefix of length $k < n$, their cylinders lie in distinct first-level descendants of that prefix, whose gap is $\beta^k \Delta_Y$. Since $k \leq n - 1$ and $0 < \beta < 1$, this is at least $\beta^{n-1} \Delta_Y$. $\square$

Lemma IV.8.14 (*exact affine renormalization identity*).

Let $g(x) = ax + c$ with $a \neq 0$. If

$$g(\psi_v(X)) \subset \phi_u(Y), \quad (17.16)$$

then

$$h_{u,v} := \phi_u^{-1} \circ g \circ \psi_v \quad (17.17)$$

is an affine embedding $X \rightarrow Y$ with slope

$$h'_{u,v} = a\alpha^{|v|}\beta^{-|u|}. \quad (17.18)$$

Proof. For $x \in X$, equation (17.16) gives $g(\psi_v(x)) \in \phi_u(Y)$, so applying ϕ_u^{-1} places the point in Y . Differentiating the affine composition gives (17.18). $\square$

Lemma IV.8.15 (*small copies are captured by one target cylinder*).

Assume $g(X) \subset Y$. Fix a source word v of length m and put

$$d_m = \text{diam}(g(\psi_v(X))) = |a|\alpha^m. \quad (17.19)$$

For all sufficiently large m there is a largest integer $n = n(m) \geq 1$ satisfying

$$d_m < \beta^{n-1} \Delta_Y. \quad (17.20)$$

Then $g(\psi_v(X))$ lies in a unique level- n cylinder $\phi_u(Y)$, and the normalized embedding $h_{u,v}$ from Lemma IV.8.14 satisfies

$$\Delta_Y \leq |h'_{u,v}| < \Delta_Y/\beta. \quad (17.21)$$

Proof. Since $d_m \rightarrow 0$, inequality (17.20) holds for $n = 1$ once m is large. As n increases its right-hand side tends to zero, so a largest such n exists. If the image met two distinct level- n cylinders of Y , its diameter would be at least the gap in (17.15), contradicting (17.20). Hence it lies in a unique level- n cylinder.

By maximality, (17.20) fails for $n + 1$:

$$d_m \geq \beta^n \Delta_Y. \quad (17.22)$$

Because $|h'_{u,v}| = |a|\alpha^m \beta^{-n} = d_m \beta^{-n}$, (17.20) and (17.22) give

$$\Delta_Y \leq |h'_{u,v}| < \Delta_Y/\beta. \quad (17.23)$$

□

Lemma IV.8.16 (*compactness of normalized embedding parameters*).

Under Lemma IV.8.15, all normalized embeddings $h_{u,v}(X) \subset Y$ belong to one compact affine-parameter window. Consequently every sequence of them has a uniformly convergent affine subsequence whose limit is again a nonsingular affine embedding $X \rightarrow Y$.

Proof. The slopes lie in the compact annulus (17.21). Since $\text{conv}(X) = [0, 1]$ and X is compact, $0, 1 \in X$. If $h(x) = sx + t$ and $h(X) \subset [0, 1]$, then $h(0) = t$ and $h(1) = s + t$ both lie in $[0, 1]$, so $|t| \leq 1$. Thus the parameter pairs (s, t) lie in a compact rectangle with $|s|$ bounded away from zero.

Take a convergent parameter subsequence. The affine maps converge uniformly on $[0, 1]$. For $x \in X$, each image lies in closed Y , so the limiting image also lies in Y . The limiting slope is nonzero. □

4. Log-scale rotation and the family-generation mechanism

The next statement is the elementary mechanism behind many rigidity arguments. It is not itself a rigidity theorem.

Lemma IV.8.17 (*exact log-scale congruence*).

Fix one infinite source address $i_1 i_2 \dots$ and let $v_m = i_1 \dots i_m$. For each large m , choose the target cylinder u_m supplied by Lemma IV.8.15 and put $h_m = h_{u_m, v_m}$. Then

$$\log |h'_m| = \log |a| + m \log \alpha - n_m \log \beta, \quad n_m = |u_m|. \quad (17.24)$$

Therefore, modulo $|\log \beta|$,

$$\log \frac{|h'_m|}{\Delta_Y} \equiv \log \frac{|a|}{\Delta_Y} + m \log \alpha \pmod{|\log \beta|}. \quad (17.25)$$

Proof. Equation (17.24) is (17.18). Subtract $\log \Delta_Y$ and observe that $n_m \log \beta$ is an integer multiple of $\log \beta$. $\square$

Lemma IV.8.18 (*irrational log-ratio generates a full slope interval*).

Under the homogeneous separated-model setup above and $g(X) \subset Y$, assume

$$\frac{\log \alpha}{\log \beta} \notin \mathbb{Q}. \quad (17.26)$$

Then for every

$$s \in [\Delta_Y, \Delta_Y/\beta] \quad (17.27)$$

(with one harmless endpoint convention) there is a nonsingular affine embedding $h_s : X \rightarrow Y$ with

$$|h'_s| = s. \quad (17.28)$$

In particular a single affine embedding generates uncountably many normalized affine embeddings in this separated homogeneous model.

Proof. By irrationality, the rotation sequence

$$\log(|a|/\Delta_Y) + m \log \alpha \pmod{|\log \beta|} \quad (17.29)$$

is dense in the circle of circumference $|\log \beta|$. By Lemma IV.8.15,

$$|h'_m| \in [\Delta_Y, \Delta_Y/\beta), \quad (17.30)$$

whose logarithmic width is exactly $|\log \beta|$. Hence (17.25) identifies the slope magnitudes with representatives of a dense rotation orbit in this single multiplicative fundamental interval.

Fix s in (17.27) and choose m_k with $|h'_{m_k}| \rightarrow s$. The orientation sign is fixed because the cylinder maps preserve orientation and g has fixed sign. By Lemma IV.8.16, after a further subsequence the translations converge, giving an affine embedding h_s with slope magnitude s . $\square$

Corollary IV.8.19 (*commensurable contrast*).

If $\log \alpha / \log \beta$ is rational, the phase sequence in (17.25) is finite. Thus this elementary renormalization argument alone produces only finitely many normalized slope magnitudes. No rigidity conclusion is asserted in either case without further information about the embedding space.

Proof. A rational rotation has a finite orbit. $\square$

Reader note. The conclusion of Lemma IV.8.18 is structure-dependent. It does not contradict Lemma IV.8.10: Lemma IV.8.10 concerns arbitrary compact sets, whereas Lemma IV.8.18 uses homogeneous cylinder self-similarity and a positive separation gap at every scale.

5. C^1 linearization and tangent embeddings

Lemma IV.8.20 (*uniform C^1 blow-up*).

Let f be C^1 on a neighborhood of x_0 and assume $f'(x_0) \neq 0$. For every $R < \infty$, define

$$F_r(y) = \frac{f(x_0 + ry) - f(x_0)}{r}. \quad (17.31)$$

Then, as $r \rightarrow 0$ through nonzero values for which the arguments stay in the domain, F_r converges uniformly on $|y| \leq R$ to

$$y \mapsto f'(x_0)y. \quad (17.32)$$

Proof. Write

$$f(x_0 + h) - f(x_0) - f'(x_0)h = h\varepsilon(h), \quad \varepsilon(h) \rightarrow 0. \quad (17.33)$$

For $|y| \leq R$,

$$|F_r(y) - f'(x_0)y| \leq R \sup_{|h| \leq R|r|} |\varepsilon(h)| \rightarrow 0. \quad (17.34)$$

□

Lemma IV.8.21 (*tangent transfer under a C^1 embedding; interior-window form*).

Let E, F be compact, let $f : E \rightarrow F$ be a C^1 embedding, let $x_0 \in E$, and let $r_n \downarrow 0$. Suppose

$$E_n = (E - x_0)/r_n \cap [-R, R] \rightarrow M_E \quad (17.35)$$

in the Hausdorff metric. Put $a = f'(x_0) \neq 0$, let

$$F_n = (F - f(x_0))/(|a|r_n) \cap [-R', R'], \quad (17.36)$$

and pass to a subsequence for which $F_n \rightarrow M_F$. Then every compact $K \subset \text{int}[-R, R]$ with $K \subset M_E$ satisfies

$$\text{sgn}(a)K \subset M_F \quad (17.37)$$

after R' is chosen large enough to contain the image. Boundary touching is therefore avoided by first working in a strictly smaller window and only then taking the microset limit.

Proof. For $y_n \in E_n$ with $y_n \rightarrow y \in K$, one has $x_0 + r_n y_n \in E$ and hence $f(x_0 + r_n y_n) \in F$. Divide the displacement from $f(x_0)$ by $|a|r_n$. By Lemma IV.8.20 the resulting coordinate converges uniformly to $\text{sgn}(a)y$. Since K lies strictly inside the source window, no point used in this argument is created or lost solely by boundary truncation. Passing to the Hausdorff limit gives (17.37). □

Corollary IV.8.22 (*self-similar local tangent consequence*).

Suppose E is a separated homogeneous self-similar set and x_0 is a fixed point of one of its cylinder maps, so that an affine copy of E occurs as a microset of E at x_0 . If f is a C^1 embedding $E \rightarrow F$, then F has a microset containing a nonsingular affine copy of E . Its coefficient depends on the normalization of the source and target windows; it is not in general $f'(x_0)$.

Proof. Write the cylinder map fixing x_0 as $\phi(x) = x_0 + c(x - x_0)$, where $0 < |c| < 1$. Use even iterates if $c < 0$. Choose $L > \sup_{x \in E} |x - x_0|$ and set $r_n = L|c|^n$ along these iterates. Then

$$K_0 = (E - x_0)/L \subset (-1, 1), \quad K_0 \subset (E - x_0)/r_n \cap [-1, 1],$$

because $\phi^n(E) \subset E$. Pass to a Hausdorff-convergent subsequence of the source minisets, whose limit therefore contains K_0 . With $a = f'(x_0) \neq 0$, pass also to a limit of

$$(F - f(x_0))/(|a|r_n) \cap [-1, 1].$$

The images of K_0 in the target blow-ups are eventually strictly inside this window, by uniform C^1 linearization and the strict choice of L . Lemma IV.8.21 gives $\text{sgn}(a)K_0$ in the target microset. This is the nonsingular affine image $x \mapsto \text{sgn}(a)(x - x_0)/L$ of E . $\square$

This is the precise local statement behind the slogan that a smooth embedding linearizes to an affine tangent embedding. It does not say that a tangent embedding extends to a global embedding.

6. The renormalization theorem**Theorem IV.8.23** (*elementary one-dimensional renormalization*).

Let X, Y satisfy the homogeneous separated-model setup above.

1. Cylinder renormalization is exact: whenever an affine image of a source cylinder lies in a target cylinder, conjugating by the two cylinder maps yields another affine embedding $X \rightarrow Y$ with slope (17.18).
2. Given an affine embedding $g : X \rightarrow Y$, sufficiently deep source cylinders are captured in unique target cylinders. Choosing the deepest capturable target cylinder normalizes every descendant to

$$\Delta_Y \leq |h'| < \Delta_Y/\beta. \quad (17.38)$$

The corresponding affine parameters lie in a compact set and every parameter limit remains an affine embedding.

3. The normalized log scales satisfy (17.25). Hence an irrational $\log \alpha / \log \beta$ makes one affine embedding generate normalized embeddings with every slope magnitude in a full multiplicative interval, while a rational ratio gives a finite elementary phase orbit.
4. For a C^1 embedding, blow-ups converge uniformly to the derivative affine map, and source microsets transfer to target microsets after the interior-window precaution in Lemma IV.8.21. In the centered normalization the derivative magnitude is divided out; the resulting affine copy need not have coefficient $f'(x_0)$.

No assertion of general affine rigidity follows from these facts alone.

Proof. The four items are Lemma IV.8.14, Lemmas IV.8.15–IV.8.16, Lemmas IV.8.17–IV.8.19, and Lemmas IV.8.20–IV.8.22, respectively. $\square$

8.2 Research-level rigidity theorems

Definition IV.8.24 (*Bedford–McMullen carpet convention*).

Fix integers $m, n \geq 2$ and a digit set

$$D \subset \{0, \dots, m-1\} \times \{0, \dots, n-1\}.$$

For $(i, j) \in D$ put

$$S_{i,j}(x, y) = ((x+i)/m, (y+j)/n). \quad (18.1)$$

The Bedford–McMullen carpet F_D is the unique compact set satisfying

$$F_D = \bigcup_{(i,j) \in D} S_{i,j}(F_D). \quad (18.2)$$

It is a *product carpet* when $D = I \times J$ for subsets I, J of the two digit sets.

Definition IV.8.25 (*similarity self-embedding*).

A similarity of $\mathbb{R}^2$ is

$$g(z) = rOz + t,$$

with $r > 0$, O orthogonal, and $t \in \mathbb{R}^2$. It self-embeds F if $g(F) \subset F$. It is an isometry when $r = 1$.

Definition IV.8.26 (*one-dimensional affine embedding space*).

For compact $X, Y \subset \mathbb{R}$, let

$$E(X, Y) = \{f(x) = ax + b : a \neq 0, f(X) \subset Y\}. \quad (18.3)$$

Definition IV.8.27 (Λ and multi-rotations).

Let X, Y be strongly separated self-similar sets, with X homogeneous in the signed sense: every map in the displayed IFS for X has the same nonzero contraction coefficient α , where $\alpha = |\alpha| < 1$. Let β_j be the absolute contraction ratios of the displayed IFS for Y . Set

$$\Lambda = \{\log(\beta_j)/\log(\alpha) : j \in J\}. \quad (18.4)$$

A Λ -multi-rotation is a circle sequence $\theta_k \in \mathbb{R}/\mathbb{Z}$ whose successive increments belong to Λ modulo one. For $U \subset \mathbb{N}$ define

$$\underline{d}(U) = \liminf_{N \rightarrow \infty} \frac{|U \cap \{1, \dots, N\}|}{N}, \quad \bar{d}(U) = \limsup_{N \rightarrow \infty} \frac{|U \cap \{1, \dots, N\}|}{N}.$$

Condition (R_*) means that for every Λ -multi-rotation and every U with $\underline{d}(U) > 0$, the selected set $\{\theta_k : k \in U\}$ has positive upper box dimension. Condition (R^*) makes the same assertion under the weaker hypothesis $\bar{d}(U) > 0$. Consequently (R^*) implies (R_*) .

1. The Algom-Hochman Bedford-McMullen theorem

The planar rigidity proof uses two quite different one-dimensional inputs. The non-principal projection input is already a consequence of the homogeneous irrational-convolution theorem proved in Volume II; we record the exact consequence so that it is not hidden inside the source citation.

Lemma IV.8.28 (*non-principal deleted-digit product projections*).

For $A \subset \{0, \dots, m-1\}$ put $D(A, m) = \{\sum_{k \geq 1} a_k m^{-k} : a_k \in A\}$. Let $K_1 = D(A, m)$ and $K_2 = D(B, n)$ be nonempty deleted-digit sets, where $m, n \geq 2$ and $\log m / \log n \notin \mathbb{Q}$. If $L(x, y) = ux + vy$ with $uv \neq 0$, then

$$\dim_H L(K_1 \times K_2) = \min\{1, \dim_H K_1 + \dim_H K_2\}. \quad (18.4a)$$

The formula also holds when either factor is replaced by a cylinder of that factor.

Proof. If one factor is a singleton, the assertion follows from invariance of Hausdorff dimension under a nonzero similarity. Otherwise let η_i be the uniform self-similar probability on K_i . The digit systems satisfy the open set condition, so their cylinders and equal weights give $D(\eta_i, q) = \dim_H K_i$ for every $q > 1$, where $D(\cdot, q)$ is the L^q dimension defined in Volume II.11. For the base- m digit law, the difference of two distinct k -prefix translations is a nonzero integer multiple of m^{-k} , hence has magnitude at least m^{-k} ; the base- n law has the analogous bound n^{-k} . Consequently the common exponential-separation hypothesis of the Volume-II theorem holds at every prefix length after choosing its exponent R large enough. The nonzero factors u, v only change these bounds by fixed multiplicative constants. The Volume-II Chapter-II.11 homogeneous irrational-convolution formula, applied to $(x, y) \mapsto ux + vy$, yields

$$D(L_*(\eta_1 \times \eta_2), q) = \min\{1, \dim_H K_1 + \dim_H K_2\} \quad (q > 1). \quad (18.4b)$$

The Hausdorff dimension of the support of a probability is at least its L^q dimension. Conversely L is Lipschitz and the product of the two Ahlfors-regular deleted-digit sets has Hausdorff dimension $\dim_H K_1 + \dim_H K_2$; the ambient line gives the cap at one. These bounds prove (18.4a). A cylinder is an affine copy of the underlying digit set. Its nonzero diagonal rescaling leaves both coefficients of L nonzero, so the same argument applies. $\square$

Corollary IV.8.29 (*no cross-base deleted-digit embedding*).

Suppose $K_1 = D(A, m)$ and $K_2 = D(B, n)$ satisfy $0 < \dim_H K_1, \dim_H K_2 < 1$ and $\log m / \log n \notin \mathbb{Q}$. There is no nonsingular affine map $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(K_1) \subset K_2$.

Proof. The fully proved Volume-II.11 affine-intersection theorem applies to the two homogeneous digit systems and gives

$$\overline{\dim}_B(K_2 \cap f(K_1)) \leq \max\{\dim_H K_1 + \dim_H K_2 - 1, 0\} < \dim_H K_1.$$

If $f(K_1) \subset K_2$, the left side is at least $\dim_H f(K_1) = \dim_H K_1$, a contradiction. $\square$

Theorem IV.8.30 (*logarithmic commensurability for deleted-digit self-embeddings*).

Let $K = D(A, m)$ with $0 < \dim_H K < 1$, and put $\rho = m^{-1}$. If $g(x) = ax + b$, $a \neq 0$, satisfies $g(K) \subset K$, then

$$\frac{\log |a|}{\log \rho} \in \mathbb{Q}. \quad (18.4c)$$

This is the deleted-digit specialization of the Feng–Wang logarithmic commensurability theorem used in the Algom–Hochman proof. We give its density argument because the conclusion is an indispensable research-level input.

Proof. Normalize $\text{diam } K = 1$ by an affine change of coordinate. Put $s = \dim_H K \in (0, 1)$ and $\mu = \mathcal{H}^s|_K$, so $H = \mu(K) \in (0, \infty)$. The open set condition for the digit IFS gives, for every length- k word u ,

$$\mu(\varphi_u(K)) = \rho^{ks}H, \quad \mu(\varphi_u(K) \cap \varphi_v(K)) = 0 \quad (u \neq v). \quad (18.4d)$$

The cylinder identities follow from similarity invariance of $\mathcal{H}^s$ and the fact that distinct digit cylinders meet, at most, at endpoints; those have zero s -measure. The Ahlfors upper estimate for this open-set-condition system makes

$$D_* := \sup_{I \text{ interval, } |I| > 0} \frac{\mu(I)}{|I|^s} \in (0, \infty). \quad (18.4e)$$

We first find an interval with a full-cylinder boundary gap and strictly excessive integer cylinder count. Choose an integer $q \geq 1$ so large that $\rho^{q-1} < 1/10$ and

$$\frac{1}{2}\rho^{qs}H > D_*[(1 + 8\rho^{q-1})^s - 1]. \quad (18.4f)$$

Such a q exists because $0 < s < 1$: the left side has order ρ^{qs} and the bracket has order ρ^q . Choose $\eta > 0$ so small that (18.4f) still holds when η is added to its right side. By the definition of D_* choose an interval $I_0 = (u, v)$ of length L with

$$\mu(I_0) > (D_* - \eta)L^s. \quad (18.4g)$$

We may require $L \leq 1$: an interval longer than $\text{diam } K = 1$ has density at most H , while an interval of length one containing K has density H . Choose n with $\rho^{n+1} < L \leq \rho^n$, set $k = n + q$ and $t = \rho^k$. Thus $\rho^q \leq t/L < \rho^{q-1}$. Equations (18.4f)–(18.4g) imply

$$\mu(I_0) + \frac{1}{2}t^sH > D_*(L + 8t)^s. \quad (18.4h)$$

Indeed, divide by L^s , use $t/L \geq \rho^q$ in the positive term and $t/L < \rho^{q-1}$ in the enlargement term.

There can be no point of K in either closed boundary band

$$[u - 3t, u - t] \quad \text{or} \quad [v + t, v + 3t]. \quad (18.4i)$$

If a point lay in such a band, the length- k digit cylinder containing it, of diameter at most t , would lie outside I_0 but within $[u - 4t, v + 4t]$. By (18.4d) it would add t^sH of measure to $\mu(I_0)$ in that enlarged interval, contradicting (18.4e) and (18.4h). In particular, for

$$I = (u - 2t, v + 2t),$$

every length- k cylinder meeting $K \cap I$ lies wholly inside I : a cylinder straddling its boundary would put a point of K in one of the forbidden bands. Let M be the number of these cylinders. They cover $K \cap I_0$, so $Mt^s H \geq \mu(I_0)$. Since $|I| = L + 4t$, (18.4h) yields

$$(M + \frac{1}{2})t^s H > D_* |I|^s. \quad (18.4j)$$

The diameter condition $g(K) \subset K$ implies $|a| \leq 1$. For $|a| = 1$ the quotient in (18.4c) is zero. Suppose $0 < |a| < 1$ and, to obtain a contradiction, that $\log |a| / \log \rho$ is irrational. Density of an irrational circle rotation gives, for every $\varepsilon > 0$, positive integers m_0, n_0 such that

$$1 - \varepsilon < \frac{\rho^{m_0}}{|a|^{n_0}} < 1. \quad (18.4k)$$

Choose ε so that $(M + 1)(1 - \varepsilon)^s > M + \frac{1}{2}$. Let $J = g^{n_0}(I)$ and $\tilde{J} = g^{n_0}(u - t, v + t)$. The M length- k cylinders inside I are also inside $(u - t, v + t)$, since (18.4i) excludes K in the strips between these intervals. Their images under g^{n_0} lie in $K \cap \tilde{J}$, because $g^{n_0}(K) \subset K$. Their s -measure is $M|a|^{n_0 s} t^s H$.

Let R count the level- $(m_0 + k)$ digit cylinders meeting $K \cap \tilde{J}$. They cover that set, and each has measure $\rho^{(m_0 + k)s} H$. Therefore

$$R\rho^{(m_0 + k)s} H \geq M|a|^{n_0 s} t^s H > M\rho^{(m_0 + k)s} H,$$

so $R \geq M + 1$. A level- $(m_0 + k)$ cylinder has diameter at most $\rho^{m_0} t < |a|^{n_0} t$, the distance from $\tilde{J}$ to either boundary of J . Thus each of the R cylinders is contained in J . Using (18.4d), (18.4k), and (18.4j) gives

$$\begin{aligned} \mu(J) &\geq (M + 1)\rho^{(m_0 + k)s} H \\ &> (M + 1)(1 - \varepsilon)^s |a|^{n_0 s} t^s H \\ &> (M + \frac{1}{2})|a|^{n_0 s} t^s H > D_* |a|^{n_0 s} |I|^s = D_* |J|^s. \end{aligned}$$

This contradicts (18.4e), proving (18.4c). $\square$

The same three one-dimensional inputs also settle the product-carpet case completely. It is worth treating separately: in a product, one coordinate projection of an affine image already detects a mixed row of its linear part, so no tangent classification is required.

Theorem IV.8.31 (*affine self-embeddings of nontrivial product carpets*).

Let $F = K_1 \times K_2$, where $K_1 = D(A, m)$ and $K_2 = D(B, n)$, $\log m / \log n \notin \mathbb{Q}$, and $0 < \dim_H K_1, \dim_H K_2 < 1$. If an invertible affine map $g(z) = Mz + t$ satisfies $g(F) \subset F$, then

$$M = \text{diag}(\varepsilon_1 m^{-p}, \varepsilon_2 n^{-q}), \quad \varepsilon_i \in \{-1, 1\}, \quad p, q \in \mathbb{Q}_{\geq 0}. \quad (18.4l)$$

In particular, if g is a similarity, then $p = q = 0$ and g is an isometry whose linear part changes only coordinate signs.

Proof. Set $s_i = \dim_H K_i$. If a row of M has two nonzero entries, the corresponding coordinate projection of $g(F)$ is a translate of a non-principal linear projection of $K_1 \times K_2$. By Lemma IV.8.28 it has dimension $\min\{1, s_1 + s_2\}$. Yet it is contained in either K_1 or K_2 ,

whose dimension is strictly smaller than that number. Consequently each row of M has at most one nonzero entry. Invertibility leaves only a diagonal or an anti-diagonal matrix.

If M is anti-diagonal, its first coordinate gives a nonsingular affine embedding $K_2 \hookrightarrow K_1$, contrary to Corollary IV.8.29. Thus M is diagonal, say $M = \text{diag}(a, d)$. Projection of $g(F) \subset F$ now gives affine self-embeddings of K_1 and K_2 with linear coefficients a and d . Theorem IV.8.30 implies $|a| = m^{-p}$ and $|d| = n^{-q}$ for rational p, q . The diameter of each non-singleton compact factor cannot increase under a self-embedding, so $|a|, |d| \leq 1$ and $p, q \geq 0$. This proves (18.4l).

If g is a similarity, $m^{-p} = n^{-q}$. If either exponent were positive, both would be positive and $\log m / \log n = q/p \in \mathbb{Q}$, a contradiction. Hence $p = q = 0$. $\square$

The remaining planar proof uses geometric information about zooms at *every* point of a carpet, not merely at typical points of a measure. The following version of the finite-piece tangent decomposition keeps the viewing-window boundary explicit. It also records the two possible endpoint limits of the scaling phase; no identification of the phases 0 and 1 is needed for this statement.

Theorem IV.8.32 (*finite-piece structure of carpet tangents*).

Let $m > n \geq 2$, let $D \subset \{0, \dots, m-1\} \times \{0, \dots, n-1\}$ be nonempty, and set $F = F_D$ and $Q = [-1, 1]^2$. Write

$$J = \{j : (i, j) \in D \text{ for some } i\}, \quad K_y = P_2 F = \left\{ \sum_{t \geq 1} j_t n^{-t} : j_t \in J \right\}.$$

For $\xi = (\xi_t)_{t \geq 1} \in J^{\mathbb{N}}$ define its horizontal symbolic fibre by

$$K_\xi = \left\{ \sum_{t \geq 1} i_t m^{-t} : (i_t, \xi_t) \in D \text{ for every } t \right\}.$$

For $f \in F$ and $\ell \geq 1$ put

$$M_\ell(f) = [m^\ell(F - f)] \cap Q, \quad k_\ell = \lfloor \ell \log_n m \rfloor, \quad s_\ell = \ell \log_n m - k_\ell \in [0, 1).$$

Suppose $\ell_r \rightarrow \infty$ and $M_{\ell_r}(f)$ converges in the Hausdorff metric to T . On a subsequence there are an integer $1 \leq N \leq 25$, a number $s \in [0, 1]$, sequences $\xi_1, \dots, \xi_N \in J^{\mathbb{N}}$, and vectors $z_1, \dots, z_N \in \mathbb{R}^2$ such that $s_{\ell_r} \rightarrow s$ and, with

$$E_j = z_j + \text{diag}(1, n^s)(K_{\xi_j} \times K_y),$$

one has

$$\left(\bigcup_{j=1}^N E_j \right) \cap \text{int } Q \subset T \subset \left(\bigcup_{j=1}^N E_j \right) \cap Q. \quad (18.4m)$$

Furthermore, for some $y \in \mathbb{R}/\mathbb{Z}$,

$$n^{\ell_r} f_2 \longrightarrow y \pmod{1}, \quad \sum_{t \geq 1} (\xi_j)_t n^{-t} = y \pmod{1} \quad (1 \leq j \leq N). \quad (18.4n)$$

Thus at most two different symbolic fibres occur among the E_j ; the two can differ only through the two base- n expansions of a base- n endpoint. The theorem makes no assertion that every choice of the displayed pieces is realized as a tangent.

Proof. For a word $c = (c_1, \dots, c_h)$ of base- b digits write $\pi_b(c) = \sum_{t=1}^h c_t b^{-t}$. For a vertical word $c \in J^h$ let

$$H(c) = \bigcup_{\substack{a=(a_1, \dots, a_h) \\ (a_t, c_t) \in D \ \forall t}} [(\pi_m(a), 0) + \text{diag}(m^{-h}, 1)F]. \quad (18.4o)$$

The union is nonempty because every $c_t \in J$ has an allowed horizontal partner. If $h_r \rightarrow \infty$ and the words $c^{(r)}$ converge coordinatewise to ξ , then

$$H(c^{(r)}) \longrightarrow K_\xi \times K_y \quad \text{in the Hausdorff metric.} \quad (18.4p)$$

To verify this, fix $p < h_r$. For large r the admissible horizontal p -prefixes for $c^{(r)}$ are exactly those for ξ . Every point of $H(c^{(r)})$ has horizontal coordinate within m^{-p} of one of their left endpoints, and its vertical coordinate belongs to K_y . Conversely, for each admissible p -prefix, each $v \in K_y$, and each extension of the horizontal word to the full $c^{(r)}$, select $(x, v) \in F$ with $P_2(x, v) = v$; the corresponding point of $H(c^{(r)})$ has horizontal coordinate within m^{-p} of the prefix endpoint. The same prefix endpoints approximate K_ξ within m^{-p} . Hence the Hausdorff distance in (18.4p) is at most $2m^{-p}$ for all large r , and then $p \rightarrow \infty$.

Fix ℓ and abbreviate $k = k_\ell$. Then $k \geq \ell$ and

$$m^{-\ell} \leq n^{-k} < nm^{-\ell}. \quad (18.4q)$$

For a horizontal word a of length ℓ and a vertical word $b = b'b''$ of length k , with $|b'| = \ell$, let $C_{a,b}$ consist of all points of F whose first ℓ horizontal digits are a and first k vertical digits are b . It is empty unless $(a_t, b_t) \in D$ for $t \leq \ell$; when nonempty, the independent continuation after level k gives the exact identity

$$m^\ell(C_{a,b} - f) = z_{\ell,a,b} + \text{diag}(1, n^{s_\ell})H(b''), \quad z_{\ell,a,b} = m^\ell((\pi_m(a), \pi_n(b)) - f). \quad (18.4r)$$

Indeed, the horizontal digits between levels $\ell + 1$ and k are precisely the admissible words in (18.4o), while all digits after level k form an independent copy of F . The vertical factor in (18.4r) is $m^\ell n^{-k} = n^{s_\ell}$.

Retain just those pairs (a, b) for which $C_{a,b} \cap (f + m^{-\ell}Q) \neq \emptyset$. Their full sets on the right of (18.4r), clipped by Q , cover $M_\ell(f)$ exactly: adding parts of one retained full set outside the viewing window does not change its intersection with Q . The first-coordinate interval of $C_{a,b}$ has length at most $m^{-\ell}$; hence a retained a has $|\pi_m(a) - f_1| \leq 2m^{-\ell}$. There are at most five such grid points. Similarly, the second-coordinate interval has length at most n^{-k} , and a retained b satisfies $|\pi_n(b) - f_2| \leq m^{-\ell} + n^{-k} \leq 2n^{-k}$. There are at most five such grid points. Therefore at most 25 pairs are retained. Their vectors $z_{\ell,a,b}$ stay in a fixed bounded set by (18.4q), and their full sets in (18.4r) stay in a fixed compact square.

Pass to a subsequence on which the number N of retained pairs is constant, label them consistently, $s_{\ell_r} \rightarrow s \in [0, 1]$, their translations converge, and their vertical suffix words

$b''_{r,j}$ converge coordinatewise to sequences ξ_j . The suffix lengths $k_{\ell_r} - \ell_r$ tend to infinity because $m > n$. Equation (18.4p), followed by (18.4r), shows that each full scaled group converges to E_j . For any convergent compact sets $A_r \rightarrow A$, limits of $A_r \cap Q$ lie in $A \cap Q$, and every point of $A \cap \text{int } Q$ is approximated by points of $A_r \cap Q$. Applying this to finitely many groups proves (18.4m). In particular T is nonempty, since every $M_\ell(f)$ contains the origin.

Finally choose a further subsequence on which $n^{\ell_r} f_2 \rightarrow y$ modulo one. For each retained word $b_{r,j}$, the preceding interval estimate and $n^{\ell_r}/m^{\ell_r} \rightarrow 0$ give

$$\left\| n^{\ell_r} \pi_n(b_{r,j}) - n^{\ell_r} f_2 \right\|_{\mathbb{R}/\mathbb{Z}} \leq 2(n/m)^{\ell_r} \rightarrow 0.$$

Modulo one, $n^{\ell_r} \pi_n(b_{r,j})$ equals $\pi_n(b''_{r,j})$, so continuity of base- n coding gives (18.4n). A point of the circle has at most two base- n expansions, proving the last assertion. $\square$

The projection bounds used in the rigidity argument follow directly from this finite-piece form. It is important to choose a *nonconstant periodic tail* when forcing a thin horizontal fibre: an eventually constant tail of all zeroes or all maximal digits can have a second base- n expansion with a different fibre.

Lemma IV.8.33 (*projection bounds and periodic-tail fibres*).

Assume the notation of Theorem IV.8.32, and put

$$A_j = \{i : (i, j) \in D\}, \quad s_* = \frac{\log \max_{j \in J} |A_j|}{\log m}, \quad t = \frac{\log |J|}{\log n}.$$

For every $f \in F$ and every m -adic tangent T at f ,

$$\overline{\dim}_B P_1 T \leq s_*, \quad \overline{\dim}_B P_2 T \leq t. \quad (18.4s)$$

Let $w = (w_1, \dots, w_p) \in J^p$ contain at least two distinct digits, and define

$$s_w = \frac{1}{p \log m} \sum_{a=1}^p \log |A_{w_a}|.$$

If f_2 has a base- n expansion consisting of a finite prefix followed by the infinite repetition w^∞ , then every tangent T at f satisfies

$$\overline{\dim}_B P_1 T \leq s_w. \quad (18.4t)$$

Moreover each full product piece E_j in the tangent decomposition has horizontal factor $K_{\sigma^a w^\infty}$ for some $a \in \{0, \dots, p-1\}$; this factor is a deleted-digit set in base m^p of Hausdorff and upper box dimension s_w .

Proof. At level q , K_ξ is covered by at most $\prod_{a=1}^q |A_{\xi_a}| \leq (\max_j |A_j|)^q$ intervals of length m^{-q} . Hence $\overline{\dim}_B K_\xi \leq s_*$. Also $K_y = D(J, n)$ has upper box dimension t by its equal-scale digit cover and the open set condition. The right inclusion in (18.4m) places each projection of T inside a finite union of affine images of these sets, proving (18.4s).

The periodic-tail expansion of f_2 is unique: a number with two base- n expansions must have one expansion eventually equal to 0^∞ and the other eventually equal to $(n-1)^\infty$, whereas w^∞ is neither. For every sufficiently large ℓ , the fractional part of $n^\ell f_2$ is the base- n value of a cyclic shift of w^∞ . In the proof of Theorem IV.8.32 pass to a subsequence with fixed ℓ modulo p . The common circle limit y in (18.4n) has the unique expansion $\sigma^a w^\infty$, so all the ξ_j equal that sequence. Grouping horizontal digits into blocks of length p identifies $K_{\sigma^a w^\infty}$ with $D(B_a, m^p)$, where B_a consists of the base- m values of the allowed horizontal p -blocks. It has $|B_a| = \prod_{b=1}^p |A_{w_b}|$, independently of a . The digit cylinders have disjoint interiors, giving both dimensions s_w . The finite-union containment in (18.4m) now gives (18.4t). $\square$

Lemma IV.8.34 (*affine covariance of m -adic tangents*).

Let $X \subset \mathbb{R}^2$ be compact and let $g(z) = Az + b$, with A invertible and $g(X) \subset X$. Suppose $x \in X$, $y = g(x)$, and $m^{\ell r}(X - x) \cap Q \rightarrow T$. Choose an integer $p \geq 0$ so large that $m^{-p}AQ \subset \text{int } Q$. After passage to a subsequence, the compact sets $m^{\ell r-p}(X - y) \cap Q$ converge to a tangent T' at y , and

$$m^{-p}AT \subset T'. \quad (18.4u)$$

Proof. Compactness of the hyperspace of nonempty compact subsets of Q gives the subsequential limit. If $v \in T$, select $x_r \in X$ with $v_r = m^{\ell r}(x_r - x) \in Q$ and $v_r \rightarrow v$. Then

$$m^{\ell r-p}(g(x_r) - y) = m^{-p}Av_r \in m^{\ell r-p}(X - y) \cap Q$$

for all large r , since $m^{-p}AQ \subset \text{int } Q$. Its limit is $m^{-p}Av$, which lies in T' . This holds for every $v \in T$. $\square$

Theorem IV.8.35 (*a self-embedding cannot mix the axes*).

Let m, n be multiplicatively independent integers and let $F = F_D$ be a non-product Bedford–McMullen carpet. If the similarity $g(z) = rOz + b$ satisfies $g(F) \subset F$, then the orthogonal matrix O is diagonal.

Proof. Interchange the coordinate axes if necessary so that $m > n$. Write $K_x = P_1 F = D(\bigcup_{j \in J} A_j, m)$ and $K_y = P_2 F = D(J, n)$, and use s_* and t from Lemma IV.8.33. Because F is not a product, $|J| \geq 2$ and $t > 0$; if $|J| = 1$, the digit set would be $A_j \times \{j\}$. We first rule out an O that is neither diagonal nor anti-diagonal. Both rows of such an orthogonal 2×2 matrix have two nonzero entries.

Choose $j \in J$ with $|A_j|$ maximal, choose $u \in J$ with $u \neq j$, and put $w = (j, u)$. The periodic vertical number $f_2 = \pi_n(w^\infty)$ has a unique expansion; choose a horizontal sequence compatible with it to obtain $f \in F$. Along a suitable sequence of m -adic scales choose a tangent T at f and, by Lemma IV.8.34, a tangent T' at $g(f)$ with $m^{-p}rOT \subset T'$. Theorem IV.8.32 gives a full product piece $E = z + \text{diag}(1, n^s)(K_\xi \times K_y)$ whose intersection with $\text{int } Q$ is nonempty and lies in T : indeed $0 \in T$ and the upper inclusion in (18.4m) puts 0 in at least one such E . By Lemma IV.8.33, K_ξ is a base- m^2 deleted-digit set of dimension

$$s_w = \frac{\log |A_j| + \log |A_u|}{2 \log m}.$$

Inside $E \cap \text{int } Q$ choose a product of a horizontal m^2 -cylinder of K_ξ and a vertical n -cylinder of K_η . Such cylinders exist at arbitrarily small scales around any point of that intersection. Since $\log(m^2)/\log n$ is irrational and both rows of O are non-principal, Lemma IV.8.28 applied to the two coordinate projections of this product gives

$$\dim_H P_i T' \geq d := \min\{1, s_w + t\}, \quad i = 1, 2. \quad (18.4v)$$

But (18.4s) gives $\dim_H P_1 T' \leq s_*$ and $\dim_H P_2 T' \leq t$. If $s_* = 0$, then $s_w = 0$ and $d = t > 0 = s_*$. If $0 < s_*$ and $t < 1$, then $s_w > 0$ and $d > t$. If $t = 1$ and $s_* < 1$, then $d = 1 > s_*$. Thus a genuinely mixed O is possible only if $s_* = t = 1$.

We next give a contradiction whenever $t = s_* = 1$ and O is not diagonal; the argument also applies to an anti-diagonal O . Here $J = \{0, \dots, n-1\}$, so $K_y = [0, 1]$, and some row j has $A_j = \{0, \dots, m-1\}$, giving $[0, 1] \times \{\pi_n(j^\infty)\} \subset F$. Because O is not diagonal, the second coordinate of g is nonconstant on this horizontal segment. Consequently $P_2 g(F)$ contains an interval I of positive length. As F is not a product and J is full, there is a row u with $1 \leq |A_u| < m$. Choose a nonconstant word w containing u ; for instance $w = (u, j)$ if $u \neq j$. Numbers whose base- n expansion is an arbitrary finite prefix followed by w^∞ are dense in $[0, 1]$; their expansion is unique because w is nonconstant. Select one such $h_2 \in I$ and a point $h = (h_1, h_2) \in g(F)$, and put $f = g^{-1}(h) \in F$. For every tangent T' at h , (18.4t) gives

$$\overline{\dim}_B P_1 T' \leq s_w = \frac{1}{|w| \log m} \sum_{a=1}^{|w|} \log |A_{w_a}| < 1. \quad (18.4w)$$

Choose a source tangent T at f and a target tangent T' as in (18.4u). A full product piece of T meets $\text{int } Q$ as above. Its vertical factor is $K_y = [0, 1]$, so T contains a nondegenerate vertical segment inside $\text{int } Q$. The first row of every non-diagonal orthogonal 2×2 matrix has a nonzero vertical entry. Thus (18.4u) makes $P_1 T'$ contain an interval, contradicting (18.4w).

It remains to rule out an anti-diagonal O when $(s_*, t) \neq (1, 1)$. Put $s_x = \dim_H K_x$ and recall $t = \dim_H K_y$. The inclusions $P_1 g(F) \subset K_x$ and $P_2 g(F) \subset K_y$ give respectively $t \leq s_x$ and $s_x \leq t$, hence $s_x = t$. Since F is not a product, $s_x = t = 0$ is impossible. If $0 < s_x = t < 1$, the first coordinate of g is a nonsingular affine embedding $K_y \hookrightarrow K_x$, contradicting Corollary IV.8.29. If $s_x = t = 1$ but $s_* < 1$, repeat the preceding tangent argument: $K_y = [0, 1]$ makes a vertical interval in a source tangent, its anti-diagonal image makes an interval in $P_1 T'$, while (18.4s) bounds $\overline{\dim}_B P_1 T' \leq s_* < 1$. The case $s_x = t = s_* = 1$ was already ruled out. Therefore O must be diagonal. $\square$

Corollary IV.8.36 (rigidity when both coordinate projections are proper fractals).

Under the hypotheses of Theorem IV.8.35, assume in addition that $\dim_H P_1 F < 1$ and $\dim_H P_2 F < 1$. Then every similarity g with $g(F) \subset F$ is an isometry whose orthogonal part is diagonal.

Proof. Both projection dimensions are positive: if one coordinate projection were a singleton, D would be a Cartesian product. By Theorem IV.8.35 the orthogonal part of g is diagonal with entries ± 1 . Its square therefore has the form $g^2(z) = r^2 z + b'$ and still maps F into F . The two coordinate projections give self-embeddings of the nontrivial proper

deleted-digit sets P_1F and P_2F with common positive contraction r^2 . Theorem IV.8.30 implies

$$\frac{\log r^2}{\log(m^{-1})} \in \mathbb{Q}, \quad \frac{\log r^2}{\log(n^{-1})} \in \mathbb{Q}.$$

If $r \neq 1$, their quotient makes $\log m / \log n$ rational, contrary to the hypothesis. Thus $r = 1$. If a diagonal entry of the orthogonal part is $+1$, repeated iteration forces the corresponding translation coordinate to vanish, since all iterates of a point of F stay in the compact set F . A diagonal entry -1 acts as a reflection about the corresponding coordinate midpoint. Thus the asserted isometry has precisely the axis-reflection form in the theorem below. $\square$

We can also complete the case in which exactly one vertical digit allows a full horizontal row. The proof uses the lengths of intervals in horizontal slices, and it avoids choosing a base- n endpoint with an ambiguous expansion.

Lemma IV.8.37 (*a long slice interval forces a full-row tail*).

Assume $m > n$ and that there is exactly one digit $j_* \in J$ with $A_{j_*} = \{0, \dots, m-1\}$. Put $y_* = \sum_{k \geq 1} j_* n^{-k} = j_*/(n-1)$, and for $y \in K_y$ write $F_y = \{x : (x, y) \in F\}$. If $p \geq 1$ and F_y contains a closed interval of length at least m^{-p} , then y has an allowed base- n expansion whose digits after position p all equal j_* . In particular,

$$y \in S_p := \left\{ \sum_{a=1}^p c_a n^{-a} + n^{-p} y_* : c_a \in J \right\}. \quad (18.4x)$$

Proof. For an allowed vertical sequence $\eta \in J^{\mathbb{N}}$, write K_η for its symbolic horizontal fibre as in Theorem IV.8.32. A real $y \in [0, 1]$ has at most two base- n expansions, and direct digit coding gives

$$F_y = \bigcup_{\eta: \pi_n(\eta)=y} K_\eta, \quad (18.4y)$$

where disallowed expansions contribute the empty set. If A_{η_k} is proper, every level- $(k-1)$ horizontal parent interval lacks at least one of its m level- k children. More explicitly, choose $d \in \{0, \dots, m-1\} \setminus A_{\eta_k}$. Every open m -adic interval

$$(am^{-k}, (a+1)m^{-k}), \quad 0 \leq a < m^k, \quad a \equiv d \pmod{m},$$

is disjoint from K_η . Their left endpoints have spacing $m^{-(k-1)}$, while each missing interval has positive length m^{-k} ; the two terminal gaps in $[0, 1]$ are also shorter than $m^{-(k-1)}$. Thus no interval of length $m^{-(k-1)}$ is contained in K_η . In particular, K_η has empty interior if $\eta_k \neq j_*$ for infinitely many k .

If y has two base- n expansions, one is eventually all zeroes and the other eventually all $(n-1)$'s. Since j_* is a single digit, at most one of these expansions can be eventually j_* . Hence at most one compact set in (18.4y) has nonempty interior; the other is nowhere dense. If an interval $I \subset F_y$ were not contained in the possible interior-bearing K_η , the nonempty relatively open set $I \setminus K_\eta$ would contain an interval belonging to the other, nowhere-dense fibre, a contradiction. Therefore the displayed interval lies in one K_η . If any $\eta_k \neq j_*$ for $k \geq p+1$, the preceding gap bound with $m^{-(k-1)} \leq m^{-p}$ contradicts its length. Thus $\eta_k = j_*$ for all $k \geq p+1$, and its first p digits give (18.4x). $\square$

Theorem IV.8.38 (*rigidity with a unique full horizontal row*).

Assume $m > n$ are multiplicatively independent and F_D is not a product carpet. If exactly one $j_* \in J$ satisfies $A_{j_*} = \{0, \dots, m-1\}$, then every similarity $g(F_D) \subset F_D$ is an isometry with diagonal orthogonal part.

Proof. Theorem IV.8.35 makes the orthogonal part diagonal. The diameter inequality gives its similarity ratio $r \leq 1$. If $r < 1$, replace g by g^2 so that its linear part is the positive scalar $r^2 I$; rename its ratio r . We shall show that no positive homothety $g(z) = rz + b$ with $0 < r < 1$ can take F into itself.

Because F is not a product, J has at least two digits. Choose $a \in J \setminus \{j_*\}$ and put

$$z = \pi_n(j_*^\infty) = y_*, \quad z' = \pi_n(a, j_*, j_*, \dots) = \frac{a + y_*}{n}.$$

The slice F_z is the full interval $[0, 1]$. Since $A_a \neq \emptyset$, the slice $F_{z'}$ contains at least one closed interval of length m^{-1} : choose a first horizontal digit paired with a and use the full row j_* thereafter. Moreover

$$|z - z'| = \frac{|j_* - a|}{n} > 0. \quad (18.4z)$$

Replace g by a sufficiently high positive power and again write its ratio as r . Choose the integer $p \geq 1$ with $m^{-p} \leq r < m^{-(p-1)}$. Since $m > n$, taking the power large makes p so large that

$$m^{-(p-1)} < n^{-(p+1)}. \quad (18.4aa)$$

Let $h = rz + b_2$ and $h' = rz' + b_2$. The slice inclusions $rF_z + b_1 \subset F_h$ and $rF_{z'} + b_1 \subset F_{h'}$ show that F_h contains an interval of length $r \geq m^{-p}$ and $F_{h'}$ one of length $r/m \geq m^{-(p+1)}$. Lemma IV.8.37 therefore gives $h \in S_p$ and $h' \in S_{p+1}$. Appending j_* to a prefix shows $S_p \subset S_{p+1}$. Every point of S_{p+1} has the form

$$n^{-(p+1)}(N + y_*), \quad N \in \mathbb{Z},$$

so the distinct numbers h, h' differ by at least $n^{-(p+1)}$. On the other hand, (18.4z), $|j_* - a| < n$, and (18.4aa) give

$$0 < |h - h'| = r \frac{|j_* - a|}{n} < r < m^{-(p-1)} < n^{-(p+1)},$$

a contradiction. Thus $r = 1$ for the original similarity. $\square$

For the no-full-row case, the fixed point of a hypothetical contracting homothety forces the horizontal projection itself to be a proper deleted-digit set. This reduction is often hidden inside the tangent argument, so we state it explicitly.

Lemma IV.8.39 (*a contracting homothety puts the whole carpet in each fixed-point tangent*).

Let $F \subset [0, 1]^2$ be compact and suppose $g(z) = rz + b$, $0 < r < 1$, satisfies $g(F) \subset F$. Let f be its fixed point; then $f \in F$. For every subsequential Hausdorff limit T of $m^\ell(F - f) \cap Q$, with $m \geq 2$ and $Q = [-1, 1]^2$, there is a $c \in [r, 1]$ such that

$$c(F - f) \subset T. \quad (18.4ab)$$

Proof. Since $g^k(x) \rightarrow f$ for every $x \in F$ and each $g^k(x)$ lies in the closed set F , the fixed point belongs to F . For each ℓ choose $k_\ell \geq 0$ so that $r^{k_\ell} \leq m^{-\ell} < r^{k_\ell - 1}$, and put $c_\ell = m^\ell r^{k_\ell} \in (r, 1]$. Because $g^{k_\ell}(F) = f + r^{k_\ell}(F - f) \subset F$ and $F - f \subset [-1, 1]^2$, we have

$$c_\ell(F - f) \subset m^\ell(F - f) \cap Q.$$

Along the scales defining T , pass to a further subsequence on which $c_\ell \rightarrow c \in [r, 1]$. The left compact sets converge to $c(F - f)$, so the inclusion passes to the Hausdorff limit. $\square$

Corollary IV.8.40 (*rational scale reduction when no row is full*).

Let $m > n$ be multiplicatively independent and let F_D be non-product, with $|A_j| < m$ for every $j \in J$. If a similarity g with ratio $r < 1$ maps F_D into itself, then its square is a positive homothety with ratio r^2 , $\dim_H R_1 F_D < 1$, and

$$\frac{\log r^2}{\log(m^{-1})} \in \mathbb{Q}_{>0}. \quad (18.4ac)$$

Consequently some positive power of g has scalar ratio m^{-L} for an integer $L \geq 1$.

Proof. Theorem IV.8.35 makes the orthogonal part diagonal with entries ± 1 , so g^2 is a positive homothety. Let f be its fixed point. Lemma IV.8.39 and the horizontal estimate (18.4s) give

$$\dim_H R_1 F_D \leq \dim_H R_1 T \leq s_* = \frac{\log \max_j |A_j|}{\log m} < 1.$$

Since F_D is non-product, $R_1 F_D$ is not a singleton, so its dimension is positive. Projection of $g^2(F_D) \subset F_D$ onto the first coordinate gives an affine self-embedding of the proper deleted-digit set $R_1 F_D$. Theorem IV.8.30 yields (18.4ac). Writing its positive rational value as L/q and taking the q th power of g^2 gives ratio m^{-L} . $\square$

Lemma IV.8.41 (*same-base embeddings have adic translation*).

Let $K = D(A, m)$ with $2 \leq |A| < m$. If $h(x) = m^{-L}x + t$ maps K into itself for an integer $L \geq 1$, then t is an m -adic rational: $t = N/m^q$ for some integers N and $q \geq 0$.

Proof. Let $d = \min A$ and $c = d/(m - 1) = \min K$. Replacing K by $K - c = D(A - d, m)$ replaces t by $t' = t - (1 - m^{-L})c$. Since

$$(1 - m^{-L})c = \frac{d(1 + m + \cdots + m^{L-1})}{m^L}$$

is m -adic, it suffices to treat $0 \in A$. Then $0 \in K$ and $t = h(0) \in K \subset [0, 1]$. If t already has a terminating base- m expansion, or has two expansions, it is m -adic. Assume instead that t is non- m -adic, so it has a unique expansion $t = \sum_{q \geq 1} \eta_q m^{-q}$ with every $\eta_q \in A$.

Fix $p \geq 1$ and $a \in A$. The point a/m^p lies in K , so $t + a/m^{L+p} \in K$. It is still non- m -adic, hence its unique base- m expansion uses only digits of A . The tail of t beyond position $L + p$ has value strictly less than one at that position, so adding a/m^{L+p} creates no carry from a lower digit. Its digit in position $L + p$ is therefore $\eta_{L+p} + a \pmod{m}$. We obtain

$$\eta_{L+p} + A = A \quad \text{in } \mathbb{Z}/m\mathbb{Z} \quad (p \geq 1), \quad (18.4ad)$$

where inclusion became equality because translation preserves cardinality.

Suppose $\eta_{L+p} = c_p > 0$ for some $p \geq 2$. Let k be the largest positive integer with $kc_p < m$ and put $d_p = kc_p$. Equation (18.4ad), applied repeatedly and using $0 \in A$, gives $d_p \in A$, while $d_p + c_p \geq m$. For any $a \in A$ the number

$$x = \frac{a}{m^{p-1}} + \frac{d_p}{m^p}$$

lies in K . In the addition of $m^{-L}x$ to t , position $L + p$ produces a carry of one into position $L + p - 1$; again the lower tail produces no carry. Because the sum is non- m -adic and belongs to K , its digit at $L + p - 1$ shows $\eta_{L+p-1} + A + 1 \subset A$ modulo m . Together with (18.4ad) at $p - 1$, this makes A invariant under translation by 1 modulo m . The only nonempty subset of the cyclic group $\mathbb{Z}/m\mathbb{Z}$ with this property is the whole group, contrary to $|A| < m$. Thus $\eta_{L+p} = 0$ for every $p \geq 2$, and t terminates after position $L + 1$, a contradiction. Hence t is m -adic after all. $\square$

Theorem IV.8.42 (*rigidity when no horizontal row is full*).

Let $m > n$ be multiplicatively independent, and let F_D be a non-product carpet with $|A_j| < m$ for every $j \in J$. Every similarity g with $g(F_D) \subset F_D$ is an isometry with diagonal orthogonal part.

Proof. Theorem IV.8.35 supplies the diagonal orthogonal part. If the similarity ratio were less than one, Corollary IV.8.40 would give a positive power, still denoted g , of the form

$$g(x, y) = (m^{-L}x + t_1, m^{-L}y + t_2), \quad L \geq 1. \quad (18.4ae)$$

The horizontal projection $K_x = P_1 F_D = D(I, m)$, $I = \bigcup_{j \in J} A_j$, has $0 < \dim_H K_x < 1$ by that corollary. Its self-embedding under the first coordinate of (18.4ae) lets us apply Lemma IV.8.41: t_1 is m -adic rational.

For a horizontal digit $i \in I$ put $B_i = \{j : (i, j) \in D\}$. Some B_i has at least two digits. Indeed, if every B_i were a singleton, the non-product condition would force at least two different singleton values, so the nonempty A_j would partition I into at least two parts. Then

$$\dim_H K_x = \frac{\log |I|}{\log m} > \frac{\log \max_j |A_j|}{\log m} = s_*,$$

contrary to the fixed-point tangent inequality in the proof of Corollary IV.8.40. Also some $B_{i'}$ is proper in $\{0, \dots, n - 1\}$; otherwise $D = I \times \{0, \dots, n - 1\}$ would be a product. Choose

distinct $i, i' \in I$ so that

$$2 \leq |B_i| |B_{i'}| < n^2. \quad (18.4af)$$

If the first chosen B_i is already proper, take any other i' ; if it is full, take a proper $B_{i'}$.

Let $w = (i, i')$ and $x = \pi_m(w^\infty)$. This number is not m -adic, so its horizontal expansion is unique. Its vertical slice $F^x = \{y : (x, y) \in F\}$ is exactly

$$K_w = D(C, n^2), \quad C = \{na + b : a \in B_i, b \in B_{i'}\}; \quad (18.4ag)$$

by grouping digit pairs. Equation (18.4af) gives $0 < \dim_H K_w < 1$. Set $x' = m^{-L}x + t_1 \in K_x$. Since t_1 is m -adic, x' is non- m -adic and has a unique horizontal expansion with a finite prefix followed by w^∞ . To see the precise tail, choose $q \geq L$ so that $m^q t_1 \in \mathbb{Z}$ and $q - L$ is even; the fractional part of $m^q x'$ equals that of $m^{q-L}x$, namely x itself. Consequently the vertical slice $F^{x'}$ is a finite union of sets

$$n^{-q}K_w + c_v, \quad c_v \in \mathbb{R}, \quad (18.4ah)$$

indexed by the allowed vertical words paired with the horizontal prefix of x' .

The vertical slice inclusion from (18.4ae) is $m^{-L}K_w + t_2 \subset F^{x'}$. Pull back the finite closed union (18.4ah) to K_w . By the Baire theorem, one preimage contains a nonempty relatively open subset of K_w , and this subset contains a cylinder $\psi(K_w)$ of the base- n^2 system, with ratio n^{-2k} for some $k \geq 0$. For one index v we therefore have

$$(n^{-q} \cdot + c_v)^{-1} \circ (m^{-L} \cdot + t_2) \circ \psi(K_w) \subset K_w.$$

This is a nonsingular affine self-embedding of the proper deleted-digit set K_w with absolute ratio $m^{-L}n^{q-2k}$. Applying Theorem IV.8.30 in base n^2 yields

$$\frac{-L \log m + (q - 2k) \log n}{-2 \log n} \in \mathbb{Q}.$$

Since $L > 0$, this forces $\log m / \log n \in \mathbb{Q}$, a contradiction. Thus the original similarity has ratio one. $\square$

Lemma IV.8.43 (*intervals force a tail of full-row digits at unique heights*).

Let $\Lambda = \{j \in J : A_j = \{0, \dots, m-1\}\}$. If $y \in K_y$ has a unique base- n expansion $\eta = (\eta_k)_{k \geq 1}$ and F_y contains a closed interval of length at least m^{-p} , then $\eta_k \in \Lambda$ for all $k \geq p+1$. Consequently, with $K_\Lambda = D(\Lambda, n)$,

$$y \in E_p := \bigcup_{v \in J^p} (\pi_n(v) + n^{-p}K_\Lambda). \quad (18.4ai)$$

Proof. Uniqueness gives $F_y = K_\eta$. If $\eta_k \notin \Lambda$, the level- k horizontal construction omits a child interval in every surviving level- $(k-1)$ parent. As in the proof of Lemma IV.8.37, no interval of length $m^{-(k-1)}$ is contained in K_η . For $k \geq p+1$ this length is at most m^{-p} , contrary to the assumed interval. Thus all digits after p are in Λ , and splitting the first p digits from the tail gives (18.4ai). $\square$

Theorem IV.8.44 (*rigidity with several full horizontal rows*).

Let $m > n$ be multiplicatively independent and let F_D be non-product. If $\Lambda = \{j \in J : A_j = \{0, \dots, m-1\}\}$ has at least two elements, then every similarity taking F_D into itself is an isometry with diagonal orthogonal part.

Proof. Since F_D is non-product, $2 \leq |\Lambda| < |J| \leq n$; thus $K_\Lambda = D(\Lambda, n)$ is a perfect deleted-digit set of dimension strictly between zero and one. Theorem IV.8.35 makes the orthogonal part diagonal. If the similarity ratio were less than one, its square would be a positive contracting homothety; write this homothety as $g(x, y) = (rx + t_1, ry + t_2)$ with $0 < r < 1$. For every $z \in K_\Lambda$ the horizontal slice F_z equals $[0, 1]$, because z has an allowed expansion using only full-row digits. Hence the slice at $rz + t_2$ contains an interval of length r . Choose $p \geq 1$ so that $m^{-p} \leq r < m^{-(p-1)}$. For every $z \in K_\Lambda$ such that $rz + t_2$ has a unique base- n expansion, Lemma IV.8.43 puts $rz + t_2$ in E_p . The excluded heights form a countable set, and the affine map $z \mapsto rz + t_2$ is injective. Removing their preimages leaves a dense subset of the perfect set K_Λ . Since E_p is closed, continuity gives

$$rK_\Lambda + t_2 \subset E_p. \quad (18.4aj)$$

The finite union in (18.4ai) covers the image in (18.4aj). Pulling the closed pieces back to K_Λ and applying Baire, some inverse image contains a relatively open subset of K_Λ , hence a cylinder $\psi_a(K_\Lambda)$ of ratio n^{-k} . For a suitable $v \in J^p$, the map

$$H = (n^{-p} \cdot + \pi_n(v))^{-1} \circ (r \cdot + t_2) \circ \psi_a$$

maps K_Λ into itself and has positive ratio rn^{p-k} . Theorem IV.8.30 in base n gives $\log r / \log n \in \mathbb{Q}$. Since $r < 1$, replacing g by a positive power lets us assume

$$r = n^{-L}, \quad L \geq 1. \quad (18.4ak)$$

Reapply (18.4aj) to this power, with its new p and t_2 , and repeat the Baire selection. The resulting self-map H has ratio $n^{p-k-L} \leq 1$, because K_Λ is non-singleton. If this ratio is n^{-M} with $M \geq 1$, Lemma IV.8.41 makes the translation of H an n -adic rational. If $M = 0$, then $H(x) = x + c$ maps a nonempty compact set into itself, forcing $c = 0$ by comparison of minimum points. In either case its translation is n -adic. But the translation of H is

$$n^p(n^{-L}\pi_n(a) + t_2 - \pi_n(v)),$$

where a and v are finite base- n words. All terms other than $n^p t_2$ are n -adic, so t_2 is n -adic.

Choose $j \in \Lambda$ and $u \in J \setminus \Lambda$. The word $w = (j, u)$ is nonconstant. Its periodic vertical value $y = \pi_n(w^\infty)$ has a unique base- n expansion, and its horizontal slice is the proper nontrivial base- m^2 deleted-digit set

$$K_w = F_y = D(C, m^2), \quad C = \{ma + b : a \in A_j, b \in A_u\}, \quad 2 \leq |C| = m|A_u| < m^2. \quad (18.4al)$$

Let $y' = n^{-L}y + t_2 \in K_y$. Since t_2 is n -adic, y' is non- n -adic and has a unique expansion consisting of a finite prefix followed by w^∞ . Explicitly, choose $q \geq L$ with $n^q t_2 \in \mathbb{Z}$ and $q - L$ even; then the fractional part of $n^q y'$ is exactly y . Digit coding therefore writes $F_{y'}$ as a finite union of translates of $m^{-q}K_w$.

The horizontal slice inclusion is $n^{-L}K_w + t_1 \subset F_{y'}$. Pull back that finite closed cover to K_w . As above, Baire gives a base- m^2 cylinder of K_w , of ratio m^{-2k} , whose image is contained in one target copy of ratio m^{-q} . After undoing the target copy we obtain an affine self-embedding of K_w with positive ratio $n^{-L}m^{q-2k}$. Applying Theorem IV.8.30 in base m^2 gives

$$\frac{-L \log n + (q - 2k) \log m}{-2 \log m} \in \mathbb{Q}.$$

This implies $\log m / \log n \in \mathbb{Q}$, a contradiction. Thus the original similarity ratio is one. $\square$

Theorem IV.8.45 (*Algom–Hochman rigidity of Bedford–McMullen self-embeddings*).

Let m, n be multiplicatively independent integers and let F be a Bedford–McMullen carpet that is not a product set. If g is a similarity with $g(F) \subset F$, then g is an isometry. More precisely, g is the identity, a reflection in a horizontal or vertical line, or a composition of two such reflections. This is the theorem of Algom and Hochman [7, Theorem 1.1].

Proof. Swap the coordinate names if necessary so that $m > n$. The number of vertical digits j for which the horizontal row $A_j = \{i : (i, j) \in D\}$ is full is zero, one, or at least two. Theorem IV.8.42, Theorem IV.8.38, and Theorem IV.8.44, respectively, show in these three exhaustive cases that g is an isometry with diagonal orthogonal part. In particular its coordinate maps have the form $x_i \mapsto \varepsilon_i x_i + b_i$ with $\varepsilon_i \in \{-1, 1\}$. If $\varepsilon_i = 1$ but $b_i \neq 0$, repeated iteration sends the i th coordinate of any point of F outside the compact coordinate projection of F , contradicting $g^k(F) \subset F$ for every k . Thus $b_i = 0$ whenever $\varepsilon_i = 1$; when $\varepsilon_i = -1$, the coordinate map is reflection in the line $x_i = b_i/2$. This is exactly the stated list. $\square$

2. Local AH examples

Proposition IV.8.46 (*applicable non-product example*).

Let $m = 2$, $n = 3$, and

$$D = \{(0, 0), (1, 0), (0, 2)\}. \quad (18.5)$$

Then D is not a Cartesian product, $\log 2 / \log 3$ is irrational, and every similarity g with $g(F_D) \subset F_D$ is an isometry. In particular there is no strictly contracting similarity self-embedding of F_D .

Proof. The coordinate projections of D are $\{0, 1\}$ and $\{0, 2\}$; their Cartesian product has four elements whereas D has three, so the carpet is non-product. If $\log 2 / \log 3$ were rational, positive integers p, q would satisfy $2^p = 3^q$, contrary to unique prime factorization. Thus the bases are multiplicatively independent. The planar self-embedding theorem stated above now gives the conclusion. A strictly contracting similarity has ratio $r < 1$ and is therefore not an isometry. $\square$

Proposition IV.8.47 (*product hypothesis is genuinely necessary*).

Take $m = 2$, $n = 3$, and the full digit set

$$D = \{0, 1\} \times \{0, 1, 2\}. \quad (18.6)$$

Then $F_D = [0, 1]^2$ and the non-isometric similarity

$$g(x, y) = (x/2, y/2) \quad (18.7)$$

satisfies $g(F_D) \subset F_D$. Thus the non-product hypothesis in the planar self-embedding theorem cannot simply be deleted.

Proof. The six basic rectangles tile the unit square, so the attractor is $[0, 1]^2$. Equation (18.7) maps it into $[0, 1/2]^2$. Its similarity ratio is $1/2$, so it is not an isometry. $\square$

3. The Algom-Hochman-Wu renormalization theorem

The first prerequisite is not just the existence of a large-dimensional tangent set. Renormalization needs an ergodic law, and that law must be approximable along one chain of consecutive cells of the original set. The compact mass-tree model of IV.1 lets us prove this stronger assertion without taking limits through discontinuous geometric restrictions.

Theorem IV.8.48 (*realizable CP law at the Assouad dimension*).

Let $E \subset \mathbb{R}^d$ be nonempty and compact, and put $s = \dim_A E$. Fix a dyadic grid. For an occupied cell D , let $H_D : D \rightarrow [0, 1]^d$ be its orientation-preserving normalization, and for a finite probability ν with $\nu(D) > 0$ put

$$\nu^D = (H_D)_*(\nu|_D/\nu(D)).$$

There is an ergodic dyadic CP distribution whose measure marginal P is supported on exact-dimensional measures of dimension s . Moreover, if d_w is any metric inducing weak convergence on $\mathcal{P}(\mathcal{P}([0, 1]^d))$, then for every $\varepsilon > 0$ and all sufficiently large integers N there are a finite probability ν supported on E and consecutive nested dyadic cells $D_1, \dots, D_N$, with $\nu(D_N) > 0$, such that

$$d_w \left(\frac{1}{N} \sum_{j=1}^N \delta_{\nu^{D_j}}, P \right) < \varepsilon. \quad (18.4ak)$$

For each such N , the initial level of D_1 can be chosen arbitrarily large. The unit-cube CP convention is the affine conjugate of the $B_1 = [-1, 1]^d$ convention in IV.1; the outer-face final-cell convention is retained in both.

Proof. *Constructing an invariant law of maximal entropy.* The dyadic characterization of Assouad dimension in Volume I.5, Lemma L-I.5.01, has two consequences. First, if $s > 0$, we can choose $\varepsilon_k \downarrow 0$, integers $n_k \rightarrow \infty$, and occupied parent cells Q_k such that the number M_k of occupied depth- n_k descendants satisfies

$$M_k \geq 2^{n_k(s-\varepsilon_k)}. \quad (18.4al)$$

The levels of Q_k have a common lower bound: the characterization only needs parents of side length at most $\text{diam } E$. Failure of an upper estimate with exponent $s - \varepsilon_k$ gives arbitrarily long depth gaps, since the number of descendants at a bounded depth is bounded by 2^{dn} . If $s = 0$, choose any occupied parents and any $n_k \rightarrow \infty$. Second, for every $u > s$ there is $C_u < \infty$ such that, below any of these parents or any of their descendants,

$$\#\{\text{occupied depth-}q \text{ descendants}\} \leq C_u 2^{uq} \quad (q \geq 1). \quad (18.4am)$$

For a singleton E these statements hold directly, with $M_k = 1$ and $C_u = 1$.

Choose one point of E in each of the M_k descendants and give these points equal mass, obtaining a finite probability ν_k . Encode $(H_{Q_k})_* \nu_k$ by its canonical dyadic addresses in the mass-tree space $\mathcal{T}$ of IV.1, using $A = \{0, 1\}^d$. Write its tree as p_k and its address law as m_{p_k} . At zero-mass nodes choose the deterministic all-zero continuation; these choices do not change m_{p_k} . For $\Omega_k \sim m_{p_k}$ set

$$\widehat{Q}_k = \frac{1}{n_k} \sum_{j=0}^{n_k-1} \text{Law}(Z^j(p_k, \Omega_k)). \quad (18.4an)$$

Every term is symbolically adapted. Compactness, the weak closure of adaptedness in Lemma IV.1.56, and the endpoint telescoping in Lemma IV.1.57 give a subsequential limit $\widehat{Q}$ that is adapted and Z -invariant. The telescoping estimate is at most $2\|f\|_\infty/n_k$, so it applies to these varying starting trees as well as to a fixed tree.

Let $h(p) = H(p_\emptyset)$, the Shannon entropy of the root probability vector, with natural logarithms and $0 \log 0 = 0$. It is continuous and bounded. The entropy chain rule gives

$$\int h d\widehat{Q}_k = \frac{H(\Omega_{k,1}, \dots, \Omega_{k,n_k})}{n_k} = \frac{\log M_k}{n_k}.$$

Consequently

$$\int h d\widehat{Q} \geq s \log 2. \quad (18.4ao)$$

All sampled subtrees in (18.4an) satisfy (18.4am), with *positive-mass words* in place of occupied cells. The deterministic completions at zero-mass nodes satisfy it as well. For fixed q , the property that at most a fixed number of length- q words have positive mass is closed in $\mathcal{T}$: each word mass is continuous, and any finite family of positive word masses stays positive in a sufficiently small neighborhood. Thus $\widehat{Q}$ satisfies (18.4am) simultaneously for every q and every rational $u > s$.

Passing to an ergodic component without losing dimension. The invariant-set saturation argument of Lemma IV.1.66 also works in the symbolic model. Here are the details needed for that use. For a Z -invariant set U , reveal the first n digits of $\Omega \sim m_p$. Invariance identifies its conditional probability with $h_U(p^{\Omega_1 \dots \Omega_n})$, where $h_U(p) = m_p\{\omega : (p, \omega) \in U\}$. This bounded martingale converges to $1_U(p, \Omega)$. Under a stationary adapted law its second moment is constant in n , so every martingale increment has zero L^2 norm. It follows that $1_U(p, \omega) = h_U(p)$ almost everywhere. Conditioning on the invariant sigma-algebra therefore changes only the tree marginal, not the kernel m_p . The standard ergodic decomposition $\widehat{Q} = \int \widehat{Q}_\zeta d\tau(\zeta)$ consequently has adapted invariant ergodic components.

For almost every component, (18.4am) bounds its conditional depth- q entropy. Adaptedness, stationarity and the entropy chain rule yield

$$q \int h d\hat{Q}_\zeta = \int H(\omega_1, \dots, \omega_q \mid p) d\hat{Q}_\zeta(p, \omega) \leq \log C_u + uq \log 2.$$

Let $q \rightarrow \infty$ and then let rational $u \downarrow s$. Each component has $\int h d\hat{Q}_\zeta \leq s \log 2$. Together with (18.4ao), this implies equality for almost every component. Choose such a component $\hat{Q}_*$ whose support is contained in $\text{supp } \hat{Q}$. This last property holds for almost every component, since the closed set $\text{supp } \hat{Q}$ has full $\hat{Q}$ -measure.

By Lemmas IV.1.60–IV.1.62, the geometric push-forward of $\hat{Q}_*$ is an ergodic CP distribution. Canonical addresses give its root-cell information mean as $\int h d\hat{Q}_* = s \log 2$. The exact-dimensionality theorem IV.1.74 therefore gives dimension s for its measure marginal P .

Realizing the component along one chain. Let $G(p) = \pi_* m_p$. Both $p \mapsto m_p$ and the coding π are continuous, so G is continuous into $\mathcal{P}([0, 1]^d)$. Choose a $\hat{Q}_*$ -generic pair (p, ω) in $\text{supp } \hat{Q}$ such that every finite prefix of ω has positive m_p -mass. Adaptedness ensures this positivity almost surely. Birkhoff's theorem, applied to a countable convergence-determining family, gives

$$\frac{1}{N} \sum_{j=1}^N \delta_{G(\text{pr}_{\mathcal{J}} Z^j(p, \omega))} \longrightarrow P. \quad (18.4ap)$$

Fix N large enough that the distance in (18.4ap) is less than $\varepsilon/2$. Continuity of the finite list $G \circ \text{pr}_{\mathcal{J}} \circ Z^j$, $1 \leq j \leq N$, gives a neighborhood V of (p, ω) on which the empirical law in (18.4ap) changes by less than $\varepsilon/2$. Shrink V to require agreement of the first N digits with ω and positivity of their cylinder masses.

For any fixed integer L , delete the first L terms from (18.4an), without renormalizing. The resulting finite subprobabilities still converge to $\hat{Q}$, because their mass discrepancy is at most L/n_k . Since $(p, \omega) \in \text{supp } \hat{Q}$, $\hat{Q}(V) > 0$. The Portmanteau inequality then shows that, for some sufficiently large k , one sampled state $Z^j(p_k, \Omega_k)$ lies in V with $j \geq L$. Following its next N digits gives consecutive cells $D_1, \dots, D_N$ of the original grid, below Q_k , with $\nu_k(D_N) > 0$. The normalized restrictions $\nu_k^{D_i}$ are exactly the geometric measures along this finite orbit, so (18.4ak) follows with $\nu = \nu_k$. The initial level is at least $\text{level}(Q_k) + L + 1$; the common lower bound on parent levels and the freedom to increase L prove the final assertion. $\square$

This is the compact-set form of the realizability theorem used in Algom–Hochman–Wu, Theorem 2.5. Its proof above uses only the local CP construction and the Assouad covering calculus, not the exceptional-direction theorem or the affine renormalization argument. Those are separate prerequisites of the source result below.

4. Preparing the Assouad projection input

The required projection theorem concerns Assouad dimension, not Hausdorff dimension. Its useful measure class is also weaker than Ahlfors–David regularity: a uniform lower

mass bound is not assumed. The following preparation isolates the precise substitute, including the estimate which lets us discard low-mass balls. It follows the setup of Orponen, Sections 1.2.1 and 3.1 [16]. These preparations do not yet prove his exceptional-direction theorem.

Definition IV.8.49 (*quasiregularity and homogeneous blow-up*).

In this subsection let $\mathbb{B}(x, r) = \{y \in \mathbb{R}^2 : |y - x| < r\}$ be an open Euclidean ball, and let $N_E(A, r)$ be the least number of such radius- r balls covering a bounded set A . For $0 \leq s \leq 2$, a nonzero Radon measure μ on $\mathbb{R}^2$ is *s-quasiregular* if

$$\mu(\mathbb{B}(x, r)) \leq r^s \quad (x \in \mathbb{R}^2, r > 0)$$

and, for every $\varepsilon > 0$, some $C_\varepsilon < \infty$ satisfies

$$N_E(\text{supp } \mu \cap \mathbb{B}(x, R), r) \leq C_\varepsilon (R/r)^{s+\varepsilon} \quad (x \in \mathbb{R}^2, 0 < r < R < \infty).$$

Here $\text{supp } \mu$ is the set of points whose every open neighborhood has positive μ -mass. For $B = \mathbb{B}(x_0, r_0)$ set

$$T_B(y) = \frac{y - x_0}{r_0}, \quad \mu^{B,s} = r_0^{-s} (T_B)_* \mu.$$

This is a homogeneous mass rescaling of the entire measure, not its normalized restriction to B . In particular, $\mu^{B,s}(\mathbb{B}(0, 1))$ need not equal one.

Lemma IV.8.50 (*quasiregularity survives blow-up*).

If μ is *s-quasiregular*, then $\mu^{B,s}$ is *s-quasiregular* with the same covering constants. For another ball $B' = \mathbb{B}(x_1, r_1)$ one also has

$$(\mu^{B,s})^{B',s} = \mu^{B'',s}, \quad B'' = \mathbb{B}(x_0 + r_0 x_1, r_0 r_1).$$

Proof. For all x and $r > 0$,

$$\mu^{B,s}(\mathbb{B}(x, r)) = r_0^{-s} \mu(\mathbb{B}(x_0 + r_0 x, r_0 r)) \leq r^s.$$

The support is $T_B(\text{supp } \mu)$, so

$$\begin{aligned} N_E(\text{supp } \mu^{B,s} \cap \mathbb{B}(x, R), r) &= N_E(\text{supp } \mu \cap \mathbb{B}(x_0 + r_0 x, r_0 R), r_0 r) \\ &\leq C_\varepsilon (R/r)^{s+\varepsilon}. \end{aligned}$$

Finally $T_{B'} T_B = T_{B''}$ and $r_0^{-s} r_1^{-s} = (r_0 r_1)^{-s}$. Applying these identities to the push-forward gives the asserted chain rule. $\square$

Suspicious source formula IV.8.51 (A sign in the source's blow-up calculation). In Orponen, arXiv:1902.04993v3, printed p. 7, the proof of Lemma 3.7 displays the stronger bound

$$N_E(\text{supp } \mu^{B,s} \cap \mathbb{B}(x, R), r) \leq C_\varepsilon (R/r)^{s-\varepsilon} \quad (x \in \mathbb{R}^2, \varepsilon > 0, 0 < r < R < \infty).$$

Here μ is s -quasiregular and $B = \mathbb{B}(x_0, r_0)$; the standing setup allows $0 < s \leq 2$ and nonzero μ supported in $\mathbb{B}(0, 1)$. The constants are to be independent of x, r, R . This displayed proof formula is false with that minus sign. The *statement* of Lemma 3.7 and its hypothesis (3.3) both have $s + \varepsilon$, which is the correct bound proved in Lemma IV.8.50. Thus the discrepancy is a local typographical sign, not a counterexample to Lemma 3.7 or the projection theorem. An independent full read of v3's Sections 1–4 and references found no standing assumption that supplies the minus exponent; no claim about a different published PDF is made here.

Example IV.8.52 (*testing every hypothesis of the sign discrepancy*).

Let $\mathcal{L}^2$ be planar Lebesgue measure and set

$$\mu = \pi^{-1} \mathcal{L}^2|_{\overline{\mathbb{B}(0, 1/2)}}, \quad s = 2, \quad B = \mathbb{B}(0, 1).$$

This measure satisfies all the preceding standing hypotheses, but not the displayed $s - \varepsilon$ bound.

Proof. The measure is a nonzero finite Radon measure with compact support strictly inside $\mathbb{B}(0, 1)$, and $\mu(\mathbb{B}(x, r)) \leq \pi^{-1} \pi r^2 = r^2$. An elementary square-grid cover of any radius- R ball gives

$$N_E(\text{supp } \mu \cap \mathbb{B}(x, R), r) \leq C(R/r)^2 \leq C(R/r)^{2+\varepsilon} \quad (0 < r < R)$$

with an absolute C , so μ is 2-quasiregular. Since B has center zero and radius one, $\mu^{B,2} = \mu$. Put $x = 0$, $R = 1/4$ and $r = R2^{-n}$. Any radius- r ball cover of $\mathbb{B}(0, R)$ contains at least $(R/r)^2 = 2^{2n}$ balls by planar area. The false bound would therefore give

$$2^{2n} \leq C_\varepsilon 2^{n(2-\varepsilon)}, \quad \text{hence} \quad 2^{n\varepsilon} \leq C_\varepsilon \quad (n \geq 1),$$

which is impossible for a fixed $\varepsilon > 0$. The invalid inference in the printed calculation is exactly the substitution of $s - \varepsilon$ for $s + \varepsilon$ when invoking (3.3). The homogeneous change of coordinates itself is valid. $\square$

Lemma IV.8.53 (*small mass occurs on a small set of centers*).

Let μ be s -quasiregular, $E \subset \mathbb{B}(0, 1)$ be Borel, $\varepsilon > 0$, and $0 < r \leq R \leq 1$. Then the Borel set

$$A = \{x \in E : \mu(E \cap \mathbb{B}(x, R)) \leq r^\varepsilon R^s\}$$

satisfies

$$\mu(A) \leq 8^{s+\varepsilon/2} C_{\varepsilon/2} r^{\varepsilon/2}.$$

Proof. The function $x \mapsto \mu(E \cap \mathbb{B}(x, R))$ is Borel by integration of the Borel indicator $1_E(y)1_{\{|x-y|<R\}}$, so A is Borel. Discard its part outside $\text{supp } \mu$, which has zero mass. Choose a maximal finite $R/2$ -separated subset $\{x_1, \dots, x_m\}$ of the remaining set. Such a choice exists: a cover of $\text{supp } \mu \cap \mathbb{B}(0, 1)$ by radius- $R/8$ balls contains at most one selected

point in each ball. Thus the possible cardinalities are bounded, and a subset of largest cardinality is maximal. In particular,

$$m \leq C_{\varepsilon/2}(8/R)^{s+\varepsilon/2}, \quad A \cap \text{supp } \mu \subset \bigcup_{j=1}^m \mathbb{B}(x_j, R).$$

Every selected center belongs to A . Since also $A \subset E$,

$$\begin{aligned} \mu(A) &\leq \sum_{j=1}^m \mu(E \cap \mathbb{B}(x_j, R)) \\ &\leq mr^\varepsilon R^s \\ &\leq 8^{s+\varepsilon/2} C_{\varepsilon/2} r^\varepsilon R^{-\varepsilon/2} \\ &\leq 8^{s+\varepsilon/2} C_{\varepsilon/2} r^{\varepsilon/2}. \end{aligned}$$

The last inequality uses $r \leq R$. □

This is not a pointwise lower Frostman bound. It bounds the mass of the bad centers at each prescribed pair of scales. Subsequent scale selections must retain this error; they cannot treat every ball as regular.

Proposition IV.8.54 (*reduction to a quasiregular tangent support*).

Let $F \subset \mathbb{R}^2$ be nonempty and compact, and $s = \dim_A F$. There is an s -quasiregular measure μ with compact support inside $\mathbb{B}(0, 1)$ such that

$$\dim_H \text{supp } \mu = \dim_A \text{supp } \mu = s, \quad \dim_A \pi_e(\text{supp } \mu) \leq \dim_A \pi_e F \quad (e \in S^1),$$

where $S^1 = \{e \in \mathbb{R}^2 : |e| = 1\}$ and $\pi_e(x) = x \cdot e$. Consequently

$$\{e : \dim_A \pi_e F < \min(s, 1)\} \subset \{e : \dim_A \pi_e(\text{supp } \mu) < \min(s, 1)\}.$$

Proof. For $s = 0$ take $\mu = \delta_0$; the two exceptional sets are empty. Suppose $s > 0$. The fully proved tangent realization theorem of Volume I, Chapter 5, gives homotheties $T_k(x) = a_k x + b_k$, $a_k \rightarrow \infty$, and a compact window W_0 such that

$$T_k F \cap W_0 \longrightarrow E, \quad \mathcal{H}^s(E) > 0, \quad \dim_H E = \dim_A E = s.$$

The convergence is in the Hausdorff metric. Fix e . Since π_e is Lipschitz, $\pi_e(T_k F \cap W_0) \rightarrow \pi_e E$ in that metric. Each approximating set is a subset of $a_k \pi_e F + \pi_e b_k$.

We spell out the covering consequence, since a projected windowed tangent need not itself be a full windowed tangent. For $q > \dim_A \pi_e F$, the local covering bound for $\pi_e F$ with exponent q supplies a constant C_q . After the homothety, the same bound holds on every fixed outer scale for all sufficiently large k , because $a_k \rightarrow \infty$. Given $0 < r < R$ and a finite $r/2$ -separated subset of $\pi_e E \cap (u - R, u + R)$, approximate its points in $\pi_e(T_k F \cap W_0)$ to within $r/16$. For large k these approximants are $r/4$ -separated and lie in $(u - 2R, u + 2R)$. A radius- $r/16$ cover therefore bounds their number by $C_q(32R/r)^q$. A maximal such separated subset covers the limiting set by radius- r balls. Hence $\dim_A \pi_e E \leq q$; let $q \downarrow \dim_A \pi_e F$.

Apply a homothety to place E strictly inside $\mathbb{B}(0, 1)$; this changes none of its dimensions or the projection inequalities. The compact Frostman lemma of Volume I, Chapter 3, supplies a probability λ on this copy of E and $C < \infty$ such that $\lambda(\mathbb{B}(x, r)) \leq Cr^s$ for every x, r . Set $\mu = C^{-1}\lambda$ (enlarging C to be positive if necessary). Its support has Hausdorff dimension at least s by the mass-distribution principle, and is a subset of E , whose Assouad dimension is s . Thus both support dimensions equal s . The Assouad covering bounds on this compact support give the defining estimate with exponent $s + \varepsilon$ for every $\varepsilon > 0$. The local estimate extends to all outer radii by boundedness: for large R and small r use one fixed outer ball containing the support, while for r above that fixed radius a bounded number of balls suffices. Therefore μ is s -quasiregular. Support inclusion and the preceding projection inequalities prove the claim about directions. $\square$

5. The finite-window inverse-theorem interface

The eventual projection argument constructs a product-like measure, not a product identity for the original measure. Its vertical factor occupies a bounded real window, whereas the inverse theorem of Volume II is stated for grid probabilities in $[0, 1)$. Restricting that factor may lose a small power of the scale. The next two lemmas retain this loss explicitly.

Lemma IV.8.55 (*normalizing one window with a controlled loss*).

Let μ, ν be finitely supported probabilities on $\mathbb{R}$. Write $\lambda\{x\} = \lambda(\{x\})$, and define the discrete norm and convolution by

$$\|\lambda\|_2^2 = \sum_x \lambda\{x\}^2, \quad (\mu * \nu)\{z\} = \sum_{x+y=z} \mu\{x\}\nu\{y\}.$$

Suppose ν is supported in at most $J \geq 1$ distinct half-open unit intervals $[k, k+1)$, $k \in \mathbb{Z}$. If $0 < a \leq 1$ and

$$\|\mu * \nu\|_2 \geq a\|\mu\|_2,$$

there is one such interval $I = [k, k+1)$, with mass $p = \nu(I) \geq a/(2J)$, whose normalized restriction $\eta = p^{-1}\nu|_I$ satisfies

$$\|\mu * \eta\|_2 \geq \frac{a}{2}\|\mu\|_2, \quad \max_x \eta\{x\} \leq \frac{2J}{a} \max_x \nu\{x\}.$$

Translating η by $-k$ preserves these two conclusions.

Proof. Write $\nu = \sum_j p_j \eta_j$ over its positive-mass unit pieces. Each η_j is a probability, and the triangle inequality applied to a convex combination of translates gives $\|\mu * \eta_j\|_2 \leq \|\mu\|_2$. The sum ν_{small} of pieces with $p_j < a/(2J)$ has mass less than $a/2$, so

$$\|\mu * \nu_{\text{small}}\|_2 \leq \nu_{\text{small}}(\mathbb{R})\|\mu\|_2 \leq \frac{a}{2}\|\mu\|_2.$$

Consequently the remaining pieces obey

$$\frac{a}{2}\|\mu\|_2 \leq \left\| \mu * \sum_{j: p_j \geq a/(2J)} p_j \eta_j \right\|_2 \leq \sum_{j: p_j \geq a/(2J)} p_j \|\mu * \eta_j\|_2.$$

Their total weight is at most one. At least one of their convolution norms is therefore at least $(a/2)\|\mu\|_2$. For this piece, division by $p_j \geq a/(2J)$ gives the atom bound. Translation only relabels the convolution output and the atoms, so neither norm nor maximal atom changes. $\square$

Lemma IV.8.56 (*flattening from a subunit covering exponent*).

Fix $0 \leq t < 1$, $C \geq 1$, $\eta_0 > 0$, $C_v \geq 1$ and an integer $J \geq 1$. There are $\chi > 0$ and an integer block length $D \geq 1$ with the following property. For all sufficiently large integers ℓ , put $m = D\ell$. Let μ be a probability on $2^{-m}\mathbb{Z} \cap [0, 1)$, with support H , and assume

$$N_1(H \cap (x - R, x + R), r) \leq C(R/r)^t \quad (x \in \mathbb{R}, 2^{-m} \leq r < R \leq 1), \quad (18.4ap)$$

where $N_1(A, r)$ is the least number of open intervals of radius r covering A . Let ν be a probability on $2^{-m}\mathbb{Z}$, supported in at most J half-open integer unit intervals, and assume

$$\max_x \nu\{x\} \leq C_v 2^{-\eta_0 m}.$$

Then

$$\|\mu * \nu\|_2 < 2^{-\chi m} \|\mu\|_2. \quad (18.4aq)$$

All norms here are real-line discrete norms. The large- ℓ threshold may depend on the displayed fixed parameters.

Proof. Choose

$$0 < b < \min\{(1-t)/4, \eta_0/4, 1/4\}$$

and an integer D_0 so large that

$$8C 2^{tD'} < 2^{(1-b)D'} \quad (D' \geq D_0). \quad (18.4ar)$$

This is possible because $1 - b - t > 0$. Apply the inverse theorem for near-nonflattening from Volume II, Chapter 11, with $q = 2$, error parameter b , and minimum block D_0 . It supplies $\varepsilon_{\text{inv}} > 0$ and $D \geq D_0$. Fix

$$0 < \chi < \min\{\varepsilon_{\text{inv}}/2, \eta_0/4\}.$$

Suppose, contrary to (18.4aq), that $\|\mu * \nu\|_2 \geq 2^{-\chi m} \|\mu\|_2$. Lemma IV.8.55 gives a normalized unit piece, translated to a probability η on $2^{-m}\mathbb{Z} \cap [0, 1)$, with

$$\|\mu * \eta\|_2 \geq 2^{-\chi m - 1} \|\mu\|_2, \quad \max_x \eta\{x\} \leq 2JC_v 2^{-(\eta_0 - \chi)m}. \quad (18.4as)$$

For large m , the norm lower bound is at least $2^{-\varepsilon_{\text{inv}} m} \|\mu\|_2$. Reducing convolution modulo one can only increase its discrete norm: masses from different integer translates are non-negative, and $(u + v)^2 \geq u^2 + v^2$. Thus the circle formulation of the inverse theorem applies.

Let A and the active block set S be its horizontal output. A preliminary circle translation of H is a union of at most two ordinary translates of subsets of H . It therefore satisfies (18.4ap) with $2C$ in place of C , uniformly in that translation. For any occupied level- jD

parent interval of A , use its midpoint and $R = 2^{-jD}$ in this covering bound; take $r = 2^{-(j+1)D} \geq 2^{-m}$. An open radius- r interval meets at most four dyadic child intervals of length r . Consequently its occupied child count obeys

$$R'_j \leq 8C2^{tD} < 2^{(1-b)D}. \quad (18.4at)$$

This applies also at the root and the last block. Pruning to A cannot increase the child count. Hence no block can be active: $S = \emptyset$.

On the other hand, since η is a probability,

$$\|\eta\|_2^2 \leq \max_x \eta\{x\}, \quad \log_2 \|\eta\|_2^{-2} \geq (\eta_0 - \chi)m - \log_2(2JC_v).$$

The inverse theorem's scale-count lower bound gives

$$D|S| \geq (\eta_0 - \chi - b)m - \log_2(2JC_v) > 0 \quad (18.4au)$$

for all sufficiently large m , since $\chi + b < \eta_0/2$. This contradicts $S = \emptyset$ and proves the lemma. $\square$

The contradiction uses only an almost-full horizontal block and a positive number of active blocks. It does not require almost-full branching of the vertical tree. What remains for the projection theorem is to construct its product comparison and diffuse vertical factor on one common stable blow-up, with the losses small enough for this lemma.

6. Selecting one blow-up for many directions

We now carry out the simultaneous selection which precedes that product comparison. Minimizing the number of occupied tubes alone would be unstable under removal of a small amount of mass. Instead, we keep tubes of comparable positive mass, and allow a change of ball only when their exponent drops by a fixed amount. Fubini's theorem makes the new ball common to a positive-mass set of directions. The finite decrease of the exponent then forces the procedure to stop.

Definition IV.8.57 (*dyadic tubes in the unit ball*).

Write $B_0 = \mathbb{B}(0, 1)$. For $e \in S^1$, a dyadic number $\rho = 2^{-j}$, $j \geq 1$, and $k \in \mathbb{Z}$, put

$$T_{\rho,e,k} = B_0 \cap \{y : k\rho \leq \pi_e(y) < (k+1)\rho\}.$$

These are the *dyadic ρ -tubes in direction e* . Their nonempty members partition B_0 and number at most $4/\rho$. They are clipped to B_0 even when the measure under consideration has support outside that ball.

Lemma IV.8.58 (*a tube family with comparable masses*).

Let λ be a Radon measure with $\lambda(B_0) \leq 1$, let $E \subset B_0$ be Borel, and suppose $\rho^C \leq M := \lambda(E) \leq 1$, where $C \geq 0$ and $\rho = 2^{-j}$, $j \geq 1$. For each $e \in S^1$ there are a nonempty family $\mathcal{T}$ of dyadic ρ -tubes and a number $v \geq 0$ such that, on writing

$$L = \lceil (C + 1) \log_2(1/\rho) \rceil + 6,$$

one has

$$\begin{aligned} \rho^v &\leq \lambda(E \cap T) \leq 2\rho^v \quad (T \in \mathcal{T}), \\ \lambda\left(E \cap \bigcup \mathcal{T}\right) &\geq \frac{M}{2L}, \quad \frac{M}{4L} \leq \rho^v |\mathcal{T}| \leq M, \\ v &\leq C + 1 + \frac{6}{\log_2(1/\rho)}. \end{aligned}$$

The family and exponent can be selected as Borel functions of e : this means that each tube index's membership and the exponent are Borel functions.

Proof. There are at most $4/\rho$ tube masses. Those smaller than $\rho^{C+1}/16$ have total at most $\rho^C/4 \leq M/4$. Partition the remaining masses into the disjoint bins $[2^{-a}, 2^{-a+1})$, $a = 0, 1, \dots$; the mass 1 belongs to the bin with $a = 0$. A bin meeting the remaining masses has

$$2^{-a} \geq \frac{\rho^{C+1}}{32}, \quad 0 \leq a \leq (C + 1) \log_2(1/\rho) + 5.$$

Thus there are at most L such bins. One contains total mass at least $3M/(4L)$, and in particular $M/(2L)$. Take its tubes and put $v = \log_\rho(2^{-a})$. The mass comparison proves the total-mass and cardinality bounds, and the displayed bound on a proves the bound on v .

For a fixed index k , the map

$$e \mapsto \int_E \mathbf{1}_{[k\rho, (k+1)\rho)}(y \cdot e) d\lambda(y)$$

is Borel by measurability of parameter integrals. Choose the first bin attaining the stated mass bound, in increasing order of a . There are only finitely many eligible bins and indices, so this selection is Borel. $\square$

The harmless $C + 1$ in this lemma is important: the number of bins need not vanish when $C = 0$. We also do not require v to be at most the quasiregularity exponent. Small multiplicative mass constants can make a tube exponent slightly larger. The bounded range $[0, 2]$ below suffices for the stopping argument.

Suspicious source formula IV.8.59 (An exact exponent range at the preparatory stage). In Orponen, arXiv:1902.04993v3, Section 3.3.2, printed pp. 10–11, the assertion that the initial objects satisfy (P4) and (P6) has the following literal consequence. Let $0 < s \leq 2$, let μ be nonzero s -quasiregular, supported inside B_0 , and let σ be an arbitrary Borel probability on S^1 . Fix the source's finite dyadic scale set $Q \subset [0, 1]$ containing 0, 1 and $0 < \alpha < 1$. Take $\tau > 0$ sufficiently small after α, Q , in particular $\tau < \alpha$, and take a sufficiently small dyadic δ satisfying the arithmetic alignment requirements of Section 3.3.1. For these

preparatory choices, there should be a set S_0 with $\sigma(S_0) > 0$, a representative $a_0 \in [0, s]$ and, for every $e \in S_0$, an exponent $v_e \in [0, s]$ and a dyadic δ -tube family $\mathcal{T}_{0,e}$ with

$$|v_e - a_0| \leq \alpha/100, \quad |\mathcal{T}_{0,e}| \geq \delta^{-v_e+\tau}, \\ \delta^{v_e} \leq \mu(T) \leq 2\delta^{v_e} \quad (T \in \mathcal{T}_{0,e}).$$

The exact range $v_e \leq s$ need not hold under these arbitrary-direction hypotheses. The example below fails for every $\tau > 0$ and every $0 < \delta < 1$, so no further smallness or alignment choice in that setup repairs this range. The later exceptional-direction counterassumption in Section 4.1 is not a premise of this preparatory claim. The next example refutes this range, not the eventual projection theorem.

Example IV.8.60 (*a small-density segment*).

Set $s = 1$ and define the Radon measure

$$\mu = \frac{1}{4} (t \mapsto (t, 0))_* (\mathcal{L}^1|_{[-1/2, 1/2]}), \quad \sigma = \delta_{(1,0)},$$

where $\mathcal{L}^1$ denotes Lebesgue measure. These satisfy every standing hypothesis of the arbitrary-direction preparatory claim, but no initial family in IV.8.59 exists.

Proof. The support is the compact horizontal segment strictly inside B_0 , and $\mu(B_0) = 1/4$. An open ball intersects its supporting line in length at most $2r$, so $\mu(\mathbb{B}(x, r)) \leq r/2 \leq r$. For $0 < r < R$, the part of the segment in $\mathbb{B}(x, R)$ has length at most $2R$ and is covered by at most $2R/r + 2 \leq 4R/r$ open radius- r balls. This is at most $4(R/r)^{1+\varepsilon}$ for every $\varepsilon > 0$. Thus μ is 1-quasiregular. Its mass is positive and at most one, and σ is an allowed Borel probability.

Positive retained σ -mass forces $e = (1, 0) \in S_0$. Every tube in this direction has $\mu(T) \leq \delta/4$, whereas

$$\delta^{v_e} \geq \delta \quad (0 \leq v_e \leq 1).$$

The lower cardinality bound is strictly positive, so the selected family must contain a tube. Its lower mass bound is therefore impossible. This holds for every $0 < \delta < 1$, including all dyadic alignment and smallness choices.

This example does not satisfy the later exceptional-direction setup: its chosen projection is an interval of Assouad dimension one, not less than a fixed $D < 1$, and its atomic direction law does not have a positive Frostman exponent. The failed inference in the preparatory initialization is the passage from comparable tube weights to an exact bound $v_e \leq s$, without accounting for a multiplicative mass constant. In this example that constant makes $\log_\delta(\delta/4) = 1 + \log(1/4)/\log \delta > 1$. Lemma IV.8.62 instead uses a bounded range with slack and does not depend on the rejected source range. $\square$

Definition IV.8.61 (*a robust bad center*).

Let ν be s -quasiregular, let $0 < \Delta < 1$, $0 \leq p < q \leq 1$, and assume $\rho = \Delta^{q-p}$ is dyadic. Fix a potential $\beta \in [0, 2]$, a decrement $\alpha > 0$, and a loss $\kappa > 0$. A center $x \in B_0$ is *bad* for (e, p, q) with these parameters if there exist $0 \leq v \leq \beta - \alpha$ and a nonempty family $\mathcal{U}$ of dyadic ρ -tubes for which

$$\begin{aligned} |\mathcal{U}| &\geq \rho^{-v+\kappa}, \\ \rho^v &\leq \nu_{\mathbb{B}(x, \Delta^p), s}(T) \leq 2\rho^v \quad (T \in \mathcal{U}). \end{aligned} \quad (18.4av)$$

Denote this set of centers by $\text{Bad}_\nu(e; p, q \mid \Delta, \beta, \alpha, \kappa)$. If $\beta < \alpha$, it is empty. The measure in this definition is the entire homogeneous blow-up, not a probability conditioned on $\mathbb{B}(x, \Delta^p)$.

Lemma IV.8.62 (*a common stable blow-up*).

Fix $0 < \alpha < 1/10$ and a finite set $Q \subset [0, 1]$ of dyadic rationals containing 0, 1. Choose $L_Q \geq 0$ so that $Q \subset 2^{-L_Q}\mathbb{Z}$, and write

$$h = \min_{\substack{p, q \in Q \\ p < q}} (q - p), \quad K = \lceil 4/\alpha \rceil, \quad A_F = \sum_{j=1}^K F(j),$$

where $F(0), \dots, F(K+1)$ is any fixed increasing sequence of positive numbers with $F(0) = 1$. Let $\tau > 0$, let μ be s -quasiregular with $\mu(B_0) > 0$, and let σ be a Borel probability on S^1 . For every sufficiently large integer M , set

$$\delta = 2^{-M2^{L_Q}(K+1)}.$$

There exist an index $0 \leq n < K$, a ball $B = \mathbb{B}(z, r)$ with $0 < r \leq 1$, a number $a \in [0, 2]$, a scale $\Delta = \delta^\theta$ with $h^K \leq \theta \leq 1$, and a Borel set $S \subset S^1$ such that $\nu := \mu^{B, s}$ is s -quasiregular and

$$\sigma(S) \geq c_{\alpha, Q} \delta^{A_F \tau}, \quad \nu(B_0) \geq \Delta^{F(n)\tau}. \quad (18.4aw)$$

For every $e \in S$ there are $v_e \in [0, 2]$ and a nonempty dyadic Δ -tube family $\mathcal{T}_e$ satisfying

$$\begin{aligned} |v_e - a| &\leq \alpha/100, \quad |\mathcal{T}_e| \geq \Delta^{-v_e + F(n)\tau}, \\ \Delta^{v_e} &\leq \nu(T) \leq 2\Delta^{v_e} \quad (T \in \mathcal{T}_e). \end{aligned} \quad (18.4ax)$$

Moreover, for every ordered pair $p < q$ in Q ,

$$\nu(\text{Bad}_\nu(e; p, q \mid \Delta, a, \alpha, F(n+1)\tau)) < \Delta^{F(n+1)\tau} \quad (e \in S). \quad (18.4ay)$$

All scales Δ^p and Δ^{q-p} appearing here are dyadic. The small-scale threshold may depend on α, Q, F, τ and $\mu(B_0)$; the constant $c_{\alpha, Q} > 0$ depends only on α, Q .

Proof. Measurability of the test. For fixed Δ, p, q , the mass of each tube in the blow-up at x is a Borel function of (x, e) :

$$\Delta^{-ps} \int \mathbf{1}_{B_0}\left(\frac{y-x}{\Delta^p}\right) \mathbf{1}_{[k\rho, (k+1)\rho)}\left(\frac{(y-x) \cdot e}{\Delta^p}\right) d\nu(y). \quad (18.4az)$$

Only $y \in \mathbb{B}(0, 2)$ can contribute, a set of finite ν -mass. There are finitely many possible tube indices and finitely many subfamilies. For one nonempty family of size J with masses $m_1, \dots, m_J$, an exponent in (18.4av) exists exactly when $\beta \geq \alpha$ and the interval

$$\left[\max \left\{ \rho^{\beta-\alpha}, \frac{\rho^\kappa}{J}, \frac{\max_i m_i}{2} \right\}, \min \left\{ 1, \min_i m_i \right\} \right] \quad (18.4ba)$$

is nonempty. Indeed its members are precisely the admissible values of ρ^ν . This makes the bad relation Borel. Choose the first admissible family and the left endpoint of this interval; then the witnesses ν and the family are Borel. Thus the applications of Fubini and exponent pigeonholing below are justified.

Initialization. Put $\mu_0 = \mu$, $\delta_0 = \delta$ and $m_0 = \mu(B_0) \in (0, 1]$. For δ small enough, Lemma IV.8.58, with $E = B_0$ and $C = \log_\delta m_0 \in [0, 1]$, selects tube families with comparable masses δ^{v_e} . The construction of that lemma gives $\delta^{v_e} \geq m_0 \delta / 32 \geq \delta^2$, once $\delta \leq m_0 / 32$, so $0 \leq v_e \leq 2$. Its cardinality lower bound and $m_0 / (4[2 \log_2(1/\delta) + 6]) \geq \delta^\tau$ give

$$|\mathcal{T}_{0,e}| \geq \delta^{-v_e + \tau}.$$

Partition $[0, 2]$ into intervals of length at most $\alpha/100$, using their left endpoints as representatives. There are at most $J_\alpha = \lceil 200/\alpha \rceil + 1$ intervals. One Borel direction set S_0 has $\sigma(S_0) \geq J_\alpha^{-1}$ and a representative $a_0 \in [0, 2]$ with $|v_{0,e} - a_0| \leq \alpha/100$.

An induction step. Suppose $\mu_n, \delta_n, S_n, a_n, v_{n,e}, \mathcal{T}_{n,e}$ have been constructed with (18.4ax), with n, μ_n, δ_n, a_n in place of n, ν, Δ, a . Let

$$P_Q = |\{(p, q) \in Q^2 : p < q\}| \geq 1.$$

Call $e \in S_n$ a bad direction if, for at least one such pair, the set of bad centers has μ_n -mass at least $\delta_n^{F(n+1)\tau}$. If the bad directions have σ -mass less than $\sigma(S_n)/2$, stop.

Otherwise one pair $p < q$ has a Borel set of bad directions of mass at least $\sigma(S_n)/(2P_Q)$. For this pair Fubini gives

$$\begin{aligned} \int_{B_0} \sigma\{e \in S_n : x \in \text{Bad}_{\mu_n}(e; p, q \mid \delta_n, a_n, \alpha, F(n+1)\tau)\} d\mu_n(x) \\ \geq \frac{\sigma(S_n)}{2P_Q} \delta_n^{F(n+1)\tau}. \end{aligned} \quad (18.4bb)$$

Since $\mu_n(B_0) \leq 1$, there is a center $x_n \in B_0$ whose direction section has mass at least half this lower bound. On that section set

$$\mu_{n+1} = \mu_n^{\mathbb{B}(x_n, \delta_n^p), s}, \quad \delta_{n+1} = \delta_n^{q-p}.$$

The badness witnesses give the tube families and exponents at level $n+1$, satisfying $v_{n+1,e} \leq a_n - \alpha$ and the required cardinality and mass bounds with loss $F(n+1)\tau$. Pigeonhole these Borel exponents into the same intervals as before. Take the left endpoint a_{n+1} and the retained set S_{n+1} . Then

$$a_{n+1} \leq a_n - \alpha, \quad \sigma(S_{n+1}) \geq \frac{\sigma(S_n)}{4P_Q J_\alpha} \delta_n^{F(n+1)\tau}. \quad (18.4bc)$$

Lemma IV.8.50 preserves quasiregularity and writes μ_{n+1} as one homogeneous blow-up of the original μ , of radius at most one.

Termination and quantitative output. After n successful steps, $0 \leq a_n \leq 2 - n\alpha$. Thus the procedure must stop before $K = \lceil 4/\alpha \rceil$ steps. The scales have the form

$$\delta_n = \delta^{\theta_n}, \quad \theta_n = \prod_{j=0}^{n-1} (q_j - p_j) \in [h^n, 1].$$

Here (p_j, q_j) is the pair selected at step j . The denominator of θ_n divides 2^{nL_Q} ; our choice of δ therefore makes δ_n , δ_n^p and δ_n^{q-p} dyadic at every stage needed. Also $\delta_n \geq \delta$, and iterating (18.4bc) yields

$$\sigma(S_n) \geq J_\alpha^{-1} (4P_Q J_\alpha)^{-n} \delta^\tau \sum_{j=1}^n F(j).$$

At the stopping index remove all bad directions, losing less than half this mass. The remaining Borel set S satisfies (18.4ay) for every pair in Q , and (18.4aw) holds, for example with $c_{\alpha,Q} = (2J_\alpha)^{-1} (4P_Q J_\alpha)^{-K}$. For any $e \in S$, disjointness of its tubes gives

$$\mu_n(B_0) \geq \sum_{T \in \mathcal{T}_{n,e}} \mu_n(T) \geq \delta_n^{v_{n,e}} |\mathcal{T}_{n,e}| \geq \delta_n^{F(n)\tau}.$$

This proves the nonzero-mass assertion and all remaining claims, on setting $\nu = \mu_n$, $\Delta = \delta_n$ and $a = a_n$. $\square$

This is the robust simultaneous-selection step of Orponen, Section 3.3 [16], with explicit retained-direction losses. The sequence F can be fixed to grow rapidly before τ is chosen. Its finite sum then remains a fixed coefficient of τ , rather than an unrecorded loss. The conclusion does not yet give the nested tube refinement or the horizontal–vertical product comparison; those are separate parts of the projection proof.

7. The mass distribution inside a fine tube

Stability now has a geometric consequence. If many fine tubes had most of their mass in unusually heavy short rectangles, a ball meeting those rectangles would contain substantial mass in too few tubes. Its homogeneous blow-up would therefore be bad. The low-mass-center lemma supplies a ball with enough mass; it does not supply a pointwise lower regularity estimate.

Lemma IV.8.63 (*nonconcentration along the fine tubes*).

Let ν be s -quasiregular, $0 \leq s \leq 2$. Fix $e \in S^1$, $0 < q < 1$, $0 < \alpha < 1/10$, $f \geq 1$, and

$$g \geq \frac{4f + 10}{1 - q}, \quad 0 < \tau < \frac{\alpha}{2(4f + 9)}.$$

For a sufficiently small Δ , assume that Δ and Δ^q are dyadic. Suppose $a, v \in [0, 2]$, $|v - a| \leq \alpha/100$, and a family $\mathcal{T}$ of dyadic Δ -tubes satisfies

$$|\mathcal{T}| \geq \Delta^{-v+f\tau}, \quad \Delta^v \leq \nu(T) \leq 2\Delta^v \quad (T \in \mathcal{T}), \quad (18.4bd)$$

while

$$\nu(\text{Bad}_\nu(e; q, 1 \mid \Delta, a, \alpha, g\tau)) < \Delta^{g\tau}. \quad (18.4be)$$

There are a subfamily $\mathcal{T}_G \subset \mathcal{T}$ and a Borel set $G \subset \text{supp } \nu \cap \bigcup \mathcal{T}_G$ such that

$$\begin{aligned} |\mathcal{T}_G| &\geq \Delta^\tau |\mathcal{T}|, & \nu(G \cap T) &\geq \Delta^\tau \nu(T) \quad (T \in \mathcal{T}_G), \\ \nu(G) &\geq \Delta^{(f+2)\tau}. \end{aligned} \quad (18.4bf)$$

$$\frac{\nu(G \cap T \cap \mathbb{B}(x, \Delta^q))}{\nu(T)} \leq 8(\Delta^q)^{s-\nu-4\alpha/q} \quad (T \in \mathcal{T}_G, x \in \mathbb{R}^2). \quad (18.4bg)$$

The small-scale threshold is uniform over measures with the same quasiregular covering constants; it may depend on s, q, α, f, g, τ and the covering constant at exponent $s + 2(f + 2)\tau$.

Proof. Put $e^\perp = (-e_2, e_1)$ and use the orthogonal coordinates $(\pi_e, \pi_{e^\perp})$. Inside a fixed $T \in \mathcal{T}$, partition into rectangles of horizontal width Δ and vertical height Δ^q , intersected with B_0 . Apply Lemma IV.8.58 with $E = T$, direction $e^\perp$, $\rho = \Delta^q$ and $C = \nu/q$. It selects rectangles $\mathcal{R}_T$ with one dyadic mass parameter m_T , such that

$$m_T \leq \nu(R) \leq 2m_T \quad (R \in \mathcal{R}_T), \quad \nu\left(\bigcup \mathcal{R}_T\right) \geq \frac{\nu(T)}{2L_0}, \quad L_0 = \lceil 3 \log_2(1/\Delta) \rceil + 6. \quad (18.4bh)$$

Indeed the lemma's number of bins is at most L_0 , and its construction gives $m_T \geq \Delta^{\nu+q}/32 \geq \Delta^3/32$. Consequently at most L_0 different dyadic values of m_T occur. Pigeonhole the tubes to a family $\mathcal{T}_G$ of size at least $|\mathcal{T}|/L_0$ on which $m_T = m$ is constant. Let G be the union of their selected rectangles, intersected with $\text{supp } \nu$. These rectangles are pairwise disjoint.

For Δ small enough, $2L_0 \leq \Delta^{-\tau}$. Thus (18.4bh) gives the first two estimates in (18.4bf), and (18.4bd) gives

$$\nu(G) \geq \frac{\Delta^{f\tau}}{2L_0^2} \geq \Delta^{(f+2)\tau}. \quad (18.4bi)$$

Set $b = f + 2$. We shall prove

$$m \leq \Delta^{qs+(1-q)\nu-4\alpha}. \quad (18.4bj)$$

First choose a ball meeting a substantial amount of G . Lemma IV.8.53, with $r = \Delta$, $R = \Delta^q$ and $\varepsilon = 4b\tau$, shows that the set

$$E_{\text{low}} = \{x \in G : \nu(G \cap \mathbb{B}(x, \Delta^q)) \leq \Delta^{qs+4b\tau}\}$$

has mass at most $8^{s+2b\tau} C_{2b\tau} \Delta^{2b\tau}$. For small Δ this is at most $\nu(G)/4$. Since $g > b$, (18.4be) also bounds the bad centers by $\nu(G)/4$, after decreasing Δ . Choose $x_0 \in G$ outside both sets and put $B = \mathbb{B}(x_0, \Delta^q)$. Then

$$\nu(G \cap B) \geq \Delta^{qs+4b\tau}, \quad x_0 \notin \text{Bad}_\nu(e; q, 1 \mid \Delta, a, \alpha, g\tau). \quad (18.4bk)$$

For every tube meeting $G \cap B$, one selected rectangle meets B . Its diameter is at most $2\Delta^q$, so it is contained in $\mathbb{B}(x_0, 4\Delta^q)$. Rectangles chosen for distinct tubes are disjoint and have mass at least m . The upper mass bound therefore gives

$$|\{T \in \mathcal{T}_G : G \cap B \cap T \neq \emptyset\}| \leq \frac{\nu(\mathbb{B}(x_0, 4\Delta^q))}{m} \leq \frac{16\Delta^{qs}}{m}. \quad (18.4bl)$$

Under the homothety T_B , these tubes have width $\rho = \Delta^{1-q}$, but their grid is translated. Every such projected half-open interval is covered by at most two standard dyadic intervals of length ρ . Thus a family $\mathcal{U}_0$ of at most $32\Delta^{qs}/m$ standard dyadic ρ -tubes covers $T_B(G \cap B)$. For $\lambda = \nu^{B,s}$ and $E_1 = \bigcup \mathcal{U}_0$, we have

$$\lambda(E_1) \geq \Delta^{4b\tau} = \rho^{4b\tau/(1-q)}.$$

Apply Lemma IV.8.58 to this entire tube union, not just to the covered image subset. It supplies a subfamily $\mathcal{U} \subset \mathcal{U}_0$ and $w \geq 0$ with full tube masses

$$\rho^w \leq \lambda(T) \leq 2\rho^w \quad (T \in \mathcal{U}), \quad |\mathcal{U}| \geq \rho^{-w+(4b+1)\tau/(1-q)}. \quad (18.4bm)$$

For the cardinality bound we have absorbed its $4L$ factor using $4L \leq \Delta^{-\tau}$, valid at small scales. Our assumption on g implies $g \geq (4b+1)/(1-q)$, so this cardinality is at least $\rho^{-w+g\tau}$.

If (18.4bj) failed, (18.4bl) and the covering construction would also give

$$|\mathcal{U}| \leq 32\rho^{-v+4\alpha/(1-q)}.$$

Combining with (18.4bm), and taking logarithms to the base $\rho < 1$, yields

$$w \leq v - \frac{4\alpha - (4b+1)\tau}{1-q} + \frac{5}{(1-q)\log_2(1/\Delta)}.$$

The hypothesis on τ gives $(4b+1)\tau < \alpha/2$. For small Δ , also $5/\log_2(1/\Delta) \leq \alpha/2$. Hence

$$w \leq v - \frac{3\alpha}{1-q} \leq v - 3\alpha < a - \alpha.$$

Together with (18.4bm), this makes x_0 bad under Definition IV.8.61, contrary to (18.4bk). This proves (18.4bj).

Finally, in a fixed tube, a ball of radius Δ^q meets at most four of the vertical grid intervals of length Δ^q , hence at most four selected rectangles. Their masses are at most $2m$. Dividing the bound $8m$ by $\nu(T) \geq \Delta^v$ and using (18.4bj) gives (18.4bg). $\square$

This proves the nonconcentration preparation corresponding to Orponen's Lemma 3.30 [16]. To apply it after Lemma IV.8.62, take $f = F(n)$ and $g = F(n+1)$ and arrange the displayed growth condition for each possible stopping index before fixing τ . It remains to preserve these slice bounds during the nested branching refinement and to construct the product comparison.

8. Nested tube families and stable branching

The fine-tube estimate must survive a multiscale organization. We first prune whole tubes, never parts of the retained set inside an individual fine tube. This gives uniform child counts without destroying the preceding slice bound. Stability then prevents a coarser level from having a substantially different exponent.

Lemma IV.8.64 (*pruning to uniform nested tube families*).

Fix $N \geq 1$, $0 = q_0 < q_1 < \dots < q_N = 1$ and $\tau > 0$, $c \geq 0$. Let all Δ^{q_i} , $1 \leq i \leq N$, be dyadic. Let ν be a Radon measure with $\nu(B_0) \leq 1$. Suppose $e \in S^1$, $v \in [0, 2]$, and a family $\mathcal{T}$ of dyadic Δ -tubes satisfies

$$|\mathcal{T}| \geq \Delta^{-v+c\tau}, \quad \Delta^v \leq \nu(T) \leq 2\Delta^v.$$

Let $G_0 \subset \text{supp } \nu \cap \bigcup \mathcal{T}$ be Borel, with $\nu(G_0 \cap T) \geq \Delta^\tau \nu(T)$ for every $T \in \mathcal{T}$. For sufficiently small Δ there are nonempty families $\mathcal{H}_i$, $1 \leq i \leq N$, of dyadic Δ^{q_i} -tubes and $G = G_0 \cap \bigcup \mathcal{H}_N$ such that:

1. $\mathcal{H}_N \subset \mathcal{T}$. Every member of $\mathcal{H}_{i+1}$ is contained in a unique member of $\mathcal{H}_i$, for $1 \leq i < N$, and each member of $\mathcal{H}_i$ has exactly $R_i \geq 1$ children in $\mathcal{H}_{i+1}$ for those indices. Here a child means a member contained in that parent.
2. On setting $M_i = |\mathcal{H}_i|$ and $t_i = \log M_i / (q_i \log(1/\Delta))$ for $1 \leq i \leq N$, and writing $C_* = c + N - 1$, one has

$$\begin{aligned} M_{i+1} &= R_i M_i \quad (1 \leq i < N), & \Delta^{-v+C_*\tau} &\leq M_N \leq \Delta^{-v}, \\ v - C_*\tau &\leq t_N \leq v, & \nu(G) &\geq \Delta^{(C_*+1)\tau}. \end{aligned} \tag{18.4bn}$$

3. For $1 \leq i \leq N$ and $T \in \mathcal{H}_i$,

$$\Delta^{q_i t_i + (C_*+1)\tau} \leq \nu(G \cap T) \leq 2\Delta^{q_i t_i}. \tag{18.4bo}$$

For a retained fine tube $T \in \mathcal{H}_N$, $G \cap T = G_0 \cap T$. Thus every previously proved upper bound for its intersections with balls is preserved.

Proof. Start with the fine family $\mathcal{T}$ at level N . We construct the coarser levels successively, upwards. Suppose level $i + 1$ has already been organized, and each of its members has the same number of fine descendants. This holds initially with one descendant per fine tube. Each occupied dyadic parent of width Δ^{q_i} has between one and $\Delta^{-(q_{i+1}-q_i)}$ occupied children. Split parents according to their child counts in bins $[2^j, 2^{j+1})$. There are at most

$$L_* = \lceil \log_2(1/\Delta) \rceil + 2$$

bins. Because all children have equal fine-descendant counts, a bin carries at least $1/L_*$ of the current fine tubes. Retain that bin, and in each retained parent keep exactly 2^j of its children and all their descendants. This loses at most half the fine tubes in the selected bin. It creates a uniform child count $R_i = 2^j$ and preserves every already established finer child count.

Repeat for $i = N-1, \dots, 1$. A later deletion removes whole parents and their descendants, so it never changes the child count inside a surviving finer parent. At each step at least $1/(2L_*)$ of the fine tubes survive. For small Δ , $2L_* \leq \Delta^{-\tau}$; hence

$$M_N \geq \Delta^{(N-1)\tau} |\mathcal{J}| \geq \Delta^{-v+C_*\tau}.$$

Disjointness and $\nu(T) \geq \Delta^v$ give $M_N \Delta^v \leq \nu(B_0) \leq 1$. This proves the cardinality and t_N bounds.

In the final families every level- i tube has exactly

$$R_i \cdots R_{N-1} = \frac{M_N}{M_i} = \Delta^{-t_N+q_i t_i}$$

fine descendants. Each has G -mass between $\Delta^{v+\tau}$ and $2\Delta^v$; no within-tube restriction was changed. Thus

$$\Delta^{q_i t_i + v - t_N + \tau} \leq \nu(G \cap T) \leq 2\Delta^{q_i t_i + v - t_N}.$$

Use $0 \leq v - t_N \leq C_*\tau$ to obtain (18.4bo). Summing the fine-tube lower bounds gives $\nu(G) \geq M_N \Delta^{v+\tau} \geq \Delta^{(C_*+1)\tau}$. The same construction proves the asserted nesting and preservation of the fine slice estimates. $\square$

Lemma IV.8.65 (*stability locks the branching exponents*).

Let the objects of Lemma IV.8.64 be given, assume ν is s -quasiregular, and put

$$Q = \{q_0, \dots, q_N\}, \quad h = \min_{i < N} (q_{i+1} - q_i), \quad B_* = C_* + 1 = c + N.$$

Suppose $0 < \alpha < 1/10$, $a \in [0, 2]$, $|v - a| \leq \alpha/100$, and

$$g \geq \frac{4B_* + 2}{h}, \quad 0 < \tau < \frac{\alpha h}{4(4B_* + 1)}. \quad (18.4bp)$$

Assume, for every $p < q$ in Q , that

$$\nu(\text{Bad}_\nu(e; p, q \mid \Delta, a, \alpha, g\tau)) < \Delta^{g\tau}. \quad (18.4bq)$$

For sufficiently small Δ , every branching exponent satisfies

$$v - 2\alpha \leq t_i \leq v + \frac{2\alpha}{q_1} \quad (1 \leq i \leq N). \quad (18.4br)$$

The threshold depends only on the displayed fixed parameters and the quasiregular covering constant at exponent $s + 2B_*\tau$.

Proof. The fine-level inequalities follow from (18.4bn), since $C_*\tau < 2\alpha$. Fix $i < N$ and put $u = q_i \in (0, 1)$. We first record a uniform way to find a good ball. For either $R = 1$ or $R = \Delta^u$, apply Lemma IV.8.53 with $E = G$, $r = \Delta$ and $\varepsilon = 4B_*\tau$. The centers in G with

$$\nu(G \cap \mathbb{B}(x, R)) \leq \Delta^{4B_*\tau} R^s$$

have mass at most $8^{s+2B_*\tau} C_{2B_*\tau} \Delta^{2B_*\tau}$. Since $\nu(G) \geq \Delta^{B_*\tau}$, this is at most $\nu(G)/4$ at small scales. The bad centers for any fixed pair in (18.4bq) also have mass at most $\nu(G)/4$, since $g > B_*$. We may therefore choose $x \in G$ outside both sets.

The lower bound. Suppose $t_i < v - 2\alpha$. Choose the preceding good center for $R = 1$ and the pair $(0, u)$. The ball $B = \mathbb{B}(x, 1)$ has $\nu(G \cap B) \geq \Delta^{4B_*\tau}$. The set G is covered by $M_i = \Delta^{-ut_i}$ coarse tubes. Translation to the blow-up at B and re-covering each shifted interval by two standard intervals gives at most $2\rho^{-t_i}$ dyadic tubes, where $\rho = \Delta^u$. Their full union has blow-up mass at least $\Delta^{4B_*\tau}$. Lemma IV.8.58 applied to that union, with its logarithmic loss absorbed by Δ^τ , gives an exponent $w \geq 0$ and full tube weights between ρ^w and $2\rho^w$, with count at least

$$\rho^{-w+(4B_*+1)\tau/u}.$$

It is at least $\rho^{-w+g\tau}$, since $u \geq h$. Comparison with the covering count gives

$$w \leq t_i + \frac{(4B_*+1)\tau}{u} + \frac{1}{u \log_2(1/\Delta)} \leq t_i + \frac{\alpha}{2} < v - \frac{3\alpha}{2} < a - \alpha.$$

The second inequality uses (18.4bp) and sufficiently small Δ , uniformly over the finitely many u . Thus x is bad for $(0, u)$, contradicting its choice.

The upper bound. Suppose $t_i > v + 2\alpha/q_1$. Choose a good center for $R = \Delta^u$ and the pair $(u, 1)$, and set $B = \mathbb{B}(x, \Delta^u)$. Its image of $G \cap B$ has homogeneous blow-up mass at least $\Delta^{4B_*\tau}$. The ball meets at most four level- i dyadic tubes. Each has M_N/M_i fine descendants. Since $t_N \leq v$,

$$\frac{M_N}{M_i} = \Delta^{-t_N+ut_i} \leq \Delta^{-v+ut_i}.$$

After blowing up and re-covering the shifted intervals, at most

$$8\Delta^{-v+ut_i} \leq 8\rho^{-v+2\alpha/(1-u)}, \quad \rho = \Delta^{1-u},$$

standard dyadic tubes cover the image subset. Here $u \geq q_1$ and the counterassumption give $u(t_i - v) > 2\alpha$. Apply Lemma IV.8.58 to the full union of these tubes. It gives full weights between ρ^w and $2\rho^w$ and count at least $\rho^{-w+(4B_*+1)\tau/(1-u)}$, which is at least $\rho^{-w+g\tau}$ because $1 - u \geq h$. Comparison with the upper count yields

$$w \leq v - \frac{2\alpha - (4B_*+1)\tau}{1-u} + \frac{3}{(1-u) \log_2(1/\Delta)} \leq v - \frac{3\alpha}{2(1-u)} < a - \alpha.$$

For the second inequality use $(4B_*+1)\tau < \alpha/4$ and $3/\log_2(1/\Delta) \leq \alpha/4$. These are again badness witnesses for the chosen pair, a contradiction. Both bounds are proved. $\square$

Together these lemmas implement Orponen's Section 3.4.2 [16] with exact child counts and explicit losses. In the application, the preceding fine-tube lemma gives $c = F(n) + 1$. Choose $F(n+1)$ large enough for both its growth requirement and (18.4bp), uniformly over $n < K$, before making τ small. The retained set is still contained in the nonconcentrated fine-tube set. The remaining projection work is the horizontal-vertical product comparison and the shorter-arc iteration, not an unproved lower AD bound.

9. Fixing the exceptional directions before the scale

The direction measure and the projection covering constant must be chosen before the small scale. Otherwise the final contradiction could compare two estimates whose constants deteriorate as that scale changes. We also need a measurable exceptional set, since the compact Frostman theorem in Volume I cannot simply be applied to an arbitrary set.

Lemma IV.8.66 (*uniform exceptional-direction data*).

Let $K \subset \mathbb{R}^2$ be nonempty and compact, let $0 < D < 1$, and put

$$E_D = \{e \in S^1 : \dim_{\mathbb{A}} \pi_e K < D\}.$$

The set E_D is Borel. If $\dim_{\mathbb{H}} E_D > 0$, there are a compact set $E \subset E_D$, constants $C_E, C_\sigma \geq 1$, an exponent $\varepsilon_0 > 0$, and a Borel probability σ supported on E such that, for every $e \in E$,

$$\begin{aligned} N_1(\pi_e K \cap (x - R, x + R), r) &\leq C_E (R/r)^D \quad (x \in \mathbb{R}, 0 < r < R < \infty), \\ \sigma(\mathbb{B}(e, u)) &\leq C_\sigma u^{\varepsilon_0} \quad (e \in \mathbb{R}^2, u > 0). \end{aligned} \quad (18.4bs)$$

The first estimate, with the same C_E , holds for $\pi_e(TK)$ for every homothety $T(y) = by + z$, $b > 0$. All these data are independent of a subsequently chosen scale. Here N_1 denotes covering by open scalar intervals of radius r .

Proof. For $x \in \mathbb{Q}$ and $R, r \in \mathbb{Q}$, $0 < r < R$, let $M_e(x, R, r)$ be the largest cardinality of a subset of $\pi_e K \cap [x - R, x + R]$ whose distinct points have distance at least r . It is finite. For each integer $j \geq 1$, the set $\{e : M_e(x, R, r) \geq j\}$ is closed: it is the projection onto S^1 of the compact set of tuples $(e, y_1, \dots, y_j) \in S^1 \times K^j$ satisfying

$$|y_i \cdot e - x| \leq R, \quad |(y_i - y_k) \cdot e| \geq r \quad (i \neq k).$$

For rational $0 < t < D$ and integer $C \geq 1$, the certificate set

$$E_{t,C} = \bigcap_{\substack{x, R, r \in \mathbb{Q} \\ 0 < r < R}} \{e : M_e(x, R, r) \leq C(R/r)^t\} \quad (18.4bt)$$

is therefore Borel.

We verify both implications, rather than replacing covering by packing without a scale comparison. If $\dim_{\mathbb{A}} \pi_e K < D$, choose rational $t < D$ larger than that dimension and an exponent $u < t$ for which the covering estimate holds. For a compact set the estimate extends to all outer radii: above a fixed diameter, use a fixed containing interval when r is small, and a bounded cover when r is large. A radius- $r/3$ cover of the closed radius- R window, enlarged to radius $2R$, contains at most one member of any r -separated subset in each covering interval. Consequently

$$M_e(x, R, r) \leq C'(6R/r)^u \leq C''(R/r)^t.$$

An integer $C \geq C''$ gives $e \in E_{t,C}$.

Conversely, suppose $e \in E_{t,C}$ and take arbitrary real x, R, r with $0 < r < R$. Choose rational x', R', r' with

$$|x' - x| < R/4, \quad 3R/2 < R' < 2R, \quad r/2 < r' < 3r/4.$$

Every r -separated subset of $\pi_e K \cap (x - R, x + R)$ is an r' -separated subset of the closed rational window at x', R' . Its cardinality is at most $C(4R/r)^t$ by (18.4bt). A maximal such subset covers the original set by open intervals of radius r : a point at distance at least r from every selected point could be added. Thus

$$N_1(\pi_e K \cap (x - R, x + R), r) \leq C4^t(R/r)^t.$$

In particular $\dim_{\mathbb{A}} \pi_e K \leq t < D$. We have proved $E_D = \bigcup_{t,C} E_{t,C}$, a countable union.

Choose $0 < \varepsilon_0 < \dim_{\mathbb{H}} E_D$. Countable stability gives one certificate with $\dim_{\mathbb{H}} E_{t,C} > \varepsilon_0$. The analytic compact-approximation theorem, already registered as a Category 2 premise in Volume III, Chapter 1, supplies compact $E \subset E_{t,C}$ with $\dim_{\mathbb{H}} E > \varepsilon_0$. The compact Euclidean Frostman lemma of Volume I, Chapter 3, then supplies the probability σ and C_σ . On this E the preceding covering estimate has the uniform constant $C4^t$; enlarging it and using $t < D$ gives (18.4bs). Finally $\pi_e(TK) = b\pi_e K + z \cdot e$. Changing x, R, r to their inverse homothetic coordinates leaves R/r unchanged, proving the last assertion. $\square$

Thus, when Lemma IV.8.62 is used with this σ , we may replace its Borel set S by $S \cap E$ without any loss of σ -mass. Every retained direction then has the same projection estimate, including for the full support of the selected homogeneous blow-up.

Lemma IV.8.67 (*the selected potential is below the projection exponent*).

Let $\nu = \mu^{B,s}$ be a blow-up of a measure with compact support K_μ , let $e \in E$ satisfy (18.4bs) for K_μ , and suppose the fine family and ν_e, a, n satisfy (18.4ax). Then

$$\nu_e \leq D + F(n)\tau + \frac{\log_2(4 \cdot 2^D C_E)}{\log_2(1/\Delta)}, \quad a \leq \nu_e + \alpha/100. \quad (18.4bu)$$

In particular, choosing $\max_{n < K} F(n)\tau < \alpha/4$ and then Δ sufficiently small gives $\nu_e < D + \alpha$ and $a < D + \alpha$, simultaneously in all retained directions.

Proof. Every positive-mass fine tube contains a point of $\text{supp } \nu$, and its scalar coordinate lies in $[-1, 1]$. Use (18.4bs) for the full homothetic support, with outer radius 2 and inner radius Δ . Each covering interval of radius Δ meets at most four standard Δ intervals. Hence

$$\Delta^{-\nu_e + F(n)\tau} \leq |\mathcal{T}_e| \leq 4C_E(2/\Delta)^D.$$

Taking logarithms gives the first assertion; the second is $|\nu_e - a| \leq \alpha/100$. The fixed constant divided by $\log_2(1/\Delta)$ is less than $\alpha/4$ at sufficiently small scales. This proves the final uniform bounds, without claiming the rejected exact range $\nu_e \leq s$. $\square$

10. Smoothing and rounding without an unstated equivalence

The geometric argument will first give an L^2 estimate for smoothed real measures. The inverse theorem uses discrete grid probabilities. The following elementary comparison fixes the normalization and, importantly, states when rounding really commutes with convolution.

Lemma IV.8.68 (*a box kernel and cell masses*).

For $\rho > 0$ and a compactly supported finite positive Borel measure λ on $\mathbb{R}$, define

$$Q_\rho \lambda = \sum_{k \in \mathbb{Z}} \lambda([k\rho, (k+1)\rho)) \delta_{k\rho}, \quad \psi_\rho = \frac{1}{2\rho} \mathbf{1}_{[-\rho, \rho]}.$$

Here $(\lambda * \psi_\rho)(x) = \int \psi_\rho(x - y) d\lambda(y)$, and the norm of a finite atomic measure is the discrete norm defined in Lemma IV.8.55. Then

$$\frac{1}{4\rho} \|Q_\rho \lambda\|_2^2 \leq \|\lambda * \psi_\rho\|_{L^2(\mathbb{R})}^2 \leq \frac{9}{4\rho} \|Q_\rho \lambda\|_2^2. \quad (18.4bv)$$

If μ is a finitely supported positive measure on $\rho\mathbb{Z}$ and ν is a compactly supported finite positive Borel measure, then

$$Q_\rho(\mu * \nu) = \mu * (Q_\rho \nu). \quad (18.4bw)$$

In particular, for such μ, ν , if $a > 0$ and

$$\|(\mu * \nu) * \psi_\rho\|_{L^2} \geq a \|\mu * \psi_\rho\|_{L^2},$$

then

$$\|\mu * (Q_\rho \nu)\|_2 \geq (a/3) \|\mu\|_2. \quad (18.4bx)$$

Proof. Write $w_k = \lambda([k\rho, (k+1)\rho))$. For x in the k th cell, every point of that same cell is at distance less than ρ from x , so $(\lambda * \psi_\rho)(x) \geq w_k/(2\rho)$. Integration over cells proves the lower bound. For almost every x in that cell, only the three cells with indices $k-1, k, k+1$ can contribute. Thus

$$|(\lambda * \psi_\rho)(x)|^2 \leq \frac{3}{4\rho^2} (w_{k-1}^2 + w_k^2 + w_{k+1}^2).$$

Each w_j^2 occurs three times after summing the cell integrals. This gives the upper constant $9/(4\rho)$. Possible contributions at a cell endpoint affect a Lebesgue-null set of x , and do not change either integral.

For $j, k \in \mathbb{Z}$, translation by $j\rho$ sends the k th half-open cell exactly to the $(k+j)$ th cell. The mass assigned to output cell k is consequently

$$\sum_j \mu\{j\rho\} \nu([(k-j)\rho, (k-j+1)\rho)),$$

which is the mass of $\mu * (Q_\rho \nu)$ at $k\rho$. This proves (18.4bw). Apply the upper bound of (18.4bv) to $\mu * \nu$ and its lower bound to μ , noting $Q_\rho \mu = \mu$. Taking square roots in the assumed inequality gives (18.4bx). $\square$

No commutation is asserted for rounding two arbitrary off-grid factors. The horizontal factor must first be placed on the grid. Constants such as 3 can later be absorbed by a strictly positive power of the scale, but the power lost in a small-mass restriction must still be retained.

11. Measurable refinements and overlap

The next averaging argument uses pairs of directions. Choosing a separate good set for each direction is not enough: their membership relation must be measurable in the direction and the point together. Our refinements involve only finite choices, which can be ordered explicitly.

Lemma IV.8.69 (*jointly measurable tube refinements*).

Fix $\nu, \Delta, a, \alpha, f, g, \tau, Q$ and a Borel direction set S . Suppose the hypotheses of Lemmas IV.8.63 and IV.8.64 hold in every direction $e \in S$, with the same parameters, and with $q = q_1$ in the former lemma. Use the output set of Lemma IV.8.63 as the input $G_{0,e}$ of Lemma IV.8.64; its fine family is also the input family of that lemma. Suppose the initial exponent ν_e and finite tube family are Borel functions of e . For a finite tube family, this means that membership of each of its possible integer tube indices is a Borel condition. The outputs of those lemmas can be chosen so that

$$\{(e, y) \in S \times \mathbb{R}^2 : y \in G_e\} \text{ is Borel,} \quad (18.4by)$$

and every refined family's index membership and branching exponent are Borel functions of e . These choices preserve all their mass, nesting and slice estimates. The common-selection outputs of Lemma IV.8.62 may serve as the input functions here.

Proof. At each fixed width ρ , the possible tube indices satisfy $|k| \leq \rho^{-1} + 1$, since tubes are intersected with B_0 . Their membership relations

$$|y| < 1, \quad k\rho \leq y \cdot e < (k+1)\rho$$

are Borel in (e, y) . The analogous rectangle relations use the additional coordinate $y \cdot e^\perp$ and are also Borel. If a previously retained set has a Borel membership relation, its mass in any such rectangle is a Borel function of e by integration against the fixed finite measure $\nu|_{B_0}$.

Order integer indices, finite index subsets and dyadic mass bins lexicographically, once and for all. In Lemma IV.8.58 choose the first bin attaining the largest retained mass. The proof uses finitely many bins; any empty bin can be assigned mass zero. All their masses are Borel functions, so the chosen bin and its index subset are Borel. The same rule applies separately in each input fine tube in Lemma IV.8.63. There the possible values of its dyadic parameter m_T lie between $\Delta^3/32$ and 1, a fixed finite list. Choose the first value attaining the largest number of input tubes. The union of the selected rectangles, intersected with the fixed closed set $\text{supp } \nu$, has a Borel membership relation. The good center used to prove its upper slice bound is not part of the definition of this union; no measurable choice of those centers is needed.

In Lemma IV.8.64, parenthood is the deterministic dyadic relation between integer indices. Child counts and the number of fine descendants are finite integer-valued Borel functions. At each step choose the first count bin carrying the largest number of fine descendants, and retain the first 2^j children in each chosen parent. All new index memberships are Borel. After the finitely many steps, intersecting the preceding Borel set with the retained fine tubes gives (18.4by). The formulas for M_i and t_i then give Borel exponents. The proofs of both lemmas require only that a maximizing bin and the stated number of children be retained, so these ordering rules change none of their estimates.

Finally, the proof of Lemma IV.8.62 already chooses badness witnesses from a finite ordered list using the interval (18.4ba). At each induction step the common center and ball are fixed, and the retained direction set is a Borel fiber. Its initialization uses the same finite bin rule. Thus its final input families and exponents have the required measurability. $\square$

Lemma IV.8.70 (*many directions overlap one retained set*).

Let ν be a positive Borel measure with $\nu(B_0) \leq 1$, let σ be a finite Borel measure on S^1 , and let S be Borel. Suppose $G_e \subset B_0$ have a jointly Borel membership relation and $\nu(G_e) \geq b$, $0 < b \leq 1$, for every $e \in S$. For any Borel arc J with $\sigma(J \cap S) > 0$, there are $e_1 \in J \cap S$ and a Borel set $S_1 \subset J \cap S$ such that

$$\sigma(S_1) \geq \frac{b^2}{4} \sigma(J \cap S), \quad \nu(G_e \cap G_{e_1}) \geq \frac{b^2}{4} \quad (e \in S_1). \quad (18.4bz)$$

Proof. Put $m = \sigma(J \cap S) > 0$ and $H(y) = \int_{J \cap S} \mathbf{1}_{G_e}(y) d\sigma(e)$. Tonelli and Cauchy–Schwarz give

$$\begin{aligned} \int_{J \cap S} \int_{J \cap S} \nu(G_e \cap G_{e'}) d\sigma(e) d\sigma(e') &= \int_{B_0} H(y)^2 d\nu(y) \\ &\geq \frac{(\int_{B_0} H d\nu)^2}{\nu(B_0)} \geq b^2 m^2. \end{aligned}$$

Here $\nu(B_0) > 0$ because $b, m > 0$. There is therefore $e_1 \in J \cap S$ with $\int_{J \cap S} \nu(G_e \cap G_{e_1}) d\sigma(e) \geq b^2 m/2$. Joint measurability makes the function being integrated Borel. Define $S_1 = \{e \in J \cap S : \nu(G_e \cap G_{e_1}) \geq b^2/4\}$. Since every intersection has mass at most one,

$$b^2 m/2 \leq \sigma(S_1) + (b^2/4)m.$$

This proves both conclusions. $\square$

For the refined sets of Lemma IV.8.64, one may take $b = \Delta^{(c+N)\tau}$. The factor 1/4 in (18.4bz) is fixed and can be absorbed by Δ^τ at sufficiently small scales. This averaging supplies a substantial intersection, not independence of its horizontal and vertical coordinates.

12. A finite product inside a substantial intersection

We now fix two directions and a pair of adjacent scales $p, 2p$. The product below is an auxiliary finite measure, not a factorization of ν . Its atoms discretize a selected part of the intersection. Stability is used once more to discard excessively heavy atoms while retaining a definite part of each vertical row.

Lemma IV.8.71 (*one-sided product domination*).

Let ν be s -quasiregular, $0 < s \leq 2$. Let e, e_1 have the refined families and sets of Lemmas IV.8.63–IV.8.65, with the same Q, α, τ, a and fine exponents $|v_e - a|, |v_{e_1} - a| \leq \alpha/100$. Write $r = q_1$, use Lemma IV.8.63 at $q = r$, and suppose $p = q_i$, $2p = q_{i+1}$, $r < p \leq 1/2$. Put

$$t_p = t_i(e_1), \quad t_{2p} = t_{i+1}(e_1), \quad \rho = \Delta^p, \quad \xi_1 = 2t_{2p} - t_p.$$

Assume, for some fixed $A \geq 0$ and $\gamma > 0$, that

$$\nu(G_e \cap G_{e_1}) \geq \Delta^{A\tau}, \quad s - t_p \geq \gamma.$$

Assume also

$$\begin{aligned} \nu(\text{Bad}_\nu(e_1; p, 2p \mid \Delta, a, \alpha, g\tau)) &< \Delta^{g\tau}, \\ g &> A + 2, \quad g \geq (A + 5)/p, \\ 0 < \tau &< \min\{\alpha p/[4(A + 5)], p\gamma/[2(A + 3)]\}. \end{aligned} \quad (18.4ca)$$

For sufficiently small Δ , use orthogonal coordinates $(x, y) = (\pi_{e_1}, \pi_{e_1^\perp})$. There are a retained p -tube T_0 , with ambient scalar interval $[j_0\rho, (j_0 + 1)\rho)$, finite sets

$$D_h \subset \rho\mathbb{Z} \cap [0, 1), \quad D_v \subset \rho\mathbb{Z} \cap [-1, 1),$$

and a number ℓ_v such that

$$|D_h| = \Delta^{-p\xi_1}, \quad |D_v| = \Delta^{-p(s-t_p)+\ell_v}, \quad (A + 2)\tau \leq \ell_v \leq (A + 3)\tau. \quad (18.4cb)$$

Let μ_h, μ_v be their uniform probabilities and $P = \mu_h \times \mu_v$. There is a nonzero positive atomic measure Γ , supported on a set $Z \subset D_h \times D_v$, with

$$\Gamma(\mathbb{R}^2) \geq \Delta^{\ell_v+(A+4)\tau}, \quad \Gamma \leq \Delta^{-C_p\alpha+\ell_v}P, \quad P(Z) \geq \Delta^{C_p\alpha+(A+4)\tau}, \quad C_p = 4p + 8p/r. \quad (18.4cc)$$

For every $(X, Y) \in Z$, there is $z = (x, y) \in G_e \cap G_{e_1} \cap T_0$ with

$$|X - (x/\rho - j_0)| < \rho, \quad |Y - y| < \rho. \quad (18.4cd)$$

Each $Y \in D_v$ is the left endpoint of an ambient ρ -square R_Y in the column of T_0 , satisfying

$$\nu(R_Y \cap G_e \cap G_{e_1}) \geq \Delta^{ps+(A+2)\tau}. \quad (18.4ce)$$

The word *ambient* is essential: the square need not be contained in the clipped tube $T_0 \subset B_0$.

Proof. Put $H = G_e \cap G_{e_1}$ and remove the bad centers for $(p, 2p, e_1)$, obtaining $H' = H \setminus \text{Bad}_\nu(e_1; p, 2p)$. For small Δ , $\nu(H') \geq 3\Delta^{A\tau}/4$. Partition the plane, in the displayed orthogonal coordinates, into half-open squares of side ρ . At most $9C_\tau 8^{s+\tau} \rho^{-(s+\tau)}$ of these squares meet $\text{supp } \nu \cap B_0$: cover that set by radius- $\rho/4$ balls using outer radius 2, and note that each ball meets at most nine grid squares. Consequently the squares having

$$\nu(R \cap H') < \Delta^{ps+(A+2)\tau}$$

carry total H' -mass at most $9C_\tau 8^{s+\tau} \Delta^{(A+2-p)\tau}$. This is less than $\Delta^{A\tau}/4$ at small scales. Call the remaining squares heavy; their union carries at least $\Delta^{A\tau}/2$ of H' .

Every heavy square has a point $x_R \in R \cap H'$ such that

$$\nu(R \cap H \cap \mathbb{B}(x_R, \rho)) \geq \Delta^{ps+(A+3)\tau}. \quad (18.4cf)$$

Indeed a maximal $\rho/2$ -separated subset of $R \cap H'$ has at most 100 points, by disjoint radius- $\rho/4$ balls in the enlarged square. Its radius- $\rho/2$ balls cover $R \cap H'$. One of the radius- ρ balls at these centers therefore carries at least 1/100 of its mass. Absorb this fixed factor using $\Delta^\tau \leq 1/100$. All these centers lie outside the bad set.

The p -tube family in direction e_1 has Δ^{-pt_p} members and covers H . One member T_0 consequently contains heavy-square intersection mass at least $\Delta^{pt_p+(A+1)\tau}$. Each ambient

square has mass at most $2\rho^s$ by the upper ball bound, since its diameter is $\sqrt{2}\rho$. Thus the number of heavy squares in its column is at least $\Delta^{-p(s-t_p)+(A+1)\tau}/2$. Select

$$V = \left\lfloor \Delta^{-p(s-t_p)+(A+2)\tau} \right\rfloor$$

of them. This is possible for small Δ . The assumption on τ ensures the quantity being rounded tends to infinity, uniformly when $s - t_p \geq \gamma$. Hence V has the form in (18.4cb), with the stated bounds on ℓ_v ; the upper bound uses $V \geq \frac{1}{2}\Delta^{-p(s-t_p)+(A+2)\tau}$ and $\Delta^\tau \leq 1/2$. Let D_v be the vertical left endpoints of these squares. They lie in $[-1, 1)$ because they meet B_0 . Their full intersections with H satisfy (18.4ce).

Every retained p -tube has the same number of retained $2p$ -tube children. Exact branching gives this number as

$$M_{i+1}(e_1)/M_i(e_1) = \Delta^{-2pt_{2p}+pt_p}.$$

Apply the affine map $\mathcal{A}(x, y) = (x/\rho - j_0, y)$ to the ambient scalar intervals of the children of T_0 . Their left endpoints form D_h . These intervals have length ρ , are dyadic in $[0, 1)$, and their number is $\Delta^{-p\xi_1}$. Write T_X for the child corresponding to $X \in D_h$.

For each selected row Y , define

$$\lambda_Y = \nu|_{R_Y \cap H \cap \mathbb{B}(x_{R_Y}, \rho)}, \quad w_{X,Y} = \rho^{-t_p} \lambda_Y(T_X).$$

Since the children cover $H \cap T_0$, (18.4cf) and the upper mass of the radius- ρ ball give

$$\Delta^{p(s-t_p)+(A+3)\tau} \leq \sum_X w_{X,Y} \leq \rho^{s-t_p}. \quad (18.4cg)$$

We retain the atoms with $0 < w_{X,Y} \leq \Lambda$, where $\Lambda = \rho^{s-4\alpha/r}$. They carry at least half of the row sum. To prove this, let $U_Y = \{X : w_{X,Y} > \Lambda\}$ and suppose its physical mass is at least $\lambda_Y(\mathbb{R}^2)/2$. The upper bound in (18.4cg) gives $|U_Y| \leq \rho^{-t_p+4\alpha/r}$. Blow up at $\mathbb{B}(x_{R_Y}, \rho)$. The selected original children have width ρ^2 , so their images have width ρ . Re-cover each shifted scalar interval by at most two standard ρ intervals. The full union of at most $2\rho^{-t_p+4\alpha/r}$ standard tubes has homogeneous blow-up mass at least $\Delta^{(A+4)\tau}$ by (18.4cf). Apply Lemma IV.8.58 to this full union, not just the restricted image of λ_Y . It produces full tube weights between ρ^z and $2\rho^z$, $z \geq 0$, and a family of size at least

$$\rho^{-z+(A+5)\tau/p},$$

where its logarithmic factor is absorbed using Δ^τ . Comparison with the preceding covering count yields

$$z \leq t_p - \frac{4\alpha}{r} + \frac{(A+5)\tau}{p} + \frac{1}{p \log_2(1/\Delta)} \leq a - \frac{2\alpha}{r} + \frac{\alpha}{100} + \frac{\alpha}{2} < a - \alpha.$$

Here $t_p \leq v_{e_1} + 2\alpha/r$ comes from Lemma IV.8.65, and (18.4ca) and a small scale bound the last two losses. Since $g \geq (A+5)/p$, the size is at least $\rho^{-z+g\tau}$. This is precisely a badness witness for the chosen center, a contradiction. Thus the retained low-weight atoms carry at least half of every row.

Let Z consist of these positive low-weight atoms and let $\Gamma = \sum_{(X,Y) \in Z} w_{X,Y} \delta_{(X,Y)}$. The row lower bound, multiplied by $V = \Delta^{-p(s-t_p)+\ell_v}$, gives $\Gamma(\mathbb{R}^2) \geq \Delta^{\ell_v+(A+4)\tau}$. Every product atom has exactly the mass

$$(|D_h||D_v|)^{-1} = \Delta^{ps+2p(t_{2p}-t_p)-\ell_v}.$$

Thus Γ/P at an atom is at most $\Delta^{-4p\alpha/r-2p(t_{2p}-t_p)+\ell_v}$. The two bounds of Lemma IV.8.65 give $t_{2p} - t_p \leq 2\alpha + 2\alpha/r$, proving the domination in (18.4cc). Dividing its mass lower bound by this domination constant proves the estimate for $P(Z)$. Finally $w_{X,Y} > 0$ gives a point of $R_Y \cap H \cap T_X$. Its horizontal error under $\mathcal{A}$ is less than ρ , and its vertical error from Y is less than ρ . This proves (18.4cd). $\square$

The proof uses the upper mass bound to count the high-weight children, and stability to prevent those children from carrying half a row. It does not impose an AD lower bound or a literal range for a *typical weight exponent*. This is the local form of the product-comparison step in Orponen's Sections 4.4–4.5 [16].

Lemma IV.8.72 (*a narrow projection prevents convolution flattening*).

Use the objects and hypotheses of Lemma IV.8.71, and assume $|e - e_1| \leq \rho$, with $\rho < 1/2$. Write

$$e_x = e \cdot e_1, \quad e_y = e \cdot e_1^\perp, \quad \theta = \frac{e_y}{\rho e_x}, \quad \xi_e = 2t_{i+1}(e) - t_i(e).$$

Then $e_x > 1/2$, $|\theta| \leq 2$, and

$$\|(\mu_h * (y \mapsto \theta y)_* \mu_v) * \psi_\rho\|_{L^2} \geq \frac{1}{12} \Delta^{(8p+12p/r)\alpha+(A+4)\tau} \|\mu_h * \psi_\rho\|_{L^2}. \quad (18.4ch)$$

If, in addition, the full projected support in direction e_1 has the uniform bound (18.4bs), then

$$N_1(D_h \cap (x - R, x + R), u) \leq 16C_E(R/u)^D \quad (x \in \mathbb{R}, \rho \leq u < R \leq 1). \quad (18.4ci)$$

Proof. For unit directions, $e_x = 1 - |e - e_1|^2/2 > 1/2$ and $|e_y| \leq |e - e_1| \leq \rho$. The scalar e -projection of T_0 lies in an interval of length at most $\rho + 2|e_y| \leq 3\rho$. It meets at most six standard p intervals. Every retained p -tube in direction e has exactly $\Delta^{-p\xi_e}$ retained $2p$ children. Thus $G_e \cap T_0$ is covered by at most $6\Delta^{-p\xi_e}$ narrow e -tubes.

Put $L_\theta(X, Y) = X + \theta Y$. For the affine map $\mathcal{A}$ in the preceding proof,

$$L_\theta \mathcal{A}(x, y) = \frac{\pi_e(x, y)}{\rho e_x} - j_0.$$

Each narrow tube has an image interval of length at most 2ρ . By (18.4cd), a point of Z and its witness have L_θ -coordinates differing by less than 3ρ . Consequently the support of $(L_\theta)_*(P|_Z) * \psi_\rho$ is contained in a union of intervals of total length at most $60\rho\Delta^{-p\xi_e}$. Its integral is $P(Z)$, so Cauchy–Schwarz gives

$$\|(L_\theta)_*(P|_Z) * \psi_\rho\|_{L^2} \geq \frac{P(Z)}{\sqrt{60\rho\Delta^{-p\xi_e}}}.$$

The full push-forward $(L_\theta)_*P = \mu_h * (y \mapsto \theta y)_* \mu_v$ dominates this restricted push-forward. All measures and the kernel are positive. Lemma IV.8.68 also gives

$$\|\mu_h * \psi_\rho\|_{L^2} \leq \frac{3}{2\sqrt{\rho|D_h|}}.$$

Dividing these estimates and using $|D_h| = \Delta^{-p\xi_1}$ therefore bounds the ratio from below by

$$\frac{2}{3\sqrt{60}}P(Z)\Delta^{p(\xi_e - \xi_1)/2}.$$

The exponent locks give

$$v_e - 4\alpha - 2\alpha/r \leq \xi_e \leq v_e + 2\alpha + 4\alpha/r.$$

The same bounds hold for ξ_1 with v_{e_1} , and $|v_e - v_{e_1}| \leq \alpha/50$. Hence $\xi_e - \xi_1 \leq (6 + 1/50)\alpha + 6\alpha/r$. Use (18.4cc) and $2/(3\sqrt{60}) > 1/12$. The resulting loss is $[(701/100)p + 11p/r]\alpha + (A + 4)\tau$, which is no larger than the displayed loss in (18.4ch).

For the covering assertion, every retained narrow child has positive G_{e_1} -mass by Lemma IV.8.64. Its rescaled left endpoint X is therefore at distance less than ρ from a point of $\tilde{K} = \rho^{-1}\pi_{e_1}(\text{supp } \nu) - j_0$. This set has the covering bound (18.4bs) with the same C_E . For $X \in D_h \cap (x - R, x + R)$ choose one such representative. All representatives lie in the radius- $2R$ window. Cover them by radius- $u/4$ intervals, using at most $C_E(8R/u)^D$ intervals. Enlarging each of these intervals by $\rho \leq u$ covers the corresponding endpoints, and two open intervals of radius u cover each enlarged interval. Thus the number is at most $2C_E 8^D (R/u)^D \leq 16C_E (R/u)^D$, since $D < 1$. $\square$

Lemma IV.8.73 (*the vertical factor has small rounded atoms*).

Use the objects of Lemmas IV.8.71 and IV.8.72. Suppose $0 < r < \varepsilon_1 < p$ and $|e - e_1| > \Delta^{2p - \varepsilon_1}$. For sufficiently small Δ , put $\eta = Q_\rho((y \mapsto \theta y)_* \mu_v)$. It is a grid probability supported in at most eight integer unit intervals, and

$$\max_x \eta\{x\} \leq 32\Delta^{r(s - v_{e_1}) - 4\alpha - (2A + 5)\tau}. \quad (18.4cj)$$

Proof. Let $I = [k\Delta^r, (k + 1)\Delta^r)$ be a standard vertical interval. Since $p > r$ and the scales are dyadic, a selected row endpoint in I has its entire ambient row interval inside I . The squares of those rows are disjoint, and (18.4ce) gives

$$\#(D_v \cap I) \Delta^{ps + (A + 2)\tau} \leq \nu(G_{e_1} \cap T_0 \cap \{y \in I\}). \quad (18.4ck)$$

The coarse tube T_0 has exactly $\Delta^{-t_N(e_1) + pt_p}$ retained fine descendants. The intersection of any one fine tube with the vertical strip over I is contained in an open ball of radius Δ^r : its ambient rectangle has width Δ and height Δ^r , whose half-diagonal is strictly smaller than Δ^r . The retained slice estimate from Lemma IV.8.63 and the full fine-tube upper mass give, for each descendant,

$$\nu(G_{e_1} \cap T \cap \{y \in I\}) \leq 16\Delta^{v_{e_1} + r(s - v_{e_1}) - 4\alpha}.$$

Sum over descendants in (18.4ck) and divide by $|D_v| = \Delta^{-p(s-t_p)+\ell_v}$. This yields

$$\mu_v(I) \leq 16\Delta^{v_{e_1}-t_N(e_1)+r(s-v_{e_1})-4\alpha-(A+2)\tau-\ell_v} \leq 16\Delta^{r(s-v_{e_1})-4\alpha-(2A+5)\tau}.$$

The last inequality uses $t_N(e_1) \leq v_{e_1}$ and $\ell_v \leq (A+3)\tau$; the first of these losses has the favorable sign.

For nearby unit directions, $|e_y| = |e - e_1| \sqrt{1 - |e - e_1|^2/4} \geq |e - e_1|/2$. Since $e_x \leq 1$,

$$|\theta| \geq \frac{1}{2}\Delta^{p-\varepsilon_1}.$$

The inverse image under $y \mapsto \theta y$ of any interval of length ρ consequently has length at most $2\Delta^{\varepsilon_1} < \Delta^r$ at small scales. It meets at most two standard Δ^r intervals. The preceding bound proves (18.4cj), including for half-open output cells used by Q_ρ . Finally $D_v \subset [-1, 1)$ and $|\theta| \leq 2$, so the rounded support lies in $[-3, 2]$. Eight integer unit intervals are more than enough to contain it. $\square$

The horizontal factor has a uniform covering exponent strictly below one, whereas the rescaled vertical factor has a positive atom-decay exponent when α and then τ are small. Together with Lemma IV.8.68, these are exactly the two inputs of Lemma IV.8.56. They are not consequences of a formal product identity for ν .

13. The exceptional-direction theorem

We can now close the projection argument. The order of choices is part of the proof: the inverse-theorem gain is fixed before the selection errors, and the original scale is chosen last. The shorter-arc iteration uses one direction measure throughout.

Theorem IV.8.74 (*Assouad projections of a compact quasiregular support*).

Let $0 < s \leq 2$, and let μ be s -quasiregular with compact support $K \subset B_0$. Then

$$\dim_{\mathbb{H}}\{e \in S^1 : \dim_{\mathbb{A}} \pi_e K < \min(s, 1)\} = 0. \quad (18.4cl)$$

Proof. By countable stability it suffices to fix $0 < D < \min(s, 1)$ and show $\dim_{\mathbb{H}} E_D = 0$. Suppose otherwise. Lemma IV.8.66 fixes $E, C_E, C_\sigma, \varepsilon_0, \sigma$ before any small scale, with (18.4bs). Write $\beta = s - D > 0$.

Parameter order and grid alignment. Choose an integer $N \geq 4$ so large that $10N2^{-N} < \varepsilon_0$, and set

$$r = 2^{-2N}, \quad \varepsilon_1 = 2^{-N}, \quad p_0 = 2^{-N+2}, \quad Q = \{0, r, 2^{-N+2}, 2^{-N+3}, \dots, 1\}.$$

Write $Q = \{q_0, \dots, q_N\}$ in increasing order and $h = \min_{i < N} (q_{i+1} - q_i)$. The pairs used below are $p = q_i$, $2p = q_{i+1}$, $2 \leq i < N$, and satisfy $p \geq p_0 > \varepsilon_1 > r$. Apply Lemma IV.8.56 with

$$t = D, \quad C = 16C_E, \quad \eta_0 = r\beta/2, \quad C_v = 1, \quad J = 8,$$

obtaining fixed $\chi > 0$ and block length D_{inv} . Choose

$$0 < \alpha < \min \left\{ \frac{1}{10}, \frac{\beta}{4(1+2/r)}, \frac{r\beta}{4(r+4)}, \frac{\chi}{4(8+12/r)} \right\}, \quad K_{\text{stop}} = [4/\alpha], \quad \gamma = \beta/2.$$

Fix an increasing sequence $F(0), \dots, F(K_{\text{stop}} + 1)$, $F(0) = 1$, so that for every relevant n , with $f = F(n)$ and $B = f + N + 1$, its next term $g = F(n + 1)$ satisfies

$$g > \max \left\{ f, \frac{4f + 10}{1 - r}, \frac{4B + 2}{h}, 2B + 3, \frac{2B + 6}{p_0} \right\}. \quad (18.4\text{cm})$$

These are finitely many elementary choices, made before τ . Put

$$\begin{aligned} f_{\max} &= \max_{n < K_{\text{stop}}} F(n), \quad B_{\max} = f_{\max} + N + 1, \quad A_{\max} = 2B_{\max} + 1, \\ A_F &= \sum_{j=1}^{K_{\text{stop}}} F(j), \quad L_{\text{dir}} = A_F/h^{K_{\text{stop}}} + 1, \quad L_{\text{arc}} = L_{\text{dir}} + 1 + (N - 2)(A_{\max} + 1). \end{aligned}$$

Choose $\tau > 0$ smaller than all eight numbers

$$\begin{aligned} &\frac{\alpha}{4f_{\max}}, \quad \frac{\alpha}{2(4f_{\max} + 9)}, \quad \frac{\alpha h}{4(4B_{\max} + 1)}, \quad \frac{\alpha p_0}{4(A_{\max} + 5)}, \\ &\frac{p_0 \gamma}{2(A_{\max} + 3)}, \quad \frac{r\beta}{4(2A_{\max} + 5)}, \quad \frac{\chi p_0}{4(A_{\max} + 5)}, \quad \frac{\varepsilon_0}{4L_{\text{arc}}}. \end{aligned} \quad (18.4\text{cn})$$

Finally apply Lemma IV.8.62 with its integer M restricted to multiples of D_{inv} and tending to infinity. Its proof gives $\Delta = \delta^{\theta_{\text{sel}}}$, where $h^{K_{\text{stop}}} \leq \theta_{\text{sel}} \leq 1$ and θ_{sel} is a product of at most $K_{\text{stop}} - 1$ differences of members of Q . If $Q \subset 2^{-L_Q} \mathbb{Z}$, its prescribed original scale is $\delta = 2^{-M 2^{L_Q(K_{\text{stop}} + 1)}}$. Consequently every $m = \log_2(1/\Delta^p)$ in our finite list is an integer multiple of M , hence of D_{inv} . This verifies the inverse theorem's block alignment, rather than rounding a scale after the geometry has been constructed.

All subsequent small-scale thresholds are uniform in the selected blow-up: its quasiregular constants are those of μ , by Lemma IV.8.50, and the projection constant is C_E . There are only finitely many stopping indices and scale pairs. Since $\Delta \leq \delta^{h^{K_{\text{stop}}}} \rightarrow 0$, take M large enough for all these thresholds and the fixed constants absorbed below.

Common refined data. The selection gives ν, a, n, S ; intersect S with E . Set $f = F(n)$, $g = F(n + 1)$. Lemma IV.8.67 gives $v_e < D + \alpha$. Apply Lemma IV.8.63 with $q = r$, then Lemma IV.8.64 with $c = f + 1$. Its mass exponent is $B = f + N + 1$. Lemma IV.8.65, justified by (18.4cm)–(18.4cn), gives for every level

$$v_e - 2\alpha \leq t_i(e) \leq v_e + 2\alpha/r.$$

In particular $s - t_i(e) > \gamma$. The sets G_e can be chosen jointly Borel by Lemma IV.8.69, and $\nu(G_e) \geq \Delta^{B\tau}$. The selection's direction loss satisfies

$$\sigma(S) \geq c_{\alpha, Q} \delta^{A_F \tau} \geq \Delta^{L_{\text{dir}} \tau}, \quad (18.4\text{co})$$

at small scales. The conversion uses $\delta^{A_F \tau} = \Delta^{A_F \tau / \theta_{\text{sel}}}$ and $\theta_{\text{sel}} \geq h^{K_{\text{stop}}}$; it is not a replacement of an original-scale loss by an equal final-scale loss.

The shorter-arc step. Fix $p = q_i$, $2 \leq i < N$, and any Borel arc J of length at most Δ^p with $\sigma(J \cap S) > 0$. Lemma IV.8.70 gives $e_1 \in J \cap S$ and $S_1 \subset J \cap S$ with

$$\sigma(S_1) \geq \Delta^{A\tau} \sigma(J \cap S), \quad \nu(G_e \cap G_{e_1}) \geq \Delta^{A\tau} \quad (e \in S_1), \quad A = 2B + 1.$$

We claim that every $e \in S_1$ satisfies $|e - e_1| \leq \Delta^{2p-\varepsilon_1}$. If not, $|e - e_1| \leq \Delta^p$ by the arc length. Lemmas IV.8.71–IV.8.73 apply: their growth and error hypotheses follow from (18.4cm)–(18.4cn) and $s - t_p > \gamma$. With $\theta = e_y/(\Delta^p e_x)$ as in Lemma IV.8.72, they give grid probabilities μ_h and $\eta = Q_{\Delta^p}((y \mapsto \theta y)_* \mu_\nu)$. Their rounded atom exponent satisfies

$$r(s - v_{e_1}) - 4\alpha - (2A + 5)\tau > r\beta/2.$$

For small Δ the factor 32 in (18.4cj) is absorbed to give $\max_x \eta\{x\} \leq \Delta^{r\beta/4} \leq \Delta^{p\eta_0} = 2^{-\eta_0 m}$. The horizontal covering bound is (18.4ci), and m has the verified block alignment. Thus Lemma IV.8.56 yields

$$\|\mu_h * \eta\|_2 < \Delta^{\chi p} \|\mu_h\|_2.$$

On the other hand, Lemma IV.8.68 applied to (18.4ch) gives, after absorbing the fixed factor $1/36$,

$$\|\mu_h * \eta\|_2 \geq \Delta^{p(8+12/r)\alpha+(A+5)\tau} \|\mu_h\|_2 > \Delta^{\chi p} \|\mu_h\|_2.$$

Indeed the loss exponent is less than $\chi p/2$ by the choices of α, τ . This contradiction proves the claim.

The ball of radius $\Delta^{2p-\varepsilon_1}$ around e_1 meets J in a set coverable by at most $C\Delta^{-\varepsilon_1}$ subarcs of J of length at most Δ^{2p} , with an absolute C . This follows, for example, by lifting the short arc to its arc-length coordinate and partitioning it into intervals of length Δ^{2p} . One such arc J' therefore satisfies

$$\sigma(J' \cap S) \geq \Delta^{\varepsilon_1+(A_{\max}+1)\tau} \sigma(J \cap S). \quad (18.4cp)$$

The extra τ absorbs C^{-1} .

Iteration against the fixed Frostman law. Cover the circle by at most $C\Delta^{-p_0}$ arcs of length at most Δ^{p_0} . By (18.4co) one of them has $\sigma(J \cap S) \geq \Delta^{p_0+(L_{\text{dir}}+1)\tau}$. Apply (18.4cp) successively for $p = q_2, \dots, q_{N-1}$, obtaining an arc J_N of length at most Δ with

$$\sigma(J_N \cap S) \geq \Delta^{(N+2)\varepsilon_1+L_{\text{arc}}\tau} > \Delta^{\varepsilon_0/2}.$$

Here $p_0 = 4\varepsilon_1$, there are $N - 2$ steps, and $10N\varepsilon_1 < \varepsilon_0$ together with (18.4cn) bounds the exponent by $\varepsilon_0/2$. But the fixed Frostman law bounds the mass of this arc by $C_\sigma \Delta^{\varepsilon_0}$. At sufficiently small Δ these inequalities are incompatible. The counterassumption was false, completing the proof. $\square$

Corollary IV.8.75 (*Orponen's compact-set projection theorem*).

For every nonempty compact $F \subset \mathbb{R}^2$,

$$\dim_{\text{H}}\{e \in S^1 : \dim_{\text{A}} \pi_e F < \min(\dim_{\text{A}} F, 1)\} = 0. \quad (18.4cq)$$

Proof. Put $s = \dim_{\text{A}} F$. If $s = 0$, the exceptional set is empty. If $s > 0$, Lemma IV.8.54 supplies a compact s -quasiregular support inside B_0 , with no larger projected Assouad dimension in any direction. The exceptional set for F is contained in that for this support. Apply Theorem IV.8.74 and monotonicity of Hausdorff dimension. $\square$

This proves the precise compact-set form of Orponen's theorem [16]. The statement is a lower bound outside a Hausdorff-zero direction set, not an assertion that Assouad dimension cannot increase under projection. The argument's research input is the fully proved inverse theorem owned by Volume II; its finite-window interface, product comparison and arc iteration have all been supplied here. The nonconformal renormalization needed for the embedding theorem below is a separate argument.

14. Nonconformal magnification in the embedding space

Ordinary magnification of a parameter square need not preserve an embedding. The alternative below shrinks the source and then undoes a target cylinder. Its horizontal coordinate remains bounded, while its vertical coordinate sees a projection of the magnified square. We keep the signed coefficients throughout; no passage to an iterated source IFS, and hence no change to Condition (R_*) , is needed.

Definition IV.8.76 (*normalized embedding windows and oriented grids*).

Write the source maps as $\varphi_i(x) = ax + \sigma_i$, with common signed coefficient a and $\alpha = |a| \in (0, 1)$, and the target maps as $\psi_j(x) = b_jx + \tau_j$, with $\beta_j = |b_j| \in (0, 1)$. Independent affine changes of source and target coordinates normalize $\min X = \min Y = 0$ and $\text{diam } X = \text{diam } Y = 1$. They preserve the contraction coefficients and conjugate $\mathcal{E} = E(X, Y)$ by an invertible linear-affine parameter map. For words $\mathbf{i}, \mathbf{j}$, write

$$\varphi_{\mathbf{i}}(x) = a^{|\mathbf{i}|}x + \sigma_{\mathbf{i}}, \quad \psi_{\mathbf{j}}(x) = b_{\mathbf{j}}x + \tau_{\mathbf{j}}, \quad \beta_{\mathbf{j}} = |b_{\mathbf{j}}|.$$

The empty word represents the identity. Set

$$\eta = \min_{j \neq l} \text{dist}(\psi_j Y, \psi_l Y) > 0, \quad \beta_- = \min_j \beta_j, \quad \beta_+ = \max_j \beta_j, \quad \rho > \max\{1, \eta^{-1}\}.$$

A compact embedding window is a compact subset $F \subset \mathcal{E}$ in the parameter plane (u, v) for $f(x) = ux + v$. It has $\inf_F |u| > 0$. Every embedding satisfies $|u| \leq 1$ and $v = f(0) \in Y \subset [0, 1]$.

An oriented generalized dyadic grid is the image of the standard grid under a homothety $H(z) = cz + t$, $c \neq 0$, with an integer level shift. Choose the shift so that its level- n side length is $\ell_n = \lambda 2^{-n}$ for one $\lambda \in (1/2, 1]$. For a cell D let z_D be the image of the standard lower corner, and let $\epsilon = \text{sign } c$. Define

$$H_D(z) = \frac{z - z_D}{\epsilon \ell_n}.$$

This sends the cell, with its transported half-open convention, to the unit cube. It also transports the outer-face final-cell convention of IV.1. It is the ordinary positive normalization when $\epsilon = 1$; when $\epsilon = -1$ it retains reflection instead of changing a CP orbit. For $\nu(D) > 0$ put $\nu^D = (H_D)_*(\nu|_D/\nu(D))$. A set or probability has constant first coordinate if it is supported on a line $\{u_0\} \times \mathbb{R}$; u_0 may be zero in the normalized unit cube. For a subset or probability in $\mathcal{E}$ we call this constant signed slope, and necessarily $u_0 \neq 0$. Saying that a probability is on $F \cap D$ means that this Borel set has full mass; its topological support may meet ∂D . The affine formulas below extend continuously to such boundary points. Here $N_d(A, r)$ is the minimum number of open Euclidean radius- r balls covering a bounded set $A \subset \mathbb{R}^d$.

Lemma IV.8.77 (*dimension range for separated sets on the line*).

A nontrivial strongly separated self-similar set $K \subset \mathbb{R}$ has $0 < \dim_H K < 1$ and a probability satisfying a positive-exponent upper Frostman bound.

Proof. Normalize its convex hull to $[0, 1]$. The SSC Moran theorem, Volume I.8, T-I.8.06, gives its dimension $s > 0$ and a canonical s -regular probability. Since $K \subset \mathbb{R}$, $s \leq 1$. We verify strictness without assuming disjoint cylinder hulls. Let $\eta_K > 0$ be the first-level set separation and r_- the minimum contraction ratio. For every word w ,

$$\text{dist}(K_w, K \setminus K_w) \geq \eta_K r_w.$$

An outside address first differs before the end of w , and its common-prefix ratio is at least r_w , proving this bound. The set K is disconnected, so a component of $[0, 1] \setminus K$ is an open interval. Choose a subinterval I adjoining one of its endpoints in K , of length $g > 0$ less than $\eta_K/2$. Every point of $S_w(I)$ is within $gr_w < \eta_K r_w$ of K_w . The separation bound shows that this image misses all of K . Given $x \in K$ and small $R > 0$, take the first address prefix with $r_w \leq R$. Then $r_w > r_- R$, its hull lies in the radius- R interval about x , and this interval contains a hole of length at least $gr_- R$. The Lebesgue density theorem, Theorem IV.2.26 for Lebesgue measure, gives $\text{Leb}(K) = 0$. If $s = 1$, upper regularity gives $\mu(B(x, R)) \leq CR$ also for arbitrary centers, by moving a nonempty ball to a point of K and doubling its radius. Interval covers give $\mu \ll \text{Leb}$, contrary to $\mu(K) = 1$. Thus $s < 1$, and upper regularity is the claimed positive-exponent Frostman bound. $\square$

Lemma IV.8.78 (*a common target cylinder and compact renormalization*).

For a sufficiently fine cell D meeting a compact window F , put $d = 2^{-n}$ and $\chi_D = \sup\{|u| : (u, v) \in \mathcal{E} \cap D\}$. Let $k = k(D)$ be the unique integer satisfying

$$\alpha d / \chi_D \leq \alpha^k < d / \chi_D. \quad (18.4da)$$

For any $\mathbf{i} \in I^k$ the set $Z_{D,\mathbf{i}} = \bigcup_{f \in \mathcal{E} \cap D} f(\varphi_{\mathbf{i}} X)$ is contained in one target cylinder. There is a unique shortest word $\mathbf{j}$ containing this set and satisfying $\beta_{\mathbf{j}} < 3\rho d$. It obeys

$$3\rho\beta_- d \leq \beta_{\mathbf{j}} < 3\rho d. \quad (18.4db)$$

The map $M_{D,\mathbf{i}} f = \psi_{\mathbf{j}}^{-1} f \varphi_{\mathbf{i}}$ sends $\mathcal{E} \cap D$ into the fixed compact set

$$\mathcal{E}_0 = \{(u, v) \in \mathcal{E} : |u| \geq c_0\}, \quad c_0 = \alpha/(6\rho) > 0. \quad (18.4dc)$$

All sufficiently fine cells meeting F satisfy these assertions uniformly over $\mathbf{i}$.

Proof. If $f(x) = ux + v$ and $g(x) = u'x + v'$ belong to the same cell, then $|u - u'|, |v - v'| \leq d$ and $X \subset [0, 1]$, so

$$|f(x) - g(y)| \leq \chi_D |x - y| + 2d.$$

Consequently $\text{diam } Z_{D,\mathbf{i}} \leq 3d$. This set has positive diameter because it contains a non-singular image of the nontrivial set X . Let w be the longest prefix common to the addresses of all its points. It is finite, and the set meets two distinct children of $\psi_w Y$. Hence $\text{diam } Z_{D,\mathbf{i}} \geq \eta\beta_w$. In particular $\beta_w \leq 3d/\eta < 3\rho d$, giving an admissible word. All cylinders containing a fixed nonempty subset of Y have comparable words, by SSC. This proves

uniqueness of the shortest admissible word $\mathbf{j}$. For $3\rho d < 1$ it is nonempty. Its parent also contains the set; minimality gives $\beta_{\mathbf{j}-} \geq 3\rho d$, proving (18.4db).

The cylinder inclusion proves $M_{D,\mathbf{i}}f \in \mathcal{E}$. For fine cells, $d \leq \chi_D/2$, and every slope in $\mathcal{E} \cap D$ has modulus at least $\chi_D - d \geq \chi_D/2$. Its renormalized modulus is thus

$$\frac{|u|\alpha^k}{\beta_{\mathbf{j}}} > \frac{(\chi_D/2)\alpha d/\chi_D}{3\rho d} = c_0. \quad (18.4dd)$$

Its upper bound is 1 by diameter normalization, and its translation is its value at 0, hence lies in Y . The set $\mathcal{E}_0$ is closed in $\{c_0 \leq |u| \leq 1, 0 \leq v \leq 1\}$: limits preserve $ux + v \in Y$ for each $x \in X$. Thus it is compact. A positive lower slope bound on F makes all smallness conditions uniform. $\square$

Lemma IV.8.79 (*projection limits and constant-slope similarities*).

Let $F \subset \mathcal{E}$ be a compact window. There is a compact family of nonsingular affine maps of the line such that, for each sufficiently fine cell D meeting F and allowed word $\mathbf{i}$, two members h_1, h_2 satisfy

$$M_{D,\mathbf{i}}(u, v) = (h_1(u), h_2\pi_{\sigma_{\mathbf{i}}}H_D(u, v)), \quad \pi_x(U, V) = xU + V. \quad (18.4de)$$

If $n_l \rightarrow \infty$, probabilities ν_l on $F \cap D_l$ have $(H_{D_l})_*\nu_l \rightarrow \nu$ and $\sigma_{\mathbf{i}_l} \rightarrow x$, then every convergent subsequence of $(M_{D_l,\mathbf{i}_l})_*\nu_l$ has the form

$$\tilde{\nu} = \delta_z \times h_*(\pi_x)_*\nu, \quad \text{supp } \tilde{\nu} \subset \mathcal{E}_0, \quad (18.4df)$$

with h nonsingular. If F has constant signed slope, there is a compact family $\mathcal{T}_F$ of nonsingular plane homotheties such that

$$(M_{D,\mathbf{i}})_*\nu = T_*(H_D)_*\nu \quad (T \in \mathcal{T}_F) \quad (18.4dg)$$

for each probability ν on $F \cap D$. Their absolute similarity ratios are bounded above and away from zero.

Proof. Direct composition, retaining signs, gives

$$M_{D,\mathbf{i}}(u, v) = \left(\frac{a^k}{b_{\mathbf{j}}}u, \frac{\sigma_{\mathbf{i}}u + v - \tau_{\mathbf{j}}}{b_{\mathbf{j}}} \right).$$

Write $z_D = (u_D, v_D)$ and $\ell_n = \lambda d$. Then (18.4de) holds with

$$h_1(t) = \frac{a^k}{b_{\mathbf{j}}}t, \quad h_2(t) = \frac{\varepsilon\ell_n}{b_{\mathbf{j}}}t + \frac{\sigma_{\mathbf{i}}u_D + v_D - \tau_{\mathbf{j}}}{b_{\mathbf{j}}}. \quad (18.4dh)$$

If $m = \inf_F |u| > 0$, then $m \leq \chi_D \leq 1$. Equations (18.4da)–(18.4db) bound the absolute slope of h_1 between $\alpha/(3\rho)$ and $1/(3\rho\beta_-m)$. The absolute slope of h_2 lies between $1/(6\rho)$ and $1/(3\rho\beta_-)$. For $(u, v) \in F \cap D$, $u\sigma_{\mathbf{i}} + v \in \psi_{\mathbf{j}}Y$ because $\sigma_{\mathbf{i}} = \varphi_{\mathbf{i}}(0) \in \varphi_{\mathbf{i}}X$. Its normalized value after subtracting $\tau_{\mathbf{j}}$ and dividing by $b_{\mathbf{j}}$ lies in Y . Replacing (u, v) by z_D changes that value by at most $2d/\beta_{\mathbf{j}} \leq 2/(3\rho\beta_-)$. This bounds the translation of h_2 . Closed bounds give the compact affine family.

Choose $(u_l, v_l) \in F \cap D_l$. On the l th image the first coordinate differs from $z_l = h_{1,l}(u_l)$ by $O(2^{-n_l})$. Pass to a subsequence with $z_l \rightarrow z$ and $h_{2,l} \rightarrow h$. For any bounded Lipschitz test, replacing the first coordinate by z_l changes its integral by $O(2^{-n_l})$. Uniform convergence of $h_{2,l}\pi_{\sigma_{j_l}}$ on the unit cube and weak convergence of the magnified measures prove (18.4df). Closedness of $\mathcal{E}_0$ gives its support assertion.

Finally let F have constant slope u_0 and put $U_0 = (u_0 - u_D)/(\epsilon \ell_n) \in [0, 1]$. With $r = \epsilon \ell_n / b_j$ define

$$T(U, V) = r(U, V) + \left(\frac{a^k u_0}{b_j} - rU_0, \frac{\sigma_i u_0 + v_D - \tau_j}{b_j} \right). \quad (18.4di)$$

Substitution on $U = U_0$ gives (18.4dg). The preceding bounds, $U_0 \in [0, 1]$ and bounded output slopes bound its translation. Its absolute ratio has the two bounds already proved for h_2 . This proves compactness, including possibly negative ratios. $\square$

Lemma IV.8.80 (*the stopping-word clock and positive-density sampling*).

Fix $f \in \mathcal{E}$, an oriented grid and an infinite source word. For sufficiently large n , form $D_n = D_n(f)$, $k(n)$, the source prefix $\mathbf{i}_n$, the shortest target word $\mathbf{j}_n$ and the renormalized slope u_n . There are integers $L, B \geq 1$, depending only on β_-, β_+ , and levels $n_1 < n_2 < \dots$ with gaps at most L , such that $\mathbf{j}_{n_i}$ strictly extend one another with length increments between 1 and B . The phases

$$t_i = \frac{\log |u_{n_i}|}{\log(1/\alpha)} \pmod{1} \quad (18.4dj)$$

are terms of a Λ -multi-rotation at indices $m_i = |\mathbf{j}_{n_i}|$, whose gaps are at most B . If $\mathbb{N} \setminus U$ has upper density less than $1/(2L)$, then $\{m_i : n_i \in U\}$ has positive lower density. Condition (R_*) consequently implies that $\{|u_{n_i}| : n_i \in U\}$ has positive upper box dimension.

Proof. For fine cells, $\chi_{D_{n+1}} \leq \chi_{D_n}$ and $\chi_{D_{n+1}} \geq \chi_{D_n} - 2^{-n} \geq \chi_{D_n}/2$. Thus $T_n = 2^{-n}/\chi_{D_n}$ satisfies $T_n/2 \leq T_{n+1} \leq T_n$. The interval rule (18.4da) implies that $k(n)$ is nondecreasing, and $k(n) \rightarrow \infty$. The source prefixes and cells are nested, hence so are $Z_{D_n, \mathbf{i}_n}$. Their target words are comparable. If $\mathbf{j}_{n+1}$ were a proper prefix of $\mathbf{j}_n$, it would contain the old set and satisfy the old, weaker size threshold, contradicting minimality. Therefore it extends $\mathbf{j}_n$.

Put $\ell = |\mathbf{j}_{n+1}| - |\mathbf{j}_n|$. By (18.4db), $\beta_{\mathbf{j}_{n+1}}/\beta_{\mathbf{j}_n} > \beta_-/2$, whereas this ratio is at most β_+^ℓ . Choose $B \geq 1$ with $\beta_+^{B+1} < \beta_-/2$; then $\ell \leq B$. If a target word stays unchanged from level n through $n+h$, (18.4db) gives $\beta_- < 2^{-h}$. Choose $L = 2 + \lceil \log_2(1/\beta_-) \rceil$. The levels after strict target-word extensions form a sequence n_i with gaps at most L . Between successive such levels there is exactly one extension, so $1 \leq m_{i+1} - m_i \leq B$. The nested words define an infinite target address $j_1 j_2 \dots$. For $A = \log(1/\alpha) > 0$, exact slopes give

$$\frac{\log |u_n|}{A} = \frac{\log |f'|}{A} - k(n) + \sum_{h=1}^{|\mathbf{j}_n|} \frac{\log \beta_{j_h}}{\log \alpha}. \quad (18.4dk)$$

Thus $\vartheta_m = \log |f'|/A + \sum_{h=1}^m \log \beta_{j_h} / \log \alpha$ modulo one is the required multi-rotation and $t_i = \vartheta_{m_i}$. Using A , rather than $\log \alpha$, gives the positive increments.

Write $\zeta < 1/(2L)$ for the upper density of the complement of U . Up to a large level N there are at least $N/L - O(1)$ extension levels, and at most $(\zeta + o(1))N$ of them lie outside

U . At least $(1/L - \zeta - o(1))N$ are retained. Given a large target index M , take i with $m_i \leq M < m_{i+1}$. From $m_i \leq m_1 + B(i-1)$ one gets $i \geq M/B - O(1)$; also $n_i \geq i - O(1)$. Apply the preceding count at $N = n_i$ to obtain

$$\#\{m_h \leq M : n_h \in U\} \geq (1/L - \zeta - o(1))M/B - O(1).$$

This proves positive lower density without asserting existence of an ordinary density for an intersection. Condition (R_*) gives positive box dimension for the retained circle phases. Reduction modulo one is Lipschitz, so their bounded real lifts have at least that dimension. The map $q \mapsto \log q/A$ is bi-Lipschitz on $[c_0, 1]$, giving the asserted dimension of slope moduli. $\square$

The clock follows target-word extensions, not changes of source length. It does not require the implication that unchanged source length implies unchanged target word. The compact homothety family in Lemma IV.8.79 will be used through uniform scale bounds, not equality of covering numbers at a fixed numerical radius.

15. From parameter tangents to constant-slope fibres

The projection limit has a useful consequence: a high-dimensional parameter tangent produces a large vertical fibre of actual embeddings. Applying CP extraction again to that fibre retains its dimension while placing typical measures, up to uniformly controlled homotheties, inside the compact window $\mathcal{E}_0$.

Lemma IV.8.81 (*amplification into a constant-slope fibre*).

For a nonempty compact $F \subset \mathcal{E}$ there is a nonempty compact constant-slope $G \subset \mathcal{E}_0$ such that

$$\dim_A G \geq \min\{1, \dim_A F\}. \quad (18.4dl)$$

Proof. Let $s = \dim_A F$ and let P be the ergodic CP measure marginal from Theorem IV.8.48, using the standard grid. Choose $\nu \in \text{supp } P$ that is exact dimensional of dimension s . Then $\dim_H \text{supp } \nu \geq s$ by the local-dimension criterion of Volume I, and $\dim_A \text{supp } \nu \geq s$. Apply Theorem IV.8.75 to the compact support. The parametrization $x \mapsto (x, 1)/\sqrt{1+x^2}$ is bi-Lipschitz on $[0, 1]$, and π_x differs from this unit-direction projection only by the nonzero scalar $\sqrt{1+x^2}$. Thus its exceptional parameters in $[0, 1]$ have Hausdorff dimension zero. Lemma IV.8.77 gives a positive-exponent Frostman law on X , so some $x \in X$ satisfies

$$\dim_A \pi_x \text{supp } \nu \geq \min\{s, 1\}.$$

We spell out the realization used with this particular ν . The CP extraction theorem gives empirical laws of normalized restrictions on F converging to P , with all initial levels tending to infinity. Each open neighborhood of ν has positive P -mass. Portmanteau therefore supplies a term of an empirical law in that neighborhood. Using neighborhoods of radius $1/l$ and initial levels at least l gives cells D_l and probabilities ν_l on $F \cap D_l$ with $(H_{D_l})_* \nu_l \rightarrow \nu$ and $n_l \rightarrow \infty$. Choose an address of x and its prefixes of length $k(D_l)$. Their translations σ_{i_l} tend to x : both x and $\varphi_{i_l}(0)$ belong to the same source cylinder, of diameter

$\alpha^{k(D_l)} \rightarrow 0$. Compactness gives a subsequential renormalized limit. By Lemma IV.8.79 it is $\tilde{\nu} = \delta_z \times h_*(\pi_x)_*\nu$, supported on $\mathcal{E}_0$. For a continuous map of a compact support, $\text{supp}(h_*(\pi_x)_*\nu) = h\pi_x \text{supp } \nu$: an open neighborhood of an image point has a positive-mass preimage neighborhood. Consequently $G = \text{supp } \tilde{\nu}$ is compact, has constant signed slope, and satisfies (18.4dl) by similarity invariance of Assouad dimension. $\square$

Lemma IV.8.82 (*a realizable CP law for a constant-slope window*).

Let $G \subset \mathcal{E}$ be nonempty, compact and of constant signed slope, and let $s = \dim_A G$. There are an ergodic dyadic CP measure marginal P of dimension s and a compact family $\mathcal{T}$ of nonsingular plane homotheties such that for P -almost every ν : ν has constant first coordinate, and for some $T \in \mathcal{T}$, $T_*\nu$ has constant signed slope and is supported on $\mathcal{E}_0$.

Proof. Use Theorem IV.8.48 on G . For every $\nu \in \text{supp } P$, the Portmanteau selection in the preceding proof gives probabilities ν_l on $G \cap D_l$, with $n_l \rightarrow \infty$ and $(H_{D_l})_*\nu_l \rightarrow \nu$. Each normalized restriction has constant first coordinate. This property is weakly closed: its first marginal is a Dirac mass at a point of $[0, 1]$, and subsequential limits of those points give the marginal of the limit. By (18.4dg), its renormalization is $(T_l)_*(H_{D_l})_*\nu_l$, with T_l in a compact family independent of l and of ν . Pass to $T_l \rightarrow T$. The limiting measure $T_*\nu$ is supported on the closed set $\mathcal{E}_0$. It has constant first coordinate, which is nonzero because of that support inclusion. The CP law is ergodic and s -dimensional by its extraction theorem. $\square$

Lemma IV.8.83 (*a maximal embedding dimension attained in a compact window*).

Assume $\mathcal{E} \neq \emptyset$ and define

$$\delta = \sup\{\dim_A F : \emptyset \neq F \subset \mathcal{E} \text{ compact}\}.$$

Then $\delta < 1$, $\dim_A \mathcal{E}_0 = \delta$, and there are an ergodic dyadic CP law P of dimension δ and a compact homothety family $\mathcal{T}$ with the properties of Lemma IV.8.82.

Proof. The value is in $[0, 2]$. Apply Lemma IV.8.81 to compact windows with dimensions approaching the supremum. It follows that $\mathcal{E}_0$ is nonempty and $\dim_A \mathcal{E}_0 \geq \min\{1, \delta\}$. Apply that lemma once more to $\mathcal{E}_0$ to obtain a compact constant-slope $G \subset \mathcal{E}_0$ with $\dim_A G \geq \min\{1, \delta\}$. If $\delta \geq 1$, then $\dim_A G = 1$, because G lies on a line. Lemma IV.8.82 gives an exact-dimensional one-dimensional probability whose image under a nonsingular homothety has support $\{u_0\} \times T \subset \mathcal{E}_0$ of Hausdorff dimension at least 1. Evaluation at $0 \in X$ gives $T \subset Y$. This contradicts $\dim_H Y < 1$ from Lemma IV.8.77. Therefore $\delta < 1$. Since $\mathcal{E}_0$ itself is compact, the definition of the supremum now implies $\dim_A \mathcal{E}_0 = \delta$. Another application of L04 gives a compact constant-slope $G \subset \mathcal{E}_0$ of dimension exactly δ . Apply L05 to it. No limit of CP laws with varying dimensions, and no unproved attainment of the original supremum, is used. $\square$

16. Uniform fibre size and the final dimension contradiction

Exact dimensionality gives lower covering bounds separately for each typical measure. We need one common error function for many measures, since the final set is a union of their supports. Finite dyadic mass counts make this uniformization measurable.

Lemma IV.8.84 (*uniform covering bounds on a large CP subset*).

Let P be a probability on probabilities on $[0, 1]^2$, almost all exact dimensional of dimension $\delta \geq 0$. For every $\zeta > 0$ there is a Borel set Ω with $P(\Omega) > 1 - \zeta$ and a function $e : (0, 1) \rightarrow [0, \infty)$, $e(r) \rightarrow 0$ as $r \downarrow 0$, such that

$$N_2(\text{supp } \nu, r) \geq r^{-\delta+e(r)} \quad (\nu \in \Omega, 0 < r < 1). \quad (18.4\text{dm})$$

For any fixed compact family $\mathcal{F}$ of nonsingular homotheties, the images $T_*\nu$, $\nu \in \Omega$, satisfy a bound of the same form with one new function tending to zero. If P is an ergodic dyadic CP measure marginal, then for P -almost every ν and ν -almost every x the set

$$U = \{n : \nu^{D_n(x)} \in \Omega\}$$

has density $P(\Omega)$.

Proof. Use the finite level- q partition of the closed cube with its outer-face final cells, and set $M_q(\nu) = \#\{D : \nu(D) > 0\}$. Evaluations on Borel cells are Borel functions of ν , so M_q is Borel. The closures of positive-mass cells cover $\text{supp } \nu$: their finite union is closed and has full mass. Each radius- 2^{-q} ball meets at most 16 such cells. Conversely, the closed cells give a cover by M_q balls of radius $2 \cdot 2^{-q}$. Thus the standard dimension inequalities of Volume I and $\dim_H \text{supp } \nu \geq \delta$ give

$$\liminf_{q \rightarrow \infty} \frac{\log_2 M_q(\nu)}{q} \geq \delta, \quad N_2(\text{supp } \nu, 2^{-q}) \geq M_q(\nu)/16.$$

The Borel functions $b_m(\nu) = \inf_{q \geq m} \log_2 M_q(\nu)/q$ increase to a limit at least δ on a conull set. For each integer $l \geq 1$, choose m_l increasing so that

$$P\{b_{m_l} < \delta - 1/l\} < \zeta/2^{l+1}.$$

Their complement intersection is a Borel set Ω of mass greater than $1 - \zeta$. For $m_l \leq q < m_{l+1}$ its measures have $M_q \geq 2^{q(\delta-1/l)}$. Absorb the factor 16 by $4/q$ in the exponent, and use monotonicity between adjacent dyadic radii. This gives (18.4dm) with an error tending to zero. At the finitely many larger radii set $e(r) \geq \delta$, so that the lower bound is at most 1.

If the absolute homothety ratios lie in $[c, C]$ with $c > 0$, then $N_2(T \text{supp } \nu, r) = N_2(\text{supp } \nu, r/|T'|)$. For small r , (18.4dm) and $r/|T'| \in [r/C, r/c]$ give a common error tending to zero, by taking the supremum of e on this shrinking interval and adding the bounded $|\log c|, |\log C|$ losses divided by $\log(1/r)$. This is uniform, not equality at radius r .

For the last assertion apply the Birkhoff theorem IV.1.33 to 1_Ω on the pointed CP system. Ergodicity makes its limit $P(\Omega)$. Adaptedness disintegrates the conull pointed set into a conull family of measures and conull sets of their points. The iterated pointed magnification is exactly $(\nu^{D_n(x)}, H_{D_n(x)}x)$ by the CP construction of IV.1. This proves the stated density for the Borel event Ω , without replacing it by a continuous test. $\square$

Lemma IV.8.85 (*adding uniform fibre size to base box dimension*).

Let $Z \subset \mathbb{R}^2$ be nonempty and bounded, and let $\pi(u, v) = u$. For $u \in \pi Z$ write $Z_u = Z \cap (\{u\} \times \mathbb{R})$. Suppose $\gamma \geq 0$ and $e(r) \rightarrow 0$ satisfy $N_2(Z_u, r) \geq r^{-\gamma+e(r)}$ for every $u \in \pi Z$ and all sufficiently small $r > 0$. Then

$$\overline{\dim}_B Z \geq \gamma + \overline{\dim}_B \pi Z. \quad (18.4dn)$$

Neither Z nor πZ is required to be closed.

Proof. Choose a maximal finite set of base points separated by more than $4r$. It covers πZ by closed radius- $4r$ intervals, hence by open radius- $5r$ intervals. Its cardinality M is at least $N_1(\pi Z, 5r)$. An open radius- r ball in the plane meets at most one of the selected fibres. Every cover of Z consequently uses at least $N_2(Z_u, r)$ balls on each of these M fibres, giving

$$N_2(Z, r) \geq N_1(\pi Z, 5r)r^{-\gamma+e(r)}.$$

Divide its logarithm by $\log(1/r)$ and take a limsup. Changing r to $5r$ does not change upper box dimension, and $e(r) \rightarrow 0$. This proves (18.4dn), including $\gamma = 0$. $\square$

One remaining bookkeeping step is essential. The CP law lives on the normalized unit cube, whereas its homothetic image consists of actual embeddings. We transport the partition together with the measure; we do not magnify the image in an unrelated grid.

Lemma IV.8.86 (*transporting the normalized CP orbit*).

Let $T(z) = cz + t$, $c \neq 0$, and let ν be a probability on the closed unit cube, with its dyadic outer-face convention. Put $\mu = T_*\nu$ and choose the integer n_0 so that $\lambda = |c|2^{-n_0} \in (1/2, 1]$. Transport the level- $(n + n_0)$ dyadic partition by T , including its half-open and outer-face conventions, and call the resulting partition $\mathcal{D}'_n$. For g in the unit cube and $f = Tg$, put $D = D_{n+n_0}(g)$ and $D'_n(f) = T(D)$. If $\nu(D) > 0$, then

$$H_{D'_n(f)} \circ T = H_D, \quad \mu(D'_n(f)) = \nu(D), \quad \mu^{D'_n(f)} = \nu^{D_{n+n_0}(g)}. \quad (18.4do)$$

These identities hold for all sufficiently large n , also if $c < 0$. Consequently a CP visitation frequency along g is preserved along f , up to a finite shift of the indices.

Proof. The image cells have side length $|c|2^{-(n+n_0)} = \lambda 2^{-n}$. If z_D is the transported lower corner of the standard cell, its image corner is $cz_D + t$ and its orientation is $\epsilon = \text{sign } c$. Definition IV.8.76 therefore gives, for every $z \in \mathbb{R}^2$,

$$H_{T(D)}(Tz) = \frac{cz + t - (cz_D + t)}{\epsilon|c|2^{-(n+n_0)}} = 2^{n+n_0}(z - z_D) = H_D(z).$$

Transported boundary conventions ensure that $g \in D$ if and only if $Tg \in T(D)$, including the final outer faces. Since T is injective, restriction commutes with push-forward: $\mu|_{T(D)} = T_*(\nu|_D)$. Dividing by the common positive mass and using the first identity proves (18.4do). Omitting finitely many levels or translating a sequence of visit indices by n_0 does not change its density, proving the last assertion. $\square$

Suspicious source formula IV.8.87 (The index in a transported-grid identity). The coordinate-change paragraph of the published AHW proof, Section 5.2, defines $\mathcal{D}'_n = T(\mathcal{D}_{n+n_0})$ but prints the normalized-measure index $n - n_0$. The universal algebraic reading of that local identity is

$$(T_*\nu)^{D'_n(Tg)} = \nu^{D_{n-n_0}(g)}, \quad D'_n(Tg) = T(D_{n+n_0}(g)), \quad |c|2^{-n_0} \in (1/2, 1]. \quad (18.4dp)$$

Here the local hypotheses are those of the preceding lemma: $T(z) = cz + t$, $c \neq 0$, a unit-cube probability ν , transported cell conventions, and positive masses for the indicated cells. This algebraic reading is false, as the next example shows. The correct index is $n + n_0$, proved in (18.4do). A complete read of the published Sections 1–6 and references finds no alternative grid convention that reverses this index while retaining the displayed definition of $\mathcal{D}'_n$. This record concerns only the local transport identity, not a counterexample to Theorem 1.5 under its embedding hypotheses. Both fixed shifts preserve visitation density, so the discrepancy does not invalidate the source's dimension-contradiction mechanism.

Example IV.8.88 (*an exact test of the transported-grid index*).

Take $T(z) = z/4$, $\nu = \delta_g$ with $g = (1/7, 0)$, $n_0 = -2$ and $n = 3$. Every local hypothesis in IV.8.87 holds, but (18.4dp) fails.

Proof. The map is a nonsingular positive homothety and $|c|2^{-n_0} = (1/4)2^2 = 1 \in (1/2, 1]$. The probability is on the unit cube. The level-one cell containing g is $D_1(g) = [0, 1/2)^2$, with positive mass one. Its image is $D'_3(Tg) = [0, 1/8)^2$, also of mass one. The point $Tg = (1/28, 0)$ normalizes there to $(2/7, 0)$, so

$$(T_*\nu)^{D'_3(Tg)} = \delta_{(2/7, 0)}.$$

The claimed other index is $3 - (-2) = 5$. Its normalized first coordinate is the fractional part of $2^5/7 = 32/7$, namely $4/7$; the second coordinate is zero. Thus

$$\nu^{D_5(g)} = \delta_{(4/7, 0)} \neq \delta_{(2/7, 0)}.$$

No boundary ambiguity is involved. In addition, the normalized Dirac measures at $(1/7, 0)$, $(2/7, 0)$ and $(4/7, 0)$ form a period-three pointed dyadic orbit. Its uniform orbit law is adapted, invariant and ergodic, so the test is compatible with being a typical measure of an ergodic zero-dimensional CP law. No assertion is made that it satisfies the contradictory no-embedding setup of the theorem. $\square$

We can now combine the two kinds of size information. The retained CP visits give a uniform amount of vertical complexity, while the target-word clock supplies a positive-dimensional set of slopes. Their union remains inside the same compact embedding window.

Theorem IV.8.89 (*renormalization obstruction to affine embeddings*).

This is the source-contract form of AHW Theorem 1.5, now proved below from the preceding local lemmas and the internally proved CP and Orponen suppliers. Let $X, Y \subset \mathbb{R}$ be nontrivial strongly separated self-similar sets. Assume X is homogeneous with a common signed contraction coefficient $a \neq 0$ and $\alpha = |a|$, form Λ as in (18.4), and suppose

- (i) Λ satisfies Condition (R_*) (or the stronger R^*),
- (ii) $1 \notin \text{span}_{\mathbb{Q}} \Lambda$.

Then

$$E(X, Y) = \emptyset. \quad (18.8)$$

Proof. Suppose $\mathcal{E} = E(X, Y) \neq \emptyset$. Normalize source and target as in Definition IV.8.76. Lemma IV.8.83 supplies

$$\delta = \sup_{\substack{\emptyset \neq F \subset \mathcal{E} \\ F \text{ compact}}} \dim_A F = \dim_A \mathcal{E}_0 < 1$$

and an ergodic dyadic CP marginal P of dimension δ , whose typical measures have constant first coordinate and admit a homothety into $\mathcal{E}_0$. Let L be the level-gap bound of Lemma IV.8.80. Apply Lemma IV.8.84 with $\zeta = 1/(4L)$. It gives a Borel set Ω , $P(\Omega) > 1 - 1/(4L)$, and one covering error tending to zero for its measures.

Choose ν simultaneously in the conull sets of L06 and L07, and choose a corresponding nonsingular homothety T such that $\mu = T_*\nu$ is supported on $\mathcal{E}_0$ and has constant signed slope. Its support $F = \text{supp } \mu$ is a nonempty compact constant-slope window. Choose a ν -typical g and set $f = Tg$. All its dyadic cells have positive ν -mass; intersecting the countably many full-mass unions of positive-mass cells ensures this. Use Lemma IV.8.86 to transport the grid by T . With its shift n_0 , the set

$$U = \{n : \mu^{D'_n(f)} \in \Omega\} = \{n : \nu^{D_{n+n_0}(g)} \in \Omega\}$$

has density $P(\Omega)$. In particular its complement has upper density less than $1/(2L)$.

Fix any infinite source address. For every sufficiently large n , let $\mathbf{i}_n$ be its prefix of the stopping length for $D'_n(f)$, and define the actual renormalized restriction

$$\eta_n = (M_{D'_n(f), \mathbf{i}_n})_* \left(\frac{\mu|_{D'_n(f)}}{\mu(D'_n(f))} \right). \quad (18.4dq)$$

By Lemmas IV.8.78 and IV.8.79, η_n is supported on $\mathcal{E}_0$, has constant signed slope u_n , and

$$\eta_n = (S_n)_* \mu^{D'_n(f)}, \quad S_n \in \mathcal{T}_F, \quad (18.4dr)$$

where $\mathcal{T}_F$ is one fixed compact family of nonsingular homotheties, independent of n . Since f belongs to the same constant-slope support, u_n is also the first coordinate of $M_{D'_n(f), \mathbf{i}_n}(f)$. The uniform transport clause of L07 therefore supplies a single $e(r) \rightarrow 0$ such that

$$N_2(\text{supp } \eta_n, r) \geq r^{-\delta+e(r)} \quad (n \in U \text{ sufficiently large, } r \text{ small}). \quad (18.4ds)$$

By L03 and Condition (R_*) , there are extension levels n_i , with bounded gaps, for which the retained modulus set

$$A = \{|u_{n_i}| : n_i \in U\}$$

has upper box dimension $\kappa > 0$. Discarding finitely many levels does not affect this conclusion: a finite union has upper box dimension equal to the maximum of its constituent dimensions. Put

$$Z = \bigcup_{\substack{i: n_i \in U \\ n_i \text{ sufficiently large}}} \text{supp } \eta_{n_i} \subset \mathcal{E}_0.$$

The set is bounded; it need not be closed. Each fibre of its first-coordinate projection contains one retained support, so (18.4ds) is the uniform fibre bound required by Lemma IV.8.85. Moreover $|\pi_1 Z| = A$ after the finite discard, and absolute value is Lipschitz. Hence

$$\overline{\dim}_B \pi_1 Z \geq \kappa > 0, \quad \overline{\dim}_B Z \geq \delta + \kappa > \delta.$$

But $Z \subset \mathcal{E}_0$ implies $\overline{\dim}_B Z \leq \dim_A \mathcal{E}_0 = \delta$, a contradiction. This also covers $\delta = 0$. Thus $\mathcal{E} = \emptyset$. $\square$

The proof uses robustness (R_*) and does not need to infer it from the rational-span hypothesis. The latter is retained to match the registered source-contract statement; no stronger theorem is asserted here. The new terminal transport-index record is not a premise of this proof.

Source theorem: Algebraic contraction ratios

Source boundary

The following statement is Theorem 1.2 of the same paper. Its proof remains in the cited source.

Under the same strong-separation setup, the Feng–Huang–Rao affine embedding conjecture holds when all defining contraction ratios of X and Y are algebraic: if some source log contraction does not belong to the $\mathbb{Q}$ -span of the target log contractions, then $E(X, Y) = \emptyset$.

17. An elementary (R_*) check for one irrational rotation

Lemma IV.8.90 (*positive-density sampling of an irrational rotation*).

Let $\lambda \in \mathbb{R} \setminus \mathbb{Q}$ and

$$\theta_k = \theta_0 + k\lambda \pmod{1}. \quad (18.9)$$

If $U \subset \mathbb{N}$ has positive lower density $\delta > 0$, then the closure K of

$$\{\theta_k : k \in U\} \quad (18.10)$$

has Lebesgue measure at least δ . In particular the selected set has upper box dimension one.

Proof. Let O be a finite union of open intervals containing K , chosen so that ∂O has Lebesgue measure zero. Every $k \in U$ satisfies $\theta_k \in O$, hence

$$\underline{d}(U) \leq \lim_{N \rightarrow \infty} \frac{1}{N} \#\{0 \leq k < N : \theta_k \in O\}. \quad (18.11)$$

Irrational rotation is uniquely ergodic, so the limit equals $\text{Leb}(O)$. Thus $\text{Leb}(O) \geq \delta$ for every such O . Outer regularity and approximation of the compact set K by finite interval unions give

$$\text{Leb}(K) \geq \delta. \quad (18.12)$$

A compact subset of the circle with positive Lebesgue measure has upper box dimension one. $\square$

Corollary IV.8.91 (*singleton irrational Λ satisfies (R_*)*).

If $\Lambda = \{\lambda\}$ with λ irrational, every Λ -multi-rotation is an irrational rotation orbit, and Condition (R_*) holds.

Proof. With one allowed increment, $\theta_{k+1} - \theta_k = \lambda \pmod{1}$ for every k . Apply Lemma IV.8.90. $\square$

18. Local AHW examples

Let

$$C_3 = \left\{ \sum_{k \geq 1} \varepsilon_k 3^{-k} : \varepsilon_k \in \{0, 2\} \right\}, \quad C_4 = \left\{ \sum_{k \geq 1} \varepsilon_k 4^{-k} : \varepsilon_k \in \{0, 3\} \right\}. \quad (18.13)$$

These are the standard two-map strongly separated homogeneous Cantor sets of ratios $1/3$ and $1/4$.

Proposition IV.8.92 (*a nontrivial AHW example: C_4 does not embed in C_3*).

There is no nonsingular affine map f with

$$f(C_4) \subset C_3. \quad (18.14)$$

Proof. Here the source contraction is $\alpha = 1/4$ and the target has the single contraction $\beta = 1/3$, so

$$\Lambda = \{\lambda\}, \quad \lambda = \frac{\log(1/3)}{\log(1/4)} = \frac{\log 3}{\log 4}. \quad (18.15)$$

The number λ is irrational: a rational relation would give $3^p = 4^q$ for positive integers p, q after clearing denominators, impossible by prime factorization. Hence Corollary IV.8.91 proves (R_*) .

For singleton Λ , membership $1 \in \text{span}_{\mathbb{Q}} \Lambda$ would force λ to be rational, which has just been excluded. Thus both hypotheses of the Algom–Hochman–Wu theorem hold and $E(C_4, C_3) = \emptyset$.

The companion algebraic-contraction theorem gives the same conclusion because $1/3$ and $1/4$ are algebraic and $\log(1/4) \notin \text{span}_{\mathbb{Q}} \{\log(1/3)\}$. This cross-check is not an additional proof premise needed above. The direction is substantive: $\dim_H C_4 = 1/2 < \log 2 / \log 3 = \dim_H C_3$, so dimension monotonicity alone does not rule out the proposed embedding. $\square$

Proposition IV.8.93 (*arithmetic hypothesis failure: identity embedding*).

Let $X = Y = C_3$. Then $E(X, Y)$ is nonempty because it contains the identity, while

$$1 \in \text{span}_{\mathbb{Q}}\{\log(1/3)/\log(1/3)\} = \text{span}_{\mathbb{Q}}\{1\}. \quad (18.16)$$

Thus the arithmetic noncommensurability hypothesis in the renormalization theorem is essential to its stated no-embedding conclusion.

Proof. Immediate from $\text{id}(C_3) = C_3$ and (18.16). $\square$

19. What the examples prove

Theorem IV.8.94 (*checked applications and hypothesis failures*).

The two cited rigidity theorems have explicit, locally verified examples.

- **Planar theorem, applicable.** The $(2, 3)$ -carpet with digits $\{(0, 0), (1, 0), (0, 2)\}$ is non-product with multiplicatively independent bases, so it has no strict contracting similarity self-embedding.
- **Planar theorem, failure of a hypothesis.** The full $(2, 3)$ product carpet is $[0, 1]^2$ and admits the strict similarity $z \mapsto z/2$; the non-product hypothesis is therefore substantive.
- **Renormalization theorem, applicable.** The set C_4 does not affinely embed in C_3 , despite having smaller Hausdorff dimension. The singleton (R_*) condition is proved locally from positive-density irrational-rotation sampling, and the $\mathbb{Q}$ -span condition is checked by prime factorization.
- **Renormalization theorem, failure of a hypothesis.** Taking $X = Y = C_3$ violates the arithmetic condition and gives the identity embedding.

Proof. Combine Propositions IV.8.46–IV.8.47, Lemma IV.8.90, Corollary IV.8.91, and Propositions IV.8.92–IV.8.93 with the first two source rigidity theorems stated above. The algebraic-ratio theorem is a comparison result and is not a premise of these examples. $\square$

20. A renormalization frontier

Question. Let X, Y be nontrivial strongly separated self-similar subsets of $\mathbb{R}$ with X homogeneous. Assume

$$1 \notin \text{span}_{\mathbb{Q}} \Lambda, \quad (18.17)$$

but do not assume algebraicity of contraction ratios and do not assume (R_*) . Determine additional verifiable conditions, weaker than (R_*) , under which the elementary renormalization orbit of one hypothetical embedding forces enough size in $E(X, Y)$ to contradict a dimension bound.

Boundary of the question. The Algom–Hochman–Wu theorem answers it under (R_*) , and the companion algebraic theorem answers the algebraic-contraction case. The elementary renormalization argument alone constructs only normalized descendants. The question remains open in the present level of generality.

21. What the examples establish

The examples verify the hypotheses and consequences of the cited rigidity theorems in concrete systems. They do not prove the source theorems themselves. Conversely, the elementary renormalization results in the first part of the chapter do not depend on those research theorems.

Chapter IV.9

Resonance, Sums, and Arithmetic Thickness

Non-resonance, uniform non-flatness, and self-conformal sumsets

Non-resonance and arithmetic thickness meet here. The higher-dimensional self-conformal continuation is developed at the original regularity. In the plane, the positive-thickness and connected analytic-curve branches are proved, but the unrestricted continuation is rejected by an explicit example satisfying the stated conformal-IFS hypotheses.

9.1 Peres-Shmerkin non-resonance

1. Definitions and the retained theorem

A self-similar set $K \subset \mathbb{R}$ is the attractor of a finite IFS

$$\mathcal{F} = \{f_i(x) = r_i x + t_i\}_{i=1}^m, \quad (19.1)$$

where $0 < |r_i| < 1$. No separation hypothesis is imposed unless explicitly stated.

Definition IV.9.1 (*scaled arithmetic sum*).

For compact $A, B \subset \mathbb{R}$ and $s \in \mathbb{R}$ put

$$A + sB = \{a + sb : a \in A, b \in B\}. \quad (19.2)$$

Definition IV.9.2 (*algebraic non-resonance*).

Two one-dimensional self-similar IFSs $\mathcal{F}, \mathcal{F}'$ are algebraically non-resonant if there are indices i, j such that

$$\frac{\log |r_i|}{\log |r'_j|} \notin \mathbb{Q}. \quad (19.3)$$

Main theorem. Let K, K' be arbitrary self-similar subsets of $\mathbb{R}$ generated by finite IFSs of the form (19.1) and its primed analogue. If (19.3) holds, then for every $s \neq 0$,

$$\dim_{\text{H}}(K + sK') = \min\{\dim_{\text{H}} K + \dim_{\text{H}} K', 1\}. \quad (19.4)$$

The proof is assembled after the homogeneous approximation lemmas below.

2. Upper dimension bounds for self-similar sums

Lemma IV.9.3 (*local self-copy property*).

Let K be a non-singleton self-similar set and normalize $\text{diam } K = 1$. Put

$$r_* := \min_i |r_i| > 0.$$

For every $x \in K$ and $0 < \delta < 1$ there is a cylinder map f_u such that

$$x \in f_u(K) \subset B(x, \delta), \quad r_* \delta \leq \text{diam } f_u(K) \leq \delta. \quad (19.5)$$

Proof. Choose a coding ω of x and let $u = \omega|n$ be the first prefix with $\text{diam } f_u(K) \leq \delta$. The parent cylinder has diameter greater than δ . Multiplication by the last contraction gives the lower bound $r_* \delta$. Since $x \in f_u(K)$ and the latter has diameter at most δ , it lies in $B(x, \delta)$. $\square$

Lemma IV.9.4 (*self-similar Hausdorff and upper box dimensions agree*).

For every self-similar set $K \subset \mathbb{R}$,

$$\overline{\dim}_B K = \dim_H K. \quad (19.6)$$

Proof. The inequality $\dim_H K \leq \overline{\dim}_B K$ is general. For the reverse direction let $d = \overline{\dim}_B K$ and choose $0 < t < d$. We show $\dim_H K \geq t$.

Take $t < t_0 < d$. Along a sequence $\delta_j \downarrow 0$, K contains a δ_j -separated set of cardinality

$$M_j \geq \delta_j^{-t_0}. \quad (19.7)$$

Fix a sufficiently small δ from this sequence and points $x_1, \dots, x_M$ with $M \geq \delta^{-t_0}$. Apply Lemma IV.9.3 at radius $\delta/3$. We obtain cylinders $f_{u_i}(K)$ satisfying

$$f_{u_i}(K) \subset B(x_i, \delta/3), \quad c\delta \leq |r_{u_i}| \leq \delta/3, \quad (19.8)$$

where $c = r_*/3$ after the diameter normalization.

The balls $B(x_i, \delta/3)$ are pairwise disjoint, so the finite subsystem $\{f_{u_i}\}$ satisfies strong separation. Let H be its attractor. For a strongly separated self-similar subsystem, the elementary Moran formula gives the dimension s_δ determined by

$$\sum_i |r_{u_i}|^{s_\delta} = 1. \quad (19.9)$$

For completeness, the upper bound follows from the cylinder cover. For the lower bound put $p_i = |r_{u_i}|^{s_\delta}$; the Bernoulli cylinder measure gives $\mu(B(x, r)) \leq Cr^{s_\delta}$ because the cylinder intervals are uniformly separated. Frostman's lemma gives $\dim_H H \geq s_\delta$.

At exponent t ,

$$\sum_i |r_{u_i}|^t \geq M(c\delta)^t \geq c^t \delta^{t-t_0}. \quad (19.10)$$

For sufficiently small δ this is greater than 1, so $s_\delta > t$. Since $H \subset K$, $\dim_H K \geq \dim_H H > t$. Let $t \uparrow d$. $\square$

Lemma IV.9.5 (*product and addition upper bound*).

For self-similar $K, K' \subset \mathbb{R}$,

$$\dim_{\mathrm{H}}(K + sK') \leq \min\{\dim_{\mathrm{H}} K + \dim_{\mathrm{H}} K', 1\} \quad (19.11)$$

for every s .

Proof. For every δ ,

$$N_{\delta}(K \times K') \leq CN_{\delta}(K)N_{\delta}(K'), \quad (19.12)$$

using product covers by intervals and squares. Hence

$$\overline{\dim}_{\mathrm{B}}(K \times K') \leq \overline{\dim}_{\mathrm{B}} K + \overline{\dim}_{\mathrm{B}} K' = \dim_{\mathrm{H}} K + \dim_{\mathrm{H}} K' \quad (19.13)$$

by Lemma IV.9.4. The map $L_s(x, y) = x + sy$ is Lipschitz, so

$$\dim_{\mathrm{H}}(K + sK') \leq \dim_{\mathrm{H}}(K \times K') \leq \overline{\dim}_{\mathrm{B}}(K \times K').$$

The ambient line contributes the additional upper bound 1. $\square$

3. A local discrete Marstrand lemma

For $\theta \in [0, \pi)$, let P_{θ} be orthogonal projection onto the line with unit vector $e_{\theta} = (\cos \theta, \sin \theta)$.

Lemma IV.9.6 (*average reciprocal projection length*).

Let μ be a compactly supported probability on $\mathbb{R}^2$ with bounded density and

$$I_1(\mu) := \iint |x - y|^{-1} d\mu(x) d\mu(y) < \infty. \quad (19.14)$$

Then

$$\int_0^{\pi} |P_{\theta}(\text{supp } \mu)|^{-1} d\theta \leq CI_1(\mu), \quad (19.15)$$

where a zero-length projection is interpreted as $+\infty$.

Proof. Let ϕ be a nonnegative smooth probability density supported in $[-1, 1]$ and write $\phi_{\eta}(t) = \eta^{-1}\phi(t/\eta)$. Let

$$\nu_{\theta} = (P_{\theta})_*\mu, \quad f_{\theta, \eta} = \nu_{\theta} * \phi_{\eta}.$$

Put $S_{\theta} = P_{\theta}(\text{supp } \mu)$ and let $S_{\theta, \eta} = \{t : \text{dist}(t, S_{\theta}) \leq \eta\}$. The support of $f_{\theta, \eta}$ is contained in this compact neighborhood. Cauchy–Schwarz gives

$$1 = \left(\int f_{\theta, \eta} \right)^2 \leq |S_{\theta, \eta}| \int f_{\theta, \eta}(t)^2 dt. \quad (19.16)$$

Put $\kappa_{\eta} = \phi_{\eta} * \widetilde{\phi_{\eta}}$, where $\widetilde{\phi_{\eta}}(t) = \phi_{\eta}(-t)$. Fubini gives

$$\int f_{\theta, \eta}^2 = \iint \kappa_{\eta}((x - y) \cdot e_{\theta}) d\mu(x) d\mu(y). \quad (19.17)$$

For $v \neq 0$,

$$\int_0^\pi \kappa_\eta(v \cdot e_\theta) d\theta \leq \frac{C}{|v|}. \quad (19.18)$$

Indeed, if $|v| \leq 2\eta$, the left side is $O(\eta^{-1}) = O(|v|^{-1})$. If $|v| > 2\eta$, the integrand vanishes except where $|v \cdot e_\theta| \leq 2\eta$, a set of angular measure $O(\eta/|v|)$, while $\|\kappa_\eta\|_\infty = O(\eta^{-1})$.

Integrating (19.17) in θ and using (19.18),

$$\int_0^\pi \int f_{\theta,\eta}(t)^2 dt d\theta \leq CI_1(\mu). \quad (19.19)$$

Combine this with (19.16) and integrate in θ . Since S_θ is compact, its decreasing closed neighborhoods satisfy $|S_{\theta,\eta}| \downarrow |S_\theta|$ as $\eta \downarrow 0$, by continuity from above of Lebesgue measure. Fatou's lemma, with the reciprocal limit interpreted as infinity when $|S_\theta| = 0$, gives (19.15). $\square$

Proposition IV.9.7 (*discrete Marstrand projection theorem*).

Fix constants $A > 1$, $A_1, A_2 > 0$ and $0 < \gamma < 1$. There is $\delta_0 > 0$, depending only on these constants and γ , with the following property.

Let $0 < \rho < 1$ and let $\mathcal{Q}$ be a finite family of pairwise disjoint closed convex sets in the unit disk such that:

- (a) every $Q \in \mathcal{Q}$ contains a disk of radius $A^{-1}\rho$ and is contained in a disk of radius $A\rho$;
- (b)

$$\#\mathcal{Q} \geq A_1^{-1}\rho^{-\gamma}; \quad (19.20)$$

- (c) every disk of radius $\ell \in [\rho, 1]$ meets at most

$$A_2(\ell/\rho)^\gamma \quad (19.21)$$

members of $\mathcal{Q}$.

Then for every $\varepsilon > 0$ there is a finite union J of open intervals in $[0, \pi)$ such that

$$\text{Leb}([0, \pi) \setminus J) \leq \varepsilon, \quad (19.22)$$

and, for each $\theta \in J$, there is $\mathcal{Q}_\theta \subset \mathcal{Q}$ with

$$\#\mathcal{Q}_\theta \geq \delta_0 \varepsilon \#\mathcal{Q} \quad (19.23)$$

whose projected sets $P_\theta(Q)$, $Q \in \mathcal{Q}_\theta$, are pairwise ρ -separated.

Proof. If $\varepsilon \geq \pi$, take $J = \emptyset$. Thus assume $0 < \varepsilon < \pi$. Let $E = \bigcup \mathcal{Q}$ and let μ be normalized planar Lebesgue measure on E . Because every Q has area comparable to ρ^2 and condition (c) with $\ell = 1$ gives $\#\mathcal{Q} \leq C\rho^{-\gamma}$, the masses of individual members satisfy

$$c\rho^\gamma \leq \mu(Q) \leq C\rho^\gamma. \quad (19.24)$$

Hence for every disk D of radius $\ell \in [\rho, 1]$,

$$\mu(D) \leq C\ell^\gamma. \quad (19.25)$$

Also

$$\frac{d\mu}{dx} \leq C\rho^{\gamma-2}. \quad (19.26)$$

Fix $w \in E$. The contribution to the potential from $|z - w| \leq \rho$ is, by (19.26),

$$C\rho^{\gamma-2} \int_{|u| \leq \rho} |u|^{-1} du \leq C\rho^{\gamma-1}. \quad (19.27)$$

For annuli $e^{j-1}\rho < |z - w| \leq e^j\rho$, (19.25) gives

$$\int_{e^{j-1}\rho < |z-w| \leq e^j\rho} |z - w|^{-1} d\mu(z) \leq C(e^j\rho)^{\gamma-1}. \quad (19.28)$$

Since $\gamma < 1$, summing over j gives

$$\int |z - w|^{-1} d\mu(z) \leq C\rho^{\gamma-1}.$$

Integrating in w ,

$$I_1(\mu) \leq C\rho^{\gamma-1}. \quad (19.29)$$

By Lemma IV.9.6,

$$\int_0^\pi |P_\theta E|^{-1} d\theta \leq C\rho^{\gamma-1}. \quad (19.30)$$

Therefore the set

$$J_* := \{\theta : |P_\theta E| > c\varepsilon\rho^{1-\gamma}\}$$

has complement of measure at most $\varepsilon/2$ after choosing c sufficiently small, independently of ρ and ε .

For $\theta \in J_*$, choose a maximal $6(A+1)\rho$ -separated set in $P_\theta E$. Its radius- $6(A+1)\rho$ intervals cover $P_\theta E$, so its cardinality is at least

$$c\varepsilon\rho^{-\gamma}. \quad (19.31)$$

For each selected point choose one Q whose projection contains it. Every projected set has diameter at most $2A\rho$. Hence the chosen sets are distinct, and their projected distance is at least $6(A+1)\rho - 4A\rho > \rho$. Condition (c) with $\ell = 1$ gives $\#Q \leq A_2\rho^{-\gamma}$, so (19.31) implies that their number is at least $\delta_0\varepsilon\#Q$ for a fixed sufficiently small δ_0 .

For a fixed chosen subcollection these strict separation inequalities persist on an open neighborhood of θ : projection of a compact set is continuous in the Hausdorff metric, as is the distance between two compact sets. Let $G \subset (0, \pi)$ be the union of the resulting open neighborhoods, chosen inside $(0, \pi)$. It contains $J_* \cap (0, \pi)$, and each direction in G has a subcollection of the required size. Thus $\text{Leb}([0, \pi] \setminus G) \leq \varepsilon/2$. By inner regularity choose a compact $H \subset G$ with $\text{Leb}(G \setminus H) < \varepsilon/2$. Cover H by finitely many open intervals contained in G and let J be their union. Then $J \subset G$, and its complement has measure at most ε . This finite extraction, rather than openness alone, proves the asserted finite-union property needed by the rotation lemma. $\square$

4. Irrational rotation frequency

Lemma IV.9.8 (*Weyl for one irrational rotation*).

Let $\beta > 0$ and $\alpha/\beta \notin \mathbb{Q}$. For every finite union F of intervals in $[0, \beta)$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \#\{0 \leq j < N : j\alpha \bmod \beta \in F\} = \frac{\text{Leb}(F)}{\beta}. \quad (19.32)$$

Proof. For every nonzero integer k ,

$$\frac{1}{N} \sum_{j=0}^{N-1} e^{2\pi i k j \alpha / \beta} = \frac{1 - e^{2\pi i k N \alpha / \beta}}{N(1 - e^{2\pi i k \alpha / \beta})} \rightarrow 0. \quad (19.33)$$

The constant Fourier mode has average one. Thus the empirical averages converge for trigonometric polynomials. Approximate interval indicators from above and below by continuous periodic functions and then by trigonometric polynomials. Boundary points have zero Lebesgue measure. Add finitely many intervals. $\square$

5. Homogeneous separated non-resonance

Theorem IV.9.9 (*homogeneous separated theorem*).

Let K, K' be attractors of

$$\{rx + a_i\}_{i=1}^N, \quad \{r'x + b_j\}_{j=1}^{N'}, \quad (19.34)$$

where $0 < r, r' < 1$, the first-level images of the convex hulls are pairwise disjoint in each system, and

$$\frac{\log r}{\log r'} \notin \mathbb{Q}. \quad (19.35)$$

Then for every $\lambda \neq 0$,

$$\dim_{\text{H}}(K + \lambda K') = \min\{\dim_{\text{H}} K + \dim_{\text{H}} K', 1\}. \quad (19.36)$$

Proof. If either K or K' is a singleton, $K + \lambda K'$ is a translate of the other set, possibly scaled by $\lambda \neq 0$, so (19.36) is immediate. Henceforth both hulls have positive length and $N, N' \geq 2$. Their disjoint closed first-level intervals have positive gaps, giving $Nr < 1$ and $N'r' < 1$; in particular $0 < s, s' < 1$ below. *Step 1: reduce to the subcritical case.* Write

$$s = \dim_{\text{H}} K = \frac{\log N}{\log(1/r)}, \quad s' = \dim_{\text{H}} K' = \frac{\log N'}{\log(1/r')}.$$

The upper bound is Lemma IV.9.5. If $s + s' \geq 1$, fix $\eta > 0$. By passing to a high iterate of the first system and keeping an appropriate number of its pairwise disjoint cylinders, obtain a homogeneous subset $K_\eta \subset K$ with contraction r^L and

$$1 - \eta < \dim_{\text{H}} K_\eta + s' < 1. \quad (19.37)$$

The possible subsystem dimensions $\log M/[L \log(1/r)]$ have mesh $O(1/L)$, so this is possible for large L . The irrationality (19.35) survives powers. A subcritical lower bound for $K_\eta + \lambda K'$ then gives $\dim_{\mathbb{H}}(K + \lambda K') \geq 1 - \eta$. Let $\eta \downarrow 0$. Hence it suffices to assume

$$\gamma := s + s' < 1. \quad (19.38)$$

Step 2: balanced rectangle families. By affine normalization take both convex hulls to have length one; write $\lambda = e^\tau$ after reflecting K' if needed. For $k \geq 1$ define

$$q_k = \left\lfloor \frac{k \log(1/r)}{\log(1/r')} \right\rfloor. \quad (19.39)$$

Then

$$\psi_k := k \log(1/r) - q_k \log(1/r') \in [0, \beta), \quad \beta := \log(1/r'). \quad (19.40)$$

At level k , the rectangles $I_u \times I'_v$, $|u| = k$, $|v| = q_k$, have side lengths r^k and

$$(r')^{q_k} = e^{\psi_k} r^k,$$

which are uniformly comparable. Their number is

$$N^k (N')^{q_k} \asymp r^{-k\gamma}. \quad (19.41)$$

Moreover, if $j < k$, a square of side comparable to r^j meets at most

$$C r^{(j-k)\gamma} \quad (19.42)$$

level- k rectangles. This follows from pairwise separation and the exact branching $N^{k-j} (N')^{q_k - q_j}$.

Step 3: one block of good projections. Fix a large block length m . Put

$$m_0 = \left\lfloor \frac{m \log(1/r)}{\beta} \right\rfloor, \quad \alpha = m \log(1/r) - m_0 \beta \in [0, \beta). \quad (19.43)$$

By (19.35),

$$\alpha/\beta \notin \mathbb{Q}. \quad (19.44)$$

At one block, the vertical depth increment is either m_0 or $m_0 + 1$. Apply Proposition IV.9.7 to the two corresponding balanced rectangle families. After intersecting the two good-direction sets and converting angle to logarithmic slope t , we obtain a finite union

$$\tilde{F} \subset [\tau, \tau + \beta) \quad (19.45)$$

with

$$\text{Leb}([\tau, \tau + \beta) \setminus \tilde{F}) \leq C \varepsilon \beta, \quad (19.46)$$

such that whenever $t \in \tilde{F}$, the appropriate block family contains at least

$$B_m := \lfloor c_1 \varepsilon r^{-m\gamma} \rfloor \geq c \varepsilon r^{-m\gamma} \quad (19.47)$$

children whose images under

$$\Pi_{e^t}(x, y) = x + e^t y$$

are pairwise cr^m -separated. Here $c_1 > 0$ is a fixed lower-bound constant from the two applications of Proposition IV.9.7; choose $c \leq c_1/2$ and take m large enough for the displayed integer lower bound to hold. Thus the later instruction to attach B_m children has an integer meaning. Put

$$F := \{u \in [0, \beta) : \tau + u \in \tilde{F}\}. \quad (19.48)$$

Step 4: recursive tree at the fixed target slope $\lambda = e^\tau$. At stage j define

$$q_j = \left\lfloor \frac{jm \log(1/r)}{\beta} \right\rfloor, \quad u_j = jm \log(1/r) - q_j \beta. \quad (19.49)$$

Then

$$u_j = j\alpha \bmod \beta. \quad (19.50)$$

A level- j parent rectangle has horizontal scale r^{jm} , vertical scale $e^{u_j} r^{jm}$, and its projection under Π_{e^τ} has the same normalized geometry as projection of a unit block at logarithmic slope $\tau + u_j$.

If $u_j \in F$, attach B_m good children supplied by Step 3. If $u_j \notin F$, attach one fixed child. The carry $q_{j+1} - q_j$ is exactly m_0 or $m_0 + 1$, so the selected child family is one of the two families already covered in Step 3.

Induction gives families $\mathcal{R}_j$ such that every child lies in its parent, all $\Pi_{e^\tau}(Q)$, $Q \in \mathcal{R}_j$, are cr^{jm} -separated, and each parent has B_m children at a good phase and one otherwise.

Step 5: frequency and a Frostman measure. By Lemma IV.9.8 and (19.46),

$$G_j := \#\{0 \leq i < j : u_i \in F\} \geq (1 - C\varepsilon)j \quad (19.51)$$

for all sufficiently large j . Hence

$$\#\mathcal{R}_j \geq B_m^{(1-C\varepsilon)j}. \quad (19.52)$$

Let E_τ be the nested intersection of the projected level unions. It is compact, nonempty, and

$$E_\tau \subset K + e^\tau K'. \quad (19.53)$$

Assign equal mass to every level- j projected interval. Because these intervals are cr^{jm} -separated, every interval of radius Cr^{jm} meets at most C' of them. Therefore

$$\mu_\tau(B(x, Cr^{jm})) \leq C' B_m^{-(1-C\varepsilon)j}. \quad (19.54)$$

The mass distribution principle yields

$$\dim_{\text{H}} E_\tau \geq (1 - C\varepsilon) \frac{m\gamma \log(1/r) + \log(c\varepsilon)}{m \log(1/r)}. \quad (19.55)$$

First let $m \rightarrow \infty$ and then $\varepsilon \rightarrow 0$. We obtain $\dim_{\text{H}}(K + e^\tau K') \geq \gamma$. Together with Lemma IV.9.5 this proves the theorem. $\square$

6. Separated subsystems inside an arbitrary self-similar set

Lemma IV.9.10 (*strongly separated inner approximation*).

Let K be any self-similar set in $\mathbb{R}$ and let $s = \dim_H K$. For every $\varepsilon > 0$ there is a self-similar $H \subset K$ generated by cylinder maps of the original IFS such that the first-level convex-hull intervals of H are pairwise disjoint and

$$\dim_H H > s - \varepsilon. \quad (19.56)$$

Proof. If $s = 0$, take the attractor $H = \{p\}$ of any one original cylinder map. Its fixed point p belongs to K : iterating the map on a point of K stays in K and converges to p . The one-map separation condition is vacuous and $\dim_H H = 0 > -\varepsilon$. Now suppose $s > 0$. Normalize $\text{diam } K = 1$ and put $I = \text{conv } K$ and $r_* := \min_i |r_i|$. Choose $0 < \eta < \min\{\varepsilon/3, s/3\}$, so the exponents $s - \eta$ and $s - 2\eta$ are positive. Since $s - \eta < \dim_H K$, for all sufficiently small δ the $(s - \eta)$ -dimensional Hausdorff content restricted to covers by intervals of diameter at most 3δ is greater than 1.

Consider the stopping cylinders $f_u(K)$ for which

$$|r_u| \leq \delta < |r_{u-}|. \quad (19.57)$$

Their convex-hull intervals have lengths between $r_*\delta$ and δ and cover K . Choose a maximal pairwise disjoint subcollection

$$f_{u_1}(I), \dots, f_{u_M}(I). \quad (19.58)$$

The interval with the same center and length enlarged by 2δ around each selected interval covers every omitted stopping interval, hence covers K . All these enlarged intervals have length at most 3δ . Therefore

$$M(3\delta)^{s-\eta} > 1, \quad M > 3^{-(s-\eta)}\delta^{-(s-\eta)}. \quad (19.59)$$

Let H be the attractor of the selected cylinder maps. They satisfy strong separation. Their contraction ratios q_j obey

$$r_*\delta \leq q_j \leq \delta. \quad (19.60)$$

At exponent $s - 2\eta$,

$$\sum_j q_j^{s-2\eta} \geq M(r_*\delta)^{s-2\eta} \geq c\delta^{-\eta}. \quad (19.61)$$

For δ sufficiently small this is greater than 1. By the Moran formula, $\dim_H H > s - 2\eta > s - \varepsilon$. $\square$

7. Homogeneous subsystems without a local limit theorem

Lemma IV.9.11 (*homogeneous inner approximation*).

Let H be a one-dimensional self-similar set satisfying strong separation, generated by similarities with absolute contraction ratios $a_1, \dots, a_M$. For every $\varepsilon > 0$ and every $L_0 > 0$ there is a homogeneous strongly separated subsystem $G \subset H$ such that

$$\dim_H G > \dim_H H - \varepsilon, \quad (19.62)$$

and its common logarithmic contraction length $A = \log(1/\rho)$ satisfies $A > L_0$. Moreover, if the original maps have pairwise disjoint images of a compact interval $I \supset H$ and map I into itself, the subsystem may be chosen with this same interval-image separation.

Proof. If there is only one map, H is its fixed point and a sufficiently high even iterate proves the claim, with positive ratio. Otherwise strong separation gives a positive distance Δ between distinct first-level cylinder sets. Here the Moran formula applies to sets, without any claim about their convex hulls. To see this, let $D = \text{diam } H > 0$ and let $t > 0$ solve $\sum_i a_i^t = 1$. Cylinder covers give the dimension upper bound t . For the lower bound, stop at words u with $a_u D \leq r < a_u D$. Two different stopping cylinders have distance at least $\Delta r/D$, by their first different symbol and their common prefix. A radius- r interval therefore meets only a bounded number of these cylinder sets. The Bernoulli measure with weights a_i^t assigns mass $a_u^t \leq (r/D)^t$ to each. Its mass on every radius- r interval is at most Cr^t , so the mass distribution principle gives the lower bound t . Thus $d = \dim_H H = t$ and

$$\sum_i a_i^d = 1.$$

Set

$$p_i = a_i^d. \quad (19.63)$$

Then

$$H(p) := -\sum_i p_i \log p_i = d \sum_i p_i \log(1/a_i). \quad (19.64)$$

For a large integer k choose integers v_i with

$$\sum_i v_i = k, \quad v_i/k \rightarrow p_i. \quad (19.65)$$

Consider all length- k words containing exactly v_i occurrences of symbol i . Every associated cylinder map has the same absolute contraction

$$\rho_k = \prod_i a_i^{v_i}. \quad (19.66)$$

The number of such words is the multinomial coefficient

$$N_k = \frac{k!}{\prod_i v_i!}. \quad (19.67)$$

Stirling's elementary estimate $\log(n!) = n \log n - n + O(\log(n+1))$ gives

$$\frac{1}{k} \log N_k \rightarrow H(p), \quad (19.68)$$

while

$$\frac{1}{k} \log(1/\rho_k) \rightarrow \sum_i p_i \log(1/a_i). \quad (19.69)$$

Distinct depth- k cylinder sets are disjoint by strong separation. The selected subsystem lies in H , so its first-level cylinder sets are disjoint as well. The same set-based Moran argument gives its dimension

$$d_k = \frac{\log N_k}{\log(1/\rho_k)} \rightarrow d. \quad (19.70)$$

Also $\log(1/\rho_k) \rightarrow \infty$. Choose k large. All selected words have the same signed ratio, since their symbol multiplicities agree. If it is negative, pass to the second iterate of this homogeneous subsystem. Its common ratio becomes ρ_k^2 , its number of maps becomes N_k^2 , and its dimension is unchanged. If the original maps have disjoint images of an invariant interval I , distinct words of a fixed depth have disjoint images of I , by their first different symbol. Selection and the possible second iterate preserve this property, proving the additional assertion. $\square$

8. Preserving one irrational ratio

Lemma IV.9.12 (*one-sided restoration of irrationality*).

Let A, B, a, b be nonzero real numbers with $B + b \neq 0$. Assume

$$A/B \in \mathbb{Q}, \quad a/b \notin \mathbb{Q}. \quad (19.71)$$

Then at least one of

$$(A + a)/B, \quad A/(B + b) \quad (19.72)$$

is irrational.

Proof. Write $A = qB$ with $q \in \mathbb{Q}$. If both quantities in (19.72) were rational, the first would imply $a/B \in \mathbb{Q}$. Since $q \neq 0$, the second implies $B/(B + b) \in \mathbb{Q}$, hence $b/B \in \mathbb{Q}$. Therefore $a/b = (a/B)/(b/B)$ would be rational, a contradiction. $\square$

Lemma IV.9.13 (*homogeneous non-resonant inner approximants*).

Let K, K' satisfy the hypothesis (19.3), and let $\varepsilon > 0$. There exist homogeneous strongly separated self-similar sets

$$G \subset K, \quad G' \subset K' \quad (19.73)$$

with positive common contraction ratios ρ, ρ' and pairwise disjoint first-level convex-hull images in each system, such that

$$\dim_H G > \dim_H K - \varepsilon, \quad \dim_H G' > \dim_H K' - \varepsilon, \quad (19.74)$$

and

$$\frac{\log \rho}{\log \rho'} \notin \mathbb{Q}. \quad (19.75)$$

Proof. Choose original contractions r_i, r'_j satisfying (19.3), and put $a = \log |r_i| < 0$, $b = \log |r'_j| < 0$. Choose $L_0 > 2 \max\{|a|, |b|\}/\varepsilon$. First use Lemma IV.9.10 to obtain strongly separated inner approximants with dimension loss less than $\varepsilon/4$. In its construction, the selected maps have disjoint images of the original hull interval $I = \text{conv } K$ (and $I' = \text{conv } K'$ in the primed system); the one-map case has the same vacuous property. These intervals are invariant under all the original cylinder maps. Apply Lemma IV.9.11, including its interval-separation clause, with this L_0 to obtain positive-ratio homogeneous subsystems with a further loss less than $\varepsilon/4$. Write

$$A = \log \rho, \quad B = \log \rho'.$$

If A/B is irrational, we are done. Otherwise apply Lemma IV.9.12; its denominator condition holds because $B < 0$ and $b < 0$. If $(A + a)/B$ is irrational, replace every homogeneous map g_k of G by $f_i \circ g_k$. Their images of I remain disjoint because f_i is injective. Each map sends K into K , so the new attractor lies in K and its hull lies in I . Its common signed ratio is $r_i \rho$, with absolute ratio $|r_i| \rho$. If $A/(B + b)$ is irrational, perform the analogous replacement on G' using I' .

Prepending one fixed map changes

$$\frac{\log N}{\log(1/\rho)} \quad \text{to} \quad \frac{\log N}{\log(1/\rho) + O(1)}. \quad (19.76)$$

More explicitly, if the old dimension is $d_0 \leq 1$ and its logarithmic length is $L > L_0$, prepending the chosen map changes it to $d_0 L / (L + |a|)$, a loss at most $|a|/L_0 < \varepsilon/2$; the primed case uses $|b|$ instead. The total loss is therefore less than ε . If the new signed ratio is negative, replace that homogeneous system by its second iterate. This preserves its attractor, dimension and interval-image separation, and makes the ratio positive. It multiplies one logarithmic length by two, which preserves irrationality of the ratio of the two logarithmic lengths. Both final systems therefore meet all the hypotheses of Theorem IV.9.9. $\square$

9. Proof of the general non-resonance theorem

Theorem IV.9.14 (*Peres–Shmerkin non-resonance*).

Under the hypotheses stated at the beginning of the section, equation (19.4) holds for every $s \neq 0$.

Proof. The upper bound is Lemma IV.9.5. Fix $\varepsilon > 0$. By Lemma IV.9.13 choose $G \subset K$ and $G' \subset K'$ with

$$\dim_{\text{H}} G > \dim_{\text{H}} K - \varepsilon, \quad \dim_{\text{H}} G' > \dim_{\text{H}} K' - \varepsilon,$$

and irrational logarithmic ratio of their homogeneous contractions. Scaling by $s \neq 0$ preserves the contraction ratios and separation, so Theorem IV.9.9 gives

$$\begin{aligned} \dim_{\text{H}}(K + sK') &\geq \dim_{\text{H}}(G + sG') \\ &= \min\{\dim_{\text{H}} G + \dim_{\text{H}} G', 1\} \\ &\geq \min\{\dim_{\text{H}} K + \dim_{\text{H}} K' - 2\varepsilon, 1\}. \end{aligned} \quad (19.77)$$

Let $\varepsilon \downarrow 0$. $\square$

10. Central Cantor sets

For $0 < a < 1/2$ define the central Cantor set

$$C_a = \left\{ (1-a) \sum_{n=0}^{\infty} \omega_n a^n : \omega_n \in \{0, 1\} \right\}. \quad (19.78)$$

It is the attractor of

$$x \mapsto ax, \quad x \mapsto ax + (1-a), \quad (19.79)$$

which satisfy strong separation. Hence

$$\dim_H C_a = \frac{\log 2}{\log(1/a)}. \quad (19.80)$$

Theorem IV.9.15 (*irrational central-Cantor formula*).

If $0 < a, b < 1/2$ and $\log a / \log b \notin \mathbb{Q}$, then for every $s \neq 0$,

$$\dim_H(C_a + sC_b) = \min \left\{ \frac{\log 2}{\log(1/a)} + \frac{\log 2}{\log(1/b)}, 1 \right\}. \quad (19.81)$$

Proof. Apply Theorem IV.9.14 to the two defining homogeneous IFSs. $\square$

Theorem IV.9.16 (*sharp rational resonance contrast*).

Assume $0 < a, b < 1/2$,

$$\frac{\log a}{\log b} \in \mathbb{Q}, \quad \dim_H C_a + \dim_H C_b \leq 1. \quad (19.82)$$

Then

$$\dim_H(C_a + C_b) < \dim_H C_a + \dim_H C_b. \quad (19.83)$$

Proof. There are $0 < r < 1$ and positive integers m, n such that

$$a = r^m, \quad b = r^n. \quad (19.84)$$

Iterate the C_a IFS n times and the C_b IFS m times. Every cylinder interval now has length

$$r^{mn}. \quad (19.85)$$

The product $C_a \times C_b$ is generated at this block scale by 2^{n+m} planar similarities with ratio r^{mn} . Under the sum projection $\Pi(x, y) = x + y$, the two block rectangles

$$[0, r^{mn}] \times [1 - r^{mn}, 1], \quad [1 - r^{mn}, 1] \times [0, r^{mn}] \quad (19.86)$$

have exactly the same projected similarity.

Thus $C_a + C_b$ is the attractor of a one-dimensional IFS obtained from the 2^{n+m} projected block maps with at least one exact duplicate. Removing one duplicate does not change the attractor. Therefore

$$\begin{aligned} \dim_{\mathrm{H}}(C_a + C_b) &\leq \frac{\log(2^{n+m} - 1)}{mn \log(1/r)} \\ &< \frac{(n+m) \log 2}{mn \log(1/r)} \\ &= \frac{\log 2}{\log(1/a)} + \frac{\log 2}{\log(1/b)}. \end{aligned} \tag{19.87}$$

This proves (19.83). □

Theorem IV.9.17 (*worked central-Cantor specialization and resonance*).

For central Cantor sets, irrational $\log(a)/\log(b)$ gives the exact formula (19.81), while rational $\log(a)/\log(b)$ in the subcritical/critical regime $\dim_{\mathrm{H}} C_a + \dim_{\mathrm{H}} C_b \leq 1$ gives the strict dimension drop (19.83). Thus the central model displays the exact algebraic/geometric resonance dichotomy retained in Volume IV.

Proof. Combine Theorems IV.9.15 and IV.9.16. □

11. What the set theorem does and does not say

1. **Set sum versus measure convolution.** The theorem computes $\dim_{\mathrm{H}}(K + sK')$. It does not assert that a chosen self-similar measure convolution is absolutely continuous or even has the same dimension as the sumset.
2. **Supercritical case.** When $\dim_{\mathrm{H}} K + \dim_{\mathrm{H}} K' > 1$, Theorem IV.9.14 gives $\dim_{\mathrm{H}}(K + sK') = 1$. It does not imply positive Lebesgue measure or nonempty interior.
3. **Separation.** No separation assumption occurs in Theorem IV.9.14. Strong separation is introduced only inside the proof by inner subsystem approximation.
4. **Uniformity over scale.** The homogeneous theorem is not an almost-everywhere projection statement. The finite-scale good directions are revisited under one irrational rotation; Weyl frequency forces good scales for the specific target slope.
5. **Volume-III ABC comparison.** Additive-combinatorial sum-product mechanisms are conceptually related but are not proof inputs here.

9.2 Uniform thickness and arithmetic sums

1. Definitions

Definition IV.9.18 (*uniform thickness*).

Throughout this section and the self-conformal continuation, $B(x, r)$ is the Euclidean closed ball; $B^\circ(x, r)$ denotes the open ball. The convex hull $\text{conv}(A)$ consists of finite convex combinations of points of A , and the affine hull is the smallest affine subspace containing A . For a nonempty compact set $E \subset \mathbb{R}^d$, its thickness $\tau(E)$ is the largest $c \in [0, 1]$ such that for every $x \in E$ and every $0 < r \leq \text{diam}(E)$ there exists $y \in \mathbb{R}^d$ with

$$B(y, cr) \subset \text{conv}(E \cap B(x, r)). \quad (20.1)$$

For a singleton the range of r is empty, so this definition gives $\tau(E) = 1$; every use of a local thickness criterion below explicitly excludes that degenerate case.

Definition IV.9.19 (*finite arithmetic sums*).

For compact sets $E_1, \dots, E_n$ write

$$E_1 + \dots + E_n := \{x_1 + \dots + x_n : x_i \in E_i\}.$$

For a positive integer n , write nE for the sum of n copies of E .

Definition IV.9.20 (*arithmetic thickness*).

A compact set $E \subset \mathbb{R}^d$ is *arithmetically thick* if $\text{int}(nE) \neq \emptyset$ for some positive integer n , where the interior is taken in the ambient space $\mathbb{R}^d$.

2. Three elementary convex-geometry lemmas**Lemma IV.9.21** (*convex hull cannot move too far inward*).

Let A be bounded, let $R > 0$, and suppose $|a| \geq R$ for all $a \in A$. Then every $z \in \text{conv}(A)$ satisfies

$$|z| \geq R - \frac{\text{diam}(A)^2}{2R}. \quad (20.2)$$

Proof. Write $z = \sum_i p_i a_i$ with $p_i \geq 0$ and $\sum_i p_i = 1$. Since

$$\langle a_i, a_j \rangle = \frac{|a_i|^2 + |a_j|^2 - |a_i - a_j|^2}{2} \geq R^2 - \frac{\text{diam}(A)^2}{2},$$

we get

$$\begin{aligned} |z|^2 &= \sum_i p_i^2 |a_i|^2 + \sum_{i \neq j} p_i p_j \langle a_i, a_j \rangle \\ &\geq R^2 - \frac{\text{diam}(A)^2}{2}. \end{aligned}$$

If $R \leq \text{diam}(A)/\sqrt{2}$, (20.2) is trivial. Otherwise use $\sqrt{1-s} \geq 1-s$ for $0 \leq s \leq 1$ to obtain (20.2). $\square$

Corollary IV.9.22 (*large ball plus a finite skeleton*).

If A is bounded and $\text{conv}(A)$ contains $B(y, r)$, then for every

$$R > \frac{\text{diam}(A)^2}{r}$$

and every z ,

$$B(z, R) + A \supset B(z, R) + B(y, r/2). \quad (20.3)$$

Proof. Translation reduces to $z = 0$. Let $X = B(0, R) + A$. Since $R > \text{diam}(A)^2/r \geq 2 \text{diam}(A)$, every point of $\text{conv}(A)$ lies in $\text{int} X$.

Take $x \in \partial X$. Then $\text{dist}(x, A) \geq R$. Apply Lemma IV.9.21 to $A - x$: every point of $\text{conv}(A)$ has distance from x exceeding

$$R - \frac{\text{diam}(A)^2}{2R} > R - r/2.$$

Since $B(y, r) \subset \text{conv}(A)$, the point of $B(y, r)$ on the ray from y toward x shows $|x - y| > R + r/2$. Thus $\text{dist}(y, \partial X) > R + r/2$, so

$$B(y, R + r/2) \subset X.$$

But $B(0, R) + B(y, r/2) = B(y, R + r/2)$, proving (20.3). $\square$

Lemma IV.9.23 (*finite approximation preserves an inner ball*).

Let A, F be compact, assume $A \subset V_\delta(F)$ and $\text{conv}(A)$ contains $B(z, r)$, with $0 \leq \delta < r$. Then

$$B(z, r - \delta) \subset \text{conv}(F). \quad (20.4)$$

Proof. For a unit vector v write

$$h_K(v) = \sup_{x \in K} \langle v, x \rangle.$$

The inclusion $A \subset V_\delta(F)$ gives $h_A(v) \leq h_F(v) + \delta$. Since $B(z, r) \subset \text{conv}(A)$,

$$\langle v, z \rangle + r \leq h_A(v),$$

and therefore

$$h_F(v) \geq \langle v, z \rangle + r - \delta$$

for every unit v . The intersection of these supporting half-spaces contains $B(z, r - \delta)$, which proves (20.4). $\square$

3. Finite skeleton of a positive-thickness set

Lemma IV.9.24 (*uniform finite skeleton*).

Assume $\tau(E) \geq c > 0$. Put

$$N = \left\lceil \left(\frac{4+c}{c} \right)^d \right\rceil.$$

For every $x \in E$ and $0 < r \leq \text{diam}(E)$, there exist $y_1, \dots, y_N \in E \cap B(x, r)$ and $z \in \mathbb{R}^d$ such that

$$B(z, cr/2) \subset \text{conv}\{y_1, \dots, y_N\}. \quad (20.5)$$

Proof. By thickness choose z with $B(z, cr) \subset \text{conv}(E \cap B(x, r))$. Take a maximal family of pairwise disjoint open balls

$$U(y_j, cr/4), \quad y_j \in E \cap B(x, r).$$

All lie in $U(x, r + cr/4)$. Comparing volumes yields at most $((4+c)/c)^d$ members, hence at most N after the harmless convention that N is chosen as an integer upper bound.

Maximality implies that the $cr/2$ -balls around the y_j cover $E \cap B(x, r)$. Apply Lemma IV.9.23 with

$$A = E \cap B(x, r), \quad F = \{y_j\}, \quad \delta = cr/2.$$

Then $\text{conv}(F)$ contains $B(z, cr/2)$. If fewer than N points occur, repeat the last point. $\square$

Remark IV.9.25 (*regularity of the parameter map*).

The precise integer convention for N is immaterial; the source's final n -threshold is independent of N .

4. Quantitative sum theorem

Theorem IV.9.26 (*positive-diameter quantitative thickness theorem*).

Let $E_1, \dots, E_n$ be compact subsets of $\mathbb{R}^d$ satisfying

$$\text{diam}(E_i) > 0, \quad \tau(E_i) \geq c > 0$$

for all i . If

$$n > 2^{11}c^{-3} + 1, \quad (20.6)$$

then $E_1 + \dots + E_n$ has nonempty interior.

Proof. Step 1: common scales. Let

$$r_0 = \min_i \text{diam}(E_i) > 0, \quad \rho = c/4.$$

Since $c \leq 1$, $\rho < 1/2$. Fix an integer N large enough for Lemma IV.9.24. For every i we construct a rooted N -ary family of points $x_j^i \in E_i$ and auxiliary points z_j^i .

Start with any $x_{\emptyset}^i \in E_i$. Given x_J^i at depth $k = |J|$, apply Lemma IV.9.24 with

$$r = \rho^k r_0 / 2.$$

This is at most $\text{diam}(E_i)$. We obtain children $x_{J_1}^i, \dots, x_{J_N}^i$ in

$$E_i \cap B(x_J^i, \rho^k r_0 / 2)$$

and a point z_J^i such that

$$B(z_J^i, \rho^k c r_0 / 4) \subset \text{conv}\{x_{J_j}^i : 1 \leq j \leq N\}. \quad (20.7)$$

Define child balls

$$B_{J_j}^i = B(x_{J_j}^i, \rho^{k+1} r_0).$$

Because $\rho < 1/2$, every child ball lies in its parent ball.

Step 2: replace child centers by next-level convex centers. Since $z_{J_j}^i$ lies in the convex hull of the grandchildren chosen at the next stage,

$$|z_{J_j}^i - x_{J_j}^i| \leq \rho^{k+1} r_0 / 2.$$

Apply Lemma IV.9.23 to the set of child points and the set of child z -points. Because

$$\rho^{k+1} r_0 / 2 = \rho^k c r_0 / 8,$$

(20.7) gives

$$B(z_J^i, \rho^k c r_0 / 8) \subset \text{conv}\{z_{J_j}^i : 1 \leq j \leq N\}. \quad (20.8)$$

Moreover

$$\text{diam}\{z_{J_j}^i : 1 \leq j \leq N\} \leq 2\rho^k r_0. \quad (20.9)$$

Step 3: one-level Minkowski replacement. Fix k and parent words $J_1, \dots, J_n$ of length k . For each i let

$$A_i = \{z_{J_{i,j}}^i : 1 \leq j \leq N\}.$$

By (20.8)–(20.9),

$$B(z_{J_i}^i, \rho^k c r_0 / 8) \subset \text{conv}(A_i), \quad \text{diam}(A_i) \leq 2\rho^k r_0. \quad (20.10)$$

We replace the child choice in the n summands one summand at a time. At an intermediate stage the remaining Minkowski summands contain a union of balls whose radius is at least

$$R_* = (n-1)\rho^{k+1} c r_0 / 16. \quad (20.11)$$

Corollary IV.9.22 applies to every A_i provided

$$R_* > \frac{\text{diam}(A_i)^2}{\rho^k c r_0 / 8}.$$

The right side is at most $32\rho^k r_0 / c$. Since $\rho = c/4$, condition (20.6) gives

$$\frac{(n-1)\rho^{k+1} c r_0}{16} = \frac{(n-1)\rho^k c^2 r_0}{64} > \frac{32\rho^k r_0}{c}.$$

Therefore every one-summand replacement is valid. Iterating over all n summands yields

$$\sum_{i=1}^n \bigcup_{j=1}^N B(z_{J_{ij}}^i, \rho^{k+1} cr_0/16) \supset \sum_{i=1}^n B(z_{J_i}^i, \rho^k cr_0/16). \quad (20.12)$$

Step 4: nested interior sets. Let

$$H_k := \sum_{i=1}^n \bigcup_{|J|=k} B(z_J^i, \rho^k cr_0/16).$$

Taking the union of (20.12) over all parent words gives

$$H_{k+1} \supset H_k. \quad (20.13)$$

Also z_J^i lies within $\rho^k r_0/2$ of $x_J^i \in E_i$, so every ball in the i th union defining H_k lies inside the $\rho^k r_0$ -neighborhood of E_i . Consequently

$$H_k \subset V_{\rho^k r_0}(E_1) + \cdots + V_{\rho^k r_0}(E_n). \quad (20.14)$$

The set H_1 contains a ball and hence has nonempty interior. For $h \in H_1$, (20.13)–(20.14) show that

$$\text{dist}(h, E_1 + \cdots + E_n) \leq n\rho^k r_0$$

for every k . The sum of compact sets is compact, hence closed. Letting $k \rightarrow \infty$ gives $h \in E_1 + \cdots + E_n$. Thus H_1 is contained in the sum, and the sum has nonempty interior. $\square$

5. Self-similar consequence

Theorem IV.9.27 (*self-similar arithmetic thickness*).

Let E be a self-similar set in $\mathbb{R}^d$ generated by finitely many similitudes. If E is not contained in an affine hyperplane, then $\tau(E) > 0$ and E is arithmetically thick.

Proof. Because E is not contained in a hyperplane, $\text{conv}(E)$ has nonempty interior. Choose

$$B(x_0, r_*) \subset \text{conv}(E), \quad r_* > 0.$$

Let ρ_i be the contraction ratios and $\rho_{\min} = \min_i \rho_i > 0$.

Fix $x \in E$ and $0 < r \leq \text{diam}(E)$. Choose a coding ω of x and the least positive integer n such that

$$\text{diam}(\varphi_{\omega|n}(E)) \leq r.$$

Then the parent cylinder has diameter at least r : it is E when $n = 1$, and has diameter greater than r when $n > 1$. Therefore

$$\rho_{\omega_1} \cdots \rho_{\omega_n} \geq \frac{\rho_{\min} r}{\text{diam}(E)}. \quad (20.15)$$

The selected cylinder contains x and has diameter at most r , hence

$$\varphi_{\omega|n}(E) \subset E \cap B(x, r).$$

Its convex hull contains the image of $B(x_0, r_*)$, a ball of radius

$$(\rho_{\omega_1} \cdots \rho_{\omega_n})r_* \geq \frac{\rho_{\min}r_*}{\text{diam}(E)} r.$$

Therefore

$$\tau(E) \geq \frac{\rho_{\min}r_*}{\text{diam}(E)} > 0.$$

Since E has positive diameter, apply Theorem IV.9.26 to sufficiently many copies of E . $\square$

6. Centered microsets—corrected local criterion

Definition IV.9.28 (*centered microset*).

A compact set F is a centered microset of a compact set E if there are $x_m \in E$ and $r_m \downarrow 0$ such that

$$K_m = r_m^{-1}((E \cap B(x_m, r_m)) - x_m)$$

converges to F in the Hausdorff metric.

The following local criterion is stated for positive diameter, excluding the vacuous singleton case of Definition IV.9.18.

Lemma IV.9.29 (*positive-diameter microset criterion*).

Let E be compact with $\text{diam}(E) > 0$. Then $\tau(E) > 0$ if and only if no centered microset of E is contained in a proper linear subspace of $\mathbb{R}^d$.

Proof. Write $\text{inrad}(K)$ for the largest radius of a Euclidean ball contained in $\text{conv}(K)$. Since every K_m contains 0 and lies in $B(0, 1)$, the thickness is the infimum of these normalized inradii over all pointed scales.

First suppose a centered microset F is contained in a proper linear subspace. Hausdorff convergence $K_m \rightarrow F$ implies Hausdorff convergence $\text{conv}(K_m) \rightarrow \text{conv}(F)$. The inradius of compact convex sets is 1-Lipschitz with respect to Hausdorff distance. Since $\text{conv}(F)$ has empty interior,

$$\text{inrad}(K_m) \rightarrow 0,$$

and hence $\tau(E) = 0$.

Conversely suppose $\tau(E) = 0$. Choose $x_m \in E$ and $0 < r_m \leq \text{diam}(E)$ such that the normalized sets

$$K_m = r_m^{-1}((E \cap B(x_m, r_m)) - x_m)$$

have inradius tending to zero. Pass to a subsequence with

$$x_m \rightarrow x \in E, \quad r_m \rightarrow r \in [0, \text{diam}(E)], \quad K_m \rightarrow K$$

in the compact Hausdorff hyperspace of $B(0, 1)$. Then $\text{conv}(K)$ has empty interior. Since $0 \in K$, its affine hull is a proper linear subspace W .

If $r = 0$, K itself is the required centered microset. If $r > 0$, let $y \in E$ with $|y - x| < r$. For all large m , $y \in B(x_m, r_m)$, hence

$$(y - x_m)/r_m \in K_m.$$

Passing to the limit gives $(y - x)/r \in K \subset W$. Thus

$$(E \cap B^\circ(x, r)) - x \subset W.$$

Now choose $0 < s_j < r$ with $s_j \downarrow 0$. Since $B(x, s_j) \subset B^\circ(x, r)$, every Hausdorff limit of

$$s_j^{-1}((E \cap B(x, s_j)) - x)$$

is a centered microset contained in W . □

7. Differentiable changes of coordinates, and a bi-Lipschitz boundary

The microset criterion explains the regularity needed for a coordinate change: an invertible derivative preserves the span of a limiting local configuration. A bi-Lipschitz map need not have such a derivative on E .

Proposition IV.9.30 (*positive thickness under C^1 diffeomorphisms*).

Let $E \subset U \subset \mathbb{R}^d$ be compact with positive diameter, where U is open, and let $f : U \rightarrow V$ be a C^1 diffeomorphism onto an open set V . Then

$$\tau(E) > 0 \iff \tau(f(E)) > 0.$$

Proof. Suppose $\tau(E) > 0$ but $\tau(f(E)) = 0$. By Lemma IV.9.29, choose $y_n = f(x_n) \in f(E)$ and $r_n \downarrow 0$ such that

$$K_n = r_n^{-1}((f(E) \cap B(y_n, r_n)) - y_n) \longrightarrow K \subset W,$$

where W is a proper linear subspace. Pass to a subsequence with $x_n \rightarrow x \in E$. There are $\rho > 0$ and $M \geq 1$ such that the closed ρ -neighborhood of E lies in U and $\|Df\| \leq M$ there. Set $s_n = r_n/(2M)$ and take a further subsequence with

$$F_n = s_n^{-1}((E \cap B(x_n, s_n)) - x_n) \longrightarrow F.$$

This centered microset F is not contained in a proper subspace. For large n , the segment from x_n to every $u \in E \cap B(x_n, s_n)$ stays in that neighborhood, and hence $|f(u) - f(x_n)| \leq Ms_n = r_n/2$. Uniform continuity of Df on a compact neighborhood also gives

$$\sup_{u \in E \cap B(x_n, s_n)} \left| \frac{f(u) - f(x_n)}{r_n} - \frac{1}{2M} Df(x) \frac{u - x_n}{s_n} \right| \longrightarrow 0.$$

For $z \in F$, choose $z_n \in F_n$ tending to z and put $u_n = x_n + s_n z_n$. Their normalized images belong to K_n , so $(2M)^{-1} Df(x) z \in K \subset W$. Thus $F \subset Df(x)^{-1} W$, a proper subspace, a contradiction. Apply this implication to the C^1 inverse on $f(E)$ for the reverse direction. □

Suspicious source formula IV.9.31. Feng–Wu, arXiv:2006.12058v1, Section 8, first paragraph, printed p. 38, makes the stronger assertion

$$\forall E \in \mathbb{R}^d : \quad \tau(E) > 0 \implies \tau(g(E)) > 0 \quad \text{for every bi-Lipschitz map } g : \mathbb{R}^d \rightarrow \mathbb{R}^d.$$

The paragraph refers directly to Definition 1.1, without imposing differentiability on g . The complete 40-page source has been checked for standing conditions. The following example refutes precisely this assertion, even for an ambient homeomorphism and a strongly separated self-similar set. It does not contradict Proposition IV.9.30.

Example IV.9.32 (*straightening a uniformly non-flat dust*).

There are a compact planar self-similar set E with $\tau(E) > 0$ and a bi-Lipschitz homeomorphism g of $\mathbb{R}^2$ such that $\tau(g(E)) = 0$ and every finite sum of $g(E)$ has empty planar interior.

Proof. Put $r = 1/10$ and let E be the attractor of

$$\varphi_j(z) = rz + a_j, \quad a_0 = (0, 0), \quad a_1 = (2/5, 2/5), \quad a_2 = (4/5, 0).$$

Its three fixed points $(0, 0), (4/9, 4/9), (8/9, 0)$ belong to E and are not collinear. Theorem IV.9.27 gives $\tau(E) > 0$. The coding series $\sum_{n \geq 0} r^n a_{\omega_{n+1}}$ puts E inside $[0, 8/9] \times [0, 4/9]$. If two codes first differ after a common prefix of length k , their points $(x, y), (x', y')$ satisfy

$$|x - x'| \geq r^k \left(\frac{2}{5} - \frac{1}{10} \frac{8}{9} \right) = r^k \frac{14}{45}, \quad |y - y'| \leq r^k \left(\frac{2}{5} + \frac{1}{10} \frac{4}{9} \right) = r^k \frac{4}{9}.$$

The first-coordinate projection is therefore injective, the first-level pieces are separated, and E is the graph of a function $F : C \rightarrow \mathbb{R}$ of Lipschitz constant $L = 10/7$, with C its compact first-coordinate projection. Define on all of $\mathbb{R}$

$$\tilde{F}(x) = \inf_{t \in C} \{F(t) + L|x - t|\}.$$

The infimum is finite, and the triangle inequality gives $|\tilde{F}(x) - \tilde{F}(x')| \leq L|x - x'|$. For $x \in C$, every term is at least $F(x)$ by the Lipschitz inequality, and $t = x$ attains $F(x)$; hence $\tilde{F}|_C = F$. Now put

$$g(x, y) = (x, y - \tilde{F}(x)), \quad g^{-1}(x, y) = (x, y + \tilde{F}(x)).$$

They are continuous inverse maps. For either sign the triangle inequality gives $|g^{\pm 1}(z) - g^{\pm 1}(z')| \leq (1 + L)|z - z'|$. Thus g is an ambient bi-Lipschitz homeomorphism, and $g(E) = C \times \{0\}$. This non-singleton set lies in a line, so its local convex hulls contain no positive-radius planar ball: $\tau(g(E)) = 0$. Every $ng(E)$ also lies in that line and has empty planar interior. The failed source inference is that distance comparability alone preserves the convex-hull ball condition; this straightening does not preserve that condition.

□

9.3 The planar self-conformal continuation

Definition IV.9.33 (*the general C^1 conformal-IFS convention*).

For $d \geq 2$, a C^1 map $\varphi : U \rightarrow \mathbb{R}^d$ on an open set is *conformal* here if, at every $x \in U$,

$$D\varphi(x) = \lambda(x)O(x), \quad \lambda(x) > 0,$$

where $O(x)$ is orthogonal. A finite C^1 conformal iterated function system consists of injective conformal maps $\varphi_i : U \rightarrow U$ on one bounded connected open set U , with a common Euclidean Lipschitz constant $q < 1$. Its attractor is a nonempty compact set $E \subseteq U$ satisfying $E = \bigcup_i \varphi_i(E)$; this compact containment is part of the convention used in this chapter. Such an attractor is called a self-conformal set. For a word $I = i_1 \cdots i_n$, put $\varphi_I = \varphi_{i_1} \circ \cdots \circ \varphi_{i_n}$. This convention requires neither a convex domain nor $C^{1+\alpha}$ regularity, unlike Definition IV.4.1.

Definition IV.9.34 (*Möbius transformations and restrictions*).

On $\mathbb{R}^d \cup \{\infty\}$, a Möbius transformation means a finite composition of similarities $x \mapsto q + sAx$, with $s \neq 0$ and A orthogonal, and the inversion $x \mapsto x/|x|^2$. Similarities fix ∞ , and inversion interchanges 0 and ∞ . A Möbius restriction is the restriction of such a transformation to an open set on which it is finite. Its pole, when finite, is the point mapped to ∞ .

1. C^1 planar conformal maps are analytic or anti-analytic

Identify $\mathbb{R}^2$ with $\mathbb{C}$.

Lemma IV.9.35 (*planar conformal maps are analytic or anti-analytic*).

Let U be connected and let $\varphi : U \rightarrow \mathbb{C}$ be C^1 with nonzero conformal differential at every point. Then φ is either holomorphic on U or anti-holomorphic on U . In either case φ is real analytic and its derivative never vanishes.

Proof. At every x ,

$$D\varphi(x) = \lambda(x)O(x), \quad \lambda(x) > 0,$$

with $O(x)$ orthogonal. Since $\det D\varphi$ is continuous and never zero, its sign is constant on connected U .

If $\det D\varphi > 0$, then O is a rotation. Writing $\varphi = u + iv$ gives the Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x.$$

Because φ is C^1 , the complex difference quotient exists and equals $u_x + iv_x$, so φ is holomorphic.

If $\det D\varphi < 0$, then $\bar{\varphi}$ satisfies the Cauchy–Riemann equations and is holomorphic; thus φ is anti-holomorphic. The nonzero conformal factor says the complex derivative is nonzero. $\square$

2. A direct finite-IFS distortion lemma

Let $\Phi = \{\varphi_1, \dots, \varphi_\ell\}$ be a planar conformal IFS in Definition IV.9.33, with attractor E . For the cylinder lemmas below assume that E is non-singleton. Let $q < 1$ be the common Euclidean Lipschitz constant for the generators.

For a compact set $L \subset U$ define its forward orbit hull

$$K_L := E \cup \overline{\bigcup_{I \text{ finite}} \varphi_I(L)}. \quad (2.1)$$

Lemma IV.9.36 (*uniform distortion for a planar conformal IFS*).

$K_L \Subset U$.

Proof. For any $z \in L$ and any $e \in E$,

$$|\varphi_I(z) - \varphi_I(e)| \leq q^{|I|}|z - e|,$$

and $\varphi_I(e) \in E$. Thus long-word images of L lie arbitrarily close to E . The finitely many word images of bounded length are compact subsets of U . Since $E \Subset U$, their union together with the long-word tail has compact closure inside U . $\square$

For a generator put

$$\lambda_i(x) := \|D\varphi_i(x)\|.$$

By Lemma IV.9.35 the generators are analytic or anti-analytic, so on K_L the functions λ_i are smooth and bounded above and below away from zero.

For $x \neq y$ define the two-point scale

$$Q_i(x, y) := \frac{|\varphi_i(x) - \varphi_i(y)|}{|x - y|},$$

and extend to the diagonal by

$$Q_i(x, x) = \lambda_i(x). \quad (2.2)$$

Lemma IV.9.37 (*one-generator logarithmic two-point error*).

There is $A_L < \infty$ such that for all generators and all $x, y \in K_L$,

$$|\log Q_i(x, y) - \log \lambda_i(x)| \leq A_L |x - y|. \quad (2.3)$$

Proof. On a fixed neighborhood of K_L , second derivatives of every generator are bounded and λ_i has a positive minimum.

For $|x - y|$ small, Taylor's formula gives

$$\varphi_i(y) - \varphi_i(x) = D\varphi_i(x)(y - x) + O(|x - y|^2).$$

Since $D\varphi_i(x)$ is a similarity,

$$Q_i(x, y) = \lambda_i(x) + O(|x - y|),$$

uniformly. The positive lower bound for λ_i converts this to (2.3).

Away from the diagonal, the continuous extension Q_i on the compact set $K_L \times K_L$ has a positive minimum because φ_i is injective. The logarithmic difference is therefore bounded, and $|x - y|$ has a positive lower bound on that region. Enlarging A_L proves (2.3) globally. $\square$

Lemma IV.9.38 (*word quasi-similarity without Koebe*).

For every compact $L \subset U$ there is $C_L \geq 1$ such that for every word I , every $x, y \in L$, and every $z \in L$,

$$C_L^{-1} \|D\varphi_I(z)\| |x - y| \leq |\varphi_I(x) - \varphi_I(y)| \leq C_L \|D\varphi_I(z)\| |x - y|. \quad (2.4)$$

In particular, if

$$\alpha_I(L) := \sup_{z \in L} \|D\varphi_I(z)\|,$$

then

$$C_L^{-1} \alpha_I(L) |x - y| \leq |\varphi_I(x) - \varphi_I(y)| \leq C_L \alpha_I(L) |x - y|. \quad (2.5)$$

Proof. Write $I = i_1 \cdots i_n$ and set

$$x_k = \varphi_{i_{k+1} \cdots i_n}(x), \quad y_k = \varphi_{i_{k+1} \cdots i_n}(y).$$

The composition identity for two-point scales is

$$\frac{|\varphi_I(x) - \varphi_I(y)|}{|x - y|} = \prod_{k=1}^n Q_{i_k}(x_k, y_k). \quad (2.6)$$

The derivative scale at x is

$$\|D\varphi_I(x)\| = \prod_{k=1}^n \lambda_{i_k}(x_k). \quad (2.7)$$

By contraction,

$$|x_k - y_k| \leq q^{n-k} |x - y|.$$

All x_k, y_k lie in K_L . Summing (2.3),

$$\left| \log \frac{|\varphi_I(x) - \varphi_I(y)|}{\|D\varphi_I(x)\| |x - y|} \right| \leq A_L |x - y| \sum_{j \geq 0} q^j \leq \frac{A_L \text{diam}(L)}{1 - q}. \quad (2.8)$$

The same geometric-series argument applied to the Lipschitz functions $\log \lambda_i$ on K_L gives a uniform bound for

$$|\log \|D\varphi_I(x)\| - \log \|D\varphi_I(z)\||.$$

Combining the two estimates gives (2.4)–(2.5). $\square$

Corollary IV.9.39 (*parent/child scale*).

Choose one compact trapping neighborhood K of E . There is $C \geq 1$ such that

$$\alpha_I \leq \alpha_{I^-} \leq C\alpha_I \quad (2.9)$$

for every nonempty word, where $\alpha_I = \sup_K \|D\varphi_I\|$.

Proof. The upper contraction gives $\alpha_I \leq q\alpha_{I^-}$. For the reverse inequality, use the uniform derivative distortion from Lemma IV.9.38 and the positive minimum of the finitely many generator derivative scales on K , exactly as in the $d \geq 3$ Möbius proof. $\square$

3. Normalized planar cylinders have analytic limits

Suppose $x_n \in E$, $r_n \downarrow 0$, and I_n is the first coding cylinder with

$$x_n \in \varphi_{I_n}(E) \subset B(x_n, r_n). \quad (3.1)$$

As in the $d \geq 3$ proof, (2.5) and (2.9) give

$$c_* \leq \alpha_{I_n}/r_n \leq C_*. \quad (3.2)$$

Define

$$f_n(z) = \frac{\varphi_{I_n}(z) - x_n}{r_n}. \quad (3.3)$$

Lemma IV.9.40 (*locally uniform analytic/anti-analytic compactness*).

After a subsequence, f_n converges locally uniformly on U to a nonconstant injective conformal map f , which is holomorphic or anti-holomorphic.

Proof. Fix compact $L \subset U$. Apply Lemma IV.9.38 on L and compare its derivative scale with the scale on E using the same orbit-hull geometric-series argument. Together with (3.2), this gives uniform two-sided Lipschitz bounds for f_n on L .

Choose $y_n \in E$ with $\varphi_{I_n}(y_n) = x_n$. After a subsequence $y_n \rightarrow y \in E$. Since $f_n(y_n) = 0$, the upper Lipschitz bound makes f_n uniformly bounded on L . Arzelà–Ascoli, followed by a diagonal exhaustion of U , gives a locally uniformly convergent subsequence.

Every cylinder map is holomorphic or anti-holomorphic. Pass to a subsequence of fixed orientation type. A locally uniform limit of holomorphic maps is holomorphic, for example by Morera’s theorem since contour integrals pass to the uniform limit; the anti-holomorphic case follows after conjugation. The lower Lipschitz bound passes to the limit, so f is nonconstant and injective. $\square$

4. Zero thickness: analytic arcs and the connected-curve boundary

Definition IV.9.41 (*simple real-analytic curves and arcs*).

A simple real-analytic curve is the image of an injective map $\gamma : J \rightarrow \mathbb{R}^2$, where J is an open interval, each coordinate of γ is locally given by a convergent real power series, and $\gamma'(t) \neq 0$ for every $t \in J$; the parametrization is required to be a homeomorphism onto its image. A compact simple real-analytic arc is the image of a compact subinterval of such a J .

Source boundary IV.9.42 (*the rejected connected-curve inference*).

The former target was that $\tau(E) = 0$ forces E into one simple real-analytic curve in Definition IV.9.41. This is not a theorem under Definition IV.9.33: the example below disproves it. The valid analytic-arc conclusion is isolated separately, so no later proof may use the rejected target as a lemma.

Suspicious source formula IV.9.43. The source examined here is Feng–Wu, *On arithmetic sums of fractal sets in $\mathbb{R}^d$* , arXiv:2006.12058v1, 22 June 2020, <https://arxiv.org/pdf/2006.12058v1>. In the notation of this chapter, its Theorem 1.4, printed p. 3, asserts the following equivalence for every $d \geq 2$ and every such attractor $E = \bigcup_{i=1}^{\ell} \varphi_i(E)$:

$$[\exists n \in \mathbb{N} : \text{int}_{\mathbb{R}^d}(nE) \neq \emptyset] \iff [\forall \text{ affine hyperplanes } H : E \not\subset H].$$

Here E is the nonempty compact attractor of a conformal IFS with the standing hypotheses specified below. Its Lemma 5.1(i), printed p. 15, additionally assumes E is not a singleton and asserts

$$d = 2, \quad \tau(E) = 0 \implies E \subset \Gamma \text{ for some simple analytic curve } \Gamma.$$

The curve-convention qualification and the failed implication in its proof are stated below; the sumset counterexample does not depend on that convention. The full 40-page version, including its standing definitions, is the scope of this comparison; no claim about a published version is made.

The source's conformal-IFS definition in Section 2, printed p. 5, and the standing assumptions of Section 5, printed p. 15, require a compact set $X \subset \mathbb{R}^d$, a bounded connected open $U \supset X$, and finitely many maps $\varphi_i : X \rightarrow X$ extending to injective contracting C^1 conformal maps $U \rightarrow \varphi_i(U) \subset U$. Conformality means

$$\|D\varphi_i(x)v\| = \|D\varphi_i(x)\| \|v\| > 0 \quad (x \in U, v \neq 0);$$

contraction is the common Euclidean Lipschitz bound strictly below 1. Section 5 additionally assumes a non-singleton attractor. These are the hypotheses checked in Example IV.9.45.

Lemma IV.9.44 (*the proved analytic-arc cover*).

If $\tau(E) = 0$, then E is covered by countably many compact simple real-analytic arcs, and some cylinder $\varphi_I(E)$ lies in one of those arcs.

Proof. Because E is non-singleton, $\text{diam}(E) > 0$. By the local microset criterion choose $x_n \in E$, $r_n \downarrow 0$, with normalized local sets converging to

$$F \subset W,$$

where W is a line through the origin.

Choose minimal cylinders I_n as in (3.1), and normalized maps f_n as in (3.3). By Lemma IV.9.40, after a subsequence $f_n \rightarrow f$ locally uniformly, where f is injective holomorphic or anti-holomorphic. Since

$$f_n(E) \subset r_n^{-1}((E \cap B(x_n, r_n)) - x_n),$$

we have

$$f(E) \subset W. \quad (4.1)$$

The inverse f^{-1} is holomorphic or anti-holomorphic on $f(U)$. The relatively open subset $f(U) \cap W$ of the line W is a countable union of open intervals. Exhaust those intervals by countably many compact subintervals. Their inverse images are compact simple real-analytic arcs and cover E .

Their intersections with E are closed in the compact, hence complete, set E . By Baire's theorem, one such compact arc contains a relatively open nonempty subset of E . Choose x in that relative interior and a coding of x . A small relative ball of E around x is contained in the arc. Since the coding cylinders contain x and their diameters tend to zero, some cylinder $\varphi_I(E)$ lies in that ball, and hence in the arc. $\square$

Source boundary

The last planar inference in the proof of source Lemma 5.1, printed pp. 16–17, passes from one cylinder in an analytic arc to the whole attractor in an analytic curve. The local inverse φ_I^{-1} is defined only on $\varphi_I(U)$, not on an arbitrary arc containing $\varphi_I(E)$. Intersecting that arc with $\varphi_I(U)$ can split it into several components. Neither a single component containing the entire cylinder nor continuation of its inverse across the intervening gaps has been established. The source does not separately define “analytic curve”; its Lemma 5.3 proof, printed p. 17, uses a parametrization on the connected interval $[0, 1]$. Example IV.9.45 rules out the connected embedded meaning fixed here, and also any one compact analytically parametrized arc. It does not rule out a disconnected analytic one-dimensional manifold or an analytic zero set. The arithmetic-sum obstruction in the same example, however, is independent of that terminology and contradicts the planar case of source Theorem 1.4 under its stated definition. Thus the unrestricted planar descendants IV.9.48 and IV.9.76 must not be used.

Example IV.9.45 (*two horizontal levels*).

There is a planar conformal IFS satisfying all the preceding local and source hypotheses whose attractor E has $\tau(E) = 0$, is not contained in a line or a simple real-analytic curve, and is not arithmetically thick.

Proof. Identify $\mathbb{R}^2$ with $\mathbb{C}$ and put

$$\begin{aligned} U &= \{x + iy : |x| < 1/2, -1/100 < y < 101/100\}, \\ X &= \{x + iy : |x| \leq 1/4, -1/200 \leq y \leq 201/200\}, \\ \psi_0(z) &= 10^{-3} \exp(\pi z), \quad \psi_1(z) = 10^{-3} \exp(\pi z) + i. \end{aligned}$$

Here the complex exponential is the entire function $\exp(w) = \sum_{n=0}^{\infty} w^n/n!$. The set U is bounded, open, connected, and convex, and X is a compact connected rectangle contained in U .

Both maps are holomorphic with derivative $10^{-3}\pi \exp(\pi z) \neq 0$, hence are C^1 conformal. If $\exp(\pi z) = \exp(\pi w)$, then $z - w = 2ki$ for an integer k . The imaginary-coordinate width of U is $51/50 < 2$, so for $z, w \in U$ only $k = 0$ is possible. Both maps are therefore injective on U . Using $\pi < 4$ and $e < 3$, for every $z \in U$ we have

$$|\psi_0(z)| < 10^{-3}e^{\pi/2} < 9/1000 < 1/100, \quad |\psi'_j(z)| < 36/1000 < 1 \quad (j = 0, 1).$$

The first bound puts $\psi_0(U)$ in U , and also puts $\psi_1(U) = \psi_0(U) + i$ in U . Integration of the derivative along the line segment between any two points of convex U gives the common Euclidean Lipschitz constant $36/1000$. Thus these are genuinely contracting maps on all of U , not only on the attractor.

The bounded-distortion convention in source Section 2 does not exclude this example either. Put $q = 36/1000$. For a word I of length n , the chain rule and $\log |\psi'_j(z)| = \log(10^{-3}\pi) + \pi \operatorname{Re} z$ give, for $z, w \in U$,

$$\left| \log \frac{|\psi'_I(z)|}{|\psi'_I(w)|} \right| \leq \pi \sum_{k=0}^{n-1} q^k |z - w| \leq \frac{\pi \operatorname{diam} U}{1 - q}.$$

Consequently the derivative ratio lies in $[L^{-1}, L]$, with $L = \exp(\pi \operatorname{diam} U/(1 - q))$, independently of I . This is stronger than the source's bound on an interior neighborhood U_1 with $X \subset U_1 \Subset U$.

For $z \in X$ the sharper estimate

$$|\psi_0(z)| \leq 10^{-3}e^{\pi/4} < 3/1000 < 1/200$$

gives $\psi_0(X), \psi_1(X) \subset X$. Let

$$Y = \{x + iy : |x| \leq 1/4, y \in \{0, 1\}\} \subset X.$$

For $z = x + iy \in Y$, the equality $\exp(\pi z) = (-1)^y e^{\pi x}$ shows that $\psi_0(z)$ is real and $\psi_1(z)$ has imaginary coordinate 1; their real coordinates have absolute value less than $3/1000$. Hence both maps preserve Y . The unique compact attractor on X is consequently also the attractor on Y , so $E \subset Y \Subset U$.

The fixed points p_0 of ψ_0 and p_1 of ψ_1 belong to E by their constant codings. Their imaginary coordinates are 0 and 1, respectively. For $n \geq 1$ let $a_n = \psi_0^n(p_1) \in E$. All a_n are real and converge to p_0 by contraction. They are pairwise distinct: if $a_n = a_m$ with $m > n$, injectivity of ψ_0^n gives $p_1 = \psi_0^{m-n}(p_1)$, impossible because the right-hand side is real. The same injectivity shows $a_n \neq p_0$. Thus E contains infinitely many points of the real axis and also p_1 off that axis; it is non-singleton and cannot lie in any line.

For $x \in E$ and $0 < r < 1/2$, the local set $B(x, r) \cap E$ lies in the single horizontal line through x . Its convex hull cannot contain a planar ball of positive radius. Since $\text{diam } E \geq 1$, these radii are within the defining range of thickness, proving $\tau(E) = 0$ in both Definition IV.9.18 and the source's Definition 1.1, printed p. 2. Moreover, for every positive integer n ,

$$nE \subset \bigcup_{k=0}^n (\mathbb{R} + ki).$$

Every open planar ball contains a point whose imaginary coordinate avoids the finitely many integers $0, \dots, n$. Thus the displayed union has empty planar interior, and E is not arithmetically thick in either definition.

Finally suppose $E \subset \gamma(J)$ for a curve as in Definition IV.9.41. Write $t_n = \gamma^{-1}(a_n)$ and $t_0 = \gamma^{-1}(p_0)$. Continuity of the inverse in the curve topology gives $t_n \rightarrow t_0$ inside the open interval J . The real-analytic function $t \mapsto \text{Im } \gamma(t)$ vanishes at the distinct t_n ; the identity theorem makes it zero on all of connected J . This contradicts $p_1 \in \gamma(J)$. The same obstruction holds for any single analytic parametrization on $[0, 1]$: choose parameters for the distinct a_n , pass to an accumulating subsequence in $[0, 1]$, and use analyticity on a neighborhood of that compact interval. By contrast, Y is contained in the disconnected real-analytic zero set $\{x + iy : y(y-1) = 0\}$, so this argument does not exclude that broader interpretation of a curve. $\square$

5. Uniform local diameter in cylinder pieces

Lemma IV.9.46 (*uniform local diameter in planar cylinders*).

There is $L_0 > 0$ such that for every word I , every $x \in \varphi_I(E)$, and every

$$0 < r < \text{diam}(\varphi_I(E)),$$

one has

$$\text{diam}(B(x, r) \cap \varphi_I(E)) \geq L_0 r. \quad (5.1)$$

Proof. The proof is identical to that of Lemma IV.9.72, using the planar quasi-similarity (2.5) and parent/child comparison (2.9): choose the first deeper cylinder contained in $B(x, r)$; its parent is not contained, so its scale is comparable to r and its diameter is bounded below by a fixed multiple of r . $\square$

6. Two pieces of a non-flat analytic curve have positive-thickness sum

Lemma IV.9.47 (*planar FYW Lemma 5.3*).

Assume E is contained in a simple real-analytic curve γ but not in a line. Then there exist words I, J such that

$$\tau(\varphi_I(E) + \varphi_J(E)) > 0. \quad (6.1)$$

Proof. Step 1: choose two genuinely different tangent directions. Because E is not contained in a line and is a non-singleton self-conformal attractor, its accumulation points cannot all have one tangent direction on the analytic curve; otherwise the analytic tangent-direction function would be constant on a set with an accumulation point and hence on the relevant analytic arc, forcing that arc, and then E , into a line.

Choose $p, q \in E$ where the tangent directions differ. Rotate coordinates so both tangent slopes are finite, say u and v . Choose

$$0 < \varepsilon < |u - v|/4.$$

Shrink neighborhoods U_p, U_q on the curve so that every chord joining two distinct points of $\gamma \cap U_p$ has slope in $(u - \varepsilon, u + \varepsilon)$, and similarly every chord in U_q has slope in $(v - \varepsilon, v + \varepsilon)$. Choose cylinder pieces

$$A = \varphi_I(E) \subset U_p, \quad B = \varphi_J(E) \subset U_q.$$

Step 2: small-scale thickness. Fix $x \in A, y \in B$ and

$$0 < r < r_* := \min\{\text{diam } A, \text{diam } B\}.$$

By Lemma IV.9.46 applied with radius $r/2$,

$$\text{diam}(B(x, r/2) \cap A) \geq L_0 r/2.$$

Hence there is $x' \in B(x, r/2) \cap A$ with

$$|x' - x| \geq L_0 r/4; \quad (6.2)$$

otherwise the diameter would be less than $L_0 r/2$. Similarly choose y' with

$$|y' - y| \geq L_0 r/4. \quad (6.3)$$

The local sum satisfies

$$B(x + y, r) \cap (A + B) \supset (B(x, r/2) \cap A) + (B(y, r/2) \cap B). \quad (6.4)$$

Therefore its convex hull contains the parallelogram with vertices

$$x + y, \quad x' + y, \quad x + y', \quad x' + y'.$$

Both generating edge lengths are at least $L_0 r/4$. Their angle is uniformly bounded away from 0 and π because the two chord-slope intervals are disjoint with a fixed gap. An elementary altitude estimate gives a constant $c_0 > 0$, independent of x, y, r , such that this

parallelogram contains a ball of radius $c_0 r$. Thus the thickness inequality holds at every scale $r < r_*$.

Step 3: extend from small scales to all scales. Let $F = A + B$ and $D = \text{diam } F$. If $r_* \leq r \leq D$, then

$$B(z, r) \cap F \supset B(z, r_*/2) \cap F$$

for every $z \in F$. The small-scale conclusion at $r_*/2$ supplies a convex ball of radius $c_0 r_*/2$, which is at least

$$\frac{c_0 r_*}{2D} r.$$

Hence $\tau(F) > 0$. □

7. Planar self-conformal arithmetic thickness

Source boundary IV.9.48 (*unrestricted planar arithmetic thickness*).

The unrestricted assertion that every planar self-conformal set not contained in a line is arithmetically thick is false under Definition IV.9.33, as Example IV.9.45 shows. The rejected source assertion is recorded in IV.9.43; it supplies no premise for the following theorem.

Theorem IV.9.49 (*the proved planar branches*).

Let E be a planar self-conformal set in Definition IV.9.33. If E is not contained in a line and either $\tau(E) > 0$ or E is contained in a simple real-analytic curve in Definition IV.9.41, then E is arithmetically thick. Conversely, arithmetic thickness always implies that E is not contained in a line.

Proof. If E lies in a line, every finite sum lies in a line and has empty interior. Assume E is not contained in a line.

If $\tau(E) > 0$, apply Theorem IV.9.26 to sufficiently many copies of E ; its diameter is positive because it is not in a line. In the other branch, containment in a simple analytic curve is an explicit hypothesis, not a consequence of zero thickness. Lemma IV.9.47 gives cylinder subsets E_1, E_2 of E with

$$\tau(E_1 + E_2) > 0.$$

The set $F = E_1 + E_2$ has positive diameter. Apply Theorem IV.9.26 to sufficiently many copies of F . Since $nF \subset (2n)E$, a finite arithmetic sum of E has nonempty interior. □

9.4 Liouville rigidity at C^1 regularity

1. Local inverse and orientation

Because Df is invertible everywhere, the inverse function theorem makes f a local C^1 diffeomorphism. Since f is injective, the local inverses agree on overlaps, so

$$V = f(\Omega)$$

is open and

$$g = f^{-1} : V \rightarrow \Omega$$

is C^1 .

The sign of $\det Df$ is continuous and nonzero on connected Ω , hence constant. Consequently the sign of $\det Dg$ is constant as well. Write the conformal scale of g as

$$Dg(y)^T Dg(y) = \mu(y)^2 I, \quad \mu(y) > 0. \quad (\text{L1.2})$$

Then, with $\text{sgn} \in \{+1, -1\}$ fixed,

$$J_g(y) = \det Dg(y) = \text{sgn } \mu(y)^n. \quad (\text{L1.3})$$

On every compact $K \Subset V$,

$$0 < m_K \leq \mu \leq M_K < \infty. \quad (\text{L1.4})$$

2. Elementary Poincaré-Piola identity for C^1 maps

For an $n \times n$ matrix A let $\text{cof } A$ be its cofactor matrix, with

$$A^T \text{cof } A = (\det A)I. \quad (\text{L1.5})$$

Lemma IV.9.50 (*Piola identity for C^1 maps*).

If $u : U \rightarrow \mathbb{R}^n$ is C^1 , then

$$\text{div}(\text{cof } Du) = 0 \quad (\text{L1.6})$$

in the distributional sense, where for a matrix field M we use the row-wise divergence

$$(\text{div } M)_i = \sum_j \partial_j M_{ij}.$$

Proof. First assume u is smooth. In alternating-symbol notation,

$$(\text{cof } Du)_{ij} = \frac{1}{(n-1)!} \varepsilon_{ii_2 \dots i_n} \varepsilon_{jj_2 \dots j_n} u_{i_2, j_2} \cdots u_{i_n, j_n}.$$

Differentiate in x_j and sum over j . Every term contains one second derivative u_{i_k, jj_k} ; interchanging j and j_k leaves that second derivative unchanged and reverses the sign of the alternating symbol. Terms cancel in pairs, giving (L1.6).

For C^1 u , mollify on $U' \Subset U$. The mollifications $u_\varepsilon \rightarrow u$ in $C^1(U')$, hence

$$\text{cof } Du_\varepsilon \rightarrow \text{cof } Du$$

uniformly on U' . Since $\text{div}(\text{cof } Du_\varepsilon) = 0$, passage against a compactly supported test function proves (L1.6) on U' . Exhaust U . $\square$

Corollary IV.9.51 (*the inverse is n -harmonic*).

The inverse g satisfies

$$\operatorname{div}(|Dg|^{n-2}Dg) = 0 \quad (\text{L1.7})$$

weakly on V , where $|\cdot|$ is the Frobenius norm.

Proof. Write $Dg = \mu O$ with O orthogonal. Then

$$\operatorname{cof} Dg = J_g Dg^{-T} = \operatorname{sgn} \mu^{n-1} O = \operatorname{sgn} \mu^{n-2} Dg.$$

Since $|Dg| = \sqrt{n}\mu$,

$$\operatorname{cof} Dg = c_n |Dg|^{n-2} Dg$$

with a nonzero constant c_n depending only on n and the orientation. Apply Lemma IV.9.50.

□

3. The only nonlinear regularity estimate needed

For $p \geq 2$ and a Euclidean vector, or matrix regarded as a vector, P define

$$A_p(P) = |P|^{p-2}P, \quad V_p(P) = |P|^{(p-2)/2}P. \quad (\text{L1.8})$$

Lemma IV.9.52 (*elementary monotonicity inequalities*).

There are constants $c_p, C_p > 0$ such that for all P, Q ,

$$\langle A_p(P) - A_p(Q), P - Q \rangle \geq c_p |V_p(P) - V_p(Q)|^2, \quad (\text{L1.9})$$

and

$$|A_p(P) - A_p(Q)| \leq C_p (|P| + |Q|)^{(p-2)/2} |V_p(P) - V_p(Q)|. \quad (\text{L1.10})$$

Proof. The Hessian of $\Phi(P) = |P|^p/p$ is

$$DA_p(R)[H, H] = |R|^{p-2}|H|^2 + (p-2)|R|^{p-4}\langle R, H \rangle^2. \quad (\text{L1.11})$$

Thus it is between

$$|R|^{p-2}|H|^2 \quad \text{and} \quad (p-1)|R|^{p-2}|H|^2.$$

Integrate (L1.11) on the segment $R_s = Q + s(P - Q)$. If $|P| \geq |Q|$, then for $s \in [3/4, 1]$,

$$|R_s| \geq (2s-1)|P| \geq |P|/2;$$

the other case is symmetric. Hence

$$\int_0^1 |R_s|^{p-2} ds \asymp_p (|P| + |Q|)^{p-2}. \quad (\text{L1.12})$$

The same segment calculation for V_p , whose derivative has size comparable to $|R|^{(p-2)/2}$, gives

$$|V_p(P) - V_p(Q)|^2 \asymp_p (|P| + |Q|)^{p-2} |P - Q|^2. \quad (\text{L1.13})$$

Equations (L1.9) and (L1.10) follow from (L1.11)–(L1.13) and the mean-value formula for A_p . □

Lemma IV.9.53 (*difference-quotient criterion, local L^2 form*).

Let $F \in L^2_{\text{loc}}(U)$. Suppose that for every $K \Subset U$ and each coordinate vector e_i ,

$$\sup_{0 < |h| < h_K} \left\| \frac{F(\cdot + he_i) - F}{h} \right\|_{L^2(K)} < \infty. \quad (\text{L1.14})$$

Then $F \in W^{1,2}_{\text{loc}}(U)$.

Proof. Fix K and i . Along $h_j \rightarrow 0$, weak compactness in L^2 gives a weakly convergent subsequence of the difference quotients, say to G_i . For $\phi \in C_c^\infty(K)$,

$$\begin{aligned} \int G_i \phi &= \lim_j \int \frac{F(x + h_j e_i) - F(x)}{h_j} \phi(x) dx \\ &= - \int F \partial_i \phi. \end{aligned}$$

Thus G_i is the distributional derivative $\partial_i F$. The bound is independent of the chosen subsequence, so $\partial_i F \in L^2(K)$. Repeat for all i and exhaust U . $\square$

Lemma IV.9.54 (*translation estimate*).

If $u \in W^{1,p}_{\text{loc}}(U)$, $K \Subset U$, and h is small, then

$$\|u(\cdot + h) - u\|_{L^p(K)} \leq |h| \|Du\|_{L^p(K_h)}, \quad (\text{L1.15})$$

where K_h is a fixed slightly larger compact set.

Proof. For smooth u ,

$$u(x + h) - u(x) = \int_0^1 Du(x + sh)h ds.$$

Jensen's inequality followed by integration gives (L1.15). Approximate a Sobolev map by smooth maps locally and pass to the limit. In the present application $u = g$ is C^1 , so the displayed identity already applies directly. $\square$

Theorem IV.9.55 (*p -harmonic difference-quotient regularity*).

Let $p \geq 2$ and $u \in W^{1,p}_{\text{loc}}(U, \mathbb{R}^m)$ satisfy

$$\text{div}(|Du|^{p-2} Du) = 0 \quad (\text{L1.16})$$

weakly. Then

$$V_p(Du) = |Du|^{(p-2)/2} Du \in W^{1,2}_{\text{loc}}(U). \quad (\text{L1.17})$$

Proof. Fix $\eta \in C_c^\infty(U)$, and let h be a small coordinate translation. Write

$$\begin{aligned} u_h(x) &= u(x + h), & \Delta u &= u_h - u, \\ \Delta A &= A_p(Du_h) - A_p(Du), & \Delta V &= V_p(Du_h) - V_p(Du). \end{aligned}$$

Subtract the weak equations for u_h and u and use

$$\psi = \eta^2 \Delta u$$

as a test function. This gives

$$\int \eta^2 \langle \Delta A, Du_h - Du \rangle = -2 \int \eta \langle \Delta A, \Delta u \otimes \nabla \eta \rangle. \quad (\text{L1.18})$$

By Lemma IV.9.52,

$$\begin{aligned} c_p \int \eta^2 |\Delta V|^2 &\leq 2 \int \eta |\Delta A| |\Delta u| |\nabla \eta| \\ &\leq C \int \eta (|Du_h| + |Du|)^{(p-2)/2} |\Delta V| |\Delta u| |\nabla \eta|. \end{aligned}$$

Absorb one half of the ΔV term by Cauchy's inequality:

$$\int \eta^2 |\Delta V|^2 \leq C \int |\Delta u|^2 |\nabla \eta|^2 (|Du_h| + |Du|)^{p-2}. \quad (\text{L1.19})$$

Hölder with exponents $p/2$ and $p/(p-2)$, followed by Lemma IV.9.54, yields

$$\begin{aligned} \text{RHS} &\leq C \left(\int |\Delta u|^p \right)^{2/p} \left(\int (|Du_h| + |Du|)^p \right)^{(p-2)/p} \\ &\leq C_K |h|^2. \end{aligned} \quad (\text{L1.20})$$

Thus the coordinate difference quotients of $V_p(Du)$ are uniformly bounded in L^2 on every compact set where $\eta = 1$. Apply Lemma IV.9.53. $\square$

Apply Theorem IV.9.55 to g with $p = n$ using Corollary IV.9.51. We obtain

$$F := V_n(Dg) = |Dg|^{(n-2)/2} Dg \in W_{\text{loc}}^{1,2}(V). \quad (\text{L1.21})$$

Lemma IV.9.56 (*local Sobolev regularity of the inverse*).

In fact

$$Dg \in W_{\text{loc}}^{1,2}(V), \quad (\text{L1.22})$$

and the conformal scale

$$\mu = |Dg|/\sqrt{n}$$

also belongs to $W_{\text{loc}}^{1,2}(V)$.

Proof. Fix $K \Subset V$. By (L1.4),

$$0 < a_K \leq |F(y)| \leq b_K < \infty.$$

The inverse of $P \mapsto V_n(P)$ is

$$\Psi(Q) = |Q|^{-(n-2)/n} Q. \quad (\text{L1.23})$$

On the annulus $a_K/2 \leq |Q| \leq 2b_K$, Ψ is Lipschitz. For small translations staying in a slightly larger compact set,

$$|Dg(y+h) - Dg(y)| = |\Psi(F(y+h)) - \Psi(F(y))| \leq C_K |F(y+h) - F(y)|.$$

The L^2 difference quotients of F are bounded, so Lemma IV.9.53 gives (L1.22).

Finally the matrix norm is 1-Lipschitz:

$$||A| - |B|| \leq |A - B|.$$

The same difference-quotient argument gives $|Dg|$, hence μ , in $W_{\text{loc}}^{1,2}$. □

Lemma IV.9.57 (*strong difference quotients*).

If $H \in W_{\text{loc}}^{1,2}(V)$, then for every coordinate vector e_i ,

$$\frac{H(\cdot + te_i) - H}{t} \longrightarrow \partial_i H \quad (\text{L1.24})$$

strongly in L_{loc}^2 as $t \rightarrow 0$.

Proof. For smooth H this follows from

$$\frac{H(y + te_i) - H(y)}{t} = \int_0^1 \partial_i H(y + ste_i) ds$$

and translation continuity in L^2 . Approximate $H \in W^{1,2}$ by smooth functions on a slightly larger compact set and use the uniform difference-quotient bounds from Lemmas IV.9.53 and IV.9.54 to pass to the limit. □

4. Translation-conjugate conformal path

Fix a coordinate vector e_i and a compact domain $A \Subset \Omega$. For $|t|$ small define

$$f_t(x) = g(f(x) + te_i), \quad x \in A. \quad (\text{L1.25})$$

Then $f_0 = \text{id}$. Each f_t is C^1 conformal. Its orientation is positive because the orientation signs of f and g are equal and cancel in the composition. Define

$$X_t(x) = \frac{f_t(x) - x}{t}. \quad (\text{L1.26})$$

Lemma IV.9.58 (*uniform first variation*).

On A ,

$$X_t \longrightarrow X_i := [Df]^{-1}e_i \quad (\text{L1.27})$$

uniformly as $t \rightarrow 0$.

Proof. Writing $y = f(x)$,

$$X_t(x) = \int_0^1 Dg(y + ste_i)e_i ds.$$

The derivative Dg is uniformly continuous on a compact neighborhood of $f(A)$. Thus the integral converges uniformly to $Dg(f(x))e_i = [Df(x)]^{-1}e_i$. $\square$

Put

$$B_t = Df_t, \quad A_t = \frac{B_t - I}{t} = DX_t. \quad (\text{L1.28})$$

Lemma IV.9.59 (*strong L^2 derivative variation*).

There is

$$A_i(x) = [\partial_i Dg](f(x))Df(x) \in L_{\text{loc}}^2(A) \quad (\text{L1.29})$$

such that

$$A_t \longrightarrow A_i$$

strongly in $L^2(A)$.

Proof. Since $Dg(f(x))Df(x) = I$,

$$A_t(x) = \frac{Dg(f(x) + te_i) - Dg(f(x))}{t} Df(x). \quad (\text{L1.30})$$

The matrix Df is bounded on A . Under $y = f(x)$, the C^1 change-of-variables formula has the locally bounded positive density $|J_g(y)|$:

$$dx = |J_g(y)| dy.$$

Hence the $L^2(A)$ norm of the difference between (L1.30) and (L1.29) is bounded by a constant times the L^2 norm on a compact subset of V of

$$\frac{Dg(y + te_i) - Dg(y)}{t} - \partial_i Dg(y).$$

Use Lemma IV.9.57. $\square$

By Lemmas IV.9.58 and IV.9.59, for every test function ϕ ,

$$\begin{aligned} \int X_i \partial_j \phi &= \lim_{t \rightarrow 0} \int X_t \partial_j \phi \\ &= - \lim_{t \rightarrow 0} \int (A_t)_{\cdot j} \phi \\ &= - \int (A_i)_{\cdot j} \phi. \end{aligned}$$

Therefore

$$X_i \in W_{\text{loc}}^{1,2}(\Omega), \quad DX_i = A_i. \quad (\text{L1.31})$$

5. Strong L^2 linearization of the conformal scale

The conformal scale of $B_t = Df_t$ is

$$\sigma_t(x) = \frac{\mu(f(x) + te_i)}{\mu(f(x))}. \quad (\text{L1.32})$$

Indeed $Dg(f(x) + te_i)$ has scale $\mu(f(x) + te_i)$, while $Df(x)$ has scale $\mu(f(x))^{-1}$. Set

$$s_t = \frac{\sigma_t - 1}{t}. \quad (\text{L1.33})$$

Lemma IV.9.60 (*strong linearization of the conformal scale*).

On A ,

$$s_t \longrightarrow s_i(x) := \frac{[\partial_i \mu](f(x))}{\mu(f(x))} \quad (\text{L1.34})$$

strongly in L^2 .

Proof. The denominator $\mu(f(x))$ is bounded away from zero on A . Apply the same change-of-variables argument as in Lemma IV.9.59 and the strong difference-quotient convergence in Lemma IV.9.57 for $\mu \in W_{\text{loc}}^{1,2}$. $\square$

6. Linearized conformality equals the Ahlfors equation

Exact conformality of f_t says

$$B_t^T B_t = \sigma_t^2 I. \quad (\text{L1.35})$$

Using

$$B_t = I + tA_t, \quad \sigma_t = 1 + ts_t,$$

we obtain the exact identity

$$A_t + A_t^T + tA_t^T A_t = (2s_t + ts_t^2)I. \quad (\text{L1.36})$$

The L^2 convergence in Lemmas IV.9.59 and IV.9.60 implies that A_t and s_t remain bounded in L^2 on compact sets. Hence

$$tA_t^T A_t \rightarrow 0, \quad ts_t^2 \rightarrow 0$$

in L^1 . Pass to the limit in (L1.36):

$$DX_i + DX_i^T = 2s_i I. \quad (\text{L1.37})$$

Taking the trace gives $2 \operatorname{div} X_i = 2ns_i$. Therefore

$$DX_i + DX_i^T = \frac{2}{n}(\operatorname{div} X_i)I \quad (\text{L1.38})$$

in distributions. This is exactly the Ahlfors conformal-Killing equation.

7. The Ahlfors deformation theorem—full local proof

Theorem IV.9.61 (*Ahlfors deformation theorem*).

Let $n \geq 3$ and let X be a distributional vector field on a connected domain Ω satisfying

$$DX + DX^T = \frac{2}{n}(\operatorname{div} X)I. \quad (\text{L1.39})$$

Then X is the restriction of a polynomial of degree at most two and has the form

$$X(x) = a + (S + \alpha I)x + 2\langle c, x \rangle x - |x|^2 c, \quad (\text{L1.40})$$

where $a, c \in \mathbb{R}^n$, $\alpha \in \mathbb{R}$, and S is skew-symmetric.

Proof. *Step 1: write the equation with one scalar field.* Put

$$q = \frac{1}{n} \operatorname{div} X.$$

Then, in distributions,

$$X_{i,j} + X_{j,i} = 2q\delta_{ij}. \quad (\text{L1.41})$$

In particular, $X_{i,j} = -X_{j,i}$ for $i \neq j$, and all diagonal derivatives $X_{i,i}$ equal q .

Step 2: recover every second derivative from ∇q . Differentiate (L1.41) in x_k and write down the two cyclic companions:

$$\begin{aligned} X_{i,jk} + X_{j,ik} &= 2\delta_{ij}q_k, \\ X_{i,kj} + X_{k,ij} &= 2\delta_{ik}q_j, \\ X_{j,ki} + X_{k,ji} &= 2\delta_{jk}q_i. \end{aligned}$$

Add the first two identities and subtract the third. Distributional derivatives commute, so the unwanted terms cancel and

$$X_{i,jk} = \delta_{ij}q_k + \delta_{ik}q_j - \delta_{jk}q_i. \quad (\text{L1.42})$$

Step 3: the Hessian of q vanishes. Differentiate (L1.42) in x_l . Since the left side is symmetric in the derivative indices k, l ,

$$\delta_{ik}q_{jl} - \delta_{jk}q_{il} = \delta_{il}q_{jk} - \delta_{jl}q_{ik}. \quad (\text{L1.43})$$

First let $j \neq l$. Choose $i = k$ distinct from both j and l ; this is possible because $n \geq 3$. Equation (L1.43) gives

$$q_{jl} = 0. \quad (\text{L1.44})$$

Thus all off-diagonal second derivatives of q vanish.

Next choose $i \neq j$ and put $k = i$, $l = j$ in (L1.43). Using (L1.44),

$$q_{jj} = -q_{ii}. \quad (\text{L1.45})$$

Given three distinct indices i, j, k ,

$$q_{ii} = -q_{jj} = q_{kk},$$

while also $q_{ii} = -q_{kk}$. Hence $q_{ii} = 0$. Therefore

$$q_{ij} = 0 \quad \text{for all } i, j \quad (\text{L1.46})$$

in distributions.

Mollifying on compact balls shows that a distribution whose Hessian vanishes is affine. Connectedness patches the local affine functions, so

$$q(x) = \alpha + 2\langle c, x \rangle \quad (\text{L1.47})$$

for constants α and c .

Step 4: integrate the constant Hessian of X . Substitute $q_j = 2c_j$ into (L1.42):

$$X_{i,jk} = 2(\delta_{ij}c_k + \delta_{ik}c_j - \delta_{jk}c_i). \quad (\text{L1.48})$$

The quadratic vector field

$$Q_c(x) = 2\langle c, x \rangle x - |x|^2 c$$

has exactly the Hessian (L1.48). Hence $Y = X - Q_c$ has zero distributional Hessian. By the same mollification argument, Y is affine on connected Ω :

$$Y = a + Lx.$$

Finally (L1.39) gives

$$L + L^T = 2\alpha I.$$

Therefore $L = S + \alpha I$ with S skew-symmetric, proving (L1.40). $\square$

8. Smoothness and reduction to the local C^2 theorem

Apply Theorem IV.9.61 to each vector field

$$X_i = [Df]^{-1}e_i, \quad i = 1, \dots, n. \quad (\text{L1.49})$$

Every X_i is a quadratic polynomial. Hence the inverse derivative matrix

$$M(x) = [Df(x)]^{-1}$$

has polynomial entries and is C^∞ . Since $\det M(x) \neq 0$ everywhere,

$$Df(x) = M(x)^{-1}$$

is C^∞ on Ω . In particular f is C^2 .

Theorem IV.9.62 (*Liouville rigidity in dimensions at least three*).

Let $n \geq 3$ and let $f : \Omega \rightarrow \mathbb{R}^n$ be a C^1 conformal diffeomorphism on a domain $\Omega \subset \mathbb{R}^n$. Then f is the restriction of a Möbius transformation.

Proof. The preceding C^1 regularity argument applies to this f : its inverse is n -harmonic by Corollary IV.9.51, the difference-quotient argument makes each inverse-derivative column $[Df]^{-1}e_i$ a distributional solution of the Ahlfors equation, and Theorem IV.9.61 makes that column a quadratic polynomial. Since Df is nonsingular, $Df = ([Df]^{-1})^{-1}$ is smooth. It remains to prove the smooth Liouville step; we do so rather than invoking a second rigidity theorem.

Write $F_i = \partial_i f$ and $Df^T Df = e^{2u}I$, where u is smooth. Taking one derivative of $F_i \cdot F_j = e^{2u}\delta_{ij}$ and solving the resulting three cyclic identities gives the exact second-derivative formula

$$\partial_i F_j = u_i F_j + u_j F_i - \delta_{ij} \sum_m u_m F_m. \quad (\text{L1.50})$$

Indeed the inner product of the right side with F_k is $e^{2u}(u_i \delta_{jk} + u_j \delta_{ik} - u_k \delta_{ij})$, which is the inner product of $\partial_i F_j$ with F_k ; the F_k form a basis.

Set

$$S_{ij} = u_{ij} - u_i u_j + \frac{1}{2} |\nabla u|^2 \delta_{ij}.$$

For an explicit commutator calculation, write the coefficient of F_m in (L1.50) as $\Gamma_{ij}^m = \delta_{mi} u_j + \delta_{mj} u_i - \delta_{ij} u_m$. Commuting $\partial_k \partial_i F_j$ and $\partial_i \partial_k F_j$ gives

$$0 = \partial_k \Gamma_{ij}^m - \partial_i \Gamma_{kj}^m + \sum_t (\Gamma_{ij}^t \Gamma_{kt}^m - \Gamma_{kj}^t \Gamma_{it}^m) = \delta_{mi} S_{kj} - \delta_{mk} S_{ij} + \delta_{kj} S_{im} - \delta_{ij} S_{km}.$$

Relabeling the indices yields

$$\delta_{ik} S_{j\ell} - \delta_{i\ell} S_{jk} + \delta_{j\ell} S_{ik} - \delta_{jk} S_{i\ell} = 0 \quad (1 \leq i, j, k, \ell \leq n). \quad (\text{L1.51})$$

For distinct j, ℓ , choose $i = k$ distinct from both; then (L1.51) says $S_{j\ell} = 0$. Setting $i = k$, $j = \ell$, $i \neq j$ shows $S_{ii} + S_{jj} = 0$. Three distinct indices now force every diagonal entry to vanish. Thus $S = 0$; here is precisely where $n \geq 3$ is used.

Put $v = e^{-u} > 0$. Direct differentiation of $S = 0$ gives

$$v_{ij} = \frac{|\nabla v|^2}{2v} \delta_{ij}. \quad (\text{L1.52})$$

Write the scalar on the right as q . If $k \neq i$, then $\partial_k q = \partial_k v_{ii} = \partial_i v_{ik} = 0$. There is such an i for each k , so q is constant on connected Ω . Consequently

$$v(x) = a + b \cdot x + c|x|^2, \quad 4ac = |b|^2, \quad (\text{L1.53})$$

where the second identity follows by substituting the polynomial into (L1.52). If $c = 0$, then $b = 0$ and the conformal scale $e^u = 1/v$ is constant. Formula (L1.50) gives $D^2 f = 0$, hence f is the restriction of a similarity.

Otherwise (L1.53) and $v > 0$ give $c > 0$ and $v(x) = c|x - p|^2$, where $p = -b/(2c) \notin \Omega$. The inversion $I_p(x) = (x - p)/|x - p|^2$ is an involution in centered coordinates and has

conformal scale $|x - p|^{-2}$. Therefore $f \circ I_p^{-1}$ on the connected domain $I_p(\Omega)$ has constant conformal scale $1/c$. By the constant scale case just proved it is a similarity. Thus f is a similarity composed with an inversion, as claimed. $\square$

The low-regularity route above follows the deformation strategy of Liu [13]; the final smooth rigidity calculation is included in full. In particular, every conformal generator used below is a Möbius restriction.

9.5 Möbius distortion and normalized cylinders

1. Möbius formulas

Every Möbius restriction needed here has one of the forms

$$M(x) = q + sAx, \quad (\text{M.1})$$

or

$$M(x) = q + sA \frac{x - a}{|x - a|^2}, \quad (\text{M.2})$$

where $s \neq 0$ and A is orthogonal. In the inversion case the pole a lies outside the domain on which M is finite.

For (M.2),

$$\|DM(x)\| = \frac{|s|}{|x - a|^2}, \quad (\text{M.3})$$

and the exact two-point identity is

$$|M(x) - M(y)| = \frac{|s| |x - y|}{|x - a| |y - a|}. \quad (\text{M.4})$$

Formula (M.4) follows from the elementary inversion identity

$$\left| \frac{x - a}{|x - a|^2} - \frac{y - a}{|y - a|^2} \right| = \frac{|x - y|}{|x - a| |y - a|}.$$

2. A compact trapping neighborhood

Let $\Phi = \{\varphi_1, \dots, \varphi_\ell\}$ be a conformal IFS as in Definition IV.9.33 in dimension $d \geq 3$, on a bounded connected open set U , with attractor $E \Subset U$. Choose $\eta > 0$ so small that

$$K = \{x : \text{dist}(x, E) \leq \eta\}$$

is compactly contained in U . If $q < 1$ is the maximum Lipschitz constant of the finite generators on U , reduce η if necessary. Since $\varphi_i(E) \subset E$,

$$\varphi_i(K) \subset \text{int } K \quad (\text{M.5})$$

for all i .

Put

$$\delta = \text{dist}(K, \mathbb{R}^d \setminus U) > 0, \quad D_0 = \text{diam } K.$$

Every pole of every cylinder Möbius map lies outside U and therefore has distance at least δ from K .

3. Uniform distortion from the Möbius formula

For a word I , write φ_I for the cylinder map and set

$$\alpha_I = \sup_{x \in K} \|D\varphi_I(x)\|. \quad (\text{M.6})$$

Lemma IV.9.63 (*uniform derivative distortion*).

There is $C_0 \geq 1$, depending only on $K \Subset U$, such that for every word I and every $x \in K$,

$$C_0^{-1}\alpha_I \leq \|D\varphi_I(x)\| \leq \alpha_I. \quad (\text{M.7})$$

Proof. For a similarity, the derivative norm is constant. For an inversion-type cylinder with pole a ,

$$\frac{\sup_{x \in K} |x - a|}{\inf_{x \in K} |x - a|} \leq 1 + \frac{D_0}{\delta}.$$

Use (M.3). One may take

$$C_0 = \left(1 + \frac{D_0}{\delta}\right)^2. \quad (\text{M.8})$$

□

Lemma IV.9.64 (*uniform quasi-similarity*).

For all $x, y \in K$ and every word I ,

$$C_0^{-1}\alpha_I|x - y| \leq |\varphi_I(x) - \varphi_I(y)| \leq \alpha_I|x - y|. \quad (\text{M.9})$$

Proof. For similarities this is exact. For inversion type, (M.4) and (M.3) show that the two-point scale lies between $|s|/r_{\max}^2$ and $|s|/r_{\min}^2 = \alpha_I$, where $r_{\min}$ and $r_{\max}$ are the minimum and maximum distances from K to the pole. Apply the same ratio bound as in Lemma IV.9.63.

□

Corollary IV.9.65 (*cylinder diameter*).

For every word I ,

$$C_0^{-1}\alpha_I \text{diam } E \leq \text{diam}(\varphi_I(E)) \leq \alpha_I \text{diam } E. \quad (\text{M.10})$$

Proof. Take the supremum of the two-sided estimate in Lemma IV.9.64 over $x, y \in E$. □

4. Parent/child scale comparability

Let I^- denote the word obtained by deleting the last symbol. Put

$$L = \max_i \sup_K \|D\varphi_i\| < 1, \quad m = \min_i \inf_K \|D\varphi_i\| > 0. \quad (\text{M.11})$$

Lemma IV.9.66 (*parent–child scale comparison*).

For every nonempty word I ,

$$\alpha_I \leq L\alpha_{I^-} < \alpha_{I^-}, \quad (\text{M.12})$$

and

$$\alpha_{I^-} \leq \frac{C_0}{m}\alpha_I. \quad (\text{M.13})$$

Proof. Write $I = I^-i$. Conformal scale factors multiply:

$$\|D\varphi_I(x)\| = \|D\varphi_{I^-}(\varphi_i(x))\| \|D\varphi_i(x)\|.$$

Because $\varphi_i(K) \subset K$, the first factor is at most α_{I^-} , giving (M.12).

By Lemma IV.9.63, at every point of K the first factor is at least $C_0^{-1}\alpha_{I^-}$; the second is at least m . Taking a supremum over x gives

$$\alpha_I \geq C_0^{-1}m\alpha_{I^-},$$

which is (M.13). Thus the FYW scale relation

$$\alpha_I \leq \alpha_{I^-} \leq C\alpha_I$$

is proved locally in $d \geq 3$. □

5. Normalized cylinders

Suppose $x_n \in E$ and $r_n \downarrow 0$. Along a coding of x_n , choose the first word I_n such that

$$\varphi_{I_n}(E) \subset B(x_n, r_n). \quad (\text{M.14})$$

Then the parent cylinder is not contained in that ball, so

$$\text{diam}(\varphi_{I_n^-}(E)) > r_n, \quad \text{diam}(\varphi_{I_n}(E)) \leq 2r_n. \quad (\text{M.15})$$

Corollary IV.9.67 (*scale normalization*).

There are constants $c_*, C_* > 0$, independent of n , such that

$$c_* \leq \frac{\alpha_{I_n}}{r_n} \leq C_*. \quad (\text{M.16})$$

Proof. Use (M.10), the parent relation (M.13), and (M.15). □

Define

$$f_n(x) = \frac{\varphi_{I_n}(x) - x_n}{r_n}. \quad (\text{M.17})$$

Then Lemma IV.9.64 and (M.16) give a uniform bi-Lipschitz estimate

$$D^{-1}|x - y| \leq |f_n(x) - f_n(y)| \leq D|x - y| \quad (x, y \in K). \quad (\text{M.18})$$

If $y_n = \varphi_{I_n}^{-1}(x_n) \in E$, then

$$f_n(y_n) = 0. \quad (\text{M.19})$$

Hence the f_n are uniformly bounded on K and equicontinuous.

6. A nondegenerate limit of these Möbius maps is Möbius: direct proof

Lemma IV.9.68 (*compactness of normalized Möbius maps*).

Let f_n be Möbius maps finite on a neighborhood of K , satisfying (M.18), and suppose y_n lies in a fixed compact subset of K with $f_n(y_n) = 0$. Then some subsequence converges uniformly on K to a Möbius map f satisfying the same bi-Lipschitz inequalities.

Proof. Arzelà–Ascoli gives a uniformly convergent subsequence, and (M.18) passes to the limit, so f is nonconstant and injective. It remains to classify the limit without invoking an external conformal-limit theorem.

Pass to a subsequence on which all f_n have the same Möbius type and $y_n \rightarrow y$.

Case A: similarities. Write

$$f_n(x) = c_n + s_n A_n x.$$

The bi-Lipschitz estimate gives $|s_n| \in [D^{-1}, D]$. Compactness of $O(d)$, boundedness of y_n , and $f_n(y_n) = 0$ make c_n bounded. The parameters converge along a subsequence, giving a similarity.

Case B: inversion type with bounded poles. Write

$$f_n(x) = b_n + s_n A_n \frac{x - a_n}{|x - a_n|^2}.$$

If (a_n) is bounded, (M.18) implies that the poles remain a positive distance from K . Indeed, if $\text{dist}(a_n, K) \rightarrow 0$, the upper derivative bound forces $|s_n| \rightarrow 0$, while the lower derivative bound at a fixed point of K away from that limiting pole forces $|s_n|$ to stay bounded below, a contradiction.

Thus, after a subsequence,

$$a_n \rightarrow a \notin K, \quad A_n \rightarrow A, \quad s_n \rightarrow s \neq 0.$$

The normalization $f_n(y_n) = 0$ also gives convergence of b_n . The limit is the inversion-type Möbius map with these parameters.

Case C: inversion type with $|a_n| \rightarrow \infty$. Write

$$a_n = R_n v_n, \quad R_n \rightarrow \infty, \quad |v_n| = 1, \quad \sigma_n = \frac{s_n}{R_n^2}.$$

At any fixed $x_0 \in K$, the derivative estimate (M.18) gives

$$0 < c \leq |\sigma_n| \leq C < \infty.$$

Pass to a subsequence so that

$$v_n \rightarrow v, \quad A_n \rightarrow A, \quad \sigma_n \rightarrow \sigma \neq 0.$$

Uniformly for $x \in K$,

$$\frac{x - a_n}{|x - a_n|^2} = -\frac{v_n}{R_n} + R_n^{-2}(I - 2v_n v_n^T)x + O_K(R_n^{-3}). \quad (\text{M.20})$$

Subtract the same formula at y_n ; the diverging constant disappears because $f_n(y_n) = 0$. Therefore

$$f_n(x) \longrightarrow \sigma A(I - 2vv^T)(x - y).$$

The matrix $A(I - 2vv^T)$ is orthogonal, so the limit is an affine similarity.

All possibilities give a Möbius limit. $\square$

Consequently, no Väisälä normal-family theorem is needed in the $d \geq 3$ self-conformal branch; the only Liouville input, if working at C^1 , is the initial finite-generator classification.

7. Hyperplanes under Möbius maps

Lemma IV.9.69 (*inverse images of hyperplanes under Möbius maps*).

The inverse image of an affine hyperplane under a Möbius transformation is contained in an affine hyperplane or a sphere, away from a possible pole.

Proof. For a similarity this is immediate.

For inversion type,

$$M(x) = q + sA \frac{x - a}{|x - a|^2},$$

let the target hyperplane be

$$\langle v, z \rangle = \beta.$$

Then $M(x)$ lies in it exactly when

$$\langle v, q \rangle + s \frac{\langle A^T v, x - a \rangle}{|x - a|^2} = \beta.$$

Multiplying by $|x - a|^2$ gives

$$\gamma |x - a|^2 + \langle w, x - a \rangle = 0, \tag{M.21}$$

for constants γ, w . If $\gamma = 0$, this is a hyperplane through a . If $\gamma \neq 0$, completing the square makes (M.21) a sphere. $\square$

9.6 The higher-dimensional self-conformal continuation

1. Setup

Let $d \geq 3$. Let $\Phi = \{\varphi_1, \dots, \varphi_\ell\}$ be a finite C^1 conformal IFS in Definition IV.9.33, on a compact invariant set $X \subset U \subset \mathbb{R}^d$. Thus each generator is an injective contracting conformal map

$$\varphi_i : U \longrightarrow \varphi_i(U) \subset U$$

on a bounded connected open set U containing X . Let E be the non-singleton attractor.

In the original theorem of Feng–Wu [9], each φ_i is C^1 . The generator classification proved locally in Section 9.4 makes every φ_i Möbius; the stronger C^2 classification there provides an independent check. Compositions φ_I are therefore Möbius. Fix the compact trapping set K constructed in the Möbius-geometry subsection. We use its constants C_0 and its parent–child scale comparison throughout.

2. Zero thickness forces a hyperplane or sphere

Lemma IV.9.70 (*zero thickness forces a hyperplane or sphere*).

If $\tau(E) = 0$, then E is contained in an affine hyperplane or a $(d - 1)$ -dimensional sphere.

Proof. The set E has positive diameter. By the locally proved centered-microset criterion, there are $x_n \in E$ and $r_n \downarrow 0$ such that

$$K_n = r_n^{-1}((E \cap B(x_n, r_n)) - x_n) \longrightarrow F \subset W, \quad (20.22)$$

where W is a proper linear subspace. Enlarge W to a hyperplane if necessary.

Choose along a coding of x_n the first cylinder I_n satisfying

$$x_n \in \varphi_{I_n}(E) \subset B(x_n, r_n). \quad (20.23)$$

The parent is not contained in the ball. By the local Möbius quasi-similarity and parent/child comparison,

$$c_* \leq \frac{\alpha_{I_n}}{r_n} \leq C_* \quad (20.24)$$

and the normalized maps

$$f_n(x) = \frac{\varphi_{I_n}(x) - x_n}{r_n} \quad (20.25)$$

obey uniform bi-Lipschitz bounds on K .

Because $x_n \in \varphi_{I_n}(E)$, choose $y_n \in E$ with $\varphi_{I_n}(y_n) = x_n$. Then $f_n(y_n) = 0$. By the local Möbius compactness lemma, after a subsequence,

$$f_n \longrightarrow f$$

uniformly on K , where f is a nondegenerate Möbius map.

Since $f_n(E) \subset K_n$ and $K_n \rightarrow F \subset W$, we have $f(E) \subset W$. Hence

$$E \subset f^{-1}(W).$$

By the local Möbius hyperplane-preimage lemma, $f^{-1}(W)$ lies in an affine hyperplane or sphere. □

Corollary IV.9.71 (*the non-flat case is spherical*).

If E is not contained in a hyperplane and $\tau(E) = 0$, then E lies in a $(d - 1)$ -sphere.

Proof. By Lemma IV.9.70, E lies in a hyperplane or a sphere. The first alternative is excluded by hypothesis. □

3. A uniform local diameter lower bound

Lemma IV.9.72 (*uniform local diameter in higher dimensions*).

There is $L_0 > 0$ such that for every word I , every $x \in \varphi_I(E)$, and every $0 < r < \text{diam}(\varphi_I(E))$,

$$\text{diam}(B(x, r) \cap \varphi_I(E)) \geq L_0 r. \quad (20.26)$$

Proof. If $\varphi_I(E) \subset B(x, r)$, the left side is $\text{diam}(\varphi_I(E)) > r$.

Otherwise follow a coding of x inside $\varphi_I(E)$ and choose the first subword J such that $\varphi_{IJ}(E) \subset B(x, r)$ but its parent is not. The same scale-normalization calculation as in (20.24), now inside the parent cylinder, gives

$$\alpha_{IJ} \geq cr$$

for a fixed $c > 0$. Cylinder diameter comparison then yields

$$\text{diam}(\varphi_{IJ}(E)) \geq C_0^{-1} \alpha_{IJ} \text{diam } E \geq L_0 r.$$

Since $\varphi_{IJ}(E)$ lies in the displayed intersection, (20.26) follows. $\square$

4. Microsets of a spherical self-conformal set

Assume now $E \subset S$, where S is a $(d-1)$ -sphere, and E is not contained in a hyperplane.

Lemma IV.9.73 (*tangent-hyperplane claim*).

Every centered microset F of E is contained in the tangent hyperplane of S at some point of E , translated to the origin. Moreover F is not contained in any $(d-2)$ -dimensional linear subspace.

The same statement holds for every cylinder $\varphi_I(E)$. Since $\varphi_I(E) \subset E \subset S$, the relevant tangent hyperplane is still the tangent hyperplane of the original sphere S at a point of $\varphi_I(E)$.

Proof. For the first assertion, take $x_n \in E$, $r_n \downarrow 0$, with normalized local sets converging to F , and pass to $x_n \rightarrow x \in E$. After translating and scaling S to the unit sphere, the identity

$$|y|^2 = |x_n|^2$$

gives

$$2\langle x_n, y - x_n \rangle + |y - x_n|^2 = 0.$$

For $|y - x_n| = O(r_n)$, divide by r_n and let $n \rightarrow \infty$. Every limit vector v satisfies

$$\langle x, v \rangle = 0,$$

the tangent-hyperplane condition.

For the second assertion, suppose instead that $F \subset H$ for a linear subspace H of dimension at most $d-2$. If F is a microset of $\varphi_I(E)$, keep the prefix I fixed and choose the

minimal further cylinders $\varphi_{IJ_n}(E)$ inside the defining balls; for a microset of E , use the same argument with empty prefix. The scale comparison and normalized-map argument from Lemma IV.9.70 then give a Möbius limit f with

$$f(E) \subset F \subset H.$$

Let $M = f^{-1}$. Write M in the local Möbius form. If M is a similarity, $M(H)$ lies in a hyperplane.

If M is inversion type with pole a in the target coordinate, choose a $(d-1)$ -dimensional linear subspace W' containing H and a . For $z \in H$ away from the pole, $z - a \in W'$, so the inversion vector $(z - a)/|z - a|^2$ lies in W' . Therefore

$$M(H) \subset AW' + b,$$

an affine hyperplane. Thus E is contained in a hyperplane, a contradiction. Hence F cannot lie in dimension at most $d-2$. $\square$

5. Two spherical pieces have positive-thickness sum

Lemma IV.9.74 (*two spherical pieces have positive-thickness sum*).

If E is contained in a $(d-1)$ -sphere but not in a hyperplane, then there are words I, J such that

$$\tau(\varphi_I(E) + \varphi_J(E)) > 0. \quad (20.27)$$

Proof. Step 1: choose pieces with distinct tangent hyperplanes. Because E is not contained in a hyperplane, it is not an antipodal two-point set. Choose distinct non-antipodal $p, q \in E$. Their tangent hyperplanes on S are distinct, and p, q lie in a common open hemisphere.

Cylinder diameters tend to zero. Choose disjoint cylinders

$$A = \varphi_I(E), \quad B = \varphi_J(E)$$

contained in small neighborhoods of p and q inside that common hemisphere. Every tangent hyperplane to S along A differs from every tangent hyperplane along B after the neighborhoods are chosen sufficiently small.

Step 2: contradiction from a flat microset of $A+B$. Suppose $\tau(A+B) = 0$. Since $A+B$ has positive diameter, the local microset criterion supplies $x_n \in A$, $y_n \in B$, $r_n \downarrow 0$, and a hyperplane W such that

$$(2r_n)^{-1}((B(x_n + y_n, 2r_n) \cap (A+B)) - (x_n + y_n)) \longrightarrow F \subset W. \quad (20.28)$$

After subsequences, the compact sets

$$F_{1,n} = r_n^{-1}((B(x_n, r_n) \cap A) - x_n), \quad F_{2,n} = r_n^{-1}((B(y_n, r_n) \cap B) - y_n)$$

converge to centered microsets F_1, F_2 .

For every n , $(F_{1,n} + F_{2,n})/2$ is contained in the left side of (20.28). Passing to the limit,

$$\frac{F_1 + F_2}{2} \subset F \subset W.$$

Because 0 belongs to both F_1 and F_2 ,

$$F_1 \subset W, \quad F_2 \subset W. \quad (20.29)$$

By Lemma IV.9.73 there are tangent hyperplanes W_1, W_2 , with $W_1 \neq W_2$, such that $F_i \subset W_i$ and neither F_i lies in a $(d-2)$ -subspace. But from (20.29),

$$F_i \subset W \cap W_i.$$

If both intersections had dimension $d-1$, then $W = W_1 = W_2$, impossible. Hence one has dimension at most $d-2$, contradicting Lemma IV.9.73. Thus (20.27) holds. $\square$

The $d \geq 3$ self-conformal arithmetic-thickness theorem

Theorem IV.9.75 (*higher-dimensional self-conformal arithmetic thickness*).

Assume $d \geq 3$ and the generators satisfy the original C^1 conformal-IFS hypotheses above. Then E is arithmetically thick if and only if E is not contained in a hyperplane.

Proof. Section 9.4 proves the local regularity upgrade from C^1 to smoothness; the locally proved Theorem IV.9.62 then supplies Liouville rigidity and makes each generator a Möbius restriction. The distortion, parent/child scale comparison, normalized-limit, microset, and hyperplane/sphere lemmas proved above therefore apply at the original regularity.

If E lies in a hyperplane, every finite arithmetic sum lies in a parallel hyperplane and has empty interior. Conversely assume E is not contained in a hyperplane. If $\tau(E) > 0$, Theorem IV.9.26 applied to sufficiently many copies of E gives interior. If $\tau(E) = 0$, Corollary IV.9.71 places E in a sphere, and Lemma IV.9.74 gives cylinder subsets $E_1, E_2 \subset E$ such that $F = E_1 + E_2$ has positive thickness. Applying Theorem IV.9.26 to sufficiently many copies of F gives nonempty interior for nF . Since $nF \subset (2n)E$, a finite arithmetic sum of E has nonempty interior. $\square$

Source boundary IV.9.76 (*the unrestricted dimensional join*).

The former unrestricted $d \geq 2$ equivalence cannot be retained: its planar implication is disproved by Example IV.9.45. This does not affect the higher-dimensional theorem, whose proof uses Möbius rigidity rather than the rejected connected-curve inference.

Corollary IV.9.77 (*the valid self-conformal continuation*).

Let $E \subset \mathbb{R}^d$ be a self-conformal set in Definition IV.9.33. If $d \geq 3$, then E is arithmetically thick if and only if it is not contained in an affine hyperplane. In dimension two the same equivalence holds with the additional hypothesis that $\tau(E) > 0$ or that E lies in a simple real-analytic curve in Definition IV.9.41.

Proof. If E lies in an affine hyperplane, every nE lies in a parallel hyperplane and has empty interior. Otherwise E is non-singleton. For $d = 2$ use Theorem IV.9.49, where an affine hyperplane is a line. For $d \geq 3$ use Theorem IV.9.75. Neither case uses the rejected assertions IV.9.42, IV.9.48, or IV.9.76. $\square$

6. A triangular-matrix proof display: a typo, not a theorem refutation

Suspicious source formula IV.9.78. In Feng–Wu arXiv:2006.12058v1, the proof of Lemma 6.10, printed p. 31, considers

$$T_i = \begin{pmatrix} c & e_i \\ 0 & d \end{pmatrix}, \quad c > d > 0, \quad T = \text{diag}(c, d),$$

$$T^{-n}T_I = \begin{pmatrix} 1 & s_I \\ 0 & 1 \end{pmatrix}, \quad s_I = \sum_{k=1}^n c^{-1}(d/c)^{n-k}e_{i_k}.$$

After fixing a uniform bound $|s_I| \leq u$, the printed estimate is

$$\|(T^{-n}T_I)^{-1}\| \leq \frac{1}{1+u}.$$

Here the norm is the Euclidean operator norm used throughout the source. The surrounding section assumes a contracting planar affine IFS with attractor not contained in a line; these extra conditions do not repair the display. The full source and the page image were checked. The next example refutes only this proof estimate, not Lemma 6.10 or the planar self-affine thickness theorem.

Example IV.9.79 (*the inverse of a nontrivial shear*).

The displayed inverse-norm estimate fails, and replacing its right-hand side by $1+u$ gives the estimate needed for the source lemma.

Proof. Take $c = 1/10$, $d = 1/20$, and $e_i = 1/20$ for every i . For every length- n word,

$$s_I = \frac{1}{2} \sum_{j=0}^{n-1} 2^{-j} = 1 - 2^{-n},$$

so $u = 1$ is a uniform bound. At $n = 1$ the inverse shear is

$$A^{-1} = \begin{pmatrix} 1 & -1/2 \\ 0 & 1 \end{pmatrix}, \quad |A^{-1}(1, 0)| = 1 > \frac{1}{1+u} = \frac{1}{2}.$$

The common original matrix $T_i = \begin{pmatrix} 1/10 & 1/20 \\ 0 & 1/20 \end{pmatrix}$ has norm at most the square root of the sum of squared entries, $\sqrt{3/200} < 1$. To realize all the surrounding IFS conditions, use this matrix with translations $(0, 0)$, $(1, 0)$, $(0, 1)$. Its attractor contains the three noncollinear fixed points

$$(0, 0), \quad (10/9, 0), \quad (10/171, 20/19).$$

These maps are invertible affine contractions, so the example also lies within the nonsingular, non-flat planar context of Section 6.3.

For a general shear $A = I + N$ with $N^2 = 0$ and $\|N\| = |s_I| \leq u$, its inverse is $I - N$. The triangle inequality gives

$$\|A\| \leq 1 + u, \quad \|A^{-1}\| \leq 1 + u.$$

Consequently, choosing any $\lambda > 1 + u$,

$$B(0, \lambda^{-1}) \subset AB(0, 1) \subset B(0, \lambda).$$

Indeed the left inclusion follows by applying A^{-1} to each point of the smaller ball, and the right inclusion by applying A to each unit-ball point. This is exactly the valid conclusion of source Lemma 6.10. The erroneous reciprocal is a proof-display typo. $\square$

Part III

Equidistribution and Diophantine Geometry

Chapter IV.10

Equidistribution from Fractal Measures

Spectral phases, normality, and graph-relative L^q flattening

Normality is obtained here by joining two kinds of information. The scenery flow records the small-scale geometry of a measure, while the map $T_\beta(x) = \beta x \pmod{1}$ records the arithmetic orbit of a point. Spectral phases connect the two. After developing this connection, we turn to a graph-relative L^q flattening argument that handles the resonant cases not settled by spectral exclusion alone.

10.1 Spectral, phase, and Pisot normality foundations

1. Flow spectrum and time-map ergodicity

In spectral and eigenfunction statements, $S_t = S_t^*$ denotes the full-space scaling flow on M_1^* . If an EFD is specified by its restricted presentation, its spectrum is computed on the canonical invariant full-space lift from Lemma IV.2.15.

Definition IV.10.1 (*pure-point spectrum*).

For a scale-invariant probability P , a number $\alpha \in \mathbb{R}$ belongs to the pure-point spectrum $\Sigma(P, S)$ if there is a nonzero $\phi \in L^2(P)$ such that

$$\phi(S_t \nu) = e(-\alpha t) \phi(\nu). \quad (21.1)$$

There is a single P -conull set of ν on which this identity holds for every $t \in \mathbb{R}$. The sign convention is chosen so that the C^1 phase rule below contains the factor $e(-\alpha \log f')$. For an ergodic flow every eigenfunction has constant modulus, and we normalize it by $|\phi| = 1$.

Lemma IV.10.2 (*one-dimensional eigenspaces*).

If P is S -ergodic, every eigenspace is one-dimensional.

Proof.

If ϕ and ψ have the same eigenvalue and are nonzero, then ϕ/ψ is S -invariant.

Ergodicity makes it constant.

□

Lemma IV.10.3 (*time-map spectral criterion*).

Let $t_0 > 0$. For an S -ergodic P , the map S_{t_0} is ergodic if and only if

$$\Sigma(P, S) \cap \left(\frac{1}{t_0} \mathbb{Z} \setminus \{0\} \right) = \emptyset. \quad (21.2)$$

Proof.

If $\alpha = k/t_0$ with $k \neq 0$, an eigenfunction satisfies

$\phi(S_{t_0}\nu) = e(-k)\phi(\nu) = \phi(\nu)$, so S_{t_0} is not ergodic.

Conversely, assume S_{t_0} is not ergodic. Choose a nonconstant

$$F \in L^2(P), \quad F \circ S_{t_0} = F.$$

For $m \in \mathbb{Z}$, put

$$F_m(\nu) = \frac{1}{t_0} \int_0^{t_0} e(mt/t_0) F(S_t \nu) dt. \quad (21.3)$$

The function $t \mapsto F(S_t \nu)$ is t_0 -periodic in L^2 . Its Fourier series has coefficients F_m . If every $m \neq 0$ coefficient vanished, F would be S -invariant and therefore constant, a contradiction. Hence some $F_m \neq 0$, with $m \neq 0$. A change of variables gives

$$F_m(S_s \nu) = e(-ms/t_0) F_m(\nu),$$

so $m/t_0 \in \Sigma(P, S)$.

□

2. Discrete generation from spectral exclusion

Let μ be a Radon measure on $\mathbb{R}$, let $x \in \text{supp } \mu$, and let $t_0 > 0$. Write

$$\mu_{x,t} = S_t^* T_x^* \mu, \quad \mu_{x,t}^\square = (\mu_{x,t})^\square. \quad (21.4)$$

The first measure is full-space; the second is its normalized restriction to $[-1, 1]$. In this subsection, empirical laws and weak limits live on the compact space of probabilities on $[-1, 1]$, and P denotes the restricted presentation of the EFD. We therefore write $S_s^\square$ explicitly for the restricted scaling maps. Projection of the full-space lift commutes with forward scaling (Lemma IV.2.14); in particular, ergodicity of a full-space time map implies ergodicity of its restricted factor.

Definition IV.10.4 (*continuous and discrete generation*).

We say that μ generates P continuously at x if

$$(1/T) \int_0^T \delta_{\mu_{x,t}^\square} dt \rightarrow P. \quad (21.5)$$

It t_0 -discretely generates P at x if

$$(1/N) \sum_{n=0}^{N-1} \delta_{\mu_{x,nt_0}^\square} \rightarrow P. \quad (21.6)$$

Lemma IV.10.5 (*phase averaging of a discrete accumulation law*).

Assume (21.5) and, with $a_n = \mu(B(x, e^{-nt_0}))$, assume

$$\limsup_{N \rightarrow \infty} \frac{-\log a_N}{N} < \infty. \quad (21.6a)$$

Let Q be a weak accumulation point of the discrete empirical laws in (21.6). Then Q -almost every ν has $0 \in \text{supp } \nu$ and, in particular, $\nu([-e^{-t_0}, e^{-t_0}]) > 0$, and

$$P = \frac{1}{t_0} \int_0^{t_0} (S_s^\square)_* Q ds. \quad (21.7)$$

Proof.

Put $r = e^{-t_0}$. Because

$$\mu_{x,nt_0}^\square([-r, r]) = \frac{a_{n+1}}{a_n},$$

the nonnegative logarithmic ratios $g_n = \log(a_n/a_{n+1})$ satisfy

$$\sum_{n < N} g_n = \log(a_0/a_N) \leq CN$$

for all sufficiently large N and some finite C , by (21.6a). Hence, for every $M > 0$, at most a fraction $C/M + o(1)$ of the first N sceneries have $\nu([-r, r]) < e^{-M}$. The map $\nu \mapsto \nu([-r, r])$ is upper semicontinuous, so the set $U_M = \{\nu : \nu([-r, r]) < e^{-M}\}$ is open. Portmanteau therefore gives $Q(U_M) \leq C/M$. Letting $M \rightarrow \infty$ proves the stated positive-mass assertion. To check the entire scaling domain, repeat this estimate with any fixed integer $k \geq 1$. Put $b_n = -\log a_n$. Then

$$\sum_{n < N} \log \frac{a_n}{a_{n+k}} = \sum_{j=N}^{N+k-1} b_j - \sum_{j=0}^{k-1} b_j \leq kC(N+k) + O_k(1).$$

The same open-set argument gives $Q\{\nu : \nu([-e^{-kt_0}, e^{-kt_0}]) < e^{-M}\} \leq kC/M$. Let $M \rightarrow \infty$ and intersect over k . Every centered neighborhood then has positive mass for Q -almost every state; thus $0 \in \text{supp } \nu$ and all restricted forward scaling maps are defined on this conull set.

Now take a bounded continuous test F on the compact restricted scenery space. For N along the convergent discrete subsequence,

$$\frac{1}{Nt_0} \int_0^{Nt_0} F(\mu_{x,t}^\square) dt = \frac{1}{N} \sum_{n=0}^{N-1} \frac{1}{t_0} \int_0^{t_0} F(S_s^\square \mu_{x,nt_0}^\square) ds. \quad (21.8)$$

The left side tends to $\int F dP$. For Q -almost every ν , the denominator $\nu([-e^{-s}, e^{-s}])$ is positive for every $0 \leq s \leq t_0$. Such a ν has at most countably many radii whose centered boundary carries mass. By Fubini, for Lebesgue-a.e. $s \in [0, t_0]$ the restricted scaling map $S_s^\square$ is Q -a.e. continuous. The almost-everywhere continuous mapping theorem passes the discrete weak limit through $S_s^\square$ for those s ; dominated convergence then passes it through the s -integral in (21.8). This gives (21.7). $\square$

Why the local growth condition is automatic at typical points. For every finite Borel measure on $\mathbb{R}$, every $\varepsilon > 0$, and each bounded interval K , cover K by $O_K(2^m)$ dyadic intervals of length 2^{-m} . The union of intervals I with $\mu(I) < 2^{-m(1+\varepsilon)}$ has μ -mass $O_K(2^{-m\varepsilon})$. Borel–Cantelli shows that, for μ -a.e. $x \in K$, the containing dyadic interval has mass at least $2^{-m(1+\varepsilon)}$ for all large m . Since it lies in $B(x, 2^{-m})$, comparison of dyadic and e^{-nt_0} radii gives (21.6a) for μ -a.e. x .

Lemma IV.10.6 (*boundary-null discrete accumulation laws are invariant*).

Under the hypotheses of Lemma IV.10.5, let Q be a discrete accumulation law. Assume additionally that Q gives full mass to measures ν with

$$\nu([-e^{-t_0}, e^{-t_0}]) > 0, \quad \nu(\{-e^{-t_0}, e^{-t_0}\}) = 0. \quad (21.9a)$$

Then

$$(S_{t_0}^\square)_* Q = Q. \quad (21.9)$$

Proof.

On the set (21.9a), restriction to $[-e^{-t_0}, e^{-t_0}]$ and division by its mass are continuous under weak convergence. Hence $F \circ S_{t_0}^\square$ is Q -a.e. continuous for every bounded continuous F on the compact restricted state space. Writing $Q_N = N^{-1} \sum_{n=0}^{N-1} \delta_{\mu_{x,nt_0}^\square}$, the exact orbit identity gives

$$\int F d(S_{t_0}^\square)_* Q_N - \int F dQ_N = \frac{F(\mu_{x,Nt_0}^\square) - F(\mu_{x,0}^\square)}{N} \rightarrow 0. \quad (21.10)$$

The Q -a.e. continuous mapping theorem passes the left side to $\int F d(S_{t_0}^\square)_* Q - \int F dQ$. It is zero for every continuous F , proving (21.9). $\square$

Why the extra hypothesis cannot be inferred from (21.7) alone. Put $t_0 = \log 2$ and

$$\mu = \sum_{k \geq 1} 2^{-k} \delta_{2^{-k(1+1/k)}}, \quad x = 0.$$

At times $n \log 2$ the normalized restricted sceneries converge to

$$\eta = \sum_{j \geq 1} 2^{-j} \delta_{2^{-j}},$$

so their empirical law tends to $Q = \delta_\eta$. But $\eta(\{e^{-t_0}\}) = 1/2$ and $S_{t_0}^\square \eta \neq \eta$: the atom at $1/2$ is retained at the boundary by the closed-window convention, while the corresponding prelimit atom is just outside the next window. Thus the unconditional invariance statement is false for a single prescribed support point. The next theorem excludes this mechanism because its limiting EFD is atomless.

Lemma IV.10.7 (*nontrivial EFD sceneries are atomless*).

Let P be the restricted presentation of a nontrivial full-space ergodic fractal distribution. Then P -almost every measure is atomless.

Proof. Work first with the full-space lift. The event that a measure has an atom somewhere is invariant under the scaling flow, so ergodicity makes its probability either zero or one. If it were one, then for some bounded neighborhood U of 0 the event that a measure has an atom in U would have positive probability. On that event the set of atoms in U has positive measure, and recentering at any such atom gives a measure with an atom at 0. The forward null-set implication in the quasi-Palm definition (5.3)–(5.4) therefore gives positive probability to the event that the recentered measure has an atom at 0. That event is scaling-invariant, so ergodicity makes it conull.

For a measure with an atom at 0, its normalized restricted zooms converge weakly to δ_0 as the zoom time tends to infinity. Scale invariance of P and dominated convergence then force its restricted presentation to be δ_{δ_0} . Uniqueness of the invariant full-space lift, Lemma IV.2.15, makes the full law the trivial atomic EFD, contrary to the hypothesis. Thus the atomic event has probability zero. $\square$

Lemma IV.10.8 (*two-sided mass at the scenery origin*).

Let P be a nontrivial EFD on the line. For P -almost every restricted state η and every $r > 0$,

$$\eta([-r, 0]) > 0 \quad \text{and} \quad \eta([0, r]) > 0. \quad (21.9b)$$

Consequently, for each $r > 0$ the masses of all closed intervals $I \subset [-1, 1]$ with $0 \in I$ and $|I| \geq r$ have a positive infimum (depending on η and r).

Proof. Use the full-space lift. The event B that a state has a left gap at the origin is invariant under scaling, hence has probability zero or one. If it had probability one, quasi-Palm would imply that for P -almost every state η , η -almost every x has a left gap in $\text{supp } \eta$. The points of a closed subset of $\mathbb{R}$ which are isolated from the left form a countable set: assign to each such point a rational number in its left gap, using disjoint gaps after retaining one point per rational interval. This contradicts atomlessness from Lemma IV.10.7. Thus $P(B) = 0$. Reflection gives the right-hand assertion. A countable sequence $r \downarrow 0$ gives the simultaneous claim. Finally every interval in the stated compact family contains either $[-r/2, 0]$ or $[0, r/2]$, proving the infimum assertion. $\square$

Theorem IV.10.9 (*spectral exclusion gives t_0 -generation*).

Suppose μ generates a nontrivial EFD P continuously at x , (21.6a) holds at x , and $S_{t_0}^\square$ is ergodic on P . Then μ t_0 -discretely generates P at x .

Proof.

Let Q be any discrete accumulation law. Lemma IV.10.5 gives positive mass to $[-e^{-t_0}, e^{-t_0}]$ for Q -almost every state and gives

$$P = \frac{1}{t_0} \int_0^{t_0} (S_s^\square)_* Q \, ds. \quad (21.10a)$$

By Lemma IV.10.7, P -almost every state is atomless. Consequently the right side of (21.10a) is supported on atomless states. If a Q -typical state ν had an atom at some $y \in (-1, 1)$, that atom would survive the restriction and scaling for every

$$0 < s < \min\{t_0, -\log |y|\}$$

(with the upper bound t_0 when $y = 0$), a set of times of positive length. Tonelli applied to (21.10a) rules this out. Thus Q -almost every ν has no atoms in $(-1, 1)$, in particular none at $\pm e^{-t_0}$. Lemma IV.10.6 now proves that Q is $S_{t_0}^\square$ -invariant.

The positive inner mass and this invariance imply successively that Q -almost every ν gives positive mass to $[-e^{-kt_0}, e^{-kt_0}]$ for every integer $k \geq 1$; hence $0 \in \text{supp } \nu$. Since a time- t_0 image can have an atom at either endpoint ± 1 only when its preimage has an atom at $\pm e^{-t_0}$, invariance also gives $Q\{\nu : \nu(\{-1, 1\}) > 0\} = 0$. Therefore the restricted scenery maps $S_s^\square$ are Q -a.e. continuous as $s \downarrow 0$, and

$$(S_s^\square)_* Q \longrightarrow Q. \quad (21.10b)$$

All $S_s^\square$ are now defined Q -a.e.; they commute with $S_{t_0}^\square$. Thus each $(S_s^\square)_* Q$ is $S_{t_0}^\square$ -invariant. As P is $S_{t_0}^\square$ -ergodic, it is an extreme point of the invariant probabilities. The barycenter identity (21.10a) forces $(S_s^\square)_* Q = P$ for Lebesgue-a.e. $s \in (0, t_0)$. Choose such $s_j \downarrow 0$ and use (21.10b) to obtain $Q = P$. Every discrete accumulation law is therefore P , so the whole empirical sequence converges. □

Theorem IV.10.10 (*spectral criterion for discrete generation*).

Let μ generate a nontrivial EFD P and let $t_0 > 0$. If

$$\Sigma(P, S) \cap \left(\frac{1}{t_0} \mathbb{Z} \setminus \{0\} \right) = \emptyset \quad (21.11)$$

then μ t_0 -discretely generates P at μ -a.e. point at which it generates P continuously.

The converse time-map statement is exact: failure of (21.11) is equivalent to non-ergodicity of S_{t_0} ; it does not by itself assert failure of normality.

Proof.

The dyadic Borel–Cantelli argument following Lemma IV.10.5 gives (21.6a) at μ -almost every point. Intersect this conull set with the set where continuous generation holds. Lemma IV.10.3 makes the full-space $S_{t_0}^*$ ergodic under (21.11). Its restricted factor $S_{t_0}^\square$ is therefore ergodic, so Theorem IV.10.9 applies at every point in that intersection. $\square$

3. Phases of discrete scenery components

Assume now that $0 \neq \alpha \in \Sigma(P, S)$, with normalized eigenfunction

$$\phi_\alpha.$$

Definition IV.10.11 (*phase at the origin*).

For a P -typical measure ν , put

$$\text{phase}_\alpha(\nu) = \phi_\alpha(\nu) \in S^1. \quad (21.12)$$

The covariance is

$$\text{phase}_\alpha(S_t \nu) = e(-\alpha t) \text{phase}_\alpha(\nu). \quad (21.13)$$

Definition IV.10.12 (*phase of a spatial point*).

For P -typical ν and ν -a.e. x define

$$\text{phase}_{\alpha, \nu}(x) = \text{phase}_\alpha(T_x^* \nu), \quad (21.14)$$

using one jointly measurable representative on the common quasi-Palm conull set.

Lemma IV.10.13 (*spatial synchronization*).

Let P be an EFD and ϕ_α a unit-modulus eigenfunction. There is a P -conull set G such that, for every $\nu \in G$,

$$\text{phase}_{\alpha, \nu}(x)$$

is defined for ν -a.e. x , and the phase of every sufficiently deep scenery at x is obtained from (21.14) by the deterministic factor $e(-\alpha t)$.

Proof.

Let f be a Borel, unit-modulus representative of the eigenfunction and write $\chi(t) = e(-\alpha t)$. For each fixed t the eigenfunction equation $f(S_t \eta) = \chi(t)f(\eta)$ holds for P -a.e. η . Joint measurability of the flow and Fubini therefore give, for P -a.e. η ,

$$f(S_s \eta) = \chi(s)f(\eta) \quad \text{for Lebesgue-a.e. } s \in \mathbb{R}. \quad (21.14a)$$

Define the Borel function

$$F(\eta) = \int_0^1 \overline{\chi(s)} f(S_s \eta) ds.$$

Equation (21.14a) shows that $F = f$ P -a.e. Let G consist of those η for which $|F(\eta)| = 1$ and

$$\int_{-n}^n |f(S_s \eta) - \chi(s)F(\eta)| ds = 0 \quad (n = 1, 2, \dots). \quad (21.14b)$$

This is a Borel, P -conull set. If $\eta \in G$ and $t \in \mathbb{R}$, then the almost-everywhere orbit identity in (21.14b), translated by t , gives

$$F(S_t \eta) = \int_0^1 \overline{\chi(s)} f(S_{s+t} \eta) ds = \chi(t)F(\eta). \quad (21.14c)$$

The same calculation shows $S_t \eta \in G$, so G is invariant under *every* flow time. Replace ϕ_α by F on G .

The quasi-Palm property applied to G implies that for P -a.e. ν , $T_x^* \nu \in G$ for ν -a.e. x . The map $(\nu, x) \mapsto F(T_x^* \nu)$ is jointly measurable on this domain. Equation (21.14c) then gives the asserted deterministic phase factor for every scenery time at each such spatial point.

□

Lemma IV.10.14 (*spatial constancy of an EFD eigenphase*).

Under the hypotheses of Lemma IV.10.13, for P -almost every full-space state ν ,

$$\phi_\alpha(T_y^* \nu) = \phi_\alpha(\nu) \quad \text{for } \nu\text{-almost every } y. \quad (21.14d)$$

Consequently, let $L > 0$ and let $\alpha \in L^{-1}\mathbb{Z}$. Here saying that ϕ_α is a *phase coordinate* means that the completed σ -algebra $\sigma(\phi_\alpha)$ equals the S_L^* -invariant σ -algebra modulo P ; resonance alone does not assert this equality. If ϕ_α is a phase coordinate, a P -typical measure L -discretely generates one S_L^* -ergodic phase component at almost every spatial point.

Proof. Use the everywhere-on-orbits representative constructed in Lemma IV.10.13. For P -almost every ν put $h_\nu(y) = \phi_\alpha(T_y^* \nu)$ on the common ν -conull domain, extending the unit-modulus Borel representative by 1 outside its domain. The exact translation–scaling identity gives, whenever $\xi = S_t^* T_x^* \nu$ and $z = e^t(y - x)$,

$$h_\xi(z) = e(-\alpha t)h_\nu(y) \quad \text{for } \xi\text{-almost every } z. \quad (21.14e)$$

Let $V(\xi)$ be the double integral of $|h_\xi(z) - h_\xi(z')|$ over the normalized restriction $\xi^\square$ on $[-1, 1]$. The recentering map is Borel (Lemma IV.2.3), so integration against its Borel measure kernel makes V a Borel function with $0 \leq V \leq 2$. At ν -almost every x , the differentiation theorem IV.2.25 applied to the bounded function h_ν gives

$$\frac{1}{\nu(B(x, e^{-t}))} \int_{B(x, e^{-t})} |h_\nu(y) - h_\nu(x)| d\nu(y) \longrightarrow 0. \quad (21.14f)$$

By (21.14e) and the triangle inequality, $V(S_t^* T_x^* \nu)$ is at most twice the left side of (21.14f) and therefore tends to zero.

Use the countable-time Borel event

$$C = \{\xi : V(S_n^* \xi) \longrightarrow 0 \text{ as } n \rightarrow \infty\}.$$

The preceding paragraph gives $T_x^* \nu \in C$ for ν -almost every x and P -almost every ν . The quasi-Palm null-class equivalence therefore gives $P(C) = 1$. Scale invariance and bounded convergence now imply

$$\int V dP = \lim_{n \rightarrow \infty} \int V(S_n^* \xi) dP(\xi) = 0. \quad (21.14g)$$

Hence $V(\nu) = 0$ almost surely, so h_ν is constant $\nu^\square$ -almost everywhere. Apply this conclusion also to $S_{-n}^* \nu$ for every positive integer n . Equation (21.14e) then extends the constancy to the whole locally finite measure ν , with one value $c(\nu) \in S^1$.

Equation (21.14e) makes c a flow eigenfunction with the same eigenvalue as ϕ_α . Ergodicity and simplicity of a one-dimensional eigenspace give $c(\nu) = c_0 \phi_\alpha(\nu)$ for one fixed $c_0 \in S^1$. The quasi-Palm property permits choosing ν and then ν -almost every y so that $T_y^* \nu$ satisfies this same identity. For such a y , recentering once more at a $T_y^* \nu$ -typical point z corresponds exactly to recentering ν at $y + z$. Since $T_y^* \nu$ is a positive scalar multiple of a translate of ν , its typical z correspond to ν -typical $y + z$. Thus the two constancy identities give respectively

$$h_{T_y^* \nu}(z) = c_0 \phi_\alpha(T_y^* \nu) = c_0^2 \phi_\alpha(\nu), \quad h_\nu(y + z) = c_0 \phi_\alpha(\nu).$$

The left sides are equal by the recentering identity; hence $c_0^2 = c_0$ and $c_0 = 1$. This proves (21.14d).

Finally, the pointwise ergodic theorem for the time- L action gives discrete generation of an ergodic component at the origin for P -almost every state. Quasi-Palm transfers this conull event to almost every recentering, and (21.14d) identifies its phase with that of the original state. The phase-coordinate hypothesis identifies the ergodic-component label with this phase: conditional expectation onto the invariant σ -algebra is precisely conditional expectation onto $\sigma(\phi_\alpha)$. $\square$

Definition IV.10.15 (*phase measure of a measure*).

Suppose a Radon measure μ t_0 -discretely generates S_{t_0} -ergodic components of an EFD P at μ -a.e. x . For an eigenvalue $\alpha = k/t_0$ define

$$\theta_\mu = \int \delta_{\text{phase}_{\alpha, \mu}(x)} d\mu(x). \quad (21.15)$$

Here μ is first normalized on the bounded region under consideration. The phase at such a point is the constant value of ϕ_α on its generated full-space component, using the common full-orbit representative of Lemma IV.10.13. This definition requires that component to give full mass to that representative's domain. It agrees with the recentering definition for P -typical states by Lemma IV.10.14. A phase is not defined by evaluating a merely Borel eigenfunction on an arbitrary limiting sequence of nearby sceneries.

For a family (μ_ω) with parameter law Q , its cumulative phase is the mixture

$$\theta = \int \theta_{\mu_\omega} dQ(\omega). \quad (21.16)$$

Lemma IV.10.16 (*absolute-continuity monotonicity*).

If $\nu \ll \mu$ and both phases are defined relative to the same generated EFD, then

$$\theta_\nu \ll \theta_\mu. \quad (21.17)$$

Proof.

Write $h = d\nu/d\mu$ on the bounded region. At ν -almost every x , $0 < h(x) < \infty$ and Radon differentiation gives

$$\frac{1}{\mu(B(x, r))} \int_{B(x, r)} |h(y) - h(x)| d\mu(y) \longrightarrow 0.$$

After normalization this says that the centered restricted sceneries of ν and μ have total-variation distance tending to zero as $r \downarrow 0$, hence also at the discrete radii e^{-nt_0} . Their generated component laws coincide. Their component-constant phases therefore coincide; no continuity of the eigenfunction is used. Push $\nu \ll \mu$ through this common measurable phase map. □

Lemma IV.10.17 (C^1 phase rule).

Let $L > 0$, $\alpha \in L^{-1}\mathbb{Z} \setminus \{0\}$, and let f be an orientation-preserving C^1 diffeomorphism of an interval. Suppose that μ L -discretely generates at μ -almost every x an S_L^* -ergodic full-space component Q_x of an EFD P . Assume each such component gives full mass to origin-supported, non-atomic states and to the common full-orbit eigenfunction domain in Lemma IV.10.13. These conditions hold for ordinary almost-everywhere components, and for their canonical flow transports. Put $\nu = f_*\mu$. Then ν discretely generates the transported component $(S_{\log f'(x)}^*)_* Q_x$ at $f(x)$, and its phase satisfies

$$\text{phase}_{\alpha, \nu}(f(x)) = e(-\alpha \log f'(x)) \text{phase}_{\alpha, \mu}(x). \quad (21.18)$$

Hence

$$\theta_{f_*\mu} = \int \delta_{e(-\alpha \log f'(x)) \text{phase}_{\alpha, \mu}(x)} d\mu(x). \quad (21.19)$$

For an orientation-reversing f , the image EFD is R_*P , where $R(y) = -y$. Use on it the conjugated eigenfunction $\phi_\alpha^R(R^*\xi) = \phi_\alpha(\xi)$. Then the same rule holds with $\log |f'|$. Reflection commutes with the flow, so it preserves the spectrum; no reflection invariance of P is assumed here.

Proof.

Fix a point satisfying the hypotheses and put $A = f'(x) > 0$. Choose an integer k so large that $a = kL > 0$ and $A^{-1}[-1, 1] \Subset [-e^a, e^a]$; also require this enlarged window to contain the preimages of the image window under the local rescaled diffeomorphisms for all sufficiently large times. The input states

$$\theta_n = \mu_{x, (n-k)L}^\square$$

have empirical law $Q_x^\square$, since deleting finitely many terms does not change the given discrete limit. The reconstruction identity (5.2b) recovers the source window at time nL from θ_n . Lemma IV.3.16, with $F_{x,nL}(z) = e^{nL}(f(x + e^{-nL}z) - f(x))$, then gives the image restricted scenery as the limiting push-forward of that reconstructed window.

All normalization boundaries have zero mass under Q_x , because its states are non-atomic; all central masses are positive by origin support. On compact subsets of this continuity set the sequential comparison is uniform. The compact-continuity-set argument in Theorem IV.3.17, now applied to the empirical sequence θ_n rather than a time integral, therefore gives the image empirical law

$$((A^*)_* Q_x)^\square = ((S_{\log A}^*)_* Q_x)^\square.$$

Here reconstruction advances the source law by $a = kL$, which leaves Q_x invariant; an arbitrary real look-back would not justify this identity. The full-space transport commutes with S_L^* , so the transported law is again an ergodic component law.

Since $\alpha L \in \mathbb{Z}$, the eigenfunction is S_L^* -invariant and is constant, say c_x , on Q_x . The full-orbit identity of Lemma IV.10.13 gives the constant $e(-\alpha \log A)c_x$ on its transport. These two component constants are exactly the two phases in (21.18), proving the rule. Integrating over x gives (21.19). For $A < 0$ add reflection to the same window comparison and use the conjugated eigenfunction just specified. This argument uses convergence of component laws and exact covariance, not continuity of a Borel eigenfunction or continuous-time density estimates at discrete samples.

□

Theorem IV.10.18 (*phase covariance and synchronization*).

The phase definitions (21.12)–(21.16) are measurable and satisfy:
 spatial synchronization at typical points;
 absolute-continuity monotonicity;
 the C^1 transformation rule (21.18);
 mixture/cumulative-phase compatibility.

These are the phase properties used in the later normality arguments.

Proof.

Lemmas IV.10.13–IV.10.17.

□

4. Finite-partition Shannon-McMillan-Breiman theorem

Let $(X, \mathcal{B}, \mu, T)$ be probability preserving and let $\mathcal{A}$ be a finite partition.

Write

$$\mathcal{A}_0^{n-1} = \bigvee_{j=0}^{n-1} T^{-j} \mathcal{A}. \quad (21.20)$$

For $x \in X$, let $\mathcal{A}_0^{n-1}(x)$ denote its atom.

The conditional information function is

$$f_m = I_\mu \left(\mathcal{A} \left| \bigvee_{k=1}^m T^{-k} \mathcal{A} \right. \right). \quad (21.21)$$

Here $m \geq 0$, and the empty join at $m = 0$ is the trivial σ -algebra, so $f_0 = I_\mu(\mathcal{A})$.

Lemma IV.10.19 (*maximal information tail*).

For all $t > 0$,

$$\mu\{\sup_m f_m > t\} \leq (\#\mathcal{A})e^{-t}. \quad (21.22)$$

Proof.

For an atom A_i of $\mathcal{A}$, let

$$p_m(x) = \mathbb{E} \left(\mathbf{1}_{A_i} \left| \bigvee_{k=1}^m T^{-k} \mathcal{A} \right. \right)(x), \quad x \in X.$$

Write $\mathcal{F}_m = \bigvee_{k=1}^m T^{-k} \mathcal{A}$ and let $B_m = \{p_m < e^{-t}, p_k \geq e^{-t} \text{ for } 0 \leq k < m\}$. These sets are disjoint and $\mathcal{F}_m$ -measurable. Hence

$$\sum_{m \geq 0} \mu(A_i \cap B_m) = \sum_{m \geq 0} \int_{B_m} p_m d\mu \leq e^{-t} \sum_{m \geq 0} \mu(B_m) \leq e^{-t}.$$

On A_i , $f_m = -\log p_m$. Summing this estimate over the at most $\#\mathcal{A}$ symbols proves (21.22), including $m = 0$. In particular $\sup_{m \geq 0} f_m$ is integrable, by integration of this exponential tail bound. □

Lemma IV.10.20 (*triangular/Maker averaging lemma*).

Suppose $g_m \rightarrow g$ a.e. and in L^1 , and $\sup_m |g_m|$ is integrable. Then

$$\frac{1}{n} \sum_{j=0}^{n-1} [g_{n-j}(T^j x) - g(T^j x)] \longrightarrow 0. \quad (21.23)$$

for μ -a.e. x .

Proof.

For fixed K , split the sum according as $n - j < K$ or $n - j \geq K$.

The at most K endpoint terms contribute $o(1)$. To justify this almost surely, for each integrable h and every $\varepsilon > 0$,

$$\sum_{n \geq 1} \mu\{|h(T^n x)| > \varepsilon n\} = \sum_{n \geq 1} \mu\{|h| > \varepsilon n\} < \infty.$$

Thus $h(T^n x)/n \rightarrow 0$ by Borel–Cantelli. Apply this to each of the finitely many $g_m - g$ with $m < K$. On the tail, dominate by

$$G_K = \sup_{m \geq K} |g_m - g|,$$

and use Birkhoff on G_K . Since $G_K \rightarrow 0$ in L^1 by dominated convergence, let $K \rightarrow \infty$. $\square$

Theorem IV.10.21 (*componentwise finite-partition SMB*).

Let $\mathcal{I}$ be the invariant σ -algebra. Then for μ -a.e. x ,

$$\begin{aligned} & -\frac{1}{n} \log \mu(\mathcal{A}_0^{n-1}(x)) \\ & \longrightarrow h_\mu(T, \mathcal{A} \mid \mathcal{I})(x), \end{aligned} \tag{21.24}$$

where the limit is the entropy of $\mathcal{A}$ in the ergodic component containing x .

If T is ergodic, this is the constant $h_\mu(T, \mathcal{A})$.

Proof.

For each symbol the conditional probabilities in (21.21) converge almost surely by the martingale theorem. Their limit is positive almost everywhere on that symbol: the integral of $\mathbf{1}_{A_i}$ over the future-measurable set where the limiting conditional probability vanishes is zero. Consequently $f_m \rightarrow f_\infty$ almost surely. Lemma IV.10.19 supplies an integrable maximal envelope, and dominated convergence gives convergence in L^1 .

The information chain rule gives

$$-\log \mu(\mathcal{A}_0^{n-1}(x)) = \sum_{j=0}^{n-1} f_{n-1-j}(T^j x). \tag{21.25}$$

This is an exact identity: the symbol at time j is conditioned on the $n-1-j$ subsequent symbols in this block. Apply Lemma IV.10.20 to $g_m = f_{m-1}$, $m \geq 1$, and then

Birkhoff to f_∞ :

$$\begin{aligned} & \frac{1}{n} \sum_{j=0}^{n-1} f_\infty(T^j x) \\ & \longrightarrow \mathbb{E}(f_\infty \mid \mathcal{I})(x). \end{aligned} \tag{21.26}$$

For a finite generator, invariant sets lie in the future tail; the standard conditional-entropy identity therefore identifies (21.26) with the entropy of the ergodic component. For a non-generating $\mathcal{A}$, it is the corresponding partition entropy. $\square$

5. Pisot Prefix Separation

Definition IV.10.22 (*Pisot*).

$\beta > 1$ is a Pisot number if it is an algebraic integer and every algebraic conjugate other than β has modulus less than 1. Integers at least 2 are included.

The greedy β -digits lie in $\{0, \dots, [\beta] - 1\}$.

Lemma IV.10.23 (*Garsia separation*).

For every Pisot number β there is $c_\beta > 0$ such that, if two distinct admissible length- n β -words u, v represent prefix sums

$$\pi_n(u) = \sum_{j=1}^n u_j \beta^{-j}, \quad (21.27)$$

then

$$|\pi_n(u) - \pi_n(v)| \geq c_\beta \beta^{-n}. \quad (21.28)$$

Proof.

Multiply the difference by β^n :

$$Z = \sum_{j=1}^n (u_j - v_j) \beta^{n-j} \in \mathbb{Z}[\beta].$$

It is nonzero. Its algebraic norm is a nonzero integer, hence has modulus at least one. At every conjugate β_l with $|\beta_l| < 1$,

$$|\sum (u_j - v_j) \beta_l^{n-j}|$$

is uniformly bounded by a geometric series depending only on β and the digit alphabet. Therefore $|Z|$ is bounded below by a positive constant.

Divide by β^n .

□

Corollary IV.10.24 (*bounded β -cylinder geometry for Pisot β*).

There is C_β such that every order- n β -cylinder has length at most β^{-n} , every nonempty full cylinder has length exactly β^{-n} , and the admissible prefix endpoints have β^{-n} -scale multiplicity bounded by C_β .

Proof.

T_β^n has slope β^n on each cylinder. Full cylinders map onto $[0, 1)$, giving exact length β^{-n} . Distinct prefix endpoints satisfy Lemma IV.10.23.

The remaining one-sided truncated cylinders can accumulate only a bounded number in any interval of length β^{-n} by (21.28).

□

6. Parry measure for a Pisot base

Let

$$T_\beta(x) = \beta x \pmod{1}, \quad 0 \leq x < 1. \quad (21.29)$$

For a point initially in $\mathbb{R}$, the same formula is used for its first iterate, which lies in $[0, 1)$; all subsequent iterates use the transformation on that interval. Thus the initial branch $[j/\beta, (j+1)/\beta)$ is mapped by $x \mapsto \beta x - j$, $j \in \mathbb{Z}$. This convention is not the fractional-part sequence $\{\beta^n x\}$ when β is nonintegral.

Write

$$r_n = T_\beta^n(1^-). \quad (21.30)$$

Lemma IV.10.25 (*eventual periodicity of the orbit of 1*).

If β is Pisot, (r_n) is eventually periodic.

Proof.

The recurrence

$$r_{n+1} = \beta r_n - d_{n+1}, \quad (21.31)$$

with integer digits d_n , shows $r_n \in \mathbb{Z}[\beta]$.

Under the Minkowski embedding of $\mathbb{Z}[\beta]$, its principal coordinate lies in $[0, 1]$, while every conjugate coordinate is uniformly bounded by the geometric series generated from (21.31), because the other conjugates have modulus < 1 .

Thus the embedded r_n lie in a bounded subset of a lattice, hence form a finite set.

□

Corollary IV.10.26 (*finite Markov partition*).

Put

$$B_\beta = \{0, 1\} \cup \{r_n : n \geq 0\} \cup \{j/\beta : j \in \mathbb{Z}, 1 \leq j < \beta\}. \quad (21.31a)$$

The ordered points of the finite set B_β partition $[0, 1]$ into intervals on whose interiors T_β is affine. The image of each interval is a union of partition intervals, up to endpoints, and the resulting directed Markov graph is irreducible.

Proof. Finiteness is Lemma IV.10.25. The points j/β are exactly the interior discontinuities of T_β ; their omission would make even the integer-base assertion false. On each interval cut out by B_β , the digit $[\beta x]$ is fixed, so $T_\beta(x) = \beta x - j$. The one-sided images of a discontinuity are 0 and 1. Every other orbit endpoint r_n maps to r_{n+1} , except when it is itself a discontinuity, in which case the preceding one-sided rule applies. Thus all one-sided images of partition endpoints belong to B_β , proving the Markov property.

For irreducibility, the quasi-greedy expansion $d_\beta(1)$ is eventually periodic by Lemma IV.10.25 and is not eventually zero. Hence there is Q larger than every run of

consecutive zeros in this expansion. The lexicographic admissibility rule implies that every admissible finite digit word w has a full extension $w0^Q$: each suffix that was already lexicographically smaller remains smaller, while a suffix that matched the extremal expansion becomes strictly smaller before the appended Q zeros end. Every nonempty partition interval contains a sufficiently small greedy digit cylinder $I(w)$ and hence the full cylinder $I(w0^Q)$. Therefore some iterate of that interval covers $[0, 1)$, modulo endpoints, and in particular reaches every partition interval. This proves strong connectivity of the finite Markov graph. $\square$

Lemma IV.10.27 (*unique maximal-entropy measure*).

T_β has a unique invariant measure of maximal entropy $\log \beta$.

Proof.

Let A be the adjacency matrix of the finite irreducible Markov graph from Corollary IV.10.26. The number of β -cylinders of order n grows as β^n up to bounded factors: their lengths are at most β^{-n} and cover $[0, 1]$, while Lemma IV.10.23 gives a matching upper count. An n -digit cylinder meets at most $|B_\beta|$ length- n Markov cylinders: an earlier Markov cut, transported by its fixed digit branch to time n , lands at a point of the forward-invariant endpoint set B_β . Since T_β^n is affine and injective on the digit cylinder, each such endpoint has at most one preimage there. Thus the Markov-path count and the digit-cylinder count differ by at most a fixed factor. Perron–Frobenius theory gives $\lim_n n^{-1} \log \#\{\text{Markov paths of length } n\} = \log \rho(A)$, so $\rho(A) = \beta$.

For a stationary Markov chain with transition q_{ij} on this graph and stationary vector π , let r, l be positive right and left Perron vectors and put

$$p_{ij} = A_{ij}r_j/(\beta r_i). \quad (21.32)$$

Then the relative-entropy inequality

$$\sum_i \pi_i \sum_j q_{ij} \log \frac{p_{ij}}{q_{ij}} \leq 0$$

rearranges to

$$h(q) \leq \log \beta, \quad (21.33)$$

with equality only when $q = p$ on all positive transitions. For an arbitrary stationary path law, the entropy rate is at most its one-step conditional entropy $h(q)$; equality forces the path law to be one-step Markov. Hence an entropy-maximizing path law has transition matrix p , whose stationary vector is unique by irreducibility. It is the Parry Markov law. Coding from paths to $[0, 1)$ is one-to-one away from the countable union of backward images of B_β ; invariant measures on that exceptional set have entropy zero. Since $\log \beta > 0$, the Parry law projects to the unique T_β -invariant measure of maximal entropy. $\square$

Lemma IV.10.28 (*explicit absolutely continuous invariant measure*).

The function

$$h_\beta(x) = c_\beta \sum_{n=0}^{\infty} \beta^{-n} \mathbf{1}_{[0, r_n)}(x). \quad (21.34)$$

With normalization c_β , this is a bounded positive density, and $h_\beta dx$ is T_β -invariant.

Proof.

The $n = 0$ term is positive on $[0, 1)$, so the density is bounded below.

The geometric coefficients give an upper bound.

Let L be the Frobenius–Perron operator

$$Lf(x) = \beta^{-1} \sum_{\substack{0 \leq j < \beta \\ (x+j)/\beta < 1}} f\left(\frac{x+j}{\beta}\right). \quad (21.35)$$

with the evident digit range for nonintegral β . Apply L to each interval indicator in (21.34). The recurrence $r_{n+1} = T_\beta r_n$ makes the terms telescope: the image of $[0, r_n)$ contributes the next indicator plus the digit-branch correction, and the $n=0$ correction exactly restores the first term. Hence $Lh_\beta = h_\beta$. □

Lemma IV.10.29 (*entropy and uniqueness of the ACIP*).

Let $\lambda_\beta = h_\beta dx$. Then

$$h_{\lambda_\beta}(T_\beta) = \log \beta, \quad (21.36)$$

λ_β is the unique absolutely continuous invariant probability, and it is the unique invariant exact-dimensional measure of dimension one.

Proof.

Because h_β is bounded above and below, λ_β is equivalent to Lebesgue with bounded Radon–Nikodym derivative. The Markov cylinders from Corollary IV.10.26 have Euclidean lengths comparable, up to bounded factors, to β^{-n} . Therefore SMB for the finite Markov generator gives

$$h_{\lambda_\beta}(T_\beta) = \log \beta. \quad (21.37)$$

By Lemma IV.10.27 it is the unique measure of maximal entropy.

If ρ is another absolutely continuous invariant probability, its ergodic components are absolutely continuous.

Every ergodic absolutely continuous invariant probability is equivalent to λ_β on a positive invariant set;

irreducibility makes that set full, and two equivalent ergodic invariant probabilities coincide. Hence $\rho = \lambda_\beta$.

Finally, for an ergodic invariant ρ , the same β -cylinder geometry and SMB argument gives

$$\dim \rho = \frac{h_\rho(T_\beta)}{\log \beta}. \quad (21.38)$$

Thus dimension one implies maximal entropy, and Lemma IV.10.27 gives $\rho = \lambda_\beta$.

The nonergodic case follows by ergodic decomposition. □

Definition IV.10.30 (*β -normality*).

A point x is β -normal if

$$(1/N) \sum_{n=0}^{N-1} \delta_{T_\beta^n x} \rightarrow \lambda_\beta. \quad (21.39)$$

For integer β , λ_β is Lebesgue measure.

Theorem IV.10.31 (*integer/Pisot normality foundation*).

For every Pisot number β :

β -prefixes have uniform Garsia separation;

T_β has the unique Parry measure λ_β described above;

λ_β is the unique absolutely continuous invariant probability, the unique measure of maximal entropy, and the unique invariant

dimension-one probability;

the finite-partition Shannon–McMillan–Breiman theorem and the entropy/dimension bridge hold.

No Bertrand–Mathis fractional-part conclusion is included.

Proof.

Lemmas IV.10.21–IV.10.29. □

10.2 Orbit distributions from local measure structure

1. A Bounded Martingale-Difference Cesàro Lemma

Lemma IV.10.32 (*bounded martingale-difference Cesàro lemma*).

Let (F_n) be an increasing filtration and let X_n be uniformly bounded random variables: there is one finite constant C such that $\|X_n\|_\infty \leq C$ for every n .

Assume also that X_n is F_{n+1} -measurable and

$$E(X_n | F_n) = 0. \quad (22.1)$$

Then

$$\frac{1}{N} \sum_{n=0}^{N-1} X_n \rightarrow 0. \quad (22.2)$$

almost surely and in L^2 .

Proof.

The partial sums $M_N = \sum_{n < N} X_n$ form a martingale with orthogonal increments. By the uniform bound,

$$E|M_N|^2 \leq C^2 N.$$

Along $N = 2^k$, Chebyshev gives

$$P(|M_{2^k}| > \varepsilon 2^k)$$

$$\leq C^2 \varepsilon^{-2} 2^{-k},$$

so Borel–Cantelli gives $M_{2^k}/2^k \rightarrow 0$. Between consecutive dyadic times,

Doob's maximal inequality applied to the increment martingale gives the same control. L^2 convergence follows directly from $N^{-2} E|M_N|^2 = O(1/N)$.

□

Corollary IV.10.33 (*finite-lookahead conditional-law replacement*).

Let Z_n take values in a compact metric space Y , let (F_n) be an increasing filtration, and let κ_n be a regular conditional law of Z_n given F_n . Assume that for every $F \in C(Y)$ and $\varepsilon > 0$ there are an integer $k \geq 1$ and uniformly bounded F_{n+k} -measurable functions $G_{n,k}$ such that

$$\sup_n \|F(Z_n) - G_{n,k}\|_\infty < \varepsilon. \quad (22.2a)$$

Then for every continuous F on Y ,

$$\frac{1}{N} \sum_{n < N} \left(F(Z_n) - \int F d\kappa_n \right) \rightarrow 0. \quad (22.3)$$

almost surely.

The conclusion holds simultaneously for a fixed countable dense family of

continuous tests, hence the empirical laws

$$\frac{1}{N} \sum_{n < N} \delta_{Z_n}, \quad \frac{1}{N} \sum_{n < N} \kappa_n \quad (22.4)$$

have the same weak accumulation points.

Proof.

Fix F, ε, k as in (22.2a) and put $X_n = G_{n,k} - E(G_{n,k} \mid F_n)$. The bound $\sup_n \|X_n\|_\infty \leq 2 \sup_n \|G_{n,k}\|_\infty < \infty$ is uniform in n for this fixed k . For each residue $p \in \{0, \dots, k-1\}$, the sequence X_{p+mk} is a uniformly bounded martingale-difference sequence for the filtration F_{p+mk} : indeed X_{p+mk} is $F_{p+(m+1)k}$ -measurable and its conditional expectation on F_{p+mk} is zero. Lemma IV.10.32 and recombination of the k residue classes show that its full Cesàro mean tends to zero almost surely. The difference between X_n and $F(Z_n) - E(F(Z_n) \mid F_n)$ is bounded by 2ε . Take a sequence $\varepsilon \downarrow 0$ and intersect the corresponding conull sets. Finally intersect over a countable dense family in $C(Y)$ to obtain the weak-limit claim. $\square$

2. A moving-partition generator criterion

Let $T : X \rightarrow X$ be Borel and let $\mathcal{A}$ be a finite Borel partition.

Definition IV.10.34 (*topological generator*).

$\mathcal{A}$ is a topological generator on a set G if, for every $x \in G$,

$$\text{diam } \mathcal{A}_0^{n-1}(x) \longrightarrow 0 \quad (22.5)$$

in a compatible metric.

Lemma IV.10.35 (*boundary-null generator*).

Suppose:

μ gives measure zero to the boundary of every atom of $\mathcal{A}$ and to all of its inverse images;

$\mathcal{A}$ is a topological generator on a μ -conull set.

Then:

- (a) the symbolic name map is injective modulo a null set;
- (b) conditional measures on $\mathcal{A}_0^{n-1}(x)$ shrink to x ;
- (c) for every bounded continuous F , the conditional expectation $\mathbb{E}(F \mid \mathcal{A}_0^{n-1})(x) \rightarrow F(x)$ for μ -a.e. x .

Proof.

The nested atoms shrink to $\{x\}$; this gives (a). Part (c) is martingale convergence because the generated σ -algebra is the full Borel σ -algebra modulo μ . Part (b) follows by testing against a countable dense family of continuous functions. $\square$

3. Abstract orbit and local-measure comparison

Let $A_n(x)$ be a nested family of positive-mass atoms around x . Suppose for each n there is an affine normalization

$$h_{n,x} : A_n(x) \rightarrow I_{n,x} \subset \mathbb{R}. \quad (22.6)$$

and write

$$\mu_{x,n} = \frac{(h_{n,x})_*(\mu|_{A_n(x)})}{\mu(A_n(x))}. \quad (22.7)$$

Let $z_n = h_{n,x}(x)$.

Theorem IV.10.36 (*orbit/local-measure comparison*).

Assume:

the σ -algebra F_n generated by A_n is increasing in the natural extension;

the observables Z_n satisfy the finite-lookahead approximation (22.2a) for this filtration;

Z_n is a point observable whose conditional law given F_n is

$$c_n((\tau_{y_n})_*\mu_{x,n}|I_{n,x})$$

for some bounded translation y_n ;

the empirical laws of $(\mu_{x,n}, y_n, I_{n,x}, c_n)$ are tight, and every accumulation law gives full mass to parameters for which $\mu_{x,n}$ has an atomless limiting state and the limiting interval has positive mass under its corresponding translated state.

Then for μ -a.e. x , every weak accumulation point ν of

$$\frac{1}{N} \sum_{n < N} \delta_{Z_n} \quad (22.8)$$

has a representation

$$\nu = \int c_\omega((\delta_{y_\omega} * \eta_\omega)|I_\omega) dQ(\omega). \quad (22.9)$$

where Q is an accumulation law of the local-scene parameters, η_ω are the corresponding local measures, I_ω are limiting restriction intervals, and c_ω normalizes the positive restriction.

Conversely, the barycentric empirical measures obtained from the right-hand conditional laws have exactly the same subsequential weak limits as (22.8).

Proof.

For every continuous test F , Corollary IV.10.33 makes the Cesàro difference between $F(Z_n)$ and its conditional expectation under the local law tend to zero. Tightness gives a subsequential joint parameter law Q . An atomless limiting state gives zero mass to the two

endpoints of a limiting interval. Positive mass there makes restriction and normalization continuous at that parameter. Thus the continuous mapping theorem identifies the limit of the conditional laws with the barycenter in (22.9). A countable dense family of continuous tests gives equality of weak limits, in both stated directions. $\square$

4. Integer base: an exact local-piece identity

Let $b \geq 2$ be an integer and $T_b(x) = bx \pmod{1}$. Let

$$I_n(x) = [kb^{-n}, (k+1)b^{-n}) \quad (22.10)$$

be the b -adic cylinder containing x .

Define

$$h_{n,x}(y) = b^n(y - kb^{-n}) \in [0, 1). \quad (22.11)$$

Then

$$h_{n,x}(x) = T_b^n x. \quad (22.12)$$

Lemma IV.10.37 (*exact integer-base local-piece identity*).

Conditioned on the b -adic past cylinder $I_n(x)$, the conditional law of $T_b^n X$ is exactly

$$\frac{(h_{n,x})_*(\mu|_{I_n(x)})}{\mu(I_n(x))}. \quad (22.13)$$

Proof.

T_b^n is the affine map (22.11) on $I_n(x)$.

Thus the orbit/local-measure comparison theorem applied to the b -adic filtration identifies every orbit accumulation law with a barycenter of normalized local pieces of μ , with no random

restriction interval. $\square$

5. Pisot bases: random restriction intervals

Let $\beta > 1$ be Pisot. On $[0, 1)$ let $I_n^\beta(x)$ be the greedy β -cylinder of order n . For an initial measure on $\mathbb{R}$ and $n \geq 1$, use the distinct cylinder family $\hat{I}_n^\beta$, defined using the initial branch and then its next $n - 1$ digits:

$$\hat{I}_n^\beta(x) = B_j \cap T_\beta^{-1} I_{n-1}^\beta(T_\beta x), \quad B_j = [j/\beta, (j+1)/\beta), \quad x \in B_j,$$

with $I_0^\beta = [0, 1)$. This is a countable initial partition followed by finite refinements; the tail I_{n-1}^β on the right is always the ordinary unit-interval greedy cylinder. In the last noninteger digit branch I_1^β can extend beyond 1, whereas I_1^β does not, so these families are not identified. For a measure supported on $[0, 1)$ the formulas below use I_n^β ; for a real-line initial measure they use $\hat{I}_n^\beta$ throughout. The latter cylinders still have slope β^n and image length in $[c_\beta, 1]$; the lower bound follows from the corresponding interval cylinder of order $n-1$. Conditional expectations and the finite-lookahead argument below apply to this increasing filtration just as to a finite initial partition. The first time index has no effect on any Cesàro limit.

On each such cylinder T_β^n is affine with slope β^n and maps the cylinder onto an interval

$$J_n(x) = [0, L_n(x)) \subset [0, 1]. \quad (22.14)$$

Put

$$\begin{aligned} h_{n,x}(y) &= T_\beta^n y, \\ \eta_{n,x} &= \frac{(h_{n,x})_*(\mu|_{I_n^\beta(x)})}{\mu(I_n^\beta(x))}. \end{aligned} \quad (22.15)$$

The point $T_\beta^n x$ is sampled from $\eta_{n,x}$ conditionally on the cylinder.

Lemma IV.10.38 (*positive restriction length in the Pisot model*).

There is $c_\beta > 0$ such that every nonempty greedy β -cylinder $I_n^\beta(x)$ satisfies

$$c_\beta \beta^{-n} \leq |I_n^\beta(x)| \leq \beta^{-n}. \quad (22.14a)$$

In particular its image interval $J_n(x)$ and every limiting restriction interval in (22.9) have length at least c_β .

Proof.

The order- n cylinders are consecutive half-open intervals whose left endpoints are the admissible prefix sums $\pi_n(u)$ in (21.27); the last right endpoint is 1. For adjacent prefix endpoints, Lemma IV.10.23 gives a gap at least $c\beta^{-n}$. For the last interval, write its left endpoint as $\pi_n(u)$. Then

$$\beta^n(1 - \pi_n(u)) = \beta^n - \sum_{j=1}^n u_j \beta^{n-j} \in \mathbb{Z}[\beta] \setminus \{0\}.$$

At every nonidentity conjugate the right side is uniformly bounded, because the digit alphabet is finite and all such conjugates have modulus less than one. Its algebraic norm is a nonzero integer, so the identity coordinate is bounded below by a positive constant. This gives the same lower bound for the final interval. The upper bound follows because T_β^n has slope β^n on a cylinder and image in $[0, 1)$. This proves (22.14a). □

Remark IV.10.39 (*the role of restriction intervals*).

Nonfull cylinders can be shorter than β^{-n} , but for Pisot β their normalized lengths cannot tend to zero. This uniform lower bound is the property needed in the normality criterion below.

6. Scenery form of the representation

Suppose μ continuously generates an EFD P at x . Every normalized β -local piece $\eta_{n,x}$ is a positive restriction of the centered scenery at the *exact* time $n \log \beta$:

$$\eta_{n,x} = c_n((\delta_{y_n} * \mu_{x,n \log \beta})|J_n). \quad (22.16)$$

where:

y_n stays in a compact interval;

for Pisot β , $|J_n|$ is bounded below by Lemma IV.10.38.

Indeed $I_n^\beta(x)$ has length at most β^{-n} , so it lies in $B(x, \beta^{-n})$. In the centered coordinate $z = \beta^n(y - x)$, the map T_β^n is $z \mapsto z + T_\beta^n x$ on that cylinder. This proves (22.16) without a time change.

Let Q be an accumulation law of the parameters in (22.16).

If the exact-time sceneries at x have empirical law Q_x , then the η -coordinate is distributed according to Q_x . Spectral exclusion is one way to obtain $Q_x = P$; in the resonant case Q_x can instead be an $S_{\log \beta}^*$ -ergodic component of P .

Theorem IV.10.40 (β -map orbit/local-scene representation).

Let β be an integer or a Pisot number and let x be martingale-generic for μ . Assume that the exact-time scenery averages at x converge to a probability Q_x on restricted states which is supported on atomless states satisfying the two-sided origin-mass property (21.9b). This includes $Q_x = P$ for a nontrivial EFD and every spectral phase component P_z of P . If

$$(1/N_j) \sum_{n < N_j} \delta_{T_\beta^n x} \rightarrow \nu, \quad (22.17)$$

then

$$\nu = \int c_\omega((\delta_{y_\omega} * \eta_\omega)|I_\omega) dQ(\omega). \quad (22.18)$$

where:

the η_ω marginal is Q_x ;

c_ω normalizes the positive restriction;

for integer β , $I_\omega = [0, 1]$;

for Pisot β , $|I_\omega| \geq c_\beta > 0$;

translations stay in a fixed compact interval. There is no uncontrolled scenery-time shift.

No equality of the orbit measure with one scenery measure is asserted; the

correct object is the barycenter (22.18).

Proof.

The greedy β -cylinders form a finite topological generator. For each $F \in C([0, 1])$, choose k so large that F oscillates by less than ε on every k -cylinder. Then $F(T_\beta^n x)$ is uniformly ε -approximated by a function constant on $(n + k)$ -cylinders; thus (22.2a) holds. The conditional law on the n -cylinder is exactly (22.13) or (22.15), and its centered-scenery form is (22.16).

The translations and interval endpoints range over a compact set. Each centered restriction interval contains 0 and has length at least c_β (take $c_\beta = 1$ in the integer case). By the two-sided property, every such limiting interval has positive η -mass for Q_x -almost every η . This also makes the normalizing constants c_n tight in empirical distribution: otherwise there would be $\epsilon > 0$ and a subsequence of initial time ranges on which at least an ϵ fraction of the interval masses tend to zero. A limit of the corresponding subprobability parameter laws has scenery marginal dominated by Q_x , while compactness of the interval parameters and atomlessness would force zero mass on a limiting interval of length at least c_β containing 0. This contradicts the two-sided property. Atomlessness also removes endpoint discontinuities. Theorem IV.10.36 now applies. The scenery marginal of the joint law is Q_x by its assumed convergence. For the phase-component assertion, disintegrate P over a unit-modulus eigenfunction. Lemmas IV.10.7 and IV.10.8 give the required properties for almost every phase. Scaling by S_t^* rotates these components transitively around the phase circle, and both properties are preserved by each positive scaling time. Hence they hold for every component in this rotation family.

□

7. Invariance of orbit limits

Lemma IV.10.41 (*invariance of non-atomic orbit limits*).

If ν is a non-atomic weak accumulation law of the T_β orbit of x , then

$$(T_\beta)_* \nu = \nu. (22.19)$$

Proof.

For a continuous F whose composition with T_β is continuous ν -a.e.,

$$\frac{1}{N} \sum_{n < N} (F(T_\beta^{n+1} x) - F(T_\beta^n x)) \rightarrow 0. \quad (22.20)$$

T_β has only finitely many discontinuities. Non-atomic ν assigns them zero mass, so approximate $F \circ T_\beta$ outside small neighborhoods of these points and pass to the weak limit.

□

10.3 Dimension growth and resonance

1. Dimension conventions and Frostman decomposition

For a probability ρ on $\mathbb{R}$, let

$$\underline{\dim}_H \rho = \inf\{\dim_H A : \rho(A) > 0\}. \quad (23.1)$$

This is the lower Hausdorff dimension of the measure. For exact-dimensional ρ , it equals its common local dimension.

Lemma IV.10.42 (*countable Frostman decomposition*).

If $0 \leq s < \underline{\dim}_H \rho$, then there are Borel sets E_j increasing to full ρ -measure and constants C_j such that the normalized restrictions ρ_j satisfy

$$\rho_j(B(x, r)) \leq C_j r^s \quad (23.2)$$

for all x and $r > 0$.

Proof.

For ρ -a.e. x ,

$$\liminf_{r \downarrow 0} \frac{\log \rho(B(x, r))}{\log r} > s. \quad (23.3)$$

For $j \geq 1$, let E_j be the points for which $\rho(B(x, r)) \leq jr^s$ for every $0 < r < 1/j$. These sets are Borel: it suffices to test rational radii, and ball mass is a Borel function of its centre. They increase and their union has full measure by (23.3). Discard any initial sets of zero mass and write $\rho_j = \rho|_{E_j}/\rho(E_j)$. If $\rho_j(B(x, r)) > 0$, choose $y \in E_j \cap B(x, r)$. Then $B(x, r) \subset B(y, 2r)$, so for $0 < r < 1/(2j)$,

$$\rho_j(B(x, r)) \leq \frac{j2^s}{\rho(E_j)} r^s.$$

For $r \geq 1/(2j)$, the bound $\rho_j(B(x, r)) \leq 1$ gives $\rho_j(B(x, r)) \leq (2j)^s r^s$, since $s \geq 0$. Taking the maximum of the two constants proves (23.2) for every centre and every radius. Negative exponents may instead be used only with a bounded radius range; they do not satisfy this universal-radius statement. □

Lemma IV.10.43 (*finite energy below a Frostman exponent*).

If $\rho(B(x, r)) \leq Cr^s$ and $0 < q < s$, then

$$I_q(\rho) = \iint |x - y|^{-q} d\rho(x) d\rho(y) < \infty. \quad (23.4)$$

Proof.

For fixed x , split into annuli

$2^{-(m+1)} < |x - y| \leq 2^{-m}$. The m th contribution is at most $C2^{mq}2^{-ms}$; the series converges.

□

2. Generic convolution dimension from projection energy

For t in a compact interval J , put

$$S_t \nu = (x \mapsto e^t x)_* \nu. \quad (23.5)$$

Lemma IV.10.44 (*one-parameter kernel estimate*).

Let $0 < q < 1$ and J compact. Then uniformly for $(u, v) \neq 0$,

$$\begin{aligned} & \int_J |u + e^t v|^{-q} dt \\ & \leq C_{J,q} |(u, v)|^{-q}. \end{aligned} \quad (23.6)$$

Proof.

Normalize $|(u, v)| = 1$. If $u + e^t v$ has no zero in a fixed enlargement of J , it is uniformly bounded away from zero unless one coordinate is already uniformly bounded below, and the integral is bounded.

If $u + e^{t_0} v = 0$, then $|v|$ and $|u|$ are bounded below by constants depending only on J . Hence $|u + e^t v| \geq c|t - t_0|$ near t_0 . Since $q < 1$, the singularity $|t - t_0|^{-q}$ is integrable. Rescale.

□

Lemma IV.10.45 (*averaged convolution energy*).

Let μ, ν be compactly supported probabilities on $\mathbb{R}$. For $0 < q < 1$,

$$\int_J I_q(\mu * S_t \nu) dt \leq C \iiint ((x - x')^2 + (y - y')^2)^{-q/2} d\mu(x) d\mu(x') d\nu(y) d\nu(y'). \quad (23.7)$$

Proof.

Expand the q -energy of the convolution and apply Fubini:

$$I_q(\mu * S_t \nu) = \iiint |x - x' + e^t(y - y')|^{-q} d\mu(x) d\mu(x') d\nu(y) d\nu(y').$$

Integrate in t and apply Lemma IV.10.44.

□

Lemma IV.10.46 (*product Frostman bound*).

Suppose

$$\mu(B(x, r)) \leq Cr^s,$$

$$\nu(B(y, r)) \leq Cr^u. (23.8)$$

Then the product $\mu \times \nu$ on $\mathbb{R}^2$ satisfies

$$(\mu \times \nu)(B((x, y), r)) \leq C'r^{s+u}, (23.9)$$

and has finite q -energy for every $q < s+u$.

Proof.

A Euclidean r -ball lies in the product of two r -intervals. Apply (23.8) and then Lemma IV.10.43 in dimension two.

□

Theorem IV.10.47 (*generic dissonance*).

Let μ, ν be exact-dimensional probabilities on $\mathbb{R}$ with dimensions s, u .

Then for Lebesgue-a.e. t ,

$$\dim_H(\mu * S_t \nu)$$

$$\geq \min\{1, s + u\}. (23.10)$$

More generally the same lower bound holds with lower measure dimensions.

Proof.

If one of s, u is zero, the required lower bound holds for every t . Indeed, for a Borel set E of dimension below the lower dimension of μ ,

$$(\mu * S_t \nu)(E) = \int \mu(E - e^t y) d\nu(y) = 0,$$

and the analogous formula exchanges the two factors. Convolution therefore has lower dimension at least the maximum of the two lower dimensions, which proves all zero-dimension cases.

Otherwise fix $0 < s' < s$ and $0 < u' < u$. Use the Frostman decomposition, subdividing also into bounded pieces, to split μ, ν into countably many positive Frostman pieces of exponents s', u' . For

$$q < \min\{1, s' + u'\},$$

Lemmas IV.10.45–IV.10.46 show that for a.e. t every pair of Frostman pieces has finite

q -energy after convolution. Hence every such convolution has lower dimension $\geq q$.

The convolution of the original measures is a positive countable sum of these piece-convolutions. A set carrying positive mass for the full convolution carries positive mass for at least one positive piece, so its dimension is

$\geq q$. Let $q \uparrow \min\{1, s' + u'\}$ and then let $s' \uparrow s$ and $u' \uparrow u$. At the endpoint $s + u = 1$, this q -up-to-one argument gives dimension one. □

3. EFD saturation

Let P be a restricted EFD of dimension δ , and let $\tilde{P}$ denote its unique invariant full-space lift, as in Lemma IV.2.15 and the convention following it. By Chapter IV.3, typical states and all their positive bounded restrictions are exact dimensional of dimension δ .

Theorem IV.10.48 (*EFD saturation by a large enough measure*).

Let τ be a probability on $\mathbb{R}$ with

$$\underline{\dim}_H \tau \geq 1 - \delta. \quad (23.11)$$

Then for P -a.e. η ,

$$\dim_H(\tau * \eta) = 1. \quad (23.12)$$

The same is true for every positive restriction of η .

Proof.

All convolution dimensions in this proof mean lower Hausdorff measure dimensions. First note a positivity fact. If ρ is a probability and $\rho' = \rho|_E / \rho(E)$ with $\rho(E) > 0$, then

$$\tau * \rho = \rho(E)(\tau * \rho') + \tau * (\rho|_{E^c}). \quad (23.12a)$$

If the left side gives zero mass to every Borel set of dimension less than one, so does each positive summand. Thus dimension one passes to every positive restriction of ρ .

For a full-space Radon state ξ , put

$$\xi_k = \frac{\xi|_{[-k, k]}}{\xi([-k, k])}, \quad k \geq 1.$$

These denominators are positive and finite. For $\tilde{P}$ -almost every ξ , each ξ_k has dimension δ . Theorem IV.10.47, the hypothesis (23.11), and a countable intersection therefore give

$$\underline{\dim}_H(\tau * S_t \xi_k) = 1 \quad \text{for every } k \geq 1 \text{ and a.e. } t \in [0, 1]. \quad (23.13)$$

The endpoint where the sum of dimensions equals one is included by taking energy exponents strictly below one. By (23.12a), the same conclusion holds for every bounded

positive restriction of $S_t \xi$. We do not need to convolve an unbounded Radon state: only its normalized bounded pieces are convolved.

Let B be the full-space bad event

$$B = \{\xi : \exists k \geq 1, \dim_{\text{H}}(\tau * \xi_k) < 1\}. \quad (23.14)$$

The lower dimension of a finite measure is universally measurable in that measure (via Hausdorff-content tests); normalization and restriction are Borel. Thus B is universally measurable, which suffices for completed-measure Fubini. It is unchanged by multiplying ξ by a positive scalar. In (23.13), the restriction of $S_t \xi$ to $[-k, k]$ is the image of a bounded restriction of ξ , so the positivity fact gives $S_t^* \xi \notin B$ for almost every t . Full-space invariance, not restricted invariance, now yields

$$\int_0^1 \tilde{P}\{\xi : S_t^* \xi \in B\} dt = \tilde{P}(B) = 0. \quad (23.15)$$

Pushing down by $\pi_{\square}$ gives (23.12) for P -almost every restricted state $\eta = \xi_1$. Finally (23.12a) proves the assertion for every positive restriction of this same η , without a new exceptional set depending on the restriction.

□

4. Entropy stability under small translations

For $\beta > 1$ Pisot let D_n^β denote the finite order- n β -cylinder partition of $[0, 1]$, extended to a bounded interval by integer translates.

Lemma IV.10.49 (*neighbor bound for partition entropy*).

There is C_β such that if $|a| \leq C\beta^{-n}$, then for every probability ρ on a fixed bounded interval,

$$\begin{aligned} & |H(T_a \rho, D_n^\beta) - H(\rho, D_n^\beta)| \\ & \leq C_\beta \log(n + 2). \end{aligned} \quad (23.16)$$

Proof.

Garsia separation and the finite-state β -cylinder geometry imply that an order- n cylinder translated by $O(\beta^{-n})$ meets at most $C_\beta(n + 1)$ order- n cylinders. If a partition $\mathcal{A}$ refines another partition $\mathcal{B}$ with every $\mathcal{B}$ -atom meeting at most M $\mathcal{A}$ -atoms, then

$$|H(\rho, \mathcal{A}) - H(\rho, \mathcal{B})| \leq \log M. \quad (23.17)$$

Apply this to D_n^β and its translate.

□

Lemma IV.10.50 (*entropy/dimension upper bound*).

Let ρ be a probability on a bounded interval. If, for some L and all k ,

$$\begin{aligned} H(\rho, D_{kL}^\beta) \\ \leq kLa + o(kL), \end{aligned} \quad (23.18)$$

then

$$\underline{\dim}_H \rho \leq \frac{a}{\log \beta}. \quad (23.19)$$

In particular, if $a < \log \beta$, then $\underline{\dim}_H \rho < 1$. The upper Hausdorff dimension of a non-ergodic mixture need not obey this bound.

Proof.

Put $n = kL$ and fix $s > a / \log \beta$. Markov's inequality for the information $-\log \rho(D_n^\beta(x))$ shows that a fixed positive fraction of the mass lies in atoms of mass at least β^{-sn} for all large k (choose s with a strict margin). There are at most β^{sn} such atoms. Their diameters are at most β^{-n} , so the limsup of their unions has positive ρ -mass and Hausdorff dimension at most s . Indeed, its t -dimensional content is bounded by $\sum_{k \geq K} C \beta^{(s-t)kL}$ for every $t > s$, and this tends to zero as $K \rightarrow \infty$. Thus $\underline{\dim}_H \rho \leq s$. Let $s \downarrow a / \log \beta$. □

5. Integer resonant test measures

Fix an integer $n \geq 2$. Let λ_n be Lebesgue measure.

For $N, M \geq 1$, put $L = N + M$. Define a random base- n digit process by independent L -blocks of the form

$$0^N W, \quad (23.20)$$

where W is an independent uniformly random word of length M . Average over the L possible starting phases to obtain a T_n^L -invariant probability

$\tau_{N,M}$ on $[0, 1]$.

Lemma IV.10.51 (*dimension of integer resonant block tests*).

$\tau_{N,M}$ is exact dimensional and

$$\dim \tau_{N,M} = \frac{M}{N + M}. \quad (23.21)$$

Proof.

At each L -block there are n^M equiprobable choices, so the block entropy is $M \log n$. The L -adic cylinders have exact length n^{-L} ; SMB for the iid block

shift gives (23.21) for the unaveraged law. Each starting phase is a finite piecewise affine image under an iterate of T_n , with the same exact dimension. Averaging finitely many measures of that same exact dimension preserves it, proving the assertion for $\tau_{N,M}$. $\square$

Lemma IV.10.52 (*integer resonance entropy*).

Let μ be T_n -invariant and ergodic with entropy $h < \log n$. There is $N_0 = N_0(\mu)$ such that for every $N \geq N_0$ and every $M \geq 1$,

$$\underline{\dim}_H(\mu * \tau_{N,M}) < 1. \quad (23.22)$$

The conclusion concerns lower Hausdorff measure dimension, as in Lemma IV.10.50.

Proof.

First use the unaveraged block law and independent random variables $X \sim \mu$, $Y \sim \tau$. Choose the usual half-open base- n digit convention. Put $L = N + M$. In each aligned L -block, the integer represented by the digits of Y lies between 0 and $n^M - 1$. Addition of two base- n expansions has an incoming carry of at most one at a block boundary. Thus the first N digits of that block of $X + Y$, read modulo n^N , are determined by the corresponding N digits of X and one binary carry. The remaining M digits have entropy at most $M \log n$. The integer part of $X + Y \in [0, 2]$ has bounded entropy. Using subadditivity over the k blocks and T_n -invariance of μ gives

$$H(\mu * \tau, D_{kL}^n) \leq k[H(\mu, D_N^n) + M \log n + \log 2] + O(1). \quad (23.23)$$

For any of the L starting phases, discard at most $L - 1$ initial digits to align the zero block. Their entropy is at most $L \log n$, and the future digits of X still have law μ . Consequently the same bound holds with an additional $O(L)$, independent of k . Finally, mixing the L phases adds at most $\log L$ to entropy. Thus (23.23), with an $O(L + \log L)$ remainder, applies to the phase-averaged $\tau_{N,M}$ of Lemma IV.10.51.

The generating digit partition has $H(\mu, D_N^n) = Nh + o(N)$. Choose N_0 so that $H(\mu, D_N^n) + \log 2 < N \log n$ for every $N \geq N_0$. This choice is independent of M . The bracket in (23.23) is then strictly smaller than $L \log n$. Let $k \rightarrow \infty$ and apply Lemma IV.10.50 to obtain (23.22). In particular, taking $M = N^2$ gives $\dim \tau_{N,N^2} = N/(N + 1) \rightarrow 1$, while (23.22) persists for every $N \geq N_0$. This is the large- N regime needed for resonant tests; no entropy deficit is asserted at an arbitrary fixed small N . $\square$

6. Pisot symbolic reset and block measures

Let β be Pisot. By eventual periodicity of the orbit of 1, the quasi-greedy β -expansion of 1 is

eventually periodic. Therefore the β -shift is sofic and has a finite irreducible Markov presentation.

Lemma IV.10.53 (*bounded zero-padding specification*).

There is Q_β with the following property. If $N, Q \geq Q_\beta$ and v is an admissible word of length M , then $0^N v 0^Q$ is a full word: every admissible infinite sequence may follow it. The associated affine maps of all such words v of one fixed length M have common ratio $\beta^{-(N+M+Q)}$ and pairwise interior-disjoint images of $[0, 1]$.

Proof.

Let $a = d_\beta(1)$ be the quasi-greedy extremal expansion. Its eventual periodicity (Lemma IV.10.25) and the fact that it is not eventually zero give a bound $Q_\beta - 1$ on the length of every zero run in a . By the lexicographic description of the β -shift, $0^N v$ is admissible for $N \geq Q_\beta$: a suffix beginning in the initial zero run is strictly smaller than a at its first digit, and a suffix beginning in v is bounded by a because v is admissible. Appending Q zeros preserves admissibility. Moreover, any suffix still matching an initial segment of a becomes strictly smaller before those Q zeros end. Thus every continuation is allowed; this is the full-word criterion. Fixed-order full cylinders have disjoint interiors and length $\beta^{-(N+M+Q)}$. □

Lemma IV.10.54 (*dimension of Pisot block tests*).

Fix $N, Q \geq Q_\beta$ and $M \geq 1$. Let $\mathcal{V}_M$ be the set of admissible M -words, let $L = N+M+Q$, and define

$$g_v(x) = \pi_L(0^N v 0^Q) + \beta^{-L}x, \quad v \in \mathcal{V}_M. \quad (23.24a)$$

The uniform Bernoulli self-similar measure $\tau_{N,M,Q}^\beta$ for these maps is regular and exact dimensional, with

$$\dim \tau_{N,M,Q}^\beta = \frac{\log |\mathcal{V}_M|}{L \log \beta}, \quad \log |\mathcal{V}_M| = M \log \beta + O_\beta(1). \quad (23.24)$$

If $M = N^2$ and Q remains bounded as $N \rightarrow \infty$, then

$$\dim \tau_{N,N^2,Q}^\beta \longrightarrow 1. \quad (23.25)$$

Proof.

Lemma IV.10.53 shows that the maps in (23.24a) are full cylinders of equal length with disjoint interiors. The Bernoulli cylinder mass and diameter at depth n are $|\mathcal{V}_M|^{-n}$ and β^{-Ln} ; the standard bounded-neighbor cylinder argument gives the stated exact dimension. The number $|\mathcal{V}_M|$ lies between β^M and $C_\beta \beta^M$. The lower bound follows because the admissible order- M cylinders cover $[0, 1]$ and each has length at most β^{-M} . The upper bound follows from the finite Markov presentation of Corollary IV.10.26 and its Perron eigenvalue β (Lemma IV.10.27). This proves (23.24) and (23.25). □

Lemma IV.10.55 (*Pisot resonance entropy*).

Let μ be a T_β -invariant probability with $h_\mu(T_\beta) < \log \beta$. Choose $M = N^2$ and a bounded $Q = Q_N \geq Q_\beta$. Then, for every sufficiently large N ,

$$\underline{\dim}_H(\mu * \tau_{N,M,Q}^\beta) < 1. \quad (23.27)$$

No ergodicity of μ is required.

Proof.

Let $B = \lceil \beta \rceil$ and lift μ to its stationary greedy β -digit law $\tilde{\mu}$ on $\{0, \dots, B-1\}^\mathbb{N}$. The test measure has the independent length- L digit-block law $\tilde{\tau}$ supported on words $0^N \cup 0^Q$. Add the two digit sequences coordinatewise, obtaining a law $\tilde{\rho}$ on $\{0, \dots, 2B-2\}^\mathbb{N}$. Its projection under $\pi((d_j)) = \sum_{j \geq 1} d_j \beta^{-j}$ is exactly $\mu * \tau_{N,M,Q}^\beta$. Because $\tilde{\mu}$ is shift-invariant and $\tilde{\tau}$ is invariant under the L -fold shift, $\tilde{\rho}$ is invariant under the L -fold shift.

For a fixed finite digit alphabet $\{0, \dots, D-1\}$ let $\mathcal{P}_k$ partition sequences according to equality of their prefix values $\pi_k(d) = \sum_{j=1}^k d_j \beta^{-j}$. Distinct such values are at least $c_{\beta,D} \beta^{-k}$ apart: multiply their difference by β^k and apply the algebraic-norm argument of Lemma IV.10.23, whose conjugate bounds remain uniform for any fixed D . All prefix values lie in one bounded interval, hence

$$|\mathcal{P}_k| \leq C_{\beta,D} \beta^k. \quad (23.28)$$

The partition $\mathcal{P}_{kL}$ is coarser than $\bigvee_{j=0}^{k-1} \sigma^{-jL} \mathcal{P}_L$. Shift invariance and entropy subadditivity therefore give

$$H(\tilde{\rho}, \mathcal{P}_{kL}) \leq kH(\tilde{\rho}, \mathcal{P}_L). \quad (23.29)$$

An atom of $\mathcal{P}_{kL}$ projects into an interval of radius $C_D \beta^{-kL}$ around its prefix value. The positive-mass entropy-cover argument in the proof of Lemma IV.10.50, applied to these intervals, proves

$$\underline{\dim}_H(\pi_* \tilde{\rho}) \leq \frac{H(\tilde{\rho}, \mathcal{P}_L)}{L \log \beta}. \quad (23.30)$$

The partition $\mathcal{P}_N \vee \sigma^{-N} \mathcal{P}_{M+Q}$ refines $\mathcal{P}_L$. The first N digits of the test sequence vanish, so the first component has exactly the $\tilde{\mu}$ law. Equation (23.28), now with $D = 2B-1$, gives

$$H(\tilde{\rho}, \mathcal{P}_L) \leq H(\tilde{\mu}, \mathcal{P}_N) + (M+Q) \log \beta + \log C_{\beta,2B-1}. \quad (23.30a)$$

The partition $\mathcal{P}_N$ on admissible β -digits is the order- N generating cylinder partition, up to a uniformly bounded endpoint ambiguity. By Corollary IV.10.24 and the finite-state entropy calculation of Lemma IV.10.27,

$$H(\tilde{\mu}, \mathcal{P}_N) = N h_\mu(T_\beta) + o_\mu(N). \quad (23.30b)$$

Since $h_\mu(T_\beta) < \log \beta$, the right side of (23.30a) is strictly less than $L \log \beta$ for all sufficiently large N ; the deficit is $N(\log \beta - h_\mu) + o_\mu(N) - O_\beta(1)$. Equation (23.30) proves (23.27). □

Theorem IV.10.56 (*high-dimensional resonant tests*).

For every Pisot number β there are regular homogeneous self-similar probabilities τ_N with

$$\dim \tau_N \rightarrow 1. \quad (23.31)$$

such that every T_β -invariant probability ν with

$$0 < \dim \nu < 1 \quad (23.32)$$

satisfies, for all sufficiently large N ,

$$\underline{\dim}_H(\nu * \tau_N) < 1. \quad (23.33)$$

Here $\dim \nu$ is the lower Hausdorff dimension of the measure. Moreover, if $0 < \gamma < 1$ and $\log \beta / \log(1/\gamma) = a/b \in \mathbb{Q}_{>0}$ in lowest terms, the common contraction of each τ_N can be chosen to equal γ^{K_N} for an integer $K_N \geq 1$.

Proof.

Choose $M = N^2$ and an integer $Q_N \in [Q_\beta, Q_\beta + b - 1]$ so that $L_N = N + M + Q_N$ is divisible by b . Set $\tau_N = \tau_{N,M,Q_N}^\beta$ from Lemma IV.10.54. It is regular, and its common ratio is $\beta^{-L_N} = \gamma^{aL_N/b}$. If no γ is specified, take $b = 1$. Equation (23.25) gives $\dim \tau_N \rightarrow 1$.

Let $\nu = \int \nu_\omega dP(\omega)$ be the ergodic decomposition and put $d = \underline{\dim}_H \nu < 1$. Fix s with $d < s < 1$. The dimension–entropy identity for ergodic T_β -measures gives $\dim \nu_\omega = h_\omega(T_\beta) / \log \beta$. The set

$$E = \{\omega : h_\omega(T_\beta) \leq s \log \beta\} \quad (23.31a)$$

has positive P -measure. Otherwise every component would give zero mass to every Borel set of dimension less than s ; integrating would imply $\underline{\dim}_H \nu \geq s$, a contradiction. The normalized invariant submixture

$$\nu_E = P(E)^{-1} \int_E \nu_\omega dP(\omega) \quad (23.31b)$$

has entropy $h_{\nu_E}(T_\beta) = P(E)^{-1} \int_E h_\omega(T_\beta) dP(\omega) \leq s \log \beta < \log \beta$. Lemma IV.10.55 therefore gives $\underline{\dim}_H(\nu_E * \tau_N) < 1$ for all sufficiently large N . Since $P(E)(\nu_E * \tau_N)$ is a positive summand of $\nu * \tau_N$, the latter has lower Hausdorff dimension at most that of $\nu_E * \tau_N$. This proves (23.33) without asserting a uniform entropy convergence rate for the individual ergodic components.

□

7. Dimension-growth consequences

Theorem IV.10.57 (*dimension growth and resonant test measures*).

The preceding results show that generic scaling gives maximal convolution dimension; an EFD of dimension δ is saturated by convolution with every measure of lower dimension at least $1 - \delta$; and a Pisot base admits resonant test measures whose dimensions tend to one.

Proof.

These are precisely Theorems IV.10.47–IV.10.56, proved above by Frostman decomposition, energy averaging, entropy stability, the zero/random block construction, finite-state Pisot specification, and the entropy resonance estimate. □

10.4 Spectral/scenery normality criterion

1. Normality criterion - statement

Let μ be a probability on $\mathbb{R}$ generating a nontrivial EFD P . Put

$$\delta = \dim P > 0. \quad (24.1)$$

Let $\beta > 1$ be Pisot. Assume the spectral exclusion

$$\Sigma(P, S) \cap \left(\frac{1}{\log \beta} \mathbb{Z} \setminus \{0\} \right) = \emptyset. \quad (24.2)$$

Theorem IV.10.58 (*Pisot spectral normality*).

Under these assumptions, μ is pointwise β -normal: for μ -a.e. x ,

$$\frac{1}{N} \sum_{n=0}^{N-1} \delta_{T_\beta^n x} \longrightarrow \lambda_\beta. \quad (24.3)$$

Proof. First reduce to a probability on $[0, 1)$. The measure μ has no atoms: scenery at an atom converges to δ_0 , which cannot generate an EFD of dimension $\delta > 0$. The countably many initial branch endpoints therefore have zero mass. For every positive-mass branch $B_j = (j/\beta, (j+1)/\beta)$, put

$$\mu_j = (x \mapsto \beta x - j)_* \frac{\mu|_{B_j}}{\mu(B_j)}.$$

At almost every interior point, positive restriction leaves the limiting scenery unchanged, by differentiation (Lemma IV.2.25). The affine covariance theorem IV.3.17 sends the generated distribution to $S_{\log \beta}^* P = P$. Thus each μ_j generates the same P and satisfies the same spectral exclusion. Once the theorem is proved on $[0, 1)$, it applies to each μ_j ; omitting the first term of an empirical orbit average changes it by at most $2/N$ against a test of norm at most one. A countable union of these branchwise conull sets proves the original real-line assertion. We may therefore assume μ is supported on $[0, 1)$.

Fix x in the common conull set where μ generates P , the martingale reconstruction of Section 10.2 holds, and all typical-measure statements from Sections 10.1 and 10.3 hold.

Take an arbitrary subsequence N_j along which the orbit empirical laws converge:

$$\nu_j = \frac{1}{N_j} \sum_{n < N_j} \delta_{T_\beta^n x} \longrightarrow \nu. \quad (24.4)$$

We prove $\nu = \lambda_\beta$.

Step 1: local-scene representation. By Theorem IV.10.40,

$$\nu = \int c_\omega \left((\delta_{y_\omega} * \eta_\omega)|_{I_\omega} \right) dQ(\omega), \quad (24.5)$$

where η_ω is distributed according to P and $|I_\omega| \geq c_\beta > 0$. Translations are bounded; the positive normalizing constants are finite and tight in the empirical parameter laws, not asserted uniformly bounded. Their limiting restrictions are positive almost surely by Theorem IV.10.40.

Step 2: positive dimension and non-atomicity. For P -a.e. η_ω , the measure η_ω is exact dimensional of dimension δ . Translation and positive restriction preserve lower Hausdorff dimension δ . Therefore every positive component in (24.5) gives zero mass to sets of Hausdorff dimension $< \delta$, and hence

$$\underline{\dim}_H \nu \geq \delta > 0. \quad (24.6)$$

In particular, ν is non-atomic.

Step 3: invariance. Lemma IV.10.41 gives

$$(T_\beta)_* \nu = \nu. \quad (24.7)$$

Step 4: rule out intermediate dimension. Suppose

$$\dim \nu < 1. \quad (24.8)$$

Choose a test τ_m from Theorem IV.10.56 with

$$\dim \tau_m \geq 1 - \delta \quad (24.9)$$

and, for the fixed invariant measure ν selected above, choose $m \geq N(\nu)$ large enough that Theorem IV.10.56 makes ν resonate with τ_m .

Theorem IV.10.48 gives, for P -a.e. η_ω ,

$$\dim(\tau_m * \eta_\omega) = 1. \quad (24.10)$$

The same holds for every positive restriction appearing in (24.5). Convolution is linear, so $\tau_m * \nu$ is a positive barycenter of these dimension-one measures. Hence every set of dimension < 1 has zero mass under $\tau_m * \nu$, and

$$\dim(\tau_m * \nu) = 1. \quad (24.11)$$

On the other hand, ν is T_β -invariant, positive dimensional, and by (24.8) sub-full. Theorem IV.10.56 gives

$$\dim(\tau_m * \nu) < 1, \quad (24.12)$$

a contradiction. Thus $\dim \nu = 1$.

Step 5: identify the invariant limit. Theorem IV.10.31 says that λ_β is the unique T_β -invariant probability of dimension one. Therefore

$$\nu = \lambda_\beta. \quad (24.13)$$

Every subsequential orbit limit is λ_β , so the full sequence converges. $\square$

2. Integer base

For an integer $n \geq 2$, λ_n is Lebesgue measure and the same proof applies with the exact n -adic versions of the orbit-reconstruction and resonance arguments in Sections 10.2 and 10.3.

Theorem IV.10.59 (*integer-base spectral normality*).

If μ generates a nontrivial EFD P and

$$\Sigma(P, S) \cap \left(\frac{1}{\log n} \mathbb{Z} \setminus \{0\} \right) = \emptyset, \quad (24.14)$$

then μ is pointwise n -normal.

Proof. Repeat Theorem IV.10.58 with $\beta = n$. The unique invariant dimension-one probability for T_n is Lebesgue measure. $\square$

3. C^1 robustness

Theorem IV.10.60 (C^1 coordinate robustness).

Let f be a C^1 diffeomorphism of $\mathbb{R}$. Under the hypotheses of Theorem IV.10.58, $f_*\mu$ is also pointwise β -normal.

Proof. The C^1 scenery-covariance theorem of Chapter IV.3 says that at μ -a.e. x , the measure $f_*\mu$ generates the derivative pushforward of P .

If $f'(x) > 0$, multiplication by $f'(x)$ is exactly a scenery time shift, so the generated EFD is

$$S_{\log f'(x)} P = P$$

because P is scale invariant. If $f'(x) < 0$, the derivative is a positive scale followed by reflection. Reflection commutes with the scenery flow, so the reflected EFD has the same pure-point spectrum as P .

In both orientation cases the spectral exclusion (24.2) is unchanged. Apply Theorem IV.10.58 to $f_*\mu$. $\square$

4. Phase-form refinement

The proof above uses only spectral exclusion. When (24.2) fails, Section 10.1 supplies a phase function for the offending eigenvalue.

Definition IV.10.61 (*full β -phase*).

For $\alpha = k/\log \beta$, $k \neq 0$, let θ_μ be the phase measure from Section 10.1. A family of local measures has *full β -phase* if its cumulative phase measure has Hausdorff dimension one on S^1 ; Haar measure is the principal example.

The phase-spread theorem used later extends this spectral-exclusion branch.

5. Normality criteria

Theorem IV.10.62 (*integer normality criterion*).

Let μ generate a nontrivial EFD P . If (24.14) holds, then μ and every C^1 diffeomorphic image are pointwise n -normal.

Proof. Apply Theorem IV.10.59 and then the coordinate-robustness Theorem IV.10.60. $\square$

Theorem IV.10.63 (*Pisot normality criterion*).

Let $\beta > 1$ be Pisot. If μ generates a nontrivial EFD P satisfying (24.2), then μ and every C^1 diffeomorphic image are pointwise β -normal with respect to the Parry measure λ_β .

Proof. Apply Theorem IV.10.58 and then the coordinate-robustness Theorem IV.10.60. $\square$

10.5 Regular-IFS normality applications

1. Regular IFSs and quasi-product measures

Let $I = [0, 1]$. A finite IFS

$$\Phi = \{f_i : I \rightarrow I\}_{i \in \Lambda} \quad (25.1)$$

is called *regular* here if every f_i is a $C^{1+\varepsilon}$ injective contraction for one $\varepsilon > 0$, there is $q < 1$ with

$$\sup_i \|f'_i\|_\infty \leq q,$$

there is $a > 0$ with $|f'_i(x)| \geq a$ for every i, x , and the intervals $f_i(I)$ are pairwise disjoint except possibly at endpoints. Its attractor is X .

For a word $u = u_1 \cdots u_n$, write

$$f_u = f_{u_1} \circ \cdots \circ f_{u_n}, \quad I(u) = f_u(I). \quad (25.2)$$

Let $\pi : \Lambda^\mathbb{N} \rightarrow X$ be the coding map.

Definition IV.10.64 (*quasi-product*).

A probability η on $\Lambda^{\mathbb{N}}$ is C -quasi-product if every cylinder has positive mass and

$$C^{-1}\eta[u]\eta[v] \leq \eta[uv] \leq C\eta[u]\eta[v] \quad (25.3)$$

for all finite words u, v . A quasi-product measure for Φ is

$$\mu = \pi_*\eta. \quad (25.4)$$

This includes Bernoulli and Hölder–Gibbs coding measures.

For a C^1 contraction f with unique fixed point p_f , define its asymptotic contraction ratio by

$$\lambda(f) = |f'(p_f)|. \quad (25.5)$$

If f is affine, this is its ordinary contraction ratio.

2. Bounded distortion from $C^{1+\varepsilon}$ regularity**Lemma IV.10.65** (*uniform log-derivative distortion*).

There is $D < \infty$ such that for every word u and $x, y \in I$,

$$|\log |f'_u(x)| - \log |f'_u(y)|| \leq D|x - y|^\varepsilon. \quad (25.6)$$

In particular,

$$D_0^{-1} \leq \frac{|f'_u(x)|}{|f'_u(y)|} \leq D_0 \quad (25.7)$$

with D_0 independent of u .

Proof. Since every f'_i is nonzero on I and there are finitely many maps, the functions $\log |f'_i|$ are uniformly ε -Hölder: for some $L < \infty$,

$$|\log |f'_i(x)| - \log |f'_i(y)|| \leq L|x - y|^\varepsilon.$$

For $u = u_1 \cdots u_n$, the chain rule gives

$$\log |f'_u(x)| = \sum_{j=1}^n \log |f'_{u_j}(f_{u_{j+1} \cdots u_n}(x))|.$$

The tail map $f_{u_{j+1} \cdots u_n}$ contracts distances by at most q^{n-j} , hence

$$|\log |f'_u(x)| - \log |f'_u(y)|| \leq L|x - y|^\varepsilon \sum_{j=1}^n q^{\varepsilon(n-j)} \leq \frac{L}{1 - q^\varepsilon} |x - y|^\varepsilon.$$

This proves (25.6), and exponentiation gives (25.7). □

Corollary IV.10.66 (*cylinder scale*).

For all u and x ,

$$D_0^{-1}|f'_u(x)| \leq |I(u)| \leq D_0|f'_u(x)|. \quad (25.8)$$

Proof. Apply the mean value theorem and Lemma IV.10.65. □

3. Normalized cylinder maps and limit geometries

For an interval J , let T_J be the orientation-preserving affine map $J \rightarrow [0, 1]$. Define the normalized cylinder map

$$f_u^* = T_{I(u)} \circ f_u : I \rightarrow [0, 1]. \quad (25.9)$$

Lemma IV.10.67 (C^1 precompactness).

The family $\{f_u^* : u \in \Lambda^*\}$ is precompact in $C^1(I)$. Every C^1 limit F is a C^1 diffeomorphism with

$$D_1^{-1} \leq |F'| \leq D_1. \quad (25.10)$$

Proof. Equation (25.8) gives uniform upper and lower derivative bounds after normalization. The Hölder-distortion calculation applied to derivatives of f_u^* gives uniform ε -Hölder control, hence equicontinuity of the derivatives. Arzelà–Ascoli yields C^1 precompactness. $\square$

For a probability ρ and a positive-mass interval J , write

$$\rho^J = \frac{(T_J)_*(\rho|_J)}{\rho(J)}. \quad (25.11)$$

Lemma IV.10.68 (*quasi-product stopping-mixture equivalence*).

Let $\mathcal{W}$ be a finite or countable prefix-free stopping antichain that meets almost every symbolic path. Define

$$M_{\mathcal{W}} = \sum_{u \in \mathcal{W}} \eta([u])(f_u)_* \mu. \quad (25.12)$$

Then, as finite measures on the geometric line,

$$C^{-1}M_{\mathcal{W}} \leq \mu \leq CM_{\mathcal{W}}. \quad (25.13)$$

For a Bernoulli symbolic law equality holds. This assertion does *not* identify $\mu|_{I(u)}$ with one individual cylinder component when cylinder images overlap. Under the present regular-IFS separation hypothesis, however, it also gives

$$\mu^{I(u)} \sim_{C_1} (f_u^*)_* \mu \quad (u \in \Lambda^*), \quad (25.13a)$$

with C_1 depending only on C ; endpoint ambiguities are μ -null by non-atomicity.

Proof. Quasi-Bernoulli gives, for every word u and tail cylinder $[v]$,

$$C^{-1}\eta([u])\eta([v]) \leq \eta([uv]) \leq C\eta([u])\eta([v]).$$

The cylinder sets generate the tail Borel sigma-algebra, so the same measure inequalities hold for every tail Borel set by the monotone-class theorem. Push each conditional tail law through $f_u \circ \Pi$, sum over the disjoint symbolic stopping cylinders $[u]$, and obtain (25.13). No geometric separation is used. For (25.13a), apply the same conditional-tail inequality to the single symbolic cylinder $[u]$. The regular geometric cylinders at level $|u|$ are disjoint except at endpoints, so $\mu|_{I(u)}$ equals the projection of the symbolic restriction to $[u]$ modulo the null endpoints. Normalize and apply $T_{I(u)}$. $\square$

Lemma IV.10.69 (*limit-geometry equivalence*).

Let $J_n = [x - r_n, x + r_n]$ be centered intervals shrinking to a μ -typical point x and suppose

$$\mu^{J_n} \longrightarrow \nu$$

weakly, with ν atomless. Then, after a subsequence, there are an affine image F of a limit diffeomorphism from Lemma IV.10.67 and an interval J with $|J| \geq c > 0$ such that

$$(T_J \circ F)_* \mu \sim_C \nu^J. \quad (25.14)$$

Conversely there is a C^1 diffeomorphism G , inverse to F on J and extended monotonically beyond it, and a positive interval J' with

$$(G_* \nu)^{J'} \sim_C \mu^{J'}. \quad (25.15)$$

Proof. At scale r_n take the stopping antichain of first cylinders whose diameters are at most r_n . Their diameters lie between cr_n and r_n for a fixed $c > 0$, by the positive lower derivative bound and bounded distortion. The cylinder intervals are disjoint in their interiors. Thus at most $N = O(c^{-1})$ of them meet J_n ; all others contribute zero to $\mu|_{J_n}$. By Lemma IV.10.68, $\mu|_{J_n}$ is boundedly equivalent to the sum of these at most N cylinder components restricted to J_n .

Normalize J_n to $[0, 1]$. The maps $T_{J_n} \circ f_u$ belonging to the retained components have uniformly bounded translations, derivatives bounded above and away from zero, and uniform Hölder derivative control. After a subsequence, the finite list of maps, coefficients, and image intervals converges; Arzelà–Ascoli gives C^1 limit maps. The normalized measure ν is therefore boundedly equivalent to a sum of at most N positive restrictions of $(F_j)_* \sigma_j$, where each limiting tail law σ_j is boundedly equivalent to μ by the quasi-product inequality. At least one component has a positive coefficient. The limiting component intervals have disjoint interiors, and ν has no mass at their finitely many endpoints. Choose an open subinterval U strictly inside one positive component interval with $\nu(U) > 0$. Because $\sigma_j \sim \mu$ and the symbolic cylinder intervals form a basis on the attractor, there is a fixed word v such that $F_j(f_v(I)) \subseteq U$ and $\sigma_j(f_v(I)) > 0$. Separation and quasi-product now give

$$\sigma_j|_{f_v(I)} \sim_C (f_v)_* \mu$$

up to normalization. Set $F = F_j \circ f_v$ and $J = F(I)$. On J all other limiting components vanish, and within the selected component the cylinder v is isolated modulo null endpoints. Hence the unscaled normalized restriction $\nu(J)^{-1} \nu|_J$ is boundedly equivalent to $F_* \mu$. Applying T_J to both sides gives (25.14); $|J| > 0$ because v is fixed and F'_j is bounded away from zero.

Extend $F^{-1} : J \rightarrow I$ to a monotone C^1 local diffeomorphism G on the ambient interval. Monotonicity makes $G^{-1}(I) = J$ up to endpoints. Applying G_* to the unscaled equivalence just proved, then normalizing the restriction to I and applying T_I to both sides, therefore gives $(G_* \nu)^I \sim_{C_1} \mu^I$, the claimed (25.15) with $J' = I$. The measure comparisons through weak limits are obtained by testing on compact subintervals away from the finite limiting boundaries; atomlessness removes those boundaries. $\square$

4. Uniform scaling of a quasi-product measure

Theorem IV.10.70 (*regular quasi-product measures generate one EFD*).

A non-atomic quasi-product measure μ for a regular IFS is a USM: there is a nontrivial EFD P such that μ generates P at μ -a.e. point. Moreover every P -typical measure ν is, locally on a positive interval, boundedly equivalent to a C^1 limit-diffeomorphic image of μ , and conversely μ is locally boundedly equivalent to a C^1 image of ν .

Proof. The one-dimensional case of Theorem IV.4.12 applies: regular maps are conformal $C^{1+\varepsilon}$ contractions and Definition IV.10.64 is the quasi-Bernoulli symbolic hypothesis. At the endpoints of I , extend each map using its first-order Taylor formula with a $C^{1+\varepsilon}$ remainder on a small common neighborhood; the strict bounds $a \leq |f'_i| \leq q < 1$ persist after shrinking that neighborhood. It supplies the common EFD P and uniform scaling. The distribution is nontrivial. Indeed every child conditional symbol probability is bounded below by $C^{-1} \min_i \eta[i] > 0$, so the maximal conditional child probability is bounded by $q = 1 - (|\Lambda| - 1)C^{-1} \min_i \eta[i] < 1$. Hence $\eta([u]) \leq q^{|u|}$. Let $a > 0$ be a uniform lower one-step contraction factor on the invariant neighborhood and use bounded distortion to obtain $\text{diam } f_u(K) \geq ca^{|u|}$. At the first stopping level with diameter at most r , every word therefore has length at least $\log(c/r)/\log(1/a)$, up to a bounded additive error, and weight at most $C_0 r^s$ for $s = -\log q/\log(1/a) > 0$. Separation and bounded distortion imply that a ball of radius r meets at most N stopping intervals of lengths in $[c_1 r, r]$. Thus $\mu(B(x, r)) \leq NC_0 r^s$ for all x, r . A trivial EFD would have dimension zero, contrary to this positive Frostman exponent.

For P -almost every state ν , atomlessness follows from Lemma IV.10.7. Since P is the scenery law generated by μ , such ν is a limit of centered sceneries at a typical point. Lemma IV.10.69 gives the stated two-way local bounded equivalence. The no-separation uniform-scaling input is proved in Chapter IV.4 as Theorem IV.4.12. $\square$

5. Spectral arithmetic from a fixed generator

Let $\alpha \neq 0$ belong to $\Sigma(P, S)$ and let ϕ be a unit eigenfunction.

Lemma IV.10.71 (*generator spectral arithmetic*).

For every orientation-preserving generator $f \in \Phi$,

$$\alpha \log \lambda(f) \in \mathbb{Z}. \quad (25.17b)$$

For an orientation-reversing generator the always-valid conclusion is

$$2\alpha \log \lambda(f) \in \mathbb{Z}. \quad (25.17a)$$

Proof. For an orientation-reversing f , apply the orientation-preserving argument below to the cylinder map f^2 , whose fixed point is the same and whose asymptotic contraction ratio is $\lambda(f)^2$; the quasi-product cylinder relation holds for f^2 . This gives (25.17a). Now fix an orientation-preserving f and its fixed point p . Since μ generates P by Theorem IV.10.70, choose a P -typical full-space state $\hat{\xi}$ to which the spatial eigenphase lemma IV.10.14 applies,

and whose unit restriction $\xi = \hat{\xi}|_{\square}$ satisfies the interior tangent-mixture representation IV.4.8. Thus

$$\xi \sim_{C_*} \rho, \quad \rho = \frac{(\int h_* \mu d\zeta(h))|_{B(0,1)}}{\int h_* \mu(B(0,1)) d\zeta(h)}, \quad (25.18)$$

and $\zeta\{h : h(V) \subset B(0,1)\} > 0$. Also require the ordinary $S_{L_\alpha}^*$ -component generation event, where $L_\alpha = 1/|\alpha|$, and the common non-atomic/full-orbit domain events. Birkhoff and quasi-Palm, as in the last paragraph of Lemma IV.10.14, supply these on a P -conull set; generation of some ordinary ergodic component does not require its entire label to be ϕ_α . Lemma IV.10.14 says that the α -phase of the discrete scenery component of ξ at ξ -almost every spatial point is the single value $c = \phi_\alpha(\hat{\xi})$. At interior spatial points the small sceneries of $\hat{\xi}$ and ξ coincide. Bounded equivalence and Radon differentiation give the same phase for ρ at ρ -almost every point, with the same qualified L_α -component generation. Let E be the fixed measurable set of points with these properties. Then $\rho(E^c) = 0$, and the mixture identity gives

$$\int (h_* \mu)(E^c \cap B(0,1)) d\zeta(h) = 0.$$

Thus for ζ -almost every h , its restricted image gives full mass to E . Intersect this conull parameter set with the positive set of interior maps and choose one such h . The phase of $\sigma = h_* \mu$ is c at σ -almost every point, and its components satisfy Lemma IV.10.17. We used Fubini on this one common conull event, not the false assertion that individual constituents of an uncountable mixture are absolutely continuous with respect to their barycenter. Here interiority makes the restriction of $h_* \mu$ to $B(0,1)$ equal to the whole measure.

Conjugate the generator by h and put $g = h \circ f \circ h^{-1}$ on a neighborhood of $h(K)$. The quasi-product inequality for the cylinder of f implies

$$g_* \sigma \ll \sigma. \quad (25.19)$$

Thus $g_* \sigma$ also has the constant phase c at almost every point. On the other hand, the C^1 phase rule IV.10.17 gives, at σ -almost every y ,

$$c = \text{phase}_{\alpha, g_* \sigma}(g(y)) = e(-\alpha \log g'(y)) \text{phase}_{\alpha, \sigma}(y) = e(-\alpha \log g'(y))c. \quad (25.20)$$

Since $c \neq 0$, continuity of g' and the fact that the fixed point $h(p)$ belongs to $\text{supp } \sigma$ extend $e(-\alpha \log g'(y)) = 1$ to $y = h(p)$. Indeed the cylinder $[f^n]$ has positive symbolic mass for every n by quasi-Bernoulli, and its image shrinks to p ; hence $p \in \text{supp } \mu$. The chain rule at the fixed point yields

$$g'(h(p)) = \frac{h'(f(p))f'(p)}{h'(p)} = f'(p) = \lambda(f). \quad (25.21)$$

Consequently $\alpha \log \lambda(f) \in \mathbb{Z}$, as claimed. The one-sided absolute continuity in (25.19) suffices; no equality between a whole overlapping measure and one cylinder image was assumed. $\square$

Corollary IV.10.72 (*spectral exclusion from contraction arithmetic*).

Let $\beta > 1$. If some generator f satisfies

$$\frac{\log \lambda(f)}{\log \beta} \notin \mathbb{Q}, \quad (25.22)$$

then

$$\Sigma(P, S) \cap \left(\frac{1}{\log \beta} \mathbb{Z} \setminus \{0\} \right) = \emptyset. \quad (25.23)$$

Proof. If $\alpha = k / \log \beta$, $k \neq 0$, were in the spectrum, Lemma IV.10.71 would give

$$2k \frac{\log \lambda(f)}{\log \beta} \in \mathbb{Z},$$

contradicting (25.22). □

Corollary IV.10.73 (*weak mixing from two contractions*).

If there are generators f, g with

$$\frac{\log \lambda(f)}{\log \lambda(g)} \notin \mathbb{Q}, \quad (25.24)$$

then P has no nonzero eigenvalue and is weak mixing.

Proof. A nonzero α in the spectrum would make both $2\alpha \log \lambda(f)$ and $2\alpha \log \lambda(g)$ integers by Lemma IV.10.71, forcing their ratio to be rational. □

6. The regular-IFS normality theorem

Theorem IV.10.74 (*regular-IFS Pisot normality*).

Let Φ be a regular $C^{1+\varepsilon}$ IFS on an interval and let μ be a non-atomic quasi-product measure for Φ . Let $\beta > 1$ be Pisot. Assume that for at least one generator f ,

$$\frac{\log \lambda(f)}{\log \beta} \notin \mathbb{Q}. \quad (25.25)$$

Then μ is pointwise β -normal, and $g_*\mu$ is pointwise β -normal for every C^1 diffeomorphism g of $\mathbb{R}$.

Proof. By Theorem IV.10.70, μ generates a nontrivial EFD P . By Corollary IV.10.72, P satisfies the β spectral exclusion. Apply Theorem IV.10.63, including its C^1 robustness. □

7. Examples and the arithmetic obstruction

Example IV.10.75 (*two incommensurable affine contractions*).

Let

$$f_0(x) = x/3, \quad f_1(x) = x/4 + 3/4. \quad (25.26)$$

The first-level intervals $[0, 1/3]$ and $[3/4, 1]$ are disjoint, so the IFS is regular. Its generator ratios are $1/3$ and $1/4$.

Let $\beta > 1$ be Pisot. They cannot both be commensurable with β : otherwise $\log(1/3)/\log(1/4)$ would be rational. Hence at least one generator satisfies (25.25). Therefore every non-atomic quasi-product measure for this IFS, and every C^1 diffeomorphic image, is pointwise β -normal for every Pisot β .

Example IV.10.76 (*middle-third obstruction*).

For

$$f_0(x) = x/3, \quad f_1(x) = x/3 + 2/3, \quad (25.27)$$

every word contraction ratio is 3^{-n} . The arithmetic hypothesis fails for $\beta = 3$.

For the Bernoulli Cantor measure, T_3 simply shifts ternary digits and every digit is 0 or 2. The middle-third interval $[1/3, 2/3]$ is never visited by a typical orbit, so the measure is not pointwise 3-normal. The failure is exact resonance, not a weakness of the proof.

Theorem IV.10.77 (*arithmetic examples*).

Examples IV.10.75–IV.10.76 respectively demonstrate a fully applicable non-homogeneous regular IFS with two independent contraction ratios, and an explicit resonant obstruction where the arithmetic hypothesis fails and normality actually fails.

No conclusion about overlapping IFSs is made: regularity includes separation.

Proof. The two assertions are exactly the conclusions established in Examples IV.10.75 and IV.10.76. □

10.6 Phase-spread refinements and analytic images

We now refine the Hochman–Shmerkin normality mechanism by keeping track of the distribution of spectral phases. This will settle the nonresonant cases directly and prepare the graph-relative L^q argument for the remaining case. Fix a Pisot number $\beta > 1$ and put

$$L = \log \beta.$$

Let P be a nontrivial ergodic fractal distribution which is not S_L -ergodic.

1. Primitive lattice phase and discrete components

Lemma IV.10.78 (*primitive lattice subgroup*).

Define

$$K_\beta(P) = \{k \in \mathbb{Z} : k/L \in \Sigma(P, S)\}.$$

Then $K_\beta(P)$ is a nonzero subgroup of $\mathbb{Z}$. Hence

$$K_\beta(P) = k_0\mathbb{Z}$$

for a unique $k_0 \geq 1$. Put

$$\alpha_0 = \frac{k_0}{L}.$$

There is a unit-circle eigenfunction ϕ with eigenvalue α_0 .

Proof. The spectrum is an additive subgroup: products and conjugates of unit-modulus eigenfunctions add and subtract eigenvalues. Intersecting with $L^{-1}\mathbb{Z}$ therefore gives a subgroup of $L^{-1}\mathbb{Z}$, equivalently a subgroup of $\mathbb{Z}$. Because P is not S_L -ergodic, the time- L map has a nonconstant invariant function, and the spectral decomposition from Section 10.1 supplies a nonzero eigenvalue in $L^{-1}\mathbb{Z}$. Thus the subgroup is nonzero and has the displayed form. $\square$

Lemma IV.10.79 (*one-dimensional eigenspaces*).

For an S -ergodic flow, every eigenvalue eigenspace is one-dimensional.

Proof. If ϕ and ψ are nonzero eigenfunctions with the same eigenvalue, then ϕ/ψ is S -invariant almost everywhere. Ergodicity makes this ratio constant. $\square$

Lemma IV.10.80 (*S_L -invariant sigma-algebra from primitive phase*).

The S_L -invariant sigma-algebra is generated, modulo P , by the primitive eigenfunction ϕ from Lemma IV.10.78.

Proof. Let $F \in L^2(P)$ satisfy $F \circ S_L = F$. For $m \in \mathbb{Z}$ define

$$F_m(\nu) = \frac{1}{L} \int_0^L e(mt/L) F(S_t \nu) dt.$$

For P -almost every ν , the function $t \mapsto F(S_t \nu)$ is L -periodic in the sense relevant to F , and its Fourier coefficients are F_m . A nonzero F_m is a flow eigenfunction with eigenvalue m/L . By Lemma IV.10.78, this can occur only when $m = qk_0$. By Lemma IV.10.79, then $F_m = c_m \phi^q$. Fourier partial sums are therefore measurable with respect to $\sigma(\phi)$ and converge to F in $L^2(P)$. Hence F is $\sigma(\phi)$ -measurable. $\square$

Let

$$P_z = P(\cdot \mid \phi = z)$$

be the S_L -ergodic component at phase z , for ϕ_*P -almost every z . Since P is flow-ergodic and ϕ is nonconstant, ϕ_*P is Haar measure on the phase circle.

Lemma IV.10.81 (*component rotation law*).

There is a Borel version $(P_z)_{z \in \mathbb{S}^1}$ of the ergodic disintegration which is equivariant at every phase. For this version, fix any phase z_0 . If $z = e(-\alpha_0 t)z_0$, then

$$P_z = (S_t^*)_* P_{z_0}.$$

The time parameter is well defined modulo

$$T_0 = \alpha_0^{-1} = L/k_0.$$

Proof. Start with any Borel conditional version $Q_z = P(\cdot \mid \phi = z)$. For each fixed s , invariance and uniqueness of disintegration give

$$(S_s^*)_* Q_z = Q_{e(-\alpha_0 s)z} \quad \text{for Haar-almost every } z. \quad (26.4a)$$

This alone does not justify a prescribed exceptional phase. Put $T_0 = 1/\alpha_0 = L/k_0 > 0$. Since $e(-\alpha_0 T_0) = 1$, (26.4a) gives $(S_{T_0}^*)_* Q_z = Q_z$ for Haar-almost every z . Define, for all z ,

$$P_z = \frac{1}{T_0} \int_0^{T_0} (S_t^*)_* Q_{e(\alpha_0 t)z} dt. \quad (26.4b)$$

The joint Borel flow and kernel make this a Borel probability kernel. For each fixed t , (26.4a) makes its integrand equal to Q_z for Haar-almost every z . Fubini and a countable determining family of Borel tests give $P_z = Q_z$ for Haar-almost every z . Thus it is a version of the same disintegration.

For a fixed z , the phase $e(\alpha_0 u)z$ runs once through Haar measure as u runs through any interval of length T_0 . Consequently

$$R_z(u) = (S_u^*)_* Q_{e(\alpha_0 u)z}$$

is T_0 -periodic for Lebesgue-almost every u , by the time- T_0 invariance of Q above. A change of variable $u = t - s$ in (26.4b) therefore gives, for every z and every s ,

$$\begin{aligned} P_{e(-\alpha_0 s)z} &= \frac{1}{T_0} \int_{-s}^{T_0-s} (S_s^*)_* R_z(u) du \\ &= (S_s^*)_* P_z. \end{aligned} \quad (26.4c)$$

Use the common full-orbit eigenfunction representative from Lemma IV.10.13. For every z , almost every time in (26.4b) samples a conditional measure supported on that conull orbit domain and on its required phase. Scaling therefore puts P_z on $\{\phi = z\}$, for every z . At Haar-almost every phase, $P_z = Q_z$ is S_L^* -ergodic by Lemma IV.10.80. Choose one such phase. Equation (26.4c) transports its ergodicity to every other phase, since the invertible flow commutes with S_L^* . The same equation with $s = T_0$ makes every P_z invariant under $S_{T_0}^*$; hence the displayed time is well defined modulo T_0 . This constructs the claimed everywhere-equivariant version rather than extending an almost-everywhere identity to a singular phase law. $\square$

Definition IV.10.82 (*component-time coordinate*).

Throughout this section, P_z means the everywhere-equivariant version constructed in Lemma IV.10.81, not arbitrary conditional-law choices on a Haar-null phase set. Choose the Borel coordinate

$$\ell : \mathbb{S}^1 \longrightarrow [0, T_0)$$

such that

$$e(-\alpha_0 \ell(z))z_0 = z.$$

On finitely many angular charts, ℓ is bi-Lipschitz.

Suppose now that a probability μ generates P and, at μ -almost every x , L -generates one S_L -ergodic component P_x . If $\phi_\mu(x)$ is the spatial phase, then

$$P_x = P_{\phi_\mu(x)} = (S_{\ell(\phi_\mu(x))}^*)_* P_{z_0}.$$

2. Projection with a full-dimensional parameter law

Lemma IV.10.83 (*Frostman parameter singular-integral estimate*).

Let ϑ be a finite measure on a compact interval J satisfying

$$\vartheta(B(t, r)) \leq Cr^\kappa$$

for some $0 < \kappa \leq 1$. If $0 < q < \kappa$, then uniformly for $(u, v) \neq (0, 0)$,

$$\int_J |u + e^t v|^{-q} d\vartheta(t) \leq C' |(u, v)|^{-q}.$$

Proof. By homogeneity normalize $|(u, v)| = 1$ and put $F(t) = u + e^t v$. The function F has at most one zero because e^t is monotone. If it has a zero t_0 in a fixed enlargement of J , compactness and the normalization imply a uniform lower bound on $|F'(t_0)| = |e^{t_0} v|$; hence near t_0 , $|F(t)| \geq c|t - t_0|$. Away from such a zero, $|F|$ is uniformly bounded below. Consequently

$$\vartheta\{|F| < r\} \leq C_1 r^\kappa.$$

Layer-cake integration yields

$$\int |F|^{-q} d\vartheta \leq C_2 \left(1 + \int_0^1 r^{\kappa-q-1} dr \right) < \infty.$$

Rescaling gives the stated bound. □

Lemma IV.10.84 (*parameter-averaged projection energy*).

Let ρ be a finite measure on $\mathbb{R}^2$, and let ϑ satisfy the hypotheses of Lemma IV.10.83. For $0 < q < \kappa$ and

$$\pi_t(x, y) = x + e^t y,$$

one has

$$\int I_q((\pi_t)_*\rho) d\vartheta(t) \leq CI_q(\rho).$$

Proof. Expand the projected q -energy as the integral over pairs of points of

$$|(x - x') + e^t(y - y')|^{-q}.$$

Apply Lemma IV.10.83 to each difference vector and then Tonelli. □

Theorem IV.10.85 (*full-dimension parameter projection*).

Let τ, η be probabilities on $\mathbb{R}$, and let ϑ be a parameter probability with $\dim \vartheta = 1$. If

$$\dim \tau + \dim \eta \geq 1,$$

then for ϑ -almost every t ,

$$\dim(\tau * S_t \eta) = 1.$$

Proof. Write $\dim$ for lower Hausdorff dimension of a measure, as in Section 10.3. If either input has dimension zero, the other has dimension one. The translation/Fubini argument in Theorem IV.10.47 gives $\dim(\tau * S_t \eta) \geq \max\{\dim \tau, \dim \eta\} = 1$ for every t ; the ambient line gives the reverse bound. Assume both dimensions are positive. Fix rational $0 < q < 1$. Choose positive numbers

$$a < \dim \tau, \quad d < \dim \eta, \quad \kappa < 1$$

with $q < a + d$ and $q < \kappa$. Use the countable Frostman-piece decomposition from Section 10.3 on τ, η , and ϑ , subdividing each piece into bounded intervals. This last subdivision is needed also for an unbounded parameter law, since Lemmas IV.10.83–IV.10.84 use a compact parameter interval. Products of bounded a -Frostman and d -Frostman pieces have finite q -energy: their ball masses are $O(r^{a+d})$, and layer-cake integration converges because $q < a + d$. Apply Lemma IV.10.84 to every triple of positive pieces. For ϑ -almost every t , every positive projected product piece has finite q -energy. Their countable sum is the convolution and gives zero mass to every set of Hausdorff dimension less than q , so its lower dimension is at least q . Intersect over rational $q < 1$ and let $q \uparrow 1$. □

3. Phase-spread convolution saturation

Put $\delta = \dim P > 0$ and let

$$\theta_\mu = (\phi_\mu)_*\mu$$

be the phase measure. Since ℓ is piecewise bi-Lipschitz,

$$\dim \theta_\mu = 1 \implies \dim(\ell_* \theta_\mu) = 1.$$

Lemma IV.10.86 (*phase-spread saturation*).

Assume $\dim \theta_\mu = 1$. If τ is a probability satisfying

$$\dim \tau \geq 1 - \delta,$$

then for μ -almost every x and P_x -almost every η ,

$$\dim \left(\tau * \frac{\eta|_A}{\eta(A)} \right) = 1 \quad \text{for every bounded Borel } A \text{ with } \eta(A) > 0.$$

In particular this applies to $\eta^\square$ and every positive local restriction used in the orbit representation. The full-space state η need not have finite total mass.

Proof. Fix $P_0 = P_{z_0}$. By Lemma IV.10.81,

$$P_x = (S_{\ell(\phi_\mu(x))}^*)^* P_0.$$

Put $\lambda = \ell_* \theta_\mu$. For P_0 -almost every full-space state ξ , every positive bounded restriction has dimension δ . Indeed the exact-dimensional, origin-supported state property is a full-flow-invariant P -conull property. It holds on almost every ordinary component and then on every canonical component by the transport in Lemma IV.10.81. For $k \geq 1$ let

$$\xi_k = \frac{\xi|_{[-k,k]}}{\xi([-k,k])}.$$

These are probabilities; the denominators are positive and finite. Apply Theorem IV.10.85 to (τ, ξ_k) and λ , then intersect over k and use Fubini. The dimension events are measurable in the completed product space by the countable ball/content criterion used in Theorem IV.10.48. Thus for $P_0 \times \lambda$ -almost every (ξ, t) ,

$$\dim(\tau * S_t \xi_k) = 1 \quad \text{for every } k. \quad (26.4d)$$

Here S_t is the unnormalized spatial dilation of a finite probability. Set $\eta = S_t^* \xi$. If A is bounded and $\eta(A) > 0$, its inverse dilation lies in $[-k, k]$ for some k . Consequently

$$\frac{\eta|_A}{\eta(A)} \ll S_t \xi_k, \quad \tau * \frac{\eta|_A}{\eta(A)} \ll \tau * S_t \xi_k.$$

A nonzero finite measure absolutely continuous with respect to a measure of lower Hausdorff dimension one also has lower dimension one. This proves the conclusion simultaneously for every such A , without convolving an unbounded Radon state. Push $P_0 \times \lambda$ through $(\xi, t) \mapsto S_t^* \xi$ and use the exact canonical rotation law to obtain the assertion for μ -almost every x and P_x -almost every state. Translations of the finite restrictions do not affect dimension. $\square$

Theorem IV.10.87 (*individual phase-spread normality; HS-N Theorem 8.1*).

Let $\beta > 1$ be Pisot and $L = \log \beta$. Let μ generate a nontrivial EFD P , and suppose P is not S_L -ergodic. Assume that for μ -almost every x , the measure μ L -generates an S_L -ergodic component P_x , and that

$$\dim \theta_\mu = 1.$$

Then μ is pointwise β -normal.

Proof. First intersect the generation and orbit-representation conull sets with the saturation sets of Lemma IV.10.86 for every member τ_N of the countable resonant test family satisfying $\dim \tau_N \geq 1 - \delta$. This includes all sufficiently large N and is still a conull set, independent of the orbit subsequence. Fix x in this set and a subsequential T_β -orbit law ν . The orbit/local-measure representation from Section 10.2 writes ν as an average of positive restrictions of translates of measures distributed according to P_x . Every such component has dimension δ , so ν is non-atomic and $\dim \nu \geq \delta$. Orbit telescoping makes ν T_β -invariant.

Choose the resonant measure τ_N from Section 10.3 with $\dim \tau_N \geq 1 - \delta$. By Lemma IV.10.86, convolution with the relevant local components has dimension one; the same remains true after the positive restrictions and translations occurring in the orbit representation. Hence

$$\dim(\tau_N * \nu) = 1.$$

If $0 < \dim \nu < 1$, the dimension-growth theorem in Section 10.3 forces ν to resonate with τ_N for large N , contradicting the preceding equality. Thus $\dim \nu = 1$, and the dimension-entropy result of Section 10.1 identifies the invariant orbit law with the Parry measure. Every subsequential orbit law is therefore the Parry measure, so the full orbit equidistributes.

□

4. Cumulative phase for a measurable family

Let $\{\mu_\omega\}_{\omega \in \Omega}$ be a measurable family over (Ω, Q) . Assume Q -almost every μ_ω generates the same nontrivial EFD P and L -generates an S_L -ergodic component at almost every spatial point. Define the cumulative phase measure

$$\Theta = \int \theta_{\mu_\omega} dQ(\omega).$$

Lemma IV.10.88 (*family phase-spread saturation*).

If $\dim \Theta = 1$, then for every probability τ with $\dim \tau \geq 1 - \delta$, for Q -almost every ω , μ_ω -almost every x , and $P_{\omega,x}$ -almost every η ,

$$\dim \left(\tau * \frac{\eta|_A}{\eta(A)} \right) = 1 \quad \text{for every bounded Borel } A \text{ with } \eta(A) > 0.$$

Proof. On the joint space (ω, x) the phase variable has law Θ . Apply the proof of Lemma IV.10.86 with the parameter law $\ell_* \Theta$: normalize every bounded piece of a full-space base state, apply Theorem IV.10.85, and transfer the conclusion to all positive finite

restrictions. The product kernel so obtained pushes forward to $(Q(d\omega)\mu_\omega(dx))P_{\omega,x}(d\eta)$ by the everywhere component-time relation in Lemma IV.10.81. Fubini proves the claim. No individual phase law θ_{μ_ω} is assumed to have full dimension. $\square$

Theorem IV.10.89 (*family phase-spread criterion; HS–N Theorem 8.2*).

Under the family hypotheses above, if

$$\dim \Theta = 1,$$

then Q -almost every μ_ω is pointwise β -normal. Consequently the mixture

$$\mu = \int \mu_\omega dQ(\omega)$$

is pointwise β -normal.

Proof. Intersect, in the joint law $Q(d\omega)\mu_\omega(dx)$, the conull saturation events of Lemma IV.10.88 for the same countable test family before choosing any orbit subsequence. For almost every (ω, x) in that common set, repeat the proof of Theorem IV.10.87 with Lemma IV.10.88 in place of Lemma IV.10.86 and with the component $P_{\omega,x}$. Fubini gives the assertion for the mixture. $\square$

5. Analytic images and the remaining resonant case

For positive numbers $a, b \neq 1$, write $a \sim b$ when $\log a / \log b \in \mathbb{Q}$; otherwise they are *multiplicatively independent*. This convention permits one of a, b to be a contraction ratio below one.

Let

$$\Phi = \{g_i(x) = r_i x + a_i\}_{i=1}^m$$

be a regular affine IFS on an interval, and let μ be a non-atomic self-similar measure with positive weights. Let f be a real-analytic diffeomorphism on one connected open interval containing $\text{supp } \mu$, and assume that f is not affine on $\text{supp } \mu$.

Lemma IV.10.90 (*easy analytic-image arithmetic cases*).

The analytic-image normality theorem holds if either

1. two generator ratios $|r_i|, |r_j|$ are multiplicatively independent; or
2. all ratios are powers of one common γ but γ is multiplicatively independent of the Pisot base β .

Proof. For the affine Bernoulli measure, Lemma IV.10.121 below gives the precise spectral arithmetic: at a nonzero eigenvalue α , every $2\alpha \log |r_i|$ is an integer. In the first case two such logarithms cannot have irrational ratio, so P is weak mixing. In the second case a spectral value $\alpha = k / \log \beta$ would make $2kp_i \log \gamma / \log \beta$ an integer, contrary to the assumed irrational ratio. Thus in both cases the spectral exclusion (24.2) holds; apply Theorem IV.10.63 to $f_*\mu$. $\square$

Thus the only remaining arithmetic case is

$$|r_i| = \gamma^{p_i}, \quad p_i \in \mathbb{N}, \quad \beta \sim \gamma.$$

Lemma IV.10.91 (*spectrum in the common-ratio lattice*).

In the resonant case, μ generates an EFD P with

$$\Sigma(P, S) \subset \frac{1}{\log \gamma} \mathbb{Q}.$$

For every eigenvalue, the phase measure of a P -typical scenery is atomic.

Proof. Generation is Theorem IV.10.70. If α is spectral, Lemma IV.10.71 gives

$$2\alpha p_i \log \gamma \in \mathbb{Z}$$

for every i , so $\alpha \log \gamma \in \mathbb{Q}$. Atomic phase for P -typical measures is the phase-synchronization statement in Section 10.1. $\square$

Lemma IV.10.92 (*analytic localization*).

If f is real analytic and non-affine on a connected open interval containing $\text{supp } \mu$, then the zero set of f'' is finite on every compact subinterval. Up to finitely many points, $\text{supp } \mu$ is therefore covered by countably many compact intervals on each of which f' and $\log |f'|$ are bi-Lipschitz onto their images.

Proof. The function f'' is analytic. An interior accumulation point of its zero set forces $f'' \equiv 0$ by the identity theorem, hence f affine. Otherwise its zeros are discrete and therefore finite on compact intervals. Since f is a diffeomorphism, f' never vanishes. Between successive zeros of f'' , the derivative f' is strictly monotone; exhaust such intervals away from their endpoints to obtain the bi-Lipschitz pieces. $\square$

Lemma IV.10.93 (*positive-dimensional nonlinear phase*).

Let $\delta = \dim \mu > 0$ and let $\nu = f_* \mu$. For every nonzero spectral value α relevant to the resonant base, the phase measure of ν has dimension

$$\dim \theta_\nu = \delta > 0.$$

Proof. The needed discrete component generation and finite-valued affine phase are proved explicitly in Lemma IV.10.121 below. Its formula (F.73q) says that the phase of $f_* \mu$ is a fixed circle rotation of $x \mapsto e(-\alpha \log |f'(x)|)$. Each positive restriction of the exact-dimensional measure μ has dimension δ . Lemma IV.10.92 partitions away a null critical set into countably many pieces on which $\log |f'|$, followed by the circle map on a short arc, is bi-Lipschitz. Countable bi-Lipschitz invariance gives $\dim \theta_\nu = \delta$. $\square$

The preceding lemma gives positive phase dimension, whereas the general phase-spread criterion requires full phase dimension. The graph-relative L^q formula below has two distinct uses. Its fixed- q consequence gives only a pressure-dependent uniform Frostman exponent. Its limit as $q \downarrow 1$ gives exact dimension and, through the zero-dimensional exceptional parameter set and the local-component comparisons below, settles the resonant analytic-image case. The latter route does not require a uniform Frostman exponent equal to Shannon dimension.

10.7 Graph-relative L^q estimates

Throughout, $q > 1$. To avoid collision between the exponent q and transition probabilities, transition weights are denoted p_w . In this section $\log$ is natural logarithm; $\log_2$ is used explicitly for dyadic q -sum exponents. In particular a symbolic entropy H measured with $\log$ becomes a dyadic exponent only after division by $\log 2$.

1. Setup

Let V be a nonempty finite state set. Fix a positive integer block length m and put

$$\lambda = \gamma^m, \quad 0 < \gamma < 1.$$

For each state s , let $W(s, t)$ be the finite family of admissible m -block edges from s to t . Their probabilities p_w are strictly positive on the retained block graph and satisfy

$$\sum_{t \in V} \sum_{w \in W(s, t)} p_w = 1.$$

Each block map is

$$\phi_w(x) = \lambda x + b_w.$$

The graph-directed family $(\mu_s)_{s \in V}$ satisfies

$$\mu_s = \sum_t \sum_{w \in W(s, t)} p_w (\phi_w)_* \mu_t. \quad (\text{GQ.1})$$

Definition IV.10.94 (*block laws and, when specified, edge-level laws*).

Let Ω_s^B be the space of infinite admissible block paths beginning at s . Define $\mathcal{N}_s^B$ by the cylinder probabilities

$$\mathcal{N}_s^B([w_1 \cdots w_n]) = \prod_{i=1}^n p_{w_i}.$$

The row-sum condition makes these probabilities consistent and defines the path law. Its block coding map is

$$\Pi_B(w_1, w_2, \dots) = \sum_{i \geq 1} \lambda^{i-1} b_{w_i}.$$

The series converges uniformly, since there are finitely many block translations. Conditioning on the first block shows that $(\Pi_B)_* \mathcal{N}_s^B$ satisfies (GQ.1). These projected probabilities equal μ_s : iterating (GQ.1) n times leaves a tail of diameter $O(\lambda^n)$, so integration against every bounded Lipschitz test converges to the same coded law.

An *edge-level presentation* is additional specified data: a finite directed edge alphabet with maps $\psi_e(x) = \gamma x + a_e$, such that each block w is an admissible word $(e_1, \dots, e_m)$ with the same endpoints and $\phi_w = \psi_{e_1} \circ \cdots \circ \psi_{e_m}$. The common-ratio towers constructed in Lemma IV.10.114 provide such data. An abstract collection of block maps alone does

not specify its intermediate edge phases. Given this presentation, let Ω_E be the space of infinite admissible edge paths, let $c : \bigsqcup_s \Omega_s^B \rightarrow \Omega_E$ concatenate the block words, let $\sigma : \Omega_E \rightarrow \Omega_E$ be the left shift, and define

$$\mathcal{N}_s = c_* \mathcal{N}_s^B, \quad \Pi(e_1, e_2, \dots) = \sum_{i \geq 1} \gamma^{i-1} a_{e_i}.$$

Then $\Pi \circ c = \Pi_B$ and $\Pi_* \mathcal{N}_s = \mu_s$. Choose a stationary probability π of the block transition matrix $M_1(s, t) = \sum_{w \in W(s, t)} p_w$, allowing a mixture of closed irreducible classes, and put $\mathcal{N} = \sum_s \pi(s) \mathcal{N}_s$. Stationarity of π and the first-block conditional law give

$$(\sigma^m)_* \mathcal{N} = \sum_t (\pi M_1)(t) \mathcal{N}_t = \sum_t \pi(t) \mathcal{N}_t = \mathcal{N}.$$

Hence $m^{-1} \sum_{j=0}^{m-1} (\sigma^j)_* \mathcal{N}$ is edge-shift invariant. In the phase notation below, $\sigma^j \mathcal{N}$ means $(\sigma^j)_* \mathcal{N}$. These edge phases are coding laws; they are not, without a separate identification, the spectral circle phases θ_μ or Θ of Section 10.6.

For an n -block admissible path $U = (w_1, \dots, w_n)$ from s , let

$$p_U = \prod_j p_{w_j},$$

$$\phi_U(x) = \lambda^n x + b_U.$$

Let $t(U)$ be its terminal state, and define the atomic translation prefix

$$A_{s,n} = \sum_{\substack{U \text{ starts at } s \\ |U|=n}} p_U \delta_{b_U}. \quad (\text{GQ.2})$$

Assume the following graph-relative exponential separation condition: there is $c \in (0, 1)$ such that, for every state s and all sufficiently large n , all distinct admissible n -block paths U, W starting at s satisfy

$$|b_U - b_W| \geq c^n. \quad (\text{GQ.3})$$

All supports $K_t = \text{supp}(\mu_t)$ are compact. Since V is finite, put

$$B = \max_t \sup\{|x| : x \in K_t\} < \infty.$$

For a finite Borel measure ν , let

$$S_k(\nu, q) = \sum_{I \in \mathcal{D}_k} \nu(I)^q$$

for the level- k dyadic partition $\mathcal{D}_k$.

Lemma IV.10.95 (*vector natural-scale q -sum comparison*).

Let k_n be any integers satisfying

$$2^{-k_n-1} < \lambda^n \leq 2^{-k_n}.$$

There is $C_q \geq 1$, depending only on q, λ, B and the finite state family, such that for every state s and every n ,

$$C_q^{-1} S_{k_n}(A_{s,n}, q) \leq S_{k_n}(\mu_s, q) \leq C_q S_{k_n}(A_{s,n}, q). \quad (\text{GQ.4})$$

In particular, the comparison constants are uniform in s and n .

Proof.

Iterating (GQ.1) n times gives

$$\mu_s = \sum_U p_U (\phi_U)_* \mu_{t(U)}. \quad (\text{GQ.5})$$

The support of the U -component lies in

$$b_U + \lambda^n K_{t(U)} \subset [b_U - B\lambda^n, b_U + B\lambda^n]. \quad (\text{GQ.6})$$

Because $\lambda^n \asymp 2^{-k_n}$, the interval in (GQ.6) meets at most L level- k_n dyadic cells, where L is independent of U, s, n .

Upper bound in (GQ.4). Fix $I \in \mathcal{D}_{k_n}$. A U -component can meet I only when b_U lies in one of at most L_1 dyadic cells J in a fixed finite neighbourhood $N(I)$. Therefore

$$\mu_s(I) \leq \sum_{J \in N(I)} A_{s,n}(J).$$

Using $(a_1 + \dots + a_r)^q \leq r^{q-1} \sum_j a_j^q$,

$$\mu_s(I)^q \leq L_1^{q-1} \sum_{J \in N(I)} A_{s,n}(J)^q.$$

After summing over I , each J occurs in only $O(1)$ such neighbourhoods, so

$$S_{k_n}(\mu_s, q) \leq C_q S_{k_n}(A_{s,n}, q).$$

Lower bound in (GQ.4). Fix $J \in \mathcal{D}_{k_n}$. If $b_U \in J$, then the whole U -component in (GQ.5) is contained in a fixed enlargement J^* , which is the union of at most L_2 level- k_n cells. Hence

$$A_{s,n}(J) = \sum_{U: b_U \in J} p_U \leq \mu_s(J^*).$$

Thus

$$\begin{aligned}
 & A_{s,n}(J)^q \\
 & \leq \left(\sum_{I \subset J^*} \mu_s(I) \right)^q \\
 & \leq L_2^{q-1} \sum_{I \subset J^*} \mu_s(I)^q.
 \end{aligned}$$

Summing over J and using bounded overlap of the enlargements proves

$$S_{k_n}(A_{s,n}, q) \leq C_q S_{k_n}(\mu_s, q).$$

This proves (GQ.4). No separation hypothesis was used. □

Notice that the proof uses the translations b_U themselves, as required by the exponential-separation hypothesis (GQ.3).

Lemma IV.10.96 (*Perron–Frobenius q -pressure, uniform in state*).

Define the q -powered transition matrix

$$M_q(s, t) = \sum_{w \in W(s, t)} p_w^q. \quad (\text{GQ.7})$$

Let

$$Z_{s,n}(q) = \sum_{\substack{U \text{ starts at } s \\ |U|=n}} p_U^q. \quad (\text{GQ.8})$$

Assume the retained block graph is irreducible. Then M_q is irreducible and, writing $\rho_q = \rho(M_q)$, there exist constants $0 < a_q \leq b_q < \infty$ such that for every s and $n \geq 0$,

$$a_q \rho_q^n \leq Z_{s,n}(q) \leq b_q \rho_q^n. \quad (\text{GQ.9})$$

Consequently

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log Z_{s,n}(q) = -\log \rho_q. \quad (\text{GQ.10})$$

uniformly over the finite state set.

Proof.

Path expansion gives

$$Z_{s,n}(q) = (M_q^n \mathbf{1})_s. \quad (\text{GQ.11})$$

The support of M_q is exactly the support of the positive block transition kernel, hence irreducibility is inherited. By finite-dimensional Perron–Frobenius theory there is a strictly positive right eigenvector r with

$$M_q r = \rho_q r.$$

Since V is finite, there are $a, b > 0$ such that

$$ar \leq \mathbf{1} \leq br$$

componentwise. Positivity of M_q^n gives

$$a\rho_q^n r \leq M_q^n \mathbf{1} \leq b\rho_q^n r.$$

Taking the minimum and maximum coordinates of r absorbs r into two positive constants and yields (GQ.9). Taking logarithms yields (GQ.10).

□

Definition. Set

$$H_q^G = -\log \rho_q. \quad (\text{GQ.12})$$

The proved state-uniform pressure estimate is exactly

$$-\frac{1}{n} \log Z_{s,n}(q) = H_q^G + O_q(1/n), \quad (\text{GQ.13})$$

uniformly in s , for each fixed $q > 1$. There is no lower bound here by a Shannon entropy that is uniform for all q . The Shannon entropy of a specified stationary block process is recovered only in the limit $q \downarrow 1$, proved classwise in Theorem IV.10.113. For the synchronized towers the entropy per edge is the explicitly defined h_G in (F.72l).

Lemma IV.10.97 (*finite-state grouping cost is subexponential in the fixed- R prefix induction*).

Let ν_t ($t \in V$) be finite positive measures and let $\nu = \sum_t \nu_t$. Then at every dyadic level k ,

$$\sum_t S_k(\nu_t, q) \leq S_k(\nu, q) \leq |V|^{q-1} \sum_t S_k(\nu_t, q). \quad (\text{GQ.14})$$

Proof.

For each dyadic cell I , positivity gives

$$\begin{aligned} \sum_t \nu_t(I)^q &\leq \left(\sum_t \nu_t(I) \right)^q \\ &\leq |V|^{q-1} \sum_t \nu_t(I)^q. \end{aligned}$$

Summing over I proves (GQ.14).

□

If the fine-prefix induction is run for a fixed integer R , terminal-state grouping at each of its R stages costs at most $|V|^{(q-1)R}$. For fixed R this is an $O(1)$ factor as the block scale tends to infinity, so its logarithm divided by that scale is $o(1)$.

Lemma IV.10.98 (*PF-weighted symbolic state accounting*).

Let ℓ be a strictly positive left PF eigenvector,

$$\ell^\top M_q = \rho_q \ell^\top.$$

For every non-negative vector $x = (x_t)$ and every $n \geq 0$,

$$\sum_s \ell_s \sum_t (M_q^n)(s, t) x_t = \rho_q^n \sum_t \ell_t x_t. \quad (\text{GQ.15})$$

Equivalently, after summing q -th powers of path weights, state transitions carry no multiplicative loss in the PF-weighted norm.

Proof.

This is exactly

$$\ell^\top M_q^n x = \rho_q^n \ell^\top x.$$

□

Equation (GQ.15), together with (GQ.14), allows the localized q -energy to be organized in an ℓ -weighted state sum, with finite-state grouping used only a bounded number of times in the fixed- R recursion.

Lemma IV.10.99 (*L^q -dimension to Frostman conversion*).

Let ν be a compactly supported probability on $\mathbb{R}$. If $q > 1$ and $D_q(\nu) > d$, then there are $C, r_0 > 0$ such that

$$\nu(B(x, r)) \leq C r^{(1-1/q)d}, \quad (\text{GQ.16})$$

for all x and $0 < r < r_0$.

Proof.

Choose $d < d_1 < D_q(\nu)$. For all sufficiently large k ,

$$S_k(\nu, q) \leq 2^{-(q-1)d_1 k}. \quad (\text{GQ.17})$$

If $2^{-k-1} < r \leq 2^{-k}$, the ball $B(x, r)$ meets at most a fixed number C_0 of level- k dyadic cells. Hölder's inequality gives

$$\nu(B(x, r)) \leq C_0^{1-1/q} S_k(\nu, q)^{1/q} \leq C 2^{-k(1-1/q)d_1} \leq C' r^{(1-1/q)d}.$$

□

For a desired positive Frostman exponent s , one must first prove a fixed- q lower bound b for every $D_q(\mu_t)$ with $(1 - 1/q)b > s$, and then choose $d < b$ retaining this strict inequality. Lemma IV.10.99 gives the result uniformly over the finite state set. Increasing q without checking how its pressure changes is not a valid parameter choice. When edge-level data are specified above, $\Pi_* \mathcal{N}_t = \mu_t$; the finite phase transfer will be proved in Lemma IV.10.109.

Remark IV.10.100 (*the target Frostman exponent*).

The former Shannon-level uniform Frostman target would require a suitably large q and a correspondingly strong pressure bound. The first condition alone does not imply the second. The valid fixed- q theorem and the counterexample to that stronger target are separated below. The applications use the fixed- q estimate or the independent limiting argument, not the refuted endpoint. In particular, q near one is sufficient for the exact-dimensional conclusion of Theorem IV.10.113.

2. The graph-directed L^q estimate

For the finite irreducible common-contraction graph family above, assume the admissible exponential-separation condition (GQ.3). We shall prove

$$D_q(\mu_s) \geq \min \left\{ 1, \frac{H_q^G}{(q-1) \log(1/\lambda)} \right\}. \quad (\text{GQ.18})$$

uniformly in s . Since $\lambda = \gamma^m$, the denominator is $(q-1)m \log(1/\gamma)$.

The preceding lemmas give the natural-scale graph-prefix comparison, the exact Perron–Frobenius q -pressure, finite-state grouping, weighted state accounting, and the conversion from L^q dimension to a Frostman estimate. The argument below replaces the scalar common tail by the finite family of state tails. Equations (GQ.14)–(GQ.15) prevent a loss at every symbolic depth; factors $|V|^{O(R)}$ are harmless when the fine-prefix multiplier R is fixed.

10.8 The inverse theorem used in the flattening argument

The argument uses the following theorem from Volume II, Chapter II.11. It is stated in full because its quantitative clauses are used separately below. Its proof appears there under the title “Inverse theorem for near-nonflattening.”

Earlier-volume theorem: inverse theorem for near-nonflattening

For integers $D, \ell \geq 1$, with $m = D\ell$, call a 2^{-m} -set $A(D, \ell, (R_s))$ -uniform if every occupied level- sD dyadic interval has exactly R_s occupied level- $(s+1)D$ descendants. A 2^{-m} -measure is a probability measure supported on $2^{-m}\mathbb{Z} \cap [0, 1)$.

Fix $q \in (1, \infty)$, $\delta > 0$, and $D_0 \in \mathbb{N}$. There are $\varepsilon > 0$ and an integer $D \geq D_0$ such that the following holds for all sufficiently large ℓ . Put $m = D\ell$, and let μ, ν be 2^{-m} -measures satisfying

$$\|\mu * \nu\|_q \geq 2^{-\varepsilon m} \|\mu\|_q. \quad (26.1)$$

After translating μ and ν by elements of $2^{-m}\mathbb{Z}$, there are sets

$$A \subset \text{supp } \mu, \quad B \subset \text{supp } \nu,$$

and branching sequences $(R'_s)_{s=0}^{\ell-1}$ and $(R''_s)_{s=0}^{\ell-1}$ such that:

1.

$$\|\mu|_A\|_q \geq 2^{-\delta m} \|\mu\|_q,$$

and $\mu(y) \leq 2\mu(x)$ for every $x, y \in A$;

2. A is $(D, \ell, (R'_s))$ -uniform, and every $x \in A$ lies in the middle half of its level- sD dyadic parent for every s ;

3.

$$\nu(B) \geq 2^{-\delta m},$$

and $\nu(y) \leq 2\nu(x)$ for every $x, y \in B$;

4. B is $(D, \ell, (R''_s))$ -uniform, and every $y \in B$ lies in the middle half of its level- sD dyadic parent for every s ;

5. for every s , either

$$R''_s = 1 \quad \text{or} \quad R'_s \geq 2^{(1-\delta)D}; \quad (26.2)$$

6. if

$$S = \{s : R'_s \geq 2^{(1-\delta)D}\}, \quad q' = \frac{q}{q-1},$$

then

$$\log_2(\|\nu\|_q^{q'}) - \delta m \leq D|S| \leq \log_2(\|\mu\|_q^{q'}) + \delta m. \quad (26.3)$$

The same exponential statement is available for compactly supported real-line discretizations after a fixed affine normalization: passing between real-line and circle convolution, or shifting the dyadic level by $O(1)$, changes the relevant ℓ^q quantities only by fixed multiplicative factors and hence does not affect the exponential conclusions used below.

10.9 Graph-relative flattening and fine-prefix recursion

1. Setup and the block-class correction

Let V be the finite state set, and let

$$\phi_w(x) = \lambda x + b_w, \quad 0 < \lambda < 1,$$

be the retained m -block similarities, and let $p_w > 0$ be the block transition weights. The

projected state-started probabilities satisfy

$$\mu_s = \sum_t \sum_{w:s \rightarrow t} p_w(\phi_w)_* \mu_t. \quad (\text{F.1})$$

For an admissible n -block path $U = (w_1, \dots, w_n)$ starting at s , write

$$p_U = \prod_i p_{w_i},$$

$$\phi_U(x) = \lambda^n x + b_U,$$

$$t(U) = \text{terminal state},$$

and

$$A_{s,n} = \sum_{\substack{U:s \rightarrow * \\ |U|=n}} p_U \delta_{b_U}. \quad (\text{F.2})$$

Because the state set is finite and all supports are compact, one fixed affine normalization places every $\text{supp}(\mu_s)$ in one bounded interval (and, if desired, in $[0, 1]$). This changes dyadic levels only by $O(1)$ and never changes an exponential q -sum rate.

Let

$$a = \log_2(1/\lambda),$$

$$k_n = \lfloor an \rfloor.$$

Then $k_n = an + O(1)$.

The admissible graph exponential-separation condition (GQ.3) says that there is $c \in (0, 1)$ such that, for every state s and all sufficiently large n ,

$$|b_U - b_W| \geq c^n \quad (\text{F.3})$$

for distinct admissible n -block paths U, W beginning at s .

The support of the m -block transition kernel need not be irreducible. If the original one-step graph has period d , the m -step graph can split into finitely many cyclic classes. By construction every state is recurrent. Hence V is a disjoint union

$$V = C_1 \dot{\cup} \dots \dot{\cup} C_J. \quad (\text{F.4})$$

of closed irreducible classes for the block kernel.

For $r > 1$, define, on a class C ,

$$M_{r,C}(s, t) = \sum_{w:s \rightarrow t} p_{w, s, t}^r, \quad t \in C, \quad (\text{F.5})$$

and

$$\rho_{r,C} = \rho(M_{r,C}),$$

$$H_{r,C} = -\log \rho_{r,C}. \quad (\text{F.6})$$

For $s \in C$, put

$$Z_{s,n}(r) = \sum_{\substack{U: s \rightarrow * \\ |U|=n}} p_U^r.$$

Perron–Frobenius theory on the irreducible matrix $M_{r,C}$ gives constants $0 < c_{r,C} \leq C_{r,C} < \infty$ with

$$\begin{aligned} c_{r,C} \rho_{r,C}^n &\leq Z_{s,n}(r) \\ &\leq C_{r,C} \rho_{r,C}^n \end{aligned} \quad (\text{F.7})$$

for $s \in C$ and $n \geq 0$. Since there are finitely many classes, constants may be made uniform after r is fixed.

For each fixed $r > 1$, the exact classwise Rényi pressure is therefore

$$\frac{H_{r,C}}{r-1} = \lim_{n \rightarrow \infty} \frac{-\log Z_{s,n}(r)}{(r-1)n}, \quad s \in C. \quad (\text{F.8})$$

The convergence is uniform in the finite state set, by (F.7). These are class-dependent entropy rates at the specified exponent, not a common Shannon lower bound for every r . The classwise formulation is necessary because the whole block graph need not be irreducible. A common Shannon entropy, when available from the tower construction, describes the limit $r \downarrow 1$; it does not replace the fixed- r pressure in (F.8).

2. A common q -spectrum within each closed block class

Lemma IV.10.101 (*classwise common spectrum*).

Fix $r > 1$ and one closed irreducible block class C . There exists a number

$$\tau_C(r) \in [0, r-1]. \quad (\text{F.9})$$

such that, uniformly for $s \in C$,

$$\lim_{k \rightarrow \infty} -k^{-1} \log_2 S_k(\mu_s, r) = \tau_C(r). \quad (\text{F.10})$$

The function $r \mapsto \tau_C(r)$ is concave on $(1, \infty)$, continuous there, and extends with $\tau_C(1) = 0$.

Proof.

Step 1: a finite-state hierarchical q-sum inequality.

For $n, l \geq 1$, iterate (F.1) for n blocks and group the prefix translations b_U by the cells I of $\mathcal{D}_{k_n}$. The contribution belonging to one such I is a convex combination, after normalization, of translated copies of the finitely many tails μ_t , $t \in C$, all at scale λ^n . At the additional scale λ^l , convexity gives

$$S\left(\sum_j \alpha_j \nu_j, r\right) \leq \sum_j \alpha_j S(\nu_j, r) \leq \max_j S(\nu_j, r). \quad (\text{F.11})$$

because $\sum_j \alpha_j = 1$. Translations, the $O(1)$ mismatch between dyadic scales and λ -powers, and the bounded overlap of the $O(\lambda^n)$ -enlarged prefix cells cost only one constant depending on r and the fixed family. Hence

$$S_{k_{n+l}}(\mu_s, r) \leq C_r S_{k_n}(A_{s,n}, r) \max_{t \in C} S_{k_l}(\mu_t, r). \quad (\text{F.12})$$

Use the natural-scale comparison in Lemma IV.10.95 to replace the prefix q -sum by the state-measure q -sum:

$$S_{k_{n+l}}(\mu_s, r) \leq C'_r S_{k_n}(\mu_s, r) \max_{t \in C} S_{k_l}(\mu_t, r). \quad (\text{F.13})$$

Put

$$Q_n = \max_{s \in C} S_{k_n}(\mu_s, r).$$

Then

$$Q_{n+l} \leq C'_r Q_n Q_l. \quad (\text{F.14})$$

Thus $\log(C'_r Q_n)$ is subadditive, so $n^{-1} \log Q_n$ has a limit.

Step 2: all states in C have the same exponent.

For $s, t \in C$ choose once and for all an admissible path $P_{s,t}$ of length $L_{s,t}$ and positive path weight $p_{s,t}$. From (F.1),

$$\mu_s \geq p_{s,t}(\phi_{P_{s,t}})_* \mu_t \quad (\text{F.15})$$

as measures. A fixed similarity changes the dyadic level by only $O(1)$, and the dyadic q -sum by only a fixed multiplicative factor. Therefore there are constants A, B depending only on the finite class and r such that

$$S_{k+A}(\mu_s, r) \geq B S_k(\mu_t, r) \quad (\text{F.16})$$

for all sufficiently large k . Applying (F.16) in both directions shows that all states have the same lower and upper exponential q -sum rates. Together with (F.14), this gives (F.10) at the natural levels k_n .

The gaps $k_{n+1} - k_n$ are uniformly bounded. Refining a dyadic cell by d levels changes an r -sum by a factor between $2^{-(r-1)d}$ and 1. Hence the same limit holds at all dyadic levels k , and the convergence is uniform over the finite class.

Step 3: concavity.

For each fixed k , $r \mapsto -k^{-1} \log_2 S_k(\mu_s, r)$ is concave by Hölder's inequality. The pointwise limit (F.10) is therefore concave. The identity $S_k(\mu_s, 1) = 1$ gives $\tau_C(1) = 0$, and concavity gives continuity on $(1, \infty)$. The ambient-cell bound gives $\tau_C(r) \leq r - 1$. $\square$

Corollary IV.10.102 (*uniform one-sided spectrum estimate*).

For every $r > 1$ and $\zeta > 0$ there is k_0 such that, for every class C , every $s \in C$ and $k \geq k_0$,

$$S_k(\mu_s, r) \leq 2^{-[\tau_C(r) - \zeta]k}. \quad (\text{F.17})$$

There are only finitely many classes, so k_0 may be chosen globally.

Proof. Apply the uniform convergence in Lemma IV.10.101 in each class and take the maximum of the finitely many thresholds. $\square$

3. Multifractal localization, uniform in state

Fix one class C and a differentiability point $r > 1$ of τ_C with

$$\tau_C(r) < r - 1. \quad (\text{F.18})$$

Put

$$\alpha = \tau'_C(r),$$

$$\theta = r\alpha - \tau_C(r). \quad (\text{F.19})$$

Concavity and $\tau_C(1) = 0$ give

$$\theta \leq \alpha < 1. \quad (\text{F.20})$$

Lemma IV.10.103 (*global localization*).

The following constants and thresholds can be chosen uniformly over $s \in C$.

(a) Factor-two localization. For every $\rho > 0$ there is $\delta_0 > 0$ such that, for all large k , if $a \in \mathbb{R}$ and $E \subset \mathcal{D}_k$ has

$$2^{-ak} \leq \mu_s(I) \leq 2^{1-ak} (I \in E) \quad (\text{F.21})$$

and

$$\begin{aligned} & \sum_{I \in E} \mu_s(I)^r \\ & \geq 2^{-[\tau_C(r) + \delta_0]k}, \end{aligned} \quad (\text{F.22})$$

then

$$|E| \leq 2^{[\theta + \rho]k}. \quad (\text{F.23})$$

(b) Sparse q -mass loss. For every $\kappa > 0$ there is $\eta > 0$ such that, for all large k ,

$$\begin{aligned} & |E| \leq 2^{[\theta - \kappa]k} \\ & \Rightarrow \\ & \sum_{I \in E} \mu_s(I)^r \\ & \leq 2^{-[\tau_C(r) + \eta]k}. \end{aligned} \quad (\text{F.24})$$

Proof.

The proof is exactly the elementary Legendre calculation used in the current Volume-II.11 Lemmas 11.4.5–11.4.7 and Proposition 11.6.4(i)–(ii); we record the two places where the state family matters.

For (a), take $r_1 = r - \sqrt{\delta_0} > 1$. From (F.21)–(F.22), comparison of the r and r_1 powers on E , followed by the uniform estimate (F.17) at r_1 , gives

$$\begin{aligned} a & \leq [\tau_C(r) - \tau_C(r_1)]/(r - r_1) + o(1) \\ & \leq \alpha + \rho/(3r) \end{aligned} \quad (\text{F.25})$$

after δ_0 is small and k is large. Applying (F.17) at r and using the lower mass bound in (F.21) then gives

$$\begin{aligned} k^{-1} \log |E| & \leq ra - \tau_C(r) + \rho/3 \\ & \leq \theta + \rho. \end{aligned} \quad (\text{F.26})$$

For (b), set $\varsigma = \kappa/(2r)$. Differentiability supplies $h > 0$ with

$$\tau_C(r + h) \geq \tau_C(r) + h\alpha - h\varsigma/4. \quad (\text{F.27})$$

By (F.17) at $r + h$, cells of mass at least $2^{-k(\alpha-\varsigma)}$ contribute at most $2^{-[\tau_C(r)+c]k}$ to the r -sum. Every other cell contributes at most $2^{-r(\alpha-\varsigma)k}$; a family of size $2^{(\theta-\kappa)k}$ therefore contributes at most

$$\begin{aligned} & 2^{(\theta-\kappa)k-r(\alpha-\varsigma)k} \\ &= 2^{-[\tau_C(r)+\kappa/2]k}. \end{aligned} \tag{F.28}$$

Decrease the positive gain to combine the two terms. Uniformity follows from (F.17) and the finiteness of C . □

Lemma IV.10.104 (*localized baseline and strict loss*).

For every $\kappa > 0$ there is $\eta > 0$ such that, for all sufficiently large k , uniformly in the class C , $s \in C$, dyadic depth $d \geq 0$ and $I \in \mathcal{D}_d$:

The same bounds, with the same thresholds up to fixed harmless constants, hold if every level of $\mathcal{D}$ is translated by one common real number. In that version I and all of its descendants belong to the translated grid. The translation may depend on d and k ; the constants do not.

(i) if $E \subset \mathcal{D}_{d+k}(I)$ and

$$|E| \leq 2^{[\theta-\kappa]k}, \tag{F.29}$$

then

$$\begin{aligned} & \sum_{J \in E} \mu_s(J)^r \\ & \leq 2^{-[\tau_C(r)+\eta]k} \mu_s(2I)^r; \end{aligned} \tag{F.30}$$

(ii) for every $\zeta > 0$, after increasing the threshold on k ,

$$\begin{aligned} & \sum_{J \in \mathcal{D}_{d+k}(I)} \mu_s(J)^r \\ & \leq 2^{-[\tau_C(r)-\zeta]k} \mu_s(2I)^r. \end{aligned} \tag{F.31}$$

Proof.

Choose an integer $n = n(d)$ so that every n -block component in the expansion of μ_s has diameter at most $|I|/2$ but the previous block scale is comparable to $|I|$. This means

$$\lambda^n \asymp 2^{-d} \tag{F.32}$$

with constants independent of s, d . Retain the admissible n -block paths U whose component

$$b_U + \lambda^n \text{supp}(\mu_{t(U)})$$

meets I . Every retained component then lies inside $2I$. If p is the total retained symbolic mass,

$$p \leq \mu_s(2I). \quad (\text{F.33})$$

For $J \subset I$,

$$\mu_s(J) = \sum_U p_U \mu_{t(U)}(\lambda^{-n}(J - b_U)), \quad (\text{F.34})$$

where only retained U contribute. Convexity gives

$$\begin{aligned} & \sum_{J \in E} \mu_s(J)^r \\ & \leq p^{r-1} \sum_U p_U \\ & \quad \sum_{J \in E} \mu_{t(U)}(\lambda^{-n}(J - b_U))^r. \end{aligned} \quad (\text{F.35})$$

By (F.32), for each U the transformed intervals have length comparable to 2^{-k} , can be covered by $O(1)$ cells of $\mathcal{D}_k$, and have bounded overlap. If (F.29) holds, the covering family has at most $2^{(\theta-\kappa/2)k}$ cells for large k . Apply Lemma IV.10.103(b), uniformly to the tail state $t(U)$, and sum over U . This gives

$$RHS(\text{F.35}) \leq Cp^r 2^{-[\tau_C(r)+\eta_0]k}. \quad (\text{F.36})$$

Absorb C into half the gain and use (F.33), proving (F.30). Replacing the sparse estimate by (F.17) proves (F.31).

The argument did not use the location of the endpoints of I , only its length and the nesting of its descendants. For a translated dyadic grid, the images of its depth- k descendants under each inverse cylinder map are still intervals of length comparable to 2^{-k} and bounded overlap; each is covered by $O(1)$ ordinary level- k dyadic cells. Thus the same proof, including the retained-component bound $p \leq \mu_s(2I)$, is uniform over all real translations of the grid. At any one level, translated-grid r -sums and ordinary-grid r -sums are comparable by a constant depending only on r : each cell of either grid meets at most two cells of the other.

□

4. Uniform Statewise Flattening - Expanded Proof

Proposition IV.10.105 (*flattening for every tail state*).

Fix the class C , a differentiability point r satisfying (F.18), $\sigma > 0$, and a fixed integer $R \geq 1$. Fix also a bounded interval K_ρ containing the supports of all perturbing probabilities used below. There are $\varepsilon > 0$ and k_0 such that, for every $s \in C$, every $k \geq k_0$, and every probability ρ supported in K_ρ ,

$$S_{Rk}(\rho, r) \leq 2^{-\sigma k} \quad (\text{F.37})$$

implies

$$\begin{aligned} & S_{Rk}(\rho * \mu_s, r) \\ & \leq C 2^{-\varepsilon k} S_{Rk}(\mu_s, r), \end{aligned} \quad (\text{F.38})$$

where C depends only on R, r, K_ρ and the common support bound for the finite state family. The exponent ε and the threshold k_0 are uniform in s . The same exponential statement is stable under fixed translations, fixed affine rescalings and dyadic discretization.

Proof.

Write

$$\tau = \tau_C(r),$$

$$\alpha = \tau'_C(r),$$

$$\theta = r\alpha - \tau,$$

$$r' = r/(r-1). \quad (\text{F.37a})$$

By (F.18)–(F.20), $\tau < r-1$ and $\theta < 1$. Put $n = Rk$ and

$$\sigma_0 = \sigma/[R(r-1)]. \quad (\text{F.37b})$$

Step 1. Dyadic discretization and the inverse-theorem norm hypothesis.

Discretize ρ and μ_s at level n by moving the mass of each half-open dyadic n -cell to one fixed grid point chosen inside that same cell. Use one consistent half-open convention at every dyadic depth. Then for every j with $0 \leq jD \leq n$, the mass of each level- jD dyadic parent is exactly unchanged by discretization, because it is the disjoint union of level- n cells and each child mass remains inside its original child. In particular, all coarser sums defining L_j below are the same for the original and discretized μ_s .

Only convolution geometry sees the within-cell displacement. Because all relevant supports lie in fixed bounded intervals, real-line convolution before and after this displacement,

and the corresponding level- n r -sums, differ by fixed multiplicative constants only (equivalently by an $O(1)$ shift of dyadic level). Such constants are $2^{o(n)}$ and will be absorbed after all positive parameters have been chosen.

The inverse theorem permits a separate $2^{-n}\mathbb{Z}$ -translation of each discretized input before producing the middle-half trees. We keep track of that translation: for the first input it replaces the original dyadic grid, when viewed back in the coordinates of μ_s , by a translated dyadic grid. The translated-grid form of Lemma IV.10.104 proved above applies uniformly, even though the translating amount depends on n . At every fixed level the two r -sums differ by at most a constant depending on r , since each cell of either grid meets at most two cells of the other. Translation of the second input has no effect on its r -norm or on the convolution norm.

For the discretized ρ , (F.37) gives

$$\begin{aligned} & \|\rho\|_r^{r'} \\ &= S_n(\rho, r)^{1/(r-1)} \\ &\leq 2^{-\sigma_0 n}. \end{aligned} \tag{F.37c}$$

The uniform convergence (F.10) gives, for every $\xi > 0$ and all sufficiently large n , uniformly in $s \in C$,

$$\begin{aligned} 2^{-(\tau+\xi)n} &\leq S_n(\mu_s, r) \\ &\leq 2^{-(\tau-\xi)n}. \end{aligned} \tag{F.37d}$$

We prove first an absolute gain

$$S_n(\rho * \mu_s, r) \leq 2^{-(\tau+c)n} \tag{F.37e}$$

for some $c > 0$ independent of s . Once (F.37e) is proved, take n still larger so that (F.37d) holds with $\xi = c/2$. Then

$$\begin{aligned} & S_n(\rho * \mu_s, r) \\ &\leq 2^{-cn/2} S_n(\mu_s, r), \end{aligned} \tag{F.37f}$$

which is (F.38) with $\varepsilon = cR/2$, up to the fixed discretization constants.

Step 2: the case $\theta \leq 0$.

Choose a number ρ_0 with

$$0 < \rho_0 < \sigma_0/4. \tag{F.37g}$$

Use Lemma IV.10.103(a) with cardinality tolerance ρ_0 and let $\delta_{\text{card}} > 0$ be its retained- r -mass threshold. Choose the inverse-theorem regularization parameter $\delta > 0$ so small that

$$\delta < \sigma_0/4, \quad r\delta < \delta_{\text{card}}/2. \quad (\text{F.37h})$$

Apply the one-dimensional L^q inverse theorem from Volume II with r, δ and a sufficiently large preliminary block size D_0 . Denote its near-nonflattening constant by ε_* and its block length by $D \geq D_0$. Choose

$$0 < \xi < \min\{\delta_{\text{card}}/2, r\varepsilon_*/4\}. \quad (\text{F.37i})$$

First assume $n = D\ell$. If the lower inequality in (F.37d) with this ξ fails, then dyadic Young immediately gives

$$\begin{aligned} S_n(\rho * \mu_s, r) &\leq CS_n(\mu_s, r) \\ &\leq 2^{-(\tau+\xi/2)n} \end{aligned} \quad (\text{F.37j})$$

for large n , proving (F.37e). Hence suppose

$$S_n(\mu_s, r) \geq 2^{-(\tau+\xi)n}. \quad (\text{F.37k})$$

By (F.37d), after increasing n ,

$$S_n(\mu_s, r) \leq 2^{-(\tau-r\varepsilon_*/4)n}. \quad (\text{F.37l})$$

Assume toward contradiction that

$$S_n(\rho * \mu_s, r) \geq 2^{-(\tau+r\varepsilon_*/4)n}. \quad (\text{F.37m})$$

Taking r -th roots and comparing (F.37l)–(F.37m) yields

$$\begin{aligned} \|\rho * \mu_s\|_r &\geq 2^{-\varepsilon_* n/2} \|\mu_s\|_r \\ &\geq 2^{-\varepsilon_* n} \|\mu_s\|_r. \end{aligned} \quad (\text{F.37n})$$

so Theorem 11.3.1 applies.

Let A be the retained set and let

$$S = \{j : R'_j \geq 2^{(1-\delta)D}\},$$

$$n_1 = D|S|. \quad (\text{F.37o})$$

The lower scale-count inequality and (F.37c) give

$$n_1 \geq (\sigma_0 - \delta)n. \quad (\text{F.37p})$$

The retained r -norm in Theorem 11.3.1 gives

$$\begin{aligned} & \sum_{I \in D_n(A)} \mu_s(I)^r \\ & \geq 2^{-r\delta n} S_n(\mu_s, r) \\ & \geq 2^{-(\tau+r\delta+\xi)n}. \end{aligned} \quad (\text{F.37q})$$

The retained level- n cell masses are factor-two comparable. By (F.37h)–(F.37i), $r\delta + \xi < \delta_{\text{card}}$, so Lemma IV.10.103(a) applies and gives

$$|A| \leq 2^{(\theta+\rho_0)n} \leq 2^{\rho_0 n}. \quad (\text{F.37r})$$

On the other hand uniform branching gives

$$\begin{aligned} |A| &= \prod_j R'_j \geq 2^{(1-\delta)n_1} \\ &\geq 2^{(1-\delta)(\sigma_0-\delta)n}. \end{aligned} \quad (\text{F.37s})$$

For the choices (F.37g)–(F.37h), the exponent in (F.37s) is strictly larger than ρ_0 . This contradicts (F.37r). Thus (F.37m) is impossible and the absolute gain (F.37e) follows when $\theta \leq 0$.

Step 3: parameter order when $\theta > 0$.

Assume now $\theta > 0$. Put

$$c_* = (1/4)(1 - \tau/(r-1)) > 0. \quad (\text{F.37t})$$

Choose

$$0 < \kappa < \min\{\theta/4, (1-\theta)\sigma_0/8\}. \quad (\text{F.37u})$$

Apply the localized strict-loss lemma with sparsity parameter $\kappa/2$, and let $\eta > 0$ be its strict local gain. Define

$$\gamma = \kappa c_*/(\theta - \kappa) > 0. \quad (\text{F.37v})$$

Choose $0 < \delta_0 < \kappa$. Let δ_{card} be the threshold in Lemma IV.10.103(a) corresponding to cardinality tolerance δ_0 . Now choose the inverse-theorem regularization parameter $\delta > 0$ sufficiently small that

$$\delta < \min\{\sigma_0/4, c_*, \delta_{\text{card}}/(2r)\}, \quad (\text{F.37w})$$

$$(1 - \theta - \delta_0 - \delta)(\sigma_0 - \delta) \geq 4\kappa, \quad (\text{F.37x})$$

$$r\delta < \frac{\eta\gamma}{16}, \quad \delta(1 + \tau + \eta) < \frac{\eta\gamma}{16}. \quad (\text{F.37y})$$

Choose D_0 so large that, for every $D \geq D_0$, the depth- $(D - 2)$ versions of the localized baseline and strict-loss estimates are valid with the already chosen parameters,

$$2^{(\theta - \kappa)D}$$

$$\leq 2^{(\theta - \kappa/2)(D-2)}, \quad (\text{F.37z})$$

all fixed per-block constants occurring in the localized recurrence below contribute less than $\eta\gamma D/16$ in the exponent.

Apply the one-dimensional L^q inverse theorem from Volume II with r, δ, D_0 . Let $\varepsilon_* > 0$ be its near-nonflattening constant and $D \geq D_0$ its block length. Finally choose

$$0 < \xi < \min\{\eta\gamma/16, r\varepsilon_*/4, (r - 1)c_*, \delta_{\text{card}} - r\delta\}. \quad (\text{F.37aa})$$

All choices were made before the scale n and uniformly over the finite state class C .

Step 4. Inverse-theorem branching bounds.

Again take first $n = D\ell$. As in Step 2, the case

$$S_n(\mu_s, r) < 2^{-(\tau + \xi)n} \quad (\text{F.37ab})$$

is finished by Young, so assume the reverse inequality. Uniform convergence also gives

$$S_n(\mu_s, r) \leq 2^{-(\tau - r\varepsilon_*/4)n}. \quad (\text{F.37ac})$$

for large n . If the absolute gain with exponent $r\varepsilon_*/4$ failed, i.e.

$$S_n(\rho * \mu_s, r) \geq 2^{-(\tau + r\varepsilon_*/4)n}. \quad (\text{F.37ad})$$

then the same root comparison as (F.37n) invokes Theorem 11.3.1. Let A and (R'_j) be the retained uniform set/branching sequence. Put

$$S = \{j : R'_j \geq 2^{(1-\delta)D}\},$$

$$n_1 = D|S|,$$

$$n_2 = n - n_1,$$

$$S_0 = \{0, \dots, \ell - 1\} \setminus S. \quad (\text{F.37ae})$$

The two scale-count inequalities give

$$(\sigma_0 - \delta)n \leq n_1$$

$$\leq ((\tau + \xi)/(r - 1) + \delta)n. \quad (\text{F.37af})$$

By (F.37t), (F.37w), and (F.37aa), the upper bound in (F.37af) implies

$$n_2 \geq 2c_* n. \quad (\text{F.37ag})$$

The retained r-mass again satisfies

$$\begin{aligned} \sum_{I \in D_n(A)} \mu_s(I)^r \\ \geq 2^{-(\tau + r\delta + \xi)n}. \end{aligned} \quad (\text{F.37ah})$$

Factor-two comparability and (F.37aa) let Lemma IV.10.103(a) apply, giving

$$|A| \leq 2^{(\theta + \delta_0)n}. \quad (\text{F.37ai})$$

Uniform branching and the definition of S give

$$\begin{aligned} |A| &= \prod_j R'_j \\ &\geq 2^{(1-\delta)n_1} \prod_{j \in S_0} R'_j. \end{aligned} \quad (\text{F.37aj})$$

Combining (F.37ai)–(F.37aj) and writing $n = n_1 + n_2$, we obtain

$$\log_2 \prod_{j \in S_0} R'_j \leq (\theta + \delta_0)n_2 - (1 - \theta - \delta_0 - \delta)n_1. \quad (\text{F.37ak})$$

By (F.37x), the lower bound for n_1 in (F.37af), $n_2 \leq n$, and $\delta_0 < \kappa$,

$$\log_2 \prod_{j \in S_0} R'_j \leq (\theta - 2\kappa)n_2. \quad (\text{F.37al})$$

Define the low-branching block set

$$S_1 = \{j \in S_0 : \log_2 R'_j \leq (\theta - \kappa)D\}. \quad (\text{F.37am})$$

Since all $R'_j \geq 1$, (F.37al) implies

$$|S_1| \geq [\kappa/(\theta - \kappa)]|S_0|. \quad (\text{F.37an})$$

By (F.37ag), $|S_0| = n_2/D \geq 2c_*\ell$, and hence

$$|S_1| \geq 2\gamma\ell \geq \gamma\ell. \quad (\text{F.37ao})$$

Step 5. Localized r -mass contradiction.

Theorem 11.3.1 also supplies the middle-half property: every point of A lies in the middle half of each occupied level- jD parent. Define

$$L_j = -\log_2 \sum_{I \in D_{jD}(A)} \mu_s(I)^r, \quad 0 \leq j \leq \ell. \quad (\text{F.37ap})$$

Fix j . If K is an occupied cell at level $jD + 2$, then $2K$ is contained in a unique occupied level- jD parent P , and each P contains at most two such doubled cells. Apply the localized baseline estimate, with $\zeta = \delta$, inside every such K at depth $D - 2$ and sum over K . The preceding containment gives a uniform per-block constant C_0 , independent of j, s, D , and therefore

$$L_{j+1} \geq L_j + (\tau - \delta)(D - 2) - C_0. \quad (\text{F.37aq})$$

If $j \in S_1$, the number of occupied level- $(j + 1)D$ descendants in any such K is at most R'_j . By (F.37z),

$$\begin{aligned} R'_j &\leq 2^{(\theta - \kappa)D} \\ &\leq 2^{(\theta - \kappa/2)(D - 2)}. \end{aligned} \quad (\text{F.37ar})$$

Hence the localized strict-loss estimate, at depth $D - 2$ with sparsity $\kappa/2$, improves (F.37aq) to

$$L_{j+1} \geq L_j + (\tau + \eta)(D - 2) - C_0. \quad (\text{F.37as})$$

Sum (F.37aq) over all j and use (F.37as) on the at least $\gamma\ell$ blocks in S_1 . By (F.37y) and the choice of D_0 absorbing the $2/D$ and C_0/D losses,

$$L_\ell \geq (\tau + \eta\gamma/2)n. \quad (\text{F.37at})$$

But (F.37ah) says

$$L_\ell \leq (\tau + r\delta + \xi)n. \quad (\text{F.37au})$$

The choices (F.37y) and (F.37aa) make $r\delta + \xi < \eta\gamma/2$, contradicting (F.37at). Therefore (F.37ad) is impossible. This proves the absolute gain (F.37e) for all n divisible by D , uniformly in s .

Step 6. Arbitrary n and the relative form.

For arbitrary n replace it by $n_0 = D \lfloor n/D \rfloor$. Coarsening a dyadic probability by at most D levels changes each relevant r -sum and the near-nonflattening comparison by a

factor depending only on D, r and the fixed support bounds. Since D is already fixed, these factors are absorbed by halving σ_0 and the absolute gain for all sufficiently large n . Thus there is $c > 0$ such that (F.37e) holds for every large $n = Rk$.

Finally use the lower half of the uniform spectrum estimate (F.37d) with $\xi = c/2$. This gives (F.37f), hence (F.38), uniformly in s . Fixed translations and fixed affine changes alter dyadic levels by $O(1)$, so the same exponential statement survives with a changed multiplicative constant.

□

The only non-elementary cross-volume input in this proof is the inverse theorem stated above: its retained r -mass and factor-two conclusion, middle-half localization, branching dichotomy, and two-sided active-scale count are used explicitly. The localized estimates and spectrum information are Lemma IV.10.101, Lemma IV.10.103, and Lemma IV.10.104; no scalar self-similar dimension formula is imported.

5. Vector Fine-Prefix Recursion

For $s \in C$ and $n \geq 1$, split the prefix by terminal state:

$$A_{s,n,t} = \sum_{\substack{U:s \rightarrow t \\ |U|=n}} p_U \delta_{b_U}, \quad A_{s,n} = \sum_{t \in C} A_{s,n,t}. \quad (\text{F.39})$$

Proposition IV.10.106 (*graph fine-prefix recursion*).

At a differentiability point r satisfying (F.18), for every fixed integer $R \geq 1$ and every $s \in C$,

$$\lim_{n \rightarrow \infty} -k_n^{-1} \log_2 S_{Rk_n}(A_{s,n}, r) = \tau_C(r), \quad (\text{F.40})$$

uniformly in s in the finite class.

Proof.

For $R = 1$ this is Lemma IV.10.95 together with (F.10). Fix $R \geq 1$; the following argument proves the $R + 1$ statement directly. Write $k = k_n$.

For $t \in C$ and $I \in \mathcal{D}_k$ put

$$a_{t,I} = A_{s,n,t}(I). \quad (\text{F.41})$$

When $a_{t,I} > 0$, let $A_{t,I}$ be the normalized restriction of $A_{s,n,t}$ to $I = [u_I, u_I + 2^{-k})$ and define

$$\rho_{t,I} = [z \mapsto (z - u_I)/\lambda^n]_* A_{t,I}. \quad (\text{F.41a})$$

Since $\lambda^n \leq 2^{-k} < 2\lambda^n$, every $\rho_{t,I}$ is supported in $[0, 2]$, independently of s, n, t, I . Thus the fixed-support hypothesis of Proposition IV.10.105 applies with $K_\rho = [0, 2]$. Iterating (F.1) gives

$$\mu_s = \sum_{t,I} a_{t,I} (A_{t,I} * S_{\lambda^n} \mu_t). \quad (\text{F.42})$$

Here $S_{\lambda^n \mu_t}$ denotes the push-forward under $x \mapsto \lambda^n x$, and the exact rescaled identity for one summand is

$$A_{t,I} * S_{\lambda^n \mu_t} = [z \mapsto u_I + \lambda^n z]_* (\rho_{t,I} * \mu_t). \quad (\text{F.42a})$$

The fine-grid estimates below use this identity and bounded dyadic overlap, with constants independent of the outer translations u_I . Because $\lambda^n \asymp 2^{-k}$, bounded overlap of neighboring coarse cells, Hölder's inequality in the finite t -sum, and the dyadic scale comparison give

$$S_{(R+1)k}(\mu_s, r) \leq C|C|^{r-1} \sum_{t,I} a_{t,I}^r S_{Rk}(\rho_{t,I} * \mu_t, r). \quad (\text{F.43})$$

On the other hand, for the atomic prefix itself, positivity gives the lower comparison

$$S_{(R+1)k}(A_{s,n}, r) \geq c \sum_{t,I} a_{t,I}^r S_{Rk}(\rho_{t,I}, r). \quad (\text{F.44})$$

Indeed different coarse cells I are separated by the refining partition. Inside one fixed I , if m_t denotes the contribution from terminal state t , then

$$\left(\sum_t m_t \right)^r \geq \sum_t m_t^r. \quad (\text{F.45})$$

Fix $\sigma > 0$ and call (t, I) flat if

$$S_{Rk}(\rho_{t,I}, r) \leq 2^{-\sigma k}. \quad (\text{F.46})$$

For flat pairs Proposition IV.10.105 gives

$$S_{Rk}(\rho_{t,I} * \mu_t, r) \leq C 2^{-\varepsilon k} S_{Rk}(\mu_t, r). \quad (\text{F.47})$$

For non-flat pairs, dyadic Young gives the weaker bound

$$S_{Rk}(\rho_{t,I} * \mu_t, r) \leq C S_{Rk}(\mu_t, r). \quad (\text{F.48})$$

By (F.10), uniformly in t ,

$$S_{Rk}(\mu_t, r) = 2^{-R\tau_C(r)k+o(k)}. \quad (\text{F.49})$$

and

$$S_{(R+1)k}(\mu_s, r) = 2^{-(R+1)\tau_C(r)k+o(k)}. \quad (\text{F.50})$$

Moreover

$$\sum_{t,I} a_{t,I}^r \leq S_k(A_{s,n}, r) = 2^{-\tau_C(r)k+o(k)}. \quad (\text{F.51})$$

The inequality in (F.51) is the left half of the finite-state grouping inequality, and the equality in exponent follows from the natural-scale comparison.

Insert (F.47)–(F.51) into (F.43). The total flat contribution is

$$\text{flat contribution} \leq 2^{-(R+1)\tau_C(r)k - \varepsilon k + o(k)}. \quad (\text{F.52})$$

and is therefore exponentially negligible relative to (F.50). By (F.48)–(F.49) the entire non-flat contribution to the right side of (F.43) is at most

$$2^{-R\tau_C(r)k + o(k)} \sum_{(t,I) \text{ nonflat}} a_{t,I}^r. \quad (\text{F.52a})$$

Comparing (F.43), (F.50), (F.52), and (F.52a), and absorbing fixed constants and $o(k)$ into σ , gives, after increasing n ,

$$\sum_{(t,I) \text{ nonflat}} a_{t,I}^r \geq 2^{-[\tau_C(r) + \sigma]k}. \quad (\text{F.53})$$

Every non-flat pair satisfies

$$S_{Rk}(\rho_{t,I}, r) > 2^{-\sigma k}. \quad (\text{F.54})$$

Using (F.44), (F.53), and (F.54),

$$S_{(R+1)k}(A_{s,n}, r) \geq 2^{-[\tau_C(r) + 2\sigma]k - O(1)}. \quad (\text{F.55})$$

Conversely refinement decreases r-sums, so

$$S_{(R+1)k}(A_{s,n}, r) \leq S_k(A_{s,n}, r) = 2^{-\tau_C(r)k + o(k)}. \quad (\text{F.56})$$

Let $\sigma \downarrow 0$. This proves (F.40) for $R + 1$. The constants are uniform over s because C is finite.

The only state grouping in (F.43) costs $|C|^{r-1}$ once. It is a fixed multiplicative constant, not $|C|^{cn}$.

□

Theorem IV.10.107 (*classwise graph L^q formula*).

For every class C , every state $s \in C$, and every $r > 1$,

$$D_r(\mu_s) = \min \left\{ 1, \frac{H_{r,C}}{(r-1) \log(1/\lambda)} \right\}. \quad (\text{F.57})$$

The convergence defining D_r is uniform over the finitely many states in each class.

Proof.

Write $\tau = \tau_C$.

For the universal upper bound, the natural-scale comparison gives

$$S_{k_n}(\mu_s, r) \asymp S_{k_n}(A_{s,n}, r). \quad (\text{F.58})$$

Merging atomic prefix masses into dyadic cells can only increase the r-sum:

$$S_{k_n}(A_{s,n}, r) \geq \sum_U p_U^r = Z_{s,n}(r). \quad (\text{F.59})$$

Using (F.7), $k_n = an + O(1)$, and (F.10),

$$\tau(r) \leq H_{r,C}/(a \log 2). \quad (\text{F.60})$$

Also $\tau(r) \leq r - 1$. Therefore

$$\tau(r) \leq \min\{H_{r,C}/(a \log 2), r - 1\}. \quad (\text{F.61})$$

Reverse inequality at differentiability points. Let r be a differentiability point. If $H_{r,C}/(a \log 2) < r - 1$, (F.61) gives

$\tau(r) < r - 1$, so Proposition IV.10.106 applies.

Choose an integer R so large that

$$Ra > \log_2(1/c) + 2. \quad (\text{F.62})$$

By (F.3), for all sufficiently large n , distinct admissible n -block paths from s have translations at least c^n apart. Since

$$2^{-Rk_n} \leq 2^R 2^{-Ran} \leq 2^{-n} c^n. \quad (\text{F.63})$$

for large n , every level- Rk_n dyadic cell contains at most one prefix atom. Hence

$$S_{Rk_n}(A_{s,n}, r) = \sum_U p_U^r = Z_{s,n}(r). \quad (\text{F.64})$$

Taking $-k_n^{-1} \log_2$ and using Proposition IV.10.106 and (F.7) yields

$$\tau(r) = H_{r,C}/(a \log 2). \quad (\text{F.65})$$

If $H_{r,C}/(a \log 2) \geq r - 1$ and $\tau(r) < r - 1$, the same argument would again give (F.65), contradicting $H_{r,C}/(a \log 2) \geq r - 1$. Therefore in this case

$$\tau(r) = r - 1. \quad (\text{F.66})$$

Thus (F.57) holds at every differentiability point of the concave τ . Differentiability points are dense. The map $r \mapsto \rho(M_{r,C})$ is continuous because $M_{r,C}$ is a finite matrix with continuous entries, hence $H_{r,C}$ is continuous. Both sides of

$$\tau(r) = \min\{H_{r,C}/(a \log 2), r - 1\}. \quad (\text{F.67})$$

are continuous, so (F.67) holds for every $r > 1$. Divide by $r - 1$.

□

Corollary IV.10.108 (*state-uniform fixed-exponent pressure bound*).

Under (GQ.1), (GQ.3), and the recurrent class decomposition (F.4), fix $r > 1$ and put

$$\beta_r = \min_C \min \left\{ 1, \frac{H_{r,C}}{(r-1) \log(1/\lambda)} \right\}. \quad D_r(\mu_s) \geq \beta_r \quad (s \in V). \quad (\text{F.68})$$

This is uniform in states for the chosen r ; its value can depend on r . Moreover (F.7)–(F.8) give the exact state-uniform symbolic rate $H_{r,C}/(r-1)$ in each class, with no state-grouping loss per depth.

Proof. For $s \in C$, expand the powered path weights to obtain $Z_{s,n}(r) = (M_{r,C}^n \mathbf{1})_s$. Choose a positive Perron eigenvector v_C and constants $a_C, b_C > 0$ with $a_C v_C \leq \mathbf{1} \leq b_C v_C$. Multiplying by the nonnegative $M_{r,C}^n$ bounds $Z_{s,n}(r)$ above and below by positive state-independent multiples of $\rho_{r,C}^n$. Taking logarithms and dividing by n proves (F.8), uniformly in $s \in C$. Since there are finitely many classes, the constants can also be made uniform in C after r is fixed. Theorem IV.10.107, whose proof uses this pressure but no Shannon lower bound, gives $D_r(\mu_s) = \min\{1, H_{r,C}/((r-1) \log(1/\lambda))\}$. Taking the minimum over the classes proves (F.68). □

6. Phase measures and the Frostman endpoint

Lemma IV.10.109 (*finite phase transfer*).

Assume the specified edge-level presentation and path laws of Definition IV.10.94. Fix $0 \leq j < m$. The projected phase measure $\Pi_*(\sigma^j \mathcal{N})$ is a finite convex combination of fixed affine images of the state-started measures μ_t . The same assertion holds with $\mathcal{N}_s$ in place of $\mathcal{N}$ for any fixed initial state s .

Proof.

For $w = (e_1, \dots, e_m)$, put $\psi_{w,j} = \psi_{e_{j+1}} \circ \dots \circ \psi_{e_m}$. Its contraction is $\gamma^{m-j} > 0$. Conditioning on the initial state s and first block w has weight $\pi(s)p_w$. By the cylinder definition, the conditional block future after w has law $\mathcal{N}_{t(w)}^B$; concatenation gives the edge law $\mathcal{N}_{t(w)}$. Thus

$$\Pi_*(\sigma^j \mathcal{N}) = \sum_{s \in V} \sum_{t \in V} \sum_{w \in W(s,t)} \pi(s)p_w(\psi_{w,j})_* \mu_{t(w)}.$$

The sum is finite and its coefficients add to one. Fixing the initial state gives the analogous formula with the coefficients p_w only. All maps and weights are specified by the edge presentation; no spectral-phase identification is used.

□

Theorem IV.10.110 (*fixed-exponent graph-relative Frostman estimate*).

Under the hypotheses of Corollary IV.10.108, fix $r > 1$. For every $0 < s < (1 - 1/r)\beta_r$, there is a finite constant $K_{r,s}$ such that, simultaneously for every state t , every $x \in \mathbb{R}$, and every $\rho > 0$,

$$\mu_t(B(x, \rho)) \leq K_{r,s} \rho^s. \quad (\text{F.68a})$$

With specified edge-level data, the same exponent holds uniformly over the finitely many projected phase laws of Lemma IV.10.109. If $\beta_r = 0$, no positive exponent is asserted; the exponent-zero bound is always valid.

Proof. Choose d with $s/(1 - 1/r) < d < \beta_r$. By (F.68), $D_r(\mu_t) > d$ for every state. Lemma IV.10.99 supplies constants and a small-radius range with exponent $(1 - 1/r)d > s$. Take the maximum of the constants and the minimum radius over the finite state set, decreasing that radius to at most one. This proves (F.68a) for small ρ . For the remaining radii use mass at most one and increase the constant; here $s > 0$ is essential for the universal-radius extension. For an affine map $A(x) = ax + b$ with $a \neq 0$, the global bound implies

$$(A_*\mu_t)(B(x, \rho)) \leq K_{r,s} |a|^{-s} \rho^s.$$

In Lemma IV.10.109 there are finitely many such maps, all with positive contraction γ^{m-j} . Taking a maximum and summing their convex weights proves the phase assertion, including all initial states and $0 \leq j < m$. $\square$

Remark IV.10.111 (*the rejected Shannon-Frostman target and its valid replacement*).

For the synchronized towers, h_G is the entropy per edge defined in (F.72l). The original proposed endpoint was

$$\beta = \min\{1, h_G / \log(1/\gamma)\}. \quad (\text{F.69})$$

Choosing m large and δ small can make the numerical quantity $\beta_0 = \min\{1, (h_G - \delta - 1/m) / \log(1/\gamma)\}$ satisfy

$$\beta_0 > \beta - \eta/4. \quad (\text{F.70})$$

One can then choose large r and $d < \beta_0$ with

$$(1 - 1/r)d > \beta - \eta. \quad (\text{F.71})$$

These numerical choices do not prove $D_r(\mu_t) > d$. The formerly promised conclusion

$$\mu_t(B(x, \rho)) \leq C_\eta \rho^{\beta-\eta} \quad (\text{F.72})$$

for every state and every sufficiently small ρ is false under the present graph hypotheses. It is retained here as a rejected endpoint, not an asserted theorem awaiting a proof. The pressure-dependent Theorem IV.10.110 is its proved replacement.

Indeed, take the one-state graph with maps $x/3$ and $x/3 + 2/3$, weights $9/10$, $1/10$, and its m -block presentation. It is recurrent, irreducible and strongly separated. At n -block depth its distinct prefix translations differ by at least $2 \cdot 3^{-mn}$, so (GQ.3) holds with $c = 3^{-m}$. The powered transition matrix is the scalar $M_{r,C} = ((9/10)^r + (1/10)^r)^m$. Its entropy per edge is $h_G = -(9/10) \log(9/10) - (1/10) \log(1/10)$, whereas

$$\frac{H_{2,C}}{m} = -\log(41/50), \quad D_2(\mu) = \frac{-\log(41/50)}{\log 3} < \frac{h_G}{\log 3}.$$

In particular $m = 100$, $\delta = 1/100$ would make the former Shannon lower bound exceed 0.277, while $D_2(\mu) < 0.181$. Moreover $\mu([0, 3^{-n}]) = (9/10)^n$; a uniform Frostman exponent therefore cannot exceed $-\log(9/10)/\log 3 < 0.096$, whereas $\beta > 0.295$. The tower definition admits this example by taking the second system to have one probability-one edge and an atomic law.

A sufficient additional pressure hypothesis for (F.72), with $0 < \eta < \beta$, would be an exponent $r > 1$ satisfying $(1 - 1/r)\beta_r > \beta - \eta$. Theorem IV.10.110 would then prove the desired bound and its edge-phase transfer. No such hypothesis follows here from Shannon entropy, and the example shows it cannot hold universally. Recovering a stronger endpoint on a restricted graph class would require an additional theorem; such a strengthening is not asserted or used in this chapter.

The consumer contracts are distinct. Lemmas IV.10.86 and IV.10.88, and their normality consequences, require lower Hausdorff dimension one for the *spectral* phase law θ_μ or Θ . A single positive exponent in (F.68a) does not supply this, and edge coding phases have not been identified with those circle phases. The resonant analytic-image route instead uses Theorem IV.10.113 as $r \downarrow 1$, Theorem IV.10.116, and Lemmas IV.10.117–IV.10.120: positive spectral phase-time dimension suffices there because each exceptional time set has dimension zero. This route uses neither the former all- r Shannon pressure bound nor (F.72), and remains subject to its earlier EFD and generation suppliers.

The zero-dimensional parameter exclusion needed later is a separate finite-path fact. It does not by itself construct the finite-state graph for a given pair of self-similar measures.

Lemma IV.10.112 (*zero-dimensional exceptional parameter set for synchronized paths*).

Let $J \subset \mathbb{R}$ be compact. For each $n \geq 1$ and each state in a fixed finite set, let Γ_n be a family of at most B^n distinct path pairs (U, V) , where $B > 1$. Associate real numbers a_U, b_V to the component paths. Suppose that, within every Γ_n , each nonzero difference of two first-component translations has absolute value at least c^n , and the same is true for second-component translations, for one $c \in (0, 1)$. Suppose also that distinct pairs never have both component translations equal. Then there is $E_J \subset J$ with $\dim_H E_J = 0$ such that, for every $t \in J \setminus E_J$, there is $c_t \in (0, 1)$ for which, at every sufficiently large n and for every two distinct pairs in the same Γ_n ,

$$|(a_U + tb_V) - (a_{U'} + tb_{V'})| \geq c_t^n. \quad (\text{F.72a})$$

Proof. Fix $K > \log(1/c)$ and let $E_{K,n} \subset J$ be the union of parameter sets where (F.72a) fails with e^{-Kn} in place of c_t^n . If $b_V = b_{V'}$, the first-component difference is at least $c^n > e^{-Kn}$, so this pair contributes nothing. Otherwise $|b_V - b_{V'}| \geq c^n$, and its bad parameters form an interval of length at most

$$2e^{-Kn}/|b_V - b_{V'}| \leq 2e^{-(K-\log(1/c))n}. \quad (\text{F.72b})$$

There are at most $|S|B^{2n}$ ordered pair comparisons, where $|S|$ is the number of states. Consequently, for every $s > 2 \log B/(K - \log(1/c))$,

$$\sum_{n \geq N} \sum_{I \text{ in the above cover of } E_{K,n}} (\text{diam } I)^s \leq |S|2^s \sum_{n \geq N} B^{2n} e^{-s(K-\log(1/c))n} \xrightarrow[N \rightarrow \infty]{} 0. \quad (\text{F.72c})$$

Hence $\dim_H(\limsup_n E_{K,n}) \leq 2 \log B/(K - \log(1/c))$. Choose any sequence $K_j \uparrow \infty$ and put $E_J = \bigcap_j \limsup_n E_{K_j,n}$. Its Hausdorff dimension is zero. If $t \notin E_J$, then $t \notin \limsup_n E_{K_j,n}$ for some j , so all sufficiently large levels satisfy (F.72a) with $c_t = e^{-K_j}$. The finite state union is already included in the cover. $\square$

Theorem IV.10.113 (*classwise exact dimension from the graph L^q formula*).

Under the finite graph-directed hypotheses (GQ.1) and (GQ.3), let C be a closed irreducible block class and let π_C be the stationary probability vector of its transition matrix $M_{1,C}$. Define its entropy per block by

$$h_C = - \sum_{s \in C} \pi_C(s) \sum_{t \in C} \sum_{w \in W(s,t)} p_w \log p_w. \quad (\text{F.72d})$$

Then every state-started measure μ_s , $s \in C$, is exact dimensional and

$$\dim \mu_s = \min \left\{ 1, \frac{h_C}{\log(1/\lambda)} \right\}. \quad (\text{F.72e})$$

Moreover $h_C = \lim_{r \downarrow 1} H_{r,C}/(r-1)$.

Proof. The matrix $M_{1,C}$ is stochastic and irreducible, so its Perron eigenvalue 1 is simple, with left eigenvector π_C and right eigenvector $\mathbf{1}$. Differentiating $M_{r,C} v_r = \rho_{r,C} v_r$ at $r = 1$ and multiplying by π_C gives

$$\left. \frac{d}{dr} \rho_{r,C} \right|_{r=1} = \pi_C M'_{1,C} \mathbf{1} = \sum_{s \in C} \pi_C(s) \sum_{t,w: s \rightarrow t} p_w \log p_w = -h_C. \quad (\text{F.72f})$$

The entries have fixed support and depend analytically on r ; finite-dimensional simple-eigenvalue perturbation justifies the differentiation. Since $\rho_{1,C} = 1$, the claimed limit follows.

Put $d_C = \min\{1, h_C/\log(1/\lambda)\}$. If $d_C = 0$, the lower local-dimension bound is automatic. Otherwise fix $0 < d < d_C$. By Theorem IV.10.107 and the limit just proved, one can choose $r > 1$ and $d' > d$ with $D_r(\mu_s) > d'$ for every $s \in C$. After an affine normalization of the finitely many supports, for every sufficiently large dyadic level k ,

$$S_k(\mu_s, r) \leq 2^{-(r-1)d'k} \quad (s \in C). \quad (\text{F.72g})$$

For any compactly supported probability ξ , the bounded number of level- k cells met by $B(x, 2^{-k})$ and discrete Hölder give

$$\int \xi(B(x, 2^{-k}))^{r-1} d\xi(x) \leq C_r S_k(\xi, r). \quad (\text{F.72h})$$

Indeed, if $a_I = \xi(I)$ and b_I is the sum of the masses of the finitely many cells neighboring I , then the left side is at most $\sum_I a_I b_I^{r-1}$. Hölder bounds this by $(\sum a_I^r)^{1/r} (\sum b_I^r)^{(r-1)/r}$, and the bounded-neighbor convolution has ℓ^r norm at most a fixed multiple of that of (a_I) . By Markov's inequality and (F.72g)–(F.72h),

$$\mu_s\{x : \mu_s(B(x, 2^{-k})) > 2^{-dk}\} \leq C_r 2^{-(r-1)(d'-d)k}. \quad (\text{F.72i})$$

Borel–Cantelli, followed by $d \uparrow d_C$, gives lower local dimension at least d_C at μ_s -almost every point.

For the upper bound, follow the finite-state Markov path law started at s . The finite-state ergodic theorem, or its martingale-information proof, gives for almost every path $U_1 U_2 \cdots$

$$-\frac{1}{n} \log p_{U_1 \dots U_n} \longrightarrow h_C. \quad (\text{F.72j})$$

The n -block image has diameter at most $2B\lambda^n$ and mass at least $p_{U_1 \dots U_n}$ in μ_s . Thus at its coded point x , $\mu_s(B(x, (1 + 2B)\lambda^n)) \geq p_{U_1 \dots U_n}$, which gives upper local dimension at most $h_C / \log(1/\lambda)$. Independently, every compactly supported probability on $\mathbb{R}$ has upper local dimension at most 1 almost everywhere: at dyadic level k , the union of cells of mass less than $2^{-(1+\varepsilon)k}$ has mass at most $O(2^{-\varepsilon k})$, and Borel–Cantelli finishes. Consequently the upper local dimension is at most d_C . Combined with the lower bound, this proves exact dimension and (F.72e) for every $s \in C$. $\square$

Lemma IV.10.114 (*common-ratio towers and synchronized convolution graph*).

Let $0 < \gamma < 1$ and let

$$g_i(x) = s_i \gamma^{p_i} x + a_i, \quad s_i \in \{-1, 1\}, \quad p_i \in \mathbb{N}, \quad \sum_i w_i = 1, \quad w_i > 0, \quad (\text{F.72k})$$

have self-similar probability μ . Put $\bar{p} = \sum_i w_i p_i$ and $H(w) = -\sum_i w_i \log w_i$. There is a finite-state graph-directed presentation of μ with common contraction γ and state space contained in $\{(\varepsilon, j) : \varepsilon \in \{-1, 1\}, 0 \leq j < \max_i p_i\}$. Its state $(1, 0)$ law is μ , and every closed irreducible class has transition entropy per common-ratio step $H(w)/\bar{p}$.

For a second system $h_j(y) = t_j \gamma^{q_j} y + c_j$ with weights v_j and self-similar law τ , the product of the two tower chains gives, for each real parameter u , a finite-state common-ratio graph whose edge translations are $b_e + u c_f$ and whose state $(1, 0; 1, 0)$ law is $\mu * (D_u)_* \tau$, where $D_u(y) = uy$. Every closed product class has entropy per step

$$h_G = \frac{H(w)}{\bar{p}} + \frac{H(v)}{\bar{q}}, \quad \bar{q} = \sum_j v_j q_j. \quad (\text{F.72l})$$

Proof. For the first system, at a decision state $(\varepsilon, 0)$ choose symbol i with probability w_i , emit the translation $b_e = \varepsilon a_i$, and move to $(\varepsilon s_i, p_i - 1)$. At a countdown state (ε, j) , $j \geq 1$, take the unique probability-one edge, emit 0, and move to $(\varepsilon, j - 1)$. Every edge acts on the represented coordinate by $z \mapsto \gamma z + b_e$. If $X \sim \mu$, the family of laws

$$\rho_{(\varepsilon, j)} = \text{Law}(\gamma^j \varepsilon X) \quad (\text{F.72m})$$

satisfies all graph-directed equations: countdown edges multiply by γ , while a decision edge gives

$$\varepsilon a_i + \gamma(\gamma^{p_i-1} \varepsilon s_i X) = \varepsilon g_i(X).$$

Contraction gives uniqueness of this finite graph-directed family, and in particular $\rho_{(1,0)} = \mu$.

At each decision epoch the next symbol has law (w_i) , independently of the orientation, and the number of common-ratio steps until the next decision is p_i . The strong law for these independent choices gives decision frequency $1/\bar{p}$ in every recurrent class. The conditional edge entropy is $H(w)$ at a decision state and 0 during countdown. Thus its stationary mean is $H(w)/\bar{p}$. The same construction and calculation apply to τ .

Take the Cartesian product of the two finite state spaces. At a state pair choose the two outgoing edges independently and assign probability $p_e p_f$, translation $b_e + u c_f$, and common contraction γ . Independent state-started variables X_s, Y_t obey the graph equation for $X_s + u Y_t$, so uniqueness identifies the projected state law and gives $\mu * (D_u)_* \tau$ at the initial state. For any stationary law on a closed irreducible product class, each coordinate projection is stationary for its own tower chain. Its one-step conditional edge entropy is the sum of the two coordinate conditional entropies, because the outgoing choices are independent. The decision-frequency calculation in each marginal therefore yields (F.72l). $\square$

Lemma IV.10.115 (*midpoint separation for regular common-ratio towers*).

Assume additionally in Lemma IV.10.114 that all g_i map one compact interval I into itself and that their images have pairwise disjoint interiors. After a fixed state-dependent translation of the tower coordinates, distinct length- n paths from any one state have translations separated by c^n for all sufficiently large n , with one $c \in (0, 1)$ independent of the state. The same assertion holds for the second regular IFS in Lemma IV.10.114.

Proof. Let x_0 be the midpoint of I . At state (ϵ, j) replace the represented coordinate $X_{\epsilon, j}$ by $X_{\epsilon, j} - c_{\epsilon, j}$, where $c_{\epsilon, j} = \gamma^j \epsilon x_0$. If an original edge $s \rightarrow t$ has translation b_e , its new translation is $b'_e = b_e + \gamma c_t - c_s$. A length- n path U therefore has

$$b'_U = b_U + \gamma^n c_{t(U)} - c_s. \quad (\text{F.72n})$$

First start at a decision state $(\epsilon, 0)$. A length- n tower path corresponds uniquely to the shortest original word $u = i_1 \cdots i_k$ for which $P(u) = p_{i_1} + \cdots + p_{i_k} \geq n$. Its exponent satisfies $n \leq P(u) \leq n + p_{\max} - 1$, and its terminal countdown is $P(u) - n$. Expanding (F.72n) gives

$$b'_U = \epsilon(g_u(x_0) - x_0). \quad (\text{F.72o})$$

The stopping words at a fixed n are prefix-free. Regularity of the original IFS implies that the intervals $g_u(I)$ for distinct stopping words have disjoint interiors. Their midpoints $g_u(x_0)$ are consequently separated by at least half the sum of their lengths, hence by at least $|I|\gamma^{n+p_{\max}-1}$. Formula (F.72o) proves the same separation for the new translations.

If the starting state has countdown $j > 0$, its first j steps are deterministic. When $n > j$, two different length- n paths first diverge after those steps, and their translation difference is γ^j times that of two paths of length $n - j$ from a decision state. The preceding bound is again at least $|I|\gamma^{n+p_{\max}-1}$. When $n \leq j$ there is only one path. Since the state set is finite, choose any $c < \gamma$; for all sufficiently large n the displayed lower bound exceeds c^n , uniformly over starting states. The proof for the second IFS is identical. $\square$

Theorem IV.10.116 (*commensurable regular convolutions outside a zero-dimensional set*).

Let μ and τ be self-similar probabilities of two regular affine IFSs with positive weights. Suppose every absolute contraction ratio in both IFSs is a positive integer power of the same $\gamma \in (0, 1)$. Then there is $E \subset \mathbb{R}$ with $\dim_H E = 0$ such that, for every $u \in \mathbb{R} \setminus E$, the convolution $\mu * (D_u)_* \tau$, $D_u(y) = uy$, is exact dimensional and

$$\dim(\mu * (D_u)_* \tau) = \min\{1, \dim \mu + \dim \tau\}. \quad (\text{F.72p})$$

Proof. Apply Lemma IV.10.114 to both IFSs and translate the tower coordinates as in Lemma IV.10.115. The conjugated product graph has edge translations affine in u ; for each fixed initial product state, its component path translations satisfy the hypotheses of Lemma IV.10.112. There are finitely many states. Apply that lemma on each compact interval $[-m, m]$ and take the countable union of its dimension-zero exceptional sets. Outside the resulting set E , every closed class of the product graph satisfies the graph-relative exponential-separation condition (GQ.3) for all large n . We apply the graph theorem to these conjugated graphs, on which separation has just been proved. The state translations used to identify their measures do not change dimensions; no invariance of raw path-translation separation under such translations is asserted.

The state pair $(1, 0; 1, 0)$ lies in a closed irreducible class: each individual reachable tower chain is finite and recurrent, and the product of their positive stationary laws is stationary with positive mass at every reachable state pair. A finite chain with a strictly positive stationary law has no transient state. Lemma IV.10.114 identifies its state-started law, up to the harmless coordinate translation, with $\mu * (D_u)_* \tau$. Its class entropy is the number h_G in (F.72l). Theorem IV.10.113 therefore gives

$$\dim(\mu * (D_u)_* \tau) = \min\left\{1, \frac{H(w)/\bar{p} + H(v)/\bar{q}}{\log(1/\gamma)}\right\}. \quad (\text{F.72q})$$

Apply Lemma IV.10.115 and Theorem IV.10.113 separately to the two component towers. Their initial laws have dimensions $\min\{1, H(w)/(\bar{p} \log(1/\gamma))\}$ and $\min\{1, H(v)/(\bar{q} \log(1/\gamma))\}$. The minima are inactive under regularity. Indeed disjoint interiors give $\sum_i \gamma^{p_i} \leq 1$; the log-sum inequality yields $H(w) \leq \bar{p} \log(1/\gamma)$, and similarly for v . Thus the numerator in (F.72q) divided by $\log(1/\gamma)$ is $\dim \mu + \dim \tau$, proving (F.72p) and exact dimensionality. $\square$

7. Consequence for analytic images

For the two regular commensurable self-similar measures in Theorem IV.10.116, define $\rho_u = \mu * (D_u)_* \tau$. The tower construction gives the precise graph entropy h_G in (F.72l), and Theorem IV.10.116 gives the actual parameter conclusion

$$\dim \rho_u = \min\{1, \dim \mu + \dim \tau\} \quad \text{for } u \notin E, \quad \dim_H E = 0. \quad (\text{F.73})$$

This removes the formerly unbound symbols in the abstract-to-application transition. Passing from (F.73) to every phase-typical scenery component of an analytic image requires a separate local-component comparison; it is not a consequence of the Frostman estimate alone.

Lemma IV.10.117 (*non-atomic affine sceneries are locally controlled by affine copies*).

Let μ be a non-atomic self-similar probability of a regular affine IFS on an interval. Let $x \in \text{supp } \mu$ and $r_n \downarrow 0$. Suppose the normalized full-space sceneries

$$\eta_n = S_{\log(1/r_n)}^* T_x^* \mu \quad (\text{F.73a})$$

converge vaguely to a non-atomic Radon measure η with $0 \in \text{supp } \eta$. Then, for every $R > 0$, $\eta|_{[-R,R]}$ is a finite sum of positive restrictions of affine images of μ . More precisely, there are affine maps $A_{R,j}(y) = a_{R,j}y + b_{R,j}$ with $a_{R,j} \neq 0$, intervals $J_{R,j}$ with $\mu(J_{R,j}) > 0$, and coefficients $c_{R,j} > 0$, for finitely many j , such that

$$\eta|_{[-R,R]} = \sum_j c_{R,j} (A_{R,j})_* (\mu|_{J_{R,j}})|_{[-R,R]}. \quad (\text{F.73b})$$

Consequently $\eta|_{[-R,R]}$ is absolutely continuous with respect to a finite sum of affine images of μ . If only the restricted sceneries $\eta_n^\square$ converge weakly to a non-atomic probability on $[-1, 1]$, the same conclusion holds for every $0 < R \leq 1$ (with restrictions interpreted in that space).

Proof. Apply a fixed affine change of coordinates so the source interval is $[0, 1]$. Put $r_* = \min_i |r_i| > 0$. At scale r_n , take the prefix-free stopping family $\mathcal{W}_n$ of minimal words u with $|r_u| \leq r_n$. Every stopping interval $g_u([0, 1])$ has length in $[r_*r_n, r_n]$, and these intervals have disjoint interiors. Hence at most $N_R = \lfloor 2R/r_* \rfloor + 3$ of them meet $B(x, Rr_n)$. The Bernoulli cylinder decomposition and non-atomicity at shared endpoints give the exact identity

$$\eta_n|_{[-R,R]} = \sum_{u: g_u([0,1]) \cap B(x, Rr_n) \neq \emptyset} \frac{w_u}{\mu(B(x, r_n))} (A_{n,u})_* (\mu|_{J_{n,u}})|_{[-R,R]}, \quad (\text{F.73c})$$

where $A_{n,u}(y) = (g_u(y) - x)/r_n$ and $J_{n,u} = A_{n,u}^{-1}([-R, R]) \cap [0, 1]$. The slopes of $A_{n,u}$ have absolute values in $[r_*, 1]$; the translations have absolute values at most $R + 1$ for the listed words.

List the at most N_R terms in spatial order and pad with zero terms. Pass to a subsequence on which every affine map converges to an affine A_j with nonzero slope, every interval $J_{n,j}$ converges by endpoints to an interval J_j , and every total term mass converges to a finite number $d_j \geq 0$. Finiteness follows from the vague convergence and a continuous cutoff equal to one on $[-R, R]$; positivity prevents cancellation.

If $\mu(J_j) > 0$, non-atomicity gives weak convergence of the normalized restrictions $\mu|_{J_{n,j}}/\mu(J_{n,j})$ to $\mu|_{J_j}/\mu(J_j)$. The corresponding term therefore tends to $d_j(A_j)_*(\mu|_{J_j})/\mu(J_j)$, restricted to $[-R, R]$. If $\mu(J_j) = 0$, any weak limit of the normalized restrictions is supported on the at most two endpoints of J_j : the interior of J_j has μ -mass zero, and the moving intervals eventually avoid every compact set outside J_j . A positive d_j would give an atom of η , which is impossible. Thus such terms have $d_j = 0$. Since η has no atoms at $\pm R$, passage to the limit in (F.73c) gives (F.73b) with $c_{R,j} = d_j/\mu(J_j)$ for the positive terms. The final absolute-continuity assertion follows because restriction preserves absolute continuity. For the restricted variant, repeat the argument with $R \leq 1$; weak convergence on $[-1, 1]$ and non-atomicity at $\pm R$ justify the same passage to the limit. $\square$

Lemma IV.10.118 (*zero-dimensional bad times for a non-atomic affine tangent*).

Let μ and τ satisfy the hypotheses of Theorem IV.10.116, and suppose $\dim \mu + \dim \tau \geq 1$. Let η be any non-atomic full-space tangent of μ obtained as in Lemma IV.10.117. There is a set $E_\eta \subset \mathbb{R}$ with $\dim_H E_\eta = 0$ such that, for every $t \notin E_\eta$, either $\epsilon \in \{-1, 1\}$, and every nonzero finite measure $\zeta \ll (D_\epsilon)_* S_t^* \eta$ supported on a bounded interval,

$$\dim(\tau * \zeta) = 1. \quad (\text{F.73d})$$

If η is only a non-atomic weak limit of restricted sceneries with $0 \in \text{supp } \eta$, the same conclusion holds for $t \geq 0$, either sign ϵ , and nonzero $\zeta \ll (D_\epsilon)_* S_t^* \eta$ supported in $[-1, 1]$.

Proof. Apply Lemma IV.10.117 for $R = 1, 2, \dots$. On each $[-R, R]$ the tangent is a finite sum of positive restrictions of affine copies $(A_{R,j})_* \mu$, with nonzero slopes $a_{R,j}$. Let E be the zero-dimensional exceptional set supplied by Theorem IV.10.116 with the order of the two measures chosen as τ, μ . For each such slope and $\epsilon \in \{-1, 1\}$ put

$$E_{R,j,\epsilon} = \{t \in \mathbb{R} : \epsilon e^t a_{R,j} \in E\}. \quad (\text{F.73e})$$

On each bounded time interval, $t \mapsto \epsilon e^t a_{R,j}$ is a bi-Lipschitz diffeomorphism onto its image. Hence $\dim_H E_{R,j,\epsilon} = 0$. The countable union $E_\eta = \bigcup_{R,j,\epsilon} E_{R,j,\epsilon}$ also has dimension zero.

Fix $t \notin E_\eta$ and a sign ϵ . Scaling and reflecting a term in (F.73b) multiplies its affine slope by ϵe^t ; normalization only multiplies the measure by a positive scalar. For every R, j , the convolution of τ with this scaled term is absolutely continuous with respect to a translate of $\tau * (D_{\epsilon e^t a_{R,j}})_* \mu$. The latter has dimension one by Theorem IV.10.116; therefore every positive restriction of the former also has dimension one. A finite sum has the same property, so $\tau * (D_\epsilon)_* S_t(\eta|_{[-R,R]})$ has dimension one whenever it is nonzero.

Given a finite $\zeta \ll (D_\epsilon)_* S_t^* \eta$ supported in a bounded interval, choose R large enough that its preimage under the scaling lies in $[-R, R]$. Then $\zeta \ll (D_\epsilon)_* S_t(\eta|_{[-R,R]})$, up to the positive normalization constant defining $S_t^* \eta$, and convolution with τ preserves absolute continuity. Thus $\tau * \zeta$ is a nonzero finite measure absolutely continuous with respect to a dimension-one measure and has dimension one, proving (F.73d). For a restricted tangent with $0 \in \text{supp } \eta$, every centered normalization window has positive mass. For $t \geq 0$, the window used by $S_t^* \eta$ on $[-1, 1]$ lies inside $[-1, 1]$ before zooming. Apply the restricted version of Lemma IV.10.117 with $R = 1$. Its finitely many affine pieces give the same exceptional-time construction and absolute-continuity argument. $\square$

Lemma IV.10.119 (*typical EFD states are affine restricted tangents*).

Let the regular affine measure μ generate a nontrivial EFD P continuously, and let $P = \int P_z dm(z)$ be its S_L^* -ergodic decomposition, with phase law m . For m -almost every z_0 and P_{z_0} -almost every full-space state η_0 , the normalized restriction $\eta_0^\square$ is a non-atomic weak limit of restricted sceneries of μ at scales tending to zero.

Proof. The restricted projection $P^\square$ is the continuous-time scenery law of μ on the compact space of probabilities on $[-1, 1]$. Choose one point x at which this generation holds. For every open set U with $P^\square(U) > 0$, the scenery orbit at x visits U at arbitrarily late times. Take a countable base of the restricted state space. A nested base-neighborhood argument shows that every element of $\text{supp } P^\square$ is a subsequential weak limit of restricted sceneries at

x with times tending to infinity. The complement of this closed support has zero $P^\square$ -mass. Since $P^\square = \int (\pi_\square)_* P_z dm(z)$, it has zero $(\pi_\square)_* P_z$ -mass for m -almost every z . Finally, a nontrivial EFD has non-atomic states almost surely by Lemma IV.10.7; the same is true in almost every component. Apply the restricted assertion of Lemma IV.10.117 to the resulting limits. $\square$

Lemma IV.10.120 (*phase-conditioned convolution saturation*).

Suppose μ and τ satisfy Theorem IV.10.116, with $\dim \mu + \dim \tau \geq 1$. Let P be the nontrivial EFD generated by μ . Fix $L > 0$ for which the S_L^* -ergodic components can be written

$$P_z = (S_{\ell(z)}^*)_* P_{z_0}, \quad 0 \leq \ell(z) < T_0, \quad (\text{F.73f})$$

as in Lemma IV.10.81. Let ν be a probability that generates P continuously and L -discretely generates $P_{z(x)}$ at ν -almost every x . If the time law $\lambda = \ell_*(z_* \nu)$ has positive Hausdorff dimension, then, for ν -almost every x and $P_{z(x)}$ -almost every η , every nonzero finite positive restriction ζ of either $\eta^\square$ or its reflection satisfies

$$\dim(\tau * \zeta) = 1. \quad (\text{F.73g})$$

The same holds after translating ζ .

Proof. Use the everywhere-equivariant version constructed in Lemma IV.10.81, not an arbitrary conditional version on Haar-null phases. Choose z_0 in the conull family of Lemma IV.10.119. If this changes the base in (F.73f), the time coordinate changes by a translation modulo T_0 ; on two intervals it is a translation, so positive lower Hausdorff dimension of its law is preserved. For P_{z_0} -almost every η_0 , its restricted projection is a non-atomic affine tangent of μ . Lemma IV.10.118 gives a set E_{η_0} of Hausdorff dimension zero outside which every positive restriction of $(S_t^* \eta_0)^\square$, and of its reflection, has dimension-one convolution with τ . Because $\dim \lambda > 0$, $\lambda(E_{\eta_0}) = 0$ for every such η_0 .

For completeness, the Fubini event can be taken to be

$$\mathcal{G}_\epsilon = \{(\eta_0, t) : \dim[\tau * (D_\epsilon)_*(S_t^* \eta_0)^\square] = 1\}, \quad \epsilon \in \{-1, 1\}. \quad (\text{F.73h})$$

It is universally measurable: convolution and normalized restriction are Borel on the origin-supported state space, and the lower Hausdorff dimension of a compactly supported finite measure is Borel when expressed through its lower local dimensions and countably many ball radii. Fubini therefore gives full $P_{z_0} \times \lambda$ -measure to both events. Every nonzero positive restriction is absolutely continuous with respect to the corresponding full scenery, so its convolution also has dimension one. Push the product law through (F.73f) and disintegrate over the phase $z(x)$ to obtain (F.73g). Translation does not change Hausdorff dimension. $\square$

Lemma IV.10.121 (*discrete component generation and finite affine phase*).

Let μ be a non-atomic Bernoulli self-similar probability for a regular affine IFS, and let P be its generated nontrivial EFD. For every $L > 0$, μ L -discretely generates an S_L^* -ergodic component of P at μ -almost every point. If $0 \neq \alpha \in \Sigma(P, S) \cap L^{-1}\mathbb{Z}$, the associated phase law θ_μ is a single atom.

If f is an orientation-preserving C^1 diffeomorphism on a connected interval containing $\text{supp } \mu$, then $\nu = f_*\mu$ generates P continuously and L -discretely generates an S_L^* -ergodic component at ν -almost every point. For the same nonzero resonant eigenvalue α , if in addition f is real analytic and not affine on that interval, then

$$\dim \theta_\nu = \dim \mu > 0. \quad (\text{F.73i})$$

Proof. By the pointwise ergodic theorem for the time- L map, for P -almost every state ξ , the discrete scenery orbit of ξ at the origin generates the S_L^* -ergodic component containing ξ . The quasi-Palm null-class equivalence transfers the Borel conull event of *generating some such component* to $T_y^*\xi$ for ξ -almost every y and P -almost every ξ . Thus a P -typical state L -discretely generates an ergodic component at almost every spatial point.

When the time- L map is not ergodic, use the canonical component family of Lemma IV.10.81; otherwise use the constant family P . On P -typical states these are the ordinary ergodic components. The canonical rotation identity transports them under every fixed real time shift, including shifts to Haar-null phases. If reflection is needed and $R_*P = P$, eigenspace simplicity gives $\phi(R\xi) = e_R\phi(\xi)$ with $e_R \in \{-1, 1\}$ for the primitive coordinate. Uniqueness of disintegration then gives $R_*P_z = P_{e_R z}$ for Haar-almost every z . Commutation with the flow and the everywhere rotation identity extend this equality to every z . Thus the affine transports below remain members of this fixed component family, not arbitrary versions on null phases.

For affine systems the limit diffeomorphisms in Lemma IV.10.69 are affine. Its converse local-equivalence assertion therefore gives a P -typical state ξ , an affine map A with nonzero slope, and a positive- μ interval J on which $\mu|_J$ is mutually absolutely continuous with a positive restriction of $A_*\xi$. Radon differentiation makes the two centered sceneries asymptotic at almost every common point, also along the discrete times nL . A fixed affine map changes the source time to $t + \log |A'|$ when comparing the image scenery at time t , and possibly reflects the states. Both operations conjugate the time- L map and take an ergodic component to an ergodic component. The continuous generated law on J is P by Theorem IV.10.70, so the affine image component is a component of P . Hence the set G of points at which μ L -generates a component of P has positive μ -measure.

For each IFS map $g_i(y) = r_i y + a_i$, the self-similar identity gives the domination

$$w_i(g_i)_*\mu \leq \mu. \quad (\text{F.73n})$$

This is an inequality, not an equality of restrictions: images may overlap. By Radon differentiation, the sceneries of $(g_i)_*\mu$ and μ are asymptotic at $(g_i)_*\mu$ -almost every point. Affine covariance therefore shows, for μ -almost every y ,

$$y \in G \implies g_i(y) \in G. \quad (\text{F.73o})$$

The fixed source-time shift $\log |r_i|$ preserves the property of generating an S_L^* -ergodic component. If $r_i < 0$, reflection is also involved. The continuous generation of μ , affine covariance, and domination (F.73n) show that $R_*P = P$, so reflection permutes the components of P .

Let $H = \pi^{-1}(G)$ in the one-sided Bernoulli coding space. Relation (F.73o) says $\sigma^{-1}H \subset H$ modulo the Bernoulli law. Shift-stationarity gives equal measures to these two sets, so they agree modulo null sets. Shift ergodicity and $\mu(G) > 0$ imply $\mu(G) = 1$.

Choose a unit-modulus flow eigenfunction ϕ_α . Since $R_*P = P$ whenever reflection is needed, $\phi_\alpha(R\xi) = c_R\phi_\alpha(\xi)$ for a constant $c_R \in \{-1, 1\}$: reflection commutes with the flow, the eigenfunction space is one-dimensional by Lemma IV.10.79, and R^2 is the identity. Put $c_+ = 1$ and $c_- = c_R$. The exact affine phase rule gives the following identity: the phase of $(g_i)_*\mu$ at $g_i(y)$ is the displayed affine transform of the phase of μ at y , and domination (F.73n) identifies it with the phase of μ at $g_i(y)$ by differentiation,

$$\phi_\mu(g_i(y)) = c_{\text{sgn } r_i} e(-\alpha \log |r_i|) \phi_\mu(y) = d_i \phi_\mu(y), \quad |d_i| = 1 \quad (\text{F.73p})$$

for μ -almost every y . Pull the phase back to the one-sided Bernoulli coding space and write $F(\omega) = \phi_\mu(\pi\omega)$. Then $F(\omega) = d_{\omega_1} F(\sigma\omega)$ almost surely. Iteration and independence of the first n symbols from the tail give

$$\mathbb{E}(F \mid \omega_1, \dots, \omega_n) = \left(\prod_{j=1}^n d_{\omega_j} \right) \mathbb{E}F. \quad (\text{F.73p1})$$

The left side converges almost surely to F by martingale convergence. Its modulus is the constant $|\mathbb{E}F|$, whereas $|F| = 1$. Hence $|\mathbb{E}F| = 1$, F is constant almost surely, and (F.73p) forces $d_i = 1$ for every positive-weight symbol. Thus θ_μ is a single atom. In particular, $\alpha \log |r_i| \in \mathbb{Z}$ for positive r_i , while $2\alpha \log |r_i| \in \mathbb{Z}$ for negative r_i . This independently verifies the affine instance of Lemma IV.10.71.

The C^1 scenery-covariance proof of Theorem IV.3.17 applies to discrete averages as well. In its larger-window construction choose the look-back parameter $a = kL$, with an integer k large enough for the fixed derivative and required windows at the point. Then the input times $nL - a = (n - k)L$ differ by only finitely many initial terms from the original discrete orbit, so have the same empirical component; an arbitrary real look-back time would not justify this. The limiting component consists of non-atomic states, so the normalization boundaries are null almost surely, and the same compact continuity-set argument works for the sequence nL . Its image is the component obtained by the fixed derivative at the point. Scalar dilation preserves P by flow invariance; no extra reflection arises because f preserves orientation. This proves both generation assertions for ν .

For the analytic conclusion, the phase rule becomes

$$\phi_\nu(f(x)) = e(-\alpha \log f'(x)) \phi_\mu(x). \quad (\text{F.73q})$$

The source phase is constant, and the exact-dimensional measure μ has dimension $\delta = \dim \mu$. By Lemma IV.10.92, away from a finite null critical set, $\log |f'|$ is covered by countably many bi-Lipschitz restrictions. Subdivide their image ranges into arcs shorter than $1/|\alpha|$. On each resulting piece, the map in (F.73q) is bi-Lipschitz up to a fixed circle

rotation. A countable positive decomposition of measures of lower Hausdorff dimension δ again has lower dimension δ ; the same holds for upper dimension here because each source restriction is exact dimensional. This proves (F.73i). $\square$

Theorem IV.10.122 (*non-affine analytic images of regular self-similar measures*).

Let

$$\Phi = \{g_i(x) = r_i x + a_i\}_{i=1}^m$$

be a regular affine IFS on an interval, and let μ be a non-atomic self-similar probability for Φ with positive weights. Let f be a real-analytic diffeomorphism on one connected open interval containing $\text{supp } \mu$ which is not affine on $\text{supp } \mu$. Then for every Pisot number $\beta > 1$,

$$f_*\mu$$

is pointwise β -normal.

Proof. If f reverses orientation, replace μ by its reflected image $R_*\mu$ and f by $f \circ R$. The new system is again regular affine, has the same absolute contraction ratios, and has the same image measure. Hence assume $f' > 0$. Lemma IV.10.90 proves the theorem in the two nonresonant arithmetic cases. It remains to consider

$$|r_i| = \gamma^{p_i}, \quad p_i \in \mathbb{N}, \quad 0 < \gamma < 1, \quad \frac{\log \beta}{\log(1/\gamma)} \in \mathbb{Q}. \quad (\text{F.73j})$$

Let P be the nontrivial EFD generated by μ and put $L = \log \beta$ and $\delta = \dim \mu = \dim P > 0$. The same P is generated by $\nu = f_*\mu$, by Lemma IV.10.121. If S_L^* is ergodic on P , the spectral-exclusion theorem IV.10.63 already proves the assertion. Assume it is not ergodic.

Choose the primitive S_L^* phase coordinate of Lemma IV.10.81. By Lemma IV.10.121, ν L -discretely generates an ergodic component $P_{z(x)}$ at ν -almost every x , and its phase-time law has Hausdorff dimension $\delta > 0$. For the commensurate γ , choose the regular homogeneous Pisot test measures τ_N of Theorem IV.10.56. They have contraction ratios that are integer powers of γ and $\dim \tau_N \rightarrow 1$. Fix N large enough that

$$\dim \tau_N + \delta > 1. \quad (\text{F.73k})$$

Theorem IV.10.116 and Lemmas IV.10.117–IV.10.120 now give, for ν -almost every x and $P_{z(x)}$ -almost every restricted scenery η ,

$$\dim(\tau_N * (\eta|_J)) = 1 \quad \text{whenever } \eta(J) > 0. \quad (\text{F.73l})$$

The assertion also holds for a translated or reflected positive restriction. Intersect these count sets over the countably many sufficiently large N .

Fix such a point x and a subsequential T_β -orbit law ρ of x . The exact-time version of the orbit/local-scene representation, Theorem IV.10.40, writes ρ as a barycenter of positive translated restrictions of sceneries with marginal $P_{z(x)}$. Each scenery is exact dimensional of dimension δ ; consequently $\underline{\dim}_H \rho \geq \delta > 0$. In particular ρ is non-atomic, so Lemma IV.10.41 makes it T_β -invariant. Convolving its barycenter with τ_N and using (F.73l) gives

$$\underline{\dim}_H(\tau_N * \rho) = 1 \quad \text{for every sufficiently large } N. \quad (\text{F.73m})$$

Indeed a Borel set of dimension less than one is null for almost every positive component convolution, hence is null for their barycenter.

If $\underline{\dim}_H \rho < 1$, the resonance conclusion of Theorem IV.10.56 contradicts (F.73m) for all sufficiently large N . Thus $\underline{\dim}_H \rho = 1$. Lemma IV.10.29 identifies ρ with the unique Parry probability λ_β . Every subsequential orbit law is λ_β , so the full orbit empirical sequence converges to it. $\square$

10.10 Host-type invariant-measure normality and smooth images

1. The Host-type conclusions

Write

$$T_\gamma(x) = \gamma x \pmod{1}. \quad (27.1)$$

For $\beta, \gamma > 1$, write $\beta \sim \gamma$ if

$$\frac{\log \beta}{\log \gamma} \in \mathbb{Q}. \quad (27.2)$$

We shall prove that if β is Pisot, $\beta \sim \gamma$, and μ is an ergodic T_γ -invariant probability of positive entropy, then μ is pointwise β -normal. The conclusion persists for every $g_*\mu$ with $g \in \text{Diff}^2(\mathbb{R})$. The two theorems are stated again, immediately with their proofs, after the necessary cylinder and scenery lemmas.

2. Arbitrary-base cylinder geometry

This section closes the dimension input for every $\gamma > 1$, not only Pisot γ .

Let $d_\gamma(1) = a_1 a_2 \dots$ be the quasi-greedy expansion of 1. A finite word $w = w_1 \dots w_n$ is γ -admissible if every suffix is lexicographically no larger than the corresponding prefix of $d_\gamma(1)$. Let $I(w)$ be the order- n cylinder of points whose greedy γ -expansion begins with w .

Definition IV.10.123 (*full γ -cylinder*).

A word w of length n is *full* if

$$T_\gamma^n I(w) = [0, 1). \quad (27.4)$$

Equivalently,

$$|I(w)| = \gamma^{-n}. \quad (27.5)$$

Lemma IV.10.124 (*full-word appendability*).

A word w is full if and only if every admissible finite word v may be appended to w .

Proof. On $I(w)$, the map T_γ^n is affine of slope γ^n . Its image is an interval starting at zero. It is all of $[0, 1)$ exactly when every admissible continuation occurs. $\square$

Lemma IV.10.125 (*full-cylinder distribution*).

For every admissible word w of length n , either w is full, or there is a maximal full prefix u and the remaining suffix is a prefix of $d_\gamma(1)$. Moreover, if a nonzero final digit of such a suffix is lowered by one, the resulting word is full.

Consequently, among every $n + 1$ consecutive order- n cylinders in left-to-right order there is a full cylinder.

Proof. Failure of fullness means that some continuation is forbidden. By the lexicographic criterion, after the last full prefix the only obstruction is following the extremal word $d_\gamma(1)$; hence the suffix is one of its prefixes.

If the last positive digit is decreased, the resulting word becomes strictly smaller than the extremal prefix at its first changed position. Every admissible continuation is then allowed, so Lemma IV.10.124 gives fullness.

Moving left through consecutive cylinders successively lowers one of the at most n suffix digits; after at most n steps a full word is reached. $\square$

Corollary IV.10.126 (*universal cylinder-neighbor bound*).

Every ball B of radius γ^{-n} satisfies

$$\#\{J : J \text{ is an order-}n \text{ } \gamma\text{-cylinder and } J \cap B \neq \emptyset\} \leq 4(n + 1). \quad (27.6)$$

Proof. Every full order- n cylinder has length γ^{-n} . By Lemma IV.10.125, in any string of $n + 1$ consecutive cylinders one is full. A ball of diameter $2\gamma^{-n}$ cannot contain more than four full cylinders, together with the blocks of at most n nonfull neighbors adjacent to each. $\square$

3. A universal neighbor-mass lemma**Lemma IV.10.127** (*universal neighbor-mass estimate*).

Let ρ be any probability on $[0, 1]$ and let $\varepsilon > 0$. For ρ -a.e. x , for all sufficiently large n , every order- n γ -cylinder J meeting $B(x, \gamma^{-n})$ satisfies

$$\rho(J) \leq e^{\varepsilon n} \rho(I_n(x)), \quad (27.7)$$

where $I_n(x)$ is the cylinder containing x .

Proof. Let E_n be the union of cylinders I for which some cylinder J meeting the γ^{-n} -neighborhood of I satisfies

$$\rho(J) > e^{\varepsilon n} \rho(I).$$

By Corollary IV.10.126 each J can be paired with at most $C(n + 1)$ such I , and conversely the neighbor graph has degree $O(n)$. Summing the smaller masses gives

$$\rho(E_n) \leq C(n + 1)e^{-\varepsilon n} \sum_J \rho(J) \leq C(n + 1)e^{-\varepsilon n}. \quad (27.8)$$

The series converges, so Borel–Cantelli proves the claim. $\square$

Exact dimension for ergodic invariant measures. **Theorem IV.10.128** (*exact dimension for T_γ -invariant measures*).

Let ρ be an ergodic T_γ -invariant probability. Then ρ is exact dimensional and

$$\dim \rho = \frac{h_\rho(T_\gamma)}{\log \gamma}. \quad (27.9)$$

Proof. The order-one γ -cylinder partition is finite and generating modulo endpoints. The finite-partition Shannon–McMillan–Breiman theorem in Section 10.1 gives

$$-\frac{1}{n} \log \rho(I_n(x)) \longrightarrow h_\rho(T_\gamma) \quad (27.10)$$

for ρ -a.e. x .

Every cylinder $I_n(x)$ has length at most γ^{-n} , so

$$I_n(x) \subset B(x, \gamma^{-n}). \quad (27.11)$$

Conversely, $B(x, \gamma^{-n})$ meets at most $4(n+1)$ order- n cylinders by Corollary IV.10.126. By Lemma IV.10.127, for every $\varepsilon > 0$ and eventually,

$$\rho(B(x, \gamma^{-n})) \leq 4(n+1)e^{\varepsilon n} \rho(I_n(x)). \quad (27.12)$$

Equations (27.10)–(27.12) squeeze the local dimension at the discrete radii γ^{-n} . Consecutive radii have fixed ratio γ , so the same limit holds for all $r \downarrow 0$. Let $\varepsilon \downarrow 0$. $\square$

4. Future conditionals in the natural extension

Let μ be T_γ -invariant and ergodic with entropy

$$h = h_\mu(T_\gamma) > 0. \quad (27.13)$$

Let $D = \{0, \dots, [\gamma] - 1\}$ and let $X_\gamma \subset D^\mathbb{N}$ be the closure of the greedy γ -expansions. The coding

$$\Pi(a_1 a_2 \dots) = \sum_{j \geq 1} a_j \gamma^{-j}$$

intertwines the one-sided shift with T_γ outside the countable set of coding endpoints. Since an ergodic invariant measure of positive entropy is non-atomic, that endpoint set is μ -null. Lift μ to the one-sided symbolic system, then take its two-sided natural extension $(\Omega, \sigma, \hat{\mu})$; thus $\omega = (\omega_j)_{j \in \mathbb{Z}}$ is a two-sided admissible digit sequence. Write

$$x(\omega) = \Pi(\omega_1 \omega_2 \dots), \quad \mathcal{P}^- = \sigma(\omega_j : j \leq 0).$$

Disintegrate $\hat{\mu}$ over the *symbolic* past $\mathcal{P}^-$. The conditional law of $(\omega_1, \omega_2, \dots)$ is ν_ω , and $\mu_\omega = \Pi_* \nu_\omega$ is the corresponding measure on $[0, 1]$. These conditionals are not taken over the past *points* $T_\gamma^{-j}x$: one such point already determines x , and would make the conditional law a Dirac mass.

Lemma IV.10.129 (*future cylinder exponent*).

For $\hat{\mu}$ -a.e. ω ,

$$-\frac{1}{n} \log \mu_\omega(I_n(x(\omega))) \longrightarrow h. \quad (27.14)$$

Proof. Let

$$p(\omega) = \hat{\mu}(\omega_1 \mid \mathcal{P}^-)(\omega)$$

be the conditional probability of the realized next digit, using a version defined on a common conull set. The chain rule for conditional probabilities gives, for every n on that set,

$$\nu_\omega([\omega_1 \cdots \omega_n]) = \prod_{j=0}^{n-1} p(\sigma^j \omega). \quad (27.15)$$

Indeed, after the first j digits have been revealed, the conditioning algebra is $\sigma(\omega_k : k \leq j)$; shifting by j identifies its next-digit conditional probability with $p(\sigma^j \omega)$. The information function $-\log p$ is integrable and

$$\int -\log p \, d\hat{\mu} = H_{\hat{\mu}}(\omega_1 \mid \mathcal{P}^-) = h_\mu(T_\gamma).$$

The first equality is the conditional-entropy identity; the second follows by taking the decreasing limit of finite-past conditional entropies, which is the entropy rate of the finite generating digit partition. Birkhoff applied to (27.15) proves the symbolic-cylinder exponent. Coding endpoints have zero conditional mass for $\hat{\mu}$ -almost every past, so the symbolic cylinder and $I_n(x(\omega))$ have the same μ_ω -mass. This gives (27.14). $\square$

Corollary IV.10.130 (*future conditionals are exact dimensional*).

For $\hat{\mu}$ -a.e. ω , the measure μ_ω is non-atomic and exact dimensional with

$$\dim \mu_\omega = \frac{h}{\log \gamma}. \quad (27.16)$$

Proof. Disintegration of the conull set in Lemma IV.10.129 shows that, for $\hat{\mu}$ -almost every symbolic past, the limit (27.14) holds for μ_ω -almost every coded future point. Since $h > 0$, its order- n cylinder masses tend to zero. Thus μ_ω has no atoms. For each such past, apply the cylinder/neighbor argument of Theorem IV.10.128 with $\rho = \mu_\omega$: the universal Lemma IV.10.127 supplies (27.12), and (27.14) supplies the cylinder exponent in place of (27.10). The two bounds yield local dimension $h / \log \gamma$ at μ_ω -almost every point. $\square$

Consequence. The original invariant measure μ is itself exact dimensional with the same dimension

$$\delta = \frac{h}{\log \gamma} > 0. \quad (27.17)$$

5. The discrete scenery factor

For ω in the symbolic natural extension, the past of $\sigma^{-n}\omega$ determines the conditional future measure before the next n digits are revealed. Define the normalized measure

$$\sigma_n(\omega) = S_{n \log \gamma}^* T_{x(\sigma^{-n}\omega)}^* \mu_{\sigma^{-n}\omega}. \quad (27.18)$$

The integer-base construction makes the consistency mechanism transparent. It is worth proving separately: the argument below uses the fact that every integer-base digit cylinder is full and therefore does not, by itself, prove the arbitrary-base statement that follows it.

Lemma IV.10.131 (*integer-base conditional Radon factor*).

Suppose $\gamma = b \in \{2, 3, \dots\}$ and $h > 0$. For $\hat{\mu}$ -almost every ω , the measures $\sigma_n(\omega)$ of (27.18) converge vaguely to a member $\vartheta(\omega)$ of M_1^* . Moreover,

$$\vartheta(\sigma\omega) = S_{\log b}^* \vartheta(\omega), \quad (T_{x(\omega)} \mu_\omega)|_{[-x(\omega), 1-x(\omega)]} = c(\omega) \vartheta(\omega)|_{[-x(\omega), 1-x(\omega)]} \quad (27.18a)$$

for a positive finite scalar $c(\omega)$. Hence the law of $\vartheta(\omega)$ is an ergodic $S_{\log b}^*$ -invariant distribution.

Proof. Work on a shift-invariant conull set on which all future conditional probabilities below are defined. Put

$$x_{-n} = x(\sigma^{-n}\omega), \quad q_n = \nu_{\sigma^{-n}\omega}([\omega_{-n+1} \cdots \omega_0]), \quad q_0 = 1.$$

The realized cylinder has positive conditional probability for almost every ω , simultaneously for all n , by disintegration. Define an *unnormalized* centered measure on $\mathbb{R}$ by

$$\tau_n(A) = q_n^{-1} \mu_{\sigma^{-n}\omega}(x_{-n} + b^{-n}A). \quad (27.18b)$$

Its support lies in the closed interval

$$D_n = [-b^n x_{-n}, b^n(1 - x_{-n})]. \quad (27.18c)$$

The block $\omega_{-m+1} \cdots \omega_{-n}$, for $m > n$, specifies one integer-base cylinder of the conditional future at time $-m$. On that cylinder the affine map T_b^{m-n} takes the point coordinate onto $[0, 1]$. Conditional consistency, first on future symbolic cylinders and then on all Borel sets, says that conditioning on the realized block gives $\mu_{\sigma^{-n}\omega}$. Its conditional probability is q_m/q_n . After the spatial rescaling in (27.18b), this proves the exact measure identity

$$\tau_m|_{D_n} = \tau_n|_{D_n}. \quad (27.18d)$$

There is no competing branch inside the interior of D_n , since distinct integer-base cylinders have disjoint interiors. Their endpoints have zero mass: by Corollary IV.10.130, every conditional here is non-atomic. Thus (27.18d) also holds on the closed interval.

Writing $x = x(\omega)$, direct expansion of the digits gives

$$b^n x_{-n} = x + \sum_{j=0}^{n-1} b^j \omega_{-j}, \quad b^n(1 - x_{-n}) = 1 - x + \sum_{j=0}^{n-1} b^j(b - 1 - \omega_{-j}). \quad (27.18e)$$

Both endpoints of D_n therefore move outward monotonically. Each sum on the right diverges almost surely: under an ergodic positive-entropy digit law, neither $\omega_0 = 0$ nor $\omega_0 = b - 1$ has probability one, and Birkhoff gives infinitely many strictly positive terms in each sum. Consequently $D_n \uparrow \mathbb{R}$. Equation (27.18d) now defines a locally finite Radon measure τ by stabilization on every compact interval. In particular $\tau|_{D_0} = T_x \mu_\omega$. As x belongs to the support of μ_ω almost surely, $0 \in \text{supp } \tau$ and $0 < \tau([-1, 1]) < \infty$.

For all sufficiently large n , the normalizing interval $[-1, 1]$ lies inside D_n . Hence $\sigma_n = \tau_n^*$ converges vaguely, indeed stabilizes on every fixed compact set after this common normalization, to $\vartheta = \tau^*$. The restriction identity in (27.18a) holds with $c(\omega) = \tau([-1, 1])$.

Finally let $p(\omega)$ be the conditional probability of the realized digit ω_1 given the past. Conditional chain consistency gives

$$q_n(\sigma\omega) = q_{n-1}(\omega) p(\omega) \quad (n \geq 1).$$

The definitions in (27.18b), with the centers at the corresponding shifted coordinates, therefore imply

$$\tau(\sigma\omega) = p(\omega)^{-1} S_{\log b} \tau(\omega).$$

Normalization removes the scalar and proves the first identity in (27.18a). The natural extension is ergodic, so its measurable factor has an ergodic invariant law. $\square$

Lemma IV.10.132 (*integer-base quasi-Palm and generation*).

Under the hypotheses of Lemma IV.10.131, let $P_0 = \vartheta_* \hat{\mu}$ and $L = \log b$. Then P_0 is quasi-Palm in the sense of Definition IV.2.17 and ergodic under S_L^* . The roof average

$$P = \frac{1}{L} \int_0^L (S_t^*)_* P_0 dt \quad (27.18f)$$

is a nontrivial EFD of dimension h/L , and for $\hat{\mu}$ -almost every ω the measure μ_ω generates P at μ_ω -almost every point.

Proof. First, $\vartheta(\omega)$ is exact dimensional of dimension h/L . Indeed, on each D_n in (27.18c) it is a positive multiple of the rescaled conditional $\mu_{\sigma^{-n}\omega}$, by (27.18b)–(27.18d). Corollary IV.10.130 applies to every one of these countably many conditionals, while $D_n \uparrow \mathbb{R}$. In particular P_0 -typical states are non-atomic and nontrivial.

We prove the quasi-Palm assertion with its quantifiers. Fix a Borel set $A \subset M_1^*$ with $P_0(A) = 1$. Intersect the conull symbolic set on which $\vartheta(\eta) \in A$ with all its integer shifts and with the conull sets used in Lemma IV.10.131; call the result E . By disintegration over the past, for $\hat{\mu}$ -almost every past the conditional future gives full mass to E .

Fix such a typical ω and $n \geq 0$, and put $\eta = \sigma^{-n}\omega$. Choose a conditional future η' with the same digits as η at indices $j \leq 0$, and put $z = x(\eta') - x(\eta)$. At every earlier time $-m$, the two paths have the same conditioning past and the same realized digit block through time 0. Their denominators q_m in (27.18b) are therefore equal, whereas their rescaled centers differ by exactly z . Taking the stabilized Radon limits gives

$$\tau(\eta') = T_z \tau(\eta), \quad \vartheta(\eta') = T_z^* \vartheta(\eta). \quad (27.18g)$$

The factor identity in (27.18a) and the commutation of translation with scaling now give

$$T_{b^n z}^* \vartheta(\omega) = S_{nL}^* T_z^* \vartheta(\eta) = \vartheta(\sigma^n \eta') \in A \quad (27.18h)$$

for conditional-almost every choice of η' . As η' varies, the law of $b^n z$ is the image of μ_η under $u \mapsto b^n(u - x(\eta))$. By (27.18b), this law is a positive multiple of $\vartheta(\omega)|_{D_n}$. Hence $T_y^* \vartheta(\omega) \in A$ for $\vartheta(\omega)$ -almost every $y \in D_n$. Since $D_n \uparrow \mathbb{R}$, the assertion holds for $\vartheta(\omega)$ -almost every $y \in \mathbb{R}$. Replacing A by $\bigcap_{k \geq 0} (S_{kL}^*)^{-1} A$ shows simultaneously that $S_{kL}^* T_y^* \vartheta(\omega) \in A$ for every $k \geq 0$ at almost every permitted y . This proves the forward full-set implication of Definition IV.2.17 and its discrete time- L version.

For the converse null-class implication, suppose the recentered sampling law from $[-1, 1]$ gives a Borel set A zero mass. At time $n = 0$, (27.18g) says that replacing the future of ω by an independent draw from its conditional law sends $\vartheta(\omega)$ to $T_y^* \vartheta(\omega)$ with $y \in D_0 \subset [-1, 1]$. The resampled two-sided sequence has law $\hat{\mu}$ by the defining property of conditional probability. The conditional law of y is proportional to $\vartheta(\omega)|_{D_0}$. Thus zero recentered mass on $[-1, 1]$ implies $\hat{\mu}\{\omega : \vartheta(\omega) \in A\} = 0$. Together with the forward implication, this gives the equivalence of null classes required by Definition IV.2.17.

The law P_0 is ergodic by Lemma IV.10.131. The constant-roof suspension of this ergodic discrete system is ergodic. Lemmas IV.2.50–IV.2.51 show that scaling and mixing preserve quasi-Palm; hence its roof law P is quasi-Palm, scale-invariant, and ergodic, so it is an EFD. Its dimension is h/L , since every P_0 -typical state is exact dimensional with that dimension, and scaling preserves local dimension.

For completeness, let $G \subset M_1^*$ be the full- P_0 -measure set of states whose S_L^* -orbit is generic for P_0 for a fixed countable convergence-determining family of bounded continuous tests. The quasi-Palm conclusion with $A = G$ shows that $T_y^* \vartheta(\omega) \in G$ for $\vartheta(\omega)$ -almost every y . On D_0 the measure $\vartheta(\omega)$ is proportional to $T_{x(\omega)} \mu_\omega$; therefore, for μ_ω -almost every $u \in (0, 1)$, $T_{u-x(\omega)}^* \vartheta(\omega) \in G$. The two measures agree up to a scalar on a neighborhood of u , so their centered sceneries agree for every sufficiently small radius. Thus μ_ω discretely generates P_0 at u .

Finally split a time interval into blocks $[jL, (j+1)L]$ and integrate each orbit test over its phase $0 \leq s < L$. For non-atomic states the map $\nu \mapsto \int_0^L F(S_s^* \nu) ds$ is continuous at every state having no mass on the relevant zoom boundaries; here this holds P_0 -almost surely. The discrete empirical convergence and the bounded, almost-everywhere-continuous mapping theorem therefore give continuous-time empirical convergence to (27.18f). This is generation of P at μ_ω -almost every u . $\square$

For a noninteger γ , a maximal-digit branch of T_γ need not map onto $[0, 1]$. Thus the *whole-support* windows (27.18c) need not give the stabilization identity (27.18d). The correct windows are the convex hulls of the supports of the *conditional* future measures. Their nesting uses only the disjointness, up to endpoints, of the one-digit γ -cylinders; no cylinder is required to be full.

Lemma IV.10.133 (*existence of the discrete scenery factor*).

For $\hat{\mu}$ -a.e. ω , $\sigma_n(\omega)$ converges vaguely to a normalized Radon measure $\vartheta(\omega) \in M_1^*$, and

$$\vartheta(\sigma\omega) = S_{\log \gamma}^* \vartheta(\omega). \quad (27.19)$$

The law

$$P_0 = \vartheta_* \hat{\mu} \quad (27.20)$$

is $S_{\log \gamma}$ -invariant and ergodic.

Proof. Work on a shift-invariant conull set on which all future conditionals are non-atomic, all realized conditional digits have positive probability, and $x(\omega)$ belongs to the support of μ_ω . The last assertion follows by disintegration. For $n \geq 0$ put

$$x_n = x(\sigma^{-n}\omega), \quad q_n = \nu_{\sigma^{-n}\omega}([\omega_{-n+1} \cdots \omega_0]), \quad q_0 = 1, \quad (27.19a)$$

and define the unnormalized centered measure

$$\tau_n(A) = q_n^{-1} \mu_{\sigma^{-n}\omega}(x_n + \gamma^{-n}A). \quad (27.19b)$$

The quantity q_n is positive almost surely, simultaneously for all n . Let $[l_n, u_n]$ be the smallest closed interval containing $\text{supp } \mu_{\sigma^{-n}\omega}$, and set

$$D_n = \gamma^n([l_n, u_n] - x_n). \quad (27.19c)$$

The endpoints are measurable, since for every rational $a \in (0, 1)$, $\{l_n < a\} = \{\mu_{\sigma^{-n}\omega}([0, a]) > 0\}$ and $\{u_n > a\} = \{\mu_{\sigma^{-n}\omega}((a, 1]) > 0\}$.

We prove exact consistency. Write $a = \omega_{-n}$, and let $A_a(y) = (a + y)/\gamma$. The branch interval $I(a)$ of points with first digit a has interior disjoint from all other first-digit branch intervals. Conditional consistency of the symbolic future, followed by the coding map, gives the Borel-measure identity

$$\mu_{\sigma^{-(n+1)}\omega}|_{I(a)} = p(\sigma^{-(n+1)}\omega)(A_a)_* \mu_{\sigma^{-n}\omega}, \quad (27.19d)$$

where p is the realized next-digit conditional probability from (27.15). Branch endpoints carry no conditional mass, by Corollary IV.10.130. Moreover $x_{n+1} = A_a(x_n)$ and $q_{n+1} = p(\sigma^{-(n+1)}\omega)q_n$. The image $A_a([l_n, u_n])$ lies in the closed branch interval $I(a)$ and, by (27.19d), in $[l_{n+1}, u_{n+1}]$. Scaling about the corresponding centers therefore shows $D_n \subset D_{n+1}$. On the smaller window the other first-digit branches have no mass, so (27.19d) yields the *exact* restriction identity

$$\tau_{n+1}|_{D_n} = \tau_n|_{D_n}. \quad (27.19e)$$

These windows exhaust the line. Conditional non-atomicity implies that $x(\omega)$ is strictly between the two endpoints of the support hull of μ_ω almost surely: conditionally on the past, either endpoint is a singleton of zero mass. Hence, for some $\varepsilon > 0$, the event

$$E_\varepsilon = \{\omega : l_0 + \varepsilon < x(\omega) < u_0 - \varepsilon\}$$

has positive $\hat{\mu}$ -measure. Ergodicity of the natural extension implies that almost every negative orbit visits E_ε infinitely often. At each such time n , $[-\varepsilon\gamma^n, \varepsilon\gamma^n] \subset D_n$. Since the D_n are nested, $D_n \uparrow \mathbb{R}$.

It follows from (27.19e) that the τ_n stabilize on every compact interval and define a locally finite Radon measure $\tau(\omega)$ with $\tau|_{D_n} = \tau_n|_{D_n}$. The point 0 is in its support because 0 is in the support of $\tau_0 = T_{x(\omega)}\mu_\omega$. Thus $0 < \tau([-1, 1]) < \infty$, and, once $[-1, 1] \subset D_n$,

$$\sigma_n(\omega) = \tau_n^* \longrightarrow \vartheta(\omega) := \tau(\omega)^*$$

vaguely, in fact by stabilization on each compact interval.

Lastly, the chain rule gives $q_n(\sigma\omega) = q_{n-1}(\omega)p(\omega)$ for $n \geq 1$. The centers satisfy $x(\sigma^{-n}\sigma\omega) = x(\sigma^{-(n-1)}\omega)$; hence (27.19b) gives $\tau_n(\sigma\omega) = p(\omega)^{-1}S_{\log \gamma}\tau_{n-1}(\omega)$. Passing to the stabilized limit and normalizing proves (27.19). The measurable factor of the ergodic natural extension has an invariant ergodic law, proving (27.20). $\square$

Lemma IV.10.134 (*future conditional versus scenery*).

For $\hat{\mu}$ -a.e. ω , with D_0 as in (27.19c) and $c(\omega) = \tau(\omega)([-1, 1])$, one has the exact Borel-measure identity

$$T_{x(\omega)}\mu_\omega = c(\omega) \vartheta(\omega)|_{D_0}. \quad (27.21)$$

In particular the centered future conditional is absolutely continuous with respect to $\vartheta(\omega)$ with a bounded density, and a positive restriction of $\vartheta(\omega)$ is proportional to it.

Proof. At $n = 0$, (27.19b) says $\tau_0 = T_{x(\omega)}\mu_\omega$. The exact stabilization (27.19e) and $D_0 \subset D_n$ give $\tau|_{D_0} = \tau_0|_{D_0}$. The measure τ_0 is supported on D_0 by definition of that interval. Divide by $c(\omega)$ to obtain (27.21). Since 0 is an interior point of D_0 almost surely, this is a positive restriction, not merely a null-set comparison. $\square$

Corollary IV.10.135 (*continuous EFD generated by future conditionals*).

Put $L = \log \gamma$ and

$$P = \frac{1}{L} \int_0^L (S_t^*)_* P_0 dt. \quad (27.22)$$

Then P is a nontrivial EFD of dimension

$$\dim P = \delta = \frac{h}{\log \gamma}. \quad (27.23)$$

For $\hat{\mu}$ -a.e. ω , μ_ω continuously generates P , and the original μ has the same exact dimension δ .

Proof. We first establish the discrete quasi-Palm property, including its null-class direction. Let $A \subset M_1^*$ be Borel with $P_0(A) = 1$. Take a shift-invariant conull set $E \subset \Omega$ on which $\vartheta(\eta) \in A$ and all the identities of Lemmas IV.10.133–IV.10.134 hold. Disintegrating $\hat{\mu}$ over the past shows that, for almost every past, its conditional future gives full mass to E .

Fix a typical ω and $n \geq 0$, and put $\eta = \sigma^{-n}\omega$. Let η' have the same digits as η at all indices $j \leq 0$, but choose its future according to the conditional law ν_η . Write $z = x(\eta') - x(\eta)$. At every time $-m$ the two sequences have the same conditioning past and the same realized block through time 0. Their numbers q_m in (27.19a) are therefore equal, while their centers differ by $\gamma^{-m}z$. From (27.19b), followed by stabilization, we obtain

$$\tau(\eta') = T_z\tau(\eta), \quad \vartheta(\eta') = T_z^*\vartheta(\eta). \quad (27.22a)$$

Using the factor identity (27.19) n times gives

$$T_{\gamma^n z}^* \vartheta(\omega) = S_{nL}^* T_z^* \vartheta(\eta) = \vartheta(\sigma^n \eta') \in A \quad (27.22b)$$

for conditional-almost every η' . Under this sampling, the law of $\gamma^n z$ is $q_n \tau(\omega)|_{D_n}$ by (27.19b) and (27.19e). Thus $T_y^* \vartheta(\omega) \in A$ for $\vartheta(\omega)$ -almost every $y \in D_n$. Since $D_n \uparrow \mathbb{R}$, this holds for $\vartheta(\omega)$ -almost every $y \in \mathbb{R}$.

Conversely, suppose the recentered-sampling law gives A zero mass: for P_0 -almost every ϑ , one has $T_y^* \vartheta \notin A$ for ϑ -almost every y . At $n = 0$ in (27.22a), resampling the future of a typical η has unconditional law $\hat{\mu}$, whereas the conditional law of z is μ_η centered at $x(\eta)$ and is proportional to $\vartheta(\eta)|_{D_0}$ by (27.21). Therefore $P_0(A) = 0$. Applying these two implications to complements gives precisely the null-class equivalence in Definition IV.2.17, with discrete time step L .

Ergodicity of P_0 under S_L^* is Lemma IV.10.133. Consequently its constant-roof suspension (27.22) is invariant and ergodic under the continuous scaling flow. Scaling and averaging preserve the quasi-Palm null classes by Lemmas IV.2.50–IV.2.51; hence P is an EFD. It is nontrivial and has dimension δ : on every D_n , $\vartheta(\omega)$ is a positive multiple of a scaled copy of $\mu_{\sigma^{-n}\omega}$ by (27.19b)–(27.19e). These intervals exhaust $\mathbb{R}$, and Corollary IV.10.130 gives exact dimension $\delta > 0$ to each conditional in this countable family. Scaling preserves local dimension, so the same is true for P -almost every measure.

It remains to prove generation rather than infer it merely from factor ergodicity. Choose a countable convergence-determining family (F_j) of bounded continuous tests on the compact restricted measure space. For $\nu \in M_1^*$ write

$$F_j^0(\nu) = F_j(\nu \square), \quad \bar{F}_j(\nu) = \frac{1}{L} \int_0^L F_j((S_s^* \nu) \square) ds. \quad (27.22c)$$

These are bounded Borel functions. Birkhoff's theorem gives a P_0 -conull Borel set G of states whose S_L^* -orbits are generic for P_0 simultaneously on every F_j^0 and $\bar{F}_j$. The quasi-Palm implication above, applied to G , shows that for $\hat{\mu}$ -almost every ω , $T_y^* \vartheta(\omega) \in G$ for $\vartheta(\omega)$ -almost every y . By (27.21), for μ_ω -almost every u the centered state $T_{u-x(\omega)}^* \vartheta(\omega)$ is therefore in G . The point u lies in the interior of the support hull of μ_ω almost surely, so (27.21) makes the sceneries of μ_ω at u and of $\vartheta(\omega)$ at $u - x(\omega)$ identical for every sufficiently small radius. Thus μ_ω discretely generates the restricted law of P_0 at μ_ω -almost every u .

For each F_j , split $[0, T]$ into blocks of length L . The averaged observable in (27.22c) gives exactly the integral of F_j over each complete block, while the unfinished last block has contribution at most $2L\|F_j\|_\infty/T$. Since the S_L^* -orbit of the recentered state is generic for $\bar{F}_j$ as well, the continuous-time average tends to

$$\int \bar{F}_j dP_0 = \int F_j(\nu \square) dP(\nu)$$

by (27.22). The family (F_j) determines the law, so μ_ω continuously generates P . Finally Theorem IV.10.128 gives the asserted exact dimension of the original μ . $\square$

6. The unsmoothed Host argument

The discrete step used in the coding construction was $\log \gamma$. The normality criterion uses the different step $\log \beta$. Continuous generation alone does not identify its individual discrete components, so we establish that change of step explicitly.

Lemma IV.10.136 (*all-step component generation by future conditionals*).

Let P_0 and P be the full-space laws constructed in Lemma IV.10.133 and Corollary IV.10.135. For every $h > 0$, almost every future conditional μ_ω h -discretely generates an S_h^* -ergodic component of P at μ_ω -almost every point. When S_h^* is not ergodic, use the everywhere canonical component family of Lemma IV.10.81, with $L = h$. For each nonzero $\alpha \in \Sigma(P, S) \cap h^{-1}\mathbb{Z}$, the spatial α -phase of μ_ω is constant and equals $\phi_\alpha(\vartheta(\omega))$. The component laws give full mass to the domain and non-atomic, origin-supported states required in Lemma IV.10.17.

Proof. Fix h . If S_h^* is ergodic, write $Q(\xi) = P$. Otherwise let $Q(\xi) = P_{\phi(\xi)}$ on the common full-orbit domain of the primitive phase ϕ , using the canonical kernel in Lemma IV.10.81. For P -almost every ξ , this is its ordinary ergodic component. In either case the full-space pointwise ergodic theorem, applied to a countable convergence-determining family on the Radon state space, gives

$$\frac{1}{N} \sum_{n=0}^{N-1} \delta_{S_{nh}^* \xi} \implies Q(\xi). \quad (27.22d)$$

The set where (27.22d) holds is Borel, since both its tests and the conditional kernel are Borel.

We need a *full-flow-invariant* version of this conull event. Require also origin support and non-atomicity of ξ , membership in the common full-orbit eigenfunction domain, and the spatial constancy in Lemma IV.10.14 for the eigenfunction under consideration. Each is a Borel, P -conull property. For the last one this follows by testing, on $[-k, k]$ for every integer k , the integrals of the phase difference and of the indicator of the recentered domain's complement. The exact translation–scaling identity shows that this property is invariant under every flow time. If α is not primitive, also include the equality $\phi_\alpha = c_\alpha \phi^q$, where $\alpha = q\alpha_0$ and $|c_\alpha| = 1$, from Lemma IV.10.79. With the full-orbit representatives this equality is a full-flow-invariant conull event. It ensures that the same canonical component has the stated α -phase, not only its primitive phase.

All ordinary component laws give full mass to non-atomic, origin-supported states; when the phase kernel is needed, canonical transport extends this fact to every phase. For any fixed ξ satisfying (27.22d) and every real t , the map S_t^* is continuous at $Q(\xi)$ -almost every state: its normalizing interval has positive mass and its boundary has zero mass. The continuous mapping theorem applied to (27.22d) gives the limit $(S_t^*)_* Q(\xi)$ for the orbit starting at $S_t^* \xi$. This is exactly $Q(S_t^* \xi)$ by canonical covariance, or by flow invariance in the ergodic case. The argument holds for every t for this fixed ξ ; it does not intersect uncountably many almost-everywhere assertions. Thus the Borel conull event G just constructed is invariant under every S_t^* .

Put $L_\gamma = \log \gamma$. The roof formula gives

$$1 = P(G) = \frac{1}{L_\gamma} \int_0^{L_\gamma} P_0((S_t^*)^{-1}G) dt = P_0(G).$$

The discrete quasi-Palm property of P_0 , proved in Corollary IV.10.135, now transfers G to $T_y^* \vartheta(\omega)$ for $\vartheta(\omega)$ -almost every y and almost every ω . By (27.21), at μ_ω -almost every interior point u , its centered sceneries and those of $\vartheta(\omega)$ at $u - x(\omega)$ agree on any fixed bounded window for all sufficiently large times. The central normalizations also agree. Hence their distance in the vague Radon-state metric tends to zero, and their empirical laws at times nh have the same limit. This proves component generation.

Spatial constancy, which also holds at P_0 -almost every state by the roof argument, identifies the component-constant eigenphase at these recenterings with $\phi_\alpha(\vartheta(\omega))$. All domain and non-atomicity properties of the generated laws follow from the canonical component construction. The claim for a different h is a separate conull assertion, as required; no simultaneous assertion over all uncountably many h is made. $\square$

Fix a Pisot β with

$$\frac{\log \beta}{\log \gamma} \notin \mathbb{Q}. \quad (27.24)$$

If

$$\Sigma(P, S) \cap \left(\frac{1}{\log \beta} \mathbb{Z} \setminus \{0\} \right) = \emptyset, \quad (27.25)$$

then the normality criteria of Section 10.4 apply immediately to every future conditional μ_ω , and then to μ by integrating the conull typical sets.

Assume instead that the intersection is nonempty. Put

$$K_\beta(P) = \left\{ k \in \mathbb{Z} : \frac{k}{\log \beta} \in \Sigma(P, S) \right\} = k_0 \mathbb{Z}, \quad (27.26)$$

and $\alpha_0 = k_0 / \log \beta$. Let ϕ be the primitive unit eigenfunction from the primitive-lattice construction in Section 10.6.

We use P_0 as the $\log \gamma$ -invariant factor law actually constructed above. No identification of a prescribed γ -phase with an arbitrary conditional-law version is needed.

Lemma IV.10.137 (*Haar β -phase on the γ -invariant factor law*).

The primitive β -phase under the constructed $S_{\log \gamma}^*$ -invariant law P_0 has Haar distribution on S^1 .

Proof. The full-orbit eigenfunction domain is P -conull and invariant under every real time, hence is P_0 -conull by the roof formula, exactly as in Lemma IV.10.136. On that domain

$$\phi(S_{\log \gamma}^* \xi) = e(-\alpha_0 \log \gamma) \phi(\xi). \quad (27.27)$$

Since P_0 is $S_{\log \gamma}^*$ -invariant, the law $\theta = \phi_* P_0$ is invariant under this rotation. Its angle is irrational, because $\alpha_0 \log \gamma = k_0 \log \gamma / \log \beta \notin \mathbb{Q}$. For every nonzero integer r ,

$$\hat{\theta}(r) = e(-r\alpha_0 \log \gamma) \hat{\theta}(r) \implies \hat{\theta}(r) = 0.$$

The zero coefficient is one; density of trigonometric polynomials identifies θ with Haar measure. This argument applies even if the constructed law P_0 sits at a Haar-null γ -phase. $\square$

Lemma IV.10.136, with $h = \log \beta$, gives the required β -time component generation of the future conditionals and identifies their constant phase with $\phi(\vartheta(\omega))$. Since $\vartheta_*\hat{\mu} = P_0$, Lemma IV.10.137 makes the cumulative β -phase distribution of the family $\{\mu_\omega\}$ Haar.

Theorem IV.10.89 is precisely the phase-spread refinement of the normality criteria in Section 10.4: a family of local measures generating P , whose cumulative β -phase has full dimension (and in particular is Haar), is pointwise β -normal.

Theorem IV.10.138 (*Host-type theorem*).

Under the hypotheses stated above, μ is pointwise β -normal. If μ is merely T_γ -invariant and almost every ergodic component has positive entropy, the conclusion holds for μ -a.e. point by applying the ergodic theorem componentwise.

Proof. Use Theorem IV.10.63 in the spectrally excluded case and Lemma IV.10.137 together with Theorem IV.10.89 in the lattice case. For a nonergodic μ , disintegrate into T_γ -ergodic components; the arithmetic hypothesis is common and the positivity assumption holds for almost every component. $\square$

7. A slicing lemma for the smooth-image phase

Lemma IV.10.139 (*dimension from one-dimensional fibers*).

Let η be a probability on $\mathbb{R} \times S^1$ and let $\delta \geq 0$. Assume that its second marginal is Haar and that, for Haar-almost every y , the conditional η_y on the first coordinate has lower Hausdorff dimension at least δ . Then

$$\underline{\dim}_H \eta \geq 1 + \delta. \quad (27.31)$$

Proof. It suffices that $\underline{\dim}_H \eta_y \geq \delta$ for Haar-almost every y ; we prove this slightly stronger version. Fix $0 < \varepsilon < \delta$, the case $\delta = 0$ following directly from the Haar marginal. The Frostman-piece Lemma IV.10.42, or its lower-local-dimension formulation, gives Borel sets

$$E_n = \{(x, y) : |x| \leq n, \eta_y(B(x, 2^{-j})) \leq 2^{-j(\delta-\varepsilon)} \text{ for every } j \geq n\}$$

whose union has full η -measure. The conditional probability kernel and countably many ball tests make these sets measurable. Let $\eta^{(n)} = \eta|_{E_n}$. For every y , the measure $\eta_y|_{\{x: (x, y) \in E_n\}}$ has ball masses at most $C_{n, \varepsilon} r^{\delta-\varepsilon}$ for all centers and radii: if a ball meets this set, recenter it at a point of the set and enlarge its radius by two; choose the next larger dyadic radius. For larger radii increase the same constant.

Disintegration over the Haar marginal consequently gives

$$\eta^{(n)}(B((x, y), r)) \leq C'_{n, \varepsilon} r^{1+\delta-\varepsilon}.$$

Each restricted measure vanishes on every set of Hausdorff dimension less than $1 + \delta - \varepsilon$ by the mass distribution principle. Their union exhausts η , so the same is true for η . Let $\varepsilon \downarrow 0$ to prove (27.31). $\square$

Lemma IV.10.140 (*product upper bound for measures*).

Let ρ be a probability on $\mathbb{R} \times S^1$ with first marginal μ and second marginal θ . If μ is exact dimensional of dimension δ , then

$$\underline{\dim}_{\mathbb{H}} \rho \leq \delta + \underline{\dim}_{\mathbb{H}} \theta. \quad (27.32)$$

Consequently, if $\underline{\dim}_{\mathbb{H}} \rho \geq 1 + \delta$, then

$$\underline{\dim}_{\mathbb{H}} \theta = 1. \quad (27.33)$$

Proof. Fix $\varepsilon > 0$ and write $s = \underline{\dim}_{\mathbb{H}} \theta$. Exact dimensionality of μ gives a full- μ set $A = \bigcup_m A_m$ with

$$\overline{\dim}_{\mathbb{B}} A_m \leq \delta + \varepsilon \quad \text{for every } m.$$

Indeed, after decomposing according to the scale from which $\mu(B(x, r)) \geq r^{\delta+\varepsilon}$ holds, a maximal r -separated subset of each piece has $O(r^{-\delta-\varepsilon})$ points.

By the definition of s , choose a Borel set $B \subset S^1$ with $\theta(B) > 0$ and

$$\dim_{\mathbb{H}} B \leq s + \varepsilon.$$

Since $\mu(A) = 1$, one has $\rho(A \times B) = \theta(B) > 0$. The elementary product covering inequality

$$\dim_{\mathbb{H}}(A_m \times B) \leq \overline{\dim}_{\mathbb{B}} A_m + \dim_{\mathbb{H}} B$$

and countable stability give

$$\underline{\dim}_{\mathbb{H}} \rho \leq \dim_{\mathbb{H}}(A \times B) \leq \delta + s + 2\varepsilon.$$

Let $\varepsilon \downarrow 0$ to obtain (27.32). If the left side is at least $1 + \delta$, then $s \geq 1$; since θ lives on S^1 , also $s \leq 1$, proving (27.33). $\square$

8. The C^2 smooth-image Host theorem

Let $g \in \text{Diff}^2(\mathbb{R})$. Localize to a compact interval carrying all but ε mass; g' is then bounded above and below and $\log |g'|$ is C^1 .

In the resonant case above define the joint position-phase measure

$$\eta = \int \mu_{\omega} \times \delta_{z(\omega)} d\hat{\mu}(\omega), \quad (27.34)$$

where $z(\omega)$ is the constant primitive β -phase provided by Lemma IV.10.136. Its phase marginal is Haar by Lemma IV.10.137. Disintegrate $\hat{\mu}$ over z . For Haar-almost every y , its conditional parameter law gives full mass to the set of ω for which μ_{ω} has lower Hausdorff dimension δ , by Corollary IV.10.130. The position fiber is the mixture of those μ_{ω} . It therefore vanishes on every set of dimension less than δ : every constituent in that full conditional-law set does so. Thus the fiber has lower Hausdorff dimension at least δ . We

do not infer exact dimensionality of an arbitrary mixture from exact dimensionality of its constituents. The stronger form proved in Lemma IV.10.139 gives

$$\underline{\dim}_{\mathbb{H}} \eta \geq 1 + \delta. \quad (27.35)$$

Define the triangular phase map

$$F_g(x, y) = (x, y - \alpha_0 \log |g'(x)| \pmod{1}). \quad (27.36)$$

Because g is C^2 , $\log |g'|$ is C^1 , so F_g is locally bi-Lipschitz with inverse

$$(x, y) \mapsto (x, y + \alpha_0 \log |g'(x)| \pmod{1}). \quad (27.37)$$

Therefore

$$\underline{\dim}_{\mathbb{H}}(F_g)_* \eta \geq 1 + \delta. \quad (27.38)$$

The first marginal remains μ , exact dimensional of dimension δ .

By the phase rule in Lemma IV.10.17, the second marginal is exactly the cumulative β -phase of the image family $\{g_*\mu_\omega\}$. Lemma IV.10.140 forces this phase marginal to have lower Hausdorff dimension one. Theorem IV.10.89 therefore applies to the image family.

Theorem IV.10.141 (*smooth Host theorem*).

Under the hypotheses of Theorem IV.10.138, for every $g \in \text{Diff}^2(\mathbb{R})$, the measure $g_*\mu$ is pointwise β -normal. The same componentwise extension holds if every T_γ -ergodic component of μ has positive entropy.

Proof. If the β spectral lattice is absent, the C^1 robustness in Theorem IV.10.63 already proves the result. Otherwise use (27.34)–(27.38) to obtain a full-dimensional cumulative phase and invoke Theorem IV.10.89. $\square$

Chapter IV.11

Diophantine Approximation on Missing-Digit Sets

Counting rational points, intrinsic approximation, and mass transference

This chapter separates three questions that are easy to conflate. First, how many *distinct* rational points of bounded standard height lie in a missing-digit set? Second, what convergence conclusions follow from such a counting bound by elementary covering and Borel–Cantelli arguments? Third, what extra distribution information is needed for divergence or Hausdorff-measure lower bounds? The improved Chow–Varjú–Yu rational-counting theorem is proved through a Fourier transfer argument and an accompanying certified finite computation; the extrinsic lower-bound theorem retains an explicit source boundary. The general mass transference principle is also proved here. The final bridge states the additional full-measure hypothesis needed for a Hausdorff lower bound.

11.1 Missing-digit sets, natural measure, and two denominator conventions

Fix an integer $b \geq 2$ and a digit set

$$D \subset \{0, 1, \dots, b-1\}, \quad m = \#D \geq 2.$$

Definition IV.11.1 (*missing-digit set*).

Define

$$K = K_{b,D} = \left\{ \sum_{j \geq 1} d_j b^{-j} : d_j \in D \right\} \subset [0, 1]. \quad (28.1)$$

Put

$$\kappa = \frac{\log m}{\log b}. \quad (28.2)$$

For $u = (u_1, \dots, u_n) \in D^n$, the level- n cylinder is

$$I_u = \sum_{j=1}^n u_j b^{-j} + b^{-n} [0, 1]. \quad (28.3)$$

Its diameter is b^{-n} . Different level- n cylinders have disjoint interiors; endpoint coincidences are allowed.

Definition IV.11.2 (*natural measure*).

Let $\nu = \nu_{b,D}$ be the distribution of

$$\sum_{j \geq 1} X_j b^{-j},$$

where the X_j are independent and uniform on D . Equivalently,

$$\nu(I_u) = m^{-n} = b^{-n\kappa}. \quad (28.4)$$

Endpoint coincidences do not change these masses. A point has at most two base- b addresses; each address has product probability at most m^{-n} for every n , hence probability zero. Thus ν has no atoms.

Lemma IV.11.3 (*Ahlfors regularity of the natural measure*).

There is $C \geq 1$, depending only on b and D , such that for $x \in K$ and $0 < r \leq 1$,

$$C^{-1}r^\kappa \leq \nu(B(x, r)) \leq Cr^\kappa. \quad (28.5)$$

Proof. Choose n so that $b^{-(n+1)} < r \leq b^{-n}$. Choose any admissible level- $(n+1)$ cylinder containing x ; at an endpoint it need not be unique. Its diameter $b^{-(n+1)}$ is strictly less than r , so it lies in $B(x, r)$ and has mass $b^{-(n+1)\kappa}$. This gives the lower bound up to the fixed factor $b^{-\kappa}$.

For the upper bound, $B(x, r)$ has diameter at most $2b^{-n}$ and therefore meets only a bounded number, depending only on b , of level- n cylinders with disjoint interiors. Each such cylinder has mass $b^{-n\kappa}$. Since $b^{-n} \leq br$, the upper bound follows. $\square$

Lemma IV.11.4 (*Hausdorff-measure regularity of K*).

There is $C' \geq 1$ such that for $x \in K$ and all sufficiently small r ,

$$C'^{-1}r^\kappa \leq \mathcal{H}^\kappa(K \cap B(x, r)) \leq C'r^\kappa. \quad (28.5a)$$

Proof. For the lower bound, let (U_j) cover $K \cap B(x, r)$. By the upper Ahlfors estimate in (28.5),

$$\nu(U_j) \leq C_1 \text{diam}(U_j)^\kappa.$$

Summing and using the lower estimate in (28.5) gives

$$\sum_j \text{diam}(U_j)^\kappa \geq C_1^{-1} \nu(K \cap B(x, r)) \geq C_2^{-1} r^\kappa.$$

Taking the infimum over arbitrarily fine covers proves the lower bound. For the upper bound, choose the level n with $b^{-n} \asymp r$. Only $O(1)$ level- n cylinders meet $B(x, r)$. For every integer $k \geq 1$, replace each of these cylinders by its m^k admissible descendants at level $n+k$. They cover its intersection with K , have diameter $b^{-(n+k)}$, and have total κ -cost

$$m^k b^{-(n+k)\kappa} = b^{-n\kappa},$$

since $m = b^\kappa$. Thus the entire cover has cost $O(r^\kappa)$ at every arbitrarily small mesh. Letting $k \rightarrow \infty$ gives the Hausdorff *measure* upper bound, not merely the one-scale content bound. $\square$

Definition IV.11.5 (*standard/reduced height*).

For $r \in \mathbb{Q}$ write $r = p/q$ in lowest terms with $q \geq 1$ and set

$$H(r) = q.$$

Equivalently, $H(r)$ is the least positive q such that qr is an integer. The distinct-rational counting function is

$$N_K(T) = \#\{r \in K \cap \mathbb{Q} : H(r) \leq T\}. \quad (28.6)$$

Thus a rational point is counted once, not once for every displayed fraction. This is the height used in the Chow–Varju–Yu theorems below.

Definition IV.11.6 (*coding denominator and intrinsic displayed height*).

Suppose an admissible base- b expansion of r is eventually periodic, with preperiod length i_0 and period length $\ell \geq 1$. The corresponding geometric series writes r with displayed denominator

$$q_{\text{int}}(\omega) = b^{i_0}(b^\ell - 1). \quad (28.7)$$

The reduced denominator $H(r)$ divides $q_{\text{int}}(\omega)$, but equality need not hold. The quantity in (28.7) also depends on a nonminimal choice of period and, at base- b endpoints, on the chosen expansion. If an intrinsic height of the point itself is needed, use

$$H_{\text{int}}(r) = \min_{\omega} q_{\text{int}}(\omega), \quad (28.8)$$

where the minimum ranges over admissible eventually periodic D -expansions of r . No theorem quoted in Section 11.4 below substitutes H_{int} for the standard height H .

Proposition IV.11.7 (*rationality and eventual periodicity*).

A point $x \in [0, 1]$ is rational if and only if every nonterminating base- b expansion of x is eventually periodic; equivalently, x has an eventually periodic base- b expansion. Consequently, every $r \in K \cap \mathbb{Q}$ has an admissible D -expansion that is eventually periodic.

Proof. For a rational $x = p/q$, the long-division remainders modulo q take only finitely many values, so the base- b digits eventually repeat. Conversely, if

$$x = 0.u_1 \cdots u_{i_0} \overline{v_1 \cdots v_\ell}_b,$$

then splitting off the preperiod and summing the geometric tail gives a rational number whose denominator divides $b^{i_0}(b^\ell - 1)$.

At a base- b endpoint, the two expansions are a terminating expansion and one with an eventual tail of $b - 1$ digits. Both are eventually periodic. Hence the proposition and the counting function (28.6) are independent of which expansion is displayed. $\square$

11.2 Elementary rational counting

Lemma IV.11.8 (*standard-height separation*).

If $r \neq s$ are rationals with $H(r), H(s) \leq T$, then

$$|r - s| \geq T^{-2}. \quad (28.9)$$

Proof. Write $r = p/q$ and $s = p'/q'$ in lowest terms. Then

$$|r - s| = \frac{|pq' - p'q|}{qq'} \geq \frac{1}{qq'} \geq T^{-2}.$$

□

Proposition IV.11.9 (*periodic-word lower bound*).

For every $n \geq 1$,

$$N_K(b^n - 1) \geq m^n. \quad (28.10)$$

Proof. For $u = (u_1, \dots, u_n) \in D^n$, let

$$x_u = 0.\overline{u_1 \cdots u_n}.$$

The geometric-series formula gives $x_u = A_u/(b^n - 1)$ for an integer A_u , so $H(x_u)$ divides $b^n - 1$.

Distinct purely periodic words give distinct points. The only two-expansion ambiguity is eventually 0 versus eventually $b - 1$; a purely periodic sequence that is eventually constant is constant from the start, and the constant-0 and constant- $(b - 1)$ sequences represent 0 and 1. Hence the m^n words yield m^n distinct rational points of standard height at most $b^n - 1$.

□

Proposition IV.11.10 (*elementary standard-height upper bound*).

There is $C = C(b, D)$ such that for all $T \geq 2$,

$$N_K(T) \leq CT^{2\kappa}. \quad (28.11)$$

Proof. Choose n minimal with $b^{-n} < T^{-2}$. By (28.9), each level- n cylinder contains at most one rational point of height at most T . Therefore

$$N_K(T) \leq m^n = b^{n\kappa} \leq b^\kappa T^{2\kappa}.$$

□

Three convention warnings are retained. First, (28.11) counts distinct rationals; unreduced pairs (p, q) may be much more numerous. Second, (28.10) uses the displayed denominator $b^n - 1$ only to certify the standard-height inequality $H(x_u) \leq b^n - 1$. Third, no lower-distribution or ubiquity assertion follows from (28.10) or (28.11).

Theorem IV.11.11 (*elementary rational counting by standard height*).

For the missing-digit set $K = K(b, D)$ with dimension $\kappa = \log |D| / \log b$, the standard-height counting function satisfies the explicit lower subsequence and global upper bounds

$$N_K(b^n - 1) \geq |D|^n, \quad N_K(T) \leq C(b, D)T^{2\kappa} \quad (T \geq 2).$$

All counts are of distinct reduced rational points; endpoint double expansions do not change the count.

Proof. This is exactly the conjunction of Proposition IV.11.9 and Proposition IV.11.10, with the endpoint and reduced-denominator conventions proved in Proposition IV.11.7 and recorded above. $\square$

11.3 Intrinsic and extrinsic limsup sets

Definition IV.11.12 (*intrinsic standard-height approximation*).

For $\alpha > 0$, let $W_K(\alpha)$ be the set of $x \in \mathbb{R}$ for which

$$|x - r| < H(r)^{-\alpha} \quad (28.12)$$

for infinitely many distinct $r \in K \cap \mathbb{Q}$. When measuring with ν , one may replace $W_K(\alpha)$ by $W_K(\alpha) \cap K$ because ν is supported on K .

The source literature sometimes writes (28.12) using fractions p/q not explicitly required to be reduced. Away from the countable set $K \cap \mathbb{Q}$ this gives the same limsup set: reducing p/q can only decrease the denominator and enlarge the right-hand side, while infinitely many multiples of one fixed rational center can approximate a different point only finitely often. The countable discrepancy has ν -measure zero and Hausdorff dimension zero.

Definition IV.11.13 (*extrinsic approximation*).

Let $W_{\mathbb{R}}(\alpha)$ be the set of $x \in \mathbb{R}$ for which

$$|x - p/q| < q^{-\alpha} \quad (28.13)$$

for infinitely many reduced rationals p/q , without requiring $p/q \in K$. Thus W_K is an intrinsic-center problem, whereas $W_{\mathbb{R}}(\alpha) \cap K$ is an extrinsic-center problem.

Lemma IV.11.14 (*counting-to-natural-measure convergence*).

Assume that for some $\beta > 0$,

$$N_K(T) \leq CT^\beta \quad (28.14)$$

for all large T . If

$$\alpha > \beta/\kappa, \quad (28.15)$$

then

$$\nu(W_K(\alpha)) = 0. \quad (28.16)$$

Proof. For $n \geq 1$, let A_n be the union of the balls in (28.12) whose reduced center r satisfies

$$e^n < H(r) \leq e^{n+1}.$$

By (28.14) there are $O(e^{\beta n})$ such centers. By Ahlfors regularity,

$$\nu(B(r, H(r)^{-\alpha})) \leq C_1 e^{-\alpha \kappa n}.$$

Hence

$$\nu(A_n) \leq C_2 e^{(\beta - \alpha \kappa)n}.$$

The series converges under (28.15), and Borel–Cantelli gives (28.16). $\square$

Lemma IV.11.15 (*counting-to-Hausdorff upper bound*).

Under (28.14), for every $\alpha > 0$,

$$\dim_{\text{H}} W_K(\alpha) \leq \beta/\alpha. \quad (28.17)$$

Consequently,

$$\dim_{\text{H}}(W_K(\alpha) \cap K) \leq \min\{\kappa, \beta/\alpha\}. \quad (28.18)$$

Proof. Use the same height annuli as in Lemma IV.11.14. For any $s > \beta/\alpha$, the s -dimensional cost of the annulus- n cover is

$$O(e^{\beta n}) O(e^{-\alpha s n}) = O(e^{(\beta - \alpha s)n}),$$

whose sum converges. Thus $\mathcal{H}^s(W_K(\alpha)) = 0$. Let $s \downarrow \beta/\alpha$. The second assertion follows by intersecting with K , whose Hausdorff dimension is κ . $\square$

Corollary IV.11.16 (*elementary intrinsic convergence*).

The local bound (28.11) gives

$$\nu(W_K(\alpha)) = 0 \quad (\alpha > 2), \quad (28.19)$$

and

$$\dim_{\text{H}} W_K(\alpha) \leq \frac{2\kappa}{\alpha}. \quad (28.20)$$

These are convergence statements only. They do not assert full measure or a matching lower dimension at a divergent exponent.

Proof. Apply Lemmas IV.11.14 and IV.11.15 with the intrinsic count exponent $\beta = 2\kappa$ from (28.11). $\square$

Proposition IV.11.17 (*elementary extrinsic upper bounds*).

For a bounded interval containing K , the number of reduced rational centers p/q with $q \leq T$ that can contribute to (28.13) is $O(T^2)$. Hence

$$\dim_H(W_{\mathbb{R}}(\alpha) \cap K) \leq \min\{\kappa, 2/\alpha\}, \quad (28.21)$$

and

$$\nu(W_{\mathbb{R}}(\alpha)) = 0 \quad \text{whenever } \alpha\kappa > 2. \quad (28.22)$$

Proof. For each $q \leq T$ there are $O(q)$ relevant numerators, so the cumulative count is $O(T^2)$. The Hausdorff argument is Lemma IV.11.15 with $\beta = 2$. For (28.22), combine the same count with the upper Ahlfors estimate in Lemma IV.11.3 and apply Borel–Cantelli. $\square$

Theorem IV.11.18 (*elementary counting-to-limsup covering consequences*).

The elementary standard-height count in Proposition IV.11.10 implies the intrinsic convergence and Hausdorff upper bounds in Corollary IV.11.16, while the elementary $O(T^2)$ count of ambient reduced rationals implies the extrinsic upper bounds in Proposition IV.11.17. Explicitly,

$$\nu(W_K(\alpha)) = 0 \quad (\alpha > 2), \quad \dim_H W_K(\alpha) \leq \frac{2\kappa}{\alpha},$$

and

$$\dim_H(W_{\mathbb{R}}(\alpha) \cap K) \leq \min\{\kappa, 2/\alpha\}, \quad \nu(W_{\mathbb{R}}(\alpha)) = 0 \quad (\alpha\kappa > 2).$$

No divergent/full-measure conclusion is asserted by this elementary package.

Proof. Apply Lemmas IV.11.14–IV.11.15 with the intrinsic exponent $\beta = 2\kappa$ supplied by Proposition IV.11.10, and apply the same covering and Borel–Cantelli arguments with the ambient rational count exponent $\beta = 2$ for the extrinsic case. This is exactly the combination of Corollary IV.11.16 and Proposition IV.11.17. $\square$

The weakness of (28.21) is instructive. Near the extrinsic critical exponent $\alpha = 2$ it gives only $\min\{\kappa, 1\}$; the sharp theorem below requires genuine arithmetic distribution information.

11.4 Effective rational counting on missing-digit sets

The arithmetic proof begins with a general Fourier-to-counting principle, independent of the particular digit set. We prove it here; the remaining digit-specific inequality is identified precisely afterward. If

$$\hat{\nu}(\xi) = \int e^{-2\pi i \xi x} d\nu(x),$$

define the Fourier ℓ^1 dimension by

$$\hat{\kappa}_1(\nu) = \sup \left\{ s \geq 0 : \sum_{0 \leq \xi \leq Q} |\hat{\nu}(\xi)| \ll Q^{1-s} \right\}. \quad (29.5)$$

The frequencies are integers and $Q \geq 1$. This convention agrees with the natural measure on $[0, 1]$ interpreted on the circle.

Theorem IV.11.19 (*Fourier ℓ^1 dimension forces a rational-counting saving*).

Let ν be a κ -Ahlfors regular probability on a compact $K \subset [0, 1]$, where $0 < \kappa < 1$. If $\hat{\kappa}_1(\nu) > 1/2$, then for every

$$\frac{1}{2} < \nu < \hat{\kappa}_1(\nu)$$

there is a constant C_ν such that

$$N_K(T) \leq C_\nu T^{2\kappa - \rho(\nu)}, \quad \rho(\nu) = \frac{(1 - \kappa)(2\nu - 1)}{1 - \nu} > 0. \quad (29.5a)$$

If an explicit estimate in (29.5) supplies ν and its constant, then the exponent and counting constant in (29.5a) are effective.

Proof. We first record that $\hat{\kappa}_1(\nu) \leq \kappa$. For $Q \geq 2$ the Fejér kernel F_Q is nonnegative, has Fourier support $\{|j| < Q\}$ with coefficients at most one, and satisfies $F_Q(t) \geq cQ$ for $|t| \leq c/Q$. Hence lower Ahlfors regularity gives

$$\sum_{|j| < Q} |\hat{\nu}(j)|^2 \geq \iint F_Q(x - y) d\nu(x) d\nu(y) \geq cQ \int \nu(B(x, c/Q)) d\nu(x) \geq c' Q^{1-\kappa}.$$

Since $|\hat{\nu}(j)| \leq 1$, the same lower bound holds for the ℓ^1 sum. Thus the chosen ν may, and below does, satisfy $1/2 < \nu < \kappa < 1$.

For $q \in \mathbb{N}$ and $0 < \delta < 1/2$ put $A(q, \delta) = \{x : \|qx\|_{\mathbb{R}/\mathbb{Z}} \leq \delta\}$. Choose the nonnegative band-limited majorant

$$\phi(t) = \frac{\pi^2}{4} \left(\frac{\sin(\pi t/2)}{\pi t/2} \right)^2, \quad \phi(0) = \frac{\pi^2}{4}.$$

It is at least one on $[-1, 1]$, is integrable, and its Fourier transform is bounded and supported on $[-1, 1]$. Periodization and Fourier-series inversion therefore give

$$\begin{aligned} \nu(A(q, \delta)) &\leq \int \sum_{a \in \mathbb{Z}} \phi\left(\frac{qx - a}{\delta}\right) d\nu(x) \\ &\leq C\delta \sum_{|k| \leq 1/\delta} |\hat{\nu}(kq)|. \end{aligned} \quad (29.5b)$$

The periodized sum converges absolutely; equivalently one may first use Fejér summation and then pass to its finite Fourier series.

Fix s with $\nu < s < \hat{\kappa}_1(\nu)$. The elementary divisor estimate $\tau(n) \ll_\epsilon n^\epsilon$ with $\epsilon = s - \nu$ and the definition (29.5) imply, for $M \geq 2$,

$$\sum_{1 \leq |j| \leq M} \tau(|j|) |\hat{\nu}(j)| \ll M^{s-\nu} \sum_{0 \leq j \leq M} |\hat{\nu}(j)| \ll M^{1-\nu}.$$

Summing (29.5b) over $Q \leq q \leq 2Q$ and grouping nonzero frequencies by the divisor q gives

$$\sum_{q=Q}^{2Q} \nu(A(q, \delta)) \ll \delta(Q + (Q/\delta)^{1-\nu}). \quad (29.5c)$$

Set $w = v/(1-v) > 1$ and $\delta = Q^{-w}$. Then $(Q/\delta)^{1-v} = Q$, so the right side is $O(\delta Q)$.

Let $\mathcal{R}_Q$ be the distinct reduced rationals $r = p/q \in K$ with $Q \leq q \leq 2Q$. Distinct members are separated by at least $(4Q^2)^{-1}$. For sufficiently large Q the balls $B(r, \delta/q)$ are disjoint, since $2\delta/Q < (4Q^2)^{-1}$. Moreover $B(r, \delta/q) \subset A(q, \delta)$ and, by the lower Ahlfors bound,

$$\nu(B(r, \delta/q)) \geq c(\delta/(2Q))^\kappa.$$

Thus (29.5c) implies

$$\#\mathcal{R}_Q \ll \delta^{1-\kappa} Q^{1+\kappa} = Q^{1+\kappa-(1-\kappa)v/(1-v)} = Q^{2\kappa-\rho(v)}.$$

The exponent is positive because $v < \kappa$. Summing over dyadic denominator shells proves (29.5a). $\square$

For the digit systems in the source boundary above, the remaining input is the strict inequality

$$\hat{\kappa}_1(\nu_{b,D}) > \frac{1}{2} \quad \text{when } b \geq 5, \#D = b-1, \quad \text{or } b = 4, D \in \{\{0, 1, 2\}, \{1, 2, 3\}\}. \quad (29.5d)$$

Here is the finite-level transfer bound that reduces (29.5d) to explicit inequalities. It is useful independently of the numerical verification.

Lemma IV.11.20 (*finite-level Fourier transfer criterion*).

For $m = \#D$ define

$$g(x) = \frac{1}{m} \left| \sum_{d \in D} e^{2\pi i d x} \right|, \quad S_L(x) = \prod_{r=0}^{L-1} g(b^r x), \quad B = b^L, \quad F_L(x) = \sum_{j=0}^{B-1} S_L(x + j/B). \quad (29.5e)$$

Then, with $M_L = \max_x F_L(x)$,

$$\hat{\kappa}_1(\nu_{b,D}) \geq 1 - \frac{\log M_L}{L \log b}. \quad (29.5f)$$

The function F_L is even and $1/B$ -periodic and has Lipschitz constant at most $2\pi B(B-1)$. Consequently, if $\delta > 0$ and

$$\max_{0 \leq k \leq \lfloor (2B\delta)^{-1} \rfloor} F_L(k\delta) + \pi\delta B(B-1) < \sqrt{B}, \quad (29.5g)$$

then $\hat{\kappa}_1(\nu_{b,D}) > 1/2$.

Proof. Independence of the base- b digits gives $|\hat{\nu}(y)| = S_L(y/B)|\hat{\nu}(y/B)|$. For $n \geq 0$ put $A_n(y) = \sum_{k=0}^{B^n-1} |\hat{\nu}(y+k)|$. The assertion $A_n(y) \leq M_L^n$ holds for $n = 0$. For the induction step write $k = r + Bq$, with $0 \leq r < B$ and $0 \leq q < B^{n-1}$. By periodicity of S_L ,

$$\begin{aligned} A_n(y) &= \sum_{r=0}^{B-1} S_L((y+r)/B) \sum_{q=0}^{B^{n-1}-1} |\hat{\nu}((y+r)/B + q)| \\ &\leq M_L^{n-1} F_L(y/B) \leq M_L^n. \end{aligned}$$

Taking $y = 0$ and placing any Q between consecutive powers of B proves (29.5f). The inequality $|\hat{v}| \leq 1$ ensures $M_L \leq B$, so this step also covers $M_L = 1$.

The modulus g is even and 1-periodic. Moreover

$$|g(x) - g(y)| \leq \frac{2\pi}{m} \sum_{d \in D} d |x - y| \leq 2\pi(b-1)|x - y|.$$

Since $0 \leq g \leq 1$, the product telescoping identity gives $\text{Lip}(S_L) \leq 2\pi(b-1) \sum_{r=0}^{L-1} b^r = 2\pi(B-1)$. Summing B translates gives the stated bound for F_L . Translating its argument by $1/B$ permutes its summands, and evenness follows by replacing j with $B-j$ modulo B . It therefore suffices to control F_L on $[0, 1/(2B)]$. Every point of this interval is within $\delta/2$ of one of the grid points in (29.5g). The Lipschitz bound proves $M_L \leq \max_k F_L(k\delta) + \pi\delta B(b-1)$. If (29.5g) holds, then $M_L < \sqrt{B}$, so (29.5f) is strictly greater than $1/2$. $\square$

Lemma IV.11.21 (*analytic Fourier bound for large omitted-digit bases*).

If $b \geq 111$ and $D = \{0, \dots, b-1\} \setminus \{a\}$, then $\hat{\kappa}_1(\nu_{b,D}) > 1/2$.

Proof. Let $e(t) = e^{2\pi it}$ and $f_b(t) = |\sum_{k=0}^{b-1} e(kt)|$. The geometric-sum formula and $|e(t) - 1| = 2 \sin(\pi \|t\|) \geq 4 \|t\|$ give

$$f_b(t) \leq b, \quad f_b(t) \leq \frac{1}{2\|t\|}, \quad f_b(t) \leq 2 \quad (1/6 \leq t \leq 5/6 \pmod{1}). \quad (29.5h)$$

For the last bound the denominator $|e(t) - 1|$ is at least one. By the $1/b$ -periodicity of $\sum_{j=0}^{b-1} f_b(x + j/b)$ we may put $x \in [0, 1/b)$. The grid has at most $b(1/b) + 1 = 2$ points in the two endpoint intervals of total length $1/b$, so their contribution is at most $2b$. On $[1/(2b), 1/6)$ the decreasing bound $1/(2t)$ in (29.5h) and the mesh $1/b$ give

$$\sum_{x+j/b \in [1/(2b), 1/6)} f_b(x + j/b) \leq b + \frac{b}{2} \int_{1/(2b)}^{1/6} \frac{dt}{t} = b + \frac{b}{2} (\log(2b) - \log 6).$$

The reflected interval $[5/6, 1 - 1/(2b))$ has the same bound. The remaining middle interval has at most $2b/3 + 1$ grid points, each contributing at most two. Adding the four regions and using $4/3 < \log 6$ yields

$$\sum_{j=0}^{b-1} f_b(x + j/b) \leq b(4 + \log(2b)) + 2. \quad (29.5i)$$

The missing-digit sum differs from the full sum by one unit complex number, hence the triangle inequality and (29.5i) imply

$$F_1(x) = \sum_{j=0}^{b-1} g(x + j/b) \leq \frac{b(5 + \log(2b)) + 2}{b-1}. \quad (29.5j)$$

This is strictly below $\sqrt{b}$ for $b \geq 111$. To check the last numerical threshold without an asymptotic convention, put $h(t) = \sqrt{t}(t-1) - t(5 + \log(2t)) - 2$. The elementary bounds

$\sqrt{111} > 10.535$ and $\log 222 < 5.403$ give $h(111) > 2.1$. Also

$$h'(t) = \frac{3}{2}\sqrt{t} - \frac{1}{2\sqrt{t}} - 6 - \log(2t), \quad h''(t) = \frac{3}{4\sqrt{t}} + \frac{1}{4t^{3/2}} - \frac{1}{t} > 0 \quad (t \geq 111),$$

and these same numerical bounds give $h'(111) > 4$. Thus $h(t) > 0$ for all $t \geq 111$. Apply Lemma IV.11.20 with $L = 1$. $\square$

The finitely many smaller bases require more than the coarse estimate (29.5j). The next lemma is computer-assisted; the strict inequalities and the complete reproducible interval-arithmetic program form its finite certificate.

Lemma IV.11.22 (*certified finite omitted-digit bound*).

If $4 \leq b \leq 110$ and $D = \{0, \dots, b-1\} \setminus \{a\}$, with the additional condition $a \in \{0, 3\}$ when $b = 4$, then $\hat{\kappa}_1(\nu_{b,D}) > 1/2$.

Proof. Reflection of digits sends a to $b-1-a$ and multiplies the Fourier polynomial by a unit complex number before conjugation. Its modulus g is unchanged. Thus it suffices to check $a = 0$ for $b = 4$ and $0 \leq a \leq \lfloor (b-1)/2 \rfloor$ otherwise.

For each pair use Lemma IV.11.20 with the following parameters. The last column is a rigorously rounded-down lower bound for the minimum, over the indicated pairs, of

$$\sqrt{b^L} - \pi\delta b^L(b^L - 1) - \max_{0 \leq k \leq \lfloor (2b^L\delta)^{-1} \rfloor} F_L(k\delta). \quad (29.5k)$$

base range	L	δ	lower margin
$b = 4, a = 0$	2	10^{-5}	0.098
$b = 5, 6$, except $(5, 1)$	2	10^{-5}	0.186
$(b, a) = (5, 1)$	4	$5 \cdot 10^{-7}$	0.186
$b = 7, 8$	2	10^{-5}	1.384
$9 \leq b \leq 110$	1	10^{-4}	0.236

The $(5, 1)$ row has an actual certified margin greater than 0.18659; it is displayed separately because its block length and mesh differ. The certificate covers all 3,075 reflection-reduced pairs, not a sampled subset.

Here are the verification rules, so the numerical output has a precise mathematical meaning. Every grid argument is stored as a rational number. For $t \notin \mathbb{Z}$ the program evaluates

$$g(t)^2 = \frac{R_b(t)^2 + 1 - 2R_b(t) \cos(\pi(b-1-2a)t)}{(b-1)^2}, \quad R_b(t) = \frac{\sin(\pi bt)}{\sin(\pi t)}, \quad (29.5l)$$

and uses $g(t) = 1$ when $t \in \mathbb{Z}$. MPFR directed rounding at 96 bits encloses π , rational multiplication/division, $\sin$, $\cos$, products, sums, and square roots. For a trigonometric argument interval $[u, v]$, the enclosure at u is widened by $v-u$, using the 1-Lipschitz property of sine and cosine; hence it remains valid even across an extremum. The program propagates upper bounds through g , S_L , and F_L , uses a lower bound for $\sqrt{b^L}$, and an upper bound for the Lipschitz correction in (29.5k). Thus a positive reported lower margin certifies the strict inequality (29.5g) without relying on ordinary floating-point tolerance.

The full source, input contract, case-by-case CSV, and validated run manifest are stored in the accompanying volume-IV/checks/cvy_fourier_certificate.cpp and volume-IV/checks/cvy-cert-2026-10-01/ files. The CSV has SHA-256 digest d314f765173cd3f7ddcc2b7b6dbfdb53ede0c888be3b7fed75c3e4f67512ee1b. The minimum certified margin is greater than 0.09825012101108735. Every row of the table is therefore positive, and the finite-level criterion proves the claim. $\square$

Theorem IV.11.23 (*improved standard-height rational counting*).

Let $K = K_{b,D}$ and $\kappa = \dim_{\mathbb{H}} K$. Assume either

$$b \geq 5 \text{ and } \#D = b - 1, \quad \text{or} \quad b = 4 \text{ and } D \in \{\{0, 1, 2\}, \{1, 2, 3\}\}.$$

Then there is an effectively computable $\rho_0 > 0$ such that

$$N_K(T) \ll T^{2\kappa - \rho_0}. \quad (29.1)$$

This is Chow–Varjú–Yu [11, Theorem 1.2]; the proof here uses a separately audited finite certificate for its numerical cases.

Proof. The natural measure is κ -Ahlfors regular by Lemma IV.11.3. Lemma IV.11.21 gives $\hat{\kappa}_1(\nu) > 1/2$ for $b \geq 111$, and Lemma IV.11.22 gives it for all the required smaller bases. Choose any explicitly computable $v \in (1/2, \hat{\kappa}_1(\nu))$ using the strict margin in (29.5g) or (29.5j) and the transfer estimate (29.5f). Theorem IV.11.19 then proves (29.1) with, for example,

$$\rho_0 = \frac{(1 - \kappa)(2v - 1)}{1 - v} > 0.$$

All quantities in this choice arise from the displayed formulas and the finite certified grid, so ρ_0 is effective. $\square$

11.5 From rational counting to intrinsic approximation

Theorem IV.11.24 (*rational counting implies an intrinsic improvement*).

Assume the hypotheses and conclusion of the improved rational-counting theorem above. Then there is an effectively computable $\rho > 0$ such that

$$\nu(W_K(2 - \rho)) = 0. \quad (29.6)$$

More generally, for every $\alpha > 0$,

$$\dim_{\mathbb{H}} W_K(\alpha) \leq \frac{2\kappa - \rho_0}{\alpha}. \quad (29.7)$$

Proof. Apply Lemma IV.11.14 with $\beta = 2\kappa - \rho_0$. Its threshold is

$$\alpha > \frac{\beta}{\kappa} = 2 - \frac{\rho_0}{\kappa}. \quad (29.8)$$

Choose, for example,

$$\rho = \min \left\{ \frac{1}{2}, \frac{\rho_0}{2\kappa} \right\} > 0.$$

Then $\alpha = 2 - \rho$ is still strictly larger than the right side of (29.8), so (29.6) follows. Formula (29.7) is Lemma IV.11.15. All constants remain effective once ρ_0 is. $\square$

Corollary IV.11.25 (*intrinsic natural-measure improvement*).

Under exactly the digit hypotheses of Theorem IV.11.23, there is an effectively computable $\rho_1 > 0$ such that

$$\nu(W_K(2 - \rho_1)) = 0. \quad (29.2)$$

This is Chow–Varjú–Yu [11, Theorem 1.4].

Proof. Use the effective exponent ρ_0 from Theorem IV.11.23 in Theorem IV.11.24, and take $\rho_1 = \min\{1/2, \rho_0/(2\kappa)\}$. $\square$

The extrinsic problem needs a different, genuinely lower-bound input. We retain the following theorem as a clearly marked source result, not as a consequence of the preceding counting argument.

Source theorem: Extrinsic dimension formula near exponent two**Chow–Varjú–Yu, Theorem 1.8**

Assume $b \geq 7$ and $\#D = b - 1$, and put $\kappa = \dim_H K$. Then there is an effectively computable $\rho_2 > 0$ such that, for every $\alpha \in [2, 2 + \rho_2)$,

$$\dim_H(W_{\mathbb{R}}(\alpha) \cap K) = \frac{2}{\alpha} + \kappa - 1. \quad (29.3)$$

In particular, if

$$\text{VWA} = \bigcup_{\alpha > 2} W_{\mathbb{R}}(\alpha),$$

then

$$\dim_H(\text{VWA} \cap K) = \kappa. \quad (29.4)$$

Indeed, the upper bound follows from $\text{VWA} \cap K \subset K$; for $\alpha_j \downarrow 2$ with $2 < \alpha_j < 2 + \rho_2$, equation (29.3) gives a lower bound tending to κ .

Source boundary

Theorem 1.8 is cited for illustration and is not used in a later proof. The middle-third Cantor set $K_{3,\{0,2\}}$ is not covered by the preceding results. The $b = 4$ exceptional digit sets occur in Theorems IV.11.23 and IV.11.25 but not in this extrinsic theorem; the difference in hypotheses is essential.

11.6 Mass transference and its additional hypothesis

Definition IV.11.26 (*dimension functions and Hausdorff gauges*).

A *dimension function* is a continuous nondecreasing function $f : [0, \infty) \rightarrow [0, \infty)$ with $f(0) = 0$ and $f(r) > 0$ for all sufficiently small $r > 0$. It is *doubling* if there are $C \geq 1$ and $r_0 > 0$ such that $f(2r) \leq Cf(r)$ for $0 < r < r_0/2$. For $E \subset X$ put

$$\mathcal{H}_\delta^f(E) = \inf \left\{ \sum_j f(r_j) : E \subset \bigcup_j B(x_j, r_j), \quad 0 < r_j \leq \delta \right\}, \quad \mathcal{H}^f(E) = \lim_{\delta \downarrow 0} \mathcal{H}_\delta^f(E). \quad (29.8a)$$

Here the centers x_j belong to X , and balls are taken in the metric space X . Multiplying radii by a fixed constant changes the displayed gauge only by a controlled constant when f is doubling. The notation $g^{-1}(u)$ below means the lower generalized inverse $\inf\{r > 0 : g(r) \geq u\}$; for sufficiently small $u > 0$, continuity gives $g(g^{-1}(u)) = u$, whether or not g is strictly increasing. In the notation B^g we retain the original ball B , as in the cited theorem: if g has a flat interval, applying the lower generalized inverse to $g(r(B))$ would otherwise shrink its radius and would not be the identity operation.

Theorem IV.11.27 (*General Mass Transference Principle*).

This is Beresnevich–Velani, “A Mass Transference Principle and the Duffin–Schaeffer conjecture for Hausdorff measures”, *Annals of Mathematics* 164 (2006), Theorem 3, p. 991. We give a full proof after three preparatory results.

Let (X, d) be locally compact. Let g be a doubling dimension function and assume there are constants $0 < c_1 < 1 < c_2 < \infty$ and $r_0 > 0$ such that

$$c_1 g(r(B)) \leq \mathcal{H}^g(B) \leq c_2 g(r(B)) \quad (29.9)$$

for every ball B centered in X with radius at most r_0 . For a dimension function f and sufficiently small $B = B(x, r)$ define

$$B^f = \begin{cases} B(x, g^{-1}(f(r))), & f \neq g, \\ B, & f = g, \end{cases} \quad (29.10)$$

Only the tail of (B_i) enters either limsup, so no convention for large radii outside the range of g is needed. Let (B_i) be balls in X with $r(B_i) \rightarrow 0$ and suppose f/g is monotone. If for every ball $B \subset X$,

$$\mathcal{H}^g(B \cap \limsup_i B_i^f) = \mathcal{H}^g(B), \quad (29.11)$$

then

$$\mathcal{H}^f(B \cap \limsup_i B_i^g) = \mathcal{H}^f(B). \quad (29.12)$$

We begin with the covering and shrinking preliminaries and the finite gauge-ratio case.

The first step in the Cantor construction turns full measure of the enlarged limsup into *finite* disjoint families at arbitrarily late indices. Its uniform lower bound will be used at every sublevel of the construction.

Lemma IV.11.28 (*late disjoint enlarged-ball extraction*).

Assume the metric and gauge hypotheses (29.9) and the full-measure hypothesis (29.11). Let $B = B(x, R)$ have $R < r_0/10$, and let $N \geq 1$. There are a constant $c > 0$, depending only on the constants in (29.9) and the doubling constant of g , and a finite set $J \subset \{i \geq N\}$ such that the balls $\{B_i^f : i \in J\}$ are pairwise disjoint, each lies in B , and

$$\mathcal{H}^g\left(\bigcup_{i \in J} B_i^f\right) \geq c \mathcal{H}^g(B). \quad (29.12a)$$

The indices may also be required to exceed any prescribed larger cutoff.

Proof. Write $\lambda = \mathcal{H}^g$ and $B' = B(x, R/2)$. By (29.9) and doubling, $\lambda(B') \geq c_0 \lambda(B)$ for a constant $c_0 > 0$. Since $r(B_i) \rightarrow 0$ and $f(r) \rightarrow 0$ as $r \downarrow 0$, the generalized-inverse radii $r(B_i^f)$ also tend to zero. Enlarge N if necessary so that every B_i^f with $i \geq N$ which meets B' is contained in B , and both $r(B_i^f)$ and $5r(B_i^f)$ are below r_0 .

By (29.11), λ -almost every point of B' belongs to infinitely many B_i^f . Thus the family of eligible enlarged balls meeting B' covers a set of λ -measure $\lambda(B')$. Apply the $5r$ covering lemma to this family: it gives a countable pairwise disjoint subfamily $\{L_j\}$ such that the eligible union is contained in $\bigcup_j 5L_j$. Therefore

$$\lambda(B') \leq \sum_j \lambda(5L_j) \leq C_1 \sum_j \lambda(L_j), \quad (29.12b)$$

where the last inequality follows from (29.9) and the doubling property of g . The L_j lie in B and are disjoint, so their λ -masses sum to at most $\lambda(B) < \infty$. Choose a finite initial subfamily whose total mass is at least half the lower bound in (29.12b). Its original indices form J , and (29.12a) follows with $c = c_0/(2C_1)$. Starting with a larger cutoff proves the last assertion. $\square$

Lemma IV.11.29 (*full-measure limsup survives substantial shrinking*).

Let λ be a locally finite Radon measure on a metric space, uniformly doubling at small radii: for some $D < \infty$ and $r_* > 0$, $\lambda(5B) \leq D\lambda(B)$ whenever $r(B) < r_*$. Suppose $B_i \subset C_i$ are concentric balls, $r(C_i) \rightarrow 0$, and there is $\theta > 0$ such that $\lambda(B_i) \geq \theta\lambda(C_i)$ for all sufficiently large i . If $\limsup_i C_i$ has full λ -measure in every ball, then $\limsup_i B_i$ also has full λ -measure in every ball.

Proof. We use the density-point property of a doubling measure: for every measurable E , almost every $x \in E$ satisfies

$$\frac{\lambda(B(x, r) \setminus E)}{\lambda(B(x, r))} \longrightarrow 0 \quad (r \downarrow 0). \quad (29.12c)$$

For clarity, this property follows from the $5r$ covering lemma and regularity. On a bounded region choose compact $K \subset E \subset U$ with U open and both $\lambda(E \setminus K)$ and $\lambda(U \setminus E)$ arbitrarily small. Fix $\varepsilon > 0$. For all sufficiently large m , every ball centered on K with radius below $1/m$ lies in U . Let $A_{\varepsilon, m}$ consist of those $x \in K$ for which some such ball $B(x, r)$ satisfies

$\lambda(B(x, r) \setminus E) > \varepsilon \lambda(B(x, r))$. The $5r$ lemma selects disjoint bad balls L_j whose fivefold dilates cover $A_{\varepsilon, m}$. If D is a doubling constant for fivefold dilation, then

$$\lambda^*(A_{\varepsilon, m}) \leq \sum_j \lambda(5L_j) \leq D \sum_j \lambda(L_j) \leq D\varepsilon^{-1} \sum_j \lambda(L_j \setminus E) \leq D\varepsilon^{-1} \lambda(U \setminus E).$$

The sets $A_{\varepsilon, m}$ decrease with m . Let $m \rightarrow \infty$, then let both regularity errors tend to zero, and finally take a countable sequence $\varepsilon \downarrow 0$. This proves (29.12c).

Suppose instead that $E = X \setminus \limsup_i B_i$ has positive measure in some bounded ball. The sets

$$E_N = X \setminus \bigcup_{i \geq N} B_i$$

increase to E , so some E_N has positive measure there. Choose a λ -density point $x \in E_N$ that also belongs to $\limsup_i C_i$. Infinitely many C_i with $i \geq N$ contain x . For each of them, $B_i \cap E_N = \emptyset$, and $C_i \subset B(x, 2r(C_i)) \subset 3C_i$. Doubling gives a constant C such that

$$\frac{\lambda(B(x, 2r(C_i)) \setminus E_N)}{\lambda(B(x, 2r(C_i)))} \geq \frac{\lambda(B_i)}{\lambda(3C_i)} \geq \frac{\theta}{C} > 0.$$

This contradicts (29.12c) because $r(C_i) \rightarrow 0$. □

Proposition IV.11.30 (*finite gauge-ratio cases of mass transference*).

Under the hypotheses of the General Mass Transference Principle, suppose in addition that the limit $c = \lim_{r \downarrow 0} f(r)/g(r)$ is finite. Then its conclusion (29.12) holds without the multilevel Cantor construction.

Proof. The full-measure hypothesis also makes X separable. Every sufficiently small ball has positive finite $\mathcal{H}^g$ -measure by (29.9), so it meets $\limsup_i B_i^f$. As $r(B_i^f) \rightarrow 0$, the countable set of centers of the B_i^f is dense in X . Hence X has a countable base, and a locally null conclusion extends to every ball by a countable small-ball cover.

If $c = 0$, any sufficiently fine ball cover has $\sum f(r_j) \leq \varepsilon \sum g(r_j)$; hence $\mathcal{H}^f(B) = 0$ for every small ball B by (29.9). The countable-cover observation gives $\mathcal{H}^f(B) = 0$ for every ball B , so the conclusion is automatic.

Suppose $0 < c < \infty$. Comparison of fine covers gives $\mathcal{H}^f = c\mathcal{H}^g$ on every Borel set. Let C_i be the larger of the two concentric balls B_i and B_i^f . Its radius tends to zero, and $\limsup_i C_i$ has full $\mathcal{H}^g$ -measure because it contains $\limsup_i B_i^f$. By (29.9) and $g(r(B_i^f)) = f(r(B_i))$, for all large i ,

$$\mathcal{H}^g(C_i) \leq c_2 \max\{g(r(B_i)), f(r(B_i))\} \leq C \mathcal{H}^g(B_i),$$

where C depends only on c, c_1, c_2 . Apply Lemma IV.11.29 with $\lambda = \mathcal{H}^g$ to conclude that $\limsup_i B_i$ is $\mathcal{H}^g$ -full in every small ball and hence, by the countable cover, in every ball. Since $\mathcal{H}^f = c\mathcal{H}^g$, it is also $\mathcal{H}^f$ -full. □

Proof. We prove Theorem IV.11.27. Put $\lambda = \mathcal{H}^g$ and $h = f/g$. Monotonicity gives a limit $\lim_{r \downarrow 0} h(r) \in [0, \infty]$. Proposition IV.11.30 settles the finite-limit case. Henceforth suppose

$$h(r) \longrightarrow \infty \quad (r \downarrow 0). \tag{29.12d}$$

In particular, after restricting all radii to a fixed small interval, $h(r) \geq 1$ and h is nonincreasing as r increases. Write $tL = B(x_L, tr_L)$ for the concentric t -dilate of a ball $L = B(x_L, r_L)$. Doubling of g and (29.9) give uniform constants $b_0, A_0, C_4, D_5 > 0$ such that, for every sufficiently small ball $P = B(x, R)$,

$$\lambda(B(x, R/2)) \geq b_0 g(R), \quad \sum_{L \in \mathcal{L}} f(r_L) \leq A_0 g(R) \quad (29.12e)$$

whenever $\{L^f : L \in \mathcal{L}\}$ is a disjoint family inside P ; also $\lambda(4L) \leq C_4 g(r_L)$ and $\lambda(5A) \leq D_5 \lambda(A)$ at the radii used below. For example, the second inequality in (29.12e) follows by summing $c_1 f(r_L) \leq \lambda(L^f)$ over disjoint enlarged balls and using $\lambda(P) \leq c_2 g(R)$. All these constants depend only on the data in (29.9) and the doubling constant of g .

We first construct a finite family of children inside a small, relatively compact parent $P = B(x, R)$. Fix a number $T_P > 0$ and put

$$m_P = \max \left\{ 2, \left\lceil \frac{T_P}{a_0 g(R)} \right\rceil \right\}, \quad (29.12f)$$

where the uniform constant $a_0 > 0$ will be chosen below. We produce m_P consecutive *sublevels* $\mathcal{K}(P, j)$, each a finite family of original balls B_i , with the following properties:

$$a_0 g(R) \leq \sum_{L \in \mathcal{K}(P, j)} f(r_L) \leq A_0 g(R), \quad \{L^f : L \in \mathcal{K}(P, j)\} \text{ is disjoint inside } P; \quad (29.12g)$$

the balls $3L$ are pairwise disjoint across *all* sublevels; and

$$\max_{L \in \mathcal{K}(P, j+1)} f(r_L) \leq \frac{1}{2} \min_{L \in \mathcal{K}(P, j)} f(r_L). \quad (29.12h)$$

Every index used can exceed a prescribed cutoff, every child radius can be made smaller than a prescribed positive bound, and the cutoff may increase from one sublevel to the next. We also impose

$$3L \subset L^f, \quad g(r_L) \leq \varepsilon_P f(r_L), \quad \varepsilon_P \leq \frac{b_0}{4C_4 A_0 m_P}. \quad (29.12i)$$

All these restrictions hold for sufficiently late indices by (29.12d), $r(B_i) \rightarrow 0$, and $f(r(B_i)) \rightarrow 0$; in particular $g^{-1}(f(r_L)) \geq 3r_L$ follows from doubling of g and $h(r_L) \rightarrow \infty$.

For the first sublevel apply Lemma IV.11.28 to P at such a late cutoff. Its lower mass bound and $\lambda(L^f) \leq c_2 f(r_L)$ give the lower inequality in (29.12g) with a uniform positive constant. The upper inequality follows from (29.12e). Suppose the first $j - 1$ sublevels have been selected, with $j \leq m_P$, and let $\mathcal{O}$ denote their union. Then

$$\lambda \left(\bigcup_{L \in \mathcal{O}} 4L \right) \leq C_4 \varepsilon_P \sum_{L \in \mathcal{O}} f(r_L) \leq C_4 \varepsilon_P m_P A_0 g(R) \leq \frac{1}{4} \lambda(B(x, R/2)). \quad (29.12j)$$

Consequently $F = B(x, R/2) \setminus \bigcup_{L \in \mathcal{O}} 4L$ has measure at least $3\lambda(B(x, R/2))/4$. Let $d = \min\{r_L : L \in \mathcal{O}\} > 0$; the earlier cutoff made $d < R/100$. The equal-radius auxiliary balls

$A_z = B(z, d/10)$, $z \in F$, lie in P and miss every old $3L$. The $5r$ covering lemma yields pairwise disjoint auxiliary balls $A_1, \dots, A_q$ for which

$$\sum_{t=1}^q \lambda(A_t) \geq D_5^{-1} \lambda(F) \geq \frac{3b_0}{4D_5} g(R). \quad (29.12k)$$

They can be chosen finite: the closure of P is totally bounded, so a maximal disjoint equal-radius family is finite and its fivefold dilates cover F . Apply Lemma IV.11.28 separately in each A_t , with one sufficiently late cutoff satisfying (29.12h)–(29.12i) and all preceding restrictions. The union of its selected original balls is $\mathcal{K}(P, j)$. Their enlarged balls are disjoint inside their respective disjoint A_t , and their $3L$ miss all old $3L$. By (29.12k) and the lower bound of Lemma IV.11.28, its total f -weight is at least $a'_0 g(R)$ for a uniform $a'_0 > 0$; the upper bound is again (29.12e). Taking a_0 no larger than the first-sublevel and subsequent-sublevel lower constants proves (29.12g)–(29.12i). The construction is finite because m_P is finite.

Now fix an arbitrary target ball $U \subset X$. Local compactness provides a small ball P_0 with $\overline{P_0}$ compact and $\overline{P_0} \subset U$. For every nonroot parent P , let r_P^{orig} denote the radius of the original ball whose contraction is P . Fix $\eta > 0$. Recursively replace each parent P by all the children in its m_P sublevels, taking

$$T_{P_0} = \eta, \quad T_P = f(r_P^{\text{orig}}) \quad (P \neq P_0), \quad (29.12l)$$

At generation n require every child radius to be less than 2^{-n} and every child index to exceed all indices used in earlier generations. Each generation is finite, so these requirements are compatible with the preceding construction. A node labelled by an original child $L = B_i$ has as its *parent ball for the next generation* the concentric contraction $L^- = B(x_i, r_i/2)$. The next children are extracted inside L^- . Since $\overline{L^-} \subset L$, the nested finite unions of the compact sets $\overline{L^-}$ have a nonempty compact intersection

$$K_\eta \subset \overline{P_0} \cap \limsup_i B_i.$$

Indeed every path point lies in each ancestral original ball, and the indices along a path tend to infinity. We write P for a contracted parent ball and $r_P^{\text{orig}} = 2r_P$ for its ancestral original radius. All sublevel estimates are those of the construction inside P and use the original radii r_L of its children. This contraction changes neither the uniform constant a_0 nor the lower bound $m_P a_0 g(r_P) \geq T_P$.

Assign mass $\nu(P_0) = 1$ and, for every child L of P , set

$$\nu(L) = \nu(P) \frac{f(r_L)}{\sum_{M \text{ child of } P} f(r_M)}. \quad (29.12m)$$

This gives a Borel probability measure on the compact path limit. Indeed the sibling compact sets are disjoint, their diameters tend to zero, and the consistent cylinder masses in (29.12m) define the push-forward of the probability measure on the finite-branching path space. By (29.12f)–(29.12g), the denominator is at least $m_P a_0 g(r_P) \geq T_P$ (for contracted parents, use the harmless adjusted a_0 just noted). Induction therefore gives

$$\nu(L) \leq \eta^{-1} f(r_L) \quad \text{for every nonroot construction ball } L. \quad (29.12n)$$

We verify the bound for an *arbitrary* test ball $A = B(x, r)$; this is the step that prevents a cylinder-only estimate from being a mass-distribution proof. First note the elementary geometric fact: if $A \cap L \neq \emptyset$ and $A \setminus cL \neq \emptyset$ for $c \geq 3$, then $r_L \leq r$ and $cL \subset 5A$ when cL denotes a concentric ball of radius cr_L and $c = 3$. More generally, if $L \subset C$ are concentric, $r_C \geq 3r_L$, and $A \setminus C \neq \emptyset$, then $r_L \leq r$ and $C \subset 5A$: for $z \in A \cap L$ and $w \in A \setminus C$, $r_C - r_L \leq d(z, w) \leq 2r$, so $r_L \leq r$ and $r_C \leq 3r$; the triangle inequality gives $C \subset 5A$.

Choose r small enough that $5r < r_0$ and that r is less than every radius of the finitely many first-generation original balls. If $\nu(A) = 0$ there is nothing to prove. Otherwise A must meet at least two original construction balls at some generation: if it met at most one at every generation, its mass would be bounded by $\eta^{-1}f(r_L) \rightarrow 0$ along the unique chain, by (29.12n). Let n be the first generation with two such balls. Since the $3L$ of the first generation are disjoint, the choice of r forces $n \geq 2$. At generation $n - 1$, A meets one parent P only. If $r \geq r_P$, then $\nu(A) \leq \nu(P) \leq \eta^{-1}f(r_P^{\text{orig}})$; this is at most $C\eta^{-1}f(r)$ if $r \geq r_P$, because $r_P^{\text{orig}} = 2r_P \leq 2r$ and, in the present case, f is locally doubling: $f(2r) = h(2r)g(2r) \leq C_g g(r)h(r) = C_g f(r)$. Suppose $r < r_P$. By (29.12n),

$$\nu(A) \leq \eta^{-1} \sum_{L \text{ child of } P: L \cap A \neq \emptyset} f(r_L). \quad (29.12o)$$

The geometric facts above apply also when the parents are contracted, because the original children L retain the disjoint $3L$ and sublevel-wise disjoint L^f properties.

For a sublevel containing at least two children that meet A , the enlarged ball L^f of each such child misses a point of A lying in another child of that sublevel. Hence $L^f \subset 5A$. The enlarged balls in this sublevel are disjoint, and (29.9) gives

$$\sum_{L \in \mathcal{K}(P, j): L \cap A \neq \emptyset} f(r_L) \leq c_1^{-1} \lambda(5A) \leq C_g g(r). \quad (29.12p)$$

There are at most m_P such sublevels. Because P is nonroot, $T_P = f(r_P^{\text{orig}})$, where r_P^{orig} is the radius of its original construction ball; since $r_P = r_P^{\text{orig}}/2$, doubling of g and monotonicity of f imply $m_P g(r) \leq Ch(r)g(r) = Cf(r)$ when $r < r_P$. Explicitly, $m_P \leq 2 + f(r_P^{\text{orig}})/(a_0 g(r_P))$; $g(r_P^{\text{orig}}) \leq Cg(r_P)$ and $h(r_P^{\text{orig}}) \leq h(r)$, so the claimed bound follows.

In each remaining sublevel at most one child meets A . Choose one child L_* in the earliest nonempty sublevel. Some other child meets A at generation n , so $A \setminus 3L_* \neq \emptyset$ by the global disjointness of the $3L$. The geometric fact gives $r_{L_*} \leq r$ and hence $f(r_{L_*}) \leq f(r)$. By (29.12h), the weight of any child in the k -th later sublevel is at most $2^{-k}f(r_{L_*})$. Consequently all one-child sublevels together contribute at most $2f(r)$. Combining this with (29.12o)–(29.12p) proves

$$\nu(B(x, r)) \leq C_* \eta^{-1} f(r) \quad (0 < r < r_*(\eta)), \quad (29.12q)$$

where C_* is independent of η .

For completeness, if a probability measure supported on E obeys $\nu(B(x, r)) \leq C_* \eta^{-1} f(r)$ for all sufficiently small balls, then every sufficiently fine ball cover of E has total f -cost at least η/C_* : sum the displayed estimate over the cover and use $\nu(E) = 1$. Thus $\mathcal{H}^f(K_\eta) \geq$

η/C_* . As $\eta > 0$ is arbitrary, $\mathcal{H}^f(U \cap \limsup_i B_i) = \infty$. Finally $\mathcal{H}^f(U) = \infty$ too: for each $M > 0$, sufficiently small radii satisfy $f(r) \geq Mg(r)$ by (29.12d), while $\mathcal{H}^g(U) > 0$ by (29.9). Since U was arbitrary, (29.12) follows. $\square$

The following proposition isolates the full-measure hypothesis that a counting estimate alone does not provide.

Proposition IV.11.31 (*checked Ahlfors-regular bridge*).

Let $K = K_{b,D}$ and $\kappa = \dim_H K$. Let (B_i) be balls in the metric space K , with centers $x_i \in K$ and radii $r_i \rightarrow 0$. Fix $0 < s < \kappa$. Assume the full-measure hypothesis

$$\mathcal{H}^\kappa\left(B \cap \limsup_i B(x_i, r_i^{s/\kappa})\right) = \mathcal{H}^\kappa(B) \quad (29.13)$$

for every sufficiently small ball $B \subset K$. Then

$$\mathcal{H}^s\left(B \cap \limsup_i B_i\right) = \mathcal{H}^s(B) \quad (29.14)$$

for every ball $B \subset K$, and in particular

$$\dim_H(\limsup_i B_i) \geq s. \quad (29.15)$$

Proof. Apply the mass-transference principle with

$$X = K, \quad g(r) = r^\kappa, \quad f(r) = r^s.$$

The Ahlfors-regularity lemma gives (29.9). Moreover $g^{-1}(f(r)) = r^{s/\kappa}$, so B_i^f is exactly the enlarged ball in (29.13), while $B_i^g = B_i$. Hence the mass-transference principle gives (29.14). Since $s < \kappa$, every nonempty K -ball has infinite $\mathcal{H}^s$ measure, so (29.15) follows. $\square$

If B_i has radius $q_i^{-\alpha}$, the enlarged radius in (29.13) is

$$q_i^{-\alpha s/\kappa}. \quad (29.16)$$

Thus a desired lower bound s for an α -approximation limsup requires a full κ -measure theorem at the looser exponent $\alpha s/\kappa$. An upper count such as (29.1) supplies neither (29.13) nor any substitute for it.

Source boundary

Logical boundary: an upper counting estimate gives convergence and dimension upper bounds; a full-measure enlarged-limsup hypothesis together with the mass-transference principle gives Hausdorff lower bounds. No implication from the first assertion to the second is claimed.

11.7 Intrinsic versus extrinsic conclusions

Intrinsic centers. The intrinsic-counting source theorem, combined with the local counting bridge, gives the natural-measure zero statement above at an exponent strictly below 2. The standard height H is the reduced denominator; the easier coding height H_{int} is not substituted.

Extrinsic centers. The quoted Fourier–arithmetic theorem for extrinsic centers gives the sharp branch

$$\frac{2}{\alpha} + \kappa - 1$$

of the predicted dimension formula for α sufficiently close to 2, for one-missing-digit systems with $b \geq 7$. This conclusion is not obtained from the intrinsic counting theorem or from mass transference alone.

Mass transference. The mass-transference principle is an implication from a full-measure enlarged-limsup hypothesis to a Hausdorff-measure conclusion. The proposition above records exactly what must be verified before it may be used on K .

Chapter IV.12

Synthesis and Frontiers

What the rigidity mechanisms prove, what the models reveal, and where the boundaries remain

The preceding chapters developed four recurring mechanisms: scenery turns local geometry into dynamics; arithmetic independence forces orbits to spread; renormalization converts a hypothetical embedding into a family of normalized copies; and counting estimates convert arithmetic information into metric consequences. This final chapter places those mechanisms side by side. The examples are elementary enough that both the conclusion and its failure modes remain visible.

For $0 < r < 1/2$, let C_r denote the central two-map Cantor set generated by

$$x \mapsto rx, \quad x \mapsto rx + (1 - r).$$

Thus $C_{1/3}$ is the middle-third Cantor set, while $C_{1/4}$ is its central one-fourth analogue.

12.1 Intersections: sharpness and hidden structure

The invariant-set theorem of Chapter IV.7 bounds affine intersections using only the two dimensions. The next example shows that its numerical constant is already sharp, even though genuinely nontrivial equality cases may require substantially more structure.

Theorem IV.12.1 (*an exact equality model for the intersection bound*).

Let

$$A = [0, 1], \quad B = C_{1/3}.$$

Regard A as a closed T_2 -invariant set and B as a closed T_3 -invariant set, where $T_m(x) = mx \pmod{1}$. Then $A \cap B = B$, and

$$\overline{\dim}_B(A \cap B) = \frac{\log 2}{\log 3} = \max\{0, \dim_H A + \dim_H B - 1\}. \quad (30.1)$$

Proof. The middle-third Cantor set satisfies

$$\dim_H B = \overline{\dim}_B B = \frac{\log 2}{\log 3},$$

while $\dim_H A = 1$. Hence the right-hand side equals $\log 2 / \log 3$, which is also the left-hand side because $A \cap B = B$. □

The simplicity of the example is its point: the bound cannot be improved from the two dimensions and multiplicative independence alone. At finite level the pair $C_{1/3}, C_{1/4}$ looks much less transparent. If depth n_3 in base 3 is paired with

$$n_4 = \text{round}\left(n_3 \frac{\log 3}{\log 4}\right),$$

exact endpoint arithmetic gives respectively 6, 8, 12, 16, 18, 20, 30, 36, 50 intersecting interval pairs for $n_3 = 4, \dots, 12$. These counts illustrate finite-scale complexity, but they neither compute $\dim_{\text{H}}(C_{1/3} \cap C_{1/4})$ nor imply convergence of a finite-scale exponent.

Frontier question. For genuinely nontrivial invariant pairs, when is the Furstenberg intersection bound an equality, and when does additional arithmetic or geometric structure force a strict drop?

12.2 Embeddings: global, positive-measure, and tangential

Chapter IV.8 separated three notions that are often blurred together. The following examples show that the distinctions are real rather than merely terminological.

Theorem IV.12.2 (*three embedding diagnostics*).

With the embedding conventions of Chapter IV.8, the following hold.

1. The set $C_{1/3}$ embeds into itself globally, on a set of positive source measure, and as a local tangent copy.
2. If F is a Smith–Volterra compact set of positive Lebesgue measure and empty interior, then $([0, 1], \mathcal{L}^1)$ partially embeds into F on a set of positive source measure, but no nonsingular affine image of $[0, 1]$ is contained in F .
3. There is a compact countable set F having $[0, 1]$ as a microset, but admitting neither a positive-Lebesgue-measure partial embedding of $[0, 1]$ nor a global affine interval inclusion.

Proof. For the first claim, $x \mapsto x/3$ maps $C_{1/3}$ into itself. Taking the whole source set gives the positive-measure version for the Cantor measure, and self-similar cylinder zooms give $C_{1/3}$ itself as a microset.

For the second, use the identity map on the positive-measure subset F of $[0, 1]$. A nonsingular affine image of $[0, 1]$ is a nondegenerate interval; if one lay in F , then F would have nonempty interior.

For the third, put, for $n \geq 2$,

$$c_n = 2^{-n}, \quad s_n = 2^{-n^3}, \quad G_n = \{j/n : 0 \leq j \leq n\},$$

and

$$F = \{0\} \cup \bigcup_{n \geq 2} (c_n + s_n G_n). \quad (30.2)$$

The set F is compact and countable. Magnification about c_n by s_n^{-1} sends the n th cluster to G_n . Every other cluster leaves each fixed bounded window because the dyadic separation of the centres dominates the superexponential scale s_n . Since $G_n \rightarrow [0, 1]$ in Hausdorff distance, $[0, 1]$ is a microset of F . A nonsingular affine image of any positive-Lebesgue-measure subset of $[0, 1]$ is uncountable, and a global affine image of $[0, 1]$ is an interval; neither can lie in the countable set F . $\square$

Here $d_H(G_n, [0, 1]) = 1/(2n)$, but this visible convergence is only one part of the microset proof: the escape of the foreign clusters is equally essential.

Frontier question. Which structural hypotheses allow a local tangent embedding to be upgraded to a positive-measure or global embedding? The countable-cluster example shows that no such upgrade holds for arbitrary compact sets.

12.3 Normality: independence and exact resonance

The same Cantor measure gives both a clean independence case and a clean resonant obstruction. This places the arithmetic hypothesis in the normality theorem in sharp relief.

Theorem IV.12.3 (*base two versus base three for the Cantor measure*).

Let μ be the $(1/2, 1/2)$ Bernoulli measure on $C_{1/3}$.

1. The measure μ is pointwise 2-normal.
2. The measure μ is not pointwise 3-normal. Indeed, every point whose ternary digits belong to $\{0, 2\}$ has a T_3 -orbit that never visits $(1/3, 2/3)$.

Proof. The middle-third IFS is regular $C^{1+\varepsilon}$, its generators have contraction ratio $1/3$, and μ is a non-atomic quasi-product measure. The integer 2 is Pisot, and

$$\frac{\log(1/3)}{\log 2} \notin \mathbb{Q}, \quad (30.3)$$

since a rational relation would give $3^q = 2^p$ for positive integers p, q . The regular-IFS normality theorem of Chapter IV.10 therefore gives pointwise 2-normality.

Every ternary tail using only 0 and 2 belongs to $C_{1/3}$, which misses the open interval $(1/3, 2/3)$. Multiplication by 3 modulo one shifts such a tail, except that a tail of all 2's represents 1 and maps to 0; that point also misses the open interval. Thus the entire T_3 -orbit avoids $(1/3, 2/3)$. The interval has Lebesgue measure $1/3$ and boundary of Lebesgue measure zero, so the orbit cannot be equidistributed. The openness is necessary: $1/3 = 0.02222 \dots_3$ is itself a Cantor point. $\square$

Finite orbit histograms can display the contrast, but cannot prove the almost-everywhere assertion: every rational truncation of a Cantor coding has a finite orbit. The base-3 obstruction, in contrast, is exact and symbolic.

Frontier question. How far can the normality theorem be extended beyond Pisot bases, regular separated IFSs, and the stated contraction arithmetic? The resonant

base-3 example shows that some exclusion is unavoidable.

12.4 Arithmetic sums: full dimension is not interior

The final comparison separates two conclusions that are frequently, and incorrectly, treated as interchangeable.

Theorem IV.12.4 (*two arithmetic-sum models*).

The non-resonant sum satisfies

$$\dim_H(C_{1/3} + C_{1/4}) = 1, \quad (30.4)$$

whereas the resonant sum is exactly

$$C_{1/3} + C_{1/3} = [0, 2]. \quad (30.5)$$

The theorem used for (30.4) yields no interior conclusion.

Proof. The ratios $1/3$ and $1/4$ have incommensurable logarithms: a rational relation would imply $3^q = 4^p$. The non-resonant sum theorem of Chapter IV.9 gives

$$\dim_H(C_{1/3} + C_{1/4}) = \min\{\dim_H C_{1/3} + \dim_H C_{1/4}, 1\} = 1,$$

because $\log 2 / \log 3 > 1/2$ and $\log 2 / \log 4 = 1/2$.

For the resonant sum $S = C_{1/3} + C_{1/3}$, the Cantor equation shows that S is the compact attractor of

$$h_0(x) = x/3, \quad h_1(x) = x/3 + 2/3, \quad h_2(x) = x/3 + 4/3.$$

These maps send $[0, 2]$ onto three intervals that tile $[0, 2]$. Uniqueness of the compact attractor gives $S = [0, 2]$. $\square$

At the first six equal-depth approximations, the sums $C_{1/3}^{(n)} + C_{1/3}^{(n)}$ have one component, while the corresponding $C_{1/3}^{(n)} + C_{1/4}^{(n)}$ approximations have respectively 3, 7, 7, 7, 79, 261 components. These finite counts display gap structure but do not decide whether the limiting non-resonant sum has interior. Full Hausdorff dimension alone implies neither positive Lebesgue measure nor interior.

Frontier question. Does the particular full-dimensional sum $C_{1/3} + C_{1/4}$ contain an interval? Answering this requires a theorem that controls interior, not merely dimension.

Taken together, the four examples identify a common lesson. Dimension is a powerful coarse invariant, but rigidity enters through the mechanism that produces it: independence of scaling flows, recurrence of renormalized copies, phase spreading, or arithmetic counting. The unresolved frontiers ask how much of that mechanism can be recovered from weaker hypotheses.

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